

Spectral Theory, Chapters 3–5

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INTRODUCTION

In Chapters III to V of this Book, we consider in detail the algebra of endomorphisms of a real or complex topological vector space. In the case of finite-dimensional spaces, the principal properties of the elements of this algebra, taken individually, follow from the Jordan decomposition (A, VII, p. 42, § 5, no. 9). Here we study their generalization to infinite-dimensional spaces, which is however more delicate.

In Chapter III, we consider, in a very general framework, compact linear maps between topological vector spaces. These are the ones whose spectral properties most closely resemble the finite-dimensional case. For example, in the particular case of Banach spaces, the nonzero elements of the spectrum of a compact endomorphism are generalized eigenvalues of finite spectral multiplicity.

In the context of Hilbert spaces, a manageable classification and structure emerge. This is the object of Chapter IV. It begins with the study of compact operators in Hilbert spaces: their description by singular values, as well as their diagonalization in the case of a normal endomorphism, are entirely analogous to the properties known for a finite-dimensional space. More generally, the spectral theorem provides simple models of normal endomorphisms of Hilbert spaces.

Certain important applications, concerning for example differential operators, but also quantum mechanics, require a more subtle theory: that of linear maps — not necessarily continuous — defined on a dense subspace of a Hilbert space, which we call partial operators. In this framework, the very definition of a self-adjoint or normal operator demands great precision if one wishes to avoid gross blunders. Once these definitions are acquired, a classification similar to that of continuous endomorphisms emerges; the functional calculus for universally measurable

maps, bounded or not, provides close links between continuous normal endomorphisms and normal partial operators.

In passing, we shall have mentioned normal endomorphisms with trace, and in particular integral endomorphisms with kernel in a Hilbert space $L^2(X, \mu)$. One knows the crucial importance, in the theory of automorphic forms and representations, of the formula establishing that the trace of such an endomorphism, when it exists, is the integral of the kernel over the diagonal. We also devote a paragraph to the fundamental definitions and properties of distributions and tempered distributions in R^n . These notions provide not only the most convenient presentation of the fundamental principles of Euclidean Fourier theory (cf. II, § 1), but they also allow one to identify the domains of many important differential operators.

Finally, in Chapter V, we relate the study of endomorphisms of Hilbert spaces to that of unitary representations of groups. After some elementary general considerations, among which Schur's lemma is the most important statement, we study the case of locally compact groups. We then consider functions of positive type, and present their first applications. One of these is the Gelfand-Naimark-Segal theorem, which asserts that every star algebra is isomorphic to a closed subalgebra of the algebra of endomorphisms of a Hilbert space. Others concern fundamental properties of the space of irreducible unitary representations of a locally compact group, in particular the Gelfand-Raikov theorem, which establishes that these representations separate points. In the last paragraph of this chapter, we prove the fundamental results concerning representations of compact groups, which served us in Chapter IX of Groups and Lie algebras.

A. BOREL, Representations of locally compact groups, Lecture Notes in Mathematics 276, Springer, 1972.

M. REED and B. SIMON, Methods of Mathematical Physics, Academic Press, Vol. 1, 1980 and Vol. 2, 1975.

K. SCHMÜDGEN, Unbounded Self-adjoint Operators on Hilbert Space, Graduate Texts in Mathematics 265, Springer, 2012.

In the preceding chapters and books, the following references to "TS, to appear", must be modified as follows:

EVT, V, p. 35, line 1. Instead of "TS, IV, § 6", read "(TS, IV, p. 160, remark)".

INT, IX, p. 94, § 6, no. 12. Instead of “One can prove a converse ... bounded”, read “The converse of prop. 11 is a particular case of TS, V, p. 455, th. 5.”

LIE, IX, p. 61, § 6, no. 4, line 6. Instead of “TS, to appear”, read “TS, III, p. 36, prop. 9”.

LIE, IX, p. 63, § 6, no. 4, line 6. Instead of “TS, to appear”, read “TS, III, p. 36, rem.”.

LIE, IX, p. 66, § 7, note 1. Instead of “to a chapter to appear”, read “to paragraph 4 of Chapter V”.

LIE, IX, p. 69, § 7, no. 2. Instead of “then according to TS”, read “then (TS, V, p. 464, prop. 6)”.

LIE, IX, p. 71, § 7, no. 3. Instead of “which, according to TS, is an isomorphism”, read “which, according to TS, V, p. 469, cor. 2, is an isomorphism.”

LIE, IX, p. 72, § 7, no. 4. Instead of “(TS)”, read “(TS, V, p. 459, cor. 3)”.

LIE, IX, p. 74, § 7, no. 4. After “of the space of central representative functions on G ”, add “(TS, V, p. 469, cor. 2, a)”; after “of the space $ZL^2(G)$ of central square-integrable functions on G ”, add “(TS, V, p. 468, prop. 9).”

LIE, IX, p. 78, § 8, no. 1. Instead of “In this number, one recalls definitions and results from TS”, read “One recalls definitions and results from TS, V, p. 386, no. 8; p. 392, no. 10; p. 470, no. 7.”

LIE, IX, p. 84, § 8, no. 3. Instead of “Recall (TS) the formulas”, read “One has (TS, V, p. 467, formulas (1) and (2)) the formulas”.

LIE, IX, p. 89, § 9, no. 1. Instead of “TS, III, §2, no. 7, prop. 16”, read “TS, III, p. 67, prop. 10”.

LIE, IX, p. 90, § 9, no. 1. Instead of “TS, III, §2, no. 6”, read “TS, III, p. 61, no. 4”.

LIE, IX, p. 91, § 9, no. 2. Instead of “TS, to appear”, read “TS, III, p. 36, prop. 9 and TS, V, p. 464, prop. 7”.

LIE, IX, p. 92, note 1. Instead of “(TS, to appear)”, read “(TS, V, p. 379, def. 3)”.

LIE, IX, p. 99, appendix 1, no. 1. Instead of “(TS, to appear)”, read “(TS, V, p. 464, remark 4)”.

LIE, IX, p. 128, exercise 3 (b) of §9. Instead of “(TS, to appear)”, read “(Use the Peter–Weyl theorem, TS, V, p. 462, th. 1)”.

TS, I, p. 103, example 6. Instead of “V, to appear”, read “V, p. 442, th. 2.”

TS, I, p. 173, exerc. 5, d). Instead of “III, to appear”, read “III, p. 82, § 6.”

TS, II, p. 237, no. 9. Instead of “IV, to appear”, read “IV, p. 196, § 3”.

TS, II, p. 251, no. 1. Instead of “V, to appear”, read “V, p. 425, prop. 10”.

Chapter III

Compact linear mappings and perturbations

In this chapter, all vector spaces considered are vector spaces over a field K equal to $\mathbb{R}$ or $\mathbb{C}$. The references to EVT, II made in this book concern in general the case $K = \mathbb{R}$; for the case $K = \mathbb{C}$, cf. EVT, II, p. 64 to 68.

A semi-normed space is a vector space E equipped with a semi-norm p and the topology defined by p (cf. EVT, II, p. 2). The unit ball of E , or of p , is the set of elements x of E such that $p(x) \leq 1$.

For every vector space E , we denote by 1_E the identity mapping of E . If E, F and G are vector spaces and $u: E \rightarrow F$, $v: F \rightarrow G$ are linear mappings, we sometimes denote by vu the linear mapping $v \circ u$ from E into G .

Let E and F be topological vector spaces. A continuous linear mapping from E into F will also be called a morphism, and an endomorphism when $F = E$. A bijective linear mapping from E into F which is continuous and whose inverse is also continuous will be called an isomorphism and, when $F = E$, an automorphism. When E and F are semi-normed spaces, the notions of endomorphism, isomorphism and automorphism refer to the underlying topological vector space structure.

Given topological vector spaces E and F , we denote by $\mathcal{L}^f(E; F)$ the vector space of continuous linear mappings of finite rank from E into F . We also denote by $\mathcal{L}^f(E)$ the vector space

$\mathcal{L}^f(E; E)$.

We recall that a continuous linear mapping u from E into F is strict if it induces, by passage to the quotient, an isomorphism of $E/\text{Ker}(u)$ onto $u(E)$ (TG, III, p. 16, def. 1); this is equivalent to saying that the image of every neighbourhood of 0 in E is a neighbourhood of 0 in $u(E)$ (TG, III, p. 16, prop. 24).

III.1 COMPACT LINEAR MAPPINGS

III.1.1 Compact linear mappings

DEFINITION 1. — Let E be a topological vector space and F a separated topological vector space. A linear mapping u from E into F is said to be compact if there exists a neighbourhood V of 0 in E such that $u(V)$ is a relatively compact subset of F .

We denote by $\mathcal{L}^c(E; F)$ the set of compact linear mappings from E into F ; we also denote by $\mathcal{L}^c(F)$ the set $\mathcal{L}^c(F; F)$.

Remarks. — 1) A neighbourhood of 0 in E absorbs every bounded subset of E ; the image of a bounded subset of E under a compact linear mapping from E into F is therefore a relatively compact subset of F .

- 2) Let E be a semi-normed space and let B be the unit ball of E . For a linear mapping u from E into a separated topological vector space F to be compact, it is necessary and sufficient that $u(B)$ is a relatively compact subset of F .
- 3) Let F be a separated topological vector space. For a vector subspace E of F to be finite-dimensional, it is necessary and sufficient that the canonical injection of E into F be compact (EVT, I, p. 15, th. 3).
- 4) For every topological vector space E over the field $\mathbf{R}$, we denote by $E_{(\mathbf{C})}$ the complexified topological vector space of E (EVT, II, p. 65). We identify E with a real topological vector subspace of $E_{(\mathbf{C})}$ by the mapping $x \mapsto 1 \otimes x$; the underlying real topological vector space of $E_{(\mathbf{C})}$ is then the topological direct sum of E and iE .

Let u be a linear mapping from E into a separated topological vector space F over $\mathbf{R}$. Denote

by $u_{(\mathbf{C})}$ the $\mathbf{C}$ -linear mapping from $E_{(\mathbf{C})}$ into $F_{(\mathbf{C})}$ which extends u . The sets of the form $V + iV$, where V is a neighbourhood of 0 in E , form a fundamental system of neighbourhoods of 0 in $E_{(\mathbf{C})}$. For the set $u_{(\mathbf{C})}(V + iV) = u(V) + iu(V)$ to be relatively compact in $F_{(\mathbf{C})}$, it is necessary and sufficient that $u(V)$ be relatively compact in F . Consequently, for the $\mathbf{C}$ -linear mapping $u_{(\mathbf{C})}$ to be compact, it is necessary and sufficient that the $\mathbf{R}$ -linear mapping u be so.

In the same vein, when v is a continuous $\mathbf{C}$ -linear mapping from $E_{(\mathbf{C})}$ into a $\mathbf{C}$ -topological vector space G , for v to be compact it is necessary and sufficient that its restriction to E be compact.

- 5) Let E be a topological vector space, and let F be a separated topological vector space. Let F_1 be a closed vector subspace of F and let j be the canonical injection of F_1 into F . For a linear map u from E into F_1 to be compact, it is necessary and sufficient that the linear map $j \circ u$ from E to F be so.
- 6) Let $(G_i)_{i \in I}$ be a family of separated topological vector spaces, and for every $i \in I$, let u_i be a compact linear map from a topological vector space E into G_i . If the set I is finite, or if E is a semi-normed space, the linear map $x \mapsto (u_i(x))_{i \in I}$ from E into $\prod_{i \in I} G_i$ is compact (Remark 2 and TG, I, p. 63, Th. 3).

PROPOSITION 1. — Let E be a topological vector space and F a separated topological vector space.

- a) Every compact linear map from E into F is continuous.
- b) Every continuous linear map of finite rank from E into F is compact;
- c) The set $\mathcal{L}^c(E; F)$ of compact linear mappings from E into F is a vector subspace of $\mathcal{L}(E; F)$.

Let us prove a). Let $v : E \rightarrow F$ be a compact linear mapping and U a neighborhood of 0 in F . Choose a neighborhood V of 0 in E such that $v(V)$ is relatively compact in F . Then $v(V)$ is bounded, and therefore there exists a real number $\lambda > 0$ such that the set $v(\lambda V)$ is contained in U ; consequently, v is continuous.

Let $u : E \rightarrow F$ be a continuous linear map of finite rank. Since the image $u(E)$ of u is

separated and finite-dimensional, there exists a compact neighborhood A of 0 in $u(E)$. The set $V = u^{-1}(A)$ is a neighborhood of 0 in E , and we have $u(V) \subset A$, so the linear map u is compact. This proves the assertion b).

Let u_1 and u_2 be compact linear mappings from E into F . Let V_1 and V_2 be neighborhoods of 0 in E such that the sets $u_1(V_1)$ and $u_2(V_2)$ are relatively compact in F . Set $V = V_1 \cap V_2$ and $u = u_1 + u_2$. Then $u(V)$ is contained in $u_1(V_1) + u_2(V_2)$ and therefore is relatively compact in F , and the linear map u is compact. It is immediate that $\mathcal{L}^c(E; F)$ is stable under homothety. This proves that $\mathcal{L}^c(E; F)$ is a vector subspace of $\mathcal{L}(E; F)$, whence assertion c).

PROPOSITION 2. — Let E be a semi-normed space and F a separated topological vector space. Suppose that every closed bounded subset of F is complete. Then the vector subspace $\mathcal{L}^c(E; F)$ of $\mathcal{L}(E; F)$ is closed for the topology of bounded convergence.

Let u be an element of $\mathcal{L}(E; F)$ adherent to $\mathcal{L}^c(E; F)$ for the topology of bounded convergence. Let B denote the unit ball of E and V a neighborhood of 0 in F . Choose a neighborhood W of 0 in F such that $W + W \subset V$. By hypothesis there exists an element v of $\mathcal{L}^c(E; F)$ such that $(u - v)(B) \subset W$. The set $v(B)$ being relatively compact in F , there exists a finite subset M of F such that $v(B) \subset M + W$, whence $u(B) \subset M + V$. Since this holds for every V , the set $u(B)$ is precompact in F , hence so is its closure (TG, II, prop. 1, p. 30). Since the latter is closed and bounded, it is complete by the hypothesis made on F , and consequently is compact (TG, II, cor., p. 30). It follows that $u(B)$ is relatively compact. This proves that u belongs to $\mathcal{L}^c(E; F)$.

If F is locally convex, the hypothesis of the proposition means that F is quasi-complete (EVT, III, p. 8, def. 6).

COROLLARY. — Let E and F be Banach spaces. The set $\mathcal{L}^c(E; F)$ is a closed vector subspace of the Banach space $\mathcal{L}(E; F)$. It contains the closure of the space $\mathcal{L}^f(E; F)$ of continuous linear mappings of finite rank from E into F .

This follows from Propositions 1 and 2.

Z

There exist Banach spaces E and F such that $\mathcal{L}^f(E; F)$ is not dense in $\mathcal{L}^c(E; F)$ (cf. Remark

6 of III, p. 16 and Theorem 4 of III, p. 19, b)).

PROPOSITION 3. — Let E_1, E, F, F_1 be topological vector spaces, with F and F_1 assumed separated. Let $v: E_1 \rightarrow E$, $u: E \rightarrow F$, and $w: F \rightarrow F_1$ be linear maps. If v and w are continuous and u is compact, then $w \circ u \circ v$ is compact.

By hypothesis, there exists a neighborhood V of 0 in E such that $u(V)$ is relatively compact in F . Set $U = v^{-1}(V)$. Then U is a neighborhood of 0 in E_1 and its image under $w \circ u \circ v$ is contained in $w(u(V))$, hence is relatively compact in F_1 . Consequently, the linear map $w \circ u \circ v$ is compact.

Let E be a separated topological vector space; by props. 1 and 3, $\mathcal{L}^c(E)$ is a two-sided ideal of the algebra $\mathcal{L}(E)$. When E is a Banach space, $\mathcal{L}(E)$ is a Banach algebra and $\mathcal{L}^c(E)$ is a closed two-sided ideal of $\mathcal{L}(E)$ (cor. of prop. 2). It is a proper ideal if E is of infinite dimension (cf. remark 3, p. 2).

COROLLARY. — Let E be a Hilbert space. The space $\mathcal{L}^c(E)$ is a closed self-adjoint two-sided ideal of $\mathcal{L}(E)$. In particular, it is a star algebra.

Indeed, the space $\mathcal{L}^c(E)$ is a closed two-sided ideal, hence self-adjoint, of $\mathcal{L}(E)$ (lemma 15 of I, p. 122).

PROPOSITION 4. — Let E be a topological vector space, $\widehat{E}$ its separated completion and j the canonical map from E into $\widehat{E}$. Let u be a compact linear map from E into a separated topological vector space F . There exists a unique compact linear map v from $\widehat{E}$ into F such that $u = v \circ j$.

Identify F with a topological vector subspace of $\widehat{F}$ and let us denote $\widehat{u}: \widehat{E} \rightarrow \widehat{F}$ the unique continuous linear map such that $\widehat{u} \circ j$ coincides with u in E . Since u is compact, there exists a neighborhood V of 0 in E and a compact subset A of F such that $u(V) \subset A$. We have $\widehat{u}(j(V)) \subset A$, whence $\widehat{u}(\overline{j(V)}) \subset A$. Now $\overline{j(V)}$ is a neighborhood of 0 in $\widehat{E}$ (TG, III, p. 24, prop. 7). Consequently, the image of $\widehat{u}$ is contained in F and the continuous linear map $v: \widehat{E} \rightarrow F$ derived from $\widehat{u}$ by passing to the subspace is compact. Since $j(E)$ is dense in $\widehat{E}$, the map v is the unique continuous linear map from $\widehat{E}$ into F such that $u = v \circ j$.

PROPOSITION 5. — Let E be a locally convex space, F a separated topological vector space, and u a compact linear map from E into F . There exist a Banach space G , a continuous

linear map v from E into G , and a compact linear map w from G into F such that $u = w \circ v$.

Let V be a neighborhood of 0 in E such that $u(V)$ is relatively compact in F . Let p be a continuous semi-norm on E such that V contains the unit ball of p (EVT, II, p. 26, cor.). Let us denote by E_p the semi-normed space obtained by endowing E with the semi-norm p . The map u is a compact linear map from E_p into F . The separated completion G of E_p is a Banach space. Denote by v the canonical linear map from E_p into G . By prop. 4, there exists a compact linear map $w : G \rightarrow F$ such that $u = w \circ v$. On the other hand, v is a continuous map from E into G because the topology of E is finer than that of E_p .

III.1.2 Compact linear maps and weak topologies

Let E be a locally convex space, E' its dual. Recall that the topology $\sigma(E, E')$ on E is called the weak topology on E (TVS, IV, p. 4). In this section, E_σ denotes the space E equipped with the weak topology.

PROPOSITION 6. — Let E be a locally convex space, F a separated locally convex space, $u : E \rightarrow F$ a compact linear map and B a bounded subset of E . The restriction of u to B is a continuous map from the set B , equipped with the topology induced by $\sigma(E, E')$, into the space F .

The map u is a continuous linear map from E into F , and also from E_σ into F_σ . Its restriction to B is therefore a continuous map from the set B , equipped with the topology induced by $\sigma(E, E')$, into the space F_σ . Now the set $u(B)$ is contained in a compact subset C of F (remark 1 of III, p. 2), and the space F_σ is Hausdorff, so that the induced topologies on C from those of F and of F_σ coincide. The proposition follows.

COROLLARY. — Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of points of E , which converges to a point x of E for the weak topology. The sequence $(u(x_n))_{n \in \mathbb{N}}$ converges in F to $u(x)$.

Indeed, the set consisting of the point x and the points of the sequence $(x_n)_{n \in \mathbb{N}}$ is a bounded subset of E (EVT, III, p. 3, cor. of prop. 2).

PROPOSITION 7. — Let E be a semi-normed space, E' its dual, B the unit ball of E' , and u a linear map from E' into a separated locally convex space F . If the restriction of u to B equipped with the topology induced by $\sigma(E', E)$ is continuous, then $u(B)$ is a compact subset of F and u

is a compact linear map from the strong dual E'_b of E into F .

Indeed, the set B , equipped with the topology induced by $\sigma(E', E)$ is compact (EVT, III, p. 17, cor. 2).

COROLLARY 1. — Let E be a semi-normed space of countable type, let E' be its dual and let B be the unit ball of E' . Let u be a linear mapping from E' into a separated locally convex space F . Suppose that for every sequence $(x_n)_{n \in \mathbb{N}}$ of elements of B that converges for $\sigma(E', E)$ to 0, the sequence $(u(x_n))_{n \in \mathbb{N}}$ converges to 0 in F . Then, $u(B)$ is a compact subset of F and u is a compact linear map from the strong dual E'_b of E into F .

Indeed, the topology induced on B by $\sigma(E', E)$ is metrizable (EVT, III, p. 19, cor. 2). The hypothesis of the corollary therefore implies that the restriction of u to B is a continuous map from B into F , when B is equipped with the topology $\sigma(E', E)$ and the corollary follows from the preceding proposition.

COROLLARY 2. — Let E be a semi-normed space and S a set of precompact subsets of E whose union is dense in E . Denote by E'_b the strong dual of E and by E'_S the dual of E equipped with the S -topology. The space E'_S is separated and the identity map of E'_b into E'_S is compact.

The space E'_S is separated according to TG, X, p. 8, prop. 7; on the unit ball of E' the S -topology coincides with the weak topology $\sigma(E', E)$ (EVT, III, p. 17, prop. 5), whence the corollary.

Example. — Let E be a semi-normed space. The identity map of E'_b into the space E' , endowed with the weak topology, the topology of compact convergence, or the topology of precompact convergence, is compact.

PROPOSITION 8. — Let E be a reflexive Banach space, B its unit ball, F a separated locally convex space, and $u: E \rightarrow F$ a linear map. The following conditions are equivalent:

- (i) The linear map u is compact;
- (ii) The set $u(B)$ is a compact subset of F ;
- (iii) The restriction of u to B is a continuous map from the set B , equipped with the topology induced by $\sigma(E, E')$ into F ;
- (iv) For any sequence $(x_n)_{n \in \mathbb{N}}$ of points of B that converges to 0 for the topology $\sigma(E, E')$ the

sequence $(u(x_n))_{n \in \mathbb{N}}$ converges to 0 in F ;

(v) From any infinite sequence of points of $u(B)$ one can extract a subsequence that converges in F ;

(vi) Any infinite sequence of points of $u(B)$ has a cluster point in F .

Since E is a reflexive space, B is a compact subset of E_σ (EVT, IV, p. 17, prop. 6), hence (iii) implies (ii). Condition (ii) implies (i) by definition, and condition (i) implies (iii) by prop. 6. It is also elementary that (iii) implies (iv) and that (v) implies (vi).

We will now show that (iv) implies (v). Let (x_n) be an infinite sequence of points of B . Since B is a compact subset of E_σ , by Šmulian's theorem (EVT, IV, p. 36, th. 2) there exists a sequence (y_n) , extracted from the sequence (x_n) , which converges in E_σ to a limit y . The sequence $(y_n - y)$ is bounded in E_σ , hence in E (EVT, IV, p. 1, prop. 1); it is therefore contained in a homothetic set of B . If condition (iv) is satisfied, the sequence $(u(y_n - y))$ converges to 0 in F , and $(u(y_n))$, which is a subsequence of $(u(x_n))$, converges to $u(y)$. This proves that (iv) implies (v).

Finally, let us show that (vi) implies (ii). Let $j : F \rightarrow \hat{F}$ be the canonical injection of F into its completion. Under hypothesis (vi), $u(B)$ is a precompact subset of F (EVT, IV, p. 32, prop. 1), hence $j(u(B))$ is a relatively compact subset of $\hat{F}$ (TG, II, no. 2, p. 29) and $j \circ u$ is a compact linear map. By the already proven equivalence of conditions (i) and (ii), $j(u(B))$ is a compact subset of $\hat{F}$. Consequently, $u(B)$ is a compact subset of F .

III.1.3 Transposition

Let E be a locally convex space and E' its dual. Recall that we denote by E'_b and E'_c the locally convex spaces obtained by endowing the vector space E' with the topology of bounded convergence and that of compact convergence respectively (EVT, III, p. 14). The space E'_b is also called the strong dual of E (loc. cit., example 4).

PROPOSITION 9. — Let E be a locally convex space, F a separated locally convex space, and u a continuous linear map from E into F .

a) If the mapping u is compact, then its transpose ${}^t u$ is a compact linear map from F'_c into

E'_c and a continuous linear map from F'_c into E'_b ;

b) Suppose that the space E is a semi-normed space and that the space F is quasi-complete.

If the transpose ${}^t u$ is a continuous map from F'_c into E'_b then the linear map u is compact.

Suppose first that the linear map u is compact. There then exists a neighborhood V of 0 in E whose image under u is contained in a compact subset C of F . By its definition (EVT, II, p. 47, def. 2 and p. 68, def. 1), the polar C° of C is a neighborhood of 0 in F'_c and we have ${}^t u(C^\circ) \subset V^\circ$. Now V° is an equicontinuous subset of E' closed for the weak topology, hence compact in E'_c (EVT, III, p. 17, cor. 2 and prop. 5). This proves that ${}^t u$ is a compact linear map from F'_c into E'_c .

Let us keep the preceding hypotheses and let U be a neighborhood of 0 in E'_b . It contains the polar A° of a bounded subset A of E . Since the linear map u is compact, the set $u(A)$ is relatively compact in F (remark 1 of III, p. 2), and $({}^t u)^{-1}(A^\circ) = u(A)^\circ$ is a neighborhood of 0 in F'_c . This proves that ${}^t u$ is a continuous linear map from F'_c into E'_b .

Let us now place ourselves under the hypotheses of b and denote by B the unit ball of E . The polar B° of B is the unit ball of E'_b and we have $({}^t u)^{-1}(B^\circ) = u(B)^\circ$. If ${}^t u$ is a continuous map from F'_c into E'_b , the set $u(B)^\circ$ is therefore a neighborhood of 0 in F'_c ; it then contains the polar C° of a compact subset C of F . By the bipolar theorem (EVT, II, p. 49, cor. 3), the set $u(B)$ is contained in the closed convex hull of $C \cup \{0\}$; the latter is compact because the space F is quasi-complete (EVT, III, p. 8). This proves that the linear map u is compact.

COROLLARY 1 (Schauder). — Let E be a seminormed space, F a Banach space, E' and F' their respective strong duals, and u a continuous linear map from E into F . The following properties are equivalent:

- (i) The linear map u from E into F is compact;
- (ii) The linear map ${}^t u$ from F' into E' is compact;
- (iii) The linear map ${}^t u$ from F'_c into E' is continuous.

The equivalence of (i) and (iii) follows from Prop. 9, a) and b); the implication (iii) $\Rightarrow$ (ii) comes from the fact that the identity map from F' into F'_c is compact (cor. of Prop. 7 of III, p.

7).

Let us conclude by proving that condition (ii) implies (i). Suppose that the linear map ${}^t u : F' \rightarrow E'$ is compact. It follows from the implication (i) $\Rightarrow$ (ii), applied to ${}^t u$, that ${}^{tt} u$ is a compact linear map from E'' into F'' . Denote by v the canonical linear map from E into E'' ; it is continuous. Since F is identified with a closed vector subspace of F'' and u coincides with ${}^t({}^t u) \circ v$ on E , it follows from Remark 5 of III, p. 3 that the linear map u is compact.

COROLLARY 2. — Let E be a semi-normed space and F a Banach space of countable type. Let u be a continuous linear map from E into F . Suppose that for every sequence (y_n) of elements of F' weakly tending to 0, the sequence $({}^t u(y_n))$ tends strongly to 0 in E' . Then the linear map u is compact.

Let B' denote the unit ball of F' , equipped with the topology induced by the topology $\sigma(F', F)$. Since F is a Banach space of countable type, B' is a compact metrizable space (EVT, III, p. 19, cor. 2). Under the hypothesis made, the restriction of ${}^t u$ to B' is a continuous map from B' into E' and ${}^t u(B')$ is a compact subset of E' . By Cor. 1, the linear map u is compact.

Remark. — Let E and F be separated locally convex spaces. The transpose of a compact linear map from E into F is not always a compact linear map from F'_b into E'_b (cf. III, p. 108, exercise 15).

III.1.4 The Leray–Schauder theorem

The following theorem will be proved in TA, forthcoming.

THEOREM 1 (Brouwer). — Let B be a nonempty compact convex subset of a finite-dimensional normed vector space. Every continuous map from B into B has a fixed point.

We shall deduce from this the following result.

THEOREM 2 (Leray–Schauder). — Let X be a nonempty compact topological space, homeomorphic to a convex subset of a locally convex space. Every continuous map from X into X has a fixed point.

We may assume that X is a convex subset of a locally convex space E . Let N be the closure of $\{0\}$ in E and $\pi : E \rightarrow E/N$ the canonical surjection. Since X is a separated space, the restriction of π to X is injective; since X is compact, it then defines a homeomorphism from X onto a

nonempty compact convex subset of the separated locally convex space E/N . This allows us to assume that the space E is separated.

Let $f : X \rightarrow X$ be a continuous map. The map $h : X \rightarrow E$ defined by $h(x) = f(x) - x$ is continuous; since X is compact, the image $h(X)$ is closed. It therefore suffices to prove that 0 is adherent to $h(X)$, that is, for every open convex neighborhood U of 0 in E , there exists a point b of X such that $f(b) - b \in U$.

Let U be a convex neighborhood of 0 in E . For every $a \in X$, we denote by V_a the set of elements x of X such that $f(x) - f(a) \in U$. It is open in X and contains a . Since X is compact and nonempty, there exists a nonempty finite subset A of X such that the sets V_a , for $a \in A$, cover X . Let $(\varphi_a)_{a \in A}$ be a continuous partition of unity subordinate to the covering $(V_a)_{a \in A}$ (TG, IX, p. 43, prop. 1 and p. 47, th. 3). Let F denote the subspace of E generated by $f(A)$. For every $x \in F \cap X$ set

$$g(x) = \sum_{a \in A} \varphi_a(x) f(a).$$

One thus defines a continuous map $g : F \cap X \rightarrow E$. Its image is contained in the convex hull of $f(A)$, hence in $F \cap X$. Since the set $F \cap X$ is a nonempty compact convex subset of a finite-dimensional vector space, the map g has a fixed point $b \in F \cap X$ by Theorem 1. We have

$$f(b) - b = f(b) - g(b) = \sum_{a \in A} \varphi_a(b) (f(b) - f(a)).$$

For every $a \in A$ such that $\varphi_a(b) \neq 0$, we have $b \in V_a$ so that $f(b) - f(a) \in U$. Since U is convex, we conclude that $f(b) - b \in U$, which completes the proof.

III.1.5 Invariant subspaces under a compact operator

The following theorem is to be compared with Schur's lemma (A, VIII, p. 43, cor. and V, p. 386, prop. 6) and with cor. 4 of I, p. 26.

THEOREM 3. — Let E be a separated locally convex space over C and let A be a subset of $\mathcal{L}(E)$. Let us make the following hypotheses:

- (i) There exists no closed vector subspace of E , distinct from $\{0\}$ and E , which is stable under

A;

- (ii) The set A contains a nonzero compact endomorphism. Then the commutant of A is reduced to scalar multiples of the identity.

By replacing A with the unital subalgebra of $\mathcal{L}(E)$ generated by A, we reduce to the case where A is a unital subalgebra of $\mathcal{L}(E)$. Let then h be a nonzero compact endomorphism of E belonging to A.

Lemma 1. — There exists an element a of A such that the kernel of $ha - 1_E$ is not reduced to 0.

There exists an element x_0 of E such that $h(x_0) \neq 0$. Let V be a closed neighborhood of $h(x_0)$ not containing 0. Since the endomorphism h is compact, one can choose an open convex neighborhood U of x_0 such that $h(U)$ is a relatively compact subset of V . The closure C of $h(U)$ is therefore a compact convex subset of E ; since it is contained in V , it does not contain 0.

Let y be a point of C . Denote by Ay the set of images of y by the elements of A. It is a vector subspace of E stable under A; it is nonzero since it contains y . By hypothesis (i) of the theorem, the set Ay is dense in E . There therefore exists an element b of A such that $b(y) \in U$. Since the set C is compact, there exists a finite family $(b_1, \dots, b_n)$ of elements of A such that the sets $b_i^{-1}(U)$ cover C . There exists a continuous partition of unity $(\varphi_1, \dots, \varphi_n)$ on C subordinate to this covering (TG, IX, p. 47, th. 3). Let us define a map $f : C \rightarrow E$ by setting

$$f(x) = \sum_{i=1}^n \varphi_i(x) b_i(x)$$

for all $x \in C$; it is a continuous map. Since U is convex and $b_i(x)$ belongs to U when $\varphi_i(x)$ is not zero, we have $f(C) \subset U$ and $h(f(C)) \subset C$. By the Leray–Schauder theorem (th. 2 of III, p. 11), there exists an element $x \in C$ such that $h(f(x)) = x$. Then set

$$a = \sum_{i=1}^n \varphi_i(x) b_i.$$

One has $h(a(x)) = h(f(x)) = x$ and x is nonzero since 0 does not belong to C , hence the kernel of $ha - 1_E$ is nonzero.

Let us finish the proof of Theorem 3. Let a be an element of A such that the kernel T of

$ha - 1_E$ is nonzero (lemma 1). Let i denote the canonical injection of T into E . The linear map i is equal to $h \circ a \circ i$, hence it is compact. Thus the dimension of T is finite (remark 3 of III, p. 2). Let u be an element of $\mathcal{L}(E)$ which commutes with A . Since u commutes with $h \circ a$, we have $u(T) \subset T$. Since T is finite-dimensional and nonzero over the algebraically closed field C , the endomorphism of T induced by u admits an eigenvalue λ .

Set $F = \text{Ker}(u - \lambda 1_E)$. This is a closed nonzero vector subspace of E ; it is stable under A , since u commutes with the elements of A . By hypothesis (i), we have $F = E$, whence $u = \lambda 1_E$.

COROLLARY 1. — Keeping the hypotheses of Theorem 3 and supposing moreover that the elements of A are pairwise commuting. Then E is one-dimensional over C .

By hypothesis (ii) of Theorem 3, the space E is nonzero. It therefore contains a vector subspace F of dimension 1. This subspace is closed since E is assumed separated. By th. 3, every element of A is a homothety, hence stabilizes F . It then follows from hypothesis (i) of th. 3 that F is equal to E .

COROLLARY 2 (Lomonosov). — Let E be a separated locally convex space of dimension at least 2 over the field C and let u be an endomorphism of E . Suppose that there exists a compact endomorphism $h \neq 0$ of E commuting with u . There then exists a closed vector subspace F of E , distinct from $\{0\}$ and from E , such that $u(F) \subset F$.

Indeed, Corollary 1 shows that the set $A = \{u, h\}$ cannot satisfy hypothesis (i) of Theorem 3.

COROLLARY 3. — Let E be a separated locally convex space over the field C and let u be a compact endomorphism of E . If E has dimension at least 2, there exists a closed vector subspace F of E , distinct from $\{0\}$ and from E , such that $u(F) \subset F$.

The case $u = 0$ is immediate. The case $u \neq 0$ follows from cor. 2 by taking $h = u$.

III.1.6 Approximation spaces

Given locally convex spaces E and F , recall (EVT, III, p. 14, example 2) that one denotes $\mathcal{L}_{pc}(E; F)$ the locally convex space obtained by equipping $\mathcal{L}(E; F)$ with the topology of pre-compact convergence. We also denote $\mathcal{L}_{pc}(E)$ the locally convex space $\mathcal{L}_{pc}(E; E)$. When the locally convex space E is Hausdorff and quasi-complete (EVT, III, p. 8, def. 6), the topology

of precompact convergence on $\mathcal{L}(E; F)$ coincides with the topology of compact convergence, since the closure of a precompact subset of E is compact (EVT, III, p. 8).

DEFINITION 2. — A locally convex space E is said to have the approximation property, or again to be an approximation space, if the identity map 1_E of E is adherent in the space $\mathcal{L}_{pc}(E)$ to the set $\mathcal{L}^f(E)$ of continuous finite-rank endomorphisms of E .

In particular, every finite-dimensional locally convex space is an approximation space.

Remarks. — 1) For the topological direct sum of locally convex spaces E and F to be an approximation space, it is necessary and sufficient that E and F be approximation spaces.

2) Let E be a locally convex space. Let N be the closure of $\{0\}$ in E and P an algebraic supplementary subspace of N in E . Then P is a topological supplement of N in E , and is isomorphic to the locally convex space E/N . The space N is an approximation space. By Remark 1, for E to be an approximation space, it is necessary and sufficient that the Hausdorff locally convex space E/N be one.

3) Let E be a locally convex space, A an equicontinuous subset of $\mathcal{L}^f(E)$ and T a total subset of E . If 1_E is adherent to A for the topology of simple convergence on T , then 1_E is adherent to A in $\mathcal{L}_{pc}(E)$ (EVT, III, p. 16, prop. 4 and p. 17, prop. 5), and E is an approximation space.

4) Let E be a locally convex space over $\mathbb{C}$. Denote by E_0 the locally convex space over $\mathbb{R}$ underlying E . For E to be an approximation space, it is necessary and sufficient that E_0 itself be one. Indeed, the condition is necessary; let us prove that it is sufficient. Suppose therefore that E_0 is an approximation space. Let C be a precompact subset of E and U a convex balanced neighborhood of 0 in E . Set $C' = C \cup iC$. There exists a continuous finite-rank $\mathbb{R}$ -linear map u from E_0 into E_0 such that $x - u(x)$ belongs to U for all $x \in C'$. Set $v(x) = \frac{1}{2}(u(x) - iu(ix))$ for all x in E . One thus defines a continuous $\mathbb{C}$ -linear map of finite rank from E into E . For every $x \in C$, we have $x \in C'$, $ix \in C'$ and $x - v(x) = \frac{1}{2}(x - u(x)) - \frac{i}{2}(ix - u(ix))$, so that $x - v(x)$ belongs to U . This proves that E is an approximation space.

5) Let E be a locally convex space over $\mathbb{R}$. For the complexified locally convex space $E_{(\mathbb{C})}$

to be an approximation space, it is necessary and sufficient that E itself be one. This follows from Remarks 1 and 4, since the underlying real locally convex space of $E_{(\mathbb{C})}$ is isomorphic to $E \times E$.

PROPOSITION 10. — Let E be a Hilbert space. Then E is an approximation space.

The set A of finite-rank orthogonal projectors on E is equicontinuous. Let $n \geq 1$ be an integer and $(x_1, \dots, x_n)$ be a family of elements of E . Let V denote the vector subspace generated by the x_i and p_V the orthogonal projector with image V . We have $p_V(x_i) = x_i$ for $1 \leq i \leq n$. This implies that 1_E is adherent to A for the topology of simple convergence, and it follows that E is an approximation space (Remark 3).

Other examples of approximation spaces will be given in No. 7 of III, p. 20.

Lemma 2. — Let E be a locally convex space. Then E is an approximation space if and only if, for every precompact subset C of E and every neighborhood U of 0 in E , there exist an integer $n \geq 0$, elements $e_1, \dots, e_n$ of E , and continuous linear forms $f_1, \dots, f_n$ on E , such that $x - \sum_{i=1}^n f_i(x)e_i$ belongs to U for all $x \in C$.

The definition of the topology of precompact convergence shows that the condition is sufficient. Conversely, suppose that E is an approximation space. Let P be a topological supplementary space in E of the closure of $\{0\}$ (cf. remark 2). The set A of $u \in \mathcal{L}^f(E)$ such that $u(E) \subset P$ is then dense in $\mathcal{L}^f(E)$. Since P is separated, every element u of A is of the form $x \mapsto \sum_{i=1}^n f_i(x)e_i$, where $n \in \mathbb{N}$, $e_1, \dots, e_n$ are elements of E and $f_1, \dots, f_n$ are continuous linear forms on E (EVT, I, p. 14, th. 2). This proves that the condition is necessary.

Remark 6. — There exist Banach spaces which do not have the approximation property, as was demonstrated by P. ENFLO (A counterexample to the approximation problem in Banach spaces, Acta Math. 130 (1973), p. 309–317; cf. Exercise 25 of III, p. 112). This answered a question of S. BANACH (Théorie des opérations linéaires, Monografie Matematyczne I, Warszawa, 1932, remarks in Chapter VI, §1). See also the remark on p. 21.

PROPOSITION 11. — Let E and F be locally convex spaces. Suppose that E is an approximation space.

a) The set $\mathcal{L}^f(E; F)$ is dense in $\mathcal{L}_{pc}(E; F)$;

b) The set $\mathcal{L}^f(F; E)$ is dense in $\mathcal{L}_{pc}(F; E)$;

c) Let $\mathfrak{S}$ be a set of bounded subsets of F and $u \in \mathcal{L}(F; E)$. Suppose that the image under u of every subset of F belonging to $\mathfrak{S}$ is precompact. Then u is adherent to $\mathcal{L}^f(F; E)$ for the $\mathfrak{S}$ -topology (EVT, III, p. 13).

Let v be an element of $\mathcal{L}(E; F)$. The map $\varphi : w \mapsto v \circ w$ from $\mathcal{L}_{pc}(E)$ into $\mathcal{L}_{pc}(E; F)$ is continuous (TG, X, p. 5, prop. 3). We have $\varphi(1_E) = v$ and $\varphi(\mathcal{L}^f(E)) \subset \mathcal{L}^f(E; F)$, hence v is adherent to $\mathcal{L}^f(E; F)$ in $\mathcal{L}_{pc}(E; F)$. This proves *a*).

Similarly, under the hypotheses of *c*), the map $\psi : w \mapsto w \circ u$ from $\mathcal{L}_{pc}(E)$ into $\mathcal{L}_{\mathfrak{S}}(F; E)$ is continuous (loc. cit.). We have $\psi(1_E) = u$ and $\psi(\mathcal{L}^f(E)) \subset \mathcal{L}^f(F; E)$, hence u is adherent to $\mathcal{L}^f(F; E)$ in $\mathcal{L}_{\mathfrak{S}}(F; E)$.

The image of a precompact subset of F under a continuous linear map from F into E is precompact, so b) follows from c).

COROLLARY. — Let E be a separated approximation space and F a locally convex space. Every compact linear map from F into E is adherent to $\mathcal{L}^f(F; E)$ in the space $\mathcal{L}(F; E)$ endowed with the topology of bounded convergence.

This follows from prop. 11, c), since the image of a bounded subset of F under a compact linear map from F into E is relatively compact in E (remark 1 of III, p. 2), hence precompact.

PROPOSITION 12. — Let E be a locally convex space, I a set, and for each $i \in I$ let F_i be a locally convex space and $v_i : E \rightarrow F_i$ a continuous linear map with dense image. Suppose that for every neighborhood U of 0 in E , there exist $i \in I$ and a neighborhood V of 0 in F_i such that $v_i^{-1}(V) \subset U$. If the F_i are approximation spaces, then E is one as well.

Let A be a precompact subset of E and U a neighborhood of 0 in E . By hypothesis there exist $i \in I$ and a continuous seminorm p on F_i such that U contains $(p \circ v_i)^{-1}([0, 1])$. Set $F = F_i$, $v = v_i$ and $B = v(A)$, and suppose that F is an approximation space. The set B is precompact in F . Hence, by Lemma 2, there exist an integer $n \geq 1$, elements $y_1, \dots, y_n$ of F and continuous linear forms $f_1, \dots, f_n$ on F , such that one has, for all $y \in B$,

$$p \left(y - \sum_{j=1}^n f_j(y) y_j \right) \leq \frac{1}{2}.$$

Since B is bounded (EVT, III, p. 3, prop. 2), there exists a real number $M > 0$ such that $|f_j(y)| \leq M$ for all j such that $1 \leq j \leq n$ and every $y \in B$. Moreover, since $v(E)$ is dense in F , there exists, for each integer j such that $1 \leq j \leq n$, an element x_j of E such that $p(y_j - v(x_j)) \leq (2nM)^{-1}$. The linear map

$$u: x \mapsto \sum_{j=1}^n f_j(v(x))x_j$$

belongs to $\mathcal{L}^f(E)$. For every $x \in A$, we have

$$v(x - u(x)) = v(x) - \sum_{j=1}^n f_j(v(x))y_j + \sum_{j=1}^n f_j(v(x))(y_j - v(x_j)),$$

whence $p(v(x - u(x))) \leq \frac{1}{2} + \frac{nM}{2nM} = 1$, and consequently $x - u(x) \in U$. The proposition follows.

COROLLARY 1. — A dense vector subspace of an approximation space is an approximation space.

COROLLARY 2. — If the separated completion of a locally convex space E is an approximation space, then E is an approximation space.

COROLLARY 3. — The product of a family of approximation spaces is an approximation space.

Indeed, let $(E_i)_{i \in I}$ be a family of approximation spaces. For every finite subset J of I , set $E_J = \prod_{i \in J} E_i$ and let us denote v_J the canonical map from $E = \prod_{i \in I} E_i$ into E_J . The locally convex space E_J is an approximation space (Remark 1). The corollary then follows from Prop. 12 applied to the locally convex space E , to the family (E_J) of locally convex spaces and to the continuous linear maps $v_J: E \rightarrow E_J$.

COROLLARY 4. — Let E be a locally convex space. If every continuous seminorm on E is majorized by a continuous pre-Hilbertian seminorm, then E is an approximation space.

Let $\mathcal{P}$ be the set of continuous pre-Hilbertian seminorms on E . For each $p \in \mathcal{P}$, the hypothesis implies that the seminormed space E_p obtained by equipping E with the seminorm p is an approximation space (Corollary 2 and Proposition 10), and the identity mapping from E into E_p is continuous. The corollary therefore follows from Proposition 12.

Lemma 3. — Let E be a metrizable locally convex space and $(x_n)_{n \in \mathbb{N}}$ a sequence of elements of E converging to 0. There exists a sequence $(y_n)_{n \in \mathbb{N}}$ of elements of E converging to 0 and a sequence $(\lambda_n)_{n \in \mathbb{N}}$ of elements of the interval $[0,1]$ converging to 0 such that we have $x_n = \lambda_n y_n$ for all $n \in \mathbb{N}$.

Since E is metrizable, there exists a fundamental system $(V_m)_{m \in \mathbb{N}}$ of balanced neighborhoods of 0 in E such that $V_0 = E$ and $2^{m+1}V_{m+1} \subset V_m$ for all $m \geq 0$. Since (x_n) converges to 0, there exists a strictly increasing sequence $(N_m)_{m \in \mathbb{N}}$ of integers such that $N_0 = 0$ and such that, for all $m \geq 0$, one has $x_n \in V_m$ for $n \geq N_m$. For every integer $n \geq 0$ there exists a unique integer $m \geq 0$ such that $N_m \leq n < N_{m+1}$, and we then set $\lambda_n = 2^{-m}$ and $y_n = 2^m x_n$. The sequence (λ_n) thus defined converges to 0. Moreover, since $y_n \in V_m$ for $n \geq N_{m+1}$ the sequence (y_n) converges to 0 in E . Finally, we have $x_n = \lambda_n y_n$ for all n .

Recall (EVT, II, p. 28) that if E is a vector space and L is a convex balanced subset of E , we denote by E_L the vector subspace of E generated by L , equipped with the seminorm whose unit ball is L . When the set L contains no line, this seminorm is a norm.

Lemma 4. — Let E be a Fréchet space and A a compact subset of E . There exists a compact convex balanced subset L of E containing A such that the topologies induced on A by those of E and of E_L coincide.

The set A is contained in the closed convex balanced hull of the set of points of a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of E converging to 0 (EVT, IV, p. 24, cor. 1). Let $(y_n)_{n \in \mathbb{N}}$ and $(\lambda_n)_{n \in \mathbb{N}}$ be sequences satisfying the conclusions of Lemma 3. The closed convex balanced hull L of the points of the sequence (y_n) contains A and is a compact subset of E (EVT, III, p. 8). Consequently, E_L is a Banach space (EVT, III, p. 8, cor.). In this Banach space, the norm of x_n is bounded above by λ_n , hence the sequence (x_n) tends to 0. The closed convex balanced hull $\tilde{A}$ of the sequence (x_n) in E_L is a compact subset of E_L (EVT, III, p. 8). As L is a bounded subset of E , the canonical injection of E_L into E is continuous. The topologies induced on $\tilde{A}$ by those of E_L and E coincide, and $\tilde{A}$ is a compact, hence closed, subset of E . Since the set $\tilde{A}$ is convex, balanced, and contains the sequence (x_n) ; it contains A . This completes the proof.

THEOREM 4. — Let E be a Fréchet space.

a) Suppose that E is an approximation space. Then, for every semi-normed space F , the

space $\mathcal{L}^f(F; E)$ is dense in $\mathcal{L}^c(F; E)$ for the topology of bounded convergence;

- b) Conversely, suppose that for every Banach space F , every compact linear map from F into E is adherent to $\mathcal{L}^f(F; E)$ for the topology of bounded convergence. Then E is an approximation space.

Let F be a semi-normed space. The closure of $\mathcal{L}^f(F; E)$ in $\mathcal{L}(F; E)$ is contained in $\mathcal{L}^c(F; E)$ (prop. 1 and 2 of III, p. 4). It is equal to $\mathcal{L}^c(F; E)$ if E is an approximation space by the cor. of prop. 11. This proves the assertion a).

Suppose that the hypothesis of b holds. Let $\varepsilon > 0$ be a real number. Let A be a compact subset of E and p a continuous seminorm on E . Let L be a compact convex balanced subset of E such that A is a compact subset of the normed space E_L (Lemma 4). The canonical injection $j : E_L \rightarrow E$ is compact and E_L is a Banach space (EVT, III, p. 8, cor.). Hence, by hypothesis, there exist an integer $n \geq 1$, elements $e_1, \dots, e_n$ of E , and continuous linear forms $\ell_1, \dots, \ell_n$ on E_L , such that the map v of E_L to E defined by $v(x) = \sum_{i=1}^n \ell_i(x) e_i$ satisfies $p(x - v(x)) \leq \frac{\varepsilon}{2}$ for $x \in A$. The image of ${}^t j : E' \rightarrow (E_L)'$ is dense in $(E_L)'$ for the weak topology (EVT, IV, p. 6, prop. 5). On $(E_L)'$ the topology of compact convergence is compatible with the duality between $(E_L)'$ and E_L (EVT, IV, p. 3, example). Hence ${}^t j(E')$ is dense in $(E_L)'$ for the topology of compact convergence (EVT, IV, p. 1, prop. 1), and there exist continuous linear forms $f_1, \dots, f_n$ on E such that

$$|\ell_i(x) - f_i(x)| p(e_i) \leq \frac{\varepsilon}{2n}$$

for $x \in A$ and $1 \leq i \leq n$. The endomorphism $u : x \mapsto \sum_{i=1}^n f_i(x) e_i$ of E belongs to $\mathcal{L}^f(E)$ and for every $x \in A$, we have

$$p(x - u(x)) \leq p(x - v(x)) + p(v(x) - u(x)) \leq \frac{\varepsilon}{2} + n \times \frac{\varepsilon}{2n} = \varepsilon.$$

It follows from this that 1_E is adherent to $\mathcal{L}^f(E)$ for the topology of compact convergence. Now this topology coincides with the topology of precompact convergence because E is complete. Therefore E is an approximation space.

III.1.7 Examples of approximation spaces

Recall that every Hilbert space is an approximation space (III, p. 15, prop. 10). This section gives other examples.

If X is a locally compact space, we denote by $\mathcal{C}_0(X; K)$, or simply $\mathcal{C}_0(X)$, the Banach space of continuous functions from X to K vanishing at infinity, equipped with the norm defined by $\|f\| = \sup_{x \in X} |f(x)|$ for $f \in \mathcal{C}_0(X)$. When X is compact, this space coincides with the space $\mathcal{C}(X)$ of continuous functions from X to K .

Lemma 5. — Let X be a compact topological space, F a finite subset of $\mathcal{C}(X)$ and $\varepsilon > 0$ a real number.

There exist a finite subset $X_0 \subset X$ and a linear map $u : \mathcal{C}(X_0) \rightarrow \mathcal{C}(X)$ of norm ≤ 1 such that $\|u(f|_{X_0}) - f\| \leq \varepsilon$ for all $f \in F$.

Since the set F is an equicontinuous subset of $\mathcal{C}(X)$ there exists a finite covering $(U_i)_{i \in I}$ of X such that, for every $i \in I$ and every $f \in F$ the diameter of $f(U_i)$ is $\leq \varepsilon$. For each $i \in I$, choose a point x_i of U_i , and denote by X_0 the finite set of x_i for $i \in I$.

Let $(\varphi_i)_{i \in I}$ be a continuous partition of unity subordinate to $(U_i)_{i \in I}$. Define a linear map u from $\mathcal{C}(X_0)$ into $\mathcal{C}(X)$ by setting

$$u(g) = \sum_{i \in I} g(x_i) \varphi_i$$

for all $g \in \mathcal{C}(X_0)$. The linear map u has norm ≤ 1 and satisfies $\|u(f|_{X_0}) - f\| \leq \varepsilon$ for all f in F , hence the assertion.

PROPOSITION 13. — Let X be a locally compact topological space. The Banach space $\mathcal{C}_0(X)$ is an approximation space.

Suppose first that X is compact. Let $A \subset \mathcal{B}^f(\mathcal{C}(X))$ be the set of maps of the form $f \mapsto u(f|_{X_0})$, where X_0 is a finite subset of X and $u : \mathcal{C}(X_0) \rightarrow \mathcal{C}(X)$ is a linear map of norm ≤ 1 . The set A is equicontinuous. By Lemma 5, the identity map of $\mathcal{C}(X)$ is adherent to A for the topology of pointwise convergence on $\mathcal{C}(X)$. The proposition then follows from Remark 3 of III, p. 14.

Let us pass to the general case. Let Y be the Alexandroff compactification of X and ω the point at infinity of Y (TG, I, p. 67). Identify $\mathcal{C}_0(X)$ with the set of elements of $\mathcal{C}(Y)$ vanishing

at ω . Then $\mathcal{C}(Y)$ is the topological direct sum of $\mathcal{C}_0(X)$ and of the vector space of constant functions on Y . Since $\mathcal{C}(Y)$ is an approximation space according to the preceding, the space $\mathcal{C}_0(X)$ is an approximation space (Remark 1 of III, p. 14).

COROLLARY. — Every commutative star algebra is an approximation space.

Indeed, such an algebra is isomorphic to the algebra of continuous functions vanishing at infinity on a locally compact space (I, p. 108, th. 1).

Remark. — If E is an infinite-dimensional Hilbert space, the star algebra $\mathcal{L}(E)$ does not have the approximation property (A. SZANKOWSKI, $\mathcal{B}(H)$ does not have the approximation property, Acta Math. 147 (1981), p. 89–108).

Lemma 6. — Let X be a locally compact topological space. Let μ be a positive measure on X and $p \in [1, +\infty[$. Let F be a finite subset of $L_K^p(X, \mu)$ and $\varepsilon > 0$ a real number.

There exists a finite-rank projector u of $L_K^p(X, \mu)$ of norm ≤ 1 such that $\|u(f) - f\| \leq \varepsilon$ for every f in F .

Let $\mathcal{P}$ be the set of finite partitions $\pi = (K_1, \dots, K_n, H)$ of X , where $n \geq 1$ is an integer and $K_1, \dots, K_n$ are integrable subsets of X with nonzero measure. For every partition $\pi = (K_1, \dots, K_n, H) \in \mathcal{P}$, one defines an endomorphism v_π of $\mathcal{L}^p(X, \mu)$ by setting

$$v_\pi(f) = \sum_{i=1}^n \frac{1}{\mu(K_i)} \left(\int_{K_i} f d\mu \right) \varphi_{K_i},$$

for $f \in \mathcal{L}_K^p(X, \mu)$, where φ_{K_i} is the characteristic function of K_i . The map v_π induces, by passing to quotients, a projector u_π of $L_K^p(X, \mu)$. One verifies easily that the image of u_π is the space of classes of functions $f \in \mathcal{L}_K^p(X, \mu)$ such that f vanishes on H and is constant on K_i for $1 \leq i \leq n$.

Let us prove that $\|u_\pi\| \leq 1$. For $f \in \mathcal{L}_K^p(X, \mu)$, we have

$$\|u_\pi(f)\|_p^p = \sum_{i=1}^n \mu(K_i)^{1-p} \left| \int_{K_i} f d\mu \right|^p.$$

By Hölder's inequality (INT, IV, p. 208, § 6, n° 4, th. 2), we have for every i the inequality

$$\left| \int_{K_i} f d\mu \right|^p \leq \mu(K_i)^{p-1} \int_{K_i} |f|^p d\mu,$$

whence $\|u_\pi(f)\|_p \leq \|f\|_p$.

Let $\mathcal{E}$ be the set of classes in $L_K^p(X, \mu)$ of integrable functions on X which take only finitely many values. Since $\mathcal{E}$ is dense in $L_K^p(X, \mu)$ (INT, IV, p. 162, §4, n° 10, cor. 1), there exists a finite set F' of $\mathcal{E}$ such that every element of F is at distance at most ε from an element of F' . Considering the finite partition π formed by the sets of non-zero measure on which the map $x \mapsto (f(x))_{f \in F'}$ takes a given value, one sees that there exists an element π of $\mathcal{P}$ such that $u_\pi(f) = f$ for all f in F' . The projector u_π therefore has the required properties.

PROPOSITION 14. — Let X be a locally compact topological space, μ a positive measure on X , and $p \in [1, +\infty]$. The space $L_K^p(X, \mu)$ is an approximation space.

If $p = +\infty$, then the space $L_C^\infty(X, \mu)$ is a commutative star algebra (example 4 of I, p. 103), hence an approximation space (cor. of prop. 13), and the same is true of $L_R^\infty(X, \mu)$ by remark 5 of III, p. 15.

Suppose that p is finite. Let $A \subset \mathcal{L}^f(L_K^p(X, \mu))$ be the set of finite-rank projectors of norm ≤ 1 . By lemma 6, the identity map of $L_K^p(X, \mu)$ is adherent to A for the topology of simple convergence, and the proposition then follows from remark 3 of III, p. 14.

III.2 EXAMPLES OF COMPACT LINEAR MAPS

III.2.1 Endomorphisms of finite trace, Hilbert–Schmidt, and nuclear p^e -power type

*Let E and F be Hilbert spaces. Every endomorphism of E of finite trace (EVT, V, p. 49, def. 7 and p. 50, def. 8) is compact (IV, p. 165, cor. 1); every Hilbert–Schmidt map from E to F (EVT, V, p. 51, def. 9) is compact (IV, p. 165, cor. 2).

For every real number $p \geq 1$, the vector space $\mathcal{L}_p(E; F)$ of linear maps from E to F of nuclear p^e -power type is contained in $\mathcal{L}^c(E; F)$ (cf. IV, p. 169, remark 2 for $p = 1$).*

III.2.2 Diagonal operators in sequence spaces

Let I be a non-empty set. Recall (EVT, I, p. 4 and p. 5) that the vector space of bounded families $x = (x_i)_{i \in I}$ of elements of $\mathbb{C}$, equipped with the norm $\|x\| = \sup_i |x_i|$, is a Banach space denoted by $\ell_{\mathbb{C}}^{\infty}(I)$.

Let p be an element of $[1, +\infty[$. The space $\ell_{\mathbb{C}}^p(I)$ of families $x = (x_i)_{i \in I}$ of elements of $\mathbb{C}$ such that the family $(|x_i|^p)$ is summable is a vector space on which the map $x \mapsto \|x\| = (\sum_i |x_i|^p)^{1/p}$ is a norm which makes it a Banach space. If one equips the space I with the discrete topology and with the measure for which $\mu(\{x\}) = 1$ for all $x \in I$, this space is none other than the Banach space $L_{\mathbb{C}}^p(I, \mu)$. When $p = 1$ or $p = 2$, this notation thus coincides with that of EVT, I, p. 4 and EVT, V, p. 18 (cf. INT, IV, p. 141, §4, n° 1, example).

For families $x = (x_i)_{i \in I}$ and $y = (y_i)_{i \in I}$ of complex numbers, one denotes by xy the family $(x_i y_i)_{i \in I}$.

PROPOSITION 1. — Let $\lambda = (\lambda_i)_{i \in I}$ be a bounded family of elements of $\mathbb{C}$. Let p be an element of $[1, +\infty]$. Let E be the Banach space $\ell_{\mathbb{C}}^p(I)$.

- a) For every $x \in E$, we have $\lambda x \in E$ and the map $u: x \mapsto \lambda x$ is a continuous endomorphism of E , whose spectrum in the Banach algebra $\mathcal{L}(E)$ is equal to the closure in $\mathbb{C}$ of the set of λ_i , and whose norm is equal to $\sup_i |\lambda_i|$;
- b) The endomorphism u is compact if and only if for every real number $\varepsilon > 0$, the set of elements $i \in I$ such that $|\lambda_i| \geq \varepsilon$ is finite.

With the notation of the proposition, the endomorphism $x \mapsto \lambda x$ is called a diagonal endomorphism of $\ell_{\mathbb{C}}^p(I)$.

Let us prove the proposition. Let $C = \sup_i |\lambda_i|$. Let $x \in E$. We have the inequalities

$$\sum_{i \in I} |\lambda_i x_i|^p \leq C^p \|x\|^p, \text{ if } p \neq +\infty$$

$$\sup_{i \in I} |\lambda_i x_i| \leq C \|x\|, \text{ if } p = +\infty.$$

This proves that $\lambda x \in E$. The map $x \mapsto \lambda x$ is therefore an endomorphism of E , and these

inequalities prove that it has norm $\leq C$.

For $j \in I$, denote by e_j the element $(x_i)_{i \in I}$ of $\ell_{\mathbf{C}}^p(I)$ such that $x_j = 1$ and $x_i = 0$ if $i \neq j$. We then have $u(e_j) = \lambda_j e_j$ for all $j \in I$, which shows that λ_j belongs to the spectrum of u . Since the spectrum of u is closed, the closure in $\mathbf{C}$ of the set of λ_i is contained in the spectrum of u . Since the spectrum of u is contained in the disk centered at 0 with radius $\|u\|$ (I, p. 24, th. 1 and formula (3) of I, p. 21), we deduce the inequality $C \leq \|u\|$, whence $\|u\| = C$.

Conversely, if $\lambda \in \mathbf{C}$ is not adherent to the set of λ_i , then the family $((\lambda_i - \lambda)^{-1})_{i \in I}$ is bounded. The preceding argument therefore shows that the linear map $x \mapsto ((\lambda_i - \lambda)^{-1} x_i)_{i \in I}$ is an endomorphism of E . It is the inverse of $u - \lambda 1_E$, and hence λ does not belong to the spectrum of u . This proves *a*).

Let us now prove *b*). Suppose that the endomorphism u is compact. Let $\varepsilon > 0$. Denote by J the set of $i \in I$ such that $|\lambda_i| \geq \varepsilon$.

The set of elements e_i for $i \in J$ is bounded in E . Its image under u , consisting of the elements $\lambda_i e_i$ for $i \in J$ is therefore relatively compact in E (Remark 1 of III, p. 2). Since $\|\lambda_i e_i - \lambda_j e_j\| \geq \varepsilon$ for every pair of elements $i \neq j$ in J , this is possible only if the set J is finite.

Let us prove the converse assertion. Let J be a finite subset of I and denote $z_J = (z_{J,i})_{i \in I}$, where $z_{J,i} = \lambda_i$ for $i \in J$ and $z_{J,i} = 0$ for $i \in I - J$; the family z_J defines a finite-rank endomorphism $u_J: x \mapsto z_J x$ of E . By the preceding, the norm of the linear map $u - u_J$ is bounded above by the supremum in R_+ of the family $(|\lambda_i|)_{i \in I - J}$. The hypothesis implies that for every $\varepsilon > 0$, there exists a finite subset $J \subset I$ such that this upper bound is $\leq \varepsilon$. It follows that the map u is adherent to $\mathcal{L}^f(E)$, and hence that it is compact (III, p. 4, cor. of prop. 2).

Remarks. — 1) The hypothesis of assertion *b*) is always valid if I is finite. When the set I is infinite, it means that the family $(\lambda_i)_{i \in I}$ tends to 0 along the filter of complements of finite subsets of I ; in particular, the set of $i \in I$ such that $\lambda_i \neq 0$ is then countable.

2) We shall see later (cf. IV, p. 149, Theorem 1) that when $p = 2$ every compact normal endomorphism of the Hilbertian space $\ell_{\mathbf{C}}^2(I)$ is of the form $u = w \circ v \circ w^{-1}$ where v is a compact diagonal endomorphism and w is a unitary endomorphism (EVT, V, p. 40) of the Hilbert space $\ell_{\mathbf{C}}^2(I)$.*

III.2.3 Linear mappings with values in a space of continuous functions defined by a kernel

Here we take $K = \mathbb{C}$. Let X and Y be locally compact topological spaces. Let μ be a complex measure on X and k a mapping from $X \times Y$ into $\mathbb{C}$ possessing the following two properties:

- (i) For every $y \in Y$, the function $k_y: x \mapsto k(x, y)$ of X into $\mathbb{C}$ is μ -integrable;
- (ii) The map $y \mapsto k_y$ from Y into $\mathcal{L}_\mathbb{C}^1(X, \mu)$ is continuous.

Equip the space $\mathcal{C}(Y)$ of complex-valued continuous functions on Y with the topology of compact convergence. For f in $\mathcal{L}_\mathbb{C}^\infty(X, \mu)$, one defines $\tilde{k}(f): Y \rightarrow \mathbb{C}$ by

$$\tilde{k}(f)(y) = \int_X k(x, y) f(x) d\mu(x) = \int_X k_y f d\mu. \quad (1)$$

The map $h \mapsto \int_X h f d\mu$ is a continuous linear form on $\mathcal{L}_\mathbb{C}^1(X, \mu)$, so condition (ii) shows that the function $\tilde{k}(f)$ is continuous.

PROPOSITION 2. — The map $\tilde{k}$ from $\mathcal{L}_\mathbb{C}^\infty(X, \mu)$ into $\mathcal{C}(Y)$ is compact.

For any y and y' in Y , we have

$$|\tilde{k}(f)(y)| \leq \|k_y\|_1 \|f\|_\infty, \quad |\tilde{k}(f)(y) - \tilde{k}(f)(y')| \leq \|k_y - k_{y'}\|_1 \|f\|_\infty.$$

By Ascoli's theorem (TG, X, p. 18, cor. 2), it follows that the image under $\tilde{k}$ of the unit ball of $\mathcal{L}_\mathbb{C}^\infty(X; \mu)$ is relatively compact in $\mathcal{C}(Y)$.

COROLLARY. — Let X be a compact topological space, Y a locally compact topological space, μ a complex measure on X and k a continuous map from $X \times Y$ into $\mathbb{C}$. Then formula (1) defines a compact linear map, still denoted $\tilde{k}$, from $\mathcal{C}(X)$ endowed with the topology of uniform convergence into $\mathcal{C}(Y)$ endowed with the topology of compact convergence.

Let us verify conditions (i) and (ii) for k . For every $y \in Y$, the map k_y is continuous on the compact space X , hence μ -integrable. Moreover, the map $y \mapsto k_y$ is continuous from Y into $\mathcal{C}(X)$ since X is compact (TG, X, p. 28, th. 3). Since $\|k_y - k_{y'}\|_1 \leq \|\mu\| \|k_y - k_{y'}\|$ for all $(y, y') \in Y^2$ this mapping from Y into $\mathcal{L}_\mathbb{C}^1(X, \mu)$ is continuous. By prop. 2, the mapping $\tilde{k}$ from $\mathcal{L}_\mathbb{C}^\infty(X, \mu)$ into $\mathcal{C}(Y)$ is compact. By composition with the canonical mapping $\mathcal{C}(X) \rightarrow$

$\mathcal{L}_C^\infty(X, \mu)$, one obtains a compact linear mapping from $\mathcal{C}(X)$ into $\mathcal{C}(Y)$.

III.2.4 Linear mappings between Lebesgue spaces defined by a kernel

Here we take $K = \mathbb{C}$. Let X be a locally compact topological space equipped with a positive measure μ . For every $r \in [1, +\infty]$, we denote $\mathcal{L}^r(X) = \mathcal{L}_C^r(X, \mu)$ and $L^r(X) = L_C^r(X, \mu)$. When $r \in [1, +\infty[$, one identifies the dual of $L^r(X)$ with $L^{r'}(X)$ where r' is the conjugate exponent of r such that $1/r + 1/r' = 1$ (INT, V, p. 61, § 5, no. 8, th. 4). We denote by $\|f\|_r$ the norm (or semi-norm) of $f \in L^r(X)$ (or $f \in \mathcal{L}^r(X)$).

Recall that a subset A of X is said to be μ -moderate (INT, V, p. 4, § 1, n° 2) if it is contained in the union of a sequence of μ -integrable sets, and that a function f defined on X with values in a vector space or in $\overline{\mathbf{R}}$ is μ -moderate if it vanishes on the complement of a μ -moderate subset of X . If $1 \leq p < +\infty$ and $f \in \mathcal{L}^p(X)$, then f is moderate (INT, V, p. 9, § 1, no. 3, cor.). If f is μ -moderate, then fg is μ -moderate for every function g .

Let p and q be elements of $]1, +\infty[$. Let X and Y be locally compact topological spaces, equipped with positive measures μ and ν , respectively. One equips the space $X \times Y$ (resp. $Y \times X$) with the product measure $\mu \otimes \nu$ (resp. $\nu \otimes \mu$).

We denote $\mathcal{N}^{p,q}(X \times Y, \mu \otimes \nu)$, or simply $\mathcal{N}^{p,q}(X \times Y)$, the set of functions k measurable with respect to $(\mu \otimes \nu)$ from $X \times Y$ into $\mathbb{C}$ such that there exists a real number $C \geq 0$ satisfying

$$\int_{X \times Y}^* |k(x, y)f(x)g(y)| d(\mu \otimes \nu)(x, y) \leq C \|f\|_p \|g\|_q \quad (2)$$

for all functions $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^q(Y)$. We then denote by $\|k\|_{p,q}$ the lower bound of the set of real numbers C having this property.

Remark. — Let $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^q(Y)$. Since $p < +\infty$ and $q < +\infty$, the functions f and g are moderate, and the function defined by $(x, y) \mapsto f(x)g(y)$ is $(\mu \otimes \nu)$ -moderate (cf. INT, V, p. 92, § 8, n°3, cor. 2). Consequently, the function $(x, y) \mapsto k(x, y)f(x)g(y)$ is $(\mu \otimes \nu)$ -moderate, and in particular the upper integral of its absolute value coincides with the essential upper integral (INT, V, p. 6, § 1, n°3, prop. 7).

The set $\mathcal{N}^{p,q}(X \times Y)$ is a vector subspace of the space of functions from $X \times Y$ into $\mathbb{C}$ and the mapping defined by $k \mapsto \|k\|_{p,q}$ is a semi-norm on $\mathcal{N}^{p,q}(X \times Y)$.

Lemma 1. — Let $k \in \mathcal{N}^{p,q}(X \times Y)$. The function k is locally $(\mu \otimes \nu)$ -integrable.

The map k is $(\mu \otimes \nu)$ -measurable by hypothesis. Let A be a compact subset of $X \times Y$. There exist compact subsets $B \subset X$ and $C \subset Y$ such that $A \subset B \times C$. Applying (2) to the characteristic functions φ_B of B and φ_C of C , we obtain

$$\begin{aligned} & \int_{X \times Y}^* |k(x, y)| \varphi_A(x, y) d(\mu \otimes \nu)(x, y) \\ & \leq \int_{X \times Y}^* |k(x, y)| \varphi_B(x) \varphi_C(y) d(\mu \otimes \nu)(x, y) < +\infty, \end{aligned}$$

whence the result (INT, V, p. 41, §5, n° 1, prop. 1 and def. 1).

PROPOSITION 3. — a) Let $k \in \mathcal{N}^{p,q}(X \times Y)$. For $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^q(Y)$, the function

$$(x, y) \mapsto k(x, y)f(x)g(y)$$

is $(\mu \otimes \nu)$ -integrable and there exists a unique map u_k from $L^p(X)$ into $L^{q'}(Y)$ such that

$$\langle g, u_k(f) \rangle = \int_{X \times Y} k(x, y)f(x)g(y) d(\mu \otimes \nu)(x, y)$$

for all $f \in L^p(X)$ and $g \in L^q(Y)$. The map u_k is linear and continuous, its norm is $\leq \|k\|_{p,q}$;

b) Let $k \in \mathcal{N}^{p,q}(X \times Y)$. We have $u_k = 0$ if and only if k is locally $(\mu \otimes \nu)$ -negligible;

c) The map which to k associates u_k is a continuous linear map from the semi-normed space

$\mathcal{N}^{p,q}(X \times Y)$ to the Banach space $\mathcal{L}(L^p(X); L^{q'}(Y))$, such that $\|u_k\| \leq \|k\|_{p,q}$ for all $k \in \mathcal{N}^{p,q}(X \times Y)$.

Let $k \in \mathcal{N}^{p,q}(X \times X)$. By definition, the function from $X \times Y$ into $\mathbf{C}$ defined by $(x, y) \mapsto k(x, y)f(x)g(y)$ is $(\mu \otimes \nu)$ -integrable when $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^q(Y)$, and the map

$$b_k: (f, g) \mapsto \int_{X \times Y} k(x, y)f(x)g(y) d(\mu \otimes \nu)(x, y)$$

is a continuous bilinear map from $L^p(X) \times L^q(Y)$ to $\mathbf{C}$. For every $f \in L^p(X)$, there therefore exists a unique $h \in L^{q'}(Y)$ such that $b_k(f, g) = \langle g, h \rangle$ for all $g \in L^q(Y)$; denote it $u_k(f) = h$.

By (2), we have $\|u_k(f)\|_{q'} \leq \|k\|_{p,q} \|f\|_p$; the map u_k is linear and continuous with norm $\leq \|k\|_{p,q}$. Moreover, the map $k \mapsto u_k$ is linear, and assertions a) and c) follow from this.

Let us prove b). Suppose that k is locally $(\mu \otimes \nu)$ -negligible. Let $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^q(Y)$; the function defined on $X \times Y$ by $(x, y) \mapsto k(x, y)f(x)g(y)$ is $(\mu \otimes \nu)$ -moderate. Since it is locally $(\mu \otimes \nu)$ -negligible, it is negligible (INT, V, p. 7, § 1, n°2, cor. 1). Thus, we have $\langle u_k(f), g \rangle = b_k(f, g) = 0$. It follows that $u_k = 0$.

Conversely, let $k \in \mathcal{N}^{p,q}(X \times Y)$ be such that $u_k = 0$. The function k is locally $(\mu \otimes \nu)$ -integrable (Lemma 1). For all functions $f \in \mathcal{H}(X)$ and $g \in \mathcal{H}(Y)$, we have

$$\langle g, u_k(f) \rangle = \int_{X \times Y} f(x)g(y)k(x, y)d(\mu \otimes \nu)(x, y),$$

therefore the measure $k \cdot (\mu \otimes \nu)$ on $X \times Y$ is zero (cf. INT, III, p. 82, § 4, no. 1, Th. 1). This means that k is locally $(\mu \otimes \nu)$ -negligible (INT, V, p. 46, §5, no. 3, Cor. 2).

DEFINITION 1. — Let $k \in \mathcal{N}^{p,q}(X \times Y)$. We say that the map u_k is the integral operator with kernel k from $L^p(X)$ into $L^{q'}(Y)$.

For $k \in \mathcal{N}^{p,q}(X \times Y)$ and $f \in \mathcal{L}^p(X)$, we will sometimes also denote by $u_k(f)$ the image under u_k of the class of f in $L^p(X)$.

PROPOSITION 4. — Let $k \in \mathcal{N}^{p,q}(X \times Y)$ and $f \in \mathcal{L}^p(X)$. The mapping defined on X by $x \mapsto k(x, y)f(x)$ is μ -integrable for every y outside a locally ν -negligible set, and $u_k(f)$ coincides with the class of the function

$$y \mapsto \int_X k(x, y)f(x)d\mu(x)$$

defined locally ν -almost everywhere.

Let Y' denote the set of $y \in Y$ such that the mapping $x \mapsto k(x, y)f(x)$ is not μ -integrable. Let $y \in Y$ and let C be a compact neighborhood of y , whose characteristic function we denote by φ . By condition (2) applied to f and φ , the function $(x, y) \mapsto k(x, y)f(x)\varphi(y)$ is integrable on $X \times Y$ with respect to the measure $\mu \otimes \nu$. By the Lebesgue–Fubini theorem (INT, V, p. 96, § 8, n°4, th. 1, a)), the function $x \mapsto k(x, y)f(x)\varphi(y)$ is μ -integrable for y outside a ν -negligible set Y_C . Since $Y' \cap C \subset Y_C$, it follows that Y' is locally ν -negligible in Y (INT, IV, p. 172, §

5, n°2, def. 3).

Let h denote the mapping defined locally ν -almost everywhere on Y by $y \mapsto \int k(x, y)f(x)d\mu$. It is locally ν -integrable (INT, V, loc.

cit.). Let $g \in \mathcal{H}(Y)$. We have

$$\begin{aligned}\langle g, u_k(f) \rangle &= \int_{X \times Y} k(x, y)f(x)g(y)d(\mu \otimes \nu)(x, y) \\ &= \int_Y \left(\int_X k(x, y)f(x)g(y)d\mu(x) \right) d\nu(y) \\ &= \int_Y g(y)h(y)d\nu(y)\end{aligned}$$

(INT, V, loc. cit.). We therefore have $u_k(f) = h$ locally ν -almost everywhere (cf. INT, V, p. 46, § 5, n°3, cor. 2).

PROPOSITION 5. — The space $\mathcal{L}^{p'}(X \times Y)$ is contained in $\mathcal{N}^{p,p}(X \times Y)$. The linear map from $\mathcal{L}^{p'}(X \times Y)$ into $\mathcal{L}(L^p(X); L^{p'}(Y))$ defined by $k \mapsto u_k$ induces, by passing to the quotient, a continuous injective linear map from $L^{p'}(X \times Y)$ into $\mathcal{L}(L^p(X); L^{p'}(Y))$.

Let $k \in \mathcal{L}^{p'}(X \times Y)$. The function k is measurable (INT, IV, p. 84, § 5, n° 6, th. 5). Let $f \in \mathcal{L}^p(X)$ and $g \in \mathcal{L}^p(Y)$. The function $h: (x, y) \mapsto f(x)g(y)$ belongs to $\mathcal{L}^p(X \times Y)$ and $\|h\|_p = \|f\|_p \|g\|_p$ (INT, V, p. 95, § 8, n° 3, cor. 2). The function hk is therefore integrable on $X \times Y$ and we have

$$(\mu \otimes \nu)^*(|hk|) \leq \|k\|_{p'} \|h\|_p = \|k\|_{p'} \|f\|_p \|g\|_p$$

by Hölder's inequality (INT, IV, p. 208, § 6, n° 4, th. 2), which proves that $k \in \mathcal{N}^{p,p}(X \times Y)$ with $\|k\|_{p,p} \leq \|k\|_{p'}$.

Since $p' \neq +\infty$, every function $k \in \mathcal{L}^{p'}(X \times Y)$ is moderate (INT, V, p. 9, § 1, n° 3, cor.), and the last assertion then follows from prop. 3, b) and c).

For any function $k: X \times Y \rightarrow \mathbb{C}$ and for every $y \in Y$, we denote by k_y the mapping from X into $\mathbb{C}$ given by $k_y(x) = k(x, y)$, and we denote by $k_\circ$ the mapping from Y into $\mathcal{F}(X; \mathbb{C})$ defined by $y \mapsto k_y$. We shall likewise denote the mapping from Y into the space of functions defined μ -almost everywhere on X that associates with y the class of k_y .

For any function k from $X \times Y$ into $\mathbb{C}$, we denote

$$N_{p',q'}(k) = \left(\int_Y N_{p'}(k_y)^{q'} d\nu(y) \right)^{1/q'}.$$

This is an element of $[0, +\infty]$.

Lemma 2. — For every $k \in \mathcal{L}^{p'}(X \times Y)$, we have $N_{p',p'}(k) = \|k\|_{p'}$.

Let $k \in \mathcal{L}^{p'}(X \times Y)$. By INT, V, p. 96, § 8, no. 4, th. 1, a), the set Y' of elements $y \in Y$ such that the function $|k_y|^{p'}$ is not μ -integrable is ν -negligible; moreover, the function defined by $y \mapsto \mu(|k_y|^{p'})$ on $Y - Y'$ and equal to zero on Y' is ν -integrable and satisfies

$$\|k\|_{p'}^{p'} = \int_{X \times Y} |k(x, y)|^{p'} d(\mu \otimes \nu)(x, y) = \int_Y \mu(|k_y|^{p'}) d\nu = N_{p',p'}(k)^{p'}.$$

We denote $\mathcal{L}^{p',q'}(X, Y, \mu, \nu)$, or simply $\mathcal{L}^{p',q'}(X, Y)$, the complex vector space of functions $(\mu \otimes \nu)$ -measurable k from $X \times Y$ into $\mathbb{C}$ such that, for ν -almost all $y \in Y$, the map k_y belongs to $\mathcal{L}^{p'}(X)$, and such that the map $k_\circ$ belongs to $\mathcal{L}_{L^{p'}(X)}^{q'}(Y, \nu)$ (INT, IV, p. 129, § 3, no. 4, def. 2). We equip $\mathcal{L}^{p',q'}(X, Y)$ with the seminorm $k \mapsto N_{p',q'}(k)$; note that this then satisfies $N_{p',q'}(k) = \|k_\circ\|_{q'}$, where $k_\circ$ is viewed as a mapping with values in the Banach space $L^{p'}(X)$.

PROPOSITION 6. — a) The space $\mathcal{L}^{p'}(X \times Y)$ is contained in $\mathcal{L}^{p',p'}(X, Y)$ and for $k \in \mathcal{L}^{p'}(X \times Y)$, we have $N_{p',p'}(k) = \|k\|_{p'}$;

b) The space $\mathcal{L}^{p',q'}(X, Y)$ is contained in $\mathcal{N}^{p,q}(X \times Y)$ and for $k \in \mathcal{L}^{p',q'}(X, Y)$, we have

$$\|k\|_{p,q} \leq N_{p',q'}(k).$$

Let us prove assertion a). Let $k \in \mathcal{L}^{p'}(X \times Y)$. By INT, V, p. 96, § 8, no. 4, th. 1, a), the function k_y belongs to $\mathcal{L}^{p'}(X)$ for ν -almost all $y \in Y$.

Let $\varepsilon > 0$. There exists $h \in \mathcal{H}(X \times Y)$ such that $\|k - h\|_{p'} < \varepsilon$. The function $h_\circ$ is a continuous function with compact support from Y into $L^{p'}(X)$ satisfying

$$\|k_\circ - h_\circ\|_{p'} = N_{p',p'}(k - h) = \|k - h\|_{p'} < \varepsilon$$

by Lemma 2. It follows that $k_\circ$ belongs to $\mathcal{L}_{L^{p'}(X)}^{p'}(Y, \nu)$, that is, k belongs to $\mathcal{L}^{p',p'}(X, Y)$. Lemma 2 then shows that $N_{p',p'}(k) = \|k\|_{p'}$.

Let us prove assertion *b*). Let $k \in \mathcal{L}^{p',q'}(X, Y)$ and let $f \in \mathcal{L}^p(X)$. For every $y \in Y$, Hölder's inequality (INT, IV, p. 208, § 6, no. 4, th. 2) applied to k_y implies

$$\int_X^* |k(x, y)f(x)|d\mu(x) \leq \|k_y\|_{p'}\|f\|_p.$$

Let $g \in \mathcal{L}^{q'}(Y)$. The maps f and g are moderate, since p and q' are finite. Consequently, the map $(x, y) \mapsto f(x)g(y)$ is $(\mu \otimes \nu)$ -moderate (cf. INT, V, p. 92, § 8, no. 3, cor. 2), and hence the map $(x, y) \mapsto k(x, y)f(x)g(y)$ is $(\mu \otimes \nu)$ -moderate. According to INT, V, p. 93, § 8, no. 3, prop. 7, a), it then follows that

$$\begin{aligned} & \int_{X \times Y}^* |k(x, y)f(x)g(y)|d(\mu \otimes \nu) \\ &= \int_Y^* |g(y)| \left(\int_X^* |k(x, y)f(x)|d\mu(x) \right) d\nu(y) \\ &\leq \|f\|_p \int_Y^* \|k_y\|_{p'} |g(y)| d\nu(y) \leq N_{p',q'}(k) \|f\|_p \|g\|_{q'}, \end{aligned}$$

using Hölder's inequality once again. This concludes the proof.

PROPOSITION 7. — Let $k \in \mathcal{L}^{p',q'}(X, Y)$. The linear map u_k from $L^p(X)$ into $L^{q'}(Y)$ is compact.

Suppose first that k is such that $k_o \in \mathcal{H}(Y; \mathcal{L}^{p'}(X))$ and denote by A the support of k_o . Let F be the subspace of $\mathcal{H}(Y, A; \mathcal{L}^{p'}(X, \mu))$ generated by the functions $y \mapsto f_2(y)f_1$, where $f_2 \in \mathcal{H}(Y, A)$ and $f_1 \in \mathcal{L}^{p'}(X)$. When $m \in \mathcal{H}(Y; \mathcal{L}^{p'}(X))$ satisfies $m_o \in F$, the linear map u_m is of finite rank, hence compact. For m_1 and m_2 in $\mathcal{H}(Y, A; \mathcal{L}^{p'}(X))$ we also have

$$\begin{aligned} \|u_{m_1} - u_{m_2}\| &\leq \|m_1 - m_2\|_{p',q'} = \left(\int_Y \|m_{1,y} - m_{2,y}\|_{p'}^{q'} d\nu(y) \right)^{1/q'} \\ &\leq \nu(A)^{1/q'} \sup_{y \in Y} N_{p'}(m_{1,y} - m_{2,y}). \end{aligned}$$

Since the space F is dense in $\mathcal{H}(Y, A; L^{p'}(X))$ for the topology of uniform convergence on A (INT, III, p. 41, § 1, n° 1, prop. 1 and INT, III, p. 46, § 1, n° 2, prop. 5), one concludes that the linear map u_k is the limit of a sequence of finite-rank linear maps. It is therefore compact (cor. 1 of III, p. 4).

Consider the general case. Let $k \in \mathcal{L}^{p',q'}(X, Y)$. For every $\varepsilon > 0$ there exists a map $k_{\varepsilon,o} \in \mathcal{H}(Y; \mathcal{L}^{p'}(X))$ such that

$$\|k_{\circ} - k_{\varepsilon, \circ}\|_{q'} < \varepsilon.$$

The corresponding function $k_{\varepsilon} : X \times Y \rightarrow \mathbb{C}$ then satisfies

$$\|u_k - u_{k_{\varepsilon}}\| \leq \|k - k_{\varepsilon}\|_{p,q} \leq N_{p',q'}(k - k_{\varepsilon}) = \|k_{\circ} - k_{\varepsilon, \circ}\|_{q'} < \varepsilon.$$

Since $u_{k_{\varepsilon}}$ is compact by the preceding, the map u_k is also compact (prop. 2 of III, p. 4).



Set $p = 1$ and suppose $q > 1$. There may exist mappings $k : X \times Y \rightarrow \mathbb{C}$ such that $k_y \in \mathcal{L}^{\infty}(X)$ for all $y \in Y$, and such that the integral

$$\int_Y^* \|k_y\|_{\infty}^{q'} d\nu(y)$$

is finite (in particular, condition (2) of III, p. 27 is valid), but such that the linear mapping u_k from $L^1(X)$ into $L^{q'}(Y)$ is not compact (exercise 2 of III, p. 119).

In particular, one deduces from the proposition the following corollary:

COROLLARY 1 (Hilbert–Schmidt). — Let $k \in \mathcal{L}^2(X \times Y)$. The linear map u_k is compact from $L^2(X)$ into $L^2(Y)$.

By prop. 5, one has $k \in \mathcal{N}^{2,2}(X \times Y)$, so the linear map u_k is defined (prop. 3). One has $k \in \mathcal{L}^{2,2}(X \times Y)$ (prop. 6, a), hence u_k is compact (prop. 7).

Remark. — In the case $p = q = 2$, it is generally more convenient to express the characterization of the integral operator u_k using the scalar product: it is the unique map from $L^2(X)$ into $L^2(Y)$ such that

$$\langle g | u_k(f) \rangle = \int_{X \times Y} k(x, y) f(x) \overline{g(y)} d(\mu \otimes \nu)(x, y)$$

for all $f \in L^2(X)$ and $g \in L^2(Y)$. Moreover, the adjoint of u_k is the linear map u_{k^*} , where $k^* \in L^2(Y \times X)$ satisfies $k^*(y, x) = \overline{k(x, y)}$ for almost every $(y, x) \in Y \times X$. Indeed, for all

$f \in L^2(X)$ and $g \in L^2(Y)$, we have

$$\langle g | u_k(f) \rangle = \int_{X \times Y} k(x, y) \overline{g(y)} f(x) d(\mu \otimes \nu)(x, y) = \langle u_{k^*}(g) | f \rangle.$$

COROLLARY 2. — Let G be a compact topological group equipped with a Haar measure μ . Let p be a real number such that $1 < p < +\infty$ and let q be the conjugate exponent of p . Let $f \in \mathcal{L}^q(G)$. Set $k(x, y) = f(x^{-1}y)$ for all $(x, y) \in G \times G$.

- a) We have $k \in \mathcal{L}^{q,p}(G, G)$;
- b) For $\varphi \in L^p(G)$ the convolution $\varphi * f$ belongs to $L^p(G)$ and the linear map $v_f : \varphi \mapsto \varphi * f$ of $L^p(G)$ into itself is continuous. It coincides with the endomorphism u_k . In particular, the linear map v_f is compact.

Since G is compact, the function f belongs to $\mathcal{L}^1(G)$. The linear map v_f is therefore defined and continuous according to INT, VIII, p. 167, § 4, n° 5, prop. 13.

The map k is $(\mu \otimes \mu)$ -measurable, and we have $\|k_y\|_q = \|f\|_q$ for all $y \in G$. Since the measure μ is bounded, it follows that

$$\int_G \|k_y\|_q^p d\mu(y) = \mu(G) \|f\|_q^p,$$

hence $k \in \mathcal{L}^{q,p}(G, G)$. The linear map u_k is therefore a compact linear map from $L^p(G)$ into $L^p(G)$ (prop. 7).

Let $\varphi \in \mathcal{K}(G)$. By INT, VIII, p. 166, § 4, no. 5, prop. 11 and prop. 4 of III, p. 29, we then have

$$\begin{aligned} v_f(\varphi)(y) &= \int_G \varphi(x) f(x^{-1}y) d\mu(x) \\ &= \int_G \varphi(x) k(x, y) d\mu(x) = u_k(\varphi)(y) \end{aligned}$$

for almost every $y \in G$. This implies that $v_f = u_k$.

III.2.5 Restriction of differentiable mappings

Let n and r be positive integers, let U be an open subset of R^n , and let F be a Banach space. Denote by $\mathcal{C}^r(U; F)$ the vector space of maps of class C^r from U into F , endowed with the topology of C^r compact convergence. Recall that this is the supremum of the topologies of C^r -uniform convergence on K (VAR, R2, 12.3.10, p. 56), as K ranges over the set of compact subsets of U . The space $\mathcal{C}^0(U; F)$ is none other than the space $\mathcal{C}(U; F)$ of continuous maps from U into F , equipped with the topology of compact convergence.

Lemma 3. — Let A be the set of multi-indices $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq r$ and let u be the linear map $f \mapsto (\partial^\alpha f)_{\alpha \in A}$ from $\mathcal{C}^r(U; F)$ into $\mathcal{C}(U; F)^A$.

- a) The map u is injective, continuous, strict, and its image is closed.
- b) The topological vector space $\mathcal{C}^r(U; F)$ is complete.

The map u is linear and injective. It is continuous and strict by definition of the topology of $\mathcal{C}^r(U; F)$.

Let $(g_\alpha)_{\alpha \in A}$ be a point of $\mathcal{C}(U; F)^A$ adherent to the image of u . There exists a filter $\mathcal{F}$ on $\mathcal{C}^r(U; F)$ such that one has $g_\alpha = \lim_{f, \mathcal{F}} \partial^\alpha f$ in $\mathcal{C}(U; F)$ for all $\alpha \in A$. Let m be an integer such that $0 \leq m \leq r$. By induction on m , it follows from th. 1 of FVR, II, p. 2, that the map g_0 is of class C^m and that $g_\alpha = \partial^\alpha g_0$ for all $\alpha \in \mathbf{N}^n$ with $|\alpha| \leq m$. Thus g_0 belongs to the space $\mathcal{C}^r(U; F)$ and its image under u is $(g_\alpha)_{\alpha \in A}$. This proves that the image of u is closed, and thus establishes assertion a).

The assertion a) implies that the space $\mathcal{C}^r(U; F)$ is isomorphic to its image in $\mathcal{C}(U; F)^A$; since this image is closed and the space $\mathcal{C}(U; F)^A$ is complete (TG, X, p. 9, cor. 3 of th. 2 and TG, II, p. 17, prop. 10), the space $\mathcal{C}^r(U; F)$ is complete.

PROPOSITION 8. — Suppose that F is finite-dimensional. Let s be an integer such that $0 \leq s < r$ and let V be a relatively compact open subset of U . The linear map $f \mapsto f|_V$ from $\mathcal{C}^r(U; F)$ into $\mathcal{C}^s(V; F)$ is compact.

Let W be the set of functions $f \in \mathcal{C}^r(U; F)$ whose partial derivatives of order $\leq r$ take at every point of $\bar{V}$ a value of norm less than 1. The set W is a neighborhood of 0 in the space $\mathcal{C}^r(U; F)$. Let α be a multi-index such that $|\alpha| \leq s$. Consider the set H of functions of the

form $(\partial^\alpha f)|V$ for f in W . The set H is an equicontinuous subset of $\mathcal{C}(V; F)$ by the mean value theorem (VAR, R, 2.2.3). Moreover, for every $x \in V$, the image of H under the map $g \mapsto g(x)$ is a bounded subset, hence relatively compact, of F . By the Ascoli theorem (TG, X, p. 18, cor. 2), the set H is relatively compact in $\mathcal{C}(V; F)$. This proves that the linear map $f \mapsto (\partial^\alpha f)|V$ from $\mathcal{C}^r(U; F)$ into $\mathcal{C}(V; F)$ is compact. The proposition then follows from Lemma 3 and Remarks 5 and 6 of III, p. 3.

III.2.6 Restriction of differentiable sections of a vector bundle

Let r be a positive integer, X a differentiable manifold of class C^r locally of finite dimension, and E a vector bundle (real or complex),

with base X and of class C^r . For every open set U of X , denote by $\mathcal{S}^r(U; E)$ the vector space (denoted $\mathcal{S}_E^r(U)$ in VAR, R1, 7.4, p. 74) of C^r sections of E over U , equipped with the topology of C^r compact convergence.

Lemma 4. — Let $\mathcal{U}$ be an open cover of X . The map $u : f \mapsto (f|U)_{U \in \mathcal{U}}$ from $\mathcal{S}^r(X; E)$ into $\prod_{U \in \mathcal{U}} \mathcal{S}^r(U; E)$ is linear, injective, continuous, and strict. Its image is closed.

The map u is linear, injective, and continuous. By virtue of TG, IX, p. 43, prop. 1 and p. 48, cor. 1, every compact subset of X has a finite cover $(C_i)_{i \in I}$ such that, for each $i \in I$, the set C_i is a compact subset of one of the open sets of the covering $\mathcal{U}$. It follows that the linear map u is strict. Finally, its image is closed, since it consists of the families $(f_U)_{U \in \mathcal{U}}$ such that f_U and f_V coincide in $U \cap V$ for all U and V in $\mathcal{U}$.

PROPOSITION 9. — The space $\mathcal{S}^r(X; E)$ is complete.

Suppose first that there exists an integer $n \geq 0$ and a Banach space F such that X is an open subset of R^n and E is the trivial vector bundle $X \times F$ with base X and fiber F . In this case, the topological vector space $\mathcal{S}^r(X; E)$ is isomorphic to $\mathcal{C}^r(X; F)$, and the result follows from lemma 3 of III, p. 34.

In the general case, let $\mathcal{U}$ be an open covering of X by coordinate domains such that for every $U \in \mathcal{U}$, the restriction of E to U is trivializable. By the lemma above, the space $\mathcal{S}^r(X; E)$ is isomorphic to the image of the linear map $f \mapsto (f|U)_{U \in \mathcal{U}}$, which is closed in the product of the spaces $\mathcal{S}^r(U; E)$ for U in $\mathcal{U}$. By the preceding case, each of the spaces $\mathcal{S}^r(U; E)$ is complete,

hence their product is complete (TG, II, p. 17, prop. 10). Consequently, $\mathcal{S}^r(X; E)$ is complete.

Remark. — Let $\mathcal{U}$ be an open covering of X by coordinate domains $c_U = (U, \varphi_U, \mathbf{R}^{n_U})$ such that for every $U \in \mathcal{U}$, the restriction of E to U is trivializable of type F_U , where F_U is a Banach space. For every $U \in \mathcal{U}$, one identifies a section f of E over U with a map f_U from $\varphi_U(U)$ into F_U . It follows from the proof of prop. 9 that the topology of the space $\mathcal{S}^r(X; E)$ is defined by the family of seminorms $p_{U,C,\alpha}$ such that

$$p_{U,C,\alpha}(f) = \sup_{x \in C} \|(\partial^\alpha f_U)(x)\|,$$

where U ranges over $\mathcal{U}$, C ranges over the set of compact subsets of $\varphi_U(U)$ and $\alpha \in \mathbf{N}^{n_U}$ ranges over the set of multi-indices such that $|\alpha| \leq r$.

PROPOSITION 10. — Suppose that the vector bundle E is locally of finite rank. Let s be an integer such that $0 \leq s < r$ and Y a relatively compact open subset of X . The linear map $f \mapsto f|Y$ from $\mathcal{S}^r(X; E)$ into $\mathcal{S}^s(Y; E)$ is compact.

Suppose first that there exists an integer $n \geq 0$ and a finite-dimensional Banach space F such that X is an open subset of $\mathbf{R}^n$ and E is the trivial vector bundle $X \times F$ with base X and fiber F . In this case, the topological vector spaces $\mathcal{S}^r(X; E)$ and $\mathcal{S}^s(Y; E)$ are isomorphic to $\mathcal{C}^r(X; F)$ and $\mathcal{C}^s(Y; F)$, respectively, and prop. 10 is a consequence of prop. 8.

Let us pass to the general case. Let C be a compact subset of X containing Y . For every point x of C , choose an open neighborhood $U(x)$ of x in X which is a coordinate domain over which the vector bundle E is trivializable. Choose moreover a relatively compact open neighborhood $V(x)$ of x in $U(x)$. Since the set C is compact, it is covered by a finite family $(V(x_1), \dots, V(x_m))$ of such open sets. For every i , set $U_i = U(x_i)$ and $Y_i = V(x_i) \cap Y$. Then Y_i is a relatively compact open subset of U_i and we have $Y = Y_1 \cup \dots \cup Y_m$. The linear maps $f \mapsto f|U_i$ from $\mathcal{S}^r(X; E)$ into $\mathcal{S}^r(U_i; E)$ are continuous and the linear maps $g \mapsto g|Y_i$ from $\mathcal{S}^r(U_i; E)$ into $\mathcal{S}^s(Y_i; E)$ are compact by the first part of the proof. The linear maps $f \mapsto f|Y_i$ from $\mathcal{S}^r(X; E)$ into $\mathcal{S}^s(Y_i; E)$ are therefore compact (prop. 3 of III, p. 5). Taking into account Lemma 4, applied to the covering of Y by the open sets Y_i , and Remarks 5 and 6 of III, p. 3, the linear map $f \mapsto f|Y$ from $\mathcal{S}^r(X; E)$ into $\mathcal{S}^s(Y; E)$ is compact, which completes the proof.

III.2.7 Restriction of analytic sections of a vector bundle

Let X be a complex analytic manifold locally of finite dimension and E an analytic vector bundle with base X (VAR, p. 35 and VAR, p. 70). Denote by $\mathcal{S}^\omega(X; E)$ the vector space of analytic sections of E over X , equipped with the topology of compact convergence.

Lemma 5. — Let X_0 be the underlying real analytic manifold of X and E_0 the complex vector bundle over X_0 underlying E .

- a) The vector subspace $\mathcal{S}^\omega(X; E)$ of $\mathcal{S}^0(X_0; E_0)$ is closed, and the injection of $\mathcal{S}^\omega(X; E)$ into $\mathcal{S}^0(X_0; E_0)$ is continuous and strict;
- b) The canonical injection of $\mathcal{S}^\omega(X; E)$ into $\mathcal{S}^1(X_0; E_0)$ is continuous.

Suppose first that there exists an integer $n \geq 0$ and a Banach space F such that X is an open subset of C^n and E is the trivial vector bundle $X \times F$. In this case, the topological vector spaces $\mathcal{S}^\omega(X; E)$, $\mathcal{S}^0(X_0; E_0)$ and $\mathcal{S}^1(X_0; E_0)$ are isomorphic to $\mathcal{C}^\omega(X; F)$, $\mathcal{C}^0(X_0; F)$ and $\mathcal{C}^1(X_0; F)$ respectively, and the space $\mathcal{S}^0(X_0; E_0)$ is metrizable (TG, X, p. 20, cor. of prop. 1). The lemma then follows from VAR, 3.3.2, p. 28.

Let $\mathfrak{F}$ be a filter on $\mathcal{S}^\omega(X; E)$ which converges in the space $\mathcal{S}^0(X_0; E_0)$ toward a limit f . We must show that f belongs to $\mathcal{S}^\omega(X; E)$ and that the filter $\mathfrak{F}$ converges to f in the space $\mathcal{S}^1(X_0; E_0)$. This statement is local in nature, and therefore follows from the first part of the proof.

PROPOSITION 11. — Suppose that the vector bundle E is locally of finite rank. Let Y be a relatively compact open subset of X . The restriction map $f \mapsto f|_Y$ from $\mathcal{S}^\omega(X; E)$ into $\mathcal{S}^\omega(Y; E)$ is compact.

With the notations of Lemma 5, we have a commutative diagram

$$\begin{array}{ccc} \mathcal{S}^\omega(X; E) & \xrightarrow{i} & \mathcal{S}^1(X_0; E_0) \\ \downarrow u & & \downarrow v \\ \mathcal{S}^\omega(Y; E) & \xrightarrow{j} & \mathcal{S}^0(Y_0; E_0) \end{array} \quad (3)$$

where i and j are the canonical injections and u, v the restriction maps. The map i is continuous (Lemma 5, b)), and the map v is compact (Prop. 10). Since the canonical injection j is

continuous, strict, and has closed image (Lemma 5, a)), the map u is compact (Remark 5 of III, p. 3).

COROLLARY. — Let X be a compact complex analytic manifold and E an analytic vector bundle over X , locally of finite rank. The vector space $\mathcal{S}^\omega(X; E)$ is of finite dimension.

Being compact, the analytic manifold X is locally of finite dimension. By prop. 11, the identity map of $\mathcal{S}^\omega(X; E)$ is a compact linear map. This implies that the space $\mathcal{S}^\omega(X; E)$ is of finite dimension (Remark 3 of III, p. 2).

III.3 FREDHOLM ENDOMORPHISMS AND RIESZ ENDOMORPHISMS

III.3.1 Strict morphisms and finite-rank linear maps

PROPOSITION 1. — Let E and F be locally convex spaces, E_1 a closed subspace of finite codimension of E and u a continuous linear map from E into F . For u to be a strict morphism with closed image, it is necessary and sufficient that $u|_{E_1}$ be one.

Suppose first that the linear map $u|_{E_1}$ is injective. We then have $E_1 \cap \text{Ker}(u) = \{0\}$ so that $\text{Ker}(u)$ is finite-dimensional. Let S be a vector subspace of E supplementary to $E_1 + \text{Ker}(u)$. The space E is the topological direct sum of E_1 , $\text{Ker}(u)$ and S (EVT, I, p. 15, cor. 4 and prop. 3). If u is a strict morphism with closed image, it defines by restriction an isomorphism from $E_1 \oplus S$ onto a closed vector subspace of F , and a fortiori an isomorphism from E_1 onto a closed vector subspace of F . Conversely, suppose that u defines by restriction an isomorphism from E_1 onto $u(E_1)$ and that $u(E_1)$ is closed in F . We have $u(E) = u(E_1) \oplus u(S)$. It follows that $u(E)$ is the topological direct sum of $u(E_1)$ and $u(S)$, and is closed in F (loc. cit.). Since $u(E_1)$ is closed and since we have $u(E_1) \cap u(S) = \{0\}$, the space $u(S)$ is separated and u defines by restriction an isomorphism from S onto $u(S)$ (EVT, I, p. 15, cor. of prop. 3), hence also of $E_1 \oplus S$ onto $u(E)$. This proves that u is a strict morphism with closed image.

Moving to the general case, set $N = E_1 \cap \text{Ker}(u)$ and $G = E/N$. The locally convex space E_1/N is identified with a closed vector subspace of finite codimension G_1 of G . Let us denote

by $v: G \rightarrow F$ the continuous linear map deduced from u by passing to the quotient. For u (resp. $u|_{E_1}$) to be a strict morphism with closed image, it is necessary and sufficient that v (resp. $v|_{G_1}$) be one. This reduces us to the case already treated.

COROLLARY 1. — Suppose F is separated. Let $v \in \mathcal{L}^f(E; F)$. If u is a strict morphism with closed image from E into F , then the same holds for $u + v$.

Since F is separated, the kernel of v is a closed vector subspace of E ; it has finite codimension in E and $u + v$ has the same restriction as u on $\text{Ker}(v)$. The corollary thus follows from the proposition.

COROLLARY 2. — Let T be a separated finite-dimensional vector subspace of F . Denote by $\pi: F \rightarrow F/T$ the canonical surjection. For u to be a strict morphism with closed image, it is necessary and sufficient that $\pi \circ u$ be one.

The identity map of T onto itself extends to a continuous map $q: F \rightarrow T$ (EVT, II, p. 26, remark). The kernel S of q is a closed topological complement of T . Let $p: F \rightarrow F$ be the projector with image S associated with the decomposition $F = T \oplus S$. For $\pi \circ u$ to be a strict morphism with closed image, it is necessary and sufficient that $p \circ u$ be one. Moreover, $p \circ u$ and u have the same restriction to $u^{-1}(S)$, which is a closed vector subspace of finite codimension in E , and the assertion follows from the proposition.

III.3.2 Fredholm Applications

Let E and F be locally convex spaces. In this section, we denote by $u \equiv v$ the congruence modulo $\mathcal{L}^f(E; F)$ of the elements u and v of $\mathcal{L}(E; F)$.

If G is a locally convex space, and if u' and v' are elements of $\mathcal{L}(F; G)$, the relations $u \equiv v$ and $u' \equiv v'$ imply the relation $u' \circ u \equiv v' \circ v$.

We say that an element w of $\mathcal{L}(F; E)$ is a quasi-inverse of the element u of $\mathcal{L}(E; F)$ if we have $w \circ u \equiv 1_E$ and $u \circ w \equiv 1_F$.

Suppose that w is a quasi-inverse of u . If u_1 is an element of $\mathcal{L}(E; F)$ and w_1 is an element of $\mathcal{L}(F; E)$ such that $u_1 \equiv u$ and $w_1 \equiv w$, then w_1 is a quasi-inverse of u_1 since $w \circ u \equiv w_1 \circ u_1$ and $u \circ w \equiv u_1 \circ w_1$.

If w and w_1 are quasi-inverses of u , then $w_1 \equiv w$ since

$$w_1 = 1_E \circ w_1 \equiv w \circ u \circ w_1 \equiv w \circ 1_F = w.$$

DEFINITION 1. — Let E and F be locally convex spaces. We say that an element u of $\mathcal{L}(E; F)$ is a Fredholm map ⁽¹⁾ if it possesses a quasi-inverse. A Fredholm map from E to E is called a Fredholm endomorphism of E .

We shall denote by $\mathcal{F}(E; F)$ the set of Fredholm maps from E to F , and $\mathcal{F}(E)$ the set of Fredholm endomorphisms of E .

Remarks. — Let E, F , and G be locally convex spaces, $u : E \rightarrow F$ and $v : F \rightarrow G$ continuous linear maps.

- 1) Suppose that u is a Fredholm map and denote by u_1 a quasi-inverse of u . Since u is a quasi-inverse of u_1 , the map u_1 is a Fredholm map.
- 2) Suppose that u and v are Fredholm maps, and let u_1 (resp. v_1) be a quasi-inverse of u (resp. v). Then $v \circ u$ is a Fredholm map from E to G and $u_1 \circ v_1$ is a quasi-inverse of $v \circ u$. Indeed, one computes

$$(u_1 \circ v_1) \circ (v \circ u) = u_1 \circ (v_1 \circ v) \circ u \equiv u_1 \circ 1_F \circ u = u_1 \circ u \equiv 1_E,$$

$$(v \circ u) \circ (u_1 \circ v_1) = v \circ (u \circ u_1) \circ v_1 \equiv v \circ 1_F \circ v_1 = v \circ v_1 \equiv 1_G.$$

- 3) Suppose that u and $v \circ u$ are Fredholm maps, and let w_1 be a quasi-inverse of $v \circ u$. Then v is a Fredholm map and $u \circ w_1$ is a quasi-inverse of v .

Indeed, w_1 is a Fredholm operator by the first remark. Let u_1 be a quasi-inverse of u ; by the second remark, $u \circ w_1$ is a Fredholm map and $(v \circ u) \circ u_1$ is a quasi-inverse. We have $u \circ u_1 \equiv 1_F$, whence $v \circ u \circ u_1 \equiv v$; this proves the assertion.

- 4) Suppose that v and $v \circ u$ are Fredholm maps, and let w_1 be a quasi-inverse of $v \circ u$. Then u is a Fredholm operator and $w_1 \circ v$ is a quasi-inverse of u .

The proof is analogous to that of the preceding remark.

Lemma 1. — Let E and F be locally convex spaces and u a Fredholm map from E into F . The kernel and cokernel of u are finite-dimensional.

Let $v: F \rightarrow E$ be a quasi-inverse of u . The kernel of u is contained in the image of the finite-rank linear map $1_E - v \circ u$, hence is finite-dimensional. The image of u contains the kernel of the finite-rank linear map $1_F - u \circ v$, hence is of finite codimension in F .

PROPOSITION 2. — Let E and F be separated locally convex spaces and let u be an element of $\mathcal{L}(E; F)$. The following properties are equivalent:

- (i) The map u is a Fredholm operator;
- (ii) The map u is a strict morphism, its kernel is finite-dimensional, its image is closed and of finite codimension in F ;
- (iii) There exist closed vector subspaces E_1 of E and F_1 of F , both of finite codimension, such that u induces, by passing to subspaces, an isomorphism from E_1 onto F_1 ;
- (iv) There exist decompositions into topological direct sums $E = E_1 \oplus E_2$ and $F = F_1 \oplus F_2$ such that E_2 and F_2 are finite-dimensional, that u vanishes on E_2 and defines by passing to subspaces an isomorphism of E_1 onto F_1 .
- (v) $\Rightarrow$ (iii): Let v be a quasi-inverse of u , E_1 the kernel of $1_E - v \circ u$ and F_1 that of $1_F - u \circ v$. Since the linear maps $1_E - v \circ u$ and $1_F - u \circ v$ are continuous and of finite rank, E_1 and F_1 are closed vector subspaces of finite codimension of E and F , respectively. Let $x \in E_1$. We have

$$(1_F - u \circ v)(u(x)) = u((1_E - v \circ u)(x)) = u(0) = 0,$$

whence $u(x) \in F_1$. We therefore have $u(E_1) \subset F_1$; we likewise have $v(F_1) \subset E_1$. The continuous linear maps $u_1: E_1 \rightarrow F_1$ and $v_1: F_1 \rightarrow E_1$ deduced from u and v are then mutually inverse isomorphisms, since $v \circ u$ and 1_E (resp. $u \circ v$ and 1_F) coincide on E_1 (resp. on F_1).

- (iii) $\Rightarrow$ (ii): Let E_1 and F_1 satisfy the hypothesis of (iii). We have $E_1 \cap \text{Ker}(u) = \{0\}$ and $F_1 \subset \text{Im}(u)$, hence $\text{Ker}(u)$ is finite-dimensional and $\text{Im}(u)$ is closed and of finite codimension

in F . It follows from prop. 1 of III, p. 39 that the map u is a strict morphism.

(iv) $\Rightarrow$ (iv): Suppose condition (ii) is satisfied. The closed vector subspace $E_2 = \text{Ker}(u)$ of E is finite-dimensional, and there exists a vector subspace E_1 of E topologically complementary to E_2 (EVT, II, p. 27, cor. 2). The vector subspace $F_1 = \text{Im}(u)$ is closed and of finite codimension, and it admits a topological complement F_2 in F . By prop. 1 of III, p. 39, the map $u|_{E_1}$ is a strict morphism, hence u induces an isomorphism of E_1 onto F_1 .

(v) $\Rightarrow$ (i): Under the hypotheses of (iv), the linear map from F to E that coincides with $u|_{F_1}^{-1}$ on F_1 and is zero on F_2 is a quasi-inverse of u .

Remark 5. — Let E, F be separated locally convex spaces and $u: E \rightarrow F$ a Fredholm operator. If u is bijective, then u is an isomorphism (indeed, u is a strict morphism by Prop. 2, (ii)).

III.3.3 Index of a Fredholm map

Let E, F , and G be locally convex spaces.

Let $u: E \rightarrow F$ be a Fredholm operator. The vector spaces $\text{Ker}(u)$ and $\text{Coker}(u)$ are of finite dimension (Lemma 1 of III, p. 41). Recall that the integer

$$\dim \text{Coker}(u) - \dim \text{Ker}(u) = \text{codim}_F \text{Im}(u) - \dim \text{Ker}(u) \quad (1)$$

is called the index of u and is denoted $\text{ind}(u)$ (A, V, p. 126).

If $u: E \rightarrow F$ and $v: F \rightarrow G$ are Fredholm maps, then so is $v \circ u$ (III, p. 41, remark 2), and one has (A, V, p. 127, lemma 2)

$$\text{ind}(v \circ u) = \text{ind}(v) + \text{ind}(u). \quad (2)$$

Suppose E and F are separated, and let $u: E \rightarrow F$ be a Fredholm operator; let us adopt the notations of condition (iv) of Prop. 2 of III, p. 42. We then have $\text{ind}(u) = \dim(F_2) - \dim(E_2)$. Equip the dual of each of these spaces with the weak topology (resp. the compact-open topology, the topology of bounded convergence). Then E' is identified with the topological direct sum of E'_1 and E'_2 , and F' is identified with that of F'_1 and F'_2 , and ${}^t u$ induces an isomorphism of F'_1

onto E'_1 and vanishes on F'_2 . Thus the transpose ${}^t u: F' \rightarrow E'$ is a Fredholm map (loc. cit.). The kernel of ${}^t u$ is F'_2 , and its dimension is that of F_2 , that is, that of the cokernel of u . Hence

$$\text{ind}(u) = \dim \text{Ker}({}^t u) - \dim \text{Ker}(u). \quad (3)$$

Moreover, the image of ${}^t u$ is E'_1 , and the dimension of the cokernel of ${}^t u$ is therefore equal to the dimension of E'_2 , which is that of the kernel E_2 of u . We deduce from this

$$\text{ind}({}^t u) = -\text{ind}(u). \quad (4)$$

Suppose E and F are separated, and let $u: E \rightarrow F$ be a Fredholm operator of index 0. Then u is a strict morphism by prop. 2 of III, p. 42. Since $\dim \text{Ker}(u) = \text{codim}_F \text{Im}(u)$, the map u is an isomorphism as soon as it is injective or surjective.

PROPOSITION 3. — Let E and F be locally convex spaces and $u \in \mathcal{L}(E; F)$. Let E_1 (resp. F_1) be a closed subspace of finite codimension of E (resp. F). Suppose that u maps E_1 into F_1 and denote by $u_1 \in \mathcal{L}(E_1; F_1)$ the map which coincides with u on E_1 . For u to be a Fredholm operator, it is necessary and sufficient that u_1 be one. We then have

$$\text{ind}(u) - \text{ind}(u_1) = \text{codim}_F(F_1) - \text{codim}_E(E_1). \quad (5)$$

Let $i: E_1 \rightarrow E$ and $j: F_1 \rightarrow F$ be the canonical injections. These are Fredholm maps, and one has

$$\text{ind}(i) = \text{codim}_E(E_1), \quad \text{ind}(j) = \text{codim}_F(F_1).$$

Since $j \circ u_1 = u \circ i$, one sees that u is a Fredholm map if and only if u_1 is one (remarks 3 and 4 of III, p. 41). If this is so, we have

$$\text{ind}(j) + \text{ind}(u_1) = \text{ind}(j \circ u_1) = \text{ind}(u \circ i) = \text{ind}(u) + \text{ind}(i),$$

whence formula (5).

PROPOSITION 4. — Let E and F be separated locally convex spaces, $u: E \rightarrow F$ a Fredholm

operator, and $\hat{u} : \hat{E} \rightarrow \hat{F}$ the extension of u to the completions. Then $\hat{u}$ is a Fredholm operator, and one has $\text{Ker}(\hat{u}) = \text{Ker}(u)$ and $\text{ind}(\hat{u}) = \text{ind}(u)$.

Let us adopt the notation of condition (iv) of Prop. 2 of III, p. 42. Since the vector spaces E_2 and F_2 are finite-dimensional, they are complete. The completion of E_1 (resp. F_1) is identified with the closure of E_1 in $\hat{E}$ (resp. of F_1 in $\hat{F}$), and $\hat{E}$ (resp. $\hat{F}$) is the topological direct sum of $\hat{E}_1$ and E_2 (resp. $\hat{F}_1$ and F_2). The linear map $\hat{u}$ defines by restriction an isomorphism of $\hat{E}_1$ onto $\hat{F}_1$ and vanishes on E_2 . The proposition then follows from implication (iv) $\Rightarrow$ (i) of loc. cit.

III.3.4 Riesz endomorphisms

Let E be a vector space and let u be an endomorphism of E .

The sequence $(\text{Ker}(u^n))_{n \in \mathbb{N}}$ of vector subspaces of E is increasing; their union is a subspace of E stable under u , called the nilspace of u .

The sequence $(\text{Im}(u^n))_{n \in \mathbb{N}}$ of vector subspaces of E is decreasing; their intersection is a subspace of E stable under u , called the conilspace of u .

Lemma 2. — Let E be a vector space and let u be an endomorphism of E which has an index (A, V, p. 126).

If two of the following properties are satisfied, then so is the third:

- (i) The endomorphism u has index 0;
- (ii) The nilspace N of u is finite-dimensional;
- (iii) The conilspace I of u has finite codimension.

For every integer $n \geq 0$, the endomorphism u^n has an index, equal to $n \text{ind}(u)$ (A, V, p. 127, lemma 2). The sequence $(\dim(\text{Ker}(u^n)))_{n \in \mathbb{N}}$ of natural integers is increasing; for it to be stationary, it is necessary and sufficient that the nilspace of u be finite-dimensional. The sequence $(\text{codim}_E(\text{Im}(u^n)))_{n \in \mathbb{N}}$ of natural integers is increasing; for it to be stationary, it is necessary and sufficient that the conilspace of u be finite-dimensional. Suppose condition (i) is satisfied; the relation

$$\dim(\text{Ker}(u^n)) - \text{codim}_E(\text{Im}(u^n)) = \text{ind}(u^n) = n \text{ind}(u) = 0$$

then implies the equivalence of conditions (ii) and (iii). Conversely, if conditions (ii) and (iii) are satisfied, this formula implies that the sequence $(n \operatorname{ind}(u))_{n \in \mathbb{N}}$ is stationary, so u has index zero.

Lemma 3. — Let E be a vector space and let u be an endomorphism of E which has an index. Suppose that the index of u is zero. Let N be the nilspace of u and I its conilspace.

- a) The space E is then the direct sum of the subspaces N and I (Weyr–Fitting decomposition, cf. A, VIII, §2, no. 2, p. 25).
- b) The endomorphism u induces, by passing to subspaces, a nilpotent endomorphism of N and an automorphism u_I of I .
- c) Let v denote the endomorphism of E which is zero on N and which coincides with u_I^{-1} on I . We have $u \circ v = v \circ u$ and the endomorphism $1_E - u \circ v$ is the projector with image N and kernel I .
- d) Every endomorphism of E that commutes with u stabilizes N and I , and commutes with v .

The endomorphism u satisfies the properties of the preceding lemma.

Let $n \in \mathbb{N}$ such that $\operatorname{Ker}(u^n) = N$. Then $\operatorname{Ker}(u^m) = \operatorname{Ker}(u^n)$ for every integer $m \geq n$; since $\operatorname{ind}(u^m) = m \operatorname{ind}(u) = 0$, this entails $\operatorname{Im}(u^m) = \operatorname{Im}(u^n)$ for $m \geq n$, hence $I = \operatorname{Im}(u^n)$. It follows from A, VIII, p. 25, prop. 2 that assertions a) and b) are satisfied.

For every $x \in N$, we have $v(x) = 0$ and $u(x) \in N$, hence $u(v(x)) = 0$ and $v(u(x)) = 0$. For every $x \in I$, we have $u(v(x)) = v(u(x)) = x$. This implies that $u \circ v = v \circ u$ and that $1_E - u \circ v$ is the projector with image N and kernel I .

Let w be an endomorphism of E that commutes with u . Let $n \in \mathbb{N}$ such that $N = \operatorname{Ker}(u^n)$ and $I = \operatorname{Im}(u^n)$. Let $x \in N$; we have $u^n(w(x)) = w(u^n(x)) = 0$, hence $w(x) \in N$. Let $x \in I$; there exists $y \in E$ such that $x = u^n(y)$; then, $w(x) = w(u^n(y)) = u^n(w(y)) \in I$. This proves that w stabilizes N and I .

Let us finally prove that w and v are permutable. For every x in N , we have $v(w(x)) = 0 = w(v(x))$. Let $x \in I$; the elements $v(w(x))$ and $w(v(x))$ of E belong to I and their images under

u are both equal to $w(x)$, hence they are equal. By linearity, this implies that v and w commute.

Remark. — Suppose there exists an integer $n \geq 0$ such that $\text{Ker}(u^n) = \text{Ker}(u^{n+1})$ (resp. $\text{Im}(u^n) = \text{Im}(u^{n+1})$); one then has $\text{Ker}(u^m) = \text{Ker}(u^n)$ (resp. $\text{Im}(u^m) = \text{Im}(u^n)$) for every integer $m \geq n$.

Suppose that u has index zero. For every integer $n \geq 0$, we have

$$n \text{ ind}(u) = \text{ind}(u^n) = \text{codim}_E(\text{Im}(u^n)) - \dim(\text{Ker}(u^n)).$$

The conditions $\text{Ker}(u^n) = \text{Ker}(u^{n+1})$ and $\text{Im}(u^n) = \text{Im}(u^{n+1})$ are therefore equivalent. When they are satisfied, the vector space E is the direct sum of $\text{Ker}(u^n)$ and of $\text{Im}(u^n)$ (A, VIII, p. 25, prop. 2), the space $\text{Ker}(u^n)$ is finite-dimensional, and u induces, by passage to subspaces, an automorphism of $\text{Im}(u^n)$.

DEFINITION 2. — Let E be a separated locally convex space. A Riesz endomorphism of E is any Fredholm endomorphism of E whose nilspace is finite-dimensional and whose conilspace is of finite codimension.

PROPOSITION 5. — Let E be a separated locally convex space. Every Riesz endomorphism of E has index zero.

This follows from the definition and Lemma 2.

Examples. — 1) Every automorphism of E is a Riesz endomorphism of E . If E is finite-dimensional, every element of $\mathcal{L}(E)$ is a Riesz endomorphism of E .

2) If $(E_i)_{i \in I}$ is a finite family of separated locally convex spaces and u_i is an element of $\mathcal{L}(E_i)$ for all i in I , the endomorphism $u = \bigoplus_{i \in I} u_i$ of the locally convex space $\bigoplus_{i \in I} E_i$ is a Riesz endomorphism if and only if u_i is one for all i .

3) Let E be a separated locally convex space over $\mathbb{R}$, and let $E_{(\mathbb{C})}$ be its complexification. Let u be an endomorphism of E . For u to be a Riesz endomorphism, it is necessary and sufficient that its complexification $u_{(\mathbb{C})}$ be a Riesz endomorphism of $E_{(\mathbb{C})}$.

PROPOSITION 6. — Let E be a separated locally convex space and let u be a Riesz endomorphism of E . Denote by N the nilspace of u and by I its conilspace.

a) The subspaces N and I of E are closed and stable under u , and the space E is their topo-

logical direct sum.

- b) The endomorphism u defines by restriction an automorphism u_I of I ;
- c) The vector space N is finite-dimensional, and u defines by restriction a nilpotent endomorphism u_N of N ;
- d) Let v be the element of $\mathcal{L}(E)$ that is zero on N and coincides with u_I^{-1} on I . It is a Riesz endomorphism of E and a quasi-inverse of u that commutes with u . The endomorphism $1_E - u \circ v$ of E is the projector with image N and kernel I . Every element of $\mathcal{L}(E)$ which commutes with u stabilizes N and I , and commutes with v .

Let n be a natural number such that $\text{Ker}(u^n) = N$ and $\text{Im}(u^n) = I$. Since E is separated and u is continuous, $N = \text{Ker}(u^n)$ is closed in E . Since u is a Fredholm endomorphism, so is u^n , and $I = \text{Im}(u^n)$ is closed (III, p. 42, prop. 2). By lemma 2, the space E is the direct sum of N and I , and assertion a) then follows from prop. 3 of EVT, I, p. 15.

Since I is closed, of finite codimension in E , and stable under u , the map $u_I : I \rightarrow I$ induced by u is a Fredholm endomorphism (III, p. 44, prop. 3); it is also bijective, hence it is an automorphism of I (III, p. 42, prop. 2). This proves b), and the remaining assertions follow from lemma 2.

The endomorphism v defined in assertion d) of the proposition is a quasi-inverse of u , which is called the canonical quasi-inverse of u .

Riesz endomorphisms enjoy certain of the stability properties of Fredholm maps.

PROPOSITION 7. — Let E be a separated locally convex space and let u be an endomorphism of E .

- a) Let E_1 be a closed subspace of finite codimension of E , stable under u . Let u_1 be the element of $\mathcal{L}(E_1)$ which coincides with u on E_1 . Then u is a Riesz endomorphism if and only if u_1 is one;
- b) Suppose that u is a Fredholm endomorphism of index 0. Let $\widehat{E}$ be the completion of E , and let $\widehat{u} \in \mathcal{L}(\widehat{E})$ be the continuous extension of u . Then u is a Riesz endomorphism if and only if $\widehat{u}$ is one;

- c) Endow the dual E' of E with the topology of bounded convergence, the topology of compact convergence, or the weak topology. If u is a Riesz endomorphism of E , then ${}^t u$ is a Riesz endomorphism of E' .
- a) For u to be a Fredholm endomorphism of index zero, it is necessary and sufficient that u_1 be one (III, p. 44, prop. 3). On the other hand, for every $n \in \mathbb{N}$, we have $\text{Ker}(u_1^n) = E_1 \cap \text{Ker}(u^n)$, whence

$$\dim \text{Ker}(u_1^n) \leq \dim \text{Ker}(u^n) \leq \dim \text{Ker}(u_1^n) + \text{codim}_E(E_1),$$

therefore the sequence $(\dim \text{Ker}(u^n))_{n \in \mathbb{N}}$ is bounded if and only if the sequence $(\dim \text{Ker}(u_1^n))_{n \in \mathbb{N}}$ is. Consequently, u is a Riesz endomorphism of E if and only if u_1 is a Riesz endomorphism of E_1 (lemma 2).

- b) By prop. 4 of III, p. 44, for every $n \in \mathbb{N}$, the map $\hat{u}^n$ is a Fredholm endomorphism of $\hat{E}$ such that $\text{Ker}(\hat{u}^n) = \text{Ker}(u^n)$ and $\text{ind}(\hat{u}^n) = \text{ind}(u^n) = n \text{ind}(u)$. Consequently, u is a Riesz endomorphism if and only if $\hat{u}$ is a Riesz endomorphism.
- c) The transpose ${}^t u$ is a Fredholm endomorphism of index 0 of E' (III, p. 43, no. 3). Since the kernel of $({}^t u)^n$ is the orthogonal complement of the image of u^n (EVT, IV, p. 27, prop. 2), the sequence $(\text{Ker}({}^t u)^n)_n$ is stationary. This proves that ${}^t u$ is a Riesz endomorphism of E' .

PROPOSITION 8. — Let E be a separated locally convex space and u an endomorphism of E . The following conditions are equivalent:

- (i) u is a Riesz endomorphism of E ;
- (ii) There exists a quasi-inverse v of u that commutes with u ;
- (iii) There exists a closed vector subspace E_1 of E , of finite codimension, stable under u , such that u induces an automorphism of E_1 .

Implication (i) $\Rightarrow$ (ii) follows from prop. 6, d).

Suppose that u has a quasi-inverse v which commutes with u . The kernel E_1 of $1_E - u \circ v$

is a closed vector subspace of E , of finite codimension, stable under u and v . Since one has $u(v(x)) = v(u(x)) = x$ for all $x \in E_1$, the maps u and v induce automorphisms of E_1 inverses of each other. Hence (ii) implies (iii).

Finally, the implication (iii) $\Rightarrow$ (i) follows from the assertion a) of Prop. 7 and from the fact that an automorphism is a Riesz endomorphism.

PROPOSITION 9. — Let E be a separated locally convex space, and let u, v be permutable elements of $\mathcal{L}(E)$. The following conditions are equivalent:

- (i) The endomorphisms u and v are Riesz endomorphisms.
- (ii) The endomorphism $u \circ v$ is a Riesz endomorphism.

Suppose that u and v are Riesz endomorphisms; denote by u' and v' their canonical quasi-inverses. The endomorphisms u, v, u' and v' of E commute (prop. 6, d)). The endomorphism $v' \circ u'$ is a quasi-inverse of $u \circ v$ which commutes with $u \circ v$, hence $u \circ v$ is a Riesz endomorphism (prop. 8).

Suppose conversely that $u \circ v$ is a Riesz endomorphism, and let w be its canonical quasi-inverse. Since u commutes with $u \circ v$, the endomorphisms u and w commute by prop. 6, d)). Likewise v and w commute. Hence the endomorphisms u, v and w of E commute; it follows that the endomorphism $v \circ w$ of E is a quasi-inverse of u and that $w \circ u$ is a quasi-inverse of v . By prop. 8, u and v are Riesz endomorphisms.



Let u be a Riesz endomorphism of E . If v is an automorphism of E , the endomorphism $u \circ v$ of E is not necessarily a Riesz endomorphism of E , even if $u \circ v - v \circ u$ has finite rank. If h is a finite-rank endomorphism of E , the endomorphism $u + h$ is not necessarily a Riesz endomorphism (cf. III, p. 120, exercise 3).

PROPOSITION 10. — Let E and F be locally convex spaces, $p \in \mathcal{L}(E; F)$ and $q \in \mathcal{L}(F; E)$. Set $u = 1_E - q \circ p$ and $v = 1_F - p \circ q$.

- a) Let $n \in \mathbb{N}$. The map p defines by restriction an isomorphism of topological vector

spaces from $\text{Ker}(u^n)$ onto $\text{Ker}(v^n)$ and defines by passage to quotients an isomorphism of topological vector spaces from $\text{Coker}(u^n)$ onto $\text{Coker}(v^n)$;

- b) If the image of u is closed in E , that of v is closed in F ;
- c) If $\text{Ker}(u)$ has a topological supplement in E , then $\text{Ker}(v)$ has one in F . If $\text{Im}(u)$ has a topological supplement in E , then $\text{Im}(v)$ has one in F ;
- d) If u is a strict morphism, then v is a strict morphism.
- e) If u is an automorphism of E , then v is an automorphism of F .
- f) If u is a Fredholm endomorphism of E , then v is a Fredholm endomorphism of F , and we have $\text{ind}(u) = \text{ind}(v)$;
- g) Suppose E and F are separated. If u is a Riesz endomorphism of E , then v is a Riesz endomorphism of F .

Let us first note the formulas

$$q \circ v = u \circ q, \quad v \circ p = p \circ u. \quad (6)$$

Let us then prove two lemmas.

Lemma 4. — Let $n \in \mathbb{N}$. Set

$$q_n = \sum_{i=1}^n (-1)^{i-1} \binom{n}{i} q \circ (p \circ q)^{i-1}.$$

One has $u^n = 1_E - q_n \circ p$ and $v^n = 1_F - p \circ q_n$.

One has $(q \circ p)^i = q \circ (p \circ q)^{i-1} \circ p$ for every integer $i \geq 1$. We then compute

$$u^n = (1_E - q \circ p)^n = 1_E - \sum_{i=1}^n (-1)^{i-1} \binom{n}{i} (q \circ p)^i = 1_E - q_n \circ p$$

$$v^n = (1_F - p \circ q)^n = 1_F - \sum_{i=1}^n (-1)^{i-1} \binom{n}{i} (p \circ q)^i = 1_F - p \circ q_n,$$

whence the result.

Lemma 5. — Let u' be an element of $\mathcal{L}(E)$. Set $v' = 1_F + p \circ u' \circ q$. If u' is an inverse of u , then v' is an inverse of v . If u' is a quasi-inverse of u , then v' is a quasi-inverse of v . If u' commutes with u , then v' commutes with v .

Using formulas (6), we compute

$$v - v' \circ v = -p \circ u' \circ q \circ v = -p \circ u' \circ u \circ q$$

$$v - v \circ v' = -v \circ p \circ u' \circ q = -p \circ u \circ u' \circ q,$$

whence

$$1_F - v' \circ v = p \circ (1_E - u' \circ u) \circ q$$

$$1_F - v \circ v' = p \circ (1_E - u \circ u') \circ q,$$

which proves the lemma.

Given Lemma 4, it suffices to prove assertion *a*) for $n = 1$. According to formulas (6), the maps p and q define, by passing to subspaces, continuous linear maps $p' : \text{Ker}(u) \rightarrow \text{Ker}(v)$ and $q' : \text{Ker}(v) \rightarrow \text{Ker}(u)$, and by passing to quotients of continuous linear maps $p'' : \text{Coker}(u) \rightarrow \text{Coker}(v)$ and $q'' : \text{Coker}(v) \rightarrow \text{Coker}(u)$.

Since $1_E - u = q \circ p$ and $1_F - v = p \circ q$, the maps p' and q' on the one hand, p'' and q'' on the other hand, are mutually inverse isomorphisms. This proves *a*).

For the image of a continuous linear map to be closed, it is necessary and sufficient that the cokernel of this map be a separated space. Thus *b*) follows from *a*).

Denote by $i : \text{Ker}(u) \rightarrow E$ and $j : \text{Ker}(v) \rightarrow F$ the canonical injections. Suppose that $\text{Ker}(u)$ has a topological supplement in E , and let $r : E \rightarrow \text{Ker}(u)$ be a continuous linear retraction of i . Set $r' = p' \circ r \circ q$. We have $r' \in \mathcal{L}(F; \text{Ker}(v))$ and

$$r' \circ j = p' \circ r \circ q \circ j = p' \circ r \circ i \circ q' = p' \circ q' = 1_{\text{Ker}(v)},$$

hence r' is a continuous linear retraction of j and $\text{Ker}(v)$ has a topological supplement in F . This proves the first assertion of c). The second is proved analogously, noting that if $s : \text{Coker}(u) \rightarrow E$ is a continuous linear section of the canonical surjection from E onto $\text{Coker}(u)$, then $p \circ s \circ q''$ is a continuous linear section of the canonical surjection of F onto $\text{Coker}(v)$.

Denote by $\bar{u} : E/\text{Ker}(u) \rightarrow \text{Im}(u)$ and $\bar{v} : F/\text{Ker}(v) \rightarrow \text{Im}(v)$ the continuous linear bijective mappings deduced from u and v . By formulas (6), the map p defines, by passing to quotients, a continuous linear map $p_1 : E/\text{Ker}(u) \rightarrow F/\text{Ker}(v)$ and p, q define, by passing to subspaces, continuous linear maps $p_0 : \text{Im}(u) \rightarrow \text{Im}(v)$ and $q_0 : \text{Im}(v) \rightarrow \text{Im}(u)$.

Denote by $t : \text{Im}(v) \rightarrow F/\text{Ker}(v)$ the composite map of the canonical injection $\text{Im}(v) \rightarrow F$ and of the canonical surjection $F \rightarrow F/\text{Ker}(v)$; it is continuous. We have $\bar{v} \circ t = 1_{\text{Im}(v)} - p_0 \circ q_0$ and $\bar{v} \circ p_1 = p_0 \circ \bar{u}$, whence

$$\bar{v} \circ (t + p_1 \circ \bar{u}^{-1} \circ q_0) = \bar{v} \circ t + p_0 \circ q_0 = 1_{\text{Im}(v)}.$$

This proves that $t + p_1 \circ \bar{u}^{-1} \circ q_0$ is the inverse bijection of $\bar{v}$. If u is strict, the map $\bar{u}^{-1}$ is continuous, and so is $\bar{v}^{-1}$ which proves that v is strict. This proves d).

By Lemma 5, if u is an automorphism of E , then v is an automorphism of F . Similarly, if u is a Fredholm endomorphism of E , then v is a Fredholm endomorphism of F , and one then has $\text{ind}(u) = \text{ind}(v)$ according to *a*). This proves *e*) and *f*). Finally, if E and F are separated and u is a Riesz endomorphism of E , then v is a Riesz endomorphism of F by Lemma 5 and condition (ii) of Prop. 8; this is assertion *g*).

III.3.5 Fredholm mappings and Riesz mappings between Fréchet spaces

PROPOSITION 11. — Let E and F be Fréchet spaces. For a continuous linear map u from E into F to be a Fredholm map, it is necessary and sufficient that its kernel and cokernel be of finite dimension.

This follows from the characterization of Fredholm maps given by condition (ii) of Prop. 2 in III, p. 42, and from the following lemma:

Lemma 6. — Let E and F be Fréchet spaces and $u \in \mathcal{L}(E; F)$. We assume that the image of u has finite codimension in F . Then the image of u is closed in F and u is a strict morphism.

Let G be a vector subspace of F supplementary to $\text{Im}(u)$; the subspace G is closed (EVT, I, p. 14, cor. 1), hence the quotient space F/G is a Fréchet space (TG, IX, p. 25, prop. 4). Let $\pi : F \rightarrow F/G$ be the canonical surjection. The map $\pi \circ u$ is surjective, hence strict (EVT, I, p. 19, cor. 3). The assertion then follows from cor. 2 of III, p. 40.

PROPOSITION 12. — Let E and F be Fréchet spaces. Endow their duals E' and F' with the weak topology, the topology of compact convergence, or the topology of bounded convergence. Let $u \in \mathcal{L}(E; F)$. For u to be a Fredholm map from E to F , it is necessary and sufficient that ${}^t u$ be a Fredholm map from F' to E' .

The condition is necessary (III, p. 43, no. 3). Let us prove that it is sufficient. If ${}^t u$ is a Fredholm map, it is a strict morphism (III, p. 42, prop. 2), hence u is a strict morphism with closed image (EVT, IV, p. 28, th. 1 and p. 29, cor. 3). Taking into account the canonical isomorphisms $\text{Coker}({}^t u) \rightarrow \text{Ker}(u)'$ and $\text{Ker}({}^t u) \rightarrow \text{Coker}(u)'$ (EVT, IV, p. 27, prop. 2), the vector spaces $\text{Ker}(u)$ and $\text{Coker}(u)$ are finite-dimensional and u is a Fredholm operator (prop. 2 of III, p. 42).

PROPOSITION 13. — Let E be a Fréchet space and $u \in \mathcal{L}(E)$.

- a) The endomorphism u is a Riesz endomorphism if and only if its nilspace N is finite-dimensional and its conilespace I is of finite codimension;
 - b) Endow E' with the topology of bounded convergence, the topology of compact convergence, or the weak topology. If ${}^t u$ is a Riesz endomorphism of E' , then u is a Riesz endomorphism of E .
- a) Suppose that the nilspace N is finite-dimensional and that the conilespace I is of finite codimension. Since $\text{Ker}(u) \subset N$ and $I \subset \text{Im}(u)$, the kernel of u is finite-dimensional and the image of u is of finite codimension. By the definition and proposition 11, the map u is a Riesz endomorphism. The converse follows from the definition.
- b) By proposition 12, the hypothesis implies that u is a Fredholm endomorphism of index $\text{ind}({}^t u) = -\text{ind}(u) = 0$ (no 3, formula (4)). Let $n \in \mathbb{N}$. Since the image of u^n is closed and has as its orthogonal the kernel of ${}^t u^n$ (EVT, IV, p. 27, prop. 2), the sequence $(\text{Im}(u^n))$ is stationary, and u is therefore a Riesz endomorphism.

III.3.6 Spectral characterization of Riesz endomorphisms

Let E be a complex Banach space and u an endomorphism of E . Recall that we denote by $\text{Sp}(u)$ the spectrum of u relative to the unital Banach algebra $\mathcal{L}(E)$ (cf. § 7 of I, p. 127).

Suppose that 0 is an isolated point of $\text{Sp}(u)$; recall that we then denote by $e_0(u)$ the spectral projector associated with u and with the open and closed subset $\{0\}$ of the spectrum of u (cf. n°3 of I, p. 131). We have $e_0(u) = f(u)$ for every germ of holomorphic function f in the neighborhood of $\text{Sp}(u)$ which equals 1 in a neighborhood of 0 and vanishes in a neighborhood of $\text{Sp}(u) \setminus \{0\}$.

By passing to subspaces, the endomorphism u induces a quasi-nilpotent endomorphism of the image of $e_0(u)$, whose spectrum is reduced to $\{0\}$, and an automorphism of the kernel of $e_0(u)$, whose spectrum is $\text{Sp}(u) \setminus \{0\}$ (loc. cit.).

PROPOSITION 14. — Let E be a complex Banach space. For an element u of $\mathcal{L}(E)$ to be a Riesz endomorphism of E , it is necessary and sufficient that one of the following two mutually exclusive conditions be satisfied:

- (i) The endomorphism u of E is an automorphism of E ;
- (ii) The point 0 is an isolated point of $\text{Sp}(u)$ and the projector $e_0(u)$ has finite rank.

When condition (ii) is satisfied, the image of $e_0(u)$ is the nilspace of u and the kernel of $e_0(u)$ is the conilespace of u .

Every automorphism of E is a Riesz endomorphism of E (III, p. 47, remark 1). If u satisfies condition (ii), the kernel F of $e_0(u)$ is a closed vector subspace of finite codimension of E , stable under u . Let u_F be the endomorphism of F deduced from u . Its spectrum is contained in $\mathbb{C} \setminus \{0\}$ (n°3 of I, p. 131), hence u_F is an automorphism of F . It follows that u is a Riesz endomorphism of E (III, p. 48, prop. 8).

Conversely, let u be a Riesz endomorphism of E . Denote by N its nilspace and by I its conilespace. By prop. 6 of III, p. 47, the space E is the topological direct sum of N and I , and the map u defines by passage to subspaces a nilpotent endomorphism u_N of N and an automorphism u_I of I . In particular, we have $\text{Sp}(u_N) \subset \{0\}$ and $0 \notin \text{Sp}(u_I)$. If N is zero, then $I = E$ and u is an automorphism of E . Otherwise, 0 is an isolated point of $\text{Sp}(u)$ and $e_0(u)$ is the projector with

image N and kernel I (prop. 2 of I, p. 129); in particular, it is of finite rank.

Remark. — Let E be a real Banach space and let u be an endomorphism of E . Suppose that 0 is an isolated point of the complex spectrum of u , that is, of the spectrum of the endomorphism $1 \otimes u$ of the complete normable complexified space $E_{(\mathbb{C})}$ of E (I, p. 85, n° 13). The subset $\{0\}$ of the complex spectrum of u is open and closed, and invariant under conjugation; denote by $e_0(u) \in \mathcal{L}(E)$ the spectral projector associated with it (n° 13 of I, p. 85). Proposition 14 holds mutatis mutandis in this setting.

III.4 PERTURBATIONS IN BANACH SPACES

III.4.1 Direct morphisms

DEFINITION 1. — Let E be a topological vector space. A vector subspace of E is said to be direct if it admits a topological complement.

For a vector subspace F of E to be direct, it is necessary and sufficient that there exist a projector of image F in $\mathcal{L}(E)$. If p is such a projector, E is the topological direct sum of F and the kernel of p . The kernel of $1_E - p$ is F ; if E is separated, then F is therefore closed.

Let F be a direct vector subspace of E and E_1 a vector subspace of E containing F ; then F is a direct vector subspace of E_1 .

Every closed vector subspace of finite codimension in E is a direct summand (EVT, I, p. 15, prop. 3), and every vector subspace F of E contained in the closure of $\{0\}$ is direct (indeed, there exists a projector with image F , and such a projector is necessarily continuous).

PROPOSITION 1. — Let E be a locally convex space. Every finite-dimensional vector subspace of E is complemented.

Let F be a finite-dimensional vector subspace of E . Let S denote a topological complement of the closure of $\{0\}$ in F , which is a direct subspace of E . The space $F_2 = F \cap S$ is separated and of finite dimension; there therefore exists an integer $n \geq 0$ such that F_2 is isomorphic to K^n (EVT, I, p. 14, th. 2). There then exists a continuous linear map $p : S \rightarrow F_2$ extending the identity map of F_2 . The kernel of p is a topological supplement of F_2 in S , and also of F in E .

DEFINITION 2. — Let E and F be locally convex spaces, and let u be a continuous linear

map from E into F . One says that u is a direct morphism if u is a strict morphism whose kernel is a direct vector subspace of E and whose image is a direct vector subspace of F .

Let $u \in \mathcal{L}(E; F)$. For the kernel and image of u to be direct vector subspaces of E and F , respectively, it is necessary and sufficient that there exist decompositions $E = E_1 \oplus E_2$ and $F = F_1 \oplus F_2$ into topological direct sums such that u be represented by a matrix $\begin{pmatrix} u_1 & 0 \\ 0 & 0 \end{pmatrix}$ where $u_1 \in \mathcal{L}(E_1; F_1)$ is bijective. Since the kernel of u is then E_2 , the canonical linear map from E_1 onto E/E_2 is an isomorphism, and the image of u is F_1 , one sees that u is a strict morphism if and only if u_1 is an isomorphism of E_1 onto F_1 . Denote by v the element of $\mathcal{L}(F; E)$ represented by the matrix $\begin{pmatrix} u_1^{-1} & 0 \\ 0 & 0 \end{pmatrix}$. Then the linear map $u \circ v$ is the projector in F with kernel F_2 and image F_1 , the linear map $v \circ u$ is the projector in E with kernel E_2 and image E_1 , and one has $u \circ v \circ u = u$.

Conversely, one has the following result:

PROPOSITION 2. — Let E and F be locally convex spaces, let $u \in \mathcal{L}(E; F)$ and $v \in \mathcal{L}(F; E)$. Suppose that $u = u \circ v \circ u$. Then u is a direct morphism. Moreover, $v \circ u$ is a continuous projector in E with kernel $\text{Ker}(u)$ and $u \circ v$ is a continuous projector in F with image $\text{Im}(u)$.

Set $p = v \circ u$ and $q = u \circ v$. We have

$$p^2 = v \circ (u \circ v \circ u) = v \circ u = p, q^2 = (u \circ v \circ u) \circ v = u \circ v = q,$$

hence p and q are projectors. They are continuous. Let E_1 and E_2 be the image and the kernel of p , and let F_1 and F_2 be the image and the kernel of q . Since $u \circ v \circ u = u$, we have

$$\text{Ker}(u) \subset \text{Ker}(v \circ u) \subset \text{Ker}(u \circ v \circ u) = \text{Ker}(u),$$

hence $\text{Ker}(u) = \text{Ker}(v \circ u) = \text{Ker}(p) = E_2$. Similarly, one deduces from the inclusions

$$\text{Im}(u) \supset \text{Im}(u \circ v) \supset \text{Im}(u \circ v \circ u) = \text{Im}(u)$$

that the image of u is F_1 . The spaces E_2 and F_1 are direct vector subspaces of E and F respectively.

The vector space E is the topological direct sum of E_1 and E_2 , and u defines by restriction a bijective linear map u_1 of E_1 onto F_1 . For every $x \in E_1$, we have $v(u(x)) = p(x) = x$, and the map u_1^{-1} of F_1 onto E_1 coincides with v on F_1 . Consequently, u_1 is an isomorphism of E_1 onto F_1 and u is a strict morphism from E into F .

Let E and F be locally convex spaces and $u \in \mathcal{L}(E; F)$. For u to be an injective (resp. surjective) direct morphism, it is necessary and sufficient that there exist a v in $\mathcal{L}(F; E)$ such that $v \circ u = 1_E$ (resp. $u \circ v = 1_F$) (cf. TG, III, p. 47 and 48).

PROPOSITION 3. — Let E and F be Banach spaces. The following sets are open subsets of the Banach space $\mathcal{L}(E; F)$:

- a) The set $\mathcal{I}(E; F)$ of isomorphisms from E onto F ;
- b) The set $\mathcal{MD}(E; F)$ of injective direct morphisms from E into F , and more precisely, for every closed vector subspace F_1 of F , the set $\mathcal{M}_{F_1}(E; F)$ of elements of $\mathcal{MD}(E; F)$ whose image is a topological supplement of F_1 ;
- c) The set $\mathcal{SD}(E; F)$ of surjective direct morphisms from E into F , and more precisely, for every closed vector subspace E_1 of E , the set $\mathcal{E}_{E_1}(E; F)$ of elements of $\mathcal{SD}(E; F)$ whose kernel is a topological supplement of E_1 .

Moreover, the map $u \mapsto u^{-1}$ of $\mathcal{I}(E; F)$ onto $\mathcal{I}(F; E)$ is analytic.

By definition, $\mathcal{I}(E; E)$ is the set of invertible elements of the complete normed algebra $\mathcal{L}(E)$. By TG, IX, p. 40, prop. 14, it is an open subset of $\mathcal{L}(E)$. The map $u \mapsto u^{-1}$ from $\mathcal{I}(E; E)$ into $\mathcal{I}(E; E)$ is analytic (LIE, III, § 1, n° 1, example 2).

If the set $\mathcal{I}(E; F)$ is empty, it is open. Otherwise, let u_0 be an isomorphism from E onto F . The map $v \mapsto u_0 \circ v$ is then an isomorphism of $\mathcal{L}(E)$ onto $\mathcal{L}(E; F)$ which transforms $\mathcal{I}(E; E)$ into $\mathcal{I}(E; F)$. The set $\mathcal{I}(E; F)$ is therefore open in $\mathcal{L}(E; F)$. Moreover, if $u = u_0 \circ v$ is an element of $\mathcal{I}(E; F)$, we have $u^{-1} = v^{-1} \circ u_0^{-1}$, and the map $u \mapsto u^{-1}$ of $\mathcal{I}(E; F)$ onto $\mathcal{I}(F; E)$ is therefore analytic.

Let F_1 be a closed vector subspace of F . For every $u \in \mathcal{L}(E; F)$, let $\bar{u}$ be the element of $\mathcal{L}(E \times F_1; F)$ defined by $\bar{u}(x, y) = u(x) + y$. The map $u \mapsto \bar{u}$ from $\mathcal{L}(E; F)$ into $\mathcal{L}(E \times F_1; F)$ is continuous, and $\mathcal{M}_{F_1}(E; F)$ is the set of elements u of $\mathcal{L}(E; F)$ such that $\bar{u}$ belongs to $\mathcal{I}(E \times$

$F_1; F)$. Since $\mathcal{I}(E \times F_1; F)$ is open in $\mathcal{L}(E \times F_1; F)$

by what precedes, the set $\mathcal{M}_{F_1}(E; F)$ is open in $\mathcal{L}(E; F)$. The same is true of $\mathcal{MD}(E; F)$, which is the union of the $\mathcal{M}_{F_1}(E; F)$ when F_1 ranges over the set of closed vector subspaces of F .

Let E_1 be a closed vector subspace of E , and let p be the canonical map of E onto the quotient Banach space E/E_1 . For an element u of $\mathcal{L}(E; F)$ to belong to $\mathcal{E}_{E_1}(E; F)$, it is necessary and sufficient that the map (u, p) from E into $F \times E/E_1$ belong to $\mathcal{I}(E; F \times E/E_1)$. As before, we deduce that $\mathcal{E}_{E_1}(E; F)$ is open in $\mathcal{L}(E; F)$; the same is true of $\mathcal{ED}(E; F)$, which is the union of the $\mathcal{E}_{E_1}(E; F)$.

III.4.2 Perturbation of Fredholm maps

THEOREM 1. — Let E and F be Banach spaces. The set $\mathcal{F}(E; F)$ of Fredholm operators from E to F is open in the Banach space $\mathcal{L}(E; F)$, and the map $u \mapsto \text{ind}(u)$ of $\mathcal{F}(E; F)$ into $\mathbf{Z}$ is locally constant.

The theorem follows from the following more precise statement:

PROPOSITION 4. — Let E and F be Banach spaces, $u_0: E \rightarrow F$ a Fredholm operator and v_0 a quasi-inverse of u_0 . There exists an open neighborhood U of u_0 in $\mathcal{L}(E; F)$ and an analytic mapping $\varphi: U \rightarrow \mathcal{L}(F; E)$ such that

- (i) One has $\varphi(u_0) = v_0$;
- (ii) For every u in U , the map $\varphi(u)$ is a quasi-inverse of u , and in particular u is a Fredholm map;
- (iii) For every u in U , we have $\text{ind}(u) = \text{ind}(u_0)$.

By prop. 2 of III, p. 42, (ii), there exist topological direct sum decompositions $E = E_1 \oplus E_2$ and $F = F_1 \oplus F_2$, and there exists $\alpha_0 \in \mathcal{I}(E_1; F_1)$ such that E_2 and F_2 are finite-dimensional and u_0 is represented by the matrix $\begin{pmatrix} \alpha_0 & 0 \\ 0 & 0 \end{pmatrix}$ relative to these decompositions.

Let U be the set of elements u of $\mathcal{L}(E; F)$ such that, in the matrix representation $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$

of u with respect to these decompositions, we have $\alpha \in \mathcal{T}(E_1; F_1)$. Since $\mathcal{T}(E_1; F_1)$ is open in $\mathcal{L}(E_1; F_1)$

(prop. 3 of III, p. 57), U is an open neighborhood of u_0 in $\mathcal{L}(E; F)$. For $u = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ in U , set

$$\varphi(u) = v_0 + \begin{pmatrix} \alpha^{-1} - \alpha_0^{-1} & 0 \\ 0 & 0 \end{pmatrix}. \quad (1)$$

One has $\varphi(u_0) = v_0$ and the map φ is analytic (loc. cit.). Modulo continuous linear maps of finite rank, we have the congruences

$$u \equiv \begin{pmatrix} \alpha & 0 \\ 0 & 0 \end{pmatrix}, v_0 \equiv \begin{pmatrix} \alpha_0^{-1} & 0 \\ 0 & 0 \end{pmatrix}, \varphi(u) \equiv \begin{pmatrix} \alpha^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore, $\varphi(u)$ is a quasi-inverse of u . Every element u from U defines by restriction an isomorphism from E_1 onto a topological complement of F_2 in F . By prop. 3 of III, p. 44, we therefore have

$$\begin{aligned} \text{ind}(u) &= \text{codim}_F(u(E_1)) - \text{codim}_E(E_1) \\ &= \dim(F_2) - \dim(E_2) = \text{ind}(u_0), \end{aligned}$$

which concludes the proof.

III.4.3 Perturbation of Riesz endomorphisms

Lemma 1. — Let E be a Banach space. Let p and q be continuous projectors on E such that $\|q - p\| < 1$. Then p induces an isomorphism of $\text{Im}(q)$ onto $\text{Im}(p)$, and $1_E - p$ induces an isomorphism of $\text{Ker}(q)$ onto $\text{Ker}(p)$.

Consider the continuous linear maps $u: x \mapsto p(x)$ from $\text{Im}(q)$ into $\text{Im}(p)$ and $v: y \mapsto q(y)$ from $\text{Im}(p)$ into $\text{Im}(q)$. For every $x \in \text{Im}(q)$, we have $x = q(x)$, whence

$$\begin{aligned} (q - p)^2(x) &= q^2(x) - q(p(x)) - p(q(x)) + p^2(x) \\ &= x - q(p(x)) = x - v(u(x)). \end{aligned}$$

It follows that we have $\|1_E - v \circ u\| \leq \|q - p\|^2 < 1$. By cor. 1 of I, p. 22, $v \circ u$ is an automorphism of $\text{Im}(q)$. One proves similarly that $u \circ v$ is an automorphism of $\text{Im}(p)$. This implies that u is an isomorphism from $\text{Im}(q)$ onto $\text{Im}(p)$, whence the first assertion of the lemma. The second follows from it by replacing p and q by $1_E - p$ and $1_E - q$ respectively.

THEOREM 2. — Let E be a Banach space. The set $\mathcal{R}(E)$ of Riesz endomorphisms of E is open in $\mathcal{L}(E)$. The map that assigns to an element of $\mathcal{R}(E)$ the dimension of its nilspace is upper semicontinuous on $\mathcal{L}(E)$.

The theorem follows from the following more precise statement:

PROPOSITION 5. — Let E be a Banach space and u_0 a Riesz endomorphism of E . Let N (resp. I) be the nilspace (resp. the conilspace) of u_0 . Let d denote the dimension of N . There exists an open neighborhood U of u_0 in $\mathcal{L}(E)$ and an analytic mapping $\pi: U \rightarrow \mathcal{L}(E)$ such that

- (i) The endomorphism $\pi(u_0)$ is the projector with image N and kernel I ;
- (ii) For every $u \in U$, the linear map $\pi(u)$ is a projector of rank d that belongs to the bicommutant of u and, in particular, commutes with u . Moreover, u induces an automorphism of $\text{Ker}(\pi(u))$;
- (iii) Every element of U is a Riesz endomorphism whose nilspace has dimension $\leq d$.

If $K = \mathbb{C}$, let $\text{Sp}(u_0)$ denote the spectrum of u_0 with respect to the algebra $\mathcal{L}(E)$. When $K = \mathbb{R}$, let $\text{Sp}(u_0)$ denote the complex spectrum $\text{Sp}_{\mathcal{L}(E_{(\mathbb{C})})}((u_0)_{(\mathbb{C})})$ of u_0 (I, p. 85, no. 13). By Prop. 14 of III, p. 54, the point 0 is isolated in $\{0\} \cup \text{Sp}(u_0)$. Let $r > 0$ be a real number such that every element of $\text{Sp}(u_0) - \{0\}$ has modulus $> r$. Let V denote the open set in $\mathbb{C}$ consisting of complex numbers of absolute value $\neq r$; let f be the holomorphic function on V defined by $f(z) = 0$ if $|z| > r$ and $f(z) = 1$ if $|z| < r$. If $K = \mathbb{C}$ (resp. $K = \mathbb{R}$), let U' denote the set of elements u of $\mathcal{L}(E)$ whose spectrum (resp. whose complex spectrum) is contained in V . The set U' is an open neighborhood of u_0 in $\mathcal{L}(E)$ and the map $u \mapsto f(u)$ from U' into $\mathcal{L}(E)$ is holomorphic (I, p. 76, prop. 10 and I, p. 85, no. 13).

Let $u \in U'$. The endomorphism $f(u)$ is the spectral projector $e_0(u)$; it commutes with u , and u induces, by passing to subspaces, an automorphism of $\text{Ker}(e_0(u))$ (cf. III, p. 53, no. 6).

The projector $e_0(u_0)$ has image N and kernel I (III, p. 54, prop. 14); its rank is d . By

Lemma 1, the set U of elements $u \in U'$ such that $e_0(u)$ is of rank d is an open neighborhood of u_0 . The set U and the map $\pi : u \mapsto e_0(u)$ from U into $\mathcal{L}(E)$ satisfy conditions (i) and (ii) of the proposition.

Let $u \in U$. Since u induces an automorphism of $\text{Ker}(e_0(u))$, which is a closed vector subspace of finite codimension of E , the endomorphism u is a Riesz endomorphism of E (III, p. 48, prop. 8); the nilspace of u is contained in the image of $\pi(u) = e_0(u)$, and has dimension $\leq d$.

III.4.4 Conorm of a continuous linear map

Let E and F be normed spaces, and let $u : E \rightarrow F$ be a continuous linear map, N the kernel of u and I its image. Let p denote the canonical surjection from E onto E/N and let i denote the canonical injection from I into F . Let $\tilde{u}$ be the bijective linear map from E/N onto I such that $u = i \circ \tilde{u} \circ p$. The vector space E/N is equipped with the quotient norm, that is,

$$\|y\| = \inf_{x \in p^{-1}(y)} \|x\| \quad (2)$$

for all $y \in E/N$. The map $\tilde{u}$ is continuous and $\|u\| = \|\tilde{u}\|$, whence

$$\|u\| = \sup_{\substack{y \in E/N \\ y \neq 0}} \frac{\|\tilde{u}(y)\|}{\|y\|}, \quad (3)$$

the supremum being taken in $\bar{R}_+$. The conorm of u is called the number

$$((u)) = \inf_{\substack{y \in E/N \\ y \neq 0}} \frac{\|\tilde{u}(y)\|}{\|y\|}, \quad (4)$$

the lower bound being taken in $\bar{R}_+$. We therefore have

$$((u))\|y\| \leq \|\tilde{u}(y)\| \leq \|u\|\|y\| \quad (5)$$

for every element y of E/N . Set $v = i \circ \tilde{u}$. We have $u = v \circ p$ and

$$((u)) = \inf_{\substack{y \in E/N \\ y \neq 0}} \frac{\|v(y)\|}{\|y\|} = \inf_{\substack{y \in E/N \\ \|y\|=1}} \|v(y)\|. \quad (6)$$

When u is the zero map, the space E/N is reduced to 0 and one has $((u)) = +\infty$ and $\|u\| = 0$. When $u \neq 0$, from (3) and (4) one deduces the inequalities

$$0 \leq ((u)) \leq \|u\| < +\infty. \quad (7)$$

When u is injective, we have

$$((u)) = \inf_{\substack{x \in E \\ x \neq 0}} \frac{\|u(x)\|}{\|x\|}, \quad (8)$$

and, for all $x \in E$, we have

$$((u))\|x\| \leq \|u(x)\| \leq \|u\|\|x\|. \quad (9)$$

Denote by j the canonical injection of the normed space F into its completion $\widehat{F}$. We have $((u)) = ((j \circ u))$.

Remark. — By definition, $((u)) > 0$ if and only if the bijective linear map $\tilde{u}$ is bicontinuous, that is, if and only if u is a strict morphism (TG, III, p. 16). We then have

$$((u)) = \|\tilde{u}^{-1}\|^{-1}. \quad (10)$$

Lemma 2. — Let c be a real number. If $c < ((u))$, then for every element $z \in \text{Im}(u)$ there exists an element x of E such that $u(x) = z$ and $c\|x\| \leq \|z\|$. Conversely, if for every $z \in \text{Im}(u)$ there exists $x \in E$ such that $u(x) = z$ and $c\|x\| \leq \|z\|$, then $c \leq ((u))$.

This is a consequence of formulas (2) and (5) and of the definition of the conorm of u .

PROPOSITION 6. — Let E and F be normed spaces and $u \in \mathcal{L}(E; F)$. Let B denote the set of elements of E with norm < 1 . Set

$$P = u(E) - u(B), \quad Q = u(E) - \overline{(u(B))} \cap u(E).$$

The conorm of u is equal to the distance from 0 to P in F . If the normed space E is complete or if u is a strict morphism, the conorm of u is equal to the distance from 0 to Q in F .

Let N be the kernel of u ; let $p : E \rightarrow E/N$ be the canonical surjection, and let $v : E/N \rightarrow F$ be the map induced by u on the quotient. The set $p(B)$ is the set of elements of E/N of norm < 1 . We have

$$P = u(E) - u(B) = v(E/N) - v(p(B)) = v((E/N) - p(B))$$

since the map v is injective. In other words, P consists of the elements of F of the form $v(y)$ with $y \in E/N$ and $\|y\| \geq 1$. Denote by d_P the distance from 0 to P in F . We have

$$d_P = \inf_{\substack{y \in E/N \\ \|y\| \geq 1}} \|v(y)\| = \inf_{\substack{y \in E/N \\ \|y\| = 1}} \|v(y)\| = ((u))$$

according to (6).

Suppose that u is a strict morphism. Let $\varepsilon > 0$. The set $\varepsilon u(B)$ is a neighborhood of 0 in $u(E)$. The closure of $u(B)$ in $u(E)$ is equal to $\overline{u(B)} \cap u(E)$. It is contained in the set $u(B) + \varepsilon u(B)$ which is equal to $(1 + \varepsilon)u(B)$ since $u(B)$ is convex. Consequently, we have $(1 + \varepsilon)P \subset Q \subset P$ and the distance d_Q from 0 to Q in F satisfies the inequalities $d_P \leq d_Q \leq (1 + \varepsilon)d_P$. Since this holds for every $\varepsilon > 0$, we have $d_Q = d_P = ((u))$.

Suppose that u is not a strict morphism, but that the normed space E is complete. We then have $((u)) = 0$. The closure of $u(B)$ in $u(E)$, which is equal to $\overline{u(B)} \cap u(E)$, is not a neighborhood of 0 in $u(E)$ (EVT, I, p. 17, th. 1). There therefore exist points of Q arbitrarily close to 0, whence $d_Q = 0 = ((u))$.

PROPOSITION 7. — Let E be a Banach space and F a normed space. The mapping $u \mapsto ((u))$ of $\mathcal{L}(E; F)$ in $\overline{\mathbf{R}}$ is upper semi-continuous.

Let $u \in \mathcal{L}(E; F)$. It suffices to prove that for every real number $c > ((u))$, the set of elements $v \in \mathcal{L}(E; F)$ such that $((v)) < c$ is a neighborhood of u . Let B denote the set of elements of E with norm < 1 . By prop. 6, there exists $y \in E$ such that $u(y) \notin \overline{u(B)}$ and $\|u(y)\| < c$. The distance d of $u(y)$ to the closed set $\overline{u(B)}$ is strictly positive. The set V of elements v of $\mathcal{L}(E; F)$ satisfying the relations $\|v(y)\| < c$ and $\|u - v\|(1 + \|y\|) < d$ is a

neighborhood of u in $\mathcal{L}(E; F)$. Let $v \in V$. For every $x \in B$, we have

$$\begin{aligned} d &\leq \|u(y) - u(x)\| \leq \|v(y) - v(x)\| + \|u - v\|(\|y\| + \|x\|) \\ &< \|v(y) - v(x)\| + d. \end{aligned}$$

Consequently $v(y)$ does not belong to $v(B)$, and one has $((v)) \leq \|v(y)\| < c$ by prop. 6.

PROPOSITION 8. — Let E be a Banach space, F a normed space. For every $u \in \mathcal{L}(E; F)$, we have $((u)) = (({}^t u))$.

Denote by j the canonical injection of the normed space F into its completion $\hat{F}$. We have $((u)) = ((j \circ u))$. Since the linear map ${}^t j : (\hat{F})' \rightarrow F'$ is bijective and isometric, one also has $(({}^t u)) = (({}^t(j \circ u)))$. It therefore suffices to prove the proposition when the normed space F is complete, which we assume in the rest of the proof.

If u is zero, one has $((u)) = (({}^t u)) = +\infty$. If u is not a strict morphism, then ${}^t u$ is not one either (EVT, IV, p. 29, cor. 3), and one has $((u)) = (({}^t u)) = 0$.

Suppose now that u is a nonzero strict morphism. Let N denote the kernel of u and I its image, and consider the canonical decomposition of u :

$$E \xrightarrow{p} E/N \xrightarrow{v} I \xrightarrow{i} F.$$

The kernel of ${}^t u$ is the orthogonal I° of I in F' (EVT, IV, p. 27, prop. 2), and ${}^t i$ induces by passage to the quotient an isometry ι from F'/I° onto I' (EVT, IV, p. 9, prop. 10). Moreover, ${}^t p$ defines an isometry from $(E/N)'$ onto the orthogonal N° of N in E' (EVT, IV, p. 8, prop. 9). The canonical decomposition of ${}^t u$ is therefore

$$F' \rightarrow F'/I^\circ \xrightarrow{v'} N^\circ \rightarrow E',$$

where $v' = \pi \circ {}^t v \circ \iota$. We then have

$$\|(v')^{-1}\| = \|({}^t v)^{-1}\| = \|({}^t(v^{-1}))\| = \|v^{-1}\|$$

(EVT, IV, p. 7, prop. 8), whence $((u)) = (({}^t u))$ according to formula (10).

III.4.5 Finite-dimensional vector subspaces of a normed space

The following statement will be proved in TA, forthcoming, as a corollary of the Borsuk–Ulam theorem.

THEOREM 3. — Let n be a positive integer, V a real normed vector space of dimension $n + 1$ and W a vector subspace of V of dimension n . Let S be the unit sphere of V . There does not exist a continuous map $f : S \rightarrow W$ such that $\|f(x) - x\| < 1$ for all $x \in S$.

THEOREM 4 (Krein, Krasnoselskii, Milman). — Let E be a normed space, and let F and G be vector subspaces of E . Suppose that the dimension of G is finite and strictly less than that of F . There exists an element of F of norm 1 whose distance to G is equal to 1.

It suffices to treat the case where the field K is equal to $\mathbb{R}$. Let n be the dimension of G . By replacing F by a vector subspace of F of dimension $n + 1$ containing G , one reduces to the case where $\dim(F) = n + 1$. Argue by contradiction by assuming that the conclusion of the theorem is not satisfied. Let S be the unit sphere of F . For every $x \in S$, the distance from x to G is then strictly less than $\|x\| = 1$ and one can choose an element $y(x)$ of G for which one has $\|x - y(x)\| < 1$. Denote by V_x the set of elements z of S such that $\|z - y(x)\| < 1$; this is an open neighborhood of x in S . There exists a locally finite continuous partition of unity $(\varphi_x)_{x \in S}$ on S subordinate to the covering $(V_x)_{x \in S}$ of S (TG, IX, p. 46, prop. 3 and p. 51, prop. 6). Let $f : S \rightarrow G$ be the continuous map given by

$$f(z) = \sum_{x \in S} \varphi_x(z) y(x).$$

For all $z \in S$, we have $\varphi_x(z) \geq 0$ for all $x \in S$, and there exists $x \in S$ such that $\varphi_x(z) > 0$ since $\sum_{x \in S} \varphi_x(z) = 1$, hence

$$\begin{aligned} \|z - f(z)\| &= \left\| \sum_{x \in S} \varphi_x(z) (z - y(x)) \right\| \leq \sum_{x \in S} \varphi_x(z) \|z - y(x)\| \\ &< \sum_{x \in S} \varphi_x(z) = 1. \end{aligned}$$

This property of f contradicts Theorem 3.

PROPOSITION 9. — Let E and F be normed spaces, $n \in \mathbb{N}$ and let u be a continuous linear map from E into F whose kernel has dimension n . The conorm of u is equal to the distance d from u to the set of maps $v \in \mathcal{L}(E; F)$ whose kernel has dimension at least $n + 1$.

When $u = 0$, we have $((u)) = +\infty$ and $\dim(E) = n$, whence $d = +\infty$. Suppose henceforth that u is nonzero, and therefore $((u)) < +\infty$. Let v be an element of $\mathcal{L}(E; F)$ such that $\|u - v\| < ((u))$ and let us prove that its kernel has dimension $\leq n$. Let x be an element of norm 1 of $\text{Ker}(v)$ and y be its image in $E / \text{Ker}(u)$. We have (formula (5) of III, p. 61)

$$((u))\|y\| \leq \|u(x)\| = \|(u - v)(x)\| \leq \|u - v\| < ((u))$$

whence $\|y\| < 1$. Now $\|y\|$ is the distance from x to $\text{Ker}(u)$. Theorem 4 then implies that we have $\dim \text{Ker}(v) \leq \dim \text{Ker}(u) = n$. This proves the inequality $((u)) \leq d$. The reverse inequality $d \leq ((u))$ follows from the more precise lemma below.

Lemma 3. — Let c be a real number such that $c > ((u))$. There exists a continuous linear map $h : E \rightarrow F$, of rank 1 and of norm $< c$ such that the kernel of $u + h$ contains that of u and is of dimension $n + 1$.

Denote by p the canonical mapping of E onto $E/\text{Ker}(u)$. There exists $a \in E$ such that $\|p(a)\| = 1$ and $\|u(a)\| < c$ (formula (6) of III, p. 61). By the Hahn–Banach theorem (EVT, II, p. 24, cor. 2), there exists a continuous linear form f on the normed space $E/\text{Ker}(u)$ such that $f(p(a)) = 1$ and $\|f\| = 1$. The linear map $h : x \mapsto -(f \circ p)(x)u(a)$ from E into F is continuous, of rank 1, and we have $\|h\| \leq \|f\|\|p\|\|u(a)\| < c$. The kernel of the map $u + h$ contains that of u and a ; since $a \notin \text{Ker}(u)$, its dimension is at least $n + 1$. On the other hand, the linear map u induces, by passing to subspaces, a linear map from $\text{Ker}(u + h)$ into $\text{Im}(h)$ with kernel $\text{Ker}(u) \cap \text{Ker}(u + h)$, so that $\dim(\text{Ker}(u + h)) \leq \dim(\text{Ker}(u)) + 1$. The conclusion follows.

COROLLARY 1. — Let E and F be normed spaces, and let u, v be nonzero continuous linear maps from E into F whose kernels have the same finite dimension. Then we have

$$|((u)) - ((v))| \leq \|u - v\|.$$

Denote by n the common dimension of the kernels of u and v . Let A be the set of continuous linear maps from E to F whose kernel has dimension $\geq n + 1$. Since $u \neq 0$, we have $\dim(E) > n$ and the set A contains 0. By Proposition 9, $((u))$ and $((v))$ are respectively the distances from u and v to the set A in $\mathcal{L}(E; F)$. It then suffices to apply the formula $|d(u, A) - d(v, A)| \leq \|u - v\|$

(cf. TG, IX, p. 13).

COROLLARY 2. — Let E and F be Banach spaces, and let u and v be nonzero continuous linear maps from E into F whose images have the same finite codimension in F . Then one has

$$|((u)) - ((v))| \leq \|u - v\|.$$

The morphism u is strict (III, p. 52, lemma 6). The kernel of the continuous linear map ${}^t u$ is the orthogonal $(\text{Im}(u))^\circ$ of $\text{Im}(u)$ (EVT, IV, p. 27, prop. 2), hence $\dim(\text{Ker}({}^t u)) = \text{codim}_F(\text{Im}(u))$, and similarly $\dim(\text{Ker}({}^t v)) = \text{codim}_F(\text{Im}(v))$. The kernels of ${}^t u$ and ${}^t v$ therefore have the same finite dimension. Since

$$\|{}^t u - {}^t v\| = \|u - v\|, \quad ({}^t u) = ((u)), \quad ({}^t v) = ((v))$$

(EVT, IV, p. 7, prop. 8 and prop. 8 of III, p. 63), the assertion follows from corollary 1, applied to ${}^t u$ and ${}^t v$.

III.4.6 Perturbations of continuous injective or surjective linear maps

In this section, we adopt the following conventions: if E is a finite-dimensional vector space, $\dim(E)$ denotes its dimension; if E is an infinite-dimensional vector space, we set $\dim(E) = +\infty \in \overline{\mathbf{R}}$. If u is a linear map whose kernel or cokernel is finite-dimensional, we set $\text{ind}(u) = \dim \text{Coker}(u) - \dim \text{Ker}(u)$, the computation being carried out in $\overline{\mathbf{R}}$.

Let E and F be normed spaces. We denote by $\mathcal{M}(E; F)$ the set of strict injective morphisms from E into F , and by $\mathcal{QM}(E; F)$ the set of strict morphisms from E into F whose kernel is finite-dimensional.

PROPOSITION 10. — Let E and F be normed spaces. The set $\mathcal{M}(E; F)$ is open in $\mathcal{L}(E; F)$. It is the interior of the set of maps in $\mathcal{L}(E; F)$ that are injective.

Let A be the set of continuous injective linear maps from E into F . For a map $u \in A$ to be a strict morphism, it is necessary and sufficient that its conorm $((u))$ be strictly positive (III, p. 62, remark). Now $((u))$ is the distance from u to the complement of A in $\mathcal{L}(E; F)$ (III, p. 65, prop. 9). The proposition follows.

PROPOSITION 11. — Let E and F be Banach spaces. The set $\mathcal{QM}(E; F)$ is open in $\mathcal{L}(E; F)$. It is the interior of the set of maps in $\mathcal{L}(E; F)$ whose kernel is finite-dimensional.

Let A be the set of continuous linear maps from E into F whose kernel is finite-dimensional.

Let u be an element of $\mathcal{QM}(E; F)$. We then have $((u)) > 0$ (III, p. 62, remark). Every element $v \in \mathcal{L}(E; F)$ whose distance from u is $< ((u))$ then belongs to A (III, p. 65, prop. 9), so that u is an interior point of A .

Conversely, let u be an element of A that is not a strict morphism. We have $((u)) = 0$ (III, p. 62, remark). Let v be an element of $\mathcal{L}(E; F)$ which differs from u by a linear map of finite rank; by Corollary 1 of III, p. 40, the morphism v cannot be strict with closed image; since a strict morphism from E into F has a closed image in F (EVT, IV, p. 28, th. 1), this means that v is not a strict morphism. We therefore have $((v)) = 0$. Let ε be a real number > 0 . By the preceding observations, Lemma 3 of III, p. 66 allows us to construct by induction a sequence $(u_m)_{m \in \mathbb{N}}$ of elements of $\mathcal{L}(E; F)$ satisfying the following conditions:

- (i) One has $u_0 = u$;
- (ii) For every $m \geq 0$, $u_{m+1} - u_m$ is a continuous linear map of rank 1 and norm $\leq 2^{-m-1} \varepsilon$;
- (iii) For every $m \geq 0$, the kernel of u_m has dimension $n + m$ and is contained in that of u_{m+1} .

The sequence (u_m) is a Cauchy sequence in the Banach space $\mathcal{L}(E; F)$. Let u' be its limit. The kernel of u' contains that of u_m for all $m \geq 0$; it is of infinite dimension. Since

$$\|u' - u\| \leq \sum_{m=0}^{\infty} \|u_{m+1} - u_m\| \leq \varepsilon \sum_{m=0}^{\infty} 2^{-m-1} = \varepsilon,$$

the distance from u to the complement of A is less than ε . Since this holds for every $\varepsilon > 0$, one concludes that u is not an interior point of A .

PROPOSITION 12. — Let E and F be Banach spaces and u an element of $\mathcal{QM}(E; F)$. Every $v \in \mathcal{L}(E; F)$ such that $\|v - u\| < ((u))$ belongs to $\mathcal{QM}(E; F)$ and satisfies the relations

$$\begin{aligned} \dim \text{Ker}(v) &\leq \dim \text{Ker}(u), \\ \dim \text{Coker}(v) &\leq \dim \text{Coker}(u), \\ \text{ind}(v) &= \text{ind}(u). \end{aligned}$$

Since u is strict, we have $((u)) > 0$. Let B denote the open ball with center v and radius $((u))$ in $\mathcal{L}(E; F)$. For every $v \in B$, we have $\dim \text{Ker}(v) \leq \dim \text{Ker}(u)$ (III, p. 65, prop. 9) and $v \in \mathcal{QM}(E; F)$ (prop. 11).

For $r \in \mathbf{Z} \cup \{+\infty\}$, denote by B_r the set of elements $v \in B$ such that $\text{ind}(v) = r$. If $r \in \mathbf{Z}$, the set B_r is the set of Fredholm maps from E to F of index r which belong to B (III, p. 52, prop. 11); it is open in B by th. 1 of III, p. 58.

Let us prove that the sets B_r are closed in B . Let $v \in B$ be a point adherent to B_r . Since the sets B_s , for $s \in \mathbf{Z}$, are open in B and pairwise disjoint, we have $v \in B_r$ or $v \in B_{+\infty}$.

Suppose $v \in B_{+\infty}$. Let n denote the dimension of $\text{Ker}(u)$. Choose a vector subspace T of F of dimension $r + 2n + 1$ whose intersection with $\text{Im}(v)$ is reduced to 0 and a topological supplement S of $\text{Ker}(v)$ in E (III, p. 55, prop. 1). Consider the continuous linear map $f : (s, t) \mapsto v(s) + t$ from $S \times T$ to F . It is injective. The linear map v is a strict morphism since v belongs to $B \subset \mathcal{QM}(E; F)$. The image of v is therefore closed (EVT, IV, p. 28, th. 1). By prop. 1 of III, p. 39, the restriction of v to S is a strict morphism with closed image from S to F , and f is a strict morphism with closed image from $S \times T$ to F .

By prop. 10, there exists a neighborhood U of v in B such that, for every $w \in U$, the linear map $(s, t) \mapsto w(s) + t$ from $S \times T$ into F is injective. Let w be an element of U . We then have $\text{Ker}(w) \cap S = \{0\}$, hence w induces by restriction and passage to the quotient an injective linear map from $\text{Ker}(w)$ into E/S , which implies that $\text{Ker}(w)$ has dimension at most n . We also have $w(S) \cap T = \{0\}$, hence

$$\text{codim}_F(\text{Im}(w)) \geq \dim(T) - \text{codim}_E(S) = r + n + 1,$$

this entails $\text{ind}(w) \geq r + 1$ and contradicts the hypothesis that v is adherent to B_r .

If r is an element of $\mathbf{Z}$ such that B_r is nonempty, we have $B_r = B$ since B_r is open and closed in B and B is connected. If B_r is empty for every $r \in \mathbf{Z}$, we have $\text{ind}(v) = +\infty$ for all $v \in B$. The map $v \mapsto \text{ind}(v)$ is therefore constant on B . Finally, for every $v \in B$, we have

$$\begin{aligned} \dim \text{Coker}(v) &= \text{ind}(v) + \dim \text{Ker}(v) \\ &\leq \text{ind}(u) + \dim \text{Ker}(u) = \dim \text{Coker}(u). \end{aligned}$$

This concludes the proof.

COROLLARY 1. — The functions defined by $u \mapsto \dim \text{Ker}(u)$ and by $u \mapsto \dim \text{Coker}(u)$ on $\mathcal{QM}(E; F)$ are upper semicontinuous. The function $u \mapsto \text{ind}(u)$ is locally constant on $\mathcal{QM}(E; F)$.

COROLLARY 2. — The function $u \mapsto \dim \text{Coker}(u)$ is locally constant on the set $\mathcal{M}(E; F)$ of strict injective morphisms from E into F .

Indeed, one has $\dim \text{Coker}(u) = \text{ind}(u)$ for $u \in \mathcal{M}(E; F)$.

Lemma 4. — Let E and F be Banach spaces. For an element u of $\mathcal{L}(E; F)$ to be a strict morphism from E into F , it is necessary and sufficient that ${}^t u$ be a strict morphism from F' into E' . In this case, one has $\dim \text{Coker}({}^t u) = \dim \text{Ker}(u)$ and $\dim \text{Ker}({}^t u) = \dim \text{Coker}(u)$.

For u to be a strict morphism, it is necessary and sufficient that ${}^t u$ be one (EVT, IV, p. 29, cor. 3); in this case, the image of u is closed (EVT, IV, p. 28, th. 1) and the vector space $\text{Ker}({}^t u)$ (resp. $\text{Coker}({}^t u)$) is canonically isomorphic to the dual of the normed space $\text{Coker}(u)$ (resp. $\text{Ker}(u)$) according to EVT, IV, p. 27, prop. 2. The lemma follows.

Let E and F be Banach spaces. We denote by $\mathcal{E}(E; F)$ the set of continuous surjective linear maps from E into F and by $\mathcal{QE}(E; F)$ the set of continuous linear maps from E into F whose image has finite codimension. Every element of $\mathcal{QE}(E; F)$ is a strict morphism with closed image (III, p. 52, lemma 6). It follows from lemma 4 that $\mathcal{E}(E; F)$ and $\mathcal{QE}(E; F)$ are respectively the inverse images of $\mathcal{M}(F'; E')$ and $\mathcal{QM}(F'; E')$ under the continuous map $u \mapsto {}^t u$ from $\mathcal{L}(E; F)$ into $\mathcal{L}(F'; E')$.

PROPOSITION 13. — Let E and F be Banach spaces. The sets $\mathcal{E}(E; F)$ and $\mathcal{QE}(E; F)$ are open in $\mathcal{L}(E; F)$. More precisely, if u is an element of $\mathcal{QE}(E; F)$ and v is an element of $\mathcal{L}(E; F)$ such that $\|v - u\| < ((u))$, we have $v \in \mathcal{QE}(E; F)$ and

$$\dim \text{Ker}(v) \leq \dim \text{Ker}(u), \quad \dim \text{Coker}(v) \leq \dim \text{Coker}(u),$$

$$\text{ind}(v) = \text{ind}(u).$$

We saw above that ${}^t u \in \mathcal{QM}(F'; E')$. Moreover, for every element v of $\mathcal{L}(E; F)$, we have $\|{}^t v - {}^t u\| = \|v - u\|$ and $(({}^t u)) = ((u))$ (EVT, IV, p. 7, prop. 8 and III, p. 63, prop. 8). By prop. 12, it follows from these relations that if $\|v - u\| < ((u))$, then ${}^t v$ belongs to $\mathcal{QM}(F'; E')$ and

one has the inequalities

$$\dim \text{Ker}({}^t v) \leq \dim \text{Ker}({}^t u), \quad \dim \text{Coker}({}^t v) \leq \dim \text{Coker}({}^t u)$$

as well as the equality $\text{ind}({}^t v) = \text{ind}({}^t u)$. The proposition then follows from lemma 4.

COROLLARY 1. — The functions defined by $u \mapsto \dim \text{Ker}(u)$ and by $u \mapsto \dim \text{Coker}(u)$ on $\mathcal{QE}(E; F)$ are upper semicontinuous. The function $u \mapsto \text{ind}(u)$ is locally constant on $\mathcal{QE}(E; F)$.

COROLLARY 2. — The function defined by $u \mapsto \dim \text{Ker}(u)$ is locally constant on $\mathcal{E}(E; F)$.

Indeed, one has $\dim \text{Ker}(u) = -\text{ind}(u)$ for all $u \in \mathcal{E}(E; F)$.

III.5 PERTURBATION BY A COMPACT LINEAR MAP

III.5.1 Strict morphisms and properness

Lemma 1. — Let E and F be topological vector spaces, let u be a continuous linear map from E into F , and let U be a neighborhood of 0 in E . Suppose that u induces a homeomorphism from U onto a closed subset of F . Then the image I of u is closed and u induces a homeomorphism from E onto I .

The set $u(U)$ is a neighborhood of 0 in I ; it is closed in F , hence the subgroup I of F is locally closed at 0 , and consequently it is closed (TG, III, p. 7, prop. 4).

Since $\text{Ker}(u)$ meets U only at 0 , one has $\text{Ker}(u) = \{0\}$ and the map u is injective. Let V be a closed neighborhood of 0 in E contained in U . Since $u(\mathring{V})$ is a neighborhood of 0 in $u(V)$, there exists a balanced neighborhood W of 0 in F such that $u(V) \cap W \subset u(\mathring{V})$. The set $u^{-1}(W)$ is balanced, hence connected. It contains 0 and is contained in $\mathring{V} \cup (E - V)$. Since $\mathring{V}$ and $E - V$ are disjoint open subsets of E , the set $u^{-1}(W)$ is contained in $\mathring{V}$, whence $W \cap I \subset u(\mathring{V})$. Consequently $u(\mathring{V})$ is a neighborhood of 0 in I . This implies that u induces a homeomorphism from E onto I .

PROPOSITION 1. — Let E be a separated locally convex space, F a locally convex space

and u a continuous linear map from E into F . The following conditions are equivalent:

- (i) The map u is a strict morphism, its kernel has finite dimension and its image is closed in F ;
- (ii) There exists a closed neighborhood V of 0 in E such that the restriction of u to V is proper (TG, I, p. 72).
- (iii) $\Rightarrow$ (ii): Suppose that u satisfies condition (i). Since the kernel of u has finite dimension, it has a topological supplement E_1 in E (III, p. 55, prop. 1), and there exists a compact neighborhood C of 0 in $\text{Ker}(u)$. Identify E with $E_1 \times \text{Ker}(u)$; the set $V = E_1 \times C$ is then a closed neighborhood of 0 in E . The restriction of u to V is the composition of the projection of $E_1 \times C$ onto E_1 which is proper (TG, I, p. 77, cor. 5) and of the restriction u_1 of u to E_1 . Now u_1 is a homeomorphism from E_1 onto a closed subspace of F , hence is proper (TG, I, p. 72, prop. 2). The composition of two proper maps is proper (TG, I, p. 73, prop. 5, a)), so the restriction of u to V is proper.
- (iv) $\Rightarrow$ (i): Let V be a closed neighborhood of 0 in E such that the restriction v of u to V is proper. The set $V \cap \text{Ker}(u) = \bar{v}(\{0\})$ is then compact (TG, I, p. 75, th. 1); consequently, the vector space $\text{Ker}(u)$ is locally compact, hence finite-dimensional (EVT, I, p. 15, th. 3). Let E_1 be a topological complement of $\text{Ker}(u)$ in E (prop. 1 of III, p. 55); set $V_1 = E_1 \cap V$. The set V_1 is closed in V . The map $u|_{V_1}$ is proper (TG, I, p. 74, cor. 1) and injective, hence is a homeomorphism from V_1 onto a closed subset of F (TG, I, p. 72, prop. 2). By Lemma 1, the restriction of u to E_1 is a homeomorphism from E_1 onto a closed subspace of F , hence u is a strict morphism with closed image.

III.5.2 Perturbation of injective or surjective linear maps

THEOREM 1. — Let E and F be separated locally convex spaces and let u be a strict morphism from E into F , whose kernel is finite-dimensional and whose image is closed. Let h be a compact linear map from E into F . The linear map $u+h$ is a strict morphism, its kernel is finite-dimensional, and its image is closed.

By prop. 1 of III, p. 71, there exists a closed neighborhood V of 0 in E such that the restriction of u to V is proper. Since h is compact, there exists a closed neighborhood W of 0 contained in V such that the set $h(W)$ is relatively compact. Set $C = \overline{h(W)}$. The restriction of u to W is proper (TG, I, p. 74, cor. 1). The map $\alpha : x \mapsto (u(x), h(x))$ from W into $F \times C$ is proper because the map $\text{pr}_1 \circ \alpha = u|_W$ from W into F is proper (TG, I, p. 73, prop. 5, d). The map $\beta : (y, z) \mapsto (y + z, z)$ from $F \times C$ into $F \times C$ is a homeomorphism, and the map pr_1 from $F \times C$ into F is proper (TG, I, p. 77, cor. 5). The composite map $\text{pr}_1 \circ \beta \circ \alpha$ from W into F is therefore proper (TG, I, p. 73, prop. 5, a). But this map is none other than the restriction of $u + h$ to W . By prop. 1 of III, p. 71, $u + h$ is a strict morphism, its kernel is finite-dimensional, and its image is closed.

Lemma 2. — Let E and F be Fréchet spaces and $u \in \mathcal{L}(E; F)$. The following conditions are equivalent:

- (i) The cokernel of u is finite-dimensional;
- (ii) ${}^t u : F'_c \rightarrow E'_c$ is a strict morphism with closed image whose kernel is finite-dimensional.
- (iii) $\Rightarrow$ (ii): Suppose that the cokernel of u is finite-dimensional. The map u is then a strict morphism (Lemma 6 of III, p. 52). By Prop. 2 of EVT, IV, p. 27, the image of ${}^t u$ is closed in E' equipped with the weak topology, and a fortiori in E'_c . The kernel of ${}^t u$ is the orthogonal complement of the image of u (loc. cit.); it is therefore finite-dimensional. Finally, ${}^t u$ is a strict morphism from F'_c into E'_c by Theorem 1 of EVT, IV, p. 28.
- (iv) $\Rightarrow$ (i): Suppose that ${}^t u : F'_c \rightarrow E'_c$ is a strict morphism. By EVT, IV, p. 28, th. 1, the image of u is closed. The kernel of ${}^t u$ is the orthogonal complement of $\text{Im}(u)$ (by EVT, IV, p. 27, prop. 2); if this kernel is finite-dimensional, the image of u is of finite codimension in F .

THEOREM 2. — Let E and F be Fréchet spaces, $u : E \rightarrow F$ a continuous linear map whose cokernel is finite-dimensional, and $h : E \rightarrow F$ a compact linear map. The linear map $u + h$ is a strict morphism, its cokernel is finite-dimensional, and its image is closed.

By lemma 6 of III, p. 52, it suffices to show that the cokernel of $u + h$ is of finite dimension. Now, by prop. 9 of III, p. 9, the map ${}^t h$ from F'_c into E'_c is compact. Theorem 2 follows from

Theorem 1 and Lemma 2.

III.5.3 Perturbation of Fredholm mappings

THEOREM 3. — Let E be a locally convex space, F a separated locally convex space, u a Fredholm map from E into F , and h a compact linear map from E into F . Then $u + h$ is a Fredholm operator, and one has $\text{ind}(u + h) = \text{ind}(u)$.

We will prove the theorem in three steps.

- A) Assume that E and F are Banach spaces. Prop. 2 of III, p. 42 shows that the linear map u is a strict morphism with closed image whose kernel and cokernel are finite-dimensional. Since E and F are Banach spaces, Ths. 1 and 2 of III, p. 73 imply that, for every $t \in [0, 1]$, the linear map $u_t = u + th$ possesses these same properties, and is therefore a Fredholm operator (III, p. 42, prop. 2). The map $t \mapsto u_t$ from $[0, 1]$ to the set $\mathcal{F}(E; F)$ of Fredholm maps from E into F is continuous. By th. 1 of III, p. 58, the map $t \mapsto \text{ind}(u_t)$ is locally constant. Since the interval $[0, 1]$ is connected, this map is constant. We therefore have $\text{ind}(u) = \text{ind}(u + h)$.
- B) We assume that $E = F$. $u = 1_E$. By prop. 5 of III, p. 5, there exists a Banach space G , a continuous linear map $p : E \rightarrow G$ and a compact linear map $q : G \rightarrow E$ such that $h = q \circ p$. The endomorphism $p \circ q$ of G is compact. By A), $1_G + p \circ q$ is a Fredholm endomorphism of G with index zero. But then $1_E + h = 1_E + q \circ p$ is a Fredholm endomorphism of E with index zero (III, p. 49, prop. 10, f)).
- C) General case. Let v be a quasi-inverse of u . The endomorphisms $u \circ v$ and $(u + h) \circ v$ of F differ from 1_F by compact linear maps. By B), these are Fredholm endomorphisms of F of index zero. It follows that $u + h$ is a Fredholm endomorphism (III, p. 40, n°2) with the same index as u (III, p. 43, no. 3, formula (2)).

COROLLARY 1. — Let E and F be separated locally convex spaces and $u \in \mathcal{L}(E; F)$. For u to be a Fredholm operator, it is necessary and sufficient that there exist a map $v \in \mathcal{L}(F; E)$ such that the linear maps $1_E - v \circ u$ and $1_F - u \circ v$ are compact.

Since a continuous linear map of finite rank is compact, the stated condition is necessary.

Let v be an element of $\mathcal{L}(F; E)$ such that the linear maps $1_E - v \circ u$ and $1_F - u \circ v$ are compact. By Theorem 3, $v \circ u$ and $u \circ v$ are Fredholm maps. Let p and q be quasi-inverses of $v \circ u$ and $u \circ v$ respectively. Set $w = p \circ v$ and $w' = v \circ q$. Using the notation of No. 2 of III, p. 40, we have $w \circ u \equiv 1_E$ and $u \circ w' \equiv 1_F$, whence $w \equiv w \circ u \circ w' \equiv w'$. It follows that w is a quasi-inverse of u .

COROLLARY 2. — Let E and F be separated locally convex spaces and $u \in \mathcal{L}(E; F)$. The following conditions are equivalent:

- (i) The map u is a Fredholm map of index zero;
- (ii) There exists an isomorphism v from E onto F such that $u - v$ has finite rank;
- (iii) There exists an isomorphism v of E onto F such that $u - v$ is compact.
- (iv) $\Rightarrow$ (ii): Suppose that u is a Fredholm operator of index zero. There exist topological direct sum decompositions $E = E_1 \oplus E_2$ and $F = F_1 \oplus F_2$ with E_2 and F_2 of finite dimension such that u vanishes on E_2 and defines by restriction an isomorphism u_1 of E_1 onto F_1 (III, p. 42, prop. 2). If $\text{ind}(u) = 0$ then the dimensions of E_2 and F_2 are equal and there exists an isomorphism v from E onto F extending u_1 . The kernel of $u - v$ contains E_1 , so $u - v$ is of finite rank.

We have (ii) $\Rightarrow$ (iii) since every continuous linear map of finite rank from E into F is compact, and (iii) $\Rightarrow$ (i) by Thm. 3.

III.5.4 Perturbation of Riesz endomorphisms

Let E be a Banach space. Recall (cf. III, p. 5, prop. 3) that the set $\mathcal{L}^c(E)$ of compact endomorphisms of E forms a closed two-sided ideal of the Banach algebra $\mathcal{L}(E)$. The quotient Banach algebra is called the Calkin algebra of E ; it is denoted $\mathcal{Calk}(E)$. We denote by π the canonical homomorphism of $\mathcal{L}(E)$ onto $\mathcal{Calk}(E)$. By cor. 1 of III, p. 74, an endomorphism u of E is a Fredholm endomorphism if and only if $\pi(u)$ is invertible in the algebra $\mathcal{Calk}(E)$.

PROPOSITION 2. — Let $u \in \mathcal{L}(E)$ such that $\|1_E - \pi(u)\| < 1$ in the algebra $\mathcal{Calk}(E)$. Then u is a Riesz endomorphism of E .

Let $r \geq 1$ be an integer such that $\|1_E - \pi(u)\|^r < \frac{1}{2}$. Let $P \in \mathbf{C}[X]$ be the polynomial $\frac{1-(1-X)^r}{X}$. Set $v = 1_E - (1_E - u)^r$. We therefore have $v = uP(u)$ and $\|1_E - \pi(v)\| < \frac{1}{2}$. Since the endomorphisms u and $P(u)$ of E commute, it suffices to prove that v is a Riesz endomorphism of E (III, p. 49, prop. 9).

Since $\|1_E - \pi(v)\| < 1/2$, there exists, by definition of the quotient norm in the space $\mathcal{C}a\ell k(E)$, a compact endomorphism h of E and an endomorphism w of E such that $v = 1_E + h + w$ and $\|w\| < \frac{1}{2}$. By corollary 1 of I, p. 22, the element $1_E + w$ is an automorphism of E . Since h is compact, $v = (1_E + w) + h$ is a Fredholm endomorphism of E of index 0 (cor. 2 of III, p. 74). For every integer $n \geq 0$, denote by N_n the kernel of v^n . To prove that v is a Riesz endomorphism of E , it suffices to show that there exists an integer $n \geq 0$ such that $N_n = N_{n+1}$ (III, p. 46, def. 2 and III, p. 46, remark).

Let us argue by contradiction by assuming the sequence (N_n) strictly increasing. For every $n \in N$, denote by p_n the canonical map from E onto the normed space E/N_n . Let c be a real number such that $2\|w\| < c < 1$. Let $n \in N$. Since N_{n+1} is different from N_n , there exists an element $x_n \in N_{n+1}$ such that $\|p_n(x_n)\| = c$ and such that $\|x_n\| < 1$ (indeed, there exists $y \in N_{n+1}/N_n$ of norm c , so for every $\varepsilon > 0$, there exist $x_n \in N_{n+1}$ such that $p_n(x_n) = y$ and $\|x_n\| \leq c + \varepsilon$).

Let m, n be two integers such that $m > n \geq 0$. We have

$$\begin{aligned} \|h(x_m) - h(x_n)\| &= \|v(x_m - x_n) - (1_E + w)(x_m - x_n)\| \\ &\geq \|(v - 1_E)(x_m - x_n)\| - \|w(x_m - x_n)\| \\ &\geq \|(v - 1_E)(x_m - x_n)\| - 2\|w\| \end{aligned} \tag{1}$$

since $\|x_m\| \leq 1$ and $\|x_n\| \leq 1$. Moreover, note that

$$(v - 1_E)(x_m - x_n) = v(x_m - x_n) + x_n - x_m.$$

But since $n < m$ and $v(N_{m+1}) \subset N_m$, we have $v(x_m - x_n) + x_n \in N_m$, whence

$$\|v(x_m - x_n) + x_n - x_m\| \geq \|p_m(x_m)\| = c.$$

Inequality (1) therefore yields the lower bound

$$\|h(x_m) - h(x_n)\| \geq c - 2\|w\| > 0. \quad (2)$$

The sequence $(h(x_m))_{m \in \mathbb{N}}$ therefore has no cluster point in E , which contradicts the fact that the image under h of the unit ball of E is relatively compact, since h is compact (remark 1 of III, p. 2).

COROLLARY 1. — Let E be a separated locally convex space and let $u \in \mathcal{L}(E)$. For u to be a Riesz endomorphism, it is necessary and sufficient that there exist an element v of $\mathcal{L}(E)$ which commutes with u and such that $1_E - u \circ v$ is compact.

Since every continuous linear map of finite rank is compact, the condition is necessary (III, p. 47, prop. 6 d)). Let us prove that it is sufficient. Let v be an element of $\mathcal{L}(E)$ that commutes with u and such that the endomorphism $h = 1_E - u \circ v$ of E is compact. There exist a Banach space G , a continuous linear map $p : E \rightarrow G$ and a compact linear map $q : G \rightarrow E$ such that $h = q \circ p$ (III, p. 5, prop. 5). The endomorphism $p \circ q$ of G is compact. By prop. 2, the linear map $1_G - p \circ q$ is a Riesz endomorphism of G . But then $u \circ v = 1_E - q \circ p$ is a Riesz endomorphism of E (III, p. 49, prop. 10, g)). Since u and v commute, u is a Riesz endomorphism of E (III, p. 49, prop. 9).

COROLLARY 2. — Let E be a separated locally convex space, u a Riesz endomorphism of E , and $h \in \mathcal{L}(E)$ a compact linear map. If u and h commute, then $u + h$ is a Riesz endomorphism of E .

Let v be the canonical quasi-inverse of u . It commutes with u and with h (III, p. 47, prop. 6), hence with $u + h$. The endomorphism $1_E - (u + h) \circ v$ of E is the sum of the continuous linear map of finite rank $1_E - u \circ v$ and the compact linear map $-h \circ v$, hence is compact. It follows that $u + h$ is a Riesz endomorphism of E (corollary 1).

PROPOSITION 3. — Let E be a Banach space. Let u be an endomorphism of E , and let A be its commutant in $\mathcal{L}(E)$. The closure B of $\pi(A)$ in the algebra $\mathcal{Calk}(E)$ is a Banach algebra. For u to be a Riesz endomorphism of E , it is necessary and sufficient that $\pi(u)$ be invertible in B .

Since $\pi(A)$ is a unital subalgebra of $\mathcal{Calk}(E)$, its closure B is a Banach algebra. If u is a Riesz endomorphism of E , the element $\pi(u)$ has an inverse in $\pi(A)$ (cor. 1), hence in B .

Conversely, suppose that $\pi(u)$ has an inverse s in B . By definition of B , there exists an element v of A such that

$$\|s - \pi(v)\| < \frac{1}{\|\pi(u)\|},$$

whence $\|1_E - \pi(u \circ v)\| < 1$. By prop. 2, the linear map $u \circ v$ is a Riesz endomorphism of E . The same is true of u since u and v commute (III, p. 49, prop. 9).

III.5.5 The theory of Frédéric Riesz

Let E be a separated locally convex space and let h be a compact endomorphism of E . Let $\lambda \in K$. The endomorphism λh is compact, hence $1_E - \lambda h$ is a Riesz endomorphism of E (cor. 2 of prop. 2 of III, p. 75); denote by N_λ and I_λ its nilspace and its conilspace. By prop. 6 of III, p. 47, we have the following properties:

- (i) The vector subspaces I_λ and N_λ of E are closed and stable under h ;
- (ii) The locally convex space E is the topological direct sum of I_λ and N_λ ;
- (iii) The map $1_E - \lambda h$ induces by restriction an automorphism of I_λ ;
- (iv) The vector space N_λ is of finite dimension, and there exists an integer $n_\lambda \geq 0$ such that $(1_E - \lambda h)^{n_\lambda}(x) = 0$ for all $x \in N_\lambda$.

These properties determine the spaces I_λ and N_λ in a unique manner (A, VIII, p. 26, remark 2).

Lemma 3. — Let X be a topological space whose topology has a countable base. Every discrete subset of X is countable.

Let $\mathcal{B}$ be a countable base of the topology of X and let D be a discrete subset of X . For each element x of D , there exists an element B_x of $\mathcal{B}$ such that $B_x \cap D = \{x\}$. The map from D to $\mathcal{B}$ defined by $x \mapsto B_x$ is injective, whence the lemma.

THEOREM 4. — Let E be a separated locally convex space and let h be a compact endomorphism of E . The set Σ of $\lambda \in K$ such that $1_E - \lambda h$ is not an automorphism of E is a countable, closed, and discrete subset of the field K .

The set Σ is also the set of $\lambda \in K$ such that $1_E - \lambda h$ is not injective and the set of $\lambda \in K$ such that $1_E - \lambda h$ is not surjective.

There exists a Banach space G , a continuous linear map $p : E \rightarrow G$ and a compact linear map $q : G \rightarrow E$ such that $h = q \circ p$ (III, p. 5, prop. 5). The endomorphism $h' = p \circ q$ of G is compact, and $1_E - \lambda h$ is an automorphism of E if and only if $1_G - \lambda h'$ is an automorphism of G (prop. 10 of III, p. 49, e)). It therefore suffices for us to prove the theorem when E is a Banach space, which we henceforth assume.

Let $\lambda \in K$. Since the index of a Riesz endomorphism is zero, it is equivalent to say that the map $1_E - \lambda h$ is an automorphism of E , or that it is injective, or again that it is surjective.

The set Σ is closed in K since the set of automorphisms of E is open in $\mathcal{L}(E)$. Let λ be an element of Σ . We have $\lambda \neq 0$. Let N denote the nilspace of h and I its conilspace, and denote by h_I and h_N the endomorphisms of I and of N induced by h on these subspaces. Then $1_I - \lambda h_I$ is an automorphism of I , and there exists a neighborhood U of λ in K such that $1_I - \mu h_I$ is an automorphism of I for all $\mu \in U$. The endomorphism $1_N - \lambda h_N$ of N is nilpotent; for all $\mu \neq \lambda$, we have

$$1_N - \mu h_N = \frac{\lambda - \mu}{\lambda} \left(1_N + \frac{\mu}{\lambda - \mu} (1_N - \lambda h_N) \right),$$

hence $1_N - \mu h_N$ is an automorphism of N . Consequently, the set $U \cap \Sigma$ is reduced to the single element λ and the set Σ is discrete. It is countable by Lemma 3.

III.5.6 Fredholm alternative

PROPOSITION 4 (Fredholm alternative). — Let E be a Banach space and let h be a compact endomorphism of E . Let λ be an element of $K - \{0\}$. Let F be the kernel of $1_E - \lambda h$ and let G be the kernel of $1_{E'} - \lambda^t h$.

- a) The spaces F and G have the same finite dimension $n \geq 0$;
- b) For $y \in E$, there exists $x \in E$ such that $x - \lambda h(x) = y$ if and only if $\langle y, \ell \rangle = 0$ for all $\ell \in G$;

- c) For $\ell \in E'$, there exists $f \in E'$ such that $f - \lambda {}^t h(f) = \ell$ if and only if $\langle x, f \rangle = 0$ for all $x \in F$.

In particular, one and only one of the following conditions holds:

- (i) The space F is nonzero;
- (ii) For every $y \in E$, there exists $x \in E$ such that $x - \lambda h(x) = y$.

By replacing h with λh we reduce to the case where $\lambda = 1$.

The endomorphism ${}^t h$ is compact by Corollary 1 of III, p. 9. The endomorphisms $w = 1_E - h$ of E and $w' = 1_{E'} - {}^t h$ of E' are therefore Riesz endomorphisms (cor. 2 of prop. 2 of III, p. 75). In particular, their kernels F and G are finite-dimensional. Since $w' = {}^t w$ and since the index of w is zero, the dimension of F is equal to that of G by formula (3) of III, p. 43.

For an element y of E , the equation $x - h(x) = y$ has a solution $x \in E$ if and only if x belongs to the image of w . Since the dual of the cokernel of w is identified with the kernel G of $\text{Ker}({}^t w)$ (EVT, IV, p. 27, prop. 2), this is the case if and only if $\langle y, \ell \rangle = 0$ for all $\ell \in G$.

For an element ℓ of E' , the equation $f - {}^t h(f) = \ell$ has a solution $f \in E'$ if and only if f belongs to the image of w' . Since the image of ${}^t w$ is the orthogonal of the kernel F of w (EVT, IV, p. 27, prop. 2), this is the case if and only if $\langle x, f \rangle = 0$ for all $x \in F$.

Example. — Let X be a compact space, μ a measure on X , real or complex depending on whether K is equal to $\mathbb{R}$ or $\mathbb{C}$, and $N: X \times X \rightarrow K$ a continuous function.

Let E denote the Banach space $\mathcal{C}(X; K)$. Suppose that a function $a \in E$ and $\lambda \in K - \{0\}$ are given. We study the problem of the existence and uniqueness of solutions $f \in E$ of the integral equation

$$f(x) - \lambda \int_X N(x, y) f(y) d\mu(y) = a(x) \quad (x \in X). \quad (3)$$

For this, let us introduce the set F_λ of solutions $f \in E$ of the associated homogeneous equation

$$f(x) - \lambda \int_X N(x, y) f(y) d\mu(y) = 0 \quad (x \in X), \quad (4)$$

and the set G_λ of solutions $g \in E$ of the homogeneous transposed equation of (4), that is,

$$g(y) - \lambda \int_X N(x, y) g(x) d\mu(x) = 0 \quad (y \in X). \quad (5)$$

If it is not empty, the set of solutions of (3) is an affine subspace of E with direction F_λ .

For $f \in E$ and $x \in X$, set

$$h(f)(x) = \int_X N(x, y) f(y) d\mu(y);$$

the map $f \mapsto h(f)$ thus defined is a compact endomorphism of the Banach space E (cor. of prop. 2 of III, p. 26). The dual E' of E consists of measures on X , real or complex depending on whether K equals R or C (INT, III, p. 47, § 1, n° 3, def. 2). The transpose ${}^t h$ of h is the endomorphism of E' characterized by the relation $\langle f, {}^t h(m) \rangle = \langle h(f), m \rangle$ for $f \in E$ and $m \in E'$, that is,

$$\begin{aligned} \int_X f(x) d({}^t h(m))(x) &= \int_X h(f)(x) dm(x) \\ &= \int_X \left(\int_X N(x, y) f(y) d\mu(y) \right) dm(x) \\ &= \int_X \left(\int_X N(x, y) dm(x) \right) f(y) d\mu(y) \end{aligned} \quad (6)$$

for $m \in E'$ and $f \in E$ (INT, III, p. 84, § 4, no. 1, th. 2).

Lemma 4. — The linear map from G_λ into E' defined by $g \mapsto g \cdot \mu$ is injective, and its image is the kernel of $1_{E'} - \lambda {}^t h$.

If $g \in G_\lambda$ satisfies $g \cdot \mu = 0$ then formula (5) implies $g = 0$, so the map $g \mapsto g \cdot \mu$ is injective.

Let $m \in E'$ be a measure belonging to the kernel of $1_{E'} - \lambda {}^t h$. By relation (6), we have

$$\int_X f(x) dm(x) = \lambda \int_X \left(\int_X N(x, y) dm(x) \right) f(y) d\mu(y) \quad (7)$$

for every continuous function $f : X \rightarrow K$. The measure m is therefore equal to the measure $g \cdot \mu$, where g is the continuous function from X into K defined by

$$g(y) = \lambda \int_X N(x, y) dm(x) \quad (8)$$

for all $y \in X$. Formula (8) then implies that the function g belongs to G_λ .

Conversely, let $g : X \rightarrow K$ be a continuous function belonging to G_λ and set $m = g \cdot \mu$. For every continuous function $f : X \rightarrow K$, we therefore have

$$\begin{aligned} \int_X f(y) dm(y) &= \int_X g(y) f(y) d\mu(y) \\ &= \lambda \int_X \left(\int_X N(x, y) g(x) d\mu(x) \right) f(y) d\mu(y) \\ &= \lambda \int_X f(x) dm(x), \end{aligned}$$

so that ${}^t h(m) = \lambda m$. This proves Lemma 4.

Applying Proposition 4 and Theorem 4 of III, p. 78 to the endomorphism h , we then obtain the following statements.

THEOREM 5. — Let $\lambda \in K$.

- a) The sets F_λ and G_λ are finite-dimensional vector subspaces of $\mathcal{C}(X; K)$ and their dimensions are equal;
- b) For equation (3) to have at least one solution $f \in \mathcal{C}(X, K)$ it is necessary and sufficient that one has $\int_X a(x) g(x) d\mu(x) = 0$ for every function $g \in G_\lambda$. The set of solutions of (3) is then an affine subspace of $\mathcal{C}(X; K)$ with direction F_λ ;
- c) One of the following mutually exclusive conditions is satisfied:
 - (i) For any function $a \in \mathcal{C}(X; K)$ there exists a unique solution $f \in \mathcal{C}(X; K)$ of the integral equation (3);
 - (ii) The homogeneous equation (4) has a nonzero solution $f \in \mathcal{C}(X; K)$.

Corollary. — The set of $\lambda \in K$ for which the space F_λ is nonzero is a countable, closed, and discrete subset of K .

III.6 SPECTRAL PROPERTIES OF ENDOMORPHISMS OF BANACH SPACES

Unless stated otherwise, the vector spaces considered in this paragraph are vector spaces over $\mathbb{C}$. The spectrum of an endomorphism of a complete normable space E is the spectrum relative to the unital algebra $\mathcal{L}(E)$ (cf. no. 7 of I, p. 127).

III.6.1 Isolated points and sensitive points of the spectrum

Let E be a complete normable space and let u be an endomorphism of E . Let λ be an isolated point of the spectrum of u . Recall that we denote by $e_\lambda(u)$ the spectral projector associated with u and the closed and open set $\{\lambda\}$ of the spectrum of u (no. 3 of I, p. 131). The endomorphism $u - \lambda 1_E$ induces an automorphism of $\text{Ker}(e_\lambda(u))$ and a quasi-nilpotent endomorphism of $\text{Im}(e_\lambda(u))$ (loc. cit.).

DEFINITION 1. — One says that λ has finite spectral multiplicity for u if the spectral projector $e_\lambda(u)$ is of finite rank; in this case, the integer $m_\lambda(u) = \dim(\text{Im}(e_\lambda(u)))$ is called the spectral multiplicity of λ for u .

DEFINITION 2. — One calls sensitive points of the spectrum of u the isolated points of $\text{Sp}(u)$ of finite spectral multiplicity. The set of sensitive points of $\text{Sp}(u)$ is called the sensitive spectrum of u and is denoted $\text{Sp}_s(u)$. ⁽²⁾ One calls essential spectrum of u and denotes by $\text{Sp}_e(u)$ the complementary set $\text{Sp}(u) - \text{Sp}_s(u)$.

As the set $\text{Sp}_s(u)$ consists of isolated points of $\text{Sp}(u)$, it is open in $\text{Sp}(u)$, discrete, and countable (III, p. 78, lemma 3); it is also bounded. The set $\text{Sp}_e(u)$ is closed in $\text{Sp}(u)$, hence compact.

By prop. 14 of III, p. 54, the sensitive points of the spectrum of u are the complex numbers λ such that $u - \lambda 1_E$ is a Riesz endomorphism of E that is not an automorphism of E .

For every complex number λ , denote by $N_\lambda(u)$ and $I_\lambda(u)$ the nilspace and the conilspace of the endomorphism $u - \lambda 1_E$ of E . Since E is a Fréchet space, $u - \lambda 1_E$ is a Riesz endomorphism if and only if $N_\lambda(u)$ is finite-dimensional and $I_\lambda(u)$ has finite codimension (prop. 13 of III, p. 53); when this is the case, $u - \lambda 1_E$ is invertible if and only if $N_\lambda(u)$ is reduced to 0 (III,

p. 47, prop. 6). The sensitive points of the spectrum of u are therefore the complex numbers λ such that $N_\lambda(u)$ is of nonzero finite dimension and $I_\lambda(u)$ is of finite codimension in E (III, p. 54, prop. 14). The spectral multiplicity $m_\lambda(u)$ of λ is then the dimension of $N_\lambda(u)$, and the endomorphism $u - \lambda 1_E$ induces by restriction an automorphism of $I_\lambda(u)$ and a nilpotent endomorphism of $N_\lambda(u)$. Since $N_\lambda(u)$ is not reduced to 0, it follows that λ is an eigenvalue of u . The smallest integer $n \geq 1$ such that $\text{Ker}((u - \lambda 1_E)^n)$ is equal to $N_\lambda(u)$ is the order of the pole of the resolvent of u at the point λ (No. 3 of I, p. 131), which is bounded above by $m_\lambda(u)$.

In particular, we have therefore proved:

PROPOSITION 1. — The sensitive points of the spectrum of u are eigenvalues of finite spectral multiplicity.

The essential spectrum of u consists of the complex numbers λ such that $u - \lambda 1_E$ is not a Riesz endomorphism of E . In particular, if u is compact, then every $\lambda \in \text{Sp}(u) - \{0\}$ is a sensitive point of the spectrum (cor. 2 of III, p. 77).

PROPOSITION 2. — Let E be a complete normable space and u an endomorphism of E . Let H be a finite subset of $\text{Sp}_s(u)$. The set H is open and closed in $\text{Sp}(u)$. The spectral projector $e_H(u)$ has finite rank, has image $\sum_{\lambda \in H} N_\lambda(u)$ and kernel $\bigcap_{\lambda \in H} I_\lambda(u)$.

This follows from no. 3 of I, p. 131 since, for every $\lambda \in \text{Sp}_s(u)$, the subspaces $N_\lambda(u)$ and $I_\lambda(u)$ are respectively the image and the kernel of the spectral projector $e_\lambda(u)$.

COROLLARY 1. — Let V be a neighborhood of $\text{Sp}_e(u)$.

- a) There exists a decomposition of E as a topological direct sum $F \oplus G$ such that F is finite-dimensional, F and G are stable under every endomorphism commuting with u , and the spectrum of the endomorphism of G induced by u is contained in V .
- b) Suppose V is nonempty. There exists an element v of the bicommutant of u in $\mathcal{L}(E)$ whose spectrum is contained in V and such that $v - u$ has finite rank.

Set $H = \text{Sp}(u) \cap (\mathbb{C} - \mathring{V})$. The set H is contained in $\text{Sp}_s(u)$, hence is discrete; since it is closed in the compact space $\text{Sp}(u)$, it is finite. One may take for F and G the image and the kernel of the spectral projector $e_H(u)$ (prop. 2).

Suppose that V is not empty. Let $\mu \in V$ and denote by v the endomorphism of E that

coincides with the homothety of ratio μ in F and with u in G . Its spectrum is contained in $\{\mu\} \cup (\text{Sp}(u) - H)$, therefore in V , and $v - u$ is of finite rank since F is finite-dimensional.

For every compact subset S of C , we denote by $\mathcal{O}(S)$ the algebra of germs of holomorphic functions in a neighborhood of S and with values in C (I, p. 49, §4, no. 1).

COROLLARY 2. — Let $f \in \mathcal{O}(\text{Sp}(u))$ and let μ be a complex number. Let H be the set of $\lambda \in \text{Sp}(u)$ such that $f(\lambda) = \mu$. Then μ is a sensitive point of the spectrum of $f(u)$ if and only if H is finite, nonempty, and contained in the sensitive spectrum of u . Under these conditions, the spectral projector $e_\mu(f(u))$ is equal to the projector $e_H(u)$ associated with u and H . We have

$$N_\mu(f(u)) = \bigoplus_{\lambda \in H} N_\lambda(u), \quad I_\mu(f(u)) = \bigcap_{\lambda \in H} I_\lambda(u)$$

and the spectral multiplicity of μ for $f(u)$ is the sum of the spectral multiplicities of the elements of H , that is,

$$m_\mu(f(u)) = \sum_{\lambda \in H} m_\lambda(u).$$

The complex number μ belongs to $f(\text{Sp}(u))$, and therefore to the spectrum of $f(u)$ (prop. 8 of I, p. 75), if and only if H is nonempty. We then have $e_\mu(f(u)) = e_H(u)$ (no. 1 of I, p. 127). The other assertions follow from this by Prop. 2.

Examples. — 1) Let E be a finite-dimensional vector space and u an endomorphism of E . The spectrum of u is finite and coincides with $\text{Sp}_s(u)$. Its elements are the roots of the characteristic polynomial χ_u of u ; these are the eigenvalues of u . The spectral multiplicity $m_\lambda(u)$ of an element $\lambda \in \text{Sp}(u)$ is the multiplicity of λ as a root of the polynomial χ_u (A, VII, p. 36, cor.). By cor. 2 above, we have

$$\chi_{f(u)}(T) = \prod_{\lambda \in \text{Sp}(u)} (T - f(\lambda))^{m_\lambda}$$

for all $f \in \mathcal{O}(\text{Sp}(u))$. This generalizes prop. 10 of A, VII, p. 36.

2) Let X be a compact subset of C and $\mathcal{C}(X)$ the Banach space of continuous functions on X with complex values. Denote by u the endomorphism of $\mathcal{C}(X)$ that sends $f \in \mathcal{C}(X)$ to the function $z \mapsto zf(z)$ in $\mathcal{C}(X)$. The spectrum of u is equal to X , its sensitive points are

the isolated points of X and their spectral multiplicities are equal to 1.

III.6.2 A partition of the spectrum

Denote by $\bar{Z}$ the subset $Z \cup \{-\infty, +\infty\}$ of $\bar{R}$. If u is a linear map whose kernel or cokernel is finite-dimensional, one calls the index of u the element $\text{ind}(u)$ of $\bar{Z}$ defined by

$$\text{ind}(u) = \dim \text{Coker}(u) - \dim \text{Ker}(u)$$

(cf. no. 6 of III, p. 67).

DEFINITION 3. — Let E be a complete normable space and u an endomorphism of E . For every $n \in \bar{Z}$, denote by $\text{Sp}_n(u)$ the set of complex numbers $\lambda \in \text{Sp}_e(u)$ such that $u - \lambda 1_E$ is a strict morphism, whose kernel or cokernel is finite-dimensional, and whose index is n . We denote $\text{Sp}_\omega(u)$ the complement in $\text{Sp}_e(u)$ of the union of the subsets $\text{Sp}_n(u)$ for $n \in \bar{Z}$.

The sets $\text{Sp}_s(u)$, $\text{Sp}_n(u)$ for $n \in \bar{Z}$ and $\text{Sp}_\omega(u)$ form a partition of the spectrum of u .

Every endomorphism of E whose cokernel is finite-dimensional is strict (III, p. 52, lemma 6). The Fredholm endomorphisms of E are the endomorphisms of E whose kernel and cokernel are finite-dimensional (III, p. 52, prop. 11). The set $\mathbf{C} - \text{Sp}_e(u)$ consists of the $\lambda \in \mathbf{C}$ such that $u - \lambda 1_E$ is a Riesz endomorphism of E , and such an endomorphism is a Fredholm endomorphism of E of index 0.

Consequently, for $\lambda \in \mathbf{C}$ and for $n \in \mathbf{Z} - \{0\}$, we have:

$$\lambda \in \mathbf{C} - \text{Sp}(u) \iff u - \lambda 1_E \text{ is an automorphism;}$$

$$\lambda \in \text{Sp}_s(u) \iff u - \lambda 1_E \text{ is a Riesz endomorphism, but is not an automorphism;}$$

$$\lambda \in \text{Sp}_0(u) \iff u - \lambda 1_E \text{ is a Fredholm endomorphism of index 0 of } E \text{ but is not a Riesz endomorphism of } E;$$

$$\lambda \in \text{Sp}_n(u) \iff u - \lambda 1_E \text{ is a Fredholm endomorphism of index } n \text{ of } E \text{ (} n \neq 0 \text{);}$$

$$\lambda \in \text{Sp}_{-\infty}(u) \iff u - \lambda 1_E \text{ is strict, its kernel is infinite-dimensional and its cokernel is finite-dimensional}$$

$$\lambda \in \text{Sp}_{+\infty}(u) \iff u - \lambda 1_E \text{ is strict, its kernel is finite-dimensional and its cokernel is}$$

infinite-dimensional;

$$\lambda \in \text{Sp}_\omega(u) \iff \text{either } u - \lambda 1_E \text{ is not strict, or its kernel and cokernel are infinite-dimensional.}$$

Remark. — Let π denote the canonical homomorphism of $\mathcal{L}(E)$ onto the Calkin algebra $\mathcal{Calk}(E)$ of E (cf. no. 4 of III, p. 75). The spectrum of $\pi(u)$ with respect to the algebra $\mathcal{Calk}(E)$ is sometimes called the stable spectrum of u . Its elements are the complex numbers λ such that $u - \lambda 1_E$ is not a Fredholm endomorphism of E (cor. 1 of th. 3 of III, p. 73). It is therefore equal to $\text{Sp}_\omega(u) \cup \text{Sp}_{-\infty}(u) \cup \text{Sp}_{+\infty}(u)$.

Let A be the set of endomorphisms of E commuting with u . The closure B of $\pi(A)$ in $\mathcal{Calk}(E)$ is a complete normable algebra and the spectrum of $\pi(u)$ with respect to B is the essential spectrum $\text{Sp}_e(u)$ of u (III, p. 77, prop. 3). By the cor. of prop. 6 of I, p. 28, applied to the subalgebra B of $\mathcal{Calk}(E)$, the set $\text{Sp}_e(u)$ is the union of $\text{Sp}_\omega(u) \cup \text{Sp}_{-\infty}(u) \cup \text{Sp}_{+\infty}(u)$ and of certain bounded connected components of its complement.

THEOREM 1. — Let E be a complete normable space and u an endomorphism of E .

- a) The set $\text{Sp}_\omega(u)$ is compact. It is nonempty if E is of infinite dimension;
- b) Let $n \in \overline{\mathbb{Z}}$. The set $\text{Sp}_n(u)$ is a union of a family of bounded connected components of $\mathbb{C} - \text{Sp}_\omega(u)$. It is open in $\mathbb{C}$, and its boundary in $\mathbb{C}$ is contained in $\text{Sp}_\omega(u)$.

The set $\mathbb{C} - \text{Sp}_\omega(u)$ consists of the complex numbers $\lambda \in \mathbb{C}$ such that $u - \lambda 1_E$ is a strict morphism whose kernel or cokernel is finite-dimensional. By Propositions 11 of III, p. 67 and 13 of III, p. 70, it is open. The set $\text{Sp}_\omega(u)$ is therefore closed. Since it is bounded, it is compact.

Let us prove b). Let $n \in \overline{\mathbb{Z}}$. The set $\text{Sp}_n(u)$ is contained in $\mathbb{C} - \text{Sp}_\omega(u)$. Let U be a connected component of $\mathbb{C} - \text{Sp}_\omega(u)$ which intersects $\text{Sp}_n(u)$. The map $\lambda \mapsto \text{ind}(u - \lambda 1_E)$ of $\mathbb{C} - \text{Sp}_\omega(u)$ in $\overline{\mathbb{Z}}$ being locally constant (cor. 1 of prop. 12 of III, p. 68 and cor. 1 of prop. 13 of III, p. 70), the index of $u - \lambda 1_E$ is equal to n for all $\lambda \in U$. If $n \neq 0$, this implies that U is contained in $\text{Sp}_n(u)$. If $n = 0$, note that the set U is a connected component of $\mathbb{C} - (\text{Sp}_\omega(u) \cup \text{Sp}_{-\infty}(u) \cup \text{Sp}_{+\infty}(u))$. Since U meets $\text{Sp}_0(u)$ and therefore $\text{Sp}_e(u)$, it follows from Remark 2 that the set U is contained in $\text{Sp}_e(u)$, and hence in $\text{Sp}_0(u)$. In all cases, we conclude that $\text{Sp}_n(u)$ is the union of the connected components of $\mathbb{C} - \text{Sp}_\omega(u)$ that meet $\text{Sp}_n(u)$. These are necessarily bounded since the set $\text{Sp}(u)$ is bounded. Consequently, $\text{Sp}_n(u)$ is open

in $\mathbf{C}$ and its boundary is contained in $\text{Sp}_\omega(u)$. This proves b).

Suppose finally that the set $\text{Sp}_\omega(u)$ is empty. By b), each of the sets $\text{Sp}_n(u)$, for $n \in \overline{\mathbf{Z}}$, is then empty. We therefore have $\text{Sp}(u) = \text{Sp}_s(u)$. The spectrum of u is consequently discrete and compact, hence finite, and since all its points have finite spectral multiplicity, the vector space E is finite-dimensional (III, p. 83, prop. 2). This completes the proof of a).

COROLLARY. — a) Let Ω be the unbounded connected component of $\mathbf{C} - \text{Sp}_\omega(u)$. We have $\Omega \cap \text{Sp}(u) \subset \text{Sp}_s(u)$;

b) Every point adherent to $\text{Sp}_s(u)$ that does not belong to $\text{Sp}_s(u)$ belongs to $\text{Sp}_\omega(u)$.

The assertion a) is a direct consequence of assertion b) of th. 1. Let λ be a point adherent to $\text{Sp}_s(u)$ that does not belong to $\text{Sp}_s(u)$. It belongs to the spectrum of u , since the latter is closed. It does not belong to any of the sets $\text{Sp}_n(u)$, for $n \in \overline{\mathbf{Z}}$, since these are open (loc. cit.) and disjoint from $\text{Sp}_s(u)$. We therefore have $\lambda \in \text{Sp}_\omega(u)$, whence b).

PROPOSITION 3. — Let E and F be complete normable spaces, $u: E \rightarrow F$ and $v: F \rightarrow E$ continuous linear maps.

- a) The traces on $\mathbf{C} - \{0\}$ of the sets $\text{Sp}(v \circ u)$ and $\text{Sp}(u \circ v)$ (resp. $\text{Sp}_s(v \circ u)$ and $\text{Sp}_s(u \circ v)$, resp. $\text{Sp}_n(v \circ u)$ and $\text{Sp}_n(u \circ v)$ for $n \in \overline{\mathbf{Z}}$, resp. $\text{Sp}_\omega(v \circ u)$ and $\text{Sp}_\omega(u \circ v)$) are equal;
- b) Let λ be an element of $\text{Sp}_s(v \circ u)$ distinct from 0. The spectral multiplicities of λ for $v \circ u$ and for $u \circ v$ are equal.

Let μ be a nonzero complex number and let $n \in \overline{\mathbf{Z}}$. For $\mu 1_E - v \circ u$ to be an automorphism (resp. a Riesz endomorphism, resp. a strict morphism whose kernel or cokernel is finite-dimensional and whose index is n), it is necessary and sufficient that $\mu 1_F - u \circ v$ be one (III, p. 49, prop. 10). The assertion a) then follows from the definitions.

Let λ be a point of $\text{Sp}_s(v \circ u)$ distinct from 0. We have

$$\dim \text{Ker}((\lambda 1_E - v \circ u)^n) = \dim \text{Ker}((\lambda 1_F - u \circ v)^n)$$

for all $n \geq 0$ (loc. cit.), hence the spectral multiplicities of λ for $v \circ u$ and $u \circ v$ are equal.

III.6.3 Spectrum of the transpose of an endomorphism

PROPOSITION 4. — Let E be a complete normable space, E' the dual space of E , and u an endomorphism of E .

- a) One has $\text{Sp}_s(u) = \text{Sp}_s({}^t u)$, $\text{Sp}_n(u) = \text{Sp}_{-n}({}^t u)$ for all $n \in \overline{\mathbf{Z}}$ and $\text{Sp}_\omega(u) = \text{Sp}_\omega({}^t u)$;
- b) Every point of $\text{Sp}_s(u)$ has the same spectral multiplicity for u and ${}^t u$.

The assertion a) of prop. 3 of I, p. 131 shows that $\text{Sp}(u) = \text{Sp}({}^t u)$, and assertion c) of loc. cit. implies that $\text{Sp}_s(u) = \text{Sp}_s({}^t u)$ and that the assertion b) is valid.

For every $\lambda \in \mathbf{C}$, Lemma 4 of III, p. 69 implies that $u - \lambda 1_E$ is a strict morphism if and only if ${}^t u - \lambda 1_{E'}$ is one, and that

$$\dim \text{Coker}({}^t(u - \lambda 1_E)) = \dim \text{Ker}(u - \lambda 1_E)$$

$$\dim \text{Ker}({}^t(u - \lambda 1_E)) = \dim \text{Coker}(u - \lambda 1_E)$$

in $\overline{\mathbf{Z}}$. Assertion a) follows from this, taking into account the definitions of the various parts of the spectrum (def. 3 of III, p. 85).

III.6.4 Perturbation by a compact operator

THEOREM 2. — Let E be a complete normable space, u an endomorphism of E , and h a compact endomorphism of E . Then $\text{Sp}_\omega(u + h) = \text{Sp}_\omega(u)$ and $\text{Sp}_n(u + h) = \text{Sp}_n(u)$ for all $n \in \overline{\mathbf{Z}} - \{0\}$.

Let $\lambda \in \mathbf{C}$. For $u + h - \lambda 1_E$ to be a strict morphism whose kernel (resp. cokernel) is finite-dimensional, it is necessary and sufficient that $u - \lambda 1_E$ have the same property, by Theorem 1 of III, p. 72 (resp. Theorem 2 of III, p. 73). The equalities

$$\begin{aligned} \text{Sp}_{-\infty}(u + h) &= \text{Sp}_{-\infty}(u), & \text{Sp}_{+\infty}(u + h) &= \text{Sp}_{+\infty}(u), \\ \text{Sp}_\omega(u + h) &= \text{Sp}_\omega(u) \end{aligned}$$

result from this. Moreover, we have $\text{Sp}_n(u + h) = \text{Sp}_n(u)$ by th. 3 of III, p. 73, for every $n \in \mathbf{Z} - \{0\}$.

COROLLARY. — Suppose that the unbounded connected component of the complement of $\text{Sp}_\omega(u)$ contains 0. Then $u + h$ is a Riesz endomorphism of E .

One has $\text{Sp}_\omega(u + h) = \text{Sp}_\omega(u)$ (by th. 2), hence 0 belongs to the unbounded connected component of $\mathbb{C} - \text{Sp}_\omega(u + h)$. By the cor. of th. 1 of III, p. 87, either 0 does not belong to the spectrum of $u + h$, or it is a sensitive point of this spectrum. In either case, $u + h$ is a Riesz endomorphism of E .

III.6.5 Spectrum of a compact operator

Lemma 1. — Let E be a separated topological vector space of dimension ≥ 2 over $\mathbb{R}$, and let X be a countable subset of E . The complement $E - X$ is connected.

First, suppose that E has dimension 2. We may assume that $E = \mathbb{R}^2$ equipped with the Euclidean norm (EVT, I, p. 14, th. 2). Since X is countable, there exists a real number $r \in \mathbb{R}_+^*$ such that the circle C with center 0 and radius r does not meet X . Let $x \in E - X$; if $x \notin C$, there exists a point $y \in C$ such that the line L_x joining x and y does not meet X , since X is countable. The set $E - X$ is the union of C , which is connected, and of the connected sets $L_x \cup C$ for $x \in E - (X \cup C)$; these sets all contain C , and therefore $E - X$ is connected (TG, I, p. 81, prop. 2).

Consider the general case. By replacing X with $X - x$ for an element $x \in E - X$ we reduce to the case where $0 \notin X$. Since $E - X$ is the union of the sets $F - (X \cap F)$ as F ranges over the set of 2-dimensional subspaces of E , and since these are connected by the preceding case and contain 0, the set $E - X$ is connected (loc. cit.).

Lemma 2. — Let $S \subset \mathbb{C}$ be an infinite, discrete, bounded, and closed set in $\mathbb{C} - \{0\}$. Then S is the set of values of a sequence $(\lambda_n)_{n \in \mathbb{N}}$ of nonzero complex numbers, pairwise distinct, such that the sequence $(|\lambda_n|)_{n \in \mathbb{N}}$ is decreasing and converges to 0.

For every integer $i \geq 1$, the set $\mathbf{A}_i$ of complex numbers $\lambda \in S$ such that $|\lambda| \geq \frac{1}{i}$ is compact and discrete in $\mathbb{C}$, hence finite. Denote its cardinal by a_i . Since S is infinite, the sequence (a_i) tends to $+\infty$. Set $\mathbf{A}_0 = \emptyset$ and $a_0 = 0$. For every $i \geq 1$, let us choose a bijection $n \mapsto \lambda_n$ from the interval $[a_{i-1}, a_i[$ of $\mathbb{N}$ onto $\mathbf{A}_i - \mathbf{A}_{i-1}$ such that the map $n \mapsto |\lambda_n|$ is decreasing on $[a_{i-1}, a_i[$. The sequence $(\lambda_n)_{n \in \mathbb{N}}$ has the required properties.

Let E be a complete normable space of infinite dimension. The algebra $\mathcal{L}^c(E)$ is a non-unital subalgebra of $\mathcal{L}(E)$. Recall that for every compact endomorphism $u \in \mathcal{L}^c(E)$ the spectrum $\text{Sp}'_{\mathcal{L}^c(E)}(u)$ is the spectrum of u relative to the unital subalgebra $\mathcal{L}^c(E) \oplus \mathbf{C}1_E$ of $\mathcal{L}(E)$ (I, p. 4, no. 4).

PROPOSITION 5. — Let E be a complete normable space and let u be a compact endomorphism of E . Every element of $\text{Sp}_s(u)$ is an eigenvalue of finite spectral multiplicity. Moreover:

- a) If E is finite-dimensional, then $\text{Sp}(u) = \text{Sp}_s(u)$;
- b) If E is of infinite dimension, then $\text{Sp}_s(u) = \text{Sp}(u) - \{0\}$ and $\text{Sp}_\omega(u) = \{0\}$;
- c) If $\text{Sp}_s(u)$ is infinite, it is the set of values of a sequence $(\lambda_n)_{n \in \mathbf{N}}$ of nonzero complex numbers, pairwise distinct, such that the sequence $(|\lambda_n|)_{n \in \mathbf{N}}$ is decreasing and converges to 0;
- d) If E is of infinite dimension, then $\text{Sp}(u) = \text{Sp}'_{\mathcal{L}^c(E)}(u)$.

Every element of $\text{Sp}_s(u)$ is an eigenvalue of finite spectral multiplicity (prop. 1 of III, p. 83).

The assertion *a*) is elementary (III, p. 85, example 1). Suppose now that E is of infinite dimension.

Let $\lambda \in \text{Sp}(u) - \{0\}$. Then $u - \lambda 1_E$ is a Riesz endomorphism of E (cor. 2 of prop. 2 of III, p. 75), hence λ belongs to $\text{Sp}_s(u)$. If E is infinite-dimensional, then $\text{Sp}_\omega(u)$ is nonempty (III, p. 87, th. 1, a)). Necessarily, $\text{Sp}_\omega(u) = \{0\}$, whence *b*).

The set $\text{Sp}_s(u)$ is discrete and bounded. Moreover, we have $\text{Sp}_s(u) = \text{Sp}(u) \cap (\mathbf{C} - \{0\})$ according to *b*), hence $\text{Sp}_s(u)$ is closed in $\mathbf{C} - \{0\}$. The assertion *c*) therefore follows from lemma 2.

The spectrum of u is countable, and its complement in $\mathbf{C}$ is therefore connected (lemma 1). By the corollary of prop. 6 of I, p. 28, applied to the unital subalgebra $\mathcal{L}^c(E) \oplus \mathbf{C}1_E$ of $\mathcal{L}(E)$, we conclude that $\text{Sp}(u) = \text{Sp}'_{\mathcal{L}^c(E)}(u)$.

PROPOSITION 6. — Let E be a complete normable space over $\mathbf{C}$ and u a compact endomorphism of E .

- a) Let $f \in \mathcal{O}(\text{Sp}(u))$ such that $f(0) = 0$. The endomorphism $f(u)$ is compact;
- b) Suppose that E is a complex Hilbert space and that u is normal. Let f be a continuous function on $\text{Sp}(u)$ such that $f(0) = 0$. The normal endomorphism $f(u)$ is compact.

Moreover, the converse assertions are valid if E is infinite-dimensional and the condition $f(0) = 0$ is not necessary if E is finite-dimensional.

We may assume that E is infinite-dimensional.

Let us prove *a*) and its converse. The endomorphism u is an element of the Banach algebra $\mathcal{L}^c(E)$. Since E is of infinite dimension, we have the equality $\text{Sp}(u) = \text{Sp}'_{\mathcal{L}^c(E)}(u)$ by prop. 5, *d*). The element $f(u)$ of the holomorphic functional calculus of the unital Banach algebra derived from $\mathcal{L}^c(E)$ by adjoining an identity element belongs to $\mathcal{L}^c(E)$ if and only if $f(0) = 0$ (I, p. 88). But moreover this element coincides with the element $f(u)$ of $\mathcal{L}(E)$ (prop. 7 of I, p. 75), hence $f(u)$ is compact if and only if $f(0) = 0$.

The proof of assertion *b*) and of its converse is entirely identical, considering the continuous functional calculus of the C^* -algebra $\mathcal{L}^c(E)$ (I, p. 110, def. 5).

COROLLARY. — Let E and F be Hilbert spaces and let u be a continuous linear map from E into F . The linear map u is compact if and only if the endomorphism $|u|$ of E is compact.

Let $(j, |u|)$ be the polar decomposition of u (def. 4 of I, p. 140). Since $u = j|u|$ and $|u| = \sqrt{u^*u}$ (prop. 10 of I, p. 139), the equivalence follows from prop. 3 of III, p. 5 and from assertion *b*) of the preceding proposition.

III.6.6 Case of Hilbert spaces

In this section, E denotes a Hilbert space over $\mathbb{C}$. We denote by π the canonical homomorphism of $\mathcal{L}(E)$ onto the Calkin algebra $\mathcal{Calk}(E)$.

PROPOSITION 7. — Let $u \in \mathcal{L}(E)$.

- a) If u is normal, then u is a Riesz endomorphism if and only if u is a Fredholm endomorphism of index 0.
- b) If u is Hermitian, then u is a Fredholm endomorphism if and only if u is a Riesz endomorphism.

Every Riesz endomorphism is a Fredholm endomorphism of index 0 (Proposition 5 of III, p. 46). Conversely, suppose that u is a Fredholm endomorphism of index 0, and that u is normal. Its nilspace then coincides with its kernel (EVT, V, p. 43, prop. 8), and is therefore finite-dimensional. It then follows from Lemma 2 of III, p. 45 and from the definition that u is a Riesz endomorphism.

Let us prove b). According to a), it suffices to check that the index of a Hermitian Fredholm endomorphism u is zero. But the orthogonal complement of the image of u (which is closed) is then equal to the kernel of u (EVT, V, p. 41, prop. 4), whence the assertion.

COROLLARY. — Let $u \in \mathcal{L}(E)$. If u is normal, then $\text{Sp}_0(u)$ is empty, and if u is Hermitian, then $\text{Sp}_e(u)$ coincides with the spectrum of $\pi(u)$ relative to the algebra $\mathcal{C}a\ell k(E)$.

Both assertions follow from the proposition and the definitions of the sets $\text{Sp}_n(u)$ for $n \in \overline{\mathbb{Z}}$ and of $\text{Sp}_\omega(u)$, which form a partition of the essential spectrum of u (def. 2 of III, p. 83).

THEOREM 3 (Weyl). — Let $u \in \mathcal{L}(E)$ be a normal endomorphism of E . The essential spectrum of u is the intersection of the sets $\text{Sp}(u + h)$ for h ranging over $\mathcal{L}^c(E)$.

Let $h \in \mathcal{L}^c(E)$. Since $\text{Sp}_0(u)$ is empty (corollary above), Theorem 2 of III, p. 89 implies that $\text{Sp}_e(u + h) = \text{Sp}_e(u)$. The intersection of the sets $\text{Sp}(u + h)$ therefore contains $\text{Sp}_e(u)$.

Let $\lambda \in \text{Sp}_s(u)$. Denote by E_λ the eigenspace of u relative to λ , and F_λ the image of the spectral projector associated with u and $\mathbf{C} = \{\lambda\}$. The space E is the topological direct sum of E_λ and F_λ . Let h be the finite-rank endomorphism of E that vanishes on F_λ and coincides on E_λ with the identity. The endomorphism $u + h$ is invertible, hence $\lambda \notin \text{Sp}(u + h)$. The theorem follows from this.

Thus, if u is a normal endomorphism of E , and if $h \in \mathcal{L}^c(E)$, the spectrum of $u + h$ can differ from the spectrum of u only by isolated points that have finite spectral multiplicity.

III.6.7 The Krein-Rutman theorem

Lemma 3. — Let $(a_n)_{n \geq 0}$ be a sequence of positive real numbers such that the power series

$$f(z) = \sum_{n \geq 0} a_n z^n$$

has a finite radius of convergence $r > 0$. Let $D \subset C$ be the open disk with center 0 and radius

r. There does not exist a holomorphic function $\tilde{f}$ defined on an open neighborhood U of r in $\mathbb{C}$ that coincides with f on $U \cap D$.

Suppose that such a holomorphic function $\tilde{f}$, defined on an open neighborhood U of r , exists. There exist real numbers $s < r$ and $\delta > 0$ such that $s + \delta > r$ and such that the open disk D' with center s and radius δ is contained in U . The power series expansion of $\tilde{f}$ at the point s (VAR, R1, p. 26, 3.2.1) then converges for every z in the disk with center 0 and radius δ (VAR, R1, p. 29, 3.3.4). This series is

$$\tilde{f}_s(z) = \sum_{n=0}^{+\infty} \frac{\tilde{f}^{(n)}(s)}{n!} z^n = \sum_{n=0}^{+\infty} \frac{f^{(n)}(s)}{n!} z^n$$

(VAR, R1, p. 27, 3.2.4). Since $s \in D$, we have

$$f^{(n)}(s) = \sum_{k=n}^{+\infty} k(k-1) \dots (k-n+1) a_k s^{k-n}$$

(VAR, R1, p. 28, 3.2.11). Take z such that $0 < z < \delta$. Since $a_k \geq 0$, we have

$$\begin{aligned} \tilde{f}_s(z) &= \sum_{n=0}^{+\infty} \left(\sum_{k=n}^{+\infty} \frac{k(k-1) \dots (k-n+1)}{n!} a_k s^{k-n} \right) z^n \\ &= \sum_{k=0}^{+\infty} \left(a_k \sum_{n=0}^k \frac{k!}{n!(k-n)!} s^{k-n} z^n \right) = \sum_{k=0}^{+\infty} a_k (s+z)^k \end{aligned}$$

(TG, III, p. 40). When $s+z > r$, the convergence of this expression contradicts the hypothesis that the radius of convergence of the power series f is equal to r .

Let E be a real Banach space. Let C be a pointed convex cone in E . The vector space generated by C is equal to $C - C$ (EVT, II, p. 12, cor. 1). In particular, the cone C is total in E if and only if $C - C$ is dense in E . Recall that C is said to be salient if $C \cap (-C)$ is reduced to 0 (EVT, II, p. 11).

The polar C° of C is the set of continuous linear forms $\ell \in E'$ such that $\ell(x) \geq 0$ for all $x \in C$ (EVT, II, p. 47, prop. 4). By the bipolar theorem (EVT, II, p. 48, th. 1), if C is a closed pointed convex cone, then $C^{\circ\circ} = C$.

Lemma 4. — Let E be a real Banach space and $u \in \mathcal{L}(E_{(C)})$ a nonzero endomorphism of $E_{(C)}$. Let C be a total convex cone in E . There exists $x \in C$ such that $u(x) \neq 0$.

Indeed, if u vanishes on C , then u vanishes on $C - C$, hence on E , and therefore on $E_{(C)}$.

THEOREM 4 (Krein–Rutman). — Let E be a complete normable space over $\mathbb{R}$. Let C be a closed total salient convex cone in E and let $u \in \mathcal{L}(E)$ be a compact linear map such that $u(C) \subset C$. If the spectral radius $\varrho(u)$ is >0 , then $\varrho(u)$ is an isolated point of the spectrum of u , and there exists a nonzero eigenvector $x \in C$ of u for the eigenvalue $\varrho(u)$.

Let $E_{(\mathbb{C})}$ be the complexification of E and $u_{(\mathbb{C})}$ the endomorphism of $E_{(\mathbb{C})}$ obtained by extension of scalars from u . The spectral radius of $u_{(\mathbb{C})}$ is equal to $\varrho(u)$ (I, p. 86); it is simply denoted by ϱ . Since $\varrho > 0$, the complex spectrum of u is not reduced to 0. There therefore exists $\lambda_0 \in \text{Sp}_s(u_{(\mathbb{C})})$ such that $|\lambda_0| = \varrho$ (prop. 5 of III, p. 90). Let e_0 be the spectral projector of $u_{(\mathbb{C})}$ associated with λ_0 .

The resolvent $\lambda \mapsto R(u_{(\mathbb{C})}, \lambda) = (\lambda - u_{(\mathbb{C})})^{-1}$ is holomorphic on the complement of the spectrum of $u_{(\mathbb{C})}$ (th. 1 of I, p. 24). The complex number λ_0 is a pole of the resolvent and its residue is the spectral projector e_0 (I, p. 131).

Let y and y' be elements of E such that $y + iy' \in E_{(\mathbb{C})}$ is an eigenvector of $u_{(\mathbb{C})}$ for the eigenvalue λ_0 . Since $e_0(y + iy') = y + iy' \neq 0$, there exists an element $x_0 \in C$ such that $e_0(x_0) \neq 0$ (lemma 4). Since C is closed and pointed, its polar C° is total (EVT, II, p. 48, cor. 1), and hence there exists a linear form $\ell_0 \in C^\circ$ such that $\langle e_0(x_0), \ell_0 \rangle \neq 0$.

Consider the function f defined by $f(0) = 0$ and by

$$f(z) = \langle R(u_{(\mathbb{C})}, z^{-1})x_0, \ell_0 \rangle$$

for $z \in \mathbb{C}$ such that $z^{-1} \notin \text{Sp}(u_{(\mathbb{C})})$. This function satisfies

$$f(z) = \ell_0 \left(\left(\sum_{n=0}^{+\infty} z^{n+1} u_{(\mathbb{C})}^n \right) x_0 \right) = \sum_{n=0}^{\infty} \langle u_{(\mathbb{C})}^n(x_0), \ell_0 \rangle z^{n+1} \quad (1)$$

for $|z| < 1/\varrho$ (Theorem 1 of I, p. 24, d)) and is therefore holomorphic in the disk centered at 0 with radius $1/\varrho$.

There exists a holomorphic function $\tilde{R}$ defined on an open neighborhood U of λ_0 and with values in $\mathcal{L}(E)$ such that for z belonging to $U - \{\lambda_0\}$, we have

$$R(u_{(\mathbb{C})}, z) = \tilde{R}(z) + \sum_{n=0}^{+\infty} (z - \lambda_0)^{-n-1} (u_{(\mathbb{C})} - \lambda_0)^n e_0$$

(prop. 17 of I, p. 83). For z such that $z^{-1} \in U$ and $z \neq 1/\lambda_0$, we therefore have

$$f(z) = \langle \tilde{R}(z^{-1})x_0, \ell_0 \rangle + \sum_{n=0}^{+\infty} (z^{-1} - \lambda_0)^{-n-1} \langle (u_{(C)} - \lambda_0)^n e_0(x_0), \ell_0 \rangle.$$

The term of the series corresponding to $n = 0$ is $(z^{-1} - \lambda_0)^{-1} \langle e_0(x_0), \ell_0 \rangle$. Since $\langle e_0(x_0), \ell_0 \rangle \neq 0$, the uniqueness of the Laurent expansion (VAR, R1, p. 30, 3.3.9) implies that f does not extend to a holomorphic function in a neighborhood of $1/\lambda_0$. In particular, the radius of convergence of the power series expansion (1) of the function f at the point 0 is equal to $1/\varrho$.

The coefficients of the power series (1) are $\langle u^n(x_0), \ell_0 \rangle \geq 0$ since $u(C) \subset C$ and $\ell_0 \in C^\circ$. By Lemma 3, the function f does not extend to a holomorphic function in a neighborhood of $1/\varrho$. The resolvent of $u_{(C)}$ therefore cannot be extended to a holomorphic function in a neighborhood of ϱ , that is, $\varrho \in \text{Sp}(u)$. This implies that ϱ is an eigenvalue of $u_{(C)}$ (prop. 5 of III, p. 90). Since ϱ is real, it is also an eigenvalue of u .

Let $d \geq 1$ be the order of the pole of the resolvent of $u_{(C)}$ at ϱ . Let e be the endomorphism

$$e = \lim_{\substack{z \rightarrow \varrho \\ z > \varrho}} (z - \varrho)^d R(u_{(C)}, z).$$

It is nonzero and commutes with $u_{(C)}$, and its image is contained in the eigenspace of $u_{(C)}$ relative to ϱ (cf. no. 3 of I, p. 131). Now let ℓ be an element of C° and x an element of C . By Theorem 1 of I, p. 24, d), we have

$$\langle e(x), \ell \rangle = \lim_{\substack{z \rightarrow \varrho \\ z > \varrho}} (z - \varrho)^d \sum_{n \geq 0} \langle u^n(x), \ell \rangle z^{-n-1} \geq 0.$$

The bipolar theorem (EVT, II, p. 48, th. 1) implies that $e(x) \in C$ for all $x \in C$. Since C is total, there exists $x \in C$ such that $e(x) \neq 0$ (Lemma 4), and then $e(x)$ belongs to C and is an eigenvector of u for the eigenvalue ϱ , as desired.

COROLLARY. — Let E be a complete normable space over R . Let C be a closed, total, salient convex cone in E , and $u \in \mathcal{L}(E)$ be a compact linear map such that $u(C) \subset C$. If the spectral radius $\varrho(u)$ of u is > 0 , then $\varrho({}^t u) = \varrho(u)$ is an eigenvalue of ${}^t u$, and ${}^t u$ admits an eigenvector relative to $\varrho(u)$ in C° .

One has $\varrho({}^t u) = \varrho(u)$ (prop. 3 of I, p. 131). By corollary 1 of III, p. 9, the endomorphism ${}^t u$ of E' is compact. Moreover, we have ${}^t u(C^\circ) \subset C^\circ$; the assertion then follows from the Krein-Rutman theorem applied to ${}^t u$, and to the closed convex cone C° since the latter is salient (because C is total) and total (because C is salient).

Remark. — The hypothesis $\varrho(u) > 0$ cannot be omitted in general in the Krein-Rutman theorem. For example, let V be the endomorphism of the Banach space $\mathcal{C}([0, 1])$ defined by

$$V(f)(x) = \int_0^x f(y) dy$$

for $f \in \mathcal{C}([0, 1])$. The map V is compact and its spectral radius is zero (exercise 1 of I, p. 187). It preserves the closed, total, salient convex cone in $\mathcal{C}([0, 1])$ consisting of positive functions, and has no eigenvalue (loc. cit.).

The following proposition describes a sufficient condition for an endomorphism preserving a cone to have a strictly positive spectral radius, and it then refines the Krein-Rutman theorem.

PROPOSITION 8. — Let E be a nonzero complete normable space over $\mathbf{R}$. Let C be a closed salient convex cone with nonempty interior $\mathring{C}$ in E , and let $u \in \mathcal{L}(E)$ be a compact linear map such that $u(C - \{0\}) \subset \mathring{C}$.

- a) One has $\varrho(u) > 0$ and there exists an eigenvector x_0 of u in $\mathring{C}$ for the eigenvalue $\varrho(u)$;
- b) The spectral projector of u corresponding to $\varrho(u)$ has rank 1;
- c) For every eigenvalue $\lambda \neq \varrho(u)$ of $u_{(C)}$, one has $|\lambda| < \varrho(u)$;
- d) Let F be a subspace of E stable under u such that $(C - \{0\}) \cap F$ is nonempty. Then $x_0 \in F$. In particular, the only eigenvectors of u in C are the multiples of x_0 .

Let us prove two preliminary lemmas.

Lemma 5. — Let E be a real Banach space, C a convex cone in E , and $u \in \mathcal{L}(E)$ such that $u(C - \{0\}) \subset \mathring{C}$. Let $\ell \in C^\circ$ be an eigenvector of ${}^t u$. Then the kernel of ℓ is stable under u and does not meet $C - \{0\}$.

Let $\lambda \in \mathbf{R}$ such that ${}^t u(\ell) = \lambda \ell$. For every $x \in \text{Ker}(\ell)$, we have

$$\langle u(x), \ell \rangle = \langle x, {}^t u(\ell) \rangle = \lambda \langle x, \ell \rangle = 0,$$

hence $\text{Ker}(\ell)$ is stable under u .

Let x be a nonzero element of $\text{Ker}(\ell)$. For every element y of E such that $\langle y, \ell \rangle < 0$, we have $\langle u(x) + y, \ell \rangle < 0$, whence it follows that $u(x) + y \notin C$ since $\ell \in C^\circ$. We deduce that $u(x)$ does not belong to $\mathring{C}$. Since $u(C - \{0\}) \subset \mathring{C}$, this implies that $x \notin C$.

Lemma 6. — Let E be a finite-dimensional real vector space and let B be a compact convex neighborhood of 0 in E . Let u be an endomorphism of E such that $u(B) \subset \mathring{B}$. The complex spectrum of u is then contained in the unit disk of $\mathbb{C}$ and, in particular, it does not meet the circle with center 0 and radius 1 in $\mathbb{C}$.

By replacing B with $B \cap (-B)$, we reduce to the case where B is balanced. The gauge p of B (EVT, II, p. 28, ex. 3) is then a norm on E , which defines its topology (EVT, I, p. 14, th. 2). The hypothesis implies that $p(u(x)) < p(x)$ for every element x of $E - \{0\}$, hence that the spectral radius of u is < 1 ; since this is also the spectral radius of $u_{(C)}$ (cf. I, p. 86), the conclusion follows.

Let us now prove the proposition.

Denote by $x \preccurlyeq y$ the order relation on E associated with the convex cone C (EVT, II, p. 13, n° 5), that is, $x \preccurlyeq y$ if and only if $y - x \in C$. We have $u(x) \preccurlyeq u(y)$ if $x \preccurlyeq y$. Since $\mathring{C}$ is nonempty, the vector space $C - C$ generated by C (EVT, II, p. 12, cor. 1) contains a neighborhood of 0, hence the cone C is total.

Let us prove assertion *a*). Let $\varrho = \varrho(u)$. Let y_0 be an element of $\mathring{C}$. We have $y_0 \neq 0$. Let $r > 0$ be such that the closed ball with center y_0 and radius r is contained in C . For every $y \in E - \{0\}$, we have $y_0 - r\|y\|^{-1}y \in C$, hence $y \preccurlyeq r^{-1}\|y\|y_0$ for all $y \in E$.

Since $y_0 \neq 0$, the hypotheses imply that $u(y_0) \in \mathring{C}$. There therefore exists $t > 0$ such that $tu(y_0) - y_0 \in C$. Set $v = tu$. This is a compact endomorphism of E such that $v(C) \subset C$ and $v(y_0) \succcurlyeq y_0$. For every integer $n \geq 1$, we have

$$y_0 \preccurlyeq v^n(y_0) \preccurlyeq r^{-1}\|v^n(y_0)\|y_0 \preccurlyeq r^{-1}\|v^n\|\|y_0\|y_0,$$

hence $(r^{-1}\|v^n\| \|y_0\| - 1)y_0 \in C$. Since C is salient, this implies that $t^n\|u^n\| = \|v^n\| \geq r/\|y_0\|$, whence $\varrho \geq t^{-1} > 0$ (I, p. 20, prop. 1).

By the Krein–Rutman theorem (th. 4), the real number ϱ is an eigenvalue of u , and there exists an eigenvector x_0 of u in C for the eigenvalue ϱ . We have $\varrho x_0 = u(x_0) \in \mathring{C}$ by hypothesis, hence $x_0 \in \mathring{C}$. This establishes assertion *a*).

Let K be the intersection of the spectrum of $u_{(C)}$ and the circle with center 0 and radius ϱ in C . Since u is compact and $\varrho > 0$, the set K is finite and the image of the spectral projector e_K is a finite-dimensional subspace of $E_{(C)}$ (prop. 2 of III, p. 83 and prop. 5 of III, p. 90). Since K is stable under complex conjugation, the image of e_K is a finite-dimensional subspace of $E_{(C)}$ rational over $\mathbb{R}$ (A, V, p.60, prop. 6); denote by F the subspace of E such that $F_{(C)}$ is equal to the image of e_K . The space F is stable under u and contains the eigenspace of u relative to ϱ , hence F is nonzero.

To prove the assertions *b*) and *c*), it suffices to prove that F is of dimension 1.

Denote by v the endomorphism of F deduced from u by passing to subspaces. Let $C_F = C \cap F$. This is a closed salient convex cone with nonempty interior in F (since $x_0 \in \mathring{C} \cap F$), hence total; it satisfies $v(C_F - \{0\}) \subset \mathring{C}_F$, and in particular is stable under v . Since $\varrho(v) \leq \varrho$ and $x_0 \in F$, we have $\varrho(v) = \varrho$. By the corollary of Theorem 4, applied to C_F and v there exists an eigenvector ℓ of ${}^t v$ in C_F° for the eigenvalue $\varrho(v) = \varrho > 0$.

The subspace $\text{Ker}(\ell)$ of F is stable under v . Let w be the endomorphism of $\text{Ker}(\ell)$ deduced from $\varrho^{-1}v$ by passing to the quotient subspaces; its spectrum is contained in the unit circle. The set B of $y \in \text{Ker}(\ell)$ such that $x_0 + y \in C_F$ is a closed convex neighborhood of 0 in $\text{Ker}(\ell)$. For every $y \in B$ and every $z \in \text{Ker}(\ell)$, we have

$$x_0 + (w(y) + z) = \varrho^{-1}(v(x_0 + y) + \varrho z)$$

which belongs to C_F if the norm of z is sufficiently small, hence $w(y) \in \mathring{B}$. The set B is bounded: indeed, if there existed a sequence $(y_n)_{n \in \mathbb{N}}$ in $\text{Ker}(\ell)$ such that $\|y_n\| \rightarrow +\infty$ and $x_0 + y_n \in C_F$ then, after passing to a subsequence, we would have $y_n/\|y_n\| \rightarrow y$ where $y \in \text{Ker}(\ell)$ is nonzero, and $y = \lim \|y_n\|^{-1}(x_0 + y_n)$ would belong to C_F , contradicting Lemma 5. The set B is therefore compact.

One then deduces from Lemma 6, applied to $\text{Ker}(\ell)$, to B , and to w , that the complex spectrum of w does not meet the circle with center 0 and radius 1; this implies that the spectrum of w is empty, which means that $\text{Ker}(\ell)$ is reduced to $\{0\}$, hence that F has dimension 1. The assertions b and c are therefore established.

The linear map ${}^t u$ is compact (Corollary 1 of III, p. 9) and ${}^t u(C^\circ) \subset C^\circ$; moreover $\varrho({}^t u) = \varrho > 0$ (Prop. 3 of I, p. 131). By the corollary of Theorem 4, there exists in C° an eigenvector $\ell \neq 0$ of ${}^t u$ for the eigenvalue ϱ . The kernel of ℓ is stable under u and does not meet $C - \{0\}$ (Lemma 5). In particular, one has $x_0 \notin \text{Ker}(\ell)$.

Let F be a subspace of E stable under u . One has the decomposition $F = (F \cap \mathbf{R}x_0) \oplus (F \cap \text{Ker}(\ell))$. If F does not contain x_0 , we therefore have $F \subset \text{Ker}(\ell)$, then F does not meet $C - \{0\}$. This establishes d) and concludes the proof of the proposition.

COROLLARY. — Let E be a nonzero complete normable space over $\mathbf{R}$. Let C be a closed convex salient cone with nonempty interior $\overset{\circ}{C}$ in E , and let u be a compact endomorphism of E such that $u(C) \subset C$. Suppose that there exists an integer $k \geq 1$ such that $u^k(C - \{0\}) \subset \overset{\circ}{C}$.

- a) One has $\varrho(u) > 0$ and there exists an eigenvector x_0 of u in $\overset{\circ}{C}$ for the eigenvalue $\varrho(u)$;
- b) The spectral projector of u corresponding to $\varrho(u)$ has rank 1;
- c) For every eigenvalue $\lambda \neq \varrho(u)$ of $u_{(C)}$, one has $|\lambda| < \varrho(u)$;
- d) The only eigenvectors of u in C are the multiples of x_0 .

Set $\varrho = \varrho(u)$. One can apply Th. 4 to u and Prop. 8 to u^k . There therefore exists an eigenvector x_0 of u in C for the eigenvalue ϱ . Since $0 < \varrho(u^k) = \varrho^k$ (formula (4) of I, p. 21), one has in particular $\varrho > 0$, and since x_0 is an eigenvector of u^k for the eigenvalue $\varrho(u^k)$, we have $x_0 \in \overset{\circ}{C}$.

One has $\text{Sp}(u_{(C)}^k) = \text{Sp}(u_{(C)})^k$ (Remark 4 of I, p. 2), hence every eigenvalue $\lambda \in C$ of $u_{(C)}$ such that $|\lambda| = \varrho$ satisfies $\lambda^k = \varrho^k$, and since every eigenvector corresponding to λ is an eigenvector of u^k for ϱ^k , hence proportional to x_0 , we have $\lambda = \varrho$.

If the spectral projector of u for the eigenvalue ϱ had rank at least 2, then so would that of u^k for the eigenvalue ϱ^k (cf. no. 2 of I, p. 129).

Finally, if $x \in C$ is an eigenvector of u , it is also one for u^k , hence is proportional to x_0 .

Let n be an integer ≥ 1 . The set R_+^n is a closed convex pointed salient cone in R^n , with nonempty interior equal to $(R_+^*)^n$.

Let $A = (a_{i,j})$ be a real matrix of type (n, n) , and u_A the endomorphism $x \mapsto Ax$ of R^n . Let $\varrho = \varrho(u_A)$ be its spectral radius.

Lemma 7. — One has $u_A(R_+^n) \subset R_+^n$ if and only if $a_{i,j} \geq 0$ for all i and j , and $u_A(R_+^n - \{0\}) \subset (R_+^*)^n$ if and only if $a_{i,j} > 0$ for all i and j .

Let $(e_1, \dots, e_n)$ be the canonical basis of R^n . The vectors e_i belong to R_+^n and $u_A(e_i) \in R_+^n$ for all i (resp. $u_A(e_i) \in (R_+^*)^n$ for all i) if and only if $a_{i,j} \geq 0$ for all i and j (resp. $a_{i,j} > 0$ for all i and j). The result follows by linearity.

THEOREM 5 (Perron–Frobenius). — a) If $a_{i,j} \geq 0$ for all i and j in $\{1, \dots, n\}$, then the real number ϱ is an eigenvalue of A ; b) If $a_{i,j} > 0$ for all i and j in $\{1, \dots, n\}$, then $\varrho > 0$ and the primary space of u_A relative to ϱ , that is, the nullspace of $A - \varrho 1_{R^n}$, has dimension 1 and is generated by a vector $x_0 \in (R_+^*)^n$. There is no other eigenvalue of A admitting an eigenvector in R_+^n , and all complex eigenvalues $\lambda \neq \varrho$ of A satisfy $|\lambda| < \varrho$.

If $\varrho = 0$, assertion a) is elementary because 0 is then an eigenvalue of A . If $\varrho > 0$, assertion a) follows from Theorem 4 in view of Lemma 7. Assertion b) is, for the same reason, a consequence of Proposition 8.

Remark. — Suppose there exists an integer $k \geq 1$ such that all coefficients of A^k are > 0 (such a matrix is sometimes called primitive). Then, by the corollary of Prop. 8, the spectral radius ϱ is an eigenvalue of A , the primary space of u_A relative to ϱ has dimension 1, and is generated by a vector $x_0 \in (R_+^*)^n$. Moreover, there is no other complex eigenvalue of A admitting an eigenvector in R_+^n and all complex eigenvalues $\lambda \neq \varrho$ of A satisfy $|\lambda| < \varrho$.

Exercises

Exercises for Section 1

In Exercises 1 to 5 exclusively, K denotes a complete non-Archimedean non-discrete valued field, whose valuation is denoted by $x \mapsto |x|$. We also denote by $G \subset \mathbf{R}_+^*$ the image of K^* under the map $x \mapsto |x|$. The Banach spaces considered are Banach spaces over K .

- 1) a) Let E be a Banach space over K . There exists a norm $x \mapsto \|x\|$ on E , defining the topology of E , such that $\|x\| \in \overline{G}$ for all $x \in E$. Such a norm will be called round. If the valuation of K is not discrete, then every norm defining the topology of E is round.
- b) Let E and F be Banach spaces over K . We equip the space $\mathcal{L}(E; F)$ with the norm $\|u\| = \sup_{x \in E - \{0\}} \|u(x)\| / \|x\|$. The space $\mathcal{L}(E; F)$ is a Banach space over K . If the norm on E is round, then $\|u\| = \sup_{\|x\| \leq 1} \|u(x)\|$. If the norm on F is also round, then the norm on $\mathcal{L}(E; F)$ is round. We denote $E' = \mathcal{L}(E; K)$ and we say that E' is the dual space of E .
- c) Let I be a set, F a Banach space over K , and $\mathcal{C}_0(I, F)$ the K -vector space of functions $f: I \rightarrow F$ such that $\|f(i)\|$ tends to 0 along the filter of complements of finite subsets of I . Equipped with the norm $\|f\| = \sup_{i \in I} \|f(i)\|$, the space $\mathcal{C}_0(I, F)$ is a Banach space over K , whose norm is round if that of F is. We denote $\mathcal{C}_0(I) = \mathcal{C}_0(I, K)$.
- d) Let I be a set. For every $i \in I$, we denote by $\varphi_i \in \mathcal{C}_0(I)$ the characteristic function of $\{i\}$. Let F be a Banach space over K . The map $u \mapsto (u(\varphi_i))_{i \in I}$ is an isomorphism from the Banach space $\mathcal{L}(\mathcal{C}_0(I); F)$ to the Banach space $\mathcal{B}(I; K)$ (EVT, I, p. 4, example; cf. EVT, I, p. 23, exercise 7).

- e) Assume that the valuation of K is discrete. Let E be a Banach space over K whose norm is round. There exists a set I such that E is isometrically isomorphic to the Banach space $\mathcal{C}_0(I)$ (cf. EVT, I, p. 23, exercise 7).
- f) Assume that the valuation of K is discrete. Let E be a Banach space over K equipped with a round norm and F a closed vector subspace of E . There exists a continuous projector of E whose image is F and which has norm ≤ 1 .
- 2) Let E and F be Banach spaces over K . We say that $u \in \mathcal{L}(E; F)$ is completely continuous if it belongs to the closure in $\mathcal{L}(E; F)$ of the space $\mathcal{L}^f(E; F)$ of finite-rank linear maps. We denote by $\mathcal{L}^c(E; F)$ the space of completely continuous linear maps from E to F , and we set $\mathcal{L}^c(E) = \mathcal{L}^c(E; E)$. a) The space $\mathcal{L}^c(E)$ is a closed two-sided ideal of $\mathcal{L}(E)$. Let $u \in \mathcal{L}(E; F)$. The map $\ell \mapsto \ell \circ u$ is a continuous linear map from F' to E' , denoted ${}^t u$. If $u \in \mathcal{L}^c(E; F)$, then ${}^t u \in \mathcal{L}^c(F'; E')$.
- b) Let J be a set and $F = \mathcal{C}_0(J)$. Let $u \in \mathcal{L}(E; F)$. For every $j \in J$, the map $\ell_j: x \mapsto u(x)(j)$ is an element of E' . We have $\|u\| = \sup_{j \in J} \|\ell_j\|$ and the map $u \mapsto (\ell_j)_{j \in J}$ induces, by passage to subspaces, an isomorphism from $\mathcal{L}^c(E; F)$ to the Banach space $\mathcal{C}_0(J; E')$.
- c) If K is locally compact, then $u \in \mathcal{L}(E; F)$ is completely continuous if and only if the image under u of every bounded subset of E is relatively compact in F . (One may compare this result with prop. 11 of III, p. 16.)
- 3) We denote by A the subring of K consisting of elements $x \in K$ such that $|x| \leq 1$; it is a local ring whose maximal ideal consists of the $x \in A$ such that $|x| < 1$.

Let I be a set and E the Banach space $\mathcal{C}_0(I)$ (cf. question c) of exercise 1). For $i \in I$, denote by $\varphi_i \in E$ the characteristic function of $\{i\}$. Let $u \in \mathcal{L}^c(E)$ such that $\|u\| \leq 1$.

- a) The subspace E_0 of elements $x \in E$ such that $\|x\| \leq 1$ is stable under u .
- b) For every nonzero ideal a of A , the linear map u defines, by passage to quotients, an endomorphism u_a of the A/a -module $E_a = E_0/aE_0$. This A/a -module is free and the image of u_a is contained in a finitely generated submodule.

c) There exists a unique formal series $f_u \in A[[t]]$ such that, for every nonzero ideal $\mathfrak{a}$ of A , the image of f_u in $(A/\mathfrak{a})[[t]]$ is equal to the polynomial $\det(1 - tu_{\mathfrak{a}})$ defined in A, VIII, p. 455.

d) Let $v \in \mathcal{L}^c(E)$ and let $c \in K^*$ such that $\|cv\| \leq 1$. The formal series f_{cv} is independent of the choice of c satisfying this property.

We shall denote by $\det(1 - tv)$ the formal series f_v for all $c \in K^*$ such that $\|cv\| \leq 1$.

¶ 4) We retain the notation from the previous exercise.

a) Let $v \in \mathcal{L}^c(E)$. The radius of convergence of the formal series $\det(1 - tv)$ is infinite.

b) Let $(v_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{L}^c(E)$ converging to $v \in \mathcal{L}^c(E)$. The sequence of formal series $\det(1 - tv_n)$ converges to $\det(1 - tv)$ for the topology of simple convergence of the coefficients of formal series.

c) Let $v \in \mathcal{L}^c(E)$. For every extension L of K and every $x \in L$, we denote by $\det(1 - xv)$ the value at x of the formal power series $\det(1 - tv)$, which is well defined by a). We have $\det(1 + u) \det(1 + v) = \det(1 + u + v + u \circ v)$.

d) Let J be a set and $F = \mathcal{C}_0(J)$. For all $u \in \mathcal{L}^c(E; F)$ and $v \in \mathcal{L}^c(F; E)$, we have $\det(1 - tu \circ v) = \det(1 - tv \circ u)$ as formal power series.

e) Let $v \in \mathcal{L}^c(E)$. We denote by $\text{Tr}(v)$ the negative of the coefficient of t in $\det(1 - tv)$. If K is of characteristic zero, then

$$\det(1 - tv) = \exp \left(- \sum_{m \geq 1} \frac{\text{Tr}(v^m)}{m} t^m \right)$$

as formal power series.

¶ 5) We keep the notations of the preceding exercise. Let $v \in \mathcal{L}^c(E)$.

a) The formal power series $\det(1 - tv) \sum_{k \geq 0} t^k v^k$ in $\mathcal{L}(E)[[t]]$ has infinite radius of convergence.

b) Let $\lambda \in K$. The endomorphism $1_E - \lambda v$ of E is invertible if and only if $\det(1 - \lambda v) \neq 0$.

c) Let $\lambda \in K$ such that $\det(1 - \lambda v) = 0$. There exist unique closed subspaces I_λ and N_λ of E , stable under v such that

- (i) The space E is the topological direct sum of I_λ and N_λ ;
- (ii) The endomorphism of I_λ deduced from $1_E - \lambda v$ by passing to subspaces is invertible;
- (iii) The endomorphism of N_λ deduced from $1_E - \lambda v$ by passing to subspaces is nilpotent.

d) Let $h \geq 1$ be the order of the zero λ of $\det(1 - tv)$. We have $\dim(N_\lambda) = h$.

The result of this exercise is to be compared with those of No. 5 of III, p. 77, which concern Banach spaces over $\mathbb{R}$ or $\mathbb{C}$.

In what follows, K denotes the field $\mathbb{R}$ or $\mathbb{C}$, and all topological vector spaces are over K .

6) Let E be the space C^N endowed with the topology associated with the sequence of seminorms $(p_n)_{n \in \mathbb{N}}$ given by $p_n(x) = |x_n|$ for every element $x = (x_n)_{n \in \mathbb{N}}$ of E .

a) The space E is a Fréchet space.

b) The set $\mathcal{L}^c(E)$ of compact endomorphisms of E is not closed in $\mathcal{L}(E)$.

7) Let E and F be locally convex topological vector spaces. If F is metrizable, then the image of every compact linear map $E \rightarrow F$ is of countable type.

8) Construct an example of a non-compact endomorphism u of a Hilbert space E such that u^2 is compact.

9) For every pair $(x, y) \in]0, 1]^2$, there exists an integer $n \geq 1$ and an integer k such that $0 \leq k \leq 2^{n-1} - 1$ so that

$$2^{-n} < x \leq 2^{-n+1}, \quad \frac{2k}{2^n} < y \leq \frac{2k+2}{2^n}.$$

We then define $N:]0, 1] \times]0, 1] \rightarrow \mathbb{C}$ by setting

$$N(x, y) = \begin{cases} 2 & \text{if } \frac{2k}{2^n} < y \leq \frac{2k+1}{2^n} \\ 0 & \text{if } \frac{2k+1}{2^n} < y \leq \frac{2k+2}{2^n}. \end{cases}$$

a) Show that the formula

$$u(f)(x) = \int_{[0,1]} N(x,y)f(y)dy$$

defines a continuous linear map u on $L^1([0,1])$ with respect to Lebesgue measure.

b) Show that the endomorphism u is not compact.

c) The endomorphism u^2 is compact.

d) Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence in $L^1([0,1])$. The sequence $(u(x_n))_{n \in \mathbb{N}}$ admits a weakly convergent subsequence.

e) Let $(x_n)_{n \in \mathbb{N}}$ be a weakly convergent sequence in $L^1([0,1])$. The sequence $(u(x_n))_{n \in \mathbb{N}}$ converges in $L^1([0,1])$.

f) Let E be the topological vector space whose elements are the functions $f: [0,1] \rightarrow \mathbb{C}$ which are bounded and measurable, equipped with the finest topology that is stronger than the topology of $L^2([0,1])$ and the weak topology relative to $L^1([0,1])$. Then the linear map v defined by

$$v(f)(x) = \int_{[0,1]} \overline{N(y,x)} f(y) dy$$

is a compact endomorphism of E . Its transpose ${}^t v$ is not compact for the strong topology on E' .

10) Let p and r be real numbers such that $1 \leq p < r$. Let u be a continuous linear map from $\ell_K^r(\mathbb{N})$ into $\ell_K^p(\mathbb{N})$.

We denote by r_n (resp. p_n) the projection

$$(x_i)_{i \in \mathbb{N}} \mapsto (x_0, \dots, x_{n-1}, 0, 0, \dots)$$

onto $\ell_K^r(\mathbb{N})$ (resp. on $\ell_K^p(\mathbb{N})$). We denote by $\|\cdot\|_p$ (resp. $\|\cdot\|_r$) the norm on $\ell_K^p(\mathbb{N})$ (resp. on $\ell_K^r(\mathbb{N})$).

- a) Show that for every $n \in \mathbb{N}$, we have $\|p_n u(1 - r_m)\| \rightarrow 0$ as $m \rightarrow +\infty$ where 1 denotes the identity morphism of $\ell_K^r(\mathbb{N})$.
- b) For $n, m \in \mathbb{N}$, let $u_{n,m} = (1 - p_n)u(1 - r_m)$. Let $\alpha = (\alpha_n)_{n \in \mathbb{N}}$ be a sequence of strictly positive real numbers such that $\|\alpha\|_r = 1$ and α does not belong to $\ell_K^p(\mathbb{N})$ and let (ε_n) be a sequence of strictly positive real numbers such that the series $\sum \varepsilon_n \alpha_n$ converges.

If u is not compact, show that there exist $\delta > 0$, strictly increasing sequences of integers $(n_i)_{i \in \mathbb{N}}$ and $(m_i)_{i \in \mathbb{N}}$, and two sequences $(x_i)_{i \in \mathbb{N}}$ of elements of $\ell_K^r(\mathbb{N})$ and $(y_i)_{i \in \mathbb{N}}$ of elements of $\ell_K^p(\mathbb{N})$ with the following properties:

- (i) One has $\|x_i\|_r = 1$ for all $i \in \mathbb{N}$;
- (ii) The sets $S_i = \{n \in \mathbb{N} \mid x_{i,n} \neq 0\}$ are pairwise disjoint and the sets $T_i = \{n \in \mathbb{N} \mid y_{i,n} \neq 0\}$ are pairwise disjoint;
- (iii) One has $\|y_i\|_p \geq \delta$
- (iv) One has $\|u(x_i)\|_q > \delta - 2\varepsilon_i$.

(Proceed by induction and use the formulas

$$u = u_{n,m} + p_n u(1 - r_m) + (1 - p_n)ur_m + p_n uq_m$$

for n and m in $\mathbb{N}$.)

- c) Deduce by contradiction that u is compact ("Pitt's theorem").
 - d) Show that the canonical injection of $\ell_K^p(\mathbb{N})$ into $\ell_K^r(\mathbb{N})$ is not compact.
 - e) Let p and r be real numbers such that $1 \leq p \leq r \leq +\infty$. Show that $\ell_K^r(\mathbb{N})$ is isomorphic to $\ell_K^p(\mathbb{N})$ if and only if $p = r$.
- 11) Let E and F be Banach spaces over K . A linear map $u \in \mathcal{L}(E; F)$ is said to be completely continuous if, for every weakly convergent sequence $(x_n)_{n \in \mathbb{N}}$ in E , the sequence $(u(x_n))_{n \in \mathbb{N}}$ converges in F .

- a) Every compact linear map is completely continuous.
 - b) If the dual space E' of E is countably generated, then every completely continuous linear map $E \rightarrow F$ is compact.
 - c) Let $(u_n)_{n \in \mathbb{N}}$ be a sequence of elements of $\ell_K^1(\mathbb{N})$. Show that (u_n) converges if and only if it converges weakly ("Schur's theorem"; cf. EVT, IV, p. 54, exercise 15).
 - d) Deduce that the injection of $\ell_K^1(\mathbb{N})$ into $\ell_K^2(\mathbb{N})$ is completely continuous but is not compact.
 - e) For every Banach space, every continuous linear map $E \rightarrow \ell_K^1(\mathbb{N})$ is completely continuous, and every continuous linear map $\ell_K^1(\mathbb{N}) \rightarrow E$ is completely continuous.
 - f) If E is reflexive or if its dual is countably generated, every continuous linear map $E \rightarrow \ell_K^1(\mathbb{N})$ is compact.
- 12) Let E be an infinite-dimensional reflexive Banach space. Let $u \in \mathcal{L}^c(E)$ be a compact endomorphism. Suppose that u is not of finite rank. Let E_σ be the space E equipped with the weak topology.

Show that the linear map ${}^t u$ is a compact endomorphism of $(E_\sigma)'$.

- 13) Let E be a countably generated Banach space over K . Recall (EVT, IV, p. 70, exercise 14) that a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of E is called a Schauder basis of E if, for every $x \in E$, there exists a unique sequence of scalars $(\lambda_n(x))_{n \in \mathbb{N}}$ such that

$$x = \sum_{n=1}^{+\infty} \lambda_n(x) x_n$$

where the series converges in E .

- a) Every separable Hilbert space admits a Banach basis; every space $\ell^p(I)$, where I is countable, admits a Banach basis.
- b) The Banach space E of bounded holomorphic functions on the open disk D of radius 1 in the complex plane, equipped with the norm $\|f\| = \sup_{z \in D} |f(z)|$, has a Schauder basis.

- c) Every Banach space E admitting a Banach basis is an approximation space. More precisely, there exists a real number $c \geq 0$ such that 1_E belongs to the closure in the space $\mathcal{L}_{pc}(E)$ of the set of finite-rank endomorphisms with norm $\leq c$ in E .
- 14) a) Let E and F be Hilbert spaces. A continuous linear map u from E into F is compact if and only if the image of u contains no closed subspace of infinite dimension.
- b) The characterization in a) of compact endomorphisms from E into F does not hold for every pair of Banach spaces (E, F) . (One may use Exercise 5, p. 121.)
- 15) Let E_0 be the complex vector space of functions $f : \mathbb{N}^* \times \mathbb{N}^* \rightarrow \mathbb{C}$. For $k \in \mathbb{N}^*$, let $\alpha_k \in E_0$ be the function such that

$$\alpha_k(n, m) = \begin{cases} m^k & \text{if } 1 \leq n \leq k-1 \\ n^k & \text{if } n \geq k. \end{cases}$$

Let E be the vector subspace of E_0 formed by the functions f such that $p_k(f)$ is finite for all $k \in \mathbb{N}^*$, where p_k is defined by

$$p_k(f) = \sum_{n, m} \alpha_k(n, m) |f(n, m)|.$$

- a) The space E equipped with the topology defined by the norms $(p_k)_{k \in \mathbb{N}^*}$ is a Fréchet space and a Montel space. Its dual E' is identified with the subspace of E_0 consisting of functions g such that, for at least one $k \geq 1$, one has

$$\sup_{m, n} \alpha_k(m, n)^{-1} |g(m, n)| < +\infty$$

(cf. EVT, IV, p. 63, exercise 8).

- b) Consider the linear map u from E into $\ell^1(\mathbb{N}^*)$ defined by

$$u(f) = \left(\sum_n f(n, m) \right)_{m \in \mathbb{N}^*}.$$

The linear map u is continuous.

- c) Identify $\ell^\infty(\mathbf{N}^*)$ with the dual of $\ell^1(\mathbf{N}^*)$. The transpose of u is the map $v: \ell^\infty(\mathbf{N}^*) \rightarrow E' \subset E_0$ determined by $(x_n) \mapsto f$ where $f(n, m) = x_m$ for all $(n, m) \in \mathbf{N}^* \times \mathbf{N}^*$.
- d) The map u induces an isomorphism $E/\text{Ker}(u) \rightarrow \ell^1(\mathbf{N}^*)$ (cf. loc. cit.).
- e) The map u is not compact.
- f) The transpose v of u is compact for the topology of convergence on bounded subsets of $\ell^\infty(\mathbf{N}^*)$.
- g) The map v is an example of a compact linear map between separated locally convex spaces whose transpose is not compact.
- 16) Let E be a locally convex space and u a compact endomorphism of E . Denote by E'_b the dual space of E equipped with the topology of bounded convergence. The linear map ${}^t u \circ {}^t u$ is a compact endomorphism of E'_b .
- 17) Let E be a Banach space of countable type. Then $\mathcal{L}^c(E)$, equipped with the topology induced by the norm of $\mathcal{L}(E)$ is of countable type if, and only if, the dual space E' is of countable type.
- 18) Let E, F, G be Banach spaces and let u be an element of $\mathcal{L}^c(E; F)$.
- a) Let v be an element of $\mathcal{L}(E; G)$. Suppose that, for every $\varepsilon > 0$, there exists $C > 0$ such that

$$\|v(x)\|_G \leq \varepsilon \|x\|_E + C \|u(x)\|_F$$

for all $x \in E$. Then $v \in \mathcal{L}^c(E; G)$.

- b) Let w be an injective element of $\mathcal{L}(F; G)$. Then, for every $\varepsilon > 0$, there exists $C > 0$ such that

$$\|u(x)\|_F \leq \varepsilon \|x\|_E + C \|(w \circ u)(x)\|_G$$

for all $x \in E$.

- 19) Let X be a locally compact, metrizable, and countable at infinity topological space, and μ a positive measure on X . We denote by A the set of atoms of μ . Let $f \in \mathcal{L}^\infty(X, \mu)$ and $p \in [1, +\infty]$.

Show that multiplication by f defines a compact endomorphism of $L^p(X, \mu)$ if and only if the following two conditions are satisfied:

- (i) The set $\{x \in X - A \mid f(x) \neq 0\}$ is μ -negligible;
- (ii) For every $\varepsilon > 0$ there exists a finite subset F of A such that $|f(a)| \leq \varepsilon$ for all $a \in A - F$.

(Consider multiplication by f on the space $L^p(X_\varepsilon, \mu)$, where X_ε is the set of $x \in X$ such that $|f(x)| \geq \varepsilon$.)

- 20) Let X, Y be compact metric spaces, T a compact operator from $\mathcal{C}(Y)$ into $\mathcal{C}(X)$.

Prove that there exists a positive measure μ on X and a measurable map N from $X \times Y$ into $\mathbb{C}$ such that

- a) For every $y \in Y$, the map $N_y: x \mapsto N(x, y)$ from X into $\mathbb{C}$ is μ -integrable;
- b) The map $y \mapsto N_y$ from Y into $\mathcal{L}^1(X, \mu)$ is continuous;
- c) For every $f \in \mathcal{C}(X)$, and every $y \in Y$, we have

$$Tf(y) = \int_X N(x, y)f(x)d\mu(x).$$

- 21) We denote by D the open disk with center 0 and radius 1 in $\mathbb{C}$, and by $H^2(D)$ the Hardy space on D (EVT, V, p. 7, example 5). We write $T = \mathbb{R}/2\pi\mathbb{Z}$, which we equip with the normalized Haar measure. Let j be the isometry from $H^2(D)$ into $L^2(T)$ defined by

$$j(f)(\theta) = \sum_{n \geq 0} \frac{f^{(n)}(0)}{n!} e^{in\theta}.$$

The image of $H^2(D)$ by j will be denoted $H^2(T)$.

Let $\tilde{P}$ be the orthogonal projector of $L^2(\mathbf{T})$ with image $H^2(\mathbf{T})$, and let P be the linear map $f \mapsto \tilde{P}(f)$ from $L^2(\mathbf{T})$ into $H^2(\mathbf{T})$. If f is an element of $L^\infty(\mathbf{T})$, let M_f be the endomorphism $g \mapsto fg$ of $L^2(\mathbf{T})$.

Show that if f is a continuous function, then the commutator $\tilde{P}M_f - M_f\tilde{P}$ is a compact linear map. (Consider the particular case where f is a trigonometric polynomial.)

22) We keep the notations from the previous exercise. Let R be the continuous linear map from $H^2(\mathbf{T})$ into $L^2(\mathbf{T})$ defined by $Rf(\theta) = f(-\theta)$. For $g \in L^\infty(\mathbf{T})$ we set $\Gamma_g = PM_gR$.

a) Prove that the linear map $\Gamma: g \mapsto \Gamma_g$ from $L^\infty(\mathbf{T})$ into $\mathcal{L}(H^2(\mathbf{T}))$ is continuous, with norm ≤ 1 , and its kernel F is equal to $H^2(\mathbf{T})^\circ \cap L^\infty(\mathbf{T})$.

b) If g is a continuous function, then Γ_g is a compact endomorphism of $H^2(\mathbf{T})$.

c) Let $g \in L^\infty(\mathbf{T})$. We denote by α_g the linear form on $H^2(\mathbf{T})$ defined by

$$\alpha_g(f) = \int_{\mathbf{T}} \Gamma_g(f)(\theta) d\theta.$$

Let $\mathcal{A}$ be the image under j of the subspace of $H^2(D)$ consisting of holomorphic functions that extend analytically to a neighborhood of $\overline{D}$. Prove that if f_1 and f_2 belong to $\mathcal{A}$, then

$$\alpha_g(f_1 f_2) = \int_{\mathbf{T}} \Gamma_g(f_1)(\theta) f_2(-\theta) d\theta.$$

d) Let φ be a holomorphic function in a neighborhood of $\overline{D}$ not vanishing on ∂D and let $f = j(\varphi|D)$. Then there exist f_1 and f_2 in $\mathcal{A}$ such that $f = f_1 f_2$ and

$$\|f\|_{L^1(\mathbf{T})} = \|f_1\|_{L^2(\mathbf{T})} \|f_2\|_{L^2(\mathbf{T})}.$$

(Let $z_1, \dots, z_n$ be the zeros of φ in D ; then there exists a function ψ holomorphic in a neighborhood of D such that

$$\varphi(z) = \prod_{k=1}^n \frac{z - z_k}{1 - \bar{z}_k z} \psi(z)^2$$

for all z .)

e) Prove that, for all $f \in H^2(\mathbf{T})$, we have the inequality

$$|\alpha_g(f)| \leq \|\Gamma_g\|_{\mathcal{L}(H^2(\mathbf{T}))} \|f\|_{L^1(\mathbf{T})}.$$

(If $\varphi \in H^2(D)$, there exists a sequence $(r_n)_{n \in \mathbb{N}}$ of $]0,1[$ such that, for every $n \in \mathbb{N}$, the function φ_n defined by $\varphi_n(z) = \varphi(r_n z)$ has no zeros on ∂D , and $\lim \varphi_n = \varphi$ in $H^2(D)$).

Deduce that there exists $h \in L^\infty(\mathbf{T})$ such that

$$\alpha_g(f) = \int_{\mathbf{T}} f(\theta) h(\theta) d\theta$$

for all $f \in H^2(\mathbf{T})$, and $\|h\|_{L^\infty(\mathbf{T})} \leq \|\Gamma_g\|_{\mathcal{L}(H^2(\mathbf{T}))}$.

f) Prove that there exists $k \in L^\infty(\mathbf{T})$ such that $\Gamma_g = \Gamma_k$ and $\|k\|_{L^\infty(\mathbf{T})} = \|\Gamma_g\|_{\mathcal{L}(H^2(\mathbf{T}))}$.

The map $\widehat{\Gamma} : L^\infty(\mathbf{T})/F \rightarrow \mathcal{L}(H^2(\mathbf{T}))$ is therefore an isometry ("Nehari's theorem").

g) Let g be a continuous function on $\mathbf{T}$. Then we have $d(g, F) = d(g, F \cap \mathcal{C}(\mathbf{T}))$. (For $R > 1$, one defines the linear map P_R from $L^\infty(\mathbf{T})$ into $\mathcal{C}(\mathbf{T})$ by

$$P_R f(\theta) = \int_{\mathbf{T}} \frac{R^2 - 1}{|Re^{i(\theta-\varphi)} - 1|^2} f(\varphi) d\varphi.$$

Then $\|P_R f\|_{\mathcal{C}(\mathbf{T})} \leq \|f\|_{L^\infty(\mathbf{T})}$ and, if $f \in \mathcal{C}(\mathbf{T})$, we have $P_R f \rightarrow f$ in $\mathcal{C}(\mathbf{T})$ as $R \rightarrow 1$.

Deduce that $\Gamma(\mathcal{C}(\mathbf{T}))$ is a closed subspace of $\mathcal{L}(H^2(\mathbf{T}))$ ("Sarason's theorem").

h) Let $g \in L^\infty(\mathbf{T})$ such that Γ_g is a compact endomorphism. For $m \in \mathbb{Z}$ we set $e_m(\theta) = e^{im\theta}$. Prove that $\|\Gamma_{ge_{-m}}\|_{\mathcal{L}(H^2(\mathbf{T}))}$ tends to zero as m tends to $+\infty$, and then that there exists a sequence $(h_m)_{m \in \mathbb{N}}$ of F such that the sequence $(e_m h_m)_{m \in \mathbb{N}}$ converges to g in $L^\infty(\mathbf{T})$. Deduce that there exists a continuous function in the class of g modulo F ("Hartman's theorem").

23) Let B be an infinite-dimensional Banach space. We denote by E the strong dual of B and by F the dual of B equipped with the topology $\sigma(B', B)$. Prove that the identity map of the dual of B is compact from E into F , but that its transpose is not compact from F'_b into E'_b .

- 24) Let E be a real or complex Banach space and $u \in \mathcal{L}(E)$. Assume that the sequence $(u^n)_{n \in \mathbb{N}}$ is bounded and that u is compact from E into E equipped with the weak topology $\sigma(E, E')$. For every integer $N \geq 1$ we set

$$v_N = \frac{1}{N}(1_E + u + \cdots + u^{N-1}) \in \mathcal{L}(E).$$

- a) Let F be the closure of the image of $1_E - u$. For every $x \in F$, we have $v_N(x) \rightarrow 0$ in E when $N \rightarrow +\infty$.
- b) Let $x \in E$. The sequence $(v_N(x))$ converges to an element $y \in E$ such that $u(y) = y$. (There exists a subsequence of $(v_N(x))$ that converges weakly to $y \in E$; show that $x - y \in F$ using the Hahn–Banach theorem.)
- c) The map $w: E \rightarrow E$ such that $w(x)$ is the element y given by the preceding question is a continuous projector whose image is the eigenspace of u corresponding to the eigenvalue 1.
- d) Let $z \in \mathbb{C}$ such that $|z| = 1$. The sequence in $\mathcal{L}(E)$ defined by

$$v_{z,N} = \frac{1}{N}(1_E + z^{-1}u + \cdots + z^{-(N-1)}u^{N-1})$$

converges in $\mathcal{L}(E)$ equipped with the topology of pointwise convergence. Let w_z be its limit. Then w_z is a projector, one has $w_z w = w w_z = z w_z$ and $w_{z_1} w_{z_2} = 0$ if $z_1 \neq z_2$.

- 25) a) Let $N \geq 1$ be an integer. We denote by G_N (resp. H_N) the set $\{-1, 1\}^N$ (resp. the set $\{-1, 2\}^N$). We denote by μ_N (resp. ν_N) the positive measure of total mass 1 on G_N (resp. on H_N) such that, for every $(t_i)_{1 \leq i \leq N}$ in G_N (resp. in H_N), one has $\mu_N((t_i)_{1 \leq i \leq N}) = \frac{1}{2^N}$, resp.

$$\nu_N((t_i)_{1 \leq i \leq N}) = \left(\frac{1}{3}\right)^j \left(\frac{2}{3}\right)^{N-j},$$

where j is the number of indices i such that $t_i = 2$.

There exists a real number $c > 0$ such that, for every family $(a_i)_{1 \leq i \leq N}$ of complex numbers,

we have

$$\mu_N\left(\left\{(t_i)_{1 \leq i \leq N} \in G_N \mid \left|\sum_{i=1}^N a_i t_i\right| > c(\log N)^{1/2} \left(\sum_{i=1}^N |a_i|^2\right)^{1/2}\right\}\right) < \frac{c}{N^3},$$

and

$$\nu_N\left(\left\{(t_i)_{1 \leq i \leq N} \in H_N \mid \left|\sum_{i=1}^N a_i t_i\right| > c(\log N)^{1/2} \left(\sum_{i=1}^N |a_i|^2\right)^{1/2}\right\}\right) < \frac{c}{N^3}.$$

(We may assume that $a_i \in \mathbf{R}$ for all i and that $\sum |a_i|^2 = 1$. For the first inequality, prove and use the upper bound.

$$\int_{G_N} \exp\left(\lambda \left|\sum_{i=1}^N a_i t_i\right|\right) d\mu_N \leq 2 \exp(\lambda^2)$$

for every real $\lambda > 0$; reason in a similar manner for the second upper bound using the inequality $\frac{1}{3}(e^{2x} + 2e^{-x}) \leq e^{2x^2}$ for all $x \in \mathbf{R}$.)

b) For $k \in N$, let $A_k = \mathbf{Z}/(3 \cdot 2^k)\mathbf{Z}$. There exists a real number $c' > 0$, not depending on k , and a function $f_k: \widehat{A}_k \rightarrow \{-1, 2\}$ such that

$$\sum_{\chi \in \widehat{A}_k} f_k(\chi) = 0$$

and, for all $x \in A_k$, one has

$$\left| \sum_{\chi \in \widehat{A}_k} \chi(x) f_k(\chi) \right| \leq c'(k+1)^{1/2} 2^{k/2}.$$

(Using a), determine the existence of a function f_k satisfying the second condition, then modify f_k to obtain the first.)

We denote by B_k (resp. C_k) the set of characters χ of A_k such that $f_k(\chi) = 2$ (resp. $f_k(\chi) = -1$). We have $\text{Card}(B_k) = 2^k$ and $\text{Card}(C_k) = 2^{k+1}$.

For every $k \geq 1$, we fix a bijection $\iota_k: B_k \rightarrow C_{k-1}$.

We denote by A the disjoint union of the A_k for $k \in N$; we equip A with the discrete topology, and we denote by $E = \mathcal{C}_b(A)$.

We fix a family $\varepsilon = (\varepsilon_k)_{k \geq \mathbf{N}}$ of functions $\varepsilon_k: \mathbf{B}_k \rightarrow \{-1, 1\}$. We then define functions $e_{k,\chi} \in E$, for $k \in \mathbf{N}$ and $\chi \in \mathbf{B}_k$ by

$$e_{k,\chi}(x) = \begin{cases} \iota_k(\chi)(x) & \text{if } k \geq 1 \text{ and } x \in A_{k-1}, \\ \varepsilon_k(\chi)\chi(x) & \text{if } x \in A_k \\ 0 & \text{otherwise.} \end{cases}$$

We denote by E_ε the closed subspace of E generated by the functions $(e_{k,\chi})_{k \in \mathbf{N}, \chi \in \mathbf{B}_k}$. This is a complex Banach space of countable type. The rest of the exercise aims to show that, for a suitable choice of ε , the space E_ε does not have the approximation property; it admits no Banach basis.

- c) For $k \in \mathbf{N}$ and $\chi \in \mathbf{B}_k$ there exists a continuous linear form $\lambda_{k,\chi} \in E'_\varepsilon$ such that, for every $f \in E_\varepsilon$,

$$\lambda_{k,\chi}(f) = \frac{1}{\text{Card}(A_k)} \sum_{x \in A_k} \varepsilon_k(\chi)\chi(x^{-1})f(x).$$

For $k \geq 1$ and $f \in E_\varepsilon$, we have

$$\lambda_{k,\chi}(f) = \frac{2}{\text{Card}(A_k)} \sum_{x \in A_k} \iota_k(\chi)(x^{-1})f(x).$$

- d) For every $k \in \mathbf{N}$, let β_k be the continuous linear form on $\mathcal{L}(E_\varepsilon)$ defined by

$$\beta_k(u) = \frac{1}{2^k} \sum_{\chi \in \mathbf{B}_k} \lambda_{k,\chi}(u(e_{k,\chi})).$$

One has $\beta_k(1_{E_\varepsilon}) = 1$

- e) Let $u \in \mathcal{L}(E_\varepsilon)$ and $k \in \mathbf{N}$. We have

$$\beta_{k+1}(u) - \beta_k(u) = \frac{1}{\text{Card}(A_k)} \sum_{x \in A_k} u(\varphi_{k,x})(x)$$

where, for $x \in A_k$, we set

$$\varphi_{k,x} = \frac{1}{2^{k+1}} \sum_{\chi \in B_{k+1}} \iota_{k+1}(\chi)(x^{-1})e_{k+1,\chi} - \frac{1}{2^k} \sum_{\chi \in B_k} \varepsilon_k(\chi)\chi(x^{-1})e_{k,\chi}.$$

One has $|\varphi_{k,x}(y)| \leq c'(k+1)^{1/2}2^{-k/2}$ for $(x,y) \in A_k^2$.

f) There exists a real number $c'' \geq c'$ such that, for every $k \geq 1$, there exists a function $\varepsilon_k: B_k \rightarrow \{-1, 1\}$ satisfying $|\varphi_{k,x}(y)| \leq c''3(k+1)^{1/2}2^{-k/2}$ for $(x,y) \in A_k \times A_{k-1}$.
(Apply the first inequality of a).) We now assume that $\varepsilon = (\varepsilon_k)$ satisfies these properties.

g) For every $k \in \mathbb{N}$ and $x \in A_k$, we have $\|\varphi_{k,x}\| \leq c''(k+1)^{1/2}2^{-k/2}$.

h) Let K be the subset of E_ε whose elements are e_{0,χ_0} , where χ_0 is the unique element of B_0 , and the functions $(k+1)^2\varphi_{k,x}$ for $k \in N$ and $x \in A_k$. The set K is relatively compact in E_ε , and we have

$$|\beta_{k+1}(u) - \beta_k(u)| \leq \frac{1}{(k+1)^2} \sup_{x \in K} \|u(x)\|$$

for all $k \in \mathbb{N}$ and every $u \in \mathcal{L}(E_\varepsilon)$.

i) The sequence $(\beta_k)_{k \in \mathbb{N}}$ converges in the dual of the space $\mathcal{L}(E_\varepsilon)$ equipped with the topology of pointwise convergence. Denote by β its limit. For every $u \in \mathcal{L}(E_\varepsilon)$, we have $|\beta(u)| \leq 3 \sup_{x \in K} \|u(x)\|$.

j) One has $\beta(1_{E_\varepsilon}) = 1$ and $\beta(u) = 0$ if u has finite rank.

k) The space E_ε does not have the approximation property.

l) Let $p \in]2, +\infty[$. We equip A with the counting measure and denote by E_ε^p the closed subspace of $L^p(A)$ generated by the $(e_{k,\chi})_{k \in \mathbb{N}, \chi \in B_k}$. The space E_ε^p does not have the approximation property. (Verify that the cardinality of the support of $\varphi_{k,x}$ is $\leq 21 \cdot 2^k$ and deduce that $\|\varphi_{k,x}\|_p = O((k+1)^{1/2}2^{-k(p-2)/2p})$ for $k \in N$ and $x \in A_k$.)

26) Let E and F be Banach spaces, and let u be a continuous linear map from E into F .

a) Prove that the following conditions are equivalent:

(i) The map u is compact;

- (ii) There exists a sequence $(x'_n)_n$ tending to 0 in the strong dual E' of E satisfying $\|u(x)\| \leq \sup |\langle x, x'_n \rangle|$ for all $x \in E$;
- (iii) There exists a closed subspace L of the Banach space c_0 and compact linear maps $f : E \rightarrow L$ and $g : L \rightarrow F$ such that $u = g \circ f$ (If u is compact, apply cor. 1 of EVT, IV, p. 24 to the image under ${}^t u$ of the polar of the unit ball of F . Under hypothesis (ii), let (λ_n) be an element of c_0 such that the sequence $(\lambda_n^{-1} x'_n)$ tends to 0; define f by $f(x) = (\langle x, \lambda_n^{-1} x'_n \rangle)_n$ and take for L the closure of the image of f .)
- b) If u is compact, we have $\|u\| = \inf(\sup_n \|x'_n\|)$ where the infimum is taken over the set of sequences (x'_n) satisfying (ii), and $\|u\| = \inf(\|f\| \|g\|)$, the infimum being taken over the set of triples (L, f, g) satisfying the conditions of (iii).
- 27) Let E and F be locally convex spaces. A mapping $u \in \mathcal{L}(E; F)$ is said to be weakly compact if it is a compact mapping from E into F equipped with the topology $\sigma(F, F')$.
- a) The weakly compact mappings form a subspace $\mathcal{L}^{fc}(E; F)$ of $\mathcal{L}(E; F)$. Let E_1, F_1 be locally convex spaces, $f \in \mathcal{L}(E_1; E)$, $g \in \mathcal{L}(F; F_1)$ and $u \in \mathcal{L}(E; F)$; if u is weakly compact, the same is true of $g \circ u \circ f$.
- b) One has $\mathcal{L}^c(E; F) \subset \mathcal{L}^{fc}(E; F)$. If E is an infinite-dimensional reflexive Banach space, the identity map of E is weakly compact but is not compact.
- 28) Let E and F be locally convex spaces and $u \in \mathcal{L}(E; F)$.
- a) Show that the following properties are equivalent:
- (i) The image under u of every bounded subset of E is relatively compact for the weak topology;
- (ii) The map ${}^{tt}u$ maps F'' into E .
- b) If F is quasi-complete, prove that these properties are equivalent to:
- (iii) The map ${}^t u$ maps every equicontinuous subset of F' to a relatively compact subset of E' for $\sigma(E', E'')$.

- c) If E and F are Banach spaces, (i) to (iii) are also equivalent to
- (i') The map u is weakly compact;
- (iii') The map ${}^t u: F'_b \rightarrow E'_b$ is weakly compact.
- 29) a) Prove that every weakly compact subset of the Banach space $\ell^1(\mathbb{N})$ is compact (cf. EVT, IV, p. 69, exerc. 11). In particular, every weakly compact linear map from a locally convex space E into $\ell^1(\mathbb{N})$ is compact.
- b) Let E be a Banach space. Deduce from a) that every weakly compact linear map from c_0 into E is compact.
- 30) a) Let E and F be Banach spaces and $u \in \mathcal{L}(E; F)$. Show that the following properties are equivalent:
- (i) The image under u of every weakly compact subset of E is (strongly) compact;
- (ii) The image under u of every weakly relatively compact subset of E is relatively compact in F ;
- (iii) The image under u of every weakly convergent sequence in E converges strongly in F .
- (Use the Šmulian theorem, EVT, IV, p. 36, th. 2.)
- One says that u is semi-compact ⁽³⁾ if it satisfies these properties.
- b) Let E and F be Banach spaces. Show that the set $\mathcal{L}^{sc}(E; F)$ of semi-compact mappings is a closed subspace of $\mathcal{L}(E; F)$.
- c) Let E and F be Banach spaces and $u \in \mathcal{L}(E; F)$. Let E_1, F_1 be Banach spaces, $f \in \mathcal{L}(E_1; E)$ and $g \in \mathcal{L}(F; F_1)$. If $u: E \rightarrow F$ is semi-compact, so is $g \circ u \circ f$.
- d) Let E and F be Banach spaces and $u \in \mathcal{L}(E; F)$. If u is compact, it is semi-compact. Conversely, if E is reflexive and u is semi-compact, then u is compact.
- e) Let E and F be Banach spaces. If the dual space E' of E is of countable type, then every completely continuous linear map $E \rightarrow F$ is compact.

- f) Let I be an infinite set; the identity map of $\ell^1(I)$ is semi-compact, but is not weakly compact (cf. EVT, IV, p. 54, exerc. 15). g) The injection of $\ell_K^1(\mathbb{N})$ into $\ell_K^2(\mathbb{N})$ is semi-compact but is not compact (cf. loc. cit.).
- g) Let E be a Banach space. Every continuous linear map $E \rightarrow \ell_K^1(\mathbb{N})$ is semi-compact, and every continuous linear map $\ell_K^1(\mathbb{N}) \rightarrow E$ is semi-compact. (3) Some authors call them completely continuous maps.
- h) Let E be a Banach space. If E is reflexive or if its dual is of countable type, then every continuous linear map $E \rightarrow \ell_K^1(\mathbb{N})$ is compact.
- 31) Let E be a Banach space.
- a) Prove that the following conditions are equivalent:
- (i) For any Banach space F , every weakly compact mapping from E into F is semi-compact.
- (ii) If (x_n) is a sequence tending to 0 in E for the topology $\sigma(E, E')$ and (x'_n) a sequence tending to 0 in E' for the topology $\sigma(E', E'')$, the sequence $\langle x_n, x'_n \rangle$ tends to 0.
- (To prove that (i) implies (ii), one takes $F = c_0$. If (i) is not satisfied, there exists a weakly compact map u from E into a Banach space F and a sequence (x_n) in E tending weakly to 0, such that $\inf(\|u(x_n)\|) > 0$; one constructs a sequence (y_n) in the unit ball of E' , convergent for $\sigma(E', E'')$ and such that $\langle u(x_n), y_n \rangle = \|u(x_n)\|$, and one takes $x'_n = y_n - \lim y_n$.)
- These conditions are satisfied in particular when $E = \mathcal{C}(T)$, where T is a compact space, and when $E = L^1(X, \mu)$, or $E = L^\infty(X, \mu)$, where X is a locally compact space and μ a measure on X (INT, VI, § 2, exerc. 23). b) Prove that a Banach space possessing the above properties admits no reflexive direct factor of infinite dimension.
- c) Let H be a reflexive Banach space, and let B be the unit ball of H' , equipped with the weak topology. Show that H can be identified with a closed non-direct subspace of $\mathcal{C}(B)$.
- 32) A Banach space of countable type admitting a Schauder basis (EVT, IV, p. 70, exerc. 14) possesses the approximation property.

33) a) Prove that the strict inductive limit of a sequence of approximation spaces is an approximation space.

b) Prove that the locally convex direct sum of a family of approximation spaces is an approximation space.

34) Let E be a Banach space. Prove that the following properties are equivalent:

- (i) The space E' is an approximation space;
- (ii) For every Banach space F , the closure of $\mathcal{L}^f(E; F)$ in $\mathcal{L}(E; F)$ is $\mathcal{L}^c(E; F)$;
- (iii) For every closed subspace F of c_0 the closure of $\mathcal{L}^f(E; F)$ in $\mathcal{L}(E; F)$ is $\mathcal{L}^c(E; F)$.

35) Let E be a Banach space. We say that E has the metric approximation property if the identity map of E belongs to the closure of the set of elements of norm ≤ 1 of $\mathcal{L}^f(E)$ in the space $\mathcal{L}(E)$ endowed with the topology of compact convergence.

Prove that the following properties are equivalent:

- (i) The space E has the metric approximation property;
- (ii) The unit ball of $\mathcal{L}(E)$ is the closure, for the topology of compact convergence, of the set of elements of norm ≤ 1 of $\mathcal{L}^f(E)$;
- (iii) For every Banach space F , the unit ball of $\mathcal{L}(E; F)$ is the closure, for the topology of compact convergence, of the set of elements of norm ≤ 1 of $\mathcal{L}^f(E; F)$;
- (iv) For every Banach space F , the unit ball of $\mathcal{L}(F; E)$ is the closure, for the topology of compact convergence, of the set of elements of norm ≤ 1 of $\mathcal{L}^f(F; E)$.

36) a) Let X be a locally compact topological space. Prove that the space $\mathcal{C}_0(X)$ has the metric approximation property (adapt the proof of Prop. 13 of III, p. 21).

b) Let X be a locally compact topological space equipped with a positive measure and $p \in [1, +\infty]$. Prove that the space $L^p(X)$ has the metric approximation property (same method).

- 37) Let X be a differentiable manifold of class C^r where $1 \leq r \leq \infty$, paracompact, locally of finite dimension, and let E be a vector bundle over X of class C^r , locally of finite dimension. Prove that the space $\mathcal{S}^r(X; E)$ has the approximation property.

Exercises for Section 2

- 1) Let $I = [a, b]$ be a compact interval of $\mathbb{R}$. For every continuous function $f : I \rightarrow \mathbb{C}$ and every $y \in I$ set

$$V(f)(y) = \int_a^y f(x) dx.$$

Show that the linear map V from $\mathcal{C}(I)$ into $\mathcal{C}(I)$ is compact (“Volterra operator”).

- 2) Let $X =]0, 1[$ and let μ be Lebesgue measure on X . One defines a sequence $(f_n)_{n \geq 1}$ of functions $X \rightarrow \mathbb{R}$ by setting

$$f_{2n}(x) = \begin{cases} \sqrt{2} \sin(2n\pi x) & \text{if } 0 < x < 1/2 \\ -2^n & \text{if } 1 - 2^{-2n} < x < 1 - 2^{-2n-1} \\ 0 & \text{otherwise,} \end{cases}$$

and

$$f_{2n-1}(x) = \begin{cases} \sqrt{2} \sin(2n\pi x) & \text{if } 0 < x < 1/2 \\ 2^n & \text{if } 1 - 2^{-2n} < x < 1 - 2^{-2n-1} \\ 0 & \text{otherwise,} \end{cases}$$

for all $n \geq 1$ and every $x \in X$.

- a) The family $(f_n)_{n \geq 1}$ is orthonormal in $\mathcal{L}^2(X, \mu)$.
- b) For every $(x, y) \in X \times X$ we set

$$N(x, y) = \sum_{n=1}^{+\infty} 2^{-n} (f_{2n-1}(x)f_{2n-1}(y) - f_{2n}(x)f_{2n}(y)).$$

Show that $|N(x, y)| \leq 2^{3/2}$ for all $(x, y) \in X \times X$. Deduce that $N_y \in \mathcal{L}^\infty(X)$ for all $y \in X$ and that

$$\int_X^* \|N_y\|_\infty^2 dy < +\infty.$$

c) Show that by setting

$$u_N(f)(y) = \int_X N(x, y)f(x)d\mu$$

for $f \in \mathcal{L}^1(X, \mu)$ and $y \in X$, by passing to the quotient one defines a continuous linear map from $L^1(X, \mu)$ into $L^2(X, \mu)$.

d) The linear map u_N is not compact. (Consider the sequence $g_n = 2^n(f_{2n-1} - f_{2n})$; show that g_n is integrable and that the sequence $(u_N(g_n))$ in $L^2(X, \mu)$ has no convergent subsequence.)

3) Let X be a locally compact topological space, μ a positive measure on X and $Y \subset X$ a locally μ -negligible subset. Let N be the characteristic function of $X \times Y$.

a) One has

$$\int_{X \times X}^* |N(x, y)f(x)g(y)|d(\mu \otimes \mu)(x, y) = 0$$

for $f, g \in L^2(X, \mu)$.

b) If $f \in L^2(X, \mu) - L^1(X, \mu)$, the set of $y \in X$ such that $x \mapsto N(x, y)f(x)$ is μ -integrable coincides with Y .

4) Let $\alpha \in]0, 1]$. Let I be a compact interval of $\mathbf{R}$. We denote by $\mathcal{C}^\alpha(I)$ the vector subspace of $\mathcal{C}(I)$ consisting of functions f such that there exists $C \geq 0$ satisfying

$$|f(x) - f(y)| \leq C|x - y|^\alpha$$

for all $(x, y) \in I \times I$.

We equip $\mathcal{C}^\alpha(I)$ with the norm

$$\|f\|_\alpha = \sup_{x \in I} |f(x)| + \sup_{\substack{(x,y) \in I \times I \\ x \neq y}} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

- a) The space $\mathcal{C}^\alpha(I)$ is a Banach space.
- b) The canonical injection $\mathcal{C}^\alpha(I) \rightarrow \mathcal{C}(I)$ is compact.
- c) If $0 < \alpha < \beta \leq 1$ then the canonical injection $\mathcal{C}^\beta(I) \rightarrow \mathcal{C}^\alpha(I)$ is compact.
- d) The Sobolev space $H^1(\mathring{I})$ (cf. IV, p. 221, no. 14) is identified with a subspace of $\mathcal{C}^{1/2}(I)$ but the canonical injection $H^1(\mathring{I}) \rightarrow \mathcal{C}^{1/2}(I)$ is not compact.

Exercises for Section 3

- 1) Let E and F be Hilbert spaces and let u be a continuous linear map from E into F . Show that u is a Fredholm map if and only if the image of u is closed and the kernels of u and u^* are finite-dimensional.
- 2) Let p be a real number such that $1 \leq p < +\infty$. Let u be a diagonal endomorphism of $\ell_K^p(\mathbb{N})$. Show that u is a Fredholm endomorphism if and only if it is a Riesz endomorphism. In particular, every diagonal Fredholm endomorphism of $\ell_K^p(\mathbb{N})$ has index zero.
- 3)
 - a) There exists a Banach space E , a continuous automorphism u of E , and a Riesz endomorphism v of E such that $u \circ v$ is not a Riesz endomorphism. There even exist such examples for which $u \circ v - v \circ u$ has finite rank.
 - b) There exists a Banach space E , an automorphism u of E , and a finite-rank endomorphism h such that $u + h$ is not a Riesz endomorphism.
- 4) Let E be an infinite-dimensional complex Hilbert space.
 - a) The group $\mathbf{U}(E)$ of unitary operators on E , equipped with the topology induced by the norm of $\mathcal{L}(E)$, is path-connected. (Show that if u is unitary, there exists a Hermitian

operator v such that $u = \exp(iv)$.)

- b) The group of linear isomorphisms of E is path-connected. (Use the polar decomposition, cf. I, p. 139, def. 3.)
 - c) For every $n \in \mathbb{Z}$ the set of Fredholm endomorphisms of E of index n is open and connected. It is nonempty.
- 5) Let E and F be infinite-dimensional Banach spaces. A continuous linear map $u : E \rightarrow F$ is said to be strictly singular if there does not exist a closed subspace $F_1 \subset E$ of infinite dimension such that u induces, by passage to subspaces, an isomorphism from F_1 onto $u(F_1)$.
- a) Let u be a continuous linear map from E into F . The following conditions are equivalent:
 - (i) The map u is strictly singular;
 - (ii) For every closed subspace F_1 of E of infinite dimension, there does not exist a real number $\delta > 0$ such that $\|u(x)\| \geq \delta\|x\|$ for all $x \in F_1$.
 - b) Let $u : E \rightarrow F$ be a continuous linear map such that the image of u is closed and the kernel of u is finite-dimensional. If $v : E \rightarrow F$ is strictly singular, then the image of $u + v$ is closed and its kernel is finite-dimensional.
 - c) The set of strictly singular endomorphisms of E is a closed two-sided ideal of $\mathcal{L}(E)$.
 - d) Every compact linear map is strictly singular. If E and F are Hilbert spaces, then every strictly singular map $E \rightarrow F$ is compact.
 - e) There exist Banach spaces E and F and a strictly singular non-compact map $E \rightarrow F$.
 - f) Let p and r be real numbers such that $1 \leq p < r$. Every continuous linear map $\ell_K^r(\mathbb{N}) \rightarrow \ell_K^p(\mathbb{N})$ is strictly singular. (Use exercise 10 of III, p. 106.)
- 6) Let E be a complex Hilbert space. Let u be a Fredholm endomorphism of E . If u is normal, then $\text{ind}(u) = 0$.

7) Let E and F be Banach spaces, and let u be an element of $\mathcal{L}(E; F)$. Prove that u is a Fredholm map if and only if the following conditions are satisfied:

- (i) There exists a continuous seminorm p on E and $C \geq 0$ such that the canonical map from E into the separated completion of (E, p) is compact, and

$$\|x\| \leq C\|u(x)\| + p(x)$$

for all $x \in E$;

- (ii) There exists a continuous seminorm q on F' and $C' \geq 0$ such that the canonical map from F' into the separated completion of (F', q) is compact, and

$$\|y'\| \leq C'\|u(y')\| + q(y')$$

for all $y' \in F'$.

8) Let U be a connected open subset of $\mathbb{C}$. We denote by $\mathcal{H}(U)$ the space of holomorphic functions on U equipped with the topology of compact convergence (EVT III, p. 10, example d).

a) Let $a \in \mathcal{H}(U)$. Multiplication by a is a Fredholm endomorphism of $\mathcal{H}(U)$ if and only if the set of zeros of a is finite. Then compute its index.

b) Let $\mathcal{U} = (U_i)_{i \in I}$ a covering of U by nonempty connected open sets. We set

$$I_2 = \{(i, j) \in I \times I \mid U_i \cap U_j \neq \emptyset\},$$

$$I_3 = \{(i, j, k) \in I \times I \times I \mid U_i \cap U_j \cap U_k \neq \emptyset\}.$$

We consider the vector spaces

$$Z^1(\mathcal{U}; \mathbb{C}) = \{(c_{i,j}) \in \mathbb{C}^{I_2} \mid c_{ij} + c_{jk} + c_{ki} = 0 \text{ for all } (i, j, k) \in I_3\}$$

$$B^1(\mathcal{U}; \mathbb{C}) = \{(c_{i,j}) \in \mathbb{C}^{I_2} \mid \text{there exists } c' \in \mathbb{C}^I \text{ such that}$$

$$c_{ij} = c'_i - c'_j \text{ for all } (i, j) \in I_2\}.$$

One has $B^1(\mathcal{U}; \mathbb{C}) \subset Z^1(\mathcal{U}; \mathbb{C})$. We set

$$H^1(\mathcal{U}; \mathbb{C}) = Z^1(\mathcal{U}; \mathbb{C}) / B^1(\mathcal{U}; \mathbb{C}).$$

Let $\mathcal{V} = (V_\alpha)_{\alpha \in A}$ be a covering of U by nonempty connected open sets, finer than U , and let $\varrho: A \rightarrow I$ be a map such that $V_\alpha \subset U_{\varrho(\alpha)}$ for all $\alpha \in A$.

The map $(c_{ij})_{i,j} \mapsto (c_{\varrho(\alpha)\varrho(\beta)})_{\alpha,\beta}$ induces a linear map, independent of the choice of ϱ , from $H^1(\mathcal{U}; \mathbb{C})$ into $H^1(\mathcal{V}; \mathbb{C})$.

We denote by $H^1(U; \mathbb{C})$ the inductive limit of vector spaces $H^1(\mathcal{U}; \mathbb{C})$ equipped with these linear maps, as U ranges over the set of coverings of U by nonempty connected open sets.

- c) Let D be the endomorphism of $\mathcal{H}(U)$ defined by $Df = f'$. Show that there exists an isomorphism of vector spaces from the cokernel of D onto $H^1(U; \mathbb{C})$. Deduce from this that D is a Fredholm endomorphism if and only if the space $H^1(U; \mathbb{C})$ is finite-dimensional. Then compute the index of D .

Exercises for Section 4

- 1) Let E and F be Banach spaces and u a continuous linear map $E \rightarrow F$. We have $((u)) > 0$ if and only if the image of u is closed in F .
- 2) Let $p \in [1, +\infty]$, and $E = \ell_C^p(\mathbb{N})$. Let u be a diagonal endomorphism of E . For every $\lambda \in \mathbb{C}$, compute $((u - \lambda 1_E))$ in terms of the spectrum of u .
- 3) Let $p \in [1, +\infty]$ and $E = \ell_K^p(\mathbb{N})$ and let u be the shift endomorphism

$$(x_0, x_1, \dots) \mapsto (0, x_0, x_1, \dots)$$

of E . Show that u is a Fredholm endomorphism of index 1 and that $((u)) = 1$.

- 4) Let E be a Banach space. The map $u \mapsto ((u))$ of $\mathcal{L}(E)$ into $\mathbb{R}$ is continuous if and only if E has dimension ≤ 1 .
- 5) Let E be a Hilbert space and $u \in \mathcal{L}(E)$. We have $((u)) = \inf(\text{Sp}(|u|) - \{0\})$ (cf. I, p. 139, def. 3 for the definition of $|u|$).
- 6) Let $E = \ell^2(\mathbb{N})$ and $(e_n)_{n \in \mathbb{N}}$ the canonical orthonormal basis of E . Let u be the diagonal endomorphism of E such that $u(e_0) = e_0$ and $u(e_n) = 2e_n$ if $n \geq 1$. Show that $((u)) = 1$ and that $u + h$ is a Fredholm endomorphism for every h such that $\|h\| < 2$.
- 7) Let E and F be Banach spaces and let u be an element of $\mathcal{QM}(E; F)$ (resp. of $\mathcal{QE}(E; F)$). Let h be an element of $\mathcal{L}(E; F)$. Suppose there exist positive real numbers a and b such that $a + b((u))^{-1} < 1$ so that

$$\|h(x)\| \leq a\|u(x)\| + b\|x\| \quad (2)$$

for all $x \in E$.

Prove that $u + h \in \mathcal{QM}(E; F)$ (resp. $u + h \in \mathcal{QE}(E; F)$) and

$$\begin{aligned} \dim \text{Ker}(u + h) &\leq \dim \text{Ker}(u), \\ \dim \text{Coker}(u + h) &\leq \dim \text{Coker}(u), \\ \text{ind}(u + h) &= \text{ind}(u). \end{aligned}$$

(Reduce to the case where a and b are > 0 and let $c > 0$ be such that $a + b/c < 1$; consider the norm p on E defined by

$$p(x) = a\|u(x)\| + b\|x\| \quad \text{for } x \in E,$$

and E_1 the Banach space E equipped with the norm p ; we have $\mathcal{QM}(E_1; F) = \mathcal{QM}(E; F)$, $\mathcal{QE}(E_1; F) = \mathcal{QE}(E; F)$ and $\mathcal{L}(E_1; F) = \mathcal{L}(E; F)$. Then apply lemma 2 of III, p. 62.)

- 8) Let E be a Hilbert space. Prove directly Theorem 4 of III, p. 64 without using the Borsuk–Ulam theorem.

- 9) Let $r \geq 1$ be an integer. If $f : M_1 \rightarrow M_2$ is a morphism of class C^r of manifolds of class C^r , one says that $x \in M_1$ is a critical point of f if f is not a submersion at x (VAR, R1, 5.9.1). Let X be the set of critical points of f . The set $f(X)$ in M_2 is called the set of critical values of f , and $f(M_1) \setminus f(X)$ is called the set of regular values of f .

If M_1 and M_2 are real manifolds of class C^r (VAR, R1, 5.1.5, p. 35), a morphism of manifolds $f : M_1 \rightarrow M_2$, of class C^r , is said to be Fredholm if, for every $x \in M_1$, the tangent linear map $T_x(f)$ is a Fredholm endomorphism of the Banach space $T_x(M_1)$ into the Banach space $T_{f(x)}(M_2)$. a) Let $f : M_1 \rightarrow M_2$ be a Fredholm morphism. If M_1 is connected, the index $\text{ind}(T_x(f))$ is independent of $x \in M_1$. It is called the index of the Fredholm morphism f and is denoted $\text{ind}(f)$.

- b) Let U be an open subset of a Banach space E such that $0 \in U$ and let F be a Banach space. Let $f : U \rightarrow F$ be a Fredholm morphism of class C^r and let $u = T_0(f)$. There exist a Banach space E_1 , a chart $(U_1, \varphi, E_1 \times \text{Ker}(u))$ of U such that $0 \in U_1$, a chart $(U_2, \psi, E_1 \times \text{Coker}(u))$ of F such that $f(0) \in U_2$ and a Fredholm morphism $g : \varphi(U_1) \rightarrow E_1 \times \text{Coker}(u)$ of class C^r such that

$$(\psi \circ f \circ \varphi^{-1})(a, b) = (a, g(a, b))$$

for all $(a, b) \in \varphi(U_1) \subset E_1 \times \text{Ker}(u)$.

- c) Let $f : M_1 \rightarrow M_2$ be a Fredholm morphism of class C^r between manifolds of class C^r . For every $y \in M_2$, there exists an open neighborhood V of y such that f defines, by passage to subspaces, a proper map $f^{-1}(V) \rightarrow V$.
- d) Let U be an open subset of a Banach space E containing 0 and let F be a Banach space. Let $f : U \rightarrow F$ be a Fredholm morphism of class C^∞ . For every $x \in U$ and every neighborhood V of $f(x)$ in F , there exists a regular value of f in V . (Use the second Sard theorem, VAR, R2, 10.1.3, p. 37.) e) Let $f : M_1 \rightarrow M_2$ be a Fredholm morphism of class C^∞ of manifolds of class C^∞ . If M_1 and M_2 admit countable bases, then the set of critical values of f is meager ("Smale's theorem").

f) Let $f: M_1 \rightarrow M_2$ be a Fredholm morphism of class C^∞ of manifolds of class C^∞ admitting countable bases. If $\text{ind}(f) < 0$, then the image of f has empty interior.

g) Let $f: M_1 \rightarrow M_2$ be a Fredholm morphism of class C^∞ of manifolds of class C^∞ admitting countable bases. For every $y \in M_2$ outside a meager set, the set $f^{-1}(y)$ is a pure submanifold of dimension $\text{ind}(f)$.

Exercises for Section 5

In the following exercises, when E is a Banach space, we denote by $\mathcal{Calk}(E)$ the Calkin algebra $\mathcal{L}(E)/\mathcal{L}^c(E)$. We denote by π the canonical projection from $\mathcal{L}(E)$ onto $\mathcal{Calk}(E)$.

If E is a complex Hilbert space, we equip $\mathcal{Calk}(E)$ with the involution induced from the involution of $\mathcal{L}(E)$ by passage to the quotient.

1) Let E be a locally convex space.

a) Suppose that every neighborhood of 0 in E contains a vector subspace of E of finite codimension. Show that the space $\mathcal{F}(E)$ of Fredholm endomorphisms of E is not closed in $\mathcal{L}(E)$ endowed with the topology of bounded convergence.

b) Give an example of an infinite-dimensional Fréchet space such that every neighborhood of 0 contains a vector subspace of E of finite codimension. c) Let E_σ be an infinite-dimensional Banach space equipped with the weak topology. The set $\mathcal{F}(E_\sigma)$ is not open in $\mathcal{L}(E_\sigma)$.

2) Let E be an infinite-dimensional Banach space and $u \in \mathcal{L}(E)$. We denote by $S(u)$ the set of complex numbers λ such that there exist a real number $\delta > 0$ and a finite-codimensional subspace F of E such that $\|u(x) - \lambda x\| \geq \delta \|x\|$ for all $x \in F$.

a) The set $S(u)$ is open in $\mathbb{C}$ and contains the complement in $\mathbb{C}$ of the spectrum of u .

b) If $\lambda \in S(u)$, the endomorphism $u - \lambda 1_E$ has a closed image and a finite-dimensional kernel.

c) The complement $F(u) = \mathbb{C} - S(u)$ is not empty. (Show that an element of $\text{Sp}_e(u)$ of

maximal modulus belongs to $F(u)$.)

3) Let E be an infinite-dimensional complex Hilbert space.

a) The involutive algebra $\mathcal{Calk}(E)$ is a star algebra.

b) Let u be an element of $\mathcal{L}(E)$ such that u^*u is compact. Show that u is compact.

c) Let G be the group of invertible elements in $\mathcal{Calk}(E)$ and G_0 its neutral component. Let $p : G \rightarrow G/G_0$ be the canonical projection. There exists a unique group isomorphism $\alpha : Z \rightarrow G/G_0$ such that $\alpha(\text{ind}(u)) = p(\pi(u))$ for every Fredholm endomorphism u of E .

4) Let E be a complex Hilbert space. For every invertible element ξ of $\mathcal{Calk}(E)$, we call the index of ξ the index of any Fredholm endomorphism u of E such that $\pi(u) = \xi$.

a) Show that for every $\xi \in \mathcal{Calk}(E)$ that is Hermitian, there exists $u \in \mathcal{L}(E)$ that is Hermitian such that $\pi(u) = \xi$.

b) Show that for every $\xi \in \mathcal{Calk}(E)$ that is unitary and has index zero, there exists $u \in \mathcal{L}(E)$ that is unitary such that $\pi(u) = \xi$. (Let v be invertible such that $\pi(v) = \xi$; consider the polar decomposition $v = j|v|$ of v and verify that $|v| - 1_E$ is compact, then deduce that $v - j$ is compact.)

c) Show that for every $\xi \in \mathcal{Calk}(E)$ that is unitary such that $\xi^2 = 1$, there exists $u \in \mathcal{L}(E)$ that is unitary such that $u^2 = 1$ and $\pi(u) = \xi$.

d) For every element $p \in \mathcal{Calk}(E)$ such that $p^2 = p$, there exists a projector $P \in \mathcal{L}(E)$ such that $\pi(P) = p$.

5) Let E be a complex Hilbert space.

a) Let n be an integer with $n \geq 2$. We set $I = \mathbf{N} \times \{0, \dots, n\}$. Let $(\sigma_i)_{1 \leq i \leq n}$ be the family of permutations of I defined by

$$\sigma_i(m, i) = (m - 1, i) \text{ for } m \geq 1$$

$$\sigma_i(0, i) = (0, i - 1)$$

$$\sigma_i(m, i - 1) = (m + 1, i - 1) \text{ for } m \geq 0$$

$$\sigma_i(m, j) = (m, j) \text{ for } m \geq 0 \text{ and } j \notin \{i, i - 1\}.$$

b) We denote by $(e_{m,i})$ the canonical orthonormal basis of $\ell^2(I)$. Let u_i be the endomorphism of $\ell^2(I)$ such that $u_i(e_{m,j}) = e_{\sigma_i(m,j)}$ for all $(m,j) \in I$. The endomorphism u_i is unitary.

c) The elements $\pi(u_i)$ are pairwise commuting.

d) There does not exist a family $(v_1, \dots, v_n)$ of unitary endomorphisms of E , pairwise commuting, such that $\pi(v_i) = \pi(u_i)$ for all i .

6) Let E be a complex Hilbert space. Recall that $\mathcal{Calk}(E)$ is a star algebra (exercise 3).

Let $\mathfrak{F}$ be a nonprincipal ultrafilter on $\mathbf{N}$. Let F be the vector space of bounded sequences in E that converge weakly to 0 along $\mathfrak{F}$ and let F_0 be the subspace of F consisting of sequences that converge in norm to 0 along $\mathfrak{F}$. We set $\widetilde{H} = F/F_0$. a) For all elements $x = (x_n)$ and $y = (y_n)$ of F , show that the sequence $(\langle x_n | y_n \rangle)_{n \in \mathbf{N}}$ converges along $\mathfrak{F}$. We denote by $\langle x | y \rangle_F$ its limit.

b) Show that $(x, y) \mapsto \langle x | y \rangle_F$ is a positive Hermitian form on F , and that $\langle x | x \rangle_F = 0$ if and only if $x \in F_0$; this Hermitian form therefore defines a separated pre-Hilbert space structure on $\widetilde{H}$.

c) Let H be the Hilbert space obtained by completing the separated pre-Hilbert space $\widetilde{H}$. Show that for every u in $\mathcal{L}(E)$, one obtains an element $\varrho(u)$ of $\mathcal{L}(H)$ by passing to the quotient of the linear map $(x_n) \mapsto (u(x_n))$ from F to F . d) The map $u \mapsto \varrho(u)$ induces an isometric morphism of C^* -algebras $\mathcal{Calk}(E) \rightarrow \mathcal{L}(H)$.

(We will see later that every C^* -algebra A has an isometric representation $A \rightarrow \mathcal{L}(E)$ for a certain Hilbert space E , cf. th. 2 of V, p. 442.)

7) We equip the unit circle $S_1 \subset \mathbf{C}$ with the normalized Haar measure. Let E be the complex Hilbert space $L^2(S_1)$. For $n \in \mathbf{Z}$, let e_n be the element $z \mapsto z^n$ of E . Let $F \subset E$ be the closed subspace generated by the e_n for $n \geq 0$ and let $p : E \rightarrow F$ be the orthogonal projector. For

- $f \in L^\infty(\mathbf{S}_1)$ we define linear maps M_f (resp. T_f) on E (resp. F) by $M_f(g) = fg$ for $g \in E$ and $T_f = p \circ M_f$. One says that T_f is the Toeplitz operator with symbol f . (a) The linear maps M_f and T_f are continuous. The spectrum of M_f is the closure of the set of Fourier coefficients $c_n = \langle f | e_n \rangle$ of f . The endomorphism M_f is a Fredholm endomorphism if and only if it is a Riesz endomorphism.
- b) The endomorphisms T_{e_n} for $n \in \mathbb{Z}$ are Fredholm endomorphisms and $\text{ind}(T_{e_n}) = -n$ for all $n \in \mathbb{Z}$.
- c) One has $T_{\bar{f}} = T_f^*$ for every function $f \in L^\infty(\mathbf{S}_1)$.
- d) If T_f is an isomorphism, then M_f is an isomorphism and f is invertible in $L^\infty(\mathbf{S}_1)$. Deduce from this that the spectrum of M_f is contained in the spectrum of T_f and that $\|T_f\| = \|f\|$.
- e) Let $f \in \mathcal{C}(\mathbf{S}_1)$ be a continuous function and $g \in L^\infty(\mathbf{S}_1)$. The endomorphisms $T_f T_g - T_{fg}$ and $T_g T_f - T_{gf}$ are compact. (First consider the case where $f = e_n$.)
- f) Let $\mathcal{Calk}(F)$ be the Calkin algebra of F and $\pi: \mathcal{L}(F) \rightarrow \mathcal{Calk}(F)$ be the canonical projection. Show that $f \mapsto \pi(T_f)$ is an injective morphism of involutive algebras from $\mathcal{C}(\mathbf{S}_1)$ into $\mathcal{Calk}(F)$ (To prove injectivity, one may use prop. 1 of I, p. 31.)
- g) Let $f \in \mathcal{C}(\mathbf{S}_1)$. Then T_f is a Fredholm endomorphism if and only if f does not vanish on $\mathbf{S}_1$. (Use prop. 5 of I, p. 106.)
- h) Let $f \in \mathcal{C}(\mathbf{S}_1)$ such that f does not vanish on $\mathbf{S}_1$. The index of T_f is equal to $n \in \mathbb{Z}$ if and only if the path $t \mapsto f(e^{2i\pi t})$ in $\mathbb{C} - \{0\}$ is homotopic to the path $t \mapsto e_n(t)$ (cf. TA, III, p. 229, def. 1).
- 8) We resume the notation of the preceding exercise.
- a) We denote by $T(\mathbf{S}_1)$ the closed subalgebra of $\mathcal{L}(F)$ generated by the Toeplitz operators T_{e_1} and $T_{e_{-1}}$. Show that $T(\mathbf{S}_1)$ is a star subalgebra of $\mathcal{L}(F)$.
- b) Let $f \in \mathcal{C}(\mathbf{S}_1)$. Show that T_f belongs to $T(\mathbf{S}_1)$.
- c) The representation of $T(\mathbf{S}_1)$ on F is irreducible; the star algebra $T(\mathbf{S}_1)$ contains $\mathcal{L}^c(F)$.

d) One has $u \in T(\mathbf{S}_1)$ if and only if there exists $f \in \mathcal{C}(\mathbf{S}_1)$ and $h \in \mathcal{L}^c(F)$ such that $u = T_f + h$. (Show that the set of endomorphisms of the form $T_f + h$ forms a closed star algebra in $\mathcal{L}(F)$.)

e) There exists an exact sequence of morphisms of involutive algebras

$$0 \rightarrow \mathcal{L}^c(F) \rightarrow T(\mathbf{S}_1) \xrightarrow{\varrho} \mathcal{C}(\mathbf{S}_1) \rightarrow 0$$

such that $\varrho(T_f) = f$ for every function $f \in \mathcal{C}(\mathbf{S}_1)$.

- 9) Let B be an infinite-dimensional Banach space. We denote by E the strong dual of B and by F the dual of B equipped with the topology $\sigma(B', B)$. Show that there exists a continuous bijective linear map u from E into F and a compact linear map h from E into F such that $u + h$ has a cokernel of infinite dimension.
- 10) Let E be a separated locally convex space, and let u and h be commuting elements of $\mathcal{L}(E)$. Suppose there exists an integer n such that h^n is a compact endomorphism.
- a) If u is a Fredholm endomorphism, then $u-h$ is a Fredholm endomorphism.
- b) If u is a Riesz endomorphism, then $u - h$ is a Riesz endomorphism.
- 11) Let $E = \ell^2(\mathbf{N})$ and let (e_n) be the canonical basis of E . We define $u \in \mathcal{L}(E)$ by $u(e_i) = e_{i+1}$ for $i \in \mathbf{N}$. Let $F = E \oplus E$ and $v = u \oplus (u^* + 2 \cdot 1_E) \in \mathcal{L}(F)$. Let $\bar{v} = \pi(v)$ be the image of v in the Calkin algebra $\mathcal{Calk}(F)$.
- a) The endomorphism v is a Fredholm endomorphism of index -1, and $v - 2 \cdot 1_E$ is a Fredholm endomorphism of index 1.
- b) Let $p = X(X - 2) \in \mathbf{C}[X]$ and let S be the exponential spectrum of $\bar{v}$ (exercise 5 of I, p. 173). We have $0 \in p(S)$ but 0 does not belong to the exponential spectrum of $p(\bar{v})$ (Use Exercise 3.)

Exercises for Section 6

1) Let A be a compact subset of $\mathbb{C}$ which is equal to the closure of its interior V . We set

$\mathcal{C}(A) = \mathcal{C}(A; \mathbb{C})$. Let $B(A)$ be the vector subspace of $\mathcal{C}(A)$ consisting of functions $f \in \mathcal{C}(A)$ whose restriction to V is holomorphic;

it is a Banach subalgebra of $\mathcal{C}(A)$. Let $u \in \mathcal{L}(B(A))$ be the linear map such that $u(f)(z) = zf(z)$ for all $z \in A$ and every $f \in B(A)$.

- a) The spectrum of u is equal to A .
 - b) For every $z \in V$ the endomorphism $u - z$ of $B(A)$ is a Fredholm endomorphism of index 1.
 - c) For every $z \in A - V$ the endomorphism $u - z$ of $B(A)$ is not a Fredholm endomorphism.
 - d) The stable spectrum of u is the boundary of A , the sensitive spectrum of u is empty, and the essential spectrum of u is A .
 - e) We assume that the complement of A in $\mathbb{C}$ is connected. For every $h \in \mathcal{L}^c(B(A))$, the essential spectrum of $u + h$ is equal to A .
- 2) We keep the notation from the previous exercise. We assume that A contains the unit circle U in $\mathbb{C}$ and that $0 \notin A$. We denote by φ_A the characteristic function of A .
- a) The map

$$\ell: f \mapsto \frac{1}{2i\pi} \int_U f(z) dz$$

is a continuous linear form on $B(A)$.

- b) Let h be the finite-rank endomorphism of $B(A)$ such that $h(f) = \ell(f)\varphi_A$. The essential spectrum of $u - h$ is distinct from that of u . (Show that it contains 0.)
- 3) Let E be a complex Hilbert space. Let U be a connected open subset of $\mathbb{C}$. Let f be a holomorphic function from U into the space $\mathcal{L}^c(E)$. Suppose that there exists $z \in U$ such that 1 belongs to the resolvent set of $f(z)$.

- a) The set D of $z \in U$ such that $1 \in \text{Sp}(f(z))$ is a discrete subset of U .
- b) The map from $U-D$ to $\mathcal{L}(E)$ defined by $z \mapsto R(f(z), 1)$ is holomorphic.

We fix a point $z_0 \in D$.

- c) There exists an integer $k \geq 1$ such that the map from $U-D$ into $\mathcal{L}(E)$ defined by $z \mapsto (z - z_0)^k R(f(z), 1)$ extends to a holomorphic map on $(U - D) \cup \{z_0\}$.
- d) There exists a unique sequence $(u_n)_{n \in \mathbb{N}}$ of endomorphisms of E such that $u_n = 0$ for n sufficiently large and a unique holomorphic function S from $(U - D) \cup \{z_0\}$ into $\mathcal{L}(E)$ such that for all $z \in U - D$, we have

$$R(f(z), 1) = \sum_{j=-k}^{-1} (z - z_0)^j u_j + S(z).$$

- e) The endomorphism u_{-1} is the spectral projector of $f(z_0)$ relative to the open and closed subset $\{1\}$ of the spectrum of $f(z_0)$.
- 4) Let E be a Banach space of countable type. Let $u \in \mathcal{L}(E)$. Assume that the sequence $(\|u^k\|)_{k \in \mathbb{N}}$ is bounded.

For every $\lambda \in \text{Sp}(u) \cap \mathbb{U}$, let $x_\lambda \in E$ be an eigenvector of u of norm 1 corresponding to λ .

- a) For all λ_1, λ_2 in $\text{Sp}(u) \cap \mathbb{U}$ and every integer $k \in \mathbb{N}$, we have

$$|\lambda_1^k - \lambda_2^k| \leq (1 + \|u^k\|) \|x_{\lambda_1} - x_{\lambda_2}\|.$$

- b) There exists a real number $c > 0$ such that $\|x_{\lambda_1} - x_{\lambda_2}\| \geq c$ for all $\lambda_1 \neq \lambda_2$ in $\text{Sp}(u) \cap \mathbb{U}$.
- c) The set $\text{Sp}(u) \cap \mathbb{U}$ is countable ("Jamison's theorem").
- d) Jamison's theorem does not hold in general if E is not of countable type.

- 5) The goal of this exercise is to give another proof of Lomonosov's theorem (cor. 2 of III, p. 13) in the case where the space E is a Banach space.

Let E be a complex Banach space of dimension at least 2. Let h be a nonzero compact endomorphism of E and let A be the commutant of h in $\mathcal{L}(E)$. a) One of the following conditions is satisfied:

- (i) There exists $y \in E$ such that Ay is a nonzero subspace that is not dense in E ;
 - (ii) For every closed bounded subset C of A with nonempty interior, there exists a finite subset I of A such that $h(C)$ is contained in the union of the sets $u^{-1}(C)$ for $u \in I$.
- b) Assume that condition (ii) of the previous question is satisfied. Let C be a closed bounded subset of E with nonempty interior. For every integer $m \geq 1$ there exists a finite sequence $(u_1, \dots, u_m)$ in A such that

$$u_1 \circ \dots \circ u_m \circ h^m(x_0) \in C.$$

(Play table tennis.)

- c) Under the same hypothesis, show that the spectral radius of h is not zero (in the contrary case, consider an element $x_0 \in E$ such that $\|h(x_0)\| > \|h\|$ and the closed ball C with center x_0 and radius $\|h\|$; noting that $\|(\lambda h)^n\|$ tends to 0 for every $\lambda \in \mathbb{R}$, show that the conclusion of the preceding question would imply that 0 belongs to the closure of C).
 - d) For every endomorphism u of E that commutes with a nonzero compact endomorphism of E , there exists a nonzero closed subspace $F \subset E$ different from E such that $u(F) \subset F$.
- 6) Let A be a unital $\mathbb{Z}$ -algebra. Assume that A is a free $\mathbb{Z}$ -module. Let B be a subset of A that is a basis of A as a $\mathbb{Z}$ -module. We say that the pair (A, B) is a natural algebra if the structure constants of A with respect to B (A, III, p. 10) belong to $\mathbb{N}$ and if the coefficients of 1_A in the basis B belong to $\mathbb{N}$. Moreover, we say that (A, B) is unital if 1_A is an element of B .

Let (A, B) be a natural algebra. Let B_0 be the set of $b \in B$ such that the coefficient of b in 1_A is nonzero. Denote by τ_A the unique homomorphism of $\mathbb{Z}$ -modules from A to $\mathbb{Z}$ such that, for every b in B , one has $\tau_A(b) = 1$ if $b \in B_0$, and $\tau_A(b) = 0$ otherwise.

a) One has $\tau_A(xy) = \tau_A(yx)$ for all $(x, y) \in A^2$.

b) For all $b \neq b'$ in B_0 , we have $b^2 = b$ and $bb' = 0$.

7) Let (A, B) be a natural algebra.

a) There exists at most one involution ι of $\mathcal{B}$ such that the unique $\mathbf{Z}$ -module homomorphism from A to A sending b to $\iota(b)$ is an anti-automorphism of A satisfying the condition $\tau_A(bb') = 1$ if $b' = \iota(b)$ and $\tau_A(bb') = 0$ otherwise.

When such an involution ι exists, we say that $(A, \mathcal{B})$ is a natural basal algebra.

We now assume that (A, B) is a natural basal algebra.

b) We then have $\iota(b) = b$ for all $b \in B_0$ as well as the formula

$$1_A = \sum_{b \in \mathcal{B}_0} b.$$

c) The structure constants $\gamma_{bb'}^{b''}$ of A satisfy $\gamma_{bb'}^{\iota(b'')} = \tau_A(bb'b'')$ for (b, b', b'') in $\mathcal{B}^3$.

8) Let (A, B) be a natural basal algebra such that A has finite rank.

a) For every $x \in A$, the element

$$\sum_{b \in \mathcal{B}} bx\iota(b)$$

is central.

In the rest of this exercise, we further assume that $(A, \mathcal{B})$ is unital.

b) For all b and b' in $\mathcal{B}$, there exist b_1 and b_2 in $\mathcal{B}$ such that the coefficient of b' in bb_1 and in b_2b is nonzero.

c) For every $x \in B$, the matrix representing, in the basis B , the map $y \mapsto xy$ from A to A is a real matrix all of whose coefficients are positive. We denote by pf the unique homomorphism from A to $\mathbf{R}$ such that $\text{pf}(b)$ is the spectral radius of this matrix when $b \in B$.

d) For every $b \in \mathcal{B}$, the real number $\alpha = \text{pf}(b)$ is an algebraic integer such that $\alpha \geq 1$; for every conjugate $\alpha' \in \mathbb{C}$ of α , we have $|\alpha'| \leq \alpha$.

e) Let $a_0 \in A$ be the sum of the elements of $\mathcal{B}$. All coefficients of the matrix representing the map $f : y \mapsto ya_0$ in the basis $\mathcal{B}$ are strictly positive. There exists an element $r \in \mathbb{C} \otimes A$ all of whose coefficients in the basis $(1 \otimes b)_{b \in \mathcal{B}}$ are strictly positive such that $f_{\mathbb{C}}(r) = \varrho(f_{\mathbb{C}})r$.

f) For every $x \in A$, there exists a unique complex number $\delta(x)$ such that $xr = \delta(x)r$ in $\mathbb{C} \otimes A$. The map $\delta : A \rightarrow \mathbb{C}$ is a character of A .

g) One has $\delta(x) = \text{pf}(x)$ for all $x \in A$. In particular, the map $x \mapsto \text{pf}(x)$ is a character of A .

h) The map pf is the unique character of A such that $\text{pf}(b) \geq 0$ for all $b \in \mathcal{B}$; moreover, one has $\text{pf}(b) > 0$ for all $b \in \mathcal{B}$.

i) For every $y \in A$, we have $ry = \delta(y)r$ in $\mathbb{C} \otimes A$.

j) The element

$$r_0 = \sum_{b \in \mathcal{B}} \text{pf}(b)b$$

possesses the properties required in question i).

9) a) Let n be an integer and M a square matrix with coefficients in $\mathbb{N}$. One assumes that

$\varrho(M^t M) = \varrho(M)^2$ and that $|\varrho(M)| < 2$. There exists an integer $m \geq 2$ such that $\varrho(M) = 2 \cos(\pi/m)$. (Use Kronecker's theorem, cf. A, V, p. 155, exercise 17.)

b) For every integer $m \geq 2$, there exists a square matrix M with coefficients in $\mathbb{N}$ such that

$$\varrho(M) = 2 \cos(\pi/m).$$

c) Let (A, B) be a unital natural basal algebra of finite rank. Let $b \in B$. If $\text{pf}(b) < 2$, then there exists an integer $m \geq 3$ such that $\text{pf}(b) = 2 \cos(\pi/m)$.

Exercises 10 to 24 provide another proof of Corollary 5 of III, p. 100, as well as certain refinements. The following notions will be used there.

Let I be a finite set and let n be its cardinality.

We denote by 1_I the real identity matrix of type (I, I) .

- A square matrix of type (I, I) with real coefficients is said to be positive if all its coefficients are ≥ 0 and strictly positive if all its coefficients are > 0 .

- A square matrix $(a_{ij})_{i,j \in I}$ of type (I, I) with real coefficients is said to be reducible if there exists a partition of I into two nonempty subsets I' and I'' such that $a_{ij} = 0$ for all $(i, j) \in I' \times I''$; it is said to be irreducible if it is not reducible.

- One calls a coordinate subspace of $\mathbf{R}^I$ any vector subspace of $\mathbf{R}^I$ generated by a subset of the vectors of the canonical basis. There are 2^n coordinate subspaces; for each integer $k \in \{0, \dots, n\}$, those among them of dimension k number $\binom{n}{k}$.

A real matrix of type (I, I) is therefore irreducible if and only if the only coordinate subspaces it stabilizes are $\{0\}$ and $\mathbf{R}^I$.

For all x and y in $\mathbf{R}^I$ we will write $x \leq y$ (and $y \geq x$) if $y - x$ is in the cone of elements of $\mathbf{R}^I$ with positive coordinates, that is, if $x_i \leq y_i$ for all $i \in I$. We will write $x < y$ (and $y > x$) if all coordinates of $y - x$ are strictly positive.

10) Let A be a real matrix of type (I, I) , positive and irreducible. For every integer n such that $n \geq \text{Card}(I)$, the matrix $(1_I + A)^{n-1}$ is strictly positive. (Let $y \in \mathbf{R}_+^n$ and $z = (1_I + A)y$; if $y \neq 0$, prove that the set of $i \in I$ such that $z_i = 0$ is strictly contained in $\{i \in I \mid y_i = 0\}$.)

11) Let A be a real matrix of type (I, I) , positive and irreducible. We denote by $\psi \in \mathbf{R}[X]$ the minimal polynomial of A and we set $m = \deg(\psi)$.

For every integer q , set $A^q = (a_{ij}^{(q)})_{i,j \in I}$.

For every $(i, j) \in I^2$, there exists an integer q such that $1 \leq q \leq m$ and $a_{ij}^{(q)} > 0$. (Let r be the remainder of the Euclidean division of $(1 + X)^{n-1}$ by ψ ; consider the matrix $r(A)$ and observe that it is strictly positive.)

12) Let A be a real matrix of type (I, I) , irreducible and positive. For every $x \in \mathbf{R}^I$ non-zero, we set

$$r_x = \inf_{\substack{i \in I \\ x_i \neq 0}} \frac{(Ax)_i}{x_i},$$

where $(Ax)_i = \sum_{k \in I} a_{ik}x_k$ is the coordinate of Ax with index i .

a) If $x \geq 0$ and $x \neq 0$, then r_x is the largest real number ϱ such that $\varrho x \leq Ax$.

b) Suppose that $x \geq 0$ and set $y = (1_I + A)x$. We then have $r_x \leq r_y$.

c) The restriction to $(\mathbf{R}_+^*)^I$ of the function $x \mapsto r_x$ is continuous.

d) Deduce that there exists $z \in \mathbf{R}^I$ such that $z \geq 0$ and such that $r_z = \sup_{x \geq 0} r_x$ (Use compact sets $M = \{x \in \mathbf{R}_+^I \mid \sum_{i \in I} x_i = 1\}$ as well as $(1_I + A)^{n-1}(M)$, then exercise 10.)

We set $r = r_z$. An element z of $\mathbf{R}^I$ such that $z \geq 0$ and $r_z = r$ will be said to be extremal.

e) We have $r > 0$.

f) If z is an extremal vector, then $Az = rz$. (If $u = Az - rz \geq 0$ were nonzero, then $(1_I + A)^{n-1}u$ would be > 0 based on Exercise 10 and, for $x = (1_I + A)^{n-1}z$, one would have $Ax > rx$.)

g) Every extremal element is > 0 .

For every vector y in $\mathbf{C}^I$, we denote by y^+ the vector $(|y_i|)_{i \in I}$; we therefore have $y^+ \in \mathbf{R}^I$ and $y^+ \geq 0$.

h) Let $\alpha \in \mathbf{C}$ be an eigenvalue of A , viewed as a complex matrix, and let y in $\mathbf{C}^I$ be a corresponding eigenvector. We have $|\alpha|y^+ \leq Ay^+$ and $|\alpha| \leq r$.

i) Deduce that if $\alpha = r$ all coordinates of y are nonzero, and then that the eigenspace (in $\mathbf{C}^I$) of A associated with the eigenvalue r has dimension 1.

We denote by $\Delta(X)$ the determinant of $X \cdot 1_I - A$ and by $B(X)$ the transpose of the matrix of cofactors of $X \cdot 1_I - A$. We write $B(X) = (B_{ij}(X))_{i,j \in I}$. We therefore have $B(X)(X \cdot 1_I - A) = (X \cdot 1_I - A)B(X) = \Delta(X) \cdot 1_E$ (A, III, p. 99, prop. 12).

j) For every $i \in I$, the column with index i of $B(r)$ is nonzero. (If this is not the case, construct an eigenvector of A associated with the eigenvalue r whose i -th coefficient is zero, in contradiction with part (f)).

k) All the coefficients of $B(r)$ have the same sign. (First show, using the relation $(r1_I - A)B(r) = 0$ that for all $i \in I$, all the coefficients of the column with index i of $B(r)$ are nonzero and of the same sign, and then reason with the transposes.)

l) One has $\Delta'(r) > 0$ and $B_{ij}(r) > 0$ for all $(i, j) \in I^2$ (Use the formula $\Delta'(X) = \sum_i B_{ii}(X)$ and observe that Δ' does not change sign on the interval $[r, +\infty]$.)

m) Conclude that r is a simple root of the characteristic polynomial of A .

13) Let A (resp. C) be a real (resp. complex) square matrix of type (I, I) . Assume that A is positive and irreducible, and that $C^+ \leq A$, where, similarly to Exercise 12, we denote by C^+ the matrix whose coefficients are the absolute values of the coefficients of C , and we write $C^+ \leq A$ to indicate that the coefficients of $A - C^+$ are positive.

We resume the notation of the preceding exercise.

a) Let $\gamma \in \mathbb{C}$ be an eigenvalue of C and let y in $\mathbb{C}^I$ be an associated eigenvector. We have $|\gamma|y^+ \leq C^+y^+ \leq Ay^+$ and $|\gamma| \leq r_{y^+} \leq r$.

b) We further assume that $|\gamma| = r$. Then y^+ is an extremal vector for A (cf. question f) of Exercise 12), and we have the relations $Ay^+ = C^+y^+ = ry^+$ and $A = C^+$.

Let u (resp. u_j , for $j \in I$) be the unique complex number such that $\gamma = |\gamma|u$ (resp. $y_j = |y_j|u_j$, for $j \in I$). Let us denote by D the square complex matrix of type (I, I) whose diagonal coefficients are $(u_j)_{j \in I}$ and whose other coefficients are zero, so that $y = Dy^+$. Let $F = u^{-1}D^{-1}CD$.

c) One has $Fy^+ = ry^+$ and $F^+y^+ = Fy^+$ (Observe that $F^+ = C^+$.)

d) One has $F = F^+$ and $C = uDAD^{-1}$.

14) Let A be a real square matrix of type (I, I) , positive and irreducible. We resume the notation of Exercise 12.

Let $(\lambda_h)_{h \in H}$ be the family of complex eigenvalues of A which have modulus r , repeated with their multiplicities; we set $v_h = \lambda_h/r$ for all $h \in H$.

Let $h \in H$; by Exercise 13 applied to $C = A$ and $\gamma = \lambda_h$, there exists a matrix $D_{(h)}$ such that $A = v_h D_{(h)} A D_{(h)}^{-1}$ and $D_{(h)}^+ = 1_I$.

- a) Let $z > 0$ be an extremal vector of A . For $h \in H$, the vector $y_{(h)} = D_{(h)} z$ is an eigenvector of A for the eigenvalue λ_h .
- b) Show that for $h \in H$, the eigenvalue λ_h of A is simple. Deduce that the matrices $D_{(h)}$ and the vectors $y_{(h)}$ are uniquely determined up to a scalar factor.
- c) Show that, for all h and k in H^2 , we have $A = u_h u_k^{-1} D_{(h)} D_{(k)}^{-1} A D_{(k)} D_{(h)}^{-1}$ and hence that the vector $D_{(h)} D_{(k)}^{-1} z$ is an eigenvector of A associated with the eigenvalue $u_h u_k^{-1} r$.
- d) Deduce that $\{u_h\}_{h \in H}$ is a subgroup of $\mathbb{C}^*$ and that there exists a family $(\mu_h)_{h \in H}$ of nonzero complex numbers such that $\{\mu_h^{-1} D_{(h)}\}_{h \in H}$ is a subgroup of the group of invertible matrices of type (I, I) with complex coefficients, and that these subgroups are cyclic of order $\text{Card}(H)$. (Choose an index $i \in I$ and normalize $D_{(h)}$ so that its coefficient with row and column index i is equal to 1.)

In the rest of the exercise, we assume that $H = \mathbb{Z}/d\mathbb{Z}$ and that there exists $\omega \in \mathbb{C}^*$ such that $\omega^d = 1$ and such that $u_h = \omega^h$ for $h \in H$. Setting $D = D_{(1)}$, we then have $D^d = 1_I$ and $D_{(h)} = D^h$ for all $h \in H$.

- e) Deduce that $A = \omega D A D^{-1}$.

We consider the partition $(I_h)_{h \in H}$ of I , so that for $i \in I$, the coefficient with indices (i, i) of D is equal to ω^h if and only if $i \in I_h$. For every h , we set $n_h = \text{Card}(I_h)$. The matrix D is written “in blocks”

$$D = \begin{pmatrix} 1_{n_0} & & & \\ & \omega 1_{n_1} & & \\ & & \ddots & \\ & & & \omega^{d-1} 1_{n_{d-1}} \end{pmatrix}$$

and we consider the corresponding decomposition of A :

$$A = \begin{pmatrix} A_{0,0} & A_{0,1} & \dots & A_{0,d-1} \\ A_{1,0} & A_{1,1} & \dots & A_{1,d-1} \\ \vdots & \vdots & \ddots & \vdots \\ A_{d-1,0} & A_{d-1,1} & \dots & A_{d-1,d-1} \end{pmatrix},$$

so that for $h, k \in H$, $A_{h,k}$ is a matrix of type (I_h, I_k) .

f) For all $(h, k) \in H$, we have $\omega A_{h,k} = \omega^{k-h} A_{h,k}$. Deduce that $A_{h,k} = 0$ if $k - h \neq 1$.

15) We resume the hypotheses and notation of Exercise 12. Let $\Delta_1(X)$ be the greatest common divisor of the coefficients of $B(X)$; the polynomial $\Delta_1(X)$ is assumed to be monic.

a) The polynomial $\Delta_1(X)$ divides $\Delta(X)$ and $\Delta_1(r) \neq 0$.

b) Deduce that the real roots of $\Delta_1(X)$ are strictly smaller than r . Conclude that $\Delta_1(r) > 0$.

c) Set $C(X) = \frac{1}{\Delta_1(X)} B(X)$. This is a matrix with coefficients in $\mathbf{R}[X]$. Show that $C(r) > 0$.

16) Let A be a real square matrix of type (I, I) , positive and irreducible. For all $j \in I$ set

$$s_j = \sum_{k \in I} a_{jk},$$

$$s = \inf_{j \in I} s_j, \quad \text{and} \quad S = \sup_{j \in I} s_j.$$

We also adopt the notation r and $B(\mathbf{X})$ of Exercise 12.

a) One has $\sum_{j \in I} (r - s_j) B_{kj}(r) = 0$

b) Deduce that $s \leq r \leq S$ and that the equality case in either of these two inequalities implies that $s = S$.

17) Let A be a real square matrix of type (I, I) , positive and irreducible. Let $y \in \mathbf{R}^I$ be an eigenvector of A associated with an eigenvalue α such that $y \geq 0$. We then have $\alpha = r$ (with the notations of Exercise 12) and $y > 0$. (Consider an extremal vector u for ${}^t A$ and compute $\langle y | {}^t A u \rangle = \langle A y | u \rangle$.)

- 18) Let A be a real square matrix of type (I, I) , positive and irreducible. Let ϱ be the spectral radius of A . For every $x \in R^I$ such that $x \geq 0$ and $x \neq 0$ we set

$$\varrho^x = \sup_{i \in I} \frac{(Ax)_i}{x_i},$$

(with the convention $\varrho^x = +\infty$ if there exists an index i such that $x_i = 0$ and $(Ax)_i \neq 0$).

Let

$$\varrho' = \inf_{\substack{x \geq 0 \\ x \neq 0}} \varrho^x.$$

Following an approach similar to that of Exercise 12, show that ϱ' is an eigenvalue of A and that $\varrho' = \varrho$.

- 19) Let A be a real square matrix of type (I, I) , positive and irreducible; let ϱ be its spectral radius. Let $z \in R^I$ such that $z \geq 0$ and $z \neq 0$.

If $\varrho z \leq Az$ (resp. $\varrho z \geq Az$), then $\varrho z = Az$ and $z > 0$.

- 20) Let A be a real square matrix of type (I, I) , positive but not necessarily irreducible; let ϱ be the spectral radius of A . We keep the notations $B(X)$ and $C(X)$ from the preceding exercises.

- a) The real number ϱ is an eigenvalue of A and there exists at least one eigenvector ≥ 0 associated with ϱ .
- b) For every real number $\lambda \geq r$, the matrices $B(\lambda)$ and $B'(\lambda)$ are positive. (For the second matrix, one may proceed by induction on the cardinality of I .)
- c) The matrix $C(\lambda)$ is positive for all $\lambda \geq r$.
- d) If A is irreducible, then for every real number $\lambda > r$, the matrices $B(\lambda)$, $C(\lambda)$ and $(\lambda 1_I - A)^{-1}$ are strictly positive.
- e) Let J be a subset of I different from I . Let ϱ_J be the spectral radius of the submatrix $(a_{ij})_{i,j \in J}$. We have $\varrho_J \leq \varrho$ and this inequality is strict if A is irreducible. Conversely, if

A is reducible, there exists a subset $J \subset I$ different from I , such that $\varrho_J = \varrho$.

f) Let J be a subset of I different from I and such that the spectral radius of the submatrix $(a_{i,j})_{i,j \in J}$ is equal to ϱ . For every subset K such that $J \subset K \subset I$, the spectral radius of the matrix $(a_{i,j})_{i,j \in K}$ is equal to ϱ . Deduce that $A > 0$ if and only if all the diagonal coefficients of $B(r)$ are strictly positive.

g) For every real number $\lambda > \varrho$ and for every subset J of I such that $J \neq I$ the determinant of the matrix $(\lambda \delta_{i,j} - a_{i,j})_{i,j \in J}$ is strictly positive.

h) Let λ be a real number; we assume that the determinant of $(\lambda \delta_{i,j} - a_{i,j})_{i,j \in J}$ is strictly positive for every subset $J \subset I$ different from I . We then have $\lambda > \varrho$.

In the following questions, we assume that I is the interval $[1, n]$ in $\mathbf{N}$. i) Let $(g_{i,j})_{1 \leq i,j \leq n}$ be a real matrix such that $g_{i,j} \leq 0$ for all $i \neq j$ and such that the determinants of the matrices $(g_{i,j})_{1 \leq i,j \leq m}$ are strictly positive for every integer m such that $1 \leq m \leq n$. Then we have $\det(g_{i,j})_{i,j \in J} > 0$ for every subset J of I .

j) Let λ be a real number; we suppose that for every integer m such that $1 \leq m \leq n$, the determinant $\det(\lambda \delta_{i,j} - a_{i,j})_{1 \leq i,j \leq m}$ is strictly positive. Then $\lambda > \varrho$.

k) Let $C = (c_{i,j})_{1 \leq i,j \leq n}$ be a real matrix such that $c_{i,j} \geq 0$ for all $i \neq j$. If $(-1)^k \det(c_{i,j})_{1 \leq i,j \leq m}$ is strictly positive for every integer m such that $1 \leq m \leq n$, then the real parts of the eigenvalues of C (viewed as a complex matrix) are negative.

21) Let A be a real square matrix of type (I, I) , positive; let ϱ be its spectral radius.

a) Prove that there exist integers p and q such that $0 \leq q \leq p$ and a partition $(I_s)_{1 \leq s \leq p}$ of I into nonempty parts such that the corresponding block decomposition $A = (A_{s,t})_{1 \leq s,t \leq p}$ has the following property:

- The matrices $A_{s,s}$ are irreducible;
- We have $A_{s,t} = 0$ if $t > s$;
- We have $A_{s,t} = 0$ if $1 \leq t < s \leq q$;
- For every integer s such that $q \leq s \leq p$, there exists t such that $A_{s,t} \neq 0$.

- b) Determine all possibilities for such a partition. (In particular, prove that the integers q and p are uniquely determined, as are the sets $\{I_1, \dots, I_q\}$ and the sets $\{I_{q+1}, \dots, I_p\}$.)
- c) There exists an eigenvector $z > 0$ of A associated with the eigenvalue ϱ if and only if, for every integer $s \in [1, p]$ the spectral radius of $A_{s,s}$ is equal to ϱ if $s \leq q$ and is strictly less than ϱ otherwise.
- d) It is necessary and sufficient for both A and tA to admit a strictly positive extremal vector that there exists a partition as above such that $A_{s,t} = 0$ for $1 \leq t < s$ and $q < s \leq p$ and such that the matrices $A_{s,s}$ for $s \leq q$ have spectral radius equal to ϱ .
- e) If A and tA both admit a strictly positive extremal vector, and if ϱ has spectral multiplicity equal to 1, then A is irreducible.
- 22) Let A be a real square matrix of type (I, I) , positive and irreducible; let ϱ be its spectral radius. Let $d \geq 1$ be the integer determined by Exercise 14. We shall say that A is primitive if $d = 1$. Let $\Delta(X)$ be the polynomial $\det(X \cdot 1_I - A)$.
- a) We write $\Delta(X) = X^n + a_1 X^{n_1} + \dots + a_t X^{n_t}$, where $a_1, \dots, a_t$ are nonzero real numbers and $n_1, \dots, n_t$ natural integers such that $n > n_1 > \dots > n_t$. Prove that d is the greatest common divisor of $n - n_1, n_1 - n_2, \dots, n_{t-1} - n_t$.
- b) There exists a nonzero integer m and a polynomial g such that $\Delta(X) = g(X^d)X^m$.
- c) If there exists an integer $p > 0$ such that A^p is strictly positive, then A is primitive.
- d) Suppose that A is primitive. Let us introduce the matrices $C(X)$ of Exercise 15, and let us set, using the notation of that exercise, $X(X) = \Delta(X)/\Delta_1(X)$. The limit of the sequence $(A^k/\varrho^k)_{k \in \mathbb{N}}$ is equal to $C(\varrho)/X'(\varrho)$ (One may use a decomposition of A into Jordan blocks.) Using the results of Exercise 15, deduce that there exists an integer $p > 0$ such that A^p is strictly positive.
- e) Suppose there exists an integer $q > 0$ such that A^q is reducible. Let p be the greatest common divisor of q and d . Prove that there exists a partition $(I_s)_{1 \leq s \leq p}$ of I into nonempty

parts for which A^q is block diagonal, with irreducible diagonal blocks and spectral radius ϱ^q .

f) If A is primitive, then A^p is irreducible for every $p > 0$.

g) There exists a partition $(I_s)_{1 \leq s \leq d}$ of I into nonempty parts for which A^d is block diagonal, with primitive diagonal blocks and spectral radius equal to ϱ^d .

23) Let P be a real square matrix of type (I, I) ; we say that P is stochastic if it is nonnegative and if the vector $(1, \dots, 1) \in \mathbf{R}^I$ is an eigenvector of P associated with the eigenvalue 1.

Let A be a real square matrix of type (I, I) . Prove that A is positive and admits a strictly positive extremal vector if and only if there exists a stochastic matrix P , a real number $r > 0$ and a diagonal matrix Z with strictly positive diagonal entries such that $A = rPZ^{-1}$. In this case the spectral radius of A is equal to r and an extremal vector of A is $Z(1, \dots, 1)$.

24) a) Let P be a real square stochastic matrix of order n . The real number 1 is a simple root of the minimal polynomial of P . (One may use Exercise 20.) Moreover, every root of modulus 1 of the minimal polynomial of P is simple.

b) Let A be a positive real matrix admitting an extremal vector $z > 0$ and of spectral radius ϱ . All roots of modulus ϱ of the minimal polynomial of A are simple.

In Exercises 25 to 30, we consider a real Fréchet space E and a closed convex cone C in E , assumed to have nonempty interior. The order relation on E induced by C is denoted $\preceq_C$. Thus, for x, y in E , we have $x \succeq_C y$ if and only if $x - y \in C$. We will write $x \succ_C y$ for the relation $x - y \in \overset{\circ}{C}$.

We consider a continuous map $u: E \rightarrow E$ satisfying the following conditions:

(i) The map u is positively homogeneous: we have $u(\lambda x) = \lambda u(x)$ for every real number $\lambda \geq 0$;

(ii) The map u is increasing with respect to C , that is, $x \succeq_C y$ implies $u(x) \succeq_C u(y)$;

(iii) For every $x \in C - \{0\}$, we have $u(x) \in \overset{\circ}{C}$.

The goal of these exercises, taken from the course of P.-L. Lions at the Collège de France, 2020/2021, is to prove, under certain conditions, that there exists an element $x \in C - \{0\}$ and $\lambda > 0$ such that $x = \lambda u(x)$; this provides in particular another proof of the Krein-Rutman theorem (III, p. 94, th. 4 and III, p. 97, prop. 8).

For $f \in \mathring{C}$ and $\lambda \in \mathbf{R}_+$, one calls an upper solution (resp. lower solution) of the equation $y = \lambda u(y) + f$ an element $y \in C$ such that $y \geq_C \lambda u(y) + f$ (resp. such that $y \leq_C \lambda u(y) + f$). We denote by U_f the set of $\lambda \in \mathbf{R}_+$ such that the set of upper solutions of the equation $y = \lambda u(y) + f$ is nonempty.

25) a) Prove that for every $x \in C$ and every $y \in \mathring{C}$, there exists a real number $\lambda \geq 0$ such that $x \leq_C \lambda y$.

b) The set U_f is an interval that contains 0.

We introduce the following conditions on the map u .

(iv) For every $\lambda \in \mathbf{R}_+$, every $f \in \mathring{C}$, every sequence (x_n) of elements of C that satisfies $x_{n+1} = \lambda u(x_n) + f$ for all n and that is either increasing and bounded above, or decreasing, is convergent;

(v) The image under u of every bounded subset of E is relatively compact.

(vi) For every map $\lambda \mapsto \varepsilon_\lambda$ from U_f into C such that $\lim_{\lambda \rightarrow \lambda_m} \varepsilon_\lambda = 0$, the set

$$\{z \in C \mid \exists \lambda \in U_f \text{ such that } z = \lambda u(z) + \varepsilon_\lambda\}$$

is relatively compact in E .

26) We resume the preceding notations and hypotheses and we suppose in addition that condition (iv) is satisfied.

a) Let $\lambda \in U_f$. Let us define a sequence $(x_n)_{n \in \mathbf{N}}$ by $x_0 = 0$ and $x_{n+1} = \lambda u(x_n) + f$ for all $n \in \mathbf{N}$. It is increasing and bounded above; we have $x_n \in \mathring{C}$ for $n \geq 1$.

b) By condition (iv), the sequence (x_n) converges; let $x(\lambda)$ denote its limit. Prove that we have $x(\lambda) \in \mathring{C}$ and $x(\lambda) = \lambda u(x(\lambda)) + f$.

c) Let $\lambda \in U_f$ and let $y \in C$ such that $y \succcurlyeq_C \lambda u(y) + f$; the sequence defined by $x_0 = y$,
 $x_{n+1} = \lambda u(x_n) + f$ is decreasing, with values in C , and its limit x satisfies the equation
 $x = \lambda u(x) + f$.

d) The upper bound of U_f in $\mathbf{R}_+$ does not belong to U_f .

e) One has $U_g = U_f$ for all $g \in \mathring{C}$ (For every $\lambda \in U_f$, let $x \in C$ be such that $x \succcurlyeq_C \lambda u(x) + f$; then there exists $M > 0$ such that $y = Mx$ satisfies $y \succcurlyeq_C \lambda u(y) + g$.) f) Let $\lambda \leq \mu$ be real numbers. Let x be a lower solution of the equation $y = \lambda u(y) + f$ and let x' be an upper solution of the equation $y = \mu u(y) + f$. We then have $x \leq_C x'$. (Show that the set of $\varepsilon \in [0, 1]$ such that $\varepsilon x \leq_C x'$ coincides with $[0, 1]$.)

f) If C is salient, then for every $\lambda \in U_f$, there exists a unique element $x \in \mathring{C}$ such that
 $x = \lambda u(x) + f$.

h) Let $\mu \in \mathbf{R}_+$ such that $f \leq_C \mu u(f)$. We then have $\mu \notin U_f$. (The sequence $(x_n)_{n \in \mathbf{N}}$ defined by $x_0 = 0$ and $x_{n+1} = \mu u(x_n) + f$ satisfies $x_{n+1} - x_n \succcurlyeq_C f$ for all $n \in \mathbf{N}$.) In particular, the interval U_f is bounded. Henceforth we denote by λ_m the supremum of U_f .

i) Let $z \in C - \{0\}$ and $\mu \in \mathbf{R}_+$ be such that $\mu u(z) \succcurlyeq_C z$. We then have $\lambda_m \leq \mu$, and hence

$$\lambda_m \leq \inf_{z \in C - \{0\}} \inf\{\mu \mid z \leq \mu u(z)\}.$$

j) The map $\lambda \mapsto x(\lambda)$ (where $x(\lambda)$ is defined in the question a) is increasing.

k) The map $\lambda \mapsto x(\lambda)$ has no limit when λ tends to λ_m . More precisely, if $(\lambda_n)_n$ is a sequence of U_f which tends to λ_m , then the sequence $(x(\lambda_n))_n$ is not convergent.

27) We assume that hypotheses (i) through (v) are satisfied.

a) Prove that the set of $x(\lambda)$, for $\lambda \in U_f$, is not bounded.

b) Let d be a metric on E that defines its topology. For $\lambda \in \mathbf{R}_+$, we define $\hat{x}(\lambda) = d(0, x(\lambda))^{-1} x(\lambda)$. Prove that the map $\hat{x}$ is bounded and that the set of values of $\hat{x}$ has a cluster value x_m such that

(i) $d(0, x_m) = 1$,

(ii) $x_m = \lambda_m u(x_m)$,

(iii) $x_m \in \overset{\circ}{C}$.

28) We assume in this exercise that conditions (i) through (iv) are satisfied. a) We fix $x_0 \in \overset{\circ}{C}$. For $\lambda \in U_f$, let $A(\lambda) = \inf\{M \in \mathbf{R}_+ \mid x(\lambda) \leq_C Mx_0\}$. Let $C_0 \in \mathbf{R}_+$ be such that $f \leq_C C_0 u(x_0)$. For every λ in U_f , the interval $[\lambda, \lambda + C_0/A(\lambda)[$ is contained in U_f (cf. exercise 26, question d)).

b) Hence deduce that the function $\lambda \mapsto A(\lambda)$ tends to $+\infty$ as $\lambda \in U_f$ tends to λ_m .

For every λ in U_f , we define $\hat{x}(\lambda) = x(\lambda)/A(\lambda)$.

We now assume that condition (vi) is satisfied.

c) Prove that the set of values of the mapping $\hat{x}$ has a cluster value x_m such that

$$0 \leq_C x_m \leq_C x_0 \quad \text{and} \quad x_m = \lambda_m u(x_m).$$

d) Prove that $x_m \neq 0$. (If one had $x_m = 0$, let U be a neighborhood of 0 in E such that $x_0 + U \subset C$; for every $\mu \geq 0$, there would exist a sequence (λ_n) in U_f converging to λ_m such that $x_0 - \mu \hat{x}(\lambda_n) \geq_C 0$, contradicting the definition of $A(\lambda)$.)

e) Let $y \in C - \{0\}$ be such that $y = \lambda_m u(y)$. Set $\theta = \sup\{\gamma \mid \gamma y \leq_C x_m\}$. We have $\theta \in \mathbf{R}_+^*$ and $x_m = \theta y$. (We have $\theta y \leq_C x_m$ and $\theta y = \lambda_m u(\theta y)$; if $x_m \neq \theta y$, then we would have $x_m - \theta y \geq_C 0$ and there would exist $\varepsilon > 0$ such that $x_m - \theta y \geq_C \varepsilon y$, contradicting the definition of θ .)

f) Let $y \in C - \{0\}$ and $\mu \in R$ be such that $y = \mu u(y)$. We then have $\mu = \lambda_m$. (Note that $0 < \mu$ and $\mu \geq \lambda_m$; if we had $\mu > \lambda_m$, let $\theta = \sup\{\gamma \mid \gamma x_m \leq_C y\}$; we would then have $\theta \in R_+^*$, then $\theta x_m \leq_C y$, whence $\theta x_m = \lambda_m u(\theta x_m) \leq_C \mu u(\theta x_m) \leq_C \mu u(y) = y$, contradicting the definition of θ .)

29) We resume the notation of the preceding exercise; in particular, we assume that C has

nonempty interior. We assume that u is a linear map and that conditions (i), (ii), and (iii) are satisfied.

a) If u is a compact endomorphism of E , then conditions (iv) and (v) are satisfied.

We assume that conditions (iv) and (vi) are satisfied. Let (λ_m, x_m) be given by the preceding exercise.

b) For every $\lambda \in \mathbf{R}_+$ such that $\lambda < \lambda_m$, the endomorphism $1_E - \lambda u$ is invertible. (First show that $1_E - \lambda u$, induced by passage to subspaces, is a bijection of $\mathring{C}$, cf. questions *a*) and *f*) of the preceding exercise; then deduce that

$1_E - \lambda u$ is surjective, and then that it is injective by writing $x_1 - x_2$ as an element of the kernel of $1_E - \lambda u$, with $x_i \in \mathring{C}$, and verifying that there exists $z \in E$ such that $x_1 + z - \lambda u(x_1 + z) \in \mathring{C}$.

c) Let y in E be such that $y = \lambda_m u(y)$. We then have $y \in C$. (There exists c such that $cx_m + y \succ_C 0$; conclude using part (e) of the preceding exercise.)

d) Let $C^\circ \subset E'$ be the polar of C . For every $x \in \mathring{C}$ and for every $\xi \in C^\circ - \{0\}$, we have $\langle x, \xi \rangle > 0$.

e) The polar C° of C is preserved by ${}^t u: E' \rightarrow E'$; for every $\lambda \in \mathbf{R}_+$ such that $\lambda < \lambda_m$, the endomorphism $1_{E'} - \lambda {}^t u$ of E' is an isomorphism.

30) We resume the hypotheses of the preceding exercise, so that u is linear, compact, and conditions (i) to (v) are assumed to be satisfied. The endomorphism ${}^t u$ is also compact.

a) Let $\eta \in C^\circ - \{0\}$. For every $\lambda \in \mathbf{R}_+$ such that $\lambda < \lambda_m$, the unique solution $\xi(\lambda)$ of the equation $\xi - \lambda {}^t u(\xi) = \eta$ belongs to C° . (If $x \in C$, let $y \in C$ be such that $y - \lambda u(y) = x$; we have $\langle y, \eta \rangle = \langle x, \xi \rangle$.)

b) Let $x_0 \in \mathring{C}$. The map $\lambda \mapsto \langle x_0, \xi(\lambda) \rangle$ tends to infinity when λ tends to λ_m . Prove that the set of values $\xi(\lambda) / \langle x_0, \xi(\lambda) \rangle$ has a cluster value ξ_m such that:

(i) $\xi_m \in C^\circ$;

(ii) $\langle x_0, \xi_m \rangle = 1$;

(iii) $\xi_m = \lambda_m^t u(\xi_m)$.

- c) Let $f \in E$ such that $\langle f, \xi_m \rangle = 0$. For $\lambda \in R_+$ such that $\lambda < \lambda_m$, we denote by $x(\lambda)$ the unique solution x in E of the equation $x - \lambda u(x) = f$. Show that the set of $x(\lambda)$ is bounded in E (let d be a metric defining the topology of E ; if there existed a sequence $\lambda_n \rightarrow \lambda_m$ such that $x(\lambda_n)$ tends to infinity, set $\hat{x}_n = x(\lambda_n)/d(0, x(\lambda_n))$, and show that the sequence $(\hat{x}_n)$ converges to an element $\hat{x}$ such that $d(0, \hat{x}) = 1$, $\hat{x} = \lambda u(\hat{x})$ and $\langle \hat{x}, \xi_m \rangle = 0$, which contradicts the fact that one must have $\hat{x} \in \overset{\circ}{C}$).
- d) There exists a unique element $x \in E$ such that $f = x - \lambda_m u(x)$ and $\langle x, \xi_m \rangle = 0$. (For existence, show using the preceding question that there exists a sequence (λ_n) converging to λ_m such that $(x(\lambda_n))$ converges to an element x satisfying the required conditions.)
- e) The endomorphism $1_{E'} - \lambda_m^t u$ of E' induces, by passage to subspaces, a bijection of the polar x_m° of x_m .
- f) The eigenspace $\text{Ker}(1_E - \lambda_m u)$ is of dimension 1 and is equal to the polar of the image of $1_E - \lambda_m u$.

CHAPTER IV

Chapter IV

Hilbertian spectral theory

For every vector space E , we denote by 1_E the identity map of E . If E , F and G are vector spaces and if $v: E \rightarrow F$ and $u: F \rightarrow G$ are linear maps, the linear map $u \circ v$ from E to G will sometimes be denoted uv .

Let E be a pre-Hilbert space. We denote by $\langle x|y \rangle_E$, or simply $\langle x|y \rangle$, the inner product on E . The orthogonal of a subset A of E is denoted A° . It is a closed subspace of E ; it is reduced to 0 if and only if A is total in E (EVT, V, p. 16, cor. 1). If $F \subset E$ is a vector subspace of E , then $F^{\circ\circ} = \overline{F}$ (cf. EVT, V, p. 13). The spectrum of an endomorphism of E is always relative to the Banach algebra $\mathcal{L}(E)$.

Let F be a Hilbert space. We denote by $\mathcal{L}_1(E)$ the space of endomorphisms of finite trace of E (EVT, V, p. 50, def. 8) and $\mathcal{L}_2(E; F)$ the space of Hilbert-Schmidt maps from E to F (EVT, V, p. 51, def. 9). The map $u \mapsto \|u\|_2 = (\text{Tr}(u^*u))^{1/2}$ is a Hilbertian norm on $\mathcal{L}_2(E; F)$ (EVT, V, p. 52, th. 1, (i)).

The space $\mathcal{L}^f(E; F)$ of continuous linear maps of finite rank is dense in $\mathcal{L}^c(E; F)$ (prop. 10 of III, p. 15 and cor. of prop. 11 of III, p. 16). In particular, the space $\mathcal{L}_2(E; F)$ is dense in $\mathcal{L}^c(E; F)$.

The Hilbertian tensor product of E and F (EVT, V, p. 28, def. 1) is denoted $E \widehat{\otimes}_2 F$.

If X is a locally compact topological space and μ a measure on X , we denote by $\mathcal{L}^p(X, \mu) = \mathcal{L}_C^p(X, \mu)$ and $L^p(X, \mu) = L_C^p(X, \mu)$ for $p \in [1, +\infty]$.

IV.1 COMPACT OPERATORS ON A HILBERT SPACE

In this section, K is a field equal to $\mathbb{R}$ or $\mathbb{C}$ and E denotes a Hilbert space over the field K .

IV.1.1 Diagonal endomorphisms

DEFINITION 1. — Let $B = (e_i)_{i \in I}$ be an orthonormal basis of E . An endomorphism u is said to be diagonal in the basis B or diagonal with respect to B if there exists a family $\lambda = (\lambda_i)_{i \in I}$ of elements of K such that $u(e_i) = \lambda_i e_i$ for all $i \in I$.

Let $B = (e_i)_{i \in I}$ be an orthonormal basis of E . We denote by $\mathcal{D}_B(E)$ the set of endomorphisms of E that are diagonal with respect to B . This is a closed, unital commutative subalgebra of $\mathcal{L}(E)$. Let $u \in \mathcal{D}_B(E)$. The family $\lambda = (\lambda_i)_{i \in I}$ such that $u(e_i) = \lambda_i e_i$ is determined uniquely by u and by the basis B , and it is called the family of eigenvalues of u relative to B .

Suppose that $K = \mathbb{R}$ and identify E with a subspace of $E_{(\mathbb{C})}$. Let $B = (e_i)_{i \in I}$ be an orthonormal basis of E . Denote by $B_{(\mathbb{C})}$ the orthonormal basis $(1 \otimes e_i)_{i \in I}$ of $E_{(\mathbb{C})}$ (cf. EVT, V, p. 29, cor. 1). The map $u \mapsto u_{(\mathbb{C})}$ induces an injective morphism of algebras from $\mathcal{D}_B(E)$ into $\mathcal{D}_{B_{(\mathbb{C})}}(E_{(\mathbb{C})})$; its image is the set of $u \in \mathcal{D}_{B_{(\mathbb{C})}}(E_{(\mathbb{C})})$ such that $u(E) \subset E$; in other words, the set of diagonal endomorphisms u in the basis $B_{(\mathbb{C})}$ whose eigenvalues are real.

For every $i \in I$, we denote by p_i the orthogonal projector of E with image Ke_i . We have $\|p_i\| = 1$ for all $i \in I$. The endomorphism p_i is diagonal in the basis B , and the family of its eigenvalues is the family $(\delta_{ij})_{j \in I}$ (Kronecker symbol, cf. A, II, p. 24).

Let $B' = (f_i)_{i \in I}$ be an orthonormal basis of E and $u : E \rightarrow E$ the isometric isomorphism such that $u(f_i) = e_i$. Then $\mathcal{D}_{B'}(E) = u^{-1} \mathcal{D}_B(E) u$.

We endow I with the discrete topology and denote by $\mathcal{B}(I) = \mathcal{C}_b(I; K)$ the unital Banach algebra of bounded functions on I with values in K (example 2 of I, p. 17); if $K = \mathbb{C}$, it is a star algebra (example 2 of I, p. 102).

Lemma 1. — Let $u \in \mathcal{D}_B(E)$ and let $\lambda = (\lambda_i)_{i \in I}$ be the family of its eigenvalues. The family λ is bounded and one has $\sup_i |\lambda_i| = \|u\|$.

One has $|\lambda_i| \leq \|u\|$ for all i , whence $\sup_{i \in I} |\lambda_i| \leq \|u\|$. Since moreover

$$\|u(x)\|^2 = \sum_{i \in I} |\lambda_i|^2 |\langle e_i | x \rangle|^2 \leq \left(\sup_{i \in I} |\lambda_i|^2 \right) \|x\|^2$$

for all $x \in E$ (EVT, V, p. 22, prop. 5), it follows that $\|u\| = \sup |\lambda_i|$.

PROPOSITION 1. — a) The map α from $\mathcal{D}_B(E)$ into $\mathcal{B}(I)$ that associates with a diagonal endomorphism u the family of eigenvalues of u is an isometric isomorphism of Banach algebras over K . If $K = \mathbb{C}$, it is a morphism of involutive algebras.

- b) For any bounded family $\lambda = (\lambda_i)_{i \in I}$ the family $(\lambda_i p_i)_{i \in I}$ is summable in the space $\mathcal{L}(E)$ endowed with the topology of pointwise convergence, and the sum of this family is the unique mapping $u \in \mathcal{D}_B(E)$ such that $\alpha(u) = \lambda$;
- c) Let $u \in \mathcal{D}_B(E)$ and let λ be the family of its eigenvalues relative to B . The spectrum of u is the closure in K of the set of values of λ ;
- d) If $K = \mathbb{C}$, then the continuous functional calculus map of $\mathcal{C}(\text{Sp}(u))$ in $\mathcal{L}(E)$ associates with a continuous function $f \in \mathcal{C}(\text{Sp}(u))$ the endomorphism $\alpha(f \circ \lambda)$ in $\mathcal{D}_B(E)$.

By Lemma 1, the family of eigenvalues of an endomorphism diagonal in B is bounded, and the map from $\mathcal{D}_B(E)$ into $\mathcal{B}(I)$ thus defined is a continuous isometric morphism of unital Banach algebras.

Let $i \in I$. For every $j \in I$, we have $\langle u^*(e_i) | e_j \rangle = \lambda_j \langle e_i | e_j \rangle = \langle \bar{\lambda}_j e_i | e_j \rangle$. It follows that $u^*(e_i) = \bar{\lambda}_i e_i$. The adjoint of u is therefore the diagonal endomorphism in the basis B for which $(\bar{\lambda}_i)_{i \in I}$ is the family of eigenvalues. Assertion a) follows from this.

Let $\lambda = (\lambda_i)_{i \in I} \in \mathcal{B}(I)$. For every $x \in E$ the family $(|\langle e_i | x \rangle|^2)_{i \in I}$ is summable, with sum $\|x\|^2$ (EVT, V, p. 22, prop. 5), whence, for every finite subset J of I :

$$\begin{aligned} \left\| \sum_{i \in J} \lambda_i p_i(x) \right\|^2 &= \sum_{i \in J} |\lambda_i|^2 |\langle e_i | x \rangle|^2 \leq \left(\sup_{i \in I} |\lambda_i|^2 \right) \sum_{i \in J} |\langle e_i | x \rangle|^2 \\ &\leq \left(\sup_{i \in I} |\lambda_i| \right)^2 \|x\|^2. \end{aligned}$$

Consequently, the family $(\lambda_i p_i)$ is summable in $\mathcal{L}(E)$ endowed with the topology of simple convergence. Its sum u_λ satisfies $u_\lambda(e_i) = \lambda_i e_i$; it is therefore a diagonal endomorphism in the basis B , with eigenvalues λ . Assertion b) follows from this.

The last assertions follow from a), from example 3 of I, p. 17 and from example 4 of I, p. 111.

Remark. — The Banach algebra $\mathcal{D}_B(E)$ is a maximal commutative closed subalgebra of $\mathcal{L}(E)$. Indeed, let u be an endomorphism of E that commutes with $\mathcal{D}_B(E)$. Let $i \in I$. Since the orthogonal projector p_i is diagonal in the basis B , we have $p_i(u(e_i)) = u(p_i(e_i)) = u(e_i)$, which implies that $u(e_i)$ is proportional to e_i . Thus u is diagonal in the basis B .

If E is infinite-dimensional, there exist maximal commutative involutive subalgebras of $\mathcal{L}(E)$ that are not isomorphic to $\mathcal{D}_B(E)$ (exercise 5 of IV, p. 314).

PROPOSITION 2. — Let $u \in \mathcal{D}_B(E)$ and $\lambda = (\lambda_i)_{i \in I}$ be the family of its eigenvalues.

a) The following conditions are equivalent:

- (i) One has $\lambda \in \mathcal{C}_0(I; K)$;
 - (ii) The endomorphism u is compact;
 - (iii) The family $(\lambda_i p_i)_{i \in I}$ is summable in the Banach space $\mathcal{L}(E)$. Its sum is then equal to u .
- b) Suppose that u is compact and let Λ denote the set of values of λ . The set of $i \in I$ such that $\lambda_i \neq 0$ is countable. If E is infinite-dimensional, we have $\text{Sp}_s(u) = \Lambda - \{0\}$ and $\text{Sp}(u) = \Lambda \cup \{0\}$. If E is finite-dimensional, then $\text{Sp}_s(u) = \text{Sp}(u) = \Lambda$.

By Lemma 1, we have $\|\sum_{j \in J} \lambda_j p_j\| = \sup_{j \in J} |\lambda_j|$ for every finite subset J of I . It follows, by the Cauchy criterion, that the family $(\lambda_i p_i)$

is summable in $\mathcal{L}(E)$ if and only if the family λ tends to 0 at infinity, which implies that conditions (i) and (iii) are equivalent.

Condition (iii) implies that u is compact (corollary of proposition 2 of III, p. 4). Conversely, suppose that the endomorphism u is compact. Since the family B is bounded in E , its image under u in E is relatively compact in E (III, p. 2), hence precompact. Let $\varepsilon > 0$ and let J be a finite subset of I such that the image of B under u is contained in the union of balls of radius ε and centers $u(e_j)$ for $j \in J$. Let $i \in I - J$. There exists $j \in J$ such that $\|u(e_i) - u(e_j)\| \leq \varepsilon$, hence

$$|\lambda_i|^2 \leq |\lambda_i|^2 + |\lambda_j|^2 = \|u(e_i) - u(e_j)\|^2 \leq \varepsilon^2.$$

Consequently, we have $\lambda \in \mathcal{C}_0(I; K)$, that is, condition (i).

Finally, the spectrum of u and its sensitive spectrum are computed in terms of λ using prop. 1, c) and proposition 5 of III, p. 90.

Remark. — When I is infinite, condition $\lambda \in \mathcal{C}_0(I; K)$ can also be stated as “the family λ tends to 0 along the filter of complements of finite subsets of I .”

IV.1.2 Diagonalization of compact endomorphisms

THEOREM 1. — Suppose $K = \mathbb{C}$. Let u be a compact and normal endomorphism of E . There exists an orthonormal basis B of E such that u is diagonal in the basis B .

The set $\text{Sp}_s(u)$ is countable, and it does not contain 0 if E is infinite-dimensional (III, p. 90, prop. 5, b). For every element $\lambda \in \text{Sp}_s(u)$, denote by N_λ the nilspace of $u - \lambda 1_E$. It is finite-dimensional (loc. cit.) and, since u is normal, it coincides with the eigenspace of u relative to λ (EVT, V, p. 43, cor. of prop. 8). The spaces N_λ are pairwise orthogonal (I, p. 132, n° 5).

Let F be the subspace of E that is the Hilbert sum of the spaces N_λ for $\lambda \in \text{Sp}_s(u) - \{0\}$. It is a space of countable type, stable under u since each subspace N_λ is stable under u . Since N_λ is also the eigenspace of u^* relative to $\bar{\lambda}$ (EVT, V, loc. cit.), the endomorphism u induces an endomorphism $\tilde{u}$ of F° by passing to subspaces. The endomorphism $\tilde{u}$ is compact (prop. 3 of III, p. 5) and normal (lemma 4 of I, p. 135). By construction, the sensitive spectrum of $\tilde{u}$ is contained in $\{0\}$ (indeed, every eigenvector for $\tilde{u}$ would be one for u , hence would belong to one of the spaces N_λ if the corresponding eigenvalue were nonzero). Thus the spectral radius of $\tilde{u}$ is zero, whence $\tilde{u} = 0$ since $\tilde{u}$ is normal (cor. 1 of I, p. 108). We therefore have $F^\circ \subset \text{Ker}(u)$.

For every $\lambda \in \text{Sp}_s(u)$, let B_λ be an orthonormal basis of N_λ and let $(e_j)_{j \in J}$ be the family formed by the union of the B_λ ; it is an orthonormal basis of F . Let B be the union of $(e_j)_{j \in J}$ and of an orthonormal basis of F° . This is an orthonormal basis of E and u is diagonal in the basis B .

COROLLARY 1. — Let u be a compact Hermitian endomorphism of E . There exists an orthonormal basis B of E such that u is diagonal in the basis B and its eigenvalues are real.

If $K = \mathbb{C}$, this follows immediately from the theorem. Suppose $K = \mathbb{R}$. The space E is an $\mathbb{R}$ -structure on $E_{(\mathbb{C})}$ (cf. A, II, p. 119). The endomorphism $u_{(\mathbb{C})}$ of $E_{(\mathbb{C})}$ is compact (remark 4 of III, p. 2) and Hermitian. Let $B = (e_j)_{j \in J}$ be an orthonormal basis of $E_{(\mathbb{C})}$ such that $u_{(\mathbb{C})} \in \mathcal{D}_B(E_{(\mathbb{C})})$ and let λ be the family of eigenvalues of $u_{(\mathbb{C})}$ (theorem 1). We have $\lambda \in R^J$ (I, p. 106, prop. 4); since the linear map $u_{(\mathbb{C})}$ is $\mathbb{R}$ -rational, the eigenspace of $u_{(\mathbb{C})}$ relative to λ_j is $\mathbb{R}$ -rational for every $j \in J$ (cf. A, V, p. 60, prop. 6). Hence there exists a basis of it belonging to E , and a fortiori, there exists an orthonormal basis in E . The union of these bases is an orthonormal basis B_R of E such that $u \in \mathcal{D}_{B_R}(E)$.

COROLLARY 2. — Let F be a Hilbert space and let u be a compact continuous linear map from E to F . There exists a countable set I , an orthonormal basis $(e_i)_{i \in I}$ of the initial space $\text{Ker}(u)^\circ$ of u , an orthonormal family $(f_i)_{i \in I}$ of F , and a family $(\alpha_i)_{i \in I} \in (\mathbf{R}_+^*)^I$ such that $u(e_i) = \alpha_i f_i$ for all $i \in I$.

Let $v = u^* \circ u$. This is a compact, positive, hence Hermitian, endomorphism of E . By corollary 1, there exists an orthonormal basis $(e_j)_{j \in J}$ of E such that v is diagonal in this basis. The family $(\lambda_j)_{j \in J}$ of its eigenvalues is contained in $\mathbf{R}_+^J$. Set $\alpha_j = \sqrt{\lambda_j}$ for all $j \in J$. Let I be the set of $j \in J$ such that $\alpha_j \neq 0$. This is a countable set since v is compact. The family $(e_i)_{i \in I}$ is an orthonormal basis of the initial space of v which is the initial space $\text{Ker}(u)^\circ$ of u (EVT, V, p. 43, prop. 8). Set $f_i = \frac{1}{\alpha_i} u(e_i)$ for $i \in I$. For any i and j in I , we have

$$\langle f_i | f_j \rangle = \frac{1}{\alpha_i \alpha_j} \langle u(e_i) | u(e_j) \rangle = \frac{1}{\alpha_i \alpha_j} \langle v(e_i) | e_j \rangle = \frac{\lambda_i}{\alpha_i \alpha_j} \langle e_i | e_j \rangle,$$

whence it follows that the family $(f_i)_{i \in I}$ is orthonormal in F . The corollary follows, since $u(e_i) = \alpha_i f_i$ for all $i \in I$.

DEFINITION 2. — With the notations of the corollary, the family $(\alpha_i)_{i \in I}$ is the family of singular values of u relative to the orthonormal basis $(e_i)_{i \in I}$ of the initial space of u .

Remarks. — 1) This corollary generalizes th. 2 of EVT, V, p. 54, which corresponds to Hilbert–Schmidt maps.

2) With the notations of the corollary, we have the formula

$$u(x) = \sum_{i \in I} \alpha_i \langle e_i | x \rangle f_i \quad (1)$$

for all $x \in E$.

- 3) Let u be a compact positive endomorphism of E . Let $B = (f_i)_{i \in I}$ be an orthonormal basis of E such that u is diagonal in the basis B (theorem 1), and let $(\lambda_i)_{i \in I}$ be the family of eigenvalues of u in this basis. Let J be the set of $i \in I$ such that $\lambda_i > 0$; the family $(e_i)_{i \in J}$ is an orthonormal basis of the space $\text{Ker}(u)^\circ$. For every $i \in J$, set $e_i = f_i$ and $\alpha_i = \lambda_i$. We obtain $u(e_i) = \alpha_i f_i$: the family $(\alpha_i)_{i \in J}$ is the family of singular values of u relative to the basis $(e_i)_{i \in J}$.

IV.1.3 Decreasing sequence of eigenvalues

In this section, we assume that $K = \mathbb{C}$.

We set $\bar{\mathbf{N}} = \mathbf{N} \cup \{+\infty\} \subset \bar{\mathbf{R}}$. In this section, we say that a vector space E has dimension $+\infty \in \bar{\mathbf{N}}$ if E is not finite-dimensional. Let $I_E \subset \mathbf{N}$ be the set of dimensions of finite-dimensional subspaces F of E such that $F \neq E$. We have $I_E = \mathbf{N}$ if E is infinite-dimensional, and otherwise $I_E = \{0, \dots, \dim(E) - 1\}$. We set $I = I_E$ when no confusion can result.

Let u be a compact positive (in particular Hermitian) endomorphism of E . The sensitive spectrum of u is the set of values of a strictly decreasing sequence $(\nu_k)_{0 \leq k < \text{Card}(\text{Sp}_s(u))}$ of positive real numbers (cf.

prop. 5 of III, p. 90). For every integer k such that $0 \leq k < \text{Card}(\text{Sp}_s(u))$, we denote by n_k the spectral multiplicity of ν_k , with $n_k \geq 1$. Let $M \in \bar{\mathbf{N}}$ be the sum of the spectral multiplicities n_k ; it is the dimension of the image of u . We have $M \leq \text{Card}(I)$.

For $0 \leq n < M$, one defines $\lambda_n(u) = \nu_k$, where $k \geq 0$ is the unique integer such that

$$n_0 + \dots + n_{k-1} \leq n < n_0 + \dots + n_k.$$

One sets $\lambda_n(u) = 0$ if $n \in I$ satisfies $n \geq M$. This case can occur only if $I = \mathbf{N}$ and if $\text{Sp}_s(u)$ is finite (or, equivalently, if E is infinite-dimensional and u is of finite rank).

The sequence $(\lambda_n(u))_{n \in I}$ is decreasing; for every $\lambda \in \text{Sp}_s(u)$, the number of integers n

such that $\lambda_n(u) = \lambda$ is equal to the spectral multiplicity of the eigenvalue λ of u .

One says, by abuse of language, that $(\lambda_n(u))_{n \in J}$ is the decreasing sequence of eigenvalues of u repeated with their multiplicities.

PROPOSITION 3. — Let u be a compact positive endomorphism of E . There exists an orthonormal family $(e_n)_{n \in I}$ in E such that, for every $x \in E$, one has

$$u(x) = \sum_{n \in I} \lambda_n(u) \langle e_n | x \rangle e_n, \quad \langle x | u(x) \rangle = \sum_{n \in I} \lambda_n(u) |\langle e_n | x \rangle|^2.$$

Let $B = (f_j)_{j \in J}$ be an orthonormal basis of E in which u is diagonal (cor. 1 of IV, p. 150) and $(\lambda_j)_{j \in J}$ the family of eigenvalues of u in the basis B . Let J' be the set of $j \in J$ such that λ_j belongs to the sensitive spectrum of u .

For each $\lambda \in \text{Sp}_s(u)$, there exists a bijection between the integers n such that $0 \leq n < M$ and $\lambda_n(u) = \lambda$ and the $j \in J'$ such that $\lambda_j = \lambda$, since these two sets have the same cardinality, namely the spectral multiplicity of λ . A choice of such bijections for every λ defines a bijection ι from the set of integers such that $0 \leq n < M$ to the set J' . One defines the sequence $(e_n)_{0 \leq n < M}$ in E by setting $e_n = f_{\iota(n)}$ for $0 \leq n < M$. It is an orthonormal family in E .

In the case where $M < \text{Card}(I) = +\infty$, the space F generated by $\{e_0, \dots, e_{M-1}\}$ is finite-dimensional and its orthogonal F° is infinite-dimensional; one chooses for $(e_n)_{n \geq M}$ an orthonormal family in F° .

For every $x \in E$, we have

$$u(x) = \sum_{j \in J'} \lambda_j \langle f_j | x \rangle f_j = \sum_{0 \leq n < M} \lambda_n(u) \langle e_n | x \rangle e_n$$

since $u(f_j) = 0$ when $j \in J - J'$. If $n \in I$ satisfies $n \geq M$, we have $\lambda_n(u) = 0$, and one obtains the first formula of the proposition. The second follows from it.

PROPOSITION 4. — Let u be a compact positive endomorphism of E . Let $f \in \mathcal{C}(\mathbf{R}_+)$ be a continuous increasing function such that $f(0) = 0$.

The endomorphism $f(u)$ is compact and positive and, for every $n \in I_E$, we have $\lambda_n(f(u)) = f(\lambda_n(u))$.

The endomorphism $f(u)$ is compact and positive by prop. 6, b) of III, p. 91 and prop. 15,

a) of I, p. 117. The spectrum of $f(u)$ is the image under f of the spectrum of u (cor. 2 of I, p. 111). If $\lambda \in \text{Sp}_s(f(u))$, the spectral multiplicity of λ is the sum of the spectral multiplicities of the $\mu \in \text{Sp}(u)$ such that $f(\mu) = \lambda$ (cor. 2 of III, p. 84). Since f is increasing, the sequence $(f(\lambda_n(u)))_{n \in \mathbb{I}_E}$ is decreasing. The assertion then follows from the definition of the sequence $(\lambda_n(f(u)))_{n \in \mathbb{I}_E}$.

IV.1.4 Variational characterizations of eigenvalues

In this section, we assume that $K = \mathbb{C}$.

Let u be a compact and positive endomorphism of E and let $(\lambda_n(u))_{n \in \mathbb{I}_E}$ be the decreasing sequence of eigenvalues of u . We set $I = \mathbb{I}_E$.

For every closed subspace F of E , we denote by

$$r_F(u) = \inf_{x \in F - \{0\}} \frac{\langle x | u(x) \rangle}{\|x\|^2}, \quad R_F(u) = \sup_{x \in F^\circ - \{0\}} \frac{\langle x | u(x) \rangle}{\|x\|^2},$$

where the lower bound (resp. the upper bound) is taken in $[0, +\infty]$.

For every $n \in \mathbb{N}$, we denote by $\mathcal{F}_n$ the set of vector subspaces $F \subset E$ of dimension n . We say that a subspace $F \in \mathcal{F}_n$ is adapted to u if it admits an orthonormal basis $(f_i)_{0 \leq i \leq n-1}$ such that $u(f_i) = \lambda_i(u)f_i$ for $0 \leq i \leq n-1$.

PROPOSITION 5. — Let $n \in I$.

a) For every subspace $F \in \mathcal{F}_{n+1}$ adapted to u , we have $\lambda_n(u) = r_F(u)$;

b) For every subspace $F \in \mathcal{F}_n$ adapted to u , we have $\lambda_n(u) = R_F(u)$.

Let $F \in \mathcal{F}_{n+1}$ be a subspace adapted to u , and $(f_i)_{0 \leq i \leq n}$ be an orthonormal basis of F such that $u(f_i) = \lambda_i(u)f_i$ for all i . For all x in F , we have

$$\langle x | u(x) \rangle = \sum_{0 \leq i \leq n} \lambda_i(u) |\langle f_i | x \rangle|^2 \geq \lambda_n(u) \sum_{0 \leq i \leq n} |\langle f_i | x \rangle|^2 = \lambda_n(u) \|x\|^2,$$

with equality if $x = f_n$. This implies that $r_F(u) = \lambda_n(u)$ whence assertion (a).

Let $F \in \mathcal{F}_n$ be a subspace adapted to u . The closed subspace F° is nonzero (since $n = \dim(F) < \dim(E)$) and stable under u ; the endomorphism $\tilde{u}$ of F° deduced from u by passing

to subspaces is compact and positive.

One has $\text{Sp}(\tilde{u}) \subset \text{Sp}(u)$. We also have $\text{Sp}(\tilde{u}) \subset [0, \lambda_n(u)]$. Indeed, it suffices to verify that $\lambda \leq \lambda_n(u)$ for all $\lambda \in \text{Sp}_s(\tilde{u})$. The number λ is then an eigenvalue of u , so there exists $j \in I$ such that $\lambda = \lambda_j(u)$. The eigenspace of u relative to $\lambda_j(u)$ is therefore not contained in F , which implies that $\lambda_j(u) \leq \lambda_n(u)$.

Suppose that $\lambda_n(u) > 0$. The eigenspace of u relative to $\lambda_n(u)$ is then not contained in F , and $\lambda_n(u)$ therefore belongs to the sensitive spectrum of $\tilde{u}$. We deduce that $\sup(\text{Sp}(\tilde{u})) = \lambda_n(u)$. If $\lambda_n(u) = 0$, we have the same result since the spectrum of $\tilde{u}$ is then reduced to $\{0\}$.

Moreover, we have by definition $R_F(u) = R_{\{0\}}(\tilde{u})$, and finally $R_{\{0\}}(\tilde{u}) = \sup(\text{Sp}(\tilde{u}))$ according to prop. 9 of I, p. 139, a). Assertion b) is proved.

PROPOSITION 6. — For all $n \in I$, we have

$$\lambda_n(u) = \sup_{F \in \mathcal{F}_{n+1}} r_F(u) = \inf_{F \in \mathcal{F}_n} R_F(u).$$

Let $(e_n)_{n \in I}$ be an orthonormal family having the property of prop. 3 of IV, p. 152. For every integer n such that $1 \leq n < M + 1$, let $F_n \in \mathcal{F}_n$ be the n -dimensional subspace of E generated by $(e_0, \dots, e_{n-1})$; by construction, the space F_n is adapted to u . According to prop. 5, we therefore have

$$\lambda_n(u) = r_{F_{n+1}}(u) = R_{F_n}(u). \quad (2)$$

Let $n \in I$. Let $F \in \mathcal{F}_{n+1}$. The restriction to F of the orthogonal projector onto F_n is not injective, so there exists $x \neq 0$ in F orthogonal to F_n .

Since $x \in F_n^\circ$, we then have (prop. 3 of IV, p. 152)

$$\begin{aligned} \langle x | u(x) \rangle &= \sum_{\substack{m \in I \\ m \geq n}} \lambda_m(u) |\langle e_m | x \rangle|^2 \\ &\leq \lambda_n(u) \sum_{\substack{m \in I \\ m \geq n}} |\langle e_m | x \rangle|^2 = \lambda_n(u) \|x\|^2. \end{aligned}$$

This proves that $r_F(u) \leq \lambda_n(u)$, whence in particular the inequality

$$\sup_{F \in \mathcal{F}_{n+1}} r_F(u) \leq \lambda_n(u). \quad (3)$$

Let $F \in F_n$. The restriction to F_{n+1} of the orthogonal projector onto F is not injective, so there exists a vector $x \neq 0$ in F_{n+1} orthogonal to F . Since $x \in F_{n+1}$, we have (loc. cit.)

$$\begin{aligned}\langle x | u(x) \rangle &= \sum_{0 \leq m \leq n} \lambda_m(u) |\langle e_m | x \rangle|^2 \\ &\geq \lambda_n(u) \sum_{0 \leq m \leq n} |\langle e_m | x \rangle|^2 = \lambda_n(u) \|x\|^2\end{aligned}$$

and therefore $R_F(u) \geq \lambda_n(u)$. In particular, it follows that

$$\inf_{F \in \mathcal{F}_n} R_F(u) \geq \lambda_n(u). \quad (4)$$

In view of formulas (2), (3) and (4), the proposition is proved.

IV.1.5 Applications of the variational characterization of eigenvalues

In this section, we consider Hilbert spaces over $\mathbb{C}$.

PROPOSITION 7. — Let u and v be positive compact endomorphisms of E .

- a) One has $|\lambda_n(u) - \lambda_n(v)| \leq \|u - v\|$ for all $n \in \mathbf{I}$;
- b) If $u \leq v$, then $\lambda_n(u) \leq \lambda_n(v)$ for all $n \in I$.

Let $n \in I$ and let F be an n -dimensional vector subspace of E . For all $x \in F$, we have

$$|\langle x | v(x) \rangle - \langle x | u(x) \rangle| \leq \|u - v\| \|x\|,$$

therefore the inequalities

$$R_F(v) - \|u - v\| \leq R_F(u) \leq R_F(v) + \|u - v\|.$$

The assertion a) follows from this according to proposition 6 of IV, p. 154.

If $u \leq v$, we have $\langle x | u(x) \rangle \leq \langle x | v(x) \rangle$ for all $x \in E$. For every $n \in I$ and every subspace F of dimension n , we therefore have $R_F(u) \leq R_F(v)$, whence $\lambda_n(u) \leq \lambda_n(v)$ (loc. cit.).

PROPOSITION 8. — Let u be a positive compact endomorphism of E . Let H be a closed subspace of E and $i_H : H \rightarrow E$ the canonical injection. Denote by u_H the endomorphism $i_H^* u i_H$ of H . It is compact and positive.

- a) One has $I_H \subset I_E$ and $\lambda_n(u_H) \leq \lambda_n(u)$ for all $n \in I_H$;
- b) If H has finite codimension $k \in \mathbb{N}$ in E , then we have $I_H + k \subset I_E$ and $\lambda_{n+k}(u) \leq \lambda_n(u_H)$ for all $n \in I_H$.

The endomorphism u_H is compact (prop. 3 of III, p. 5). It is positive because $\langle x | u_H(x) \rangle = \langle i_H(x) | u(i_H(x)) \rangle \geq 0$ for all $x \in H$.

Let $n \in I_H \subset I_E$. Let F be a subspace of dimension $n + 1$ of H adapted to u_H . We therefore have $\lambda_n(u_H) = r_F(u_H)$ (prop 5 of IV, p. 153, a)), and since moreover $r_F(u_H) = r_F(u) \leq \lambda_n(u)$ (prop. 6 of IV, p. 154), we obtain assertion a).

Suppose that H has codimension $k \in \mathbb{N}$ in E and that $n \in I_H$. Let F be a subspace of H of dimension n adapted to u_H . Its orthogonal in H is equal to $H \cap F^\circ$, and it is the orthogonal in E of the subspace $F + H^\circ$ of dimension $n + k$. Hence $n + k \in I_E$ and (prop 5 of IV, p. 153, b))

$$\lambda_n(u_H) = \sup_{\substack{x \in H \cap F^\circ \\ x \neq 0}} \frac{\langle x | u_H(x) \rangle}{\|x\|^2} = R_{F+H^\circ}(u)$$

whence $\lambda_n(u_H) \leq \lambda_{n+k}(u)$ (prop. 6 of IV, p. 154).

DEFINITION 3. — Let F be a Hilbert space and let u be a compact linear map from E into F .

For every integer $n \in I_E$ we set $\alpha_n(u) = \lambda_n(u^* \circ u)$. Let J be the set of $n \in I_E$ such that $\alpha_n(u) > 0$. The family $(\alpha_n(u))_{n \in J}$ is called the sequence of singular values of u repeated with multiplicity.

We say that the sequence $(\alpha_n(u))_{n \in I_E}$ is the extended sequence of singular values of u .

The sequence $(\alpha_n(u))_{n \in I_E}$ is well defined since the endomorphism $u^* \circ u$ of E is compact (III, p. 5, prop. 3); it is a decreasing family of positive real numbers since $u^* \circ u$ is positive.

PROPOSITION 9. — Let F be a Hilbert space and let u be a compact linear map from E into F .

- a) For $n \in I_E$, we have

$$\alpha_n(u) = \sup_{F \in \mathcal{F}_{n+1}} \inf_{x \in F - \{0\}} \frac{\|u(x)\|}{\|x\|} = \inf_{F \in \mathcal{F}_n} \sup_{x \in F^\circ - \{0\}} \frac{\|u(x)\|}{\|x\|}.$$

- b) Let J be the set of $n \in I_E$ such that $\alpha_n(u) \neq 0$. There exist orthonormal families $(e_n)_{n \in J}$ in E and $(f_n)_{n \in J}$ in F such that for every $x \in E$, we have

$$u(x) = \sum_{n \in J} \alpha_n(u) \langle e_n | x \rangle f_n.$$

Since $\langle x | u^* u(x) \rangle = \|u(x)\|^2$ for all $x \in E$, the definitions and prop. 6 of IV, p. 154, a), imply the equality of the first assertion. Let $(e_n)_{n \in I_E}$ be an orthonormal family of E satisfying the conclusions of prop. 3 of IV, p. 152 applied to the compact positive endomorphism $u^* \circ u$ of E . Set $f_n = \alpha_n^{-1} u(e_n)$ for $n \in J$. Reasoning as in the proof of corollary 2 of IV, p. 150, we obtain assertion c).

COROLLARY. — Let F be a Hilbert space and u a compact linear map from E into F . Let $w \in \mathcal{L}(E)$ and $v \in \mathcal{L}(F)$. For every $n \in I_E$, we have $\alpha_n(w \circ u \circ v) \leq \|v\| \|w\| \alpha_n(u)$.

This is a consequence of assertion a) of the preceding proposition.

Remarks. — 1) The sequence $(\alpha_n(u))_{n \in I_E}$ being decreasing, the set J is either equal to I_E , or equal to a segment $\{0, \dots, m\}$ in I_E where $m \in \mathbb{N}$. This latter case holds if and only if u has finite rank.

- 2) Since $\|u(x)\| = \| |u|(x) \|$ for all $x \in E$ (I, p. 139, prop. 10), we have $\alpha_n(u) = \alpha_n(|u|)$ for all $n \in I_E$.

- 3) If u is positive, then $\alpha_n(u) = \lambda_n(u)$ for all $n \in I_E$ (indeed, we then have $\alpha_n(u)^2 = \lambda_n(u^* u) = \lambda_n(u^2) = \lambda_n(u)^2$ according to prop. 4 of IV, p. 153).

- 4) It is possible that $\alpha_n(v \circ u \circ w) = 0$ even if $\alpha_n(u)$ is nonzero; in this case, $\alpha_n(v \circ u \circ w)$ is not a singular value of $v \circ u \circ w$.

IV.1.6 Weyl inequalities

In this issue, we consider Hilbert spaces over $K = \mathbb{C}$.

Let E be a Hilbert space. For every $n \in \mathbb{N}$, we recall that the Hilbertian exterior power $\hat{\Lambda}^n E$ was defined in EVT, V, p. 34. For every Hilbert space F and for $u \in \mathcal{L}(E; F)$, we have also defined the linear map $\hat{\Lambda}^n u \in \mathcal{L}(\hat{\Lambda}^n E; \hat{\Lambda}^n F)$ (loc. cit.). These constructions are functorial: for

every Hilbert space G and for every linear map $v \in \mathcal{L}(F; G)$, the formula

$$\widehat{\Lambda}^n v \circ \widehat{\Lambda}^n u = \widehat{\Lambda}^n (v \circ u)$$

is valid (loc. cit., formula (28)).

Let H be a closed subspace of E and let i_H be the canonical injection from H into E . For every endomorphism u of E , we denote by u_H the endomorphism $i_H^* u i_H$ of H .

Lemma 2. — Let $n \in \mathbb{N}$. The map $\widehat{\Lambda}^n i_H$ is a linear isometric mapping from $\widehat{\Lambda}^n H$ into $\widehat{\Lambda}^n E$.

Let $(e_j)_{j \in J}$ be an orthonormal basis of H and $(e_j)_{j \in J'}$ an orthonormal basis of E , where $J \subset J'$. Equip J' with a total order. The elements $e_{j_1} \wedge \cdots \wedge e_{j_n}$ for $j_1 < \cdots < j_n$ in J' (resp. in J) form an orthonormal basis of $\widehat{\Lambda}^n E$ (resp. of $\widehat{\Lambda}^n H$) by prop. 5 of EVT, V, p. 34, prop. 5. The lemma follows.

In what follows, we shall identify $\widehat{\Lambda}^n H$ with a closed subspace of $\widehat{\Lambda}^n E$ by means of the mapping $\widehat{\Lambda}^n i_H$.

Lemma 3. — Let F be a Hilbert space and $u \in \mathcal{L}(E; F)$. Let $n \in \mathbb{N}$. a) We have $(\widehat{\Lambda}^n u)^* = \widehat{\Lambda}^n (u^*)$;

b) If $F = E$ and u is Hermitian (resp. positive, normal, unitary), then $\widehat{\Lambda}^n u$ is Hermitian (resp. positive, normal, unitary);

c) We have $|\widehat{\Lambda}^n u| = \widehat{\Lambda}^n |u|$;

d) Let H be a closed subspace of E . The restriction $(\widehat{\Lambda}^n u)|_{\widehat{\Lambda}^n H}$ of $\widehat{\Lambda}^n u$ to $\widehat{\Lambda}^n H$ is equal to $\widehat{\Lambda}^n (u|_H)$ (equality in $\mathcal{L}(\widehat{\Lambda}^n H; \widehat{\Lambda}^n F)$);

e) Suppose $F = E$. Let H be a closed subspace of E . We have $(\widehat{\Lambda}^n u)_{\widehat{\Lambda}^n H} = \widehat{\Lambda}^n (u_H)$ in $\mathcal{L}(\widehat{\Lambda}^n H)$.

The assertion a) follows from EVT, V, p. 39, formula (13). It immediately implies that $\widehat{\Lambda}^n u$ is Hermitian (resp. unitary) when u is Hermitian (resp. unitary). If u is positive, then $u^{1/2}$ is Hermitian, hence so is $\widehat{\Lambda}^n (u^{1/2})$; the relation $\widehat{\Lambda}^n u = (\widehat{\Lambda}^n u^{1/2})^2$ implies that $\widehat{\Lambda}^n u$ is positive (I, p. 138, prop. 8). This proves b).

What precedes allows one to compute that

$$\begin{aligned} \left(\widehat{\Lambda}^n |u| \right)^2 &= \widehat{\Lambda}^n (|u|^2) = \widehat{\Lambda}^n (u^* u) \\ &= \widehat{\Lambda}^n u^* \widehat{\Lambda}^n u = \left(\widehat{\Lambda}^n u \right)^* \left(\widehat{\Lambda}^n u \right) = |\widehat{\Lambda}^n u|^2. \end{aligned}$$

Since $\widehat{\Lambda}^n |u|$ is positive by b), it follows from prop. 16 of I, p. 118 that $\widehat{\Lambda}^n |u| = |\widehat{\Lambda}^n u|$, whence c).

Let i_H be the canonical injection of H into E . The space $\widehat{\Lambda}^n H$ is identified with a closed subspace of $\widehat{\Lambda}^n E$ by the mapping $\widehat{\Lambda}^n i_H$, whence

$$(\widehat{\Lambda}^n u)|_{\widehat{\Lambda}^n H} = \widehat{\Lambda}^n u \circ \widehat{\Lambda}^n i_H = \widehat{\Lambda}^n (u \circ i_H) = \widehat{\Lambda}^n (u|_H),$$

and

$$(\widehat{\Lambda}^n u)_{\widehat{\Lambda}^n H} = (\widehat{\Lambda}^n i_H)^* \circ \widehat{\Lambda}^n u \circ \widehat{\Lambda}^n i_H = \widehat{\Lambda}^n (i_H^* u i_H) = \widehat{\Lambda}^n (u_H),$$

which proves d) and e).

Let F be a Hilbert space, and let $u \in \mathcal{L}^c(E; F)$. As in number 3 of IV, p. 151, we denote by I_E the set of dimensions of finite-dimensional subspaces F of E such that $F \neq E$. We recall that $(\alpha_n(u))_{n \in I_E}$ denotes the extended sequence of singular values of u (def. 3 of IV, p. 156).

PROPOSITION 10. — One has the equality

$$\prod_{i=0}^n \alpha_i(u) = \|\widehat{\Lambda}^{n+1} u\|$$

for all $n \in I_E$.

Lemma 3, c) shows that $\|\widehat{\Lambda}^{n+1} u\| = \|\widehat{\Lambda}^{n+1} |u|\|$; moreover, since $\alpha_i(u) = \alpha_i(|u|)$ for all $i \in I_E$ (remark 5 of IV, p. 157, (2)), it suffices to prove the assertion for $|u|$. This allows us to assume that $F=E$ and that u is positive.

Let then $B = (e_j)_{j \in J}$ be an orthonormal basis of E such that u is diagonalizable in the basis B (theorem 1 of IV, p. 149), and let $(\lambda_j)_{j \in J}$ be the family of eigenvalues of u in the basis B . Endow J with a total order, and denote by J_n the set of strictly increasing families $(j_0, \dots, j_n) \in \mathbf{J}^{n+1}$ of elements of J . The vectors

$$e_{\iota} = e_{j_0} \wedge \cdots \wedge e_{j_n}$$

for $\iota = (j_0, \dots, j_n) \in J_n$ form an orthonormal basis B_n of $\widehat{\Lambda}^{n+1}E$ (EVT, V, p. 34, prop. 5).

For every $\iota = (j_0, \dots, j_n) \in J_n$, denote

$$\lambda_{\iota} = \prod_{j=0}^n \lambda_{i_j}.$$

It follows that $(\widehat{\Lambda}^{n+1}u)e_{\iota} = \lambda_{\iota}e_{\iota}$, hence $\widehat{\Lambda}^{n+1}u$ is diagonal in the basis B_n . Consequently, one has

$$\|\widehat{\Lambda}^{n+1}u\| = \sup_{\iota \in I_n} \lambda_{\iota}$$

(lemma 1 of IV, p. 147), which is equal to the product $\lambda_0(u) \cdots \lambda_n(u)$ of the $n+1$ largest eigenvalues of u . The desired formula follows since $\lambda_i(u) = \alpha_i(u)$ for all $i \in I_E$ when u is positive (remark 5 of IV, p. 157, (3)).

In particular, if $\alpha_1(u) < \alpha_0(u) = \|u\|$, one sees that the inequality $\|\widehat{\Lambda}^n(u)\| \leq \|u\|^n$ (EVT, V, p. 34, formula (29)) is not an equality in general if $n \geq 2$.

COROLLARY. — Let G be a Hilbert space and $v \in \mathcal{L}^c(F; G)$. We have

$$\prod_{i=0}^n \alpha_i(v \circ u) \leq \prod_{i=0}^n \alpha_i(u) \alpha_i(v)$$

for all $n \in I_E$.

It suffices to note that

$$\|\widehat{\Lambda}^{n+1}(vu)\| = \|\widehat{\Lambda}^{n+1}v \circ \widehat{\Lambda}^{n+1}u\| \leq \|\widehat{\Lambda}^{n+1}v\| \|\widehat{\Lambda}^{n+1}u\|$$

and to apply proposition 10.

Lemma 4. — Let A be a ring. Let $n \in \mathbb{N}$ and $(a_i)_{0 \leq i \leq n}$ and $(b_i)_{0 \leq i \leq n}$ be families of elements of A . For $0 \leq j \leq n$ set

$$A_j = \sum_{i=0}^j a_i.$$

One has

$$\sum_{i=0}^n a_i b_i = A_n b_n - \sum_{i=0}^{n-1} A_i (b_{i+1} - b_i).$$

Set $A_{-1} = 0$. We have $a_i = A_i - A_{i-1}$ for $0 \leq i \leq n$, hence

$$\sum_{i=0}^n a_i b_i = \sum_{i=0}^n (A_i - A_{i-1}) b_i = \sum_{i=0}^{n-1} A_i (b_i - b_{i+1}) + A_n b_n,$$

as desired.

Let I be an interval of $\bar{\mathbf{R}}$ not containing $+\infty$. Recall (FVR, I, p. 38, remark) that a continuous function $f : I \rightarrow [-\infty, +\infty[$ is said to be convex if its restriction to the interior of I is a convex function with values in $\mathbf{R}$; it then has a limit (finite or infinite) on the right at $\inf I$. In the statements below, the product of 0 and an element of $\{-\infty, +\infty\}$ is by convention equal to 0.

Lemma 5. — Let $I \subset \mathbf{R}$ be an interval not containing $+\infty$ and f an increasing convex function from I into $[-\infty, +\infty[$.

Let n be a natural integer and

$$a_0 \geq a_1 \geq \dots \geq a_n, \quad b_0 \geq b_1 \geq \dots \geq b_n$$

elements of I .

Let $\varrho_0 \geq \varrho_1 \geq \dots \geq \varrho_n$ be positive real numbers. Suppose that

$$\sum_{i=0}^j a_i \leq \sum_{i=0}^j b_i. \quad (5)$$

for every integer j such that $0 \leq j \leq n$. We then have

$$\sum_{i=0}^n \varrho_i f(a_i) \leq \sum_{i=0}^n \varrho_i f(b_i). \quad (6)$$

Suppose first that $a_i \in \mathring{I}$, and hence $f(a_i) \in \mathbf{R}$, for all i . Let i be such that $0 \leq i \leq n$, and let (α_i, β_i) be real numbers such that the line with equation $y = \alpha_i x + \beta_i$ is a supporting line of the graph of f at the point $(a_i, f(a_i))$ (FVR, I, p. 37). One consequently has $f(a_i) = \alpha_i a_i + \beta_i$ and $f(b_i) \geq \alpha_i b_i + \beta_i$ since the graph of f lies above the supporting line, whence

$$f(b_i) - f(a_i) \geq \alpha_i(b_i - a_i). \quad (7)$$

Moreover, if $i < n$, we have

$$\alpha_i \geq f'_g(a_i) \geq f'_d(a_{i+1}) \geq \alpha_{i+1}$$

(loc. cit. and FVR, I, p. 36, cor. 1); furthermore $\alpha_i \geq 0$ since f is increasing (FVR, I, p. 22, corollary), whence $\varrho_i \alpha_i \geq \varrho_{i+1} \alpha_{i+1} \geq 0$ for $0 \leq i < n$.

Let j be an integer such that $0 \leq j \leq n$. Set

$$A_j = \sum_{i=0}^j (b_i - a_i),$$

so that $A_j \geq 0$ by hypothesis (5). Applying inequality (7) and then Lemma 4, we deduce

$$\begin{aligned} \sum_{i=0}^n \varrho_i (f(b_i) - f(a_i)) &\geq \sum_{i=0}^n \varrho_i \alpha_i (b_i - a_i) \\ &= \varrho_n \alpha_n A_n + \sum_{j=0}^{n-1} (\varrho_j \alpha_j - \varrho_{j+1} \alpha_{j+1}) A_j \geq 0. \end{aligned}$$

Consider the general case, and reason by induction on n . If one of the a_i does not belong to the interior of I , we necessarily have $a_0 = \sup I$ or $a_n = \inf I$.

Suppose first that $a_n = \inf I$. We then have $a_n \leq b_n$ and therefore $\varrho_n f(a_n) \leq \varrho_n f(b_n)$. The induction hypothesis, applied to the families $(a_0, \dots, a_{n-1})$, $(b_0, \dots, b_{n-1})$ and $(\varrho_0, \dots, \varrho_{n-1})$, implies

$$\sum_{i=0}^{n-1} \varrho_i f(a_i) \leq \sum_{i=0}^{n-1} \varrho_i f(b_i),$$

whence the desired inequality follows by adding $\varrho_n f(a_n)$. The case where $a_0 = \sup I$ is treated in a similar manner.

PROPOSITION 11 (Weyl inequalities). — Let G be a Hilbert space and $\nu \in \mathcal{L}^c(F; G)$. Let $g : \mathbf{R}_+ \rightarrow [-\infty, +\infty[$ be a nondecreasing function such that the function $g \circ \exp$ is convex. We have

$$\sum_{i=0}^n g(\alpha_i(v \circ u)) \leq \sum_{i=0}^n g(\alpha_i(v)\alpha_i(u))$$

for all $n \in I_E \cap I_F$.

Set $I = [-\infty, +\infty[$ and $f = g \circ \exp$. It is permissible to apply Lemma 5 with

$$a_i = \log(\alpha_i(v \circ u)) \in I, \quad b_i = \log(\alpha_i(v)\alpha_i(u)) \in I$$

and $\varrho_i = 1$ for $0 \leq i \leq n$ since

$$\sum_{i=0}^j a_i = \log\left(\prod_{i=0}^j \alpha_i(v \circ u)\right) \leq \log\left(\prod_{i=0}^j \alpha_i(v)\alpha_i(u)\right) = \sum_{i=0}^j b_i$$

for $0 \leq j \leq n$ by the corollary of prop. 10. Inequality (6) is then the desired conclusion.

Lemma 6. — Let g and h be convex functions defined on intervals I and J of $\mathbb{R}$, respectively.

If g is increasing and defined on the image of h , then the function $g \circ h$ is convex on J .

Indeed, for $t \in [0, 1]$ and $(x, y) \in J \times J$, we have

$$\begin{aligned} g(h(tx + (1-t)y)) &\leq g(th(x) + (1-t)h(y)) \\ &\leq tg(h(x)) + (1-t)g(h(y)). \end{aligned}$$

COROLLARY. — Let G be a Hilbert space and $v \in \mathcal{L}^c(F; G)$. Let $n \in I_E \cap I_F$.

a) Let $r \in \mathbf{R}_+^*$. We have

$$\sum_{i=0}^n \alpha_i(v \circ u)^r \leq \sum_{i=0}^n \alpha_i(v)^r \alpha_i(u)^r.$$

b) Let $p, q, r \in \mathbf{R}_+^*$ such that $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$. Then

$$\left(\sum_{i=0}^n \alpha_i(v \circ u)^r \right)^{1/r} \leq \left(\sum_{i=0}^n \alpha_i(v)^p \right)^{1/p} \left(\sum_{i=0}^n \alpha_i(u)^q \right)^{1/q}.$$

c) Suppose that $F = E$. For every integer $m \geq 2$, we have

$$\sum_{i=0}^n \alpha_i(u^m) \leq \left(\sum_{i=0}^n \alpha_i(u)^2 \right)^{m/2}.$$

Let $r \in R_+^*$ and let g be the function defined on $\mathbf{R}_+$ by $g(x) = x^r$. The function $g \circ \exp$ is convex (lemma 6), so assertion a) follows from prop. 11 applied to the function g .

The assertion b) follows from a) and from Hölder's inequality (INT, I, prop. 4).

Let $m \geq 2$ be an integer. Let us apply b) with $r = 1$, $p = q = 2$ and $v = u^{m-1}$. We find

$$\sum_{i=0}^n \alpha_i(u^m) \leq \left(\sum_{i=0}^n \alpha_i(u^{m-1})^2 \right)^{1/2} \left(\sum_{i=0}^n \alpha_i(u)^2 \right)^{1/2}.$$

Let us prove c) by induction on $m \geq 2$. The preceding inequality establishes the assertion when $m = 2$. Suppose that $m \geq 3$ and that the assertion is valid for $m - 1$; since we have

$$\sum_{i=0}^n \alpha_i(u^{m-1})^2 \leq \left(\sum_{i=0}^m \alpha_i(u^{m-1}) \right)^2,$$

the above inequality implies

$$\sum_{i=0}^n \alpha_i(u^m) \leq \left(\sum_{i=0}^n \alpha_i(u^{m-1}) \right)^{1/2} \left(\sum_{i=0}^n \alpha_i(u)^2 \right)^{1/2} \leq \left(\sum_{i=0}^n \alpha_i(u)^2 \right)^{m/2}$$

by applying the induction hypothesis.

IV.1.7 Endomorphisms of finite trace

We recall that a positive endomorphism u of E is of finite trace if and only if there exists an orthonormal basis $(e_i)_{i \in I}$ of E such that

$$\sum_{i \in I} \langle e_i | u(e_i) \rangle < +\infty$$

(EVT, V, p. 48, lemma 3 and p. 49, def. 7). If $K = \mathbb{C}$, the space $\mathcal{L}_1(E)$ of endomorphisms of finite trace of E is the vector space generated by the set of positive endomorphisms of finite trace (EVT, V, p. 50, def. 8); if $K = \mathbb{R}$, the space $\mathcal{L}_1(E)$ is defined as the intersection $\mathcal{L}(E) \cap \mathcal{L}_1(E_{(\mathbb{C})})$ (EVT, V, p. 50).

If $u \in \mathcal{L}_1(E)$, then the series

$$\sum_{i \in I} \langle e_i | u(e_i) \rangle$$

converges for every orthonormal basis $(e_i)_{i \in I}$ of E and its sum is independent of the orthonormal basis; this sum is called the trace $\text{Tr}(u)$ of u (EVT, V, p. 50). If $K = \mathbb{R}$, we have $\text{Tr}(u) = \text{Tr}(u_{(\mathbb{C})})$.

Let $u \in \mathcal{L}_1(E)$. We have $u^* \in \mathcal{L}_1(E)$ and $\text{Tr}(u^*) = \overline{\text{Tr}(u)}$ (loc. cit.).

PROPOSITION 12. — Let $B = (e_i)_{i \in I}$ be an orthonormal basis of E . Let $u \in \mathcal{D}_B(E)$ and denote by $\lambda = (\lambda_i)_{i \in I}$ the family of its eigenvalues. The endomorphism u is of finite trace if and only if the family λ is summable in K . We then have $\text{Tr}(u) = \sum \lambda_i$.

Replacing u by $u_{(\mathbb{C})}$ if necessary, we may assume that $K = \mathbb{C}$.

Suppose first that u is of finite trace. According to EVT, V, p. 50 and p. 49, formula (25), the family $(\langle e_i | u(e_i) \rangle)_{i \in I} = (\lambda_i)_{i \in I}$ is summable.

Conversely, suppose that the family λ is summable.

Each of the families $(\mathcal{R}(\lambda_i)^+)$, $(\mathcal{R}(\lambda_i)^-)$, $(\mathcal{I}(\lambda_i)^+)$, $(\mathcal{I}(\lambda_i)^-)$ is then summable. The endomorphism u is a linear combination of the elements of $\mathcal{D}_B(E)$ whose eigenvalues are these families. These elements of $\mathcal{D}_B(E)$ are positive. Since, by definition, the space of finite-trace endomorphisms is generated by positive finite-trace endomorphisms, we may therefore assume that $\lambda_i \geq 0$ for all i . Since $\langle e_i | u(e_i) \rangle = \lambda_i$, the family $(\langle e_i | u(e_i) \rangle)_{i \in I}$ is summable, hence u is of finite trace (EVT, V, p. 48, lemma 3).

Finally, if u has finite trace, then by EVT, V, p. 50, we have

$$\text{Tr}(u) = \sum_{i \in I} \langle e_i | u(e_i) \rangle = \sum_{i \in I} \lambda_i.$$

COROLLARY 1. — Let u be an endomorphism of E with finite trace.

- a) The endomorphism u is compact;
- b) Let $B = (e_i)_{i \in I}$ be an orthonormal basis of the initial space $\text{Ker}(u)^\circ$ of u , $C = (f_i)_{i \in I}$ be an orthonormal family of E and $(\alpha_i)_{i \in I}$ be a family in $(\mathbb{R}_+^*)^I$ such that $u(e_i) = \alpha_i f_i$ for all $i \in I$ (cor. 2 of IV, p. 150). We have

$$\text{Tr}(u) = \sum_{i \in I} \alpha_i \langle e_i | f_i \rangle.$$

To prove *a*), one may assume that $\mathbf{K} = \mathbf{C}$ (EVT, V, p. 50 and remark 4 of III, p. 2) and that u is positive (EVT, V, p. 50, def. 8).

By EVT, V, p. 56, cor. 1, there exists an orthonormal basis $B = (e_i)_{i \in I}$ of E such that u is diagonal in the basis B and moreover the family λ of the eigenvalues of u belongs to $\mathbf{R}_+^I$ and is summable. The family λ therefore belongs to $\mathcal{C}_0(I; \mathbf{K})$ (TG, III, p. 38, prop. 1), and consequently the endomorphism u is compact (prop. 2 of IV, p. 148).

Let us prove *b*). Let $(e_i)_{i \in \mathbf{J}}$ be an orthonormal basis of E extending the family B , so that $u(e_i) = 0$ if $i \in \mathbf{J} - \mathbf{I}$. We obtain

$$\text{Tr}(u) = \sum_{i \in \mathbf{J}} \langle e_i | u(e_i) \rangle = \sum_{i \in \mathbf{I}} \langle e_i | u(e_i) \rangle = \sum_{i \in \mathbf{I}} \alpha_i \langle e_i | f_i \rangle.$$

COROLLARY 2. — Let F be a Hilbert space. Let $u \in \mathcal{L}(E; F)$ be a Hilbert–Schmidt operator. Then u is a compact linear operator.

Let $(j, |u|)$ be the polar decomposition of u (I, p. 140, def. 4). By definition (EVT, V, p. 50, def. 9) the endomorphism u^*u has finite trace, hence is compact (corollary 1, a)); the same is true of $|u| = \sqrt{u^*u}$ (prop. 6 of III, p. 91, b)) and of $u = j|u|$ (prop. 3 of III, p. 5).

COROLLARY 3. — Suppose that $\mathbf{K} = \mathbf{C}$. Let u be a positive compact endomorphism of E . The endomorphism u is of finite trace if and only if the decreasing family of its eigenvalues $(\lambda_n(u))_{n \in \mathbf{I}_E}$ is summable; the trace of u is then the sum of this family.

By th. 1 of IV, p. 149, this follows from the preceding proposition and the definition of the sequence $(\lambda_n(u))_{n \in \mathbf{I}_E}$ (no. 3 of IV, p. 151), taking into account formula (28) of EVT, V, p. 49.

COROLLARY 4. — Let F be a Hilbert space. Let $u \in \mathcal{L}(E; F)$ be a compact linear map. Let $B = (e_i)_{i \in \mathbf{I}}$ be an orthonormal basis of the initial space $\text{Ker}(u)^\circ$ of u , $C = (f_i)_{i \in \mathbf{I}}$ be an orthonormal family of F and $(\alpha_i)_{i \in \mathbf{I}}$ be a family in $(\mathbf{R}_+^*)^{\mathbf{I}}$ such that $u(e_i) = \alpha_i f_i$ for all $i \in \mathbf{I}$ (cor. 2 of IV, p. 150).

The endomorphism $|u|$ of E has finite trace if and only if the family $(\alpha_i)_{i \in \mathbf{I}}$ is summable. We then have $u \in \mathcal{L}_2(E; F)$ and

$$\text{Tr}(|u|) = \sum_{i \in \mathbf{I}} \alpha_i, \quad \|u\|_2^2 = \text{Tr}(u^*u) = \sum_{i \in \mathbf{I}} \alpha_i^2, \quad (8)$$

and in particular $\|u\|_2 \leq \text{Tr}(|u|)$.

The family of nonzero eigenvalues of $|u|$ is $(\alpha_i)_{i \in I}$, hence $|u|$ has finite trace if and only if the family $(\alpha_i)_{i \in I}$ is summable (prop. 12). If that is the case, the family (α_i^2) of nonzero eigenvalues of $u^*u = |u|^2$ is summable, and formulas (8) result from loc. cit.

Lemma 7. — Let $\lambda = (\lambda_i)_{i \in I}$ be a family of complex numbers. For every $t \in \mathbf{C}^*$, let n_t be the cardinality of the set of $i \in I$ such that $\lambda_i = t$. The family $(\lambda_i)_{i \in I}$ is summable if and only if n_t is finite for all $t \in \mathbf{C}^*$ and if the family $(n_t t)_{t \in \mathbf{C}^*}$ is summable. In this case, the sums of these two families are equal.

Suppose that the family $(\lambda_i)_{i \in I}$ is summable. For every $t \in \mathbf{C}$, let I_t be the set of $i \in I$ such that $\lambda_i = t$. By TG, III, p. 39, th. 2, applied to the partition of I by the sets I_t , the set I_t is finite for all $t \in \mathbf{C}^*$, and moreover the family $(n_t t)_{t \in \mathbf{C}^*}$ is summable with sum equal to that of the family $(\lambda_i)_{i \in I}$.

Conversely, suppose that n_t is finite for all $t \in \mathbf{C}^*$ and that the family $(n_t t)_{t \in \mathbf{C}^*}$ is summable. Let J be a finite subset of I and Λ the set of λ_i for $i \in J$. We have

$$\left| \sum_{i \in J} \lambda_i \right| \leq \sum_{t \in \Lambda - \{0\}} n_t |t| \leq \sum_{t \in \mathbf{C}^*} n_t |t|,$$

therefore the family $(\lambda_i)_{i \in I}$ is summable (TG, VII, p. 17, corollary).

PROPOSITION 13. — Suppose that $K = \mathbf{C}$. Let u be a compact normal endomorphism of E . For $t \in \text{Sp}_s(u)$, let $n_t \geq 1$ be the spectral multiplicity of the eigenvalue t of u . For u to have finite trace, it is necessary and sufficient that the family $(n_t t)_{t \in \text{Sp}_s(u)}$ be summable. The trace of u is then the sum of this family.

Let $B = (e_i)_{i \in I}$ be an orthonormal basis of E such that u is diagonal in the basis B (th. 1 of IV, p. 149), and let $\lambda = (\lambda_i)_{i \in I}$ be the family of its eigenvalues. Since, for $t \in \text{Sp}_s(u)$, the multiplicity n_t is equal to the number of elements $i \in I$ such that $\lambda_i = t$, and since the nonzero elements of λ belong to $\text{Sp}_s(u)$, the assertion then follows from prop. 12 and the preceding lemma.

IV.1.8 Nuclear maps

In this section, F denotes a Hilbert space over K . When $K = \mathbb{C}$, we recall (I, p. 139, def. 3) that for $u \in \mathcal{L}(E; F)$, we denote by $|u|$ the positive endomorphism $\sqrt{u^* \circ u}$ of E . When $K = \mathbb{R}$, the element $|u_{(\mathbb{C})}|$ of $\mathcal{L}(E_{(\mathbb{C})})$ is of the form $v_{(\mathbb{C})}$ for a unique endomorphism $v \in \mathcal{L}(E)$, which is still denoted $|u|$ (cf. I, p. 87).

We denote by $(u, v) \mapsto \langle u|v \rangle = \text{Tr}(u^*v)$ the inner product in the Hilbert space $\mathcal{L}_2(E; F)$ (EVT, V, p. 53, remarks 1 and 2).

For every $u \in \mathcal{L}(E; F)$, we denote by $\|u\|_1$ the quantity $\text{Tr}(|u|)$ if $|u|$ has finite trace and $\|u\|_1 = +\infty$ if this is not the case. We therefore have $\|u\|_1 \in \mathbb{R}_+ \cup \{+\infty\}$. Since $\|u\| = \| |u| \|$ (prop. 10, a) of I, p. 139) and since, if v is positive and of finite trace, then $\text{Tr}(v) \geq \|v\|$ (EVT, V, p. 49, formula (24bis) and p. 44, prop. 9), it follows that

$$\|u\| \leq \|u\|_1. \quad (9)$$

If $\|u\|_1$ is finite, then u is a Hilbert–Schmidt operator and $\|u\|_2 \leq \|u\|_1$ (cor. 4 of IV, p. 165).

PROPOSITION 14. — Let $u \in \mathcal{L}_2(E; F)$. We have

$$\|u\|_1 = \sup_{\substack{v \in \mathcal{L}_2(E; F) \\ \|v\| \leq 1}} |\langle v|u \rangle|,$$

the supremum being computed in $\mathbb{R}_+ \cup \{+\infty\}$.

Let $B = (e_i)_{i \in I}$ be an orthonormal basis of the initial space $\text{Ker}(u)^\circ$ of u , $C = (f_i)_{i \in I}$ be an orthonormal family of F and $(\alpha_i)_{i \in I}$ be a family in $(\mathbb{R}_+^*)^I$ such that $u(e_i) = \alpha_i f_i$ for all $i \in I$ (cor. 2 of IV, p. 150). The family $(\alpha_i)_{i \in I}$ is square-integrable since u belongs to $\mathcal{L}_2(E; F)$ (cor. 4 of IV, p. 165). Let $(e_j)_{j \in J}$ be an orthonormal basis of E extending $(e_i)_{i \in I}$.

Let L be a finite subset of I . Let v_L be the continuous finite-rank linear map from E into F defined by

$$v_L(x) = \sum_{i \in L} \langle e_i | x \rangle f_i$$

for all x in E . We have $\|v_L\| \leq 1$ and $v_L \in \mathcal{L}_2(E; F)$. Moreover, for every $j \in J$, we have

$v_L(e_j) = 0$ if $j \notin L$ and $v_L(e_j) = f_j$ if $j \in L$. Thus

$$|\langle v_L | u \rangle| = |\text{Tr}(v_L^* u)| = \left| \sum_{j \in J} \langle v_L(e_j) | u(e_j) \rangle \right| = \sum_{j \in L} \alpha_j.$$

From loc. cit., we deduce that

$$\|u\|_1 = \sup_L \sum_{j \in L} \alpha_j \leq \sup_{\substack{v \in \mathcal{L}_2(E;F) \\ \|v\| \leq 1}} |\langle v | u \rangle| \quad (10)$$

in $\mathbf{R}_+ \cup \{+\infty\}$.

This implies the equality of the proposition when $\|u\|_1 = +\infty$.

Suppose that $\|u\|_1$ is finite. For every $i \in I$, let p_i be the linear map from E to F such that $p_i(e_i) = f_i$ and $p_i(x) = 0$ for all $x \in e_i^\circ$. For all j and k in I , we have

$$\langle p_j | p_k \rangle = \text{Tr}(p_j^* p_k) = \sum_{i \in J} \langle p_j(e_i) | p_k(e_i) \rangle,$$

which is zero unless $j = k$, in which case this quantity equals $\|f_k\|^2 = 1$. The family $(p_i)_{i \in I}$ is therefore orthonormal in $\mathcal{L}_2(E;F)$. Consequently, the family $(\alpha_i p_i)_{i \in I}$ is summable in $\mathcal{L}_2(E;F)$; its sum is equal to u since these two continuous linear maps coincide on the elements e_i for all $i \in J$.

Let $v \in \mathcal{L}_2(E;F)$. For every $i \in I$, it follows that

$$\langle v | p_i \rangle = \text{Tr}(v^* p_i) = \sum_{j \in J} \langle v(e_j) | p_i(e_j) \rangle = \langle v(e_i) | f_i \rangle.$$

If $\|v\| \leq 1$, we have

$$\begin{aligned} |\langle v | u \rangle| &= |\langle v | \sum_{i \in I} \alpha_i p_i \rangle| \leq \sum_{i \in I} \alpha_i |\langle v | p_i \rangle| \\ &= \sum_{i \in I} \alpha_i |\langle v(e_i) | f_i \rangle| \leq \sum_{i \in I} \alpha_i = \text{Tr}(|u|), \end{aligned}$$

whence the inequality

$$\sup_{\substack{v \in \mathcal{L}_2(E;F) \\ \|v\| \leq 1}} |\langle v | u \rangle| \leq \|u\|_1,$$

which, combined with formula (10), concludes the proof of the proposition.

It follows from this proposition that the set $\mathcal{L}_1(E; F)$ of continuous linear maps u from E to F such that $\|u\|_1$ is finite is a vector subspace of $\mathcal{L}_2(E; F)$ and that the map $u \mapsto \|u\|_1$ is a semi-norm on $\mathcal{L}_1(E; F)$; inequality (9) shows that it is a norm.

DEFINITION 4. — We say that the vector space $\mathcal{L}_1(E; F)$ equipped with the norm $u \mapsto \|u\|_1$ is the space of nuclear maps from E to F . If $u \in \mathcal{L}_1(E; F)$, one says that u is nuclear.

Remarks. — 1) Suppose that $K = \mathbb{R}$. Let $u \in \mathcal{L}(E; F)$. We have $u \in \mathcal{L}_1(E; F)$ if and only if $u_{(\mathbb{C})} \in \mathcal{L}_1(E_{(\mathbb{C})}; F_{(\mathbb{C})})$; in this case, we have $\|u\|_1 = \|u_{(\mathbb{C})}\|_1$.

2) Let u be a nuclear map from E to F . The map u being Hilbert–Schmidt, it is compact (cor. 2 of IV, p. 165). Moreover, proposition 14 implies that for every $v \in \mathcal{L}_2(E; F)$, we have

$$|\langle v|u \rangle| \leq \|v\| \|u\|_1. \quad (11)$$

3) Suppose that $K = \mathbb{C}$. Let $u \in \mathcal{L}_1(E; E)$ such that u is positive. The norm $\|u\|_1$ of u is the sum of the sequence $(\lambda_n(u))_{n \in \mathbb{I}_E}$ (cor. 3 of IV, p. 165).

4) The canonical inclusion of $\mathcal{L}_1(E; F)$ in $\mathcal{L}(E; F)$ has norm ≤ 1 (inequality (9), p. 167).

PROPOSITION 15. — Let G be a Hilbert space over K . The map $(u, v) \mapsto v \circ u$ from $\mathcal{L}(E; F) \times \mathcal{L}(F; G)$ into $\mathcal{L}(E; G)$ defines by passage to quotients a continuous bilinear map of norm ≤ 1 from $\mathcal{L}_2(E; F) \times \mathcal{L}_2(F; G)$ into $\mathcal{L}_1(E; G)$.

Let $u \in \mathcal{L}_2(E; F)$ and $v \in \mathcal{L}_2(F; G)$. Let $w \in \mathcal{L}_2(E; G)$. We obtain $\langle uv|w \rangle = \text{Tr}(v^* u^* w) = \langle v|u^* w \rangle$ whence

$$|\langle uv|w \rangle| \leq \|v\|_2 \|u^* w\|_2 \leq \|v\|_2 \|u\|_2 \|w\|$$

by the Cauchy–Schwarz inequality and formula (37) of EVT, V, p. 52. The result therefore follows from prop. 14.

Lemma 8. — The map $u \mapsto u^*$ from $\mathcal{L}(E; F)$ into $\mathcal{L}(F; E)$ defines by passage to quotients an isometric linear map from $\mathcal{L}_1(E; F)$ into $\mathcal{L}_1(F; E)$.

Let $u \in \mathcal{L}_1(E; F)$. Since u is a Hilbert–Schmidt map, so is u^* (EVT, V, p. 54). The map $v \mapsto v^*$ is a bijection from the set of $v \in \mathcal{L}_2(E; F)$ such that $\|v\| \leq 1$ onto the set of $w \in \mathcal{L}_2(F; E)$

such that $\|w\| \leq 1$; for every $v \in \mathcal{L}_2(E; F)$

with $\|v\| \leq 1$, we have $\langle v|u \rangle = \langle u^*|v^* \rangle$ (EVT, V, p. 54, formula (42)), whence the result by prop. 14.

PROPOSITION 16. — Let E_1 and F_1 be Hilbert spaces. Let u be in $\mathcal{L}_1(E; F)$, v be in $\mathcal{L}(E_1; E)$ and v_2 be in $\mathcal{L}(F; F_1)$. We then have $v_2 u v_1 \in \mathcal{L}_1(E_1; F_1)$ and $\|v_2 u v_1\|_1 \leq \|v_2\| \|v_1\| \|u\|_1$.

Let $w \in \mathcal{L}_2(E; F_1)$ such that $\|w\| \leq 1$. We have $v_2 u \in \mathcal{L}_2(E; F_1)$ (EVT, V, p. 52, formula (36)). Since $v_2^* w \in \mathcal{L}_2(E; F)$ (loc. cit.), it follows

$$|\langle w|v_2 u \rangle| = |\langle v_2^* w|u \rangle| \leq \|v_2\| \|u\|_1$$

(formula (11)), whence $v_2 u \in \mathcal{L}_1(E; F_1)$ and $\|v_2 u\|_1 \leq \|v_2\| \|u\|_1$ (prop. 14). Let $v_1 \in \mathcal{L}(E_1; E)$; since $u v_1 = (v_1^* u^*)^*$, we have $u v_1 \in \mathcal{L}_1(E_1; F)$ and $\|u v_1\|_1 \leq \|v_1^*\| \|u^*\|_1 = \|v_1\| \|u\|_1$ (lemma 8).

The proposition follows immediately from these inequalities.

PROPOSITION 17. — The space $\mathcal{L}_1(E; E)$ coincides with the space $\mathcal{L}_1(E)$ of endomorphisms of finite trace of E , and we have $|\text{Tr}(u)| \leq \|u\|_1$ for every endomorphism u of E with finite trace.

We may assume that $K = \mathbb{C}$. The space $\mathcal{L}_1(E; E)$ contains by definition the set of positive endomorphisms of finite trace, and hence $\mathcal{L}_1(E) \subset \mathcal{L}_1(E; E)$ (EVT, p. 50, def. 8). Conversely, let us show that $\mathcal{L}_1(E; E)$ is contained in $\mathcal{L}_1(E)$.

Let $u \in \mathcal{L}_1(E; E)$. We have $u^* \in \mathcal{L}_1(E; E)$ (lemma 8); it therefore suffices to show that the Hermitian elements of $\mathcal{L}_1(E; E)$ are of finite trace (lemma 2 of I, p. 96).

Let u be such an endomorphism. It is compact (remark 2, p. 169). Let B be an orthonormal basis of E such that u is diagonal in the basis B (th. 1 of IV, p. 149), and let λ be the family of eigenvalues of u relative to B . The endomorphism $|u|$ is diagonalizable in the basis B and the family of its eigenvalues is $|\lambda|$ (prop. 1, d) of IV, p. 147). Since $|u|$ is of finite trace, this latter family is summable (prop. 12 of IV, p. 164), hence the family λ is summable. Consequently, u is of finite trace (loc. cit.), and we have $|\text{Tr}(u)| \leq \text{Tr}(|u|) = \|u\|_1$.

Consequently, the space $\mathcal{L}_1(E)$ is a self-adjoint two-sided ideal of the C^* -algebra $\mathcal{L}(E)$. If

E is of infinite dimension, this ideal is not closed in $\mathcal{L}(E)$.

Propositions 17 and 16 show that the map b from $\mathcal{L}_1(E; F) \times \mathcal{L}(F; E)$ into $\mathbb{C}$ defined by $b(u, v) = \text{Tr}(vu)$ is a continuous bilinear form which puts the spaces $\mathcal{L}_1(E; F)$ and $\mathcal{L}(F; E)$ in duality.

Lemma 9. — Let $u \in \mathcal{L}_1(E; F)$. We have

$$\|u\|_1 = \sup_{\substack{v \in \mathcal{L}^c(F; E) \\ \|v\| \leq 1}} |b(u, v)|.$$

Since every Hilbert–Schmidt map is compact (cor. 2 of IV, p. 165), we have

$$\|u\|_1 \leq \sup_{\substack{v \in \mathcal{L}^c(E; F) \\ \|v\| \leq 1}} |\text{Tr}(vu)| = \sup_{\substack{v \in \mathcal{L}^c(F; E) \\ \|v\| \leq 1}} |b(u, v)|$$

by prop. 14.

Let $v \in \mathcal{L}^c(E; F)$ be of norm ≤ 1 . Since $\mathcal{L}_2(E; F)$ is dense in $\mathcal{L}^c(E; F)$, there exists a sequence $(v_n)_{n \in \mathbb{N}}$ in $\mathcal{L}_2(E; F)$ which converges to v in $\mathcal{L}(E; F)$. The sequence $(b(u, v_n))_{n \in \mathbb{N}}$ converges to $b(u, v)$; since $|b(u, v_n)| = |\text{Tr}(v_n u)| = |\langle v_n^* | u \rangle| \leq \|u\|_1$ (prop. 14) for all $n \in \mathbb{N}$, it follows that $|b(u, v)| \leq \|u\|_1$.

Denote by θ the continuous linear map

$$\theta: \mathcal{L}_1(E; F) \rightarrow \mathcal{L}^c(F; E)'$$

such that $\theta(u)(v) = b(u, v)$.

PROPOSITION 18. — The map θ is an isometric isomorphism.

By lemma 9, the map θ is isometric, and it suffices to show that θ is surjective.

Let $\lambda \in \mathcal{L}^c(F; E)'$. Since $\|v\| \leq \|v\|_2$ for all $v \in \mathcal{L}_2(F; E)$ (EVT, V, p. 52, formula (33)), the restriction of λ to the subspace $\mathcal{L}_2(F; E)$ is a continuous linear form on the Hilbert space $\mathcal{L}_2(F; E)$. There therefore exists an element u of $\mathcal{L}_2(F; E)$ such that $\lambda(v) = \langle u | v \rangle$ for all $v \in \mathcal{L}_2(F; E)$ (EVT, V, p. 15, th. 3).

For every $w \in \mathcal{L}_2(E; F)$ of norm ≤ 1 , we have $|\langle w | u \rangle| = |\lambda(w)| \leq \|\lambda\|$. Consequently, u is a nuclear map from E into F (prop. 14).

The continuous linear forms λ and $\theta(u)$ are equal on the dense subspace $\mathcal{L}_2(F, E)$ of $\mathcal{L}^c(F; E)$. It follows that $\lambda = \theta(u)$, which concludes the proof.

COROLLARY. — The normed space $\mathcal{L}_1(E; F)$ is a Banach space.

The assertion follows from the proposition and from EVT, III, p. 24, cor. 2, since the space $\mathcal{L}^c(F; E)$ is a normed space.

IV.1.9 Hilbert–Schmidt integral operators

In this issue, X and Y are locally compact topological spaces. Let μ be a positive measure on X and ν be a positive measure on Y . We denote by $L^2(X)$, $L^2(Y)$ and $L^2(X \times Y)$ the spaces $L^2(X, \mu)$, $L^2(Y, \nu)$ and $L^2(X \times Y, \mu \otimes \nu)$ respectively, and likewise for $\mathcal{L}^2(X)$, $\mathcal{L}^2(Y)$ and $\mathcal{L}^2(X \times Y)$.

We recall (cf. No. 4 of III, p. 26) that for every $N \in \mathcal{L}^2(X \times Y)$ there exists a unique continuous linear map u_N from $L^2(X)$ into $L^2(Y)$ such that

$$\langle g | u_N(f) \rangle = \int_{X \times Y} \overline{g(y)} N(x, y) f(x) d(\mu \otimes \nu)(x, y) \quad (12)$$

for all $f \in L^2(X)$ and every $g \in L^2(Y)$. The map u_N is compact (cor. 1 of III, p. 33). The map $N \mapsto u_N$ induces, by passage to quotients, a continuous and injective linear map from $L^2(X \times Y)$ into $\mathcal{L}(L^2(X); L^2(Y))$ (prop. 5 of III, p. 30).

We denote by θ the unique linear map from $\mathcal{L}^2(X) \otimes \mathcal{L}^2(Y)$ into $\mathcal{L}^2(X \times Y)$ which, for all $f \in \mathcal{L}^2(X)$ and $g \in \mathcal{L}^2(Y)$, sends $f \otimes g$ to the function defined by $(x, y) \mapsto f(x)g(y)$.

Lemma 10. — The map θ defines by passage to quotients a linear map $\tilde{\theta}$ from $L^2(X) \otimes L^2(Y)$ into $L^2(X \times Y)$ and there exists a unique isometric isomorphism from $L^2(X) \widehat{\otimes}_2 L^2(Y)$ onto $L^2(X \times Y)$ which coincides with it on $L^2(X) \otimes L^2(Y)$.

The first assertion is elementary. Let us prove the second.

Let $(f_i)_{i \in I}$ and $(g_j)_{j \in J}$ be orthonormal bases of $L^2(X)$ and $L^2(Y)$ respectively. The family $(\bar{f}_i \otimes g_j)_{(i,j) \in I \times J}$ is an orthonormal basis of $L^2(X) \widehat{\otimes}_2 L^2(Y)$ (EVT, V, p. 29, cor. 1) and the family $(\tilde{\theta}(\bar{f}_i \otimes g_j))_{(i,j) \in I \times J}$ is orthonormal in $L^2(X \times Y)$. The map $\tilde{\theta}$ extends therefore by continuity to an isometric linear map from $L^2(X) \widehat{\otimes}_2 L^2(Y)$ into $L^2(X \times Y)$. It remains to show that this extension is surjective.

For this, it suffices to prove that the image of $\tilde{\theta}$ is dense in $L^2(X \times Y)$. Let N be an element of $L^2(X \times Y)$ orthogonal to the image of $\tilde{\theta}$. For all $i \in I$ and $j \in J$, we have $\langle g_j | u_N(f_i) \rangle = \langle \tilde{\theta}(\bar{f}_i \otimes g_j) | N \rangle = 0$. From formula (12), it follows that $u_N = 0$ and therefore $N=0$.

The preceding lemma establishes the assertion stated in EVT, V, p. 29, example 2.

In what follows, we will identify $L^2(X) \widehat{\otimes}_2 L^2(Y)$ and $L^2(X \times Y)$ by the isometric isomorphism of Lemma 10.

PROPOSITION 19. — The map $N \mapsto u_N$ is an isometric isomorphism from $L^2(X \times Y)$ onto the space $\mathcal{L}_2(L^2(X); L^2(Y))$ of Hilbert–Schmidt maps from $L^2(X)$ into $L^2(Y)$.

The linear map from $L^2(Y) \otimes \overline{L^2(X)}$ into $\mathcal{L}_2(L^2(X); L^2(Y))$ that sends $g \otimes f$ to the Hilbert–Schmidt map $h \mapsto \langle f | h \rangle g$ extends to an isometric isomorphism θ_1 of $L^2(Y) \widehat{\otimes}_2 \overline{L^2(X)}$ onto $\mathcal{L}_2(L^2(X); L^2(Y))$ (EVT, V, p. 52, th. 1).

Additionally, let us denote by θ_2 the isometric isomorphism of $L^2(X) \widehat{\otimes}_2 L^2(Y)$ onto $L^2(Y) \widehat{\otimes}_2 \overline{L^2(X)}$ which sends $f \otimes g$ to $g \otimes \bar{f}$ for all $f \in L^2(X)$ and every $g \in L^2(Y)$.

The linear map $\theta_3 = \theta_1 \circ \theta_2$ is identified with an isometric isomorphism of $L^2(X \times Y)$ onto $\mathcal{L}_2(L^2(X); L^2(Y))$.

Let $f \in L^2(X)$ and $g \in L^2(Y)$ and let N be the element of $L^2(X \times Y)$ identified with $f \otimes g$. By INT, V, p. 95, § 8, no. 3, cor. 2 and formula (12), we have

$$\langle g_1 | u_N(f_1) \rangle = \langle \bar{f} | f_1 \rangle \langle g_1 | g \rangle$$

for all $f_1 \in L^2(X)$ and $g_1 \in L^2(Y)$. The map $u = \theta_3(N)$ is the linear map $\theta_1(g \otimes \bar{f})$; it therefore satisfies $u(h) = \langle \bar{f} | h \rangle g$ for all $h \in L^2(X)$. Consequently, for all $f_1 \in L^2(X)$ and $g_1 \in L^2(Y)$, it follows that

$$\langle g_1 | u(h_1) \rangle = \langle \bar{f} | h_1 \rangle \langle g_1 | g \rangle = \langle g_1 | u_N(h_1) \rangle,$$

whence $\theta_3(N) = u_N$. Since θ_3 and $N \mapsto u_N$ are continuous, we conclude that $\theta_3(N) = u_N$ for all $N \in L^2(X \times Y)$, which concludes the proof.

COROLLARY. — For every $N \in L^2(X \times Y)$, the linear map u_N is a Hilbert–Schmidt operator and we have $\text{Tr}(u_N^* u_N) = \|N\|^2$.

Indeed, one has $\|u\|_2 = \sqrt{\text{Tr}(u^*u)}$ for all $u \in \mathcal{L}_2(L^2(X); L^2(Y))$.

Note. — Let $N \in L^2(X \times Y)$. By Corollary 2 of IV, p. 150, there exist a countable set I , orthonormal families $(f_i)_{i \in I}$ in $L^2(X)$ and $(g_i)_{i \in I}$ in $L^2(Y)$, and a family $(\alpha_i)_{i \in I}$ in $\mathbf{R}_+^*$ such that

$$u_N(f) = \sum_{i \in I} \alpha_i \langle f_i | f \rangle g_i$$

for all $f \in L^2(X)$, where the series converges in $L^2(Y)$. By cor. 4 of IV, p. 165 and the corollary above, we have

$$\sum_{i \in I} \alpha_i^2 = \text{Tr}(u_N^* u_N) = \|u_N\|_2^2 = \int_{X \times Y} |N(x, y)|^2 d(\mu \otimes \nu)(x, y).$$

Additionally, let us note that $h_{i,j} = \bar{f}_i \otimes g_j \in L^2(X \times Y)$ for all $(i, j) \in I \times I$. We then have

$$N = \sum_{i \in I} \alpha_i h_{i,i}$$

in $L^2(X \times Y)$.

Indeed, let $(f_j)_{j \in J}$ and $(g_k)_{k \in K}$ be orthonormal bases of $L^2(X)$ and $L^2(Y)$, respectively, extending the families $(f_i)_{i \in I}$ and $(g_i)_{i \in I}$. Set $h_{j,k} = \bar{f}_j \otimes g_k$ for all $(j, k) \in J \times K$. By Lemma 10, the family $(h_{j,k})_{(j,k) \in J \times K}$ is an orthonormal basis of $L^2(X \times Y)$. We have $\langle h_{j,k} | N \rangle = \langle g_k | u_N(f_j) \rangle$ for all $(j, k) \in J \times K$. If $j \notin I$ this quantity is zero. If $j \in I$, it is equal to $\alpha_j \langle g_k | g_j \rangle$, hence is zero unless $k=j$, in which case $\langle h_{j,j} | N \rangle = \alpha_j$. Consequently

$$N = \sum_{(j,k) \in J \times K} \langle h_{j,k} | N \rangle h_{j,k} = \sum_{i \in I} \alpha_i h_{i,i}.$$

IV.1.10 Trace of integral operators with continuous kernel

In this section, we keep the conventions of the preceding section with $Y = X$ and $\nu = \mu$. We assume that X is a locally compact topological space that is countable at infinity (TG, I, p. 68, def. 5).

In particular, we identify the spaces $L^2(X \times X)$ and $L^2(X) \widehat{\otimes}_2 L^2(X)$ (lemma 10 of IV, p. 172). We shall denote by $\tilde{f}$ the class in $L^2(X)$ (resp. in $L^2(X \times X)$) of a function $f \in \mathcal{L}^2(X)$ (resp. in $\mathcal{L}^2(X \times X)$).

PROPOSITION 20. — Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable maps from X into a metrizable space F . Suppose that the limit of $f_n(x)$ exists outside a μ -negligible subset of X . There exists a sequence $(C_m)_{m \in \mathbb{N}}$ of compact subsets of X whose union $\tilde{X}$ satisfies $|\mu|(X - \tilde{X}) = 0$, such that the functions f_n are continuous on C_m for all $n \in \mathbb{N}$ and every $m \in \mathbb{N}$, and the sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f on C_m for all $m \in \mathbb{N}$.

It follows from th. 2 of INT, IV, p. 175, § 5, no. 4 and from prop. 12, b) of INT, IV, p. 188, § 5, no. 8, that the set of compact subsets C of X such that the functions f_n are continuous on C for every n , and such that (f_n) converges uniformly to f on C , is μ -dense in X (INT, IV, p. 189, § 5, no. 8, def. 6). The assertion then follows from the remark of INT, IV, p. 189, § 5, no. 8.

Let N be a function belonging to $\mathcal{L}^2(X \times X)$ and u_N the Hilbert-Schmidt map from $L^2(X)$ into $L^2(X)$ with kernel N (Prop. 19 of IV, p. 173). Since the map u_N is compact, there exist, by Prop. 9 of IV, p. 156, an element M of $\bar{\mathbb{N}}$ and, denoting by I the set of integers $n \in \mathbb{N}$ such that $n \leq M$, orthonormal families $(f_i)_{i \in I}$ and $(g_i)_{i \in I}$ in $\mathcal{L}^2(X)$ and a family $(\alpha_i)_{i \in I}$ in $\mathbf{R}_+^*$ such that

$$u_N(f) = \sum_{i \in I} \alpha_i \langle \tilde{f}_i | f \rangle \tilde{g}_i \quad (13)$$

for all $f \in L^2(X)$, where the series converges in $L^2(X)$.

For every $i \in I$, we denote by $h_i \in \mathcal{L}^2(X \times X)$ the function defined by $h_i(x, y) = \overline{f_i(x)} g_i(y)$, so that $\tilde{h}_i$ is the class of $\tilde{f}_i \otimes g_i$. By Remark 9 of IV, p. 173, the series with general term $\alpha_i h_i$ converges to N in $L^2(X \times X)$.

In the rest of this issue, we assume that N is a continuous function.

PROPOSITION 21. — Suppose that u_N has finite trace. There then exists a set $\tilde{X} \subset X$ whose complement $X - \tilde{X}$ is μ -negligible and a function $H \in \mathcal{L}^2(X \times X)$ satisfying the following conditions:

- (i) For every $(x, y) \in \tilde{X} \times \tilde{X}$ the family $(\alpha_i \overline{f_i(x)} g_i(y))_{i \in I}$ is summable in $\mathbf{C}$ and its sum is $N(x, y)$;
- (ii) For every finite subset J of I and every $(x, y) \in \tilde{X} \times \tilde{X}$, we have

$$\left| \sum_{i \in J} \alpha_i h_i(x, y) \right| \leq H(x, y); \quad (14)$$

(iii) The function $x \mapsto H(x, x)$ belongs to $\mathcal{L}^1(X)$.

Since u_N has finite trace, the family (α_i) is summable (cor. 4 of IV, p. 165). The series

$$\sum_{i \in I} \alpha_i |f_i|^2, \quad \sum_{i \in I} \alpha_i |g_i|^2$$

therefore converge in $\mathcal{L}^1(X)$ and consequently μ -almost everywhere. Let F and G denote, respectively, the functions defined μ -almost everywhere by the sum of these series, setting $F(x) = 0$ and $G(x) = 0$ for all x such that the corresponding series does not converge. Since F and G belong to $\mathcal{L}^1(X)$, the function H defined by $H(x, y) = \sqrt{F(x)G(y)}$ belongs to $\mathcal{L}^2(X \times X)$ (INT, V, p. 95, § 8, no. 3, cor. 2).

Let us apply Prop. 20 to the space X , with $F = C^2$ and with the measurable maps

$$s_n : x \mapsto \left(\sum_{\substack{i \in I \\ i \leq n}} \alpha_i |f_i(x)|^2, \sum_{\substack{i \in I \\ i \leq n}} \alpha_i |g_i(x)|^2 \right)$$

defined on X . There therefore exists a sequence $(C_m)_{m \in \mathbb{N}}$ of compact subsets of X satisfying the following conditions:

- (1) the union $\tilde{X}$ satisfies $\mu(X - \tilde{X}) = 0$;
- (2) for all $m \in \mathbb{N}$ and every $n \in \mathbb{N}$, the functions s_n are continuous on C_m ;
- (3) for all $m \in \mathbb{N}$, the series $\sum \alpha_i |f_i|^2$ and $\sum \alpha_i |g_i|^2$ converge uniformly on C_m to F and G , respectively;
- (4) the support of the measure induced by μ on C_m is equal to C_m (possibly after replacing C_m by the support of this measure).

Let us prove that the set $\tilde{X}$ and the function H satisfy conditions (i), (ii), and (iii).

Let $(x, y) \in \tilde{X} \times \tilde{X}$. For every finite subset $J \subset I$, we have

$$\left| \sum_{i \in J} \alpha_i \overline{f_i(x)} g_i(y) \right| \leq \left(\sum_{i \in J} \alpha_i |f_i(x)|^2 \right)^{1/2} \left(\sum_{i \in J} \alpha_i |g_i(y)|^2 \right)^{1/2} \quad (15)$$

whence also

$$\left| \sum_{i \in J} \alpha_i \overline{f_i(x)} g_i(y) \right| \leq H(x, y), \quad (16)$$

which already establishes property (ii).

By inequality (15), the series $\sum_i \alpha_i h_i$ converges uniformly on $K_m \times K_n$ for all $(m, n) \in \mathbb{N}^2$.

Let $\tilde{N}$ be the function on $X \times X$ defined by

$$\tilde{N}(x, y) = \sum_{i \in I} \alpha_i h_i(x, y) = \sum_{i \in I} \alpha_i \overline{f_i(x)} g_i(y)$$

for all $(x, y) \in \tilde{X} \times \tilde{X}$ and by $\tilde{N}(x, y) = 0$ otherwise. The function $\tilde{N}$ is measurable (INT, IV, p. 175, § 5, n°4, th. 2), and it is continuous on $K_m \times K_n$ for all $(m, n) \in \mathbb{N}^2$.

From (16), we have $|\tilde{N}(x, y)| \leq H(x, y)$ for all $(x, y) \in X \times X$, and in particular $\tilde{N}$ belongs to $\mathcal{L}^2(X \times X)$ (INT, IV, p. 84, § 5, no. 6, th. 5).

Let us prove that $N = \tilde{N}$ on $\tilde{X} \times \tilde{X}$, which will establish property (i). Let f and g be elements of $\mathcal{L}^2(X)$. We have

$$\langle g | u_{\tilde{N}}(f) \rangle = \int_{X \times X} \tilde{N}(x, y) f(x) \overline{g(y)} d(\mu \otimes \mu)(x, y)$$

(formula (12) of IV, p. 172). For every $(x, y) \in \tilde{X} \times \tilde{X}$ and every finite subset J of I , we have

$$\left| \sum_{i \in J} \alpha_i \overline{f_i(x)} g_i(y) f(x) \overline{g(y)} \right| \leq |f(x) g(y)| H(x, y)$$

by formula (16). Since the right-hand side of this inequality is integrable on $X \times X$ (INT, V, p. 95, § 8, n° 3, cor. 2), one can apply Lebesgue's theorem (INT, IV, p. 137, § 3, n° 7, th. 6) and formula (13) to deduce that

$$\begin{aligned} \langle g | u_{\tilde{N}}(f) \rangle &= \sum_{i \in I} \alpha_i \int_{X \times X} \overline{f_i(x)} g_i(y) f(x) \overline{g(y)} d(\mu \otimes \mu)(x, y) \\ &= \sum_{i \in I} \alpha_i \langle f_i | f \rangle \langle g | g_i \rangle = \langle g | u_N(f) \rangle. \end{aligned}$$

Therefore, it follows that $u_{\tilde{N}} = u_N$, whence $N = \tilde{N}$ in $L^2(X \times X)$ (prop. 3 of III, p. 28, b)).

For every $(m, n) \in \mathbb{N}^2$, the functions N and $\tilde{N}$ are continuous on $K_m \times K_n$, therefore are equal on $K_m \times K_n$ (prop. 9 of INT, III, p. 69, § 2, n°2) since the support of the measure induced

by $\mu \otimes \mu$ onto $K_m \times K_n$ is equal to $K_m \times K_n$. Therefore N coincides with $\tilde{N}$ on $\tilde{X} \times \tilde{X}$.

Finally, the upper bound $\sqrt{FG} \leq (F + G)/2$ implies that the function $x \mapsto \sqrt{F(x)G(x)} = H(x, x)$ is integrable on X , whence property (iii).

In the remainder of this issue, we keep the notation of the proposition.

Theorem 2. — Suppose that u_N has finite trace. Then the function $x \mapsto N(x, x)$ belongs to $\mathcal{L}^1(X)$ and we have

$$\text{Tr}(u_N) = \int_X N(x, x) d\mu(x).$$

By conditions (i) and (ii), we have $|N(x, x)| \leq H(x, x)$ for $x \in \tilde{X}$, hence the function $x \mapsto N(x, x)$ belongs to $\mathcal{L}^1(X)$ by condition (iii).

Condition (i) proves the equality.

$$N(x, x) = \sum_{i \in I} \alpha_i \overline{f_i(x)} g_i(x)$$

for all $x \in \tilde{X}$. By conditions (ii) and (iii), one may apply Lebesgue's theorem (INT, IV, p. 137, § 3, no. 7, th. 6), from which it follows that

$$\begin{aligned} \int_X N(x, x) d\mu(x) &= \sum_{i \in I} \alpha_i \int_X \overline{f_i(x)} g_i(x) d\mu(x) \\ &= \sum_{i \in I} \alpha_i \langle f_i | g_i \rangle = \text{Tr}(u_N), \end{aligned}$$

the last equality resulting from the corollary of Proposition 17 of IV, p. 170.

Lemma 11. — If the endomorphism u_N of $L^2(X)$ is positive, then $N(x, x) \geq 0$ for all x in the support of μ .

Since u_N is positive, the families (f_i) and (g_i) can be chosen such that $f_i = g_i$ for all $i \in I$ (for remark 3 of IV, p. 151). For every $x \in \tilde{X}$, we then have

$$N(x, x) = \sum_{i \in I} \alpha_i |f_i(x)|^2 \geq 0.$$

Since N is continuous and $\mu(X - \tilde{X}) = 0$, the function $x \mapsto N(x, x)$ is nonnegative on the support of μ .

Remark. — The converse assertion of the lemma is not valid (exercise 14 of IV, p. 316).

PROPOSITION 22. — Suppose that u_N is a positive endomorphism of $L^2(X)$. Then the endomorphism u_N has finite trace if and only if the function $x \mapsto N(x, x)$ is μ -integrable. In this case, we have

$$\text{Tr}(u_N) = \int_X N(x, x) d\mu(x).$$

If u_N has finite trace, Theorem 2 implies that $x \mapsto N(x, x)$ is integrable on X and that its integral is the trace of u_N .

Conversely, suppose that the function $x \mapsto N(x, x)$ is integrable. Since u_N is positive, the orthonormal families $(f_i)_{i \in I}$ and $(g_i)_{i \in I}$ can be chosen such that $f_i = g_i$ for all $i \in I$ (remark 3 of IV, p. 151). For every $x \in \tilde{X}$, we have

$$N(x, x) = \sum_{i \in I} \alpha_i |f_i(x)|^2,$$

whence, for every finite subset J of I , we deduce

$$\begin{aligned} \sum_{i \in J} \alpha_i &= \sum_{i \in J} \alpha_i \int_X |f_i(x)|^2 d\mu(x) \\ &= \int_X \sum_{i \in J} \alpha_i |f_i(x)|^2 d\mu(x) \leq \int_X N(x, x) d\mu(x). \end{aligned}$$

The family $(\alpha_i)_{i \in I}$ is therefore summable, which implies that the endomorphism u_N has finite trace (prop. 12 of IV, p. 164).

Remark. — Even when X is compact and N is continuous, the endomorphism u_N of $L^2(X)$ is not always of finite trace (cf. exercise 8 of IV, p. 314).

IV.2 NORMAL ENDOMORPHISMS

Throughout this paragraph, the Hilbert spaces considered are complex, unless stated otherwise.

IV.2.1 Complements on spaces $L^p(X, \mu)$

PROPOSITION 1. — Let X be a locally compact topological space. Let μ be a positive measure on X and $p \in [1, +\infty[$. Let $(X_i)_{i \in I}$ be a locally countable family (INT, IV, p. 190, § 5, no. 9, def.

7) of pairwise disjoint locally compact subsets of X such that the complement of the union of the X_i is locally μ -negligible. Let φ_i be the characteristic function of X_i and μ_i be the measure induced by μ on X_i (INT, IV, p. 186, §5, n° 7, def. 4).

- a) The map $p_i : f \mapsto f\varphi_i$ is a projector of $\mathcal{L}^p(X, \mu)$ and, by passage to quotients, defines a projector $\tilde{p}_i$ of $L^p(X, \mu)$. The latter is an orthogonal projector of $L^2(X, \mu)$ if $p = 2$;
- b) The map $f \mapsto f|_{X_i}$ induces an isometric isomorphism from the image of p_i onto the space $\mathcal{L}^p(X_i, \mu_i)$ and, by passage to quotients, defines an isometric isomorphism from the image of $\tilde{p}_i$ onto $L^p(X_i, \mu_i)$, by means of which these two spaces are identified;
- c) The sum of the spaces $L^p(X_i, \mu_i)$ is total in the space $L^p(X, \mu)$. In the case $p = 2$, the space $L^2(X, \mu)$ is the Hilbert sum of the spaces $L^2(X_i, \mu_i)$.

The assertion *a*) is elementary. The assertion *b*) follows from the scholium of INT, V, p. 84, § 7, n° 1.

Let $q \in]1, +\infty]$ be the conjugate exponent of p . One identifies the dual of $L^p(X, \mu)$ with $L^q(X, \mu)$ (INT, V, p. 61, § 5, n° 8, th. 4). Let f be a function in $\mathcal{L}^q(X, \mu)$ whose class $\tilde{f}$ in $L^q(X, \mu)$ satisfies $\langle \tilde{f}, \tilde{p}_i(\varphi) \rangle = 0$ for all $i \in I$ and every $\varphi \in L^p(X, \mu)$. Since the image of p_i contains $\mathcal{K}(X_i)$, it follows that the measure $(f|_{X_i}) \cdot \mu_i$ on X_i is zero, which means that the restriction of f to X_i is locally μ_i -negligible on X_i (INT, V, p. 46, § 5, n° 3, cor. 2). Since the complement in X of the union of the X_i is locally μ -negligible, one concludes that the function f is locally μ -negligible on X (INT, IV, p. 190, § 5, n° 9 and p. 172, n° 2, prop. 5). The class of f is then zero in $L^q(X, \mu)$ (using INT, V, p. 8, § 1, n° 3, lemma 1 and the corollary of proposition 9 when $p \neq 1$). By the Hahn–Banach theorem (EVT, II, p. 46, cor. 1), the first part of assertion *c*) follows.

If $p = 2$, the image of p_i is orthogonal to that of p_j for all $i \neq j$ since X_i and X_j are then disjoint, whence the last assertion.

PROPOSITION 2. — Let X be a locally compact space that is countable at infinity. Let μ be a positive measure on X . For every $p \in [1, +\infty[$, the space $L^p(X, \mu)$ is of countable type.

Let $(U_n)_{n \in \mathbb{N}}$ be a sequence of relatively compact open sets of X whose union is equal to X and which satisfy $\overline{U_n} \subset U_{n+1}$ for all $n \in \mathbb{N}$ (TG, I, p. 68, prop. 15). For every $n \in \mathbb{N}$, the space

$\mathcal{H}(X, \overline{U}_n)$ is identified with a closed subspace of the Banach space $\mathcal{C}(\overline{U}_n)$ (INT, III, p. 40, § 1, n° 1); since the latter space is of countable type (TG, X, p. 25, corollary), the same is true of $\mathcal{H}(X, \overline{U}_n)$ (TG, IX, p. 19, cor., (i)). Let F_n be a countable dense subset of $\mathcal{H}(X, \overline{U}_n)$.

Let $f \in \mathcal{L}^p(X, \mu)$ and let $\varepsilon > 0$. There exists an integer $n \in \mathbb{N}$ such that $\int_{X-U_n} |f|^p < \varepsilon/2$, and there exists $g \in \mathcal{F}_n$ such that $\int_{\overline{U}_n} |f - g|^p < \varepsilon/2$. The union of the classes in $L^p(X, \mu)$ of the elements of the sets $\mathcal{F}_n$ is therefore dense in $L^p(X, \mu)$, which concludes the proof (TG, IX, p. 18, prop. 12).

IV.2.2 Essential image of a measurable function

In this issue, we denote by X a locally compact topological space, and by μ a positive measure on X .

DEFINITION 1. — Let Y be a topological space. For every measurable function g from X to Y , the μ -essential image of g is the subset of the $y \in Y$ such that, for every open neighborhood U of y , the set $g^{-1}(U)$ is not locally μ -negligible in X (INT, IV, p. 172, § 5, no. 2, def. 3).

Lemma 1. — Let g be a μ -measurable function from X into a topological space Y and let S be its μ -essential image. The set of elements $x \in X$ such that $g(x)$ does not belong to S is locally μ -negligible in X .

Let $Z = g^{-1}(Y - S)$ be the set in question.

Suppose first that X is compact and g is continuous. In this case, the pushforward measure $g(\mu)$ is defined (INT, V, p. 69, § 6, no. 1, def. 1) since μ is a bounded measure. It follows from the definitions that the μ -essential image of g is the support of the measure $g(\mu)$ (cf. INT, V, p. 70, § 6, no. 2, cor. 2), whence $\mu(Z) = g(\mu)(Y - S) = 0$ (INT, IV, p. 118, § 2, no. 2, prop. 5).

Now consider the general case. Let C be a compact subset of X , and let $\varepsilon > 0$ be a real number. Since g is measurable, there exists a compact subset $C_1 \subset C$ such that $\mu(C - C_1) \leq \varepsilon$ and such that $g|_{C_1}$ is continuous (INT, IV, p. 169, § 5, no. 1, prop. 1). We then have

$$\mu(Z \cap C) \leq \mu(C - C_1) + \mu(Z \cap C_1) \leq \varepsilon + \mu(Z \cap C_1).$$

Let μ_1 be the measure induced by μ on C_1 (INT, IV, p. 186, § 5, no. 7, def. 4). Let S_1 be the μ_1 -essential image of $g|_{C_1}$. We have $Z \cap C_1 \subset (g|_{C_1})^{-1}(Y - S_1)$. By the first case, the

set $Z \cap C_1$ is therefore μ_1 -negligible, and consequently μ -negligible (INT, IV, p. 187, § 5, no. 7, lemma 2, (i)). The above inequality becomes $\mu(Z \cap C) \leq \varepsilon$; since ε and C are arbitrary, the set Z is locally μ -negligible (INT, IV, p. 172, § 5, no. 2, prop. 5).

Lemma 2. — Let g be a continuous function from X into $\mathbb{C}$. The μ -essential image of g is the closure of the image of the support of μ under g .

Let Y be the support of μ . If $z \in \mathbb{C}$ is not adherent to $g(Y)$, then there exists an open neighborhood U of z such that the open set $g^{-1}(U)$ does not meet Y and is therefore locally μ -negligible; this means that z does not belong to the μ -essential image of g .

Conversely, if $z \in \mathbb{C}$ is adherent to $g(Y)$, then for every open neighborhood U of z , the set $g^{-1}(U)$ is an open set in X which meets Y . It is therefore not μ -negligible, and since it is open, it is not locally μ -negligible (INT, IV, p. 172, § 5, n°2, cor. 2). Hence z belongs to the μ -essential image of g .

Lemma 3. — Let g be a continuous function from X into $\mathbb{C}$ and let S be its μ -essential image. Suppose that S is nonempty and that $0 \notin S$, and denote by δ the distance from 0 to S . Then $\delta > 0$.

Set $h(x) = 1/g(x)$ if $g(x) \neq 0$ and $h(x) = 0$ otherwise. The function h belongs to $\mathcal{L}^\infty(X, \mu)$. Let $\tilde{h}$ be the class of h in $L^\infty(X, \mu)$. We then have the formula $\delta^{-1} = \|\tilde{h}\|_\infty$.

Let U be an open neighborhood of 0 such that the open set $Z = g^{-1}(U)$ is locally μ -negligible, hence negligible (INT, IV, p. 172, § 5, no. 2, cor. 2). Let Y be the support of μ ; we have $Y \subset X - Z$. The restriction of the function h to $X - Z$ is continuous and bounded, hence $h \in \mathcal{L}^\infty(X, \mu)$ and the norm of $\tilde{h}$ in $\mathcal{L}^\infty(X, \mu)$ is equal to the norm of its restriction to $X - Z$. Moreover, for every $\alpha \in \mathbb{R}_+$ the set of $x \in X - Z$ such that $|h(x)| > \alpha$ is open in $X - Z$; it is therefore locally μ -negligible if and only if it does not meet Y (INT, IV, loc. cit. and INT, III, p. 66, § 3, no. 2, def. 1). Consequently, we have

$$\|\tilde{h}\|_\infty = \sup_{x \in Y} \frac{1}{|g(x)|},$$

whence

$$\frac{1}{\|\tilde{h}\|_\infty} = \inf_{x \in Y} |g(x)| = \inf_{\lambda \in g(Y)} |\lambda| = \inf_{\lambda \in g(\bar{Y})} |\lambda|$$

which is equal to δ by the preceding lemma.

IV.2.3 Universally measurable functions

In this issue, X is a locally compact topological space. We recall (INT, V, p. 28, § 3, n° 4, def. 2) that a map f from X into a topological space Y is universally measurable if it is μ -measurable for every positive measure μ on X . It suffices for this that f be μ -measurable for every positive measure μ with compact support on X (INT, V, p. 28, § 4, no. 3, prop. 6).

Lemma 4. — Let Y and Z be locally compact topological spaces, $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ universally measurable mappings. The mapping $g \circ f: X \rightarrow Z$ is universally measurable.

Let μ be a positive measure with compact support on X , and let C be its support. The restriction of f to C is $(\mu|C)$ -proper. The map g is measurable with respect to the image measure $(f|C)(\mu|C)$, so the map $(g \circ f)|C = g \circ (f|C)$ is $(\mu|C)$ -measurable. Consequently, the map $g \circ f$ is μ -measurable. The lemma follows.

We denote by $\mathcal{L}_u(X)$ the vector space of complex-valued functions that are universally measurable on X and by $\mathcal{L}_u^\infty(X)$ the subspace of functions $f \in \mathcal{L}_u(X)$ which are bounded on X . For $f \in \mathcal{L}_u^\infty(X)$, we denote by $\|f\|_\infty = \sup_{x \in X} |f(x)|$.

Proposition 3. — a) The set $\mathcal{L}_u(X)$ is an involutive subalgebra of the involutive algebra of functions from X to $\mathbb{C}$;

b) The set $\mathcal{L}_u^\infty(X)$ is a subalgebra of $\mathcal{L}_u(X)$ and the mapping defined by $f \mapsto \|f\|_\infty$ is a norm on $\mathcal{L}_u^\infty(X)$ for which $\mathcal{L}_u^\infty(X)$ is a unital star algebra;

c) The algebra $\mathcal{L}_u(X)$ contains $\mathcal{C}(X)$ and $\mathcal{L}_u^\infty(X)$ contains $\mathcal{C}_b(X)$;

d) Every Borel function from X to $\mathbb{C}$ (TG, IX, p. 61, def. 5) belongs to $\mathcal{L}_u(X)$;

e) For any sequence $(f_n)_{n \in \mathbb{N}}$ in $\mathcal{L}_u(X)$ which converges pointwise to a function f from X into $\mathbb{C}$, we have $f \in \mathcal{L}_u(X)$.

The assertions a) and c) follow from the definitions (cf. INT, IV, p. 175, § 5, no. 3, cor. 3). Assertion d) is a consequence of INT, IV, p. 179, § 5, no. 5, th. 4, since the inverse image under a Borel function of every Borel subset of $\mathbb{C}$ is a Borel subset of X , which is universally

measurable (INT, V, p. 28, § 3, no. 4). Finally, assertion e) follows from Egoroff's theorem (INT, IV, p. 175, § 5, no. 4, th. 2).

To prove assertion b), it suffices to note that assertion e) implies, *a fortiori*, that a uniform limit of universally measurable functions is universally measurable.

There may exist universally measurable functions on X that are not Borel. Indeed, if X is metrizable, the characteristic function of a Souslin subset of X is universally measurable (cf. INT, IV, p. 171, § 5, no. 1, cor. 2); however, in every uncountable Polish space, there exists a Souslin subset that is not Borel ("Souslin's theorem"; this follows from TG, IX, p. 120, exerc. 8 and from the fact that for every uncountable Polish space X , there exists a Borel bijection $X \rightarrow \mathbb{N}^{\mathbb{N}}$ whose inverse is Borel).

Lemma 5. — Let μ be a positive measure on X . Let g be a μ -measurable function from X into $\mathbb{C}$ and let S be its μ -essential image. For every function $f \in \mathcal{L}_u(S)$, the function h from X into $\mathbb{C}$ such that $h(x) = 0$ if $g(x) \notin S$ and $h(x) = f(g(x))$ if $g(x) \in S$ is μ -measurable.

Let $x \in X$ and U be an open relatively compact neighborhood of x . The set $N = U - (U \cap g^{-1}(S))$ is $(\mu|_U)$ -negligible. For every integer $n \geq 1$, let V_n be an open set such that $N \subset V_n \subset U$ and $\mu(V_n - N) < 1/n$ (INT, IV, p. 116, § 1, no. 4, prop. 19).

The restriction $\tilde{g}$ of g to $U - V_n$ is $\mu|(U - V_n)$ -proper and its image is contained in S . Since f is universally measurable, the map $x \mapsto f(g(x))$ from $U - V_n$ into $\mathbb{C}$ is measurable with respect to the measure $\mu|(U - V_n)$ (INT, V, p. 71, § 6, no. 2, prop. 3). The map h_n from X into $\mathbb{C}$ such that $h_n(x) = 0$ if $x \notin U - V_n$ and $h_n(x) = f(g(x))$ if $x \in U - V_n$ is therefore μ -measurable (INT, IV, p. 193, § 5, no. 10, prop. 16).

Since $h_n(x) \rightarrow h(x)$ for μ -almost every $x \in U$, the restriction of h to U is μ -measurable (INT, IV, p. 175, § 5, no. 4, th. 2). One concludes that h is μ -measurable by the localization principle (INT, IV, p. 171, § 4, no. 2, prop. 4).

Remark. — Under the hypotheses of the lemma, let us denote, by abuse of language, $f \circ g$ the function h defined in the lemma.

If $g = g_1$ locally μ -almost everywhere on X , the functions g and g_1 have the same μ -essential image and one has $f \circ g = f \circ g_1$.

If g is a μ -measurable function defined μ -almost everywhere on X , we shall also denote by

$f \circ g$ the function $f \circ g_1$, where g_1 is a μ -measurable function defined on X equal to g locally μ -almost everywhere.

IV.2.4 The star algebra $L^\infty(X, \mu)$

Let X be a locally compact topological space and let μ be a positive measure on X . We consider the commutative star algebra $L^\infty(X, \mu)$ (example 4 of I, p. 103).

Lemma 6. — Let $g \in \mathcal{L}^\infty(X, \mu)$. The spectrum of the class of g in $L^\infty(X, \mu)$ is equal to the μ -essential image of g .

Denote by $\tilde{g}$ the class of g in $L^\infty(X, \mu)$ and S the μ -essential image of g . Let $z \in \mathbf{C} - S$. By definition, there exists an open neighborhood U of z such that $Y = g^{-1}(U)$ is locally μ -negligible. The function h defined by $h(x) = (g(x) - z)^{-1}$ if $x \notin Y$, and $h(x) = 0$ if $x \in Y$, belongs to $\mathcal{L}^\infty(X, \mu)$. Its class $\tilde{h}$ in $L^\infty(X, \mu)$ satisfies $(\tilde{g} - z)\tilde{h} = 1$ since $(g(x) - z)h(x) = 1$ for all x not belonging to the locally μ -negligible set Y . Therefore $z \in \mathbf{C} - \text{Sp}(\tilde{g})$.

Conversely, let $z \in \mathbf{C} - \text{Sp}(\tilde{g})$. Let $h \in \mathcal{L}^\infty(X, \mu)$ be a function whose class is the inverse of $\tilde{g} - z$ in $L^\infty(X, \mu)$. There exists a real number $M > 0$ such that $|h(x)| \leq M$ locally μ -almost everywhere, and moreover we have $(g(x) - z)h(x) = 1$ locally μ -almost everywhere. Let U be the open ball with center z and radius M^{-1} in $\mathbf{C}$; then $g^{-1}(U)$ is contained in the locally μ -negligible set

$$h^{-1}(]M, +\infty[) \cup \{x \in X \mid (g(x) - z)h(x) \neq 1\},$$

hence $z \in S$. We have proved the lemma.

PROPOSITION 4. — Let $g \in \mathcal{L}^\infty(X, \mu)$. Denote by $\tilde{g}$ the class of g in $L^\infty(X, \mu)$ and S the spectrum of $\tilde{g}$. The map $f \mapsto f \circ g$ (remark, p. 184) is the functional calculus mapping from $\mathcal{C}(S)$ into $L^\infty(X, \mu)$ associated with $\tilde{g}$.

Let $f \in \mathcal{C}(S)$. The function $h = f \circ g$ is μ -measurable (Lemma 5 of IV, p. 184) and bounded. Denote by $\widetilde{f \circ g}$ its class in $L^\infty(X, \mu)$. The map $f \mapsto \widetilde{f \circ g}$ is a continuous unital involutive algebra homomorphism from $\mathcal{C}(S)$ into $L^\infty(X, \mu)$. Since $g(x)$ belongs to S locally μ -almost everywhere (Lemma 1 of IV, p. 181 and Lemma 6), this homomorphism sends the identity map on S to the class $\tilde{g}$ of the function g . It therefore coincides with the functional

calculus mapping of $\tilde{g}$ (Prop. 7 of I, p. 111).

IV.2.5 Multiplication endomorphisms

Let X be a locally compact topological space and let μ be a positive measure on X . Let $p \in [1, +\infty[$.

Let $g \in \mathcal{L}^\infty(X, \mu)$. Denote by m_g the map $f \mapsto g \cdot f$ of $\mathcal{L}^p(X, \mu)$ into itself. The map m_g is linear and continuous since

$$N_p(m_g(f)) \leq N_\infty(g)N_p(f) \quad (1)$$

for every function $f \in \mathcal{L}^p(X, \mu)$. In particular, the function $m_g(f)$ is μ -negligible if f is μ -negligible. The map m_g therefore induces, by passing to quotients, an endomorphism of $L^p(X, \mu)$, which will be denoted by $\tilde{m}_g$.

Lemma 7. — Let $g \in \mathcal{L}^\infty(X, \mu)$. The endomorphism $\tilde{m}_g$ of $L^p(X, \mu)$ is injective if and only if the set of $x \in X$ such that $g(x) = 0$ is locally μ -negligible.

Let A denote the set of $x \in X$ such that $g(x) = 0$.

Suppose that A is locally μ -negligible. Let $f \in \mathcal{L}^p(X, \mu)$ and $\tilde{f}$ its class in $L^p(X, \mu)$. Suppose that $\tilde{m}_g(\tilde{f}) = 0$. By definition, this is the case if and only if the function fg is μ -negligible, so that the set B of $x \in X$ such that $f(x)g(x) \neq 0$ is μ -negligible. The function f vanishes outside $A \cup B$, hence is μ -negligible. This implies that the endomorphism $\tilde{m}_g$ is injective.

Conversely, suppose that A is not locally μ -negligible. Let C be a compact subset of X such that $A \cap C$ is not μ -negligible, and let φ be the characteristic function of $A \cap C$. The class of the function φ in $L^p(X, \mu)$ is not zero, but that of $m_g(\varphi)$ is, so m_g is not injective.

Since the product of a locally μ -negligible function and a μ -negligible function is μ -negligible, the endomorphism $\tilde{m}_g$ depends only on the class of g in $L^\infty(X, \mu)$. For $\tilde{g} \in L^\infty(X, \mu)$, one also denotes by $\tilde{m}_{\tilde{g}}$ the endomorphism $\tilde{m}_g$ of $L^p(X, \mu)$ for every function $g \in \mathcal{L}^\infty(X, \mu)$ of which $\tilde{g}$ is the class. One says that $\tilde{m}_{\tilde{g}}$ is the multiplication endomorphism by $\tilde{g}$ in $L^p(X, \mu)$.

PROPOSITION 5. — Let $p \in [1, +\infty[$. The map $m : g \mapsto \tilde{m}_g$ from $L^\infty(X, \mu)$ into $\mathcal{L}(L^p(X, \mu))$ is an isometric morphism of unital Banach algebras.

For a, b in $\mathbb{C}$ and g_1, g_2 in $L^\infty(X, \mu)$, one immediately verifies that $\widetilde{m}_{ag_1+bg_2} = a\widetilde{m}_{g_1} + b\widetilde{m}_{g_2}$, so the map $\widetilde{m}$ is linear. Moreover, one has $\widetilde{m}_1 = 1_{L^p(X, \mu)}$ and $\widetilde{m}_{g_1g_2} = \widetilde{m}_{g_1}\widetilde{m}_{g_2}$, so the map $\widetilde{m}$ is a unital algebra morphism from $L^\infty(X, \mu)$ into $\mathcal{L}(L^p(X, \mu))$.

Formula (1) above shows that $\widetilde{m}$ has norm ≤ 1 .

Let $g \in \mathcal{L}^\infty(X, \mu)$ and $\tilde{g}$ be its class in $L^\infty(X, \mu)$. Let $\varepsilon > 0$. The set Y of $x \in X$ such that $|g(x)| > \|\tilde{g}\|_\infty - \varepsilon$ is not locally μ -negligible. There therefore exists a compact subset C of X such that $\mu(Y \cap C) > 0$. Let φ be the characteristic function of $Y \cap C$. Since

$$\begin{aligned} \|\widetilde{m}_{\tilde{g}}(\varphi)\|_p &= \left(\int_X |\varphi g|^p d\mu \right)^{1/p} \geq \left(\int_Y |\varphi g|^p d\mu \right)^{1/p} \\ &\geq (\|\tilde{g}\|_\infty - \varepsilon) \mu(Y \cap C)^{1/p} = (\|\tilde{g}\|_\infty - \varepsilon) \|\varphi\|_p, \end{aligned}$$

it follows that $\|\widetilde{m}_{\tilde{g}}\| \geq \|\tilde{g}\|_\infty - \varepsilon$. Since ε is arbitrary, we deduce that $\|\widetilde{m}_{\tilde{g}}\| \geq \|\tilde{g}\|_\infty$, which proves that $\widetilde{m}$ is isometric.

Lemma 8. — Let (g_n) be a bounded sequence in $\mathcal{L}^\infty(X, \mu)$ converging simply μ -almost everywhere. Let $\tilde{m} \in L^\infty(X, \mu)$ be the class of its limit. Then $\widetilde{m}_{g_n}$ converges to $\widetilde{m}_{\tilde{g}}$ in the space $\mathcal{L}(L^p(X, \mu))$ endowed with the topology of simple convergence.

Let $f \in \mathcal{L}^p(X, \mu)$. The sequence $(g_n f)$ is bounded in $\mathcal{L}^p(X, \mu)$ and converges simply μ -almost everywhere to gf . By the Lebesgue theorem (INT, IV, p. 137, § 3, no. 7, th. 6), the sequence $(g_n f)$ converges to gf in $\mathcal{L}^p(X, \mu)$, and the assertion follows.

We now consider the case $p = 2$.

PROPOSITION 6. — The map $m: g \mapsto \widetilde{m}_g$ is a unital isometric morphism of the stellar algebra $L^\infty(X, \mu)$ into the stellar algebra $\mathcal{L}(L^2(X, \mu))$.

In particular, for every $g \in L^\infty(X, \mu)$ the multiplication endomorphism $\widetilde{m}_g$ is normal; it is Hermitian if and only if g is locally real-valued μ -almost everywhere.

By prop. 5, the map $\tilde{m}$ is an injective, isometric morphism of unital Banach algebras from $L^\infty(X, \mu)$ into $\mathcal{L}(L^2(X, \mu))$. Let $g \in \mathcal{L}^\infty(X, \mu)$. For f_1 and $f_2 \in \mathcal{L}^2(X, \mu)$, we have

$$\langle f_1 | \widetilde{m}_g(f_2) \rangle = \int_X \overline{f_1(x)} g(x) f_2(x) d\mu(x) = \langle \widetilde{m}_{\tilde{g}}(f_1) | f_2 \rangle,$$

from which it follows that $\widetilde{m}_g^* = \widetilde{m}_{\tilde{g}}$, which proves that m is an involutive morphism. The last assertions follow from this (cf. I, p. 106, prop. 5).

COROLLARY. — Let $g \in L^\infty(X, \mu)$.

- a) The spectrum of $\tilde{m}_g$ in $\mathcal{L}(L^2(X, \mu))$ is the μ -essential image of g ;
- b) The endomorphism $\tilde{m}_g$ is positive if and only if g is locally positive μ -almost everywhere;
- c) For any function $f \in \mathcal{C}(\text{Sp}(\tilde{m}_g))$, we have $f(\tilde{m}_g) = \tilde{m}_{f \circ g}$.

Since $\tilde{m}$ is injective, we have $\text{Sp}(\tilde{m}_g) = \text{Sp}(g)$ by prop. 5 of I, p. 106; the result then follows from lemma 6 of IV, p. 185, proposition 4 of IV, p. 185, and definition 6 of I, p. 115.

PROPOSITION 7. — The image of the morphism $\tilde{m}$ from $L^\infty(X, \mu)$ into $\mathcal{L}(L^2(X, \mu))$ is a maximal commutative subalgebra of the algebra $\mathcal{L}(L^2(X, \mu))$.

It suffices to prove that if $u \in \mathcal{L}(L^2(X, \mu))$ is an endomorphism that commutes with $\tilde{m}_g$ for every function $g \in \mathcal{L}^\infty(X, \mu)$, then u belongs to the image of $\tilde{m}$. Let u be such an endomorphism.

We denote $\tilde{f}$ the class in $L^2(X, \mu)$ of a function $f \in \mathcal{L}^2(X, \mu)$. Denote by v the linear map from $\mathcal{L}^2(X, \mu)$ into $L^2(X, \mu)$ defined by $f \mapsto u(\tilde{f})$. We have $v \circ m_g = \tilde{m}_g \circ v$ for all $g \in \mathcal{L}^\infty(X, \mu)$.

For any subset μ -measurable Y of X , we denote φ_Y its characteristic function; the endomorphism $\tilde{m}_{\varphi_Y}$ is an orthogonal projector of $L^2(X, \mu)$.

Suppose first that X is compact, so that the class of the constant function 1 belongs to $L^2(X, \mu)$. Let g be a function in $\mathcal{L}^2(X, \mu)$ whose class in $L^2(X, \mu)$ is $v(1)$.

Let c be a positive real number and Y the set of $x \in X$ such that $|g(x)| \geq c$; it is a μ -measurable set. We have in $L^2(X, \mu)$ the equalities $\tilde{\varphi}_Y g = \tilde{m}_{\varphi_Y}(v(1)) = v(m_{\varphi_Y}(1)) = v(\varphi_Y)$.

Since moreover $|c\varphi_Y| \leq |g\varphi_Y|$, we obtain

$$c^2 \mu(Y) = \int_X (c\varphi_Y)^2 d\mu \leq \int_X |\varphi_Y g|^2 d\mu = \|\tilde{u}(\varphi_Y)\|^2 \leq \|u\|^2 \mu(Y).$$

This inequality implies that $\mu(Y) = 0$ if $c > \|u\|$, so that the function g is bounded μ -almost everywhere by $\|u\|$. For every function $f \in \mathcal{C}(X)$, we finally have

$$u(\tilde{f}) = v(m_f(1)) = \tilde{m}_f(v(1)) = \tilde{m}_f(\tilde{g}) = \tilde{m}_g(\tilde{f}),$$

whence the equality $u = \tilde{m}_g$, and in particular $N_\infty(g) = \|u\|$.

Consider now the general case. There exists a locally countable family $(X_i)_{i \in I}$ of pairwise disjoint compact subsets of X such that $Z = X - \bigcup_{i \in I} X_i$ is locally μ -negligible (INT, IV, p.190, § 5, no. 9, prop. 14). Let μ_i be the measure induced by μ on X_i (INT, IV, p. 186, § 5, no. 7, def. 4).

By prop. 1 of IV, p. 179, for every $i \in I$ the image E_i of $\tilde{m}_{\varphi_{X_i}}$ is identified with $L^2(X_i, \mu_i)$ by $f \mapsto f|_{X_i}$. For every function $g \in \mathcal{L}^\infty(X, \mu)$ the multiplication endomorphism $\tilde{m}_g$ commutes with $\tilde{m}_{\varphi_{X_i}}$, hence leaves stable E_i , and the endomorphism of E_i deduced from $\tilde{m}_g$ by passage to the subspaces coincides with the endomorphism of multiplication by $g|_{X_i}$ on $L^2(X_i, \mu_i)$.

Since u commutes with $m_{\varphi_{X_i}}$, it also leaves stable the subspace E_i . Denote by u_i the endomorphism of $L^2(X_i, \mu_i)$ deduced from u by passage to the subspaces.

The map from $\mathcal{L}^\infty(X, \mu)$ into $\mathcal{L}^\infty(X_i, \mu_i)$ defined by $g \mapsto g|_{X_i}$ being surjective, the hypothesis implies that u_i commutes with $\tilde{m}_g$ for every function $g \in \mathcal{L}^\infty(X_i, \mu_i)$. By the first part of the proof, there exists a function $g_i \in \mathcal{L}^\infty(X_i, \mu_i)$ such that $u_i = \tilde{m}_{g_i}$ and $N_\infty(g_i) \leq \|u_i\| \leq \|u\|$.

The function g on X which coincides with g_i on X_i and which is zero on Z is bounded and μ -measurable (INT, IV, p. 193, § 5, no. 10, prop. 16), hence $g \in \mathcal{L}^\infty(X, \mu)$. For every $i \in I$, the restriction of u to E_i coincides with that of $\tilde{m}_g$; consequently, we have $u = m_g$ by prop. 1, c) of IV, p. 179.

COROLLARY. — Let u be an endomorphism of the Hilbert space $L^2(X, \mu)$ commuting with $\tilde{m}_g$ for every function $g \in \mathcal{H}(X)$. Then there exists a unique element $f \in L^\infty(X, \mu)$ such that $u = \tilde{m}_f$.

By prop. 7, it suffices to prove that u commutes with $\tilde{m}_g$ for all $g \in \mathcal{L}^\infty(X, \mu)$. Let h_1 and h_2 be elements of $\mathcal{L}^2(X, \mu)$ with classes $\tilde{h}_1$ and $\tilde{h}_2$ in $L^2(X, \mu)$. Let k_1 (resp. k_2) be a function in $\mathcal{L}^2(X, \mu)$ whose class is $u(\tilde{h}_1)$ (resp. $u^*(\tilde{h}_2)$).

Denote by $h = h_1 \bar{k}_2 - k_1 \bar{h}_2$; we have $h \in \mathcal{L}^1(X, \mu)$. Define the measure $\nu = h \cdot \mu$ on X ; it is bounded. For every $g \in \mathcal{L}^\infty(X, \mu)$, we have

$$\begin{aligned} \langle \tilde{h}_2 | u(\tilde{m}_g(\tilde{h}_1)) - \tilde{m}_g(u(\tilde{h}_1)) \rangle &= \langle u^*(\tilde{h}_2) | \tilde{m}_g(\tilde{h}_1) \rangle - \langle \tilde{h}_2 | \tilde{m}_g(u(\tilde{h}_1)) \rangle \\ &= \int_X g \cdot h_1 \cdot \bar{k}_2 d\mu - \int_X g \cdot k_1 \cdot \bar{h}_2 d\mu = \nu(g). \end{aligned}$$

We therefore have by hypothesis $\nu(g) = 0$ for every function $g \in \mathcal{K}(X)$, that is, $\nu = 0$. Consequently, it follows that

$$\langle \tilde{h}_2 | u(\tilde{m}_g(\tilde{h}_1)) - \tilde{m}_g(u(\tilde{h}_1)) \rangle = \nu(g) = 0$$

for all $g \in \mathcal{L}^\infty(X, \mu)$. Since this is the case for all h_1 and h_2 in $\mathcal{L}^2(X, \mu)$, we have $u \circ \tilde{m}_g = \tilde{m}_g \circ u$.

Remark. — In what follows, we will often simply denote by m_g the multiplication operator by g on $L^p(X, \mu)$.

IV.2.6 Spectral measures

In this section, we fix a complex Hilbert space E .

Recall that if A is a commutative unital stellar algebra, we denote by $X(A)$ the compact topological space of its unital characters (cor. 1 of I, p. 29) and $\mathcal{G}_A$ the Gelfand transform of A (def. 5 of I, p. 7), which is an isometric isomorphism of unital involutive algebras from A onto $\mathcal{C}(X(A))$ (th. 1 of I, p. 108). We will denote by $\mathcal{H}_A$ the inverse isomorphism.

Lemma 9. — Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. For all x and y in E , the map μ from $X(A)$ into $\mathbb{C}$ defined by $\mu(f) = \langle x | \mathcal{H}_A(f)y \rangle$ for all $f \in \mathcal{C}(X(A))$ is a bounded measure on the compact space $X(A)$. It is positive if $x = y$. Its norm is $\leq \|x\| \|y\|$, with equality if $x = y$.

The map μ is linear. For every function $f \in \mathcal{C}(X(A))$, we have

$$|\mu(f)| \leq \|x\| \|y\| \|\mathcal{H}_A(f)\| = \|x\| \|y\| \|f\|,$$

hence μ is a bounded measure on $X(A)$ of norm $\leq \|x\| \|y\|$.

Suppose that $x = y$. For every positive function $f \in \mathcal{C}(X(A))$, the element $\mathcal{H}_A(f)$ is a positive element of A (example 1 of I, p. 115), hence a positive element of $\mathcal{L}(E)$, whence $\mu(f) = \langle x | \mathcal{H}_A(f)(x) \rangle \geq 0$ (prop. 8 of I, p. 138). This shows that μ is a positive measure. Since $\mu(1) = \|x\|^2$ the total mass of μ is $\|x\|^2$ (INT, III, p. 58, § 1, no. 8, cor. 2).

DEFINITION 2. — Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. For all x

and y in E , the linear form $f \mapsto \langle x | \mathcal{H}_A(f)y \rangle$ on $\mathcal{C}(X(A))$ is called the spectral measure of (x, y) relative to A . If $x = y$, we say that it is the spectral measure of x relative to A .

Remark. — Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. For all x and y in E , denote $\mu_{x,y}$ the spectral measure of (x, y) relative to A . The map defined by $(x, y) \mapsto \mu_{x,y}$ from $E \times E$ into $\mathcal{M}^1(X(A))$ is sesquilinear.

Let u be a normal endomorphism of E and A the unital stellar subalgebra of $\mathcal{L}(E)$ generated by u . It is commutative. The space $X(A)$ is identified with $\text{Sp}(u)$ by the mapping $\chi \mapsto \chi(u)$ (lemma 10 of I, p. 109). The spectral measure $\mu_{x,y}$ of (x, y) relative to A is therefore identified with a measure on $\text{Sp}(u)$, called the spectral measure of (x, y) relative to u . For every function $f \in \mathcal{C}(\text{Sp}(u))$, we then have

$$\int_{\text{Sp}(u)} f d\mu_{x,y} = \langle x | f(u)y \rangle$$

according to def. 5 of I, p. 110.

IV.2.7 Commutative stellar algebras of endomorphisms of a Hilbert space

DEFINITION 3. — Let A be an algebra over $\mathbb{C}$ and E a complex topological vector space. Let π be a representation of A in E . An element x of E is said to be a cyclic vector for π if the set of elements $\pi(a)x$ for $a \in A$ is total in E .

Let $u \in \mathcal{L}(E)$ be an endomorphism of E . We say that $x \in E$ is a cyclic vector for u if it is a cyclic vector for the identity representation of the stellar subalgebra generated by u in $\mathcal{L}(E)$.

PROPOSITION 8. — Let E be a complex Hilbert space. Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. Let x be an element of E . Set $E_x = \overline{A \cdot x}$ and let us denote μ_x the spectral measure of x relative to A .

There exists a unique isometric isomorphism θ_x of $L^2(X(A), \mu_x)$ onto E_x such that $\theta_x(f) = \mathcal{H}_A(f)(x)$ for every function $f \in \mathcal{C}(X(A))$. For every element $a \in A$, the space E_x is stable under a and if we denote by a_x the endomorphism of E_x deduced from a by passing to subspaces, then we have $a_x \circ \theta_x = \theta_x \circ m_{\mathcal{G}_A(a)}$.

Let $\tilde{\theta}_x$ be the linear map from $\mathcal{C}(X(A))$ into E_x defined by

$$\tilde{\theta}_x(f) = \mathcal{H}_A(f)(x).$$

For any function $f \in \mathcal{C}(X(A))$, we have

$$\|\tilde{\theta}_x(f)\|^2 = \langle \mathcal{H}_A(f)x | \mathcal{H}_A(f)x \rangle = \langle x | \mathcal{H}_A(|f|^2)x \rangle = \int_{X(A)} |f|^2 d\mu_x.$$

Since $\mathcal{C}(X(A))$ is dense in $\mathcal{L}^2(X(A), \mu_x)$, there exists a unique isometric linear map from $\mathcal{L}^2(X(A), \mu_x)$ into E_x which extends $\tilde{\theta}_x$; it is denoted by θ_x . The map θ_x is surjective since its image is closed (Lemma 8 of I, p. 107) and contains $A \cdot x$. It is therefore an isometric isomorphism.

Let $a \in A$. Denote by $g = \mathcal{G}_A(a) \in \mathcal{C}(X(A))$. The space E_x is stable under a . For $f \in \mathcal{C}(X(A))$, we then have

$$\tilde{\theta}_x(m_g(f)) = \mathcal{H}_A(\mathcal{G}_A(a)f)x = (a \circ \mathcal{H}_A(f))x = a(\tilde{\theta}_x(f)),$$

and it follows that $\theta_x \circ m_g = a_x \circ \theta_x$.

COROLLARY. — Let E be a complex Hilbert space. Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$ admitting a cyclic vector x . Let μ_x be the spectral measure of x with respect to A . There exists a unique isometric isomorphism

$$\theta_x : \mathcal{L}^2(X(A), \mu_x) \rightarrow E$$

such that $\theta_x(f) = \mathcal{H}_A(f)$ for every function $f \in \mathcal{C}(X(A))$. For every a in A , we have $a \circ \theta_x = \theta_x \circ m_{\mathcal{G}_A(a)}$.

Indeed, we then have $E_x = E$ and $a_x = a$ for all $a \in A$.

Let E be a complex Hilbert space. Let $x \in E$ and let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. Denote by E_x the closed subspace $\overline{A \cdot x}$ of E . For every $a \in A$, we then have $a(E_x) \subset E_x$. Denote by A_x the commutative unital stellar subalgebra of $\mathcal{L}(E_x)$ consisting of the endomorphisms of E_x deduced from the elements of A by passing to subspaces. The vector x is a cyclic vector for $A_x \subset \mathcal{L}(E_x)$.

PROPOSITION 9. — There exists a subset C of E such that E is the Hilbert sum of the spaces E_x for $x \in C$. If E is countably generated, the set C is countable.

Let $\mathcal{O}$ the set of subsets C of E such that the subspaces E_x for $x \in C$ are pairwise orthogonal. The set $\mathcal{O}$, ordered by inclusion, is of finite character (E, III, p. 34, def. 2) since C belongs to $\mathcal{O}$ if and only if the sets formed of two elements of C belong to $\mathcal{O}$. By E, III, p. 35, th. 1, there exists a maximal element C of $\mathcal{O}$.

Let F be the closed subspace of E generated by the subspaces E_x for $x \in C$. It suffices to show that F° is reduced to 0 to complete the proof of the proposition. Let y be an element of F° . For every $x \in C$ and all a and b in A , we have $\langle a(y)|b(x) \rangle = \langle y|a^*b(x) \rangle = 0$ since $a^*b(x)$ belongs to E_x , hence in F . Since the elements $a(y)$ (resp. $b(x)$) generate a dense subspace of E_y (resp. E_x), we therefore have $E_y \subset E_x^\circ$. Since C is maximal in $\mathcal{O}$, this means that E_y is reduced to 0, hence that $y = 0$.

Suppose that E is countably generated. Since E_x is nonzero for all $x \in C$ non-zero, any set C such that E is the Hilbert sum of the spaces E_x for $x \in C$ is countable.

THEOREM 1. — Let A be a commutative unital stellar subalgebra of $\mathcal{L}(E)$. There exists a locally compact topological space X , a positive measure μ on X , an isometric isomorphism θ of $L^2(X, \mu)$ onto E and an isometric morphism of stellar algebras π from A into $\mathcal{C}_b(X)$ such that for every $a \in A$, one has

$$a \circ \theta = \theta \circ m_{\pi(a)}.$$

By prop. 9, there exists a subset C of E such that E is the Hilbert sum of the subspaces E_x for $x \in C$. Let X be the locally compact topological space $X(A) \times C$, where C is equipped with the discrete topology. There exists a unique measure μ on X such that $\mu|(X(A) \times \{x\})$ is identified with the spectral measure μ_x of x for every $x \in C$ (INT, III, p. 65, § 2, n°1, prop. 1); this measure is positive (cf. loc. cit.). One then has a Hilbert sum decomposition

$$L^2(X, \mu) = \bigoplus_{x \in C} L^2(X(A), \mu_x)$$

(prop. 1 of IV, p. 179, c) applied to the sets $X(A) \times \{x\}$ for $x \in C$). There therefore

exists a unique isometry $\theta : L^2(X, \mu) \rightarrow E$ which coincides with $\theta_x : L^2(X(A), \mu_x) \rightarrow E_x$ on $L^2(X(A), \mu_x)$ for all $x \in C$.

Let $p: X \rightarrow X(A)$ be the canonical projection; it is surjective, hence the linear map $p^*: \mathcal{C}_b(X(A)) \rightarrow \mathcal{C}(X)$ defined by $f \mapsto f \circ p$ is injective. Denote $\pi = p^* \circ \mathcal{G}_A$; this is an injective morphism of stellar algebras from A into $\mathcal{C}_b(X)$; it is therefore isometric (prop. 9 of I, p. 112).

Let $a \in A$. For every $x \in C$, we have $a_x \circ \theta_x = \theta_x \circ m_{\mathcal{G}_A(a)}$. The continuous linear maps $a \circ \theta$ and $\theta \circ m_{\pi(a)}$ therefore coincide on the subspaces $L^2(X(A), \mu_x)$, and are consequently equal.

COROLLARY (Spectral Theorem). — Let E be a complex Hilbert space. Let $u \in \mathcal{L}(E)$ be a normal endomorphism of E . There exist a locally compact topological space X , a positive measure μ on X , an isometric isomorphism θ of $L^2(X, \mu)$ onto E and a continuous bounded function g on X such that $u = \theta \circ m_g \circ \theta^{-1}$.

If u admits a cyclic vector x , one may take $X = \text{Sp}(u)$ and for g the identity function of X .

It suffices to apply Theorem 1 (resp. the corollary of Proposition 8) to the stellar subalgebra A of $\mathcal{L}(E)$ generated by u , and to set $g = \pi(u)$.

This statement reduces every question concerning a single normal endomorphism of a Hilbert space to a similar question for a multiplication operator, which often simplifies its study (cf. for example Exercise 19 of IV, p. 325).

IV.2.8 Continuity of the functional calculus

In this section, we consider the continuity properties of the functional calculus with respect to the two variables.

For a stellar algebra A and a normal element a of A , and for a function f continuous on a subset of C containing $\text{Sp}_A(a)$, we denote $f(a)$ the element obtained by the continuous functional calculus of a applied to the restriction of f to the spectrum of a .

PROPOSITION 10. — Let A be a unital stellar algebra. Let U be a relatively compact open set in C . Denote by Ω_n the set of normal elements of A such that $\text{Sp}_A(a)$ is contained in U . The map $(f, a) \mapsto f(a)$ of $\mathcal{C}(\overline{U}) \times \Omega_n$ into A is continuous.

Denote by q the map $(f, a) \mapsto f(a)$ of $\mathcal{C}(\bar{U}) \times \Omega_n$ into A . The set of functional calculus maps $f \mapsto f(a)$ for $a \in \Omega_n$ is equicontinuous in $\mathcal{L}(\mathcal{C}(\bar{U}); A)$, since each is a continuous linear map of norm ≤ 1 . To prove the assertion, it therefore suffices to verify that, for every $f \in \mathcal{C}(\bar{U})$, the map from Ω_n into A defined by $a \mapsto f(a)$ is continuous (TG, X, p. 13, cor. 3).

Let $\mathcal{A}$ be the set of $f \in \mathcal{C}(\bar{U})$ such that the map $a \mapsto f(a)$ of Ω_n into A is continuous; it must be proved that $\mathcal{A} = \mathcal{C}(\bar{U})$.

The set $\mathcal{A}$ is a unital involutive subalgebra of $\mathcal{C}(\bar{U})$. It contains the identity function on $\bar{U}$, hence it separates points. Consequently, it is dense in $\mathcal{C}(\bar{U})$ (TG, X, p. 39, prop. 7). Let us prove that it is closed.

Let (f_n) be a sequence in $\mathcal{A}$ converging to $f \in \mathcal{C}(\bar{U})$. Let $\varepsilon > 0$ and choose $n \in \mathbb{N}$ such that $\|f - f_n\|_\infty \leq \varepsilon/4$. For every $(a_1, a_2) \in \Omega_n^2$ we have

$$\begin{aligned} \|f(a_1) - f(a_2)\| &\leq 2\|f - f_n\|_\infty + \|f_n(a_1) - f_n(a_2)\| \\ &\leq \frac{\varepsilon}{2} + \|f_n(a_1) - f_n(a_2)\|. \end{aligned}$$

Since f_n belongs to $\mathcal{A}$, there exists a neighborhood V of a_1 in Ω_n such that $\|f_n(a_1) - f_n(a_2)\| \leq \varepsilon/2$ for all $a_2 \in V$. We therefore have $\|f(a_1) - f(a_2)\| \leq \varepsilon$ for all $a_2 \in V$. The map $a \mapsto f(a)$ is therefore continuous at a_1 ; the assertion is proved.

COROLLARY 1. — Let A be a unital stellar algebra and let A_n be the set of normal elements of A . Endow the space $\mathcal{C}(C)$ with the topology of compact convergence. The map $(f, a) \mapsto f(a)$ of $\mathcal{C}(C) \times A_n$ into A is continuous.

Let $(f_0, a_0) \in \mathcal{C}(C) \times A_n$. Let U be a relatively compact neighborhood of the spectrum of a_0 . Let V be the set of normal elements a of A such that $\text{Sp}_A(a) \subset U$; this is an open neighborhood of a_0 in A_n (I, p. 76, prop. 10). For every $a \in V$ and every function $f \in \mathcal{C}(C)$, we have $f(a) = (f|_{\bar{U}})(a)$. Since the map $f \mapsto \|f|_{\bar{U}}\|_\infty$ is a continuous seminorm on $\mathcal{C}(C)$ endowed with the topology of compact convergence, the continuity of the map $(f, a) \mapsto f(a)$ at (f_0, a_0) follows from prop. 10.

COROLLARY 2. — Let E be a complex Hilbert space and let $\mathcal{L}(E)_n$ be the set of normal endomorphisms of E . Endow the space $\mathcal{C}(C)$ with the topology of compact convergence. The map from $\mathcal{C}(C) \times \mathcal{L}(E)_n$ into $\mathcal{L}(E)$ defined by $(f, u) \mapsto f(u)$ is continuous.

IV.3 DISTRIBUTIONS AND TEMPERED DISTRIBUTIONS

In this section, n denotes a natural integer. We denote by μ the Lebesgue measure on $\mathbf{R}^n$, as well as its restriction to every locally compact subset of $\mathbf{R}^n$.

We denote

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

the scalar product on Euclidean space $\mathbf{R}^n$; the Euclidean norm is denoted $x \mapsto \|x\|$ (TG, VI, p. 7). We recall that the group $\mathbf{R}^n$ is in duality with itself with respect to the map $(x, y) \mapsto \exp(2i\pi x \cdot y)$ and that the dual measure of the Lebesgue measure is then identified with the Lebesgue measure (corollary 3 of II, p. 236). We denote by F (resp. $\bar{F}$) the Fourier transform (resp. the inverse Fourier transform) on $\mathbf{R}^n$ (cf. no. 9 of II, p. 237).

For every $\alpha = (\alpha_i)_{1 \leq i \leq n} \in \mathbf{N}^n$, we denote by X^α the function from $\mathbf{R}^n$ into $\mathbf{R}$ defined by $x = (x_i)_{1 \leq i \leq n} \mapsto x^\alpha = \prod_{i=1}^n x_i^{\alpha_i}$.

Let α and β be elements of $\mathbf{N}^n$. We denote

$$|\alpha| = \sum_{i=1}^n \alpha_i, \quad \binom{\alpha}{\beta} = \prod_{i=1}^n \binom{\alpha_i}{\beta_i}.$$

Lemma 1. — Let U be an open subset of $\mathbf{R}^n$.

- a) Let K be a compact subset of U and V an open neighborhood of K in U . There exists a function $\varphi \in \mathcal{C}^\infty(U)$ with compact support contained in V such that $0 \leq \varphi \leq 1$ and such that $\varphi(x) = 1$ for all $x \in K$;
- b) For every locally finite open cover of U , there exists a partition of unity consisting of functions of class C^∞ subordinate to it.

This follows from VAR, R1, p. 40, 5.3.6.

PROPOSITION 1. — Let U be an open subset of $\mathbf{R}^n$ and $k \in \mathbf{N}$. Let E , F , and G be topological vector spaces and $b: E \times F \rightarrow G$ a continuous bilinear map. Let f and g be functions

of class C^k from U into E and F , respectively. Then the map $h: x \mapsto b(f(x), g(x))$ is of class C^k on U ; moreover, for every $\alpha \in \mathbb{N}^n$ such that $|\alpha| \leq k$ and

for all $x \in U$, we have

$$\partial^\alpha h(x) = \sum_{\substack{\beta \in \mathbb{N}^n \\ \beta \leq \alpha}} \binom{\alpha}{\beta} b(\partial^\beta f(x), \partial^{\alpha-\beta} g(x)).$$

The map h is the composition of the map $(f, g) : U \rightarrow E \times F$, which is of class C^k , and of the map $b : E \times F \rightarrow G$ which is of class C^∞ . It is therefore of class C^k . The expression for its partial derivatives results from Leibniz's formula (FVR, I, p. 28, prop. 2, cf. A, III, p. 122, corollary).

We recall that a separated locally convex topological vector space is a Montel space if it is barrelled and if every bounded subset is relatively compact (EVT, IV, p. 18, def. 4).

We also recall that if E and F are locally convex spaces, and if E is bornological (EVT, III, p. 12, def. 1), a linear mapping $u : E \rightarrow F$ is continuous if and only if the image under u of every bounded subset of E is bounded in F (EVT, III, p. 11, prop. 1, (iiibis), when F is semi-normed, the general case following formally, cf. EVT, II, p. 7, prop. 5, b)).

IV.3.1 Differentiation under the integral sign

We fix a locally compact topological space X and a measure ν on X . Let E be a Banach space.

PROPOSITION 2. — Let I be an interval of $\mathbf{R}$ and f a mapping from $X \times I$ into E such that

- (i) For every $t \in I$, the map $x \mapsto f(x, t)$ of X into E is ν -integrable;
- (ii) For every $x \in X$, the map $t \mapsto f(x, t)$ from I into E admits a derivative, denoted $t \mapsto f'(x, t)$;
- (iii) There exists a positive function ν -integrable g on X such that $\|f'(x, t)\| \leq g(x)$ for all $(x, t) \in X \times I$.

Then the map F from I into E defined by

$$F(t) = \int_X f(x, t) d\nu(x)$$

is differentiable in I and for every $t \in I$, we have

$$F'(t) = \int_X f'(x, t) d\nu(x).$$

Let $t_0 \in I$ and let J be an interval of $\mathbf{R}$ contained in I which is a neighborhood of t_0 in I . Let $h : X \times J \rightarrow E$ be the map defined by $(x, t) \mapsto (f(x, t) - f(x, t_0)) / (t - t_0)$ for $(x, t) \in X \times (J - \{t_0\})$ and $(x, t_0) \mapsto f'(x, t_0)$ for $x \in X$. Let $x \in X$. We have $\|h(x, t_0)\| \leq g(x)$ and, for all $t \neq t_0$, it follows that $\|h(x, t)\| \leq g(x)$. By FVR, I, p. 23, th. 2. By definition of the derivative, the proposition follows from Corollary 1 of INT, IV, p. 144, § 4, no. 3, applied to the mapping h .

Let us take up the notations of VAR, R1, 2.4, p. 19 concerning partial derivatives.

COROLLARY 1. — Let U be an open subset of $\mathbf{R}^n$. Let $k \in \mathbf{N}$ and f be a map from $X \times U$ into E satisfying the following conditions:

- (i) For every $t \in U$, the map $x \mapsto f(x, t)$ of X into E is ν -integrable;
- (ii) For every $x \in X$, the map $t \mapsto f(x, t)$ from U into E is of class C^k , with partial derivatives denoted $\partial^\alpha f(x, t)$ for all $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq k$;
- (iii) There exists a function ν -integrable g on X such that for every $\alpha \in \mathbf{N}^n$ satisfying $|\alpha| \leq k$ and for every $(x, t) \in X \times U$, we have $\|\partial^\alpha f(x, t)\| \leq g(x)$.

Then the map F from U to E defined by

$$F(t) = \int_X f(x, t) d\nu(x)$$

is of class C^k in U and for all $\alpha \in \mathbf{N}^n$ satisfying $|\alpha| \leq k$ and every $t \in U$, we have

$$\partial^\alpha F(t) = \int_X \partial^\alpha f(x, t) d\nu(x).$$

This follows from the proposition by induction on k .

COROLLARY 2. — Let k be a natural number. Let $f \in \mathcal{L}^1(\mathbf{R}^n, \mu)$ and $g \in \mathcal{C}^k(\mathbf{R}^n)$. Assume that for every $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq k$, the function $\partial^\alpha g$ is bounded. Then the convolution product $f * g$ belongs to $\mathcal{C}^k(\mathbf{R}^n)$ and for every α such that $|\alpha| \leq k$, we have $\partial^\alpha (f * g) = f * \partial^\alpha g$.

We can apply Corollary 1 to the space $X = \mathbf{R}^n$ with Lebesgue measure and to the mapping h defined by $(x, t) \mapsto f(x)g(t - x)$ from

$\mathbf{R}^n \times \mathbf{R}^n$ into $\mathbf{C}$; indeed, for every $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq k$, we have the inequality

$$|\partial^\alpha h(x, t)| \leq \left(\sup_{|\beta| \leq k} \sup_{y \in \mathbf{R}^n} |\partial^\beta g(y)| \right) |f(x)|$$

whose right-hand side is an integrable function on $\mathbf{R}^n$.

IV.3.2 Integrability criteria in $\mathbf{R}^n$ and $\mathbf{Z}^n$

PROPOSITION 3. — Let N be a norm on $\mathbf{R}^n$, r a real number and $p \in [1, +\infty[$.

- a) The function $(1 + N)^r$ belongs to $\mathcal{L}^p(\mathbf{R}^n, \mu)$ if and only if $rp < -n$;
- a') The restriction to $\mathbf{Z}^n$ of the function $(1 + N)^r$ belongs to $\ell^p(\mathbf{Z}^n)$ if and only if $rp < -n$;
- b) Let V be a bounded measurable neighborhood of 0 in $\mathbf{R}^n$. The function N^r belongs to $\mathcal{L}^p(\mathbf{R}^n - V, \mu)$ if and only if $rp < -n$.

By virtue of the equivalence of norms on $\mathbf{R}^n$ (EVT I, p. 14, th. 2 and TG, IX, p. 32, prop. 8), one may assume that the norm N is given by $N(x) = \sup |x_i|$ for $x = (x_i) \in \mathbf{R}^n$. Let B denote the unit ball of $\mathbf{R}^n$ for this norm.

Let V be a bounded measurable neighborhood of 0 in $\mathbf{R}^n$. The function N^r is continuous and bounded on $(B - V) \cup (V - B)$, which shows that the assertion $b)$ is valid if and only if it is when $V = B$, which we will assume from now on. The function $(1 + N)^r$ being continuous and bounded on B , and satisfying the inequality $N^r \leq (1 + N)^r \leq 2^r N^r$ on $\mathbf{R}^n - B$ one observes that the assertions $a)$ and $b)$ are equivalent.

Let us prove $b)$ when $V = B$. We may assume that $p = 1$. The case where $r > 0$ being elementary, suppose that $r \leq 0$. For every integer $j \geq 1$ the integrable set

$$C_j = \{x \in \mathbf{R}^n \mid 2^{j-1} \leq N(x) < 2^j\}$$

has measure $2^{nj}(2^n - 1)$. The sets $(C_j)_{j \geq 1}$ form a partition of $\mathbf{R}^n - B$. By INT, V, p. 4, § 1, no. 1, cor., we therefore have

$$\begin{aligned} \int_{\mathbf{R}^n - B}^* N^r d\mu &= \sum_{j \geq 1} \int_{C_j}^* N^r d\mu \\ &\leq \sum_{j \geq 1} 2^{n+nj} 2^{r(j-1)} = 2^{n-r} \sum_{j \geq 1} 2^{(n+r)j} \end{aligned}$$

which is finite if $r < -n$. On the other hand, we have (loc. cit.)

$$\int_{\mathbf{R}^n - B}^* N^r d\mu \geq \sum_{j \geq 1} 2^{rj} 2^{nj} (2^n - 1) = (2^n - 1) \sum_{j \geq 1} 2^{(r+n)j}$$

which is infinite if $r \geq -n$.

One proves a') in a similar manner by considering the sets $C_j \cap \mathbf{Z}^n$ which cover $\mathbf{Z}^n - \{0\}$ and satisfy

$$\text{Card}(C_j \cap \mathbf{Z}^n) = (2^{j+1} - 1)^n - (2^j - 1)^n,$$

which belongs to the interval $[(1 - 2^{-n})(2^{j+1} - 1)^n, 2^{n(j+1)}]$.

IV.3.3 Test functions

In this issue, U denotes an open subset of $\mathbf{R}^n$. For $p \in [1, +\infty]$ we will write $\mathcal{L}^p(U)$ and $L^p(U)$ rather than $\mathcal{L}^p(U, \mu)$ and $L^p(U, \mu)$. We identify the continuous functions belonging to $\mathcal{L}^p(U)$ with their images in $L^p(U)$.

Equip the space $\mathcal{C}^\infty(U)$ of infinitely differentiable functions in U with complex values, with the topology defined by the family of seminorms $p_{\alpha, K}$ defined for $\alpha \in \mathbf{N}^n$ and $K \subset U$ by

$$p_{\alpha, K}(\varphi) = \sup_{x \in K} |\partial^\alpha \varphi(x)|.$$

This space is complete (cf. EVT, III, p. 9, example b)).

For every compact subset K of U , we denote by $\mathcal{C}_K^\infty(U)$ the subspace of $\mathcal{C}^\infty(U)$ formed by functions with support in K . The space $\mathcal{C}_K^\infty(U)$ is equipped with the topology induced by that of $\mathcal{C}^\infty(U)$. It is a Fréchet space, a countable family of seminorms defining its topology being the family $(p_{\alpha, K})_{\alpha \in \mathbf{N}^n}$. In particular, it is a bornological space (EVT, III, p. 12, prop. 2).

We denote by $\mathcal{D}(U)$ the vector space $\mathcal{H}(U) \cap \mathcal{C}^\infty(U)$ of functions of class C^∞ with compact support in U . We say that $\mathcal{D}(U)$ is the space of test functions in U . The space $\mathcal{D}(U)$ is the

inductive limit of the spaces $\mathcal{C}_K^\infty(U)$ and it is equipped with the corresponding locally convex inductive limit topology (EVT, II, p. 31).

This space is denoted $\mathcal{C}_\infty^\infty(U)$ in EVT, III, p. 9.

Let (K_m) be an increasing sequence of compact subsets of U whose interiors form a covering of U . The space $\mathcal{D}(U)$ is then the strict inductive limit of the spaces $\mathcal{C}_{K_m}^\infty(U)$ (EVT, III, p. 9). It is therefore a complete space (EVT, II, p. 35, prop. 9). Every bounded subset of $\mathcal{D}(U)$ is contained in one of the subspaces $\mathcal{C}_{K_m}^\infty(U)$ (EVT, III, p. 5, prop. 6). This means that $B \subset \mathcal{D}(U)$ is bounded if and only if there exists a compact subset K of U and a family $(M_\alpha)_{\alpha \in \mathbb{N}^n}$ in $\mathbb{R}_+$ such that B is contained in the set of functions $\varphi \in \mathcal{C}_K^\infty(U)$ satisfying

$$p_{\alpha,K}(\varphi) \leq M_\alpha$$

for all $\alpha \in \mathbb{N}^n$.

The space $\mathcal{D}(U)$ is a bornological space (EVT, III, p. 12, example 3) and a Montel space (EVT, IV, p. 18, example 4).

Let V be an open subset of $\mathbb{R}^n$ contained in U . If $K \subset V$ is compact, then the restriction of functions to V defines a continuous and surjective linear map from $\mathcal{C}_K^\infty(U)$ onto $\mathcal{C}_K^\infty(V)$. The extension by zero of a function defined on V induces an injective continuous linear map, called canonical, from $\mathcal{D}(V)$ into $\mathcal{D}(U)$.

Remark. — One defines similarly the space $\mathcal{D}_{\mathbb{R}}(U)$ of test functions in U with real values. The linear map such that $z \otimes \varphi \mapsto z\varphi$ for all $z \in \mathbb{C}$ and every $\varphi \in \mathcal{D}_{\mathbb{R}}(U)$ is an isomorphism of $\mathbb{C} \otimes \mathcal{D}_{\mathbb{R}}(U)$ into $\mathcal{D}(U)$.

Lemma 2. — Let $\mathcal{U}$ be an open covering of U . There exists a locally finite open covering of U finer than $\mathcal{U}$ and consisting of relatively compact sets.

Since U is locally compact, there exists a covering $\mathcal{V}$ of U finer than $\mathcal{U}$ consisting of open sets relatively compact in U . Since U is paracompact (TG, IX, p. 51, th. 4), there exists a locally finite open covering $\mathcal{W}$ finer than $\mathcal{V}$. Then $\mathcal{W}$ is finer than $\mathcal{U}$ and consists of relatively compact open sets.

Lemma 3. — Let $f \in \mathcal{H}(\mathbb{R}^n)$ and let V be an open neighborhood of the support K of f . There exists an integer $m_0 \geq 1$ such that, for every $x \in K$, the set V contains the ball of radius

m_0^{-1} centered at x . For every integer $m > m_0$, let V_m be the closed ball of radius m^{-1} centered at 0 in $\mathbf{R}^n$ and let φ_m be a positive test function with support in V_m and integral 1. Set $f_m = \varphi_m * f$.

- a) One has $f_m \in \mathcal{D}(\mathbf{R}^n)$ and the support of f_m is contained in V ;
- b) Let $p \in [1, +\infty]$. The sequence $(f_m)_{m>m_0}$ converges to f in $L^p(\mathbf{R}^n)$.

According to TG, IX, p. 14, remark, we have $d(K, \mathbf{R}^n - V) > 0$. Let $m_0 \geq 1$ be an integer such that $m_0^{-1} < d(K, \mathbf{R}^n - V)$; it satisfies the required condition. For all $m > m_0$, the function f_m is continuous (INT, VIII, p. 166, § 4, n° 5, prop. 11) and has support contained in $K + V_m$ (INT, VIII, p. 126, § 1, n° 4, prop. 5), hence in V . By Corollary 2 of IV, p. 198, we have $f_m \in \mathcal{D}(\mathbf{R}^n)$. If $p \neq +\infty$, then the sequence (f_m) converges to f in $L^p(\mathbf{R}^n)$ (INT, VIII, p. 172, § 4, no. 7, prop. 20). ⁽¹⁾

Suppose that $p = +\infty$. Let $\varepsilon > 0$. The function f is uniformly continuous on $\mathbf{R}^n$; therefore there exists an integer $m_1 > m_0$ such that, for every $m \geq m_1$ and for all $x \in \mathbf{R}^n$ and $y \in V_m$, one has $|f(x) - f(x - y)| < \varepsilon$.

For every $m \geq m_1$, the function φ_m vanishes outside V_m , is positive, and has integral 1. We therefore deduce the inequality

$$|f(x) - f_m(x)| = \left| \int_{\mathbf{R}^n} (f(x) - f(x - y)) \varphi_m(y) d\mu(y) \right| \leq \varepsilon$$

for all $x \in \mathbf{R}^n$, whence the result.

Proposition 4. — a) The canonical injection of $\mathcal{D}(U)$ into $\mathcal{K}(U)$ is continuous and $\mathcal{D}(U)$ is dense in $\mathcal{K}(U)$;

- b) For every $p \in [1, +\infty[$, the canonical injection of $\mathcal{D}(U)$ into $\mathcal{L}^p(U)$ is continuous and $\mathcal{D}(U)$ is dense in $L^p(U)$.

Every continuous seminorm on $\mathcal{K}(U)$ is continuous on $\mathcal{D}(U)$, hence the canonical injection of $\mathcal{D}(U)$ into $\mathcal{K}(U)$ is continuous.

Let $f \in \mathcal{K}(U)$. There exists a sequence $(f_m)_{m \in \mathbf{N}}$ in $\mathcal{D}(U)$ which converges uniformly to f on U and such that the supports of the f_m are contained in a fixed relatively compact neighbor-

hood of the support of f (Lemma 3). The sequence (f_m) therefore converges to f in $\mathcal{K}(U)$. This concludes the proof of a).

Let $p \in [1, +\infty[$. Let $f \in \mathcal{D}(U)$ and let K be the support of f . We then have the inequality $N_p(f) \leq \mu(K)^{1/p} \sup_{x \in K} |f(x)|$, hence the canonical injection of $\mathcal{D}(U)$ into $\mathcal{L}^p(U)$ is continuous. The last part of assertion b) then follows from a).

The product of two test functions being again a test function, the Leibniz formula (FVR, I, p. 28, prop. 2) shows that the space $\mathcal{D}(U)$ is a topological algebra.

More generally, let $f \in \mathcal{C}^\infty(U)$. The linear map $\varphi \mapsto f\varphi$ from $\mathcal{D}(U)$ into $\mathcal{D}(U)$ is then continuous. Indeed, for every compact subset K of U and every $\alpha \in \mathbb{N}^n$, we have

$$p_{\alpha, K}(f\varphi) \leq \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} p_{\beta, K}(f) p_{\alpha-\beta, K}(\varphi)$$

(cf. loc. cit.).

IV.3.4 Distributions

We keep the notations of the preceding number; U therefore denotes an open subset of $\mathbb{R}^n$.

DEFINITION 1. — The dual space of $\mathcal{D}(U)$, endowed with the topology of bounded convergence, is called the space of distributions on U . It is denoted by $\mathcal{D}'(U)$.

A distribution on U is therefore a continuous linear form on $\mathcal{D}(U)$.

If f is a linear form on $\mathcal{D}(U)$ (and in particular if f is a distribution) and if φ belongs to $\mathcal{D}(U)$, we denote by $\langle f, \varphi \rangle$ the evaluation of f at φ .

Since $\mathcal{D}(U)$ is bornological, the space $\mathcal{D}'(U)$ is complete (TVS, III, p. 24, cor. 1). Since $\mathcal{D}(U)$ is a Montel space, the same holds for $\mathcal{D}'(U)$ (EVT, IV, p. 19, prop. 9). In particular, the space $\mathcal{D}'(U)$ is reflexive (EVT, IV, p. 16, th. 2).

Lemma 4. — Let f be a linear map from $\mathcal{D}(U)$ into $\mathbb{C}$. Then f is a distribution if and only if, for every compact subset K of U and for every family $(M_\alpha)_{\alpha \in \mathbb{N}^n}$ in $\mathbb{R}_+$, the linear form f is bounded on the set of functions $\varphi \in \mathcal{C}_K^\infty(U)$ such that, for every $\alpha \in \mathbb{N}^n$, we have

$$\sup_{x \in K} |\partial^\alpha \varphi(x)| \leq M_\alpha.$$

Indeed, the space $\mathcal{D}(U)$ is bornological and every bounded subset of $\mathcal{D}(U)$ is contained in one of the bounded sets described in the statement.

Lemma 5. — Let F be a filter on $\mathcal{D}'(U)$ having a countable base or containing a simply bounded set. Then F converges to a distribution if and only if $\langle f, \varphi \rangle$ converges according to F for every test function φ on U .

In particular, a sequence $(f_m)_{m \in \mathbb{N}}$ of distributions converges to a distribution f if and only if, for every $\varphi \in \mathcal{D}(U)$ the sequence $(\langle f_m, \varphi \rangle)_{m \in \mathbb{N}}$ converges to $\langle f, \varphi \rangle$.

Since $\mathcal{D}'(U)$ is a Montel space, hence barrelled; this follows from the Banach–Steinhaus theorem (EVT, III, p. 26, cor. 3 of th. 1).

Let V be an open set contained in U . The transpose of the canonical linear map from $\mathcal{D}(V)$ into $\mathcal{D}(U)$ is a continuous linear map from $\mathcal{D}'(U)$ into $\mathcal{D}'(V)$, called the restriction of the distributions from U to V and denoted $r_{V,U}$, or sometimes $r_{V,U}$.

One has $r_{V,V} = 1_{\mathcal{D}'(V)}$. If $W \subset V \subset U$ are open subsets of U , then we have $r_{W,V} \circ r_{V,U} = r_{W,U}$. In other words, if $\mathcal{T}$ denotes the topology on U , the projective system $\underline{\mathcal{D}'}(U) = ((\mathcal{D}'(V))_{V \in \mathcal{T}}, (r_{W,V}))$ is a presheaf on U with values in the species of structure of locally convex topological vector spaces (TA, I, p. 42, def. 1 and p. 66, §10).

PROPOSITION 5. — The presheaf $\underline{\mathcal{D}'}(U)$ is a sheaf.

Recall (TA, I, p. 43, def. 2) that this means that for every open subset V of U and every open covering $(V_i)_{i \in I}$ of V , the following conditions are satisfied:

- (i) The map $(r_{V_i,V})_{i \in I} : \mathcal{D}'(V) \rightarrow \prod_{i \in I} \mathcal{D}'(V_i)$ is injective;
- (ii) For every family $(f_i) \in \prod_{i \in I} \mathcal{D}'(V_i)$ such that

$$r_{(V_i \cap V_{i'})V_i}(f_i) = r_{(V_i \cap V_{i'})V_{i'}}(f_{i'})$$

for every pair $(i, i') \in I \times I$, there exists a distribution $f \in \mathcal{D}'(V)$ such that for every $i \in I$, one has $r_{V_i,V}(f) = f_i$.

Let V be an open subset of U and $\mathcal{V} = (V_i)_{i \in I}$ an open covering of V . Let $(W_j)_{j \in J}$ be an open covering of V that is locally finite, finer than $\mathcal{V}$, and consisting of relatively compact parts (lemma 2 of IV, p. 201). Fix a partition of unity $(\varphi_j)_{j \in J}$ subordinate to the covering $(W_j)_{j \in J}$

such that the support of φ_j is contained in W_j for all $j \in J$ (lemma 1 of IV, p. 196).

Let $\varphi \in \mathcal{D}(V)$. Since the set of $j \in J$ such that W_j meets the support of φ is finite (TG, I, §9, no. 1, p. 59), we have

$$\varphi = \sum_{j \in J} \varphi \varphi_j$$

where the sum has only finitely many nonzero terms.

Let us prove (i). Suppose that $f \in \mathcal{D}'(V)$ satisfies $r_{V_i V}(f) = 0$ for all $i \in I$. This means that $\langle f, \varphi \rangle = 0$ for every test function φ whose support is contained in one of the open sets V_i . This is a fortiori true if the support of φ is contained in one of the open sets W_j . But then, for every $\varphi \in \mathcal{D}(V)$, we have

$$\langle f, \varphi \rangle = \langle f, \sum_{j \in J} \varphi \varphi_j \rangle = \sum_{j \in J} \langle f, \varphi \varphi_j \rangle = 0,$$

hence $f = 0$.

Let us prove (ii). Let $(f_i)_{i \in I}$ be a family such that $f_i \in \mathcal{D}'(V_i)$ for all $i \in I$ and $r_{V_i \cap V_{i'}, V_i}(f_i) = r_{V_i \cap V_{i'}, V_{i'}}(f_{i'})$ for all i and i' in I . Let $\iota : J \rightarrow I$ be a map such that $W_j \subset V_{\iota(j)}$ for all $j \in J$. For every $j \in J$ set $\tilde{f}_j = r_{W_j, V_{\iota(j)}}(f_{\iota(j)})$. We then have

$$\begin{aligned} r_{W_j \cap W_{j'}, W_j}(\tilde{f}_j) &= r_{W_j \cap W_{j'}, W_j}(r_{W_j, V_{\iota(j)}}(f_{\iota(j)})) \\ &= r_{W_j \cap W_{j'}, V_{\iota(j)}}(f_{\iota(j)}) \\ &= r_{W_j \cap W_{j'}, V_{\iota(j)} \cap V_{\iota(j')}} \circ r_{V_{\iota(j)} \cap V_{\iota(j')}, V_{\iota(j)}}(f_{\iota(j)}). \end{aligned}$$

By exchanging the roles of j and j' and noting that

$$r_{V_{\iota(j)} \cap V_{\iota(j')}, V_{\iota(j)}}(f_{\iota(j)}) = r_{V_{\iota(j)} \cap V_{\iota(j')}, V_{\iota(j')}}(f_{\iota(j')}),$$

by hypothesis, we deduce that

$$r_{W_j \cap W_{j'}, W_j}(\tilde{f}_j) = r_{W_j \cap W_{j'}, W_j}(\tilde{f}_{j'}) \quad (1)$$

for all $(j, j') \in J^2$.

For $\varphi \in \mathcal{D}(V)$, set

$$\lambda(\varphi) = \sum_{j \in J} \langle \tilde{f}_j, \varphi \varphi_j | W_j \rangle,$$

where the sum is finite since only finitely many terms can be nonzero.

The map λ is a linear form on $\mathcal{D}(V)$. Let us prove that it is a distribution. Let B be a bounded subset of $\mathcal{D}(V)$ and let K be a compact subset of V such that $B \subset \mathcal{C}_K^\infty(V)$. By TG, I, §9, no. 1, p. 59, there exists a finite subset J' of J such that

$$\lambda(\varphi) = \sum_{j \in J'} \langle \tilde{f}_j, \varphi \varphi_j | W_j \rangle$$

for all $\varphi \in B$. Since $\tilde{f}_j$ is a distribution for every $j \in J'$ and that $\mathcal{D}(V)$ is a topological algebra, it follows that λ is bounded on B , whence the result (Lemma 4).

Let $j \in J$ and let $\varphi \in \mathcal{D}(V)$ have support contained in W_j . We then have

$$\langle \lambda, \varphi \rangle = \sum_{j' \in J} \langle \tilde{f}_{j'}, \varphi \varphi_{j'} | W_{j'} \rangle = \sum_{j' \in J} \langle \tilde{f}_{j'}, \varphi \varphi_{j'} | W_j \rangle$$

according to (1), since $\varphi \varphi_{j'}$ has support contained in $W_j \cap W_{j'}$. Consequently

$$\langle \lambda, \varphi \rangle = \langle \tilde{f}_j, \sum_{j' \in J} \varphi \varphi_{j'} | W_j \rangle = \langle \tilde{f}_j, \varphi | W_j \rangle,$$

whence $r_{W_j V}(\lambda) = \tilde{f}_j$ for all $j \in J$.

Let $i \in I$. Let us finally prove that the restriction of λ to V_i coincides with f_i . By condition (i), applied to the covering of V_i by the open sets W_j , it suffices to verify that for every $j \in J$ the restriction of λ to $V_i \cap W_j$ coincides with that of f_i . By the preceding, it remains to verify that the restriction of f_i to $V_i \cap W_j$ is that of $\tilde{f}_j$. Now, we have

$$\begin{aligned} r_{V_i \cap W_j, V_i}(f_i) &= r_{V_i \cap W_j, V_i \cap V_{i(j)}}(r_{V_i \cap V_{i(j)}, V_i}(f_i)) \\ &= r_{V_i \cap W_j, V_i \cap V_{i(j)}}(r_{V_i \cap V_{i(j)}, V_{i(j)}}(f_{i(j)})) = r_{V_i \cap W_j, V_{i(j)}}(f_{i(j)}), \end{aligned}$$

where we used the hypothesis concerning the family (f_i) . But then

$$r_{V_i \cap W_j, V_{i(j)}}(f_{i(j)}) = r_{V_i \cap W_j, W_j}(r_{W_j, V_{i(j)}}(f_{i(j)})) = r_{V_i \cap W_j, W_j}(\tilde{f}_j),$$

which allows us to conclude.

IV.3.5 Interpretation of functions as distributions

PROPOSITION 6. — Let ν be a measure on U . The restriction of ν to $\mathcal{D}(U)$ is a distribution, which is zero if and only if the measure ν is zero.

Let K be a compact subset of U . For every function $\varphi \in \mathcal{C}_K^\infty(U)$, we have $|\langle \nu, \varphi \rangle| \leq p_{0,K}(\varphi)|\nu|(K)$, hence the restriction of ν to $\mathcal{D}(U)$ is continuous. Since $\mathcal{D}(U)$ is dense in $\mathcal{H}(U)$, by proposition 4(a) of IV, p. 202, the restriction of ν to $\mathcal{D}(U)$ is zero if and only if ν is zero.

We will identify the space $\mathcal{M}(U; \mathbb{C})$ of complex measures on U with a subspace of $\mathcal{D}'(U)$. We will also identify the space $L_{\text{loc}}^1(U)$ with a subspace of $\mathcal{D}'(U)$ by the map induced by passing to quotients of the map that assigns to $f \in \mathcal{L}_{\text{loc}}^1(U)$ the measure $f \cdot \mu$ (INT, V, p. 44, § 5, no. 2, def. 2). In other words, for $f \in L_{\text{loc}}^1(U)$ and $\varphi \in \mathcal{D}(U)$, we have

$$\langle f, \varphi \rangle = \int_U f \varphi d\mu.$$

In particular, for $p \in [1, +\infty]$ this allows one to identify the space $L^p(U)$ with a subspace of $\mathcal{D}'(U)$ (cf. INT, V, p. 43, § 5, no. 1).

PROPOSITION 7. — Let $p \in [1, +\infty]$. The injection of $L^p(U)$ into $\mathcal{D}'(U)$ is continuous.

Let $f \in L^p(U)$. Let K be a compact subset of U , and denote by φ_K its characteristic function. For every test function φ with support contained in K , Hölder's inequality implies

$$|\langle f, \varphi \rangle| = \left| \int_U f \varphi d\mu \right| \leq N_p(f) N_q(\varphi) \leq N_p(f) N_q(\varphi_K) p_{0,K}(\varphi)$$

where q is the conjugate exponent of p .

In particular, one can identify $\mathcal{D}(U)$ with a subspace of $\mathcal{D}'(U)$.

PROPOSITION 8. — The space $\mathcal{D}(U)$ is dense in $\mathcal{D}'(U)$.

Let λ be a linear form on $\mathcal{D}'(U)$ vanishing on $\mathcal{D}(U)$. Since $\mathcal{D}(U)$ is reflexive, there exists a test function $\varphi \in \mathcal{D}(U)$ such that $\lambda(f) = \langle f, \varphi \rangle$ for all $f \in \mathcal{D}'(U)$. We obtain

$$0 = \lambda(\overline{\varphi}) = \langle \overline{\varphi}, \varphi \rangle = \int_U |\varphi|^2 d\mu,$$

whence $\varphi = 0$. The proposition then follows from the Hahn–Banach theorem (EVT, II, p. 49, cor. 3 (ii)).

For $h \in \mathcal{C}^\infty(U)$, the transpose of the continuous linear map $\varphi \mapsto h\varphi$ of $\mathcal{D}(U)$ into itself is a continuous linear map from $\mathcal{D}'(U)$ into itself, which is denoted $f \mapsto hf$. This definition is justified because if f is the distribution associated with a measure ν on U , then hf is associated with the measure $h \cdot \nu$. Indeed, for every test function $\varphi \in \mathcal{D}(U)$, one computes

$$\langle hf, \varphi \rangle = \langle f, h\varphi \rangle = \int_U h\varphi d\nu = \int_U \varphi d(h \cdot \nu).$$

IV.3.6 Differentiation of distributions

Let $\alpha \in \mathbf{N}^n$. The linear map $\varphi \mapsto \partial^\alpha \varphi$ is continuous from $\mathcal{D}(U)$ into $\mathcal{D}(U)$. Its transpose ${}^t\partial^\alpha$ is a continuous linear map from $\mathcal{D}'(U)$ into $\mathcal{D}'(U)$ (EVT, IV, p. 6, cor., b)).

We denote ∂^α the continuous linear map $(-1)^{|\alpha|} {}^t\partial^\alpha$ from $\mathcal{D}'(U)$ into itself.

DEFINITION 2. — If $f \in \mathcal{D}'(U)$ is a distribution, one says that $\partial^\alpha f$ is the iterated partial derivative of order α of f .

Thus, by definition

$$\langle \partial^\alpha f, \varphi \rangle = (-1)^{|\alpha|} \langle f, \partial^\alpha \varphi \rangle$$

for every function $\varphi \in \mathcal{D}(U)$. We have $\partial^{\alpha+\beta} = \partial^\alpha \circ \partial^\beta$ for all α and β in $\mathbf{N}^n$.

If $n = 1$, one also denotes f' the derivative of a distribution $f \in \mathcal{D}'(U)$. The definition is justified by the following lemma.

Lemma 6. — Let k be a natural integer. Let $f \in \mathcal{C}^k(U)$ and let λ be the distribution associated with f . For every $\beta \in \mathbf{N}^n$ such that $|\beta| \leq k$, the distribution $\partial^\beta \lambda$ is the distribution associated with the function $\partial^\beta f$.

By induction on k , it suffices to prove this property when β satisfies $|\beta| = 1$, and one may even assume that $\beta = (0, \dots, 0, 1)$. Since distributions form a sheaf (prop. 5 of IV, p. 204), it suffices to verify the assertion when there exists an open set $V \subset \mathbf{R}^{n-1}$ and an open interval $I \subset \mathbf{R}$ such that $U = V \times I$.

For every test function $\varphi \in \mathcal{D}(U)$, one has by definition

$$\langle \partial^\beta \lambda, \varphi \rangle = - \int_U f(x) \partial^\beta \varphi(x) dx = - \int_V \left(\int_I f(y, t) \partial^\beta \varphi(y, t) dt \right) dy$$

by the Lebesgue–Fubini theorem (INT, V, p. 96, § 8, n° 4, th. 1). By integration by parts (FVR, II, p. 10), one has

$$- \int_I f(y, t) \partial^\beta \varphi(y, t) dt = \int_I \partial^\beta f(y, t) \varphi(y, t) dt,$$

since $t \mapsto \varphi(y, t)$ is compactly supported in I . One therefore obtains

$$\langle \partial^\beta \lambda, \varphi \rangle = \int_V \left(\int_I \partial^\beta f(y, t) \varphi(y, t) dt \right) dy = \langle \partial^\beta f, \varphi \rangle.$$

PROPOSITION 9 (Leibniz formula). — Let f be a distribution on U and g an infinitely differentiable function on U . Let $\alpha \in \mathbf{N}^n$. One has the relation

$$\partial^\alpha (fg) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta f \partial^{\alpha-\beta} g.$$

Proceeding by induction on $|\alpha|$ as in the proof of FVR, I, p. 28, prop. 2, it suffices to consider the case where $|\alpha| = 1$. The result then follows from the computation

$$\begin{aligned} \langle \partial^\alpha (fg), \varphi \rangle &= \langle fg, -\partial^\alpha \varphi \rangle = \langle f, -g \partial^\alpha \varphi \rangle \\ &= \langle f, -\partial^\alpha (g\varphi) + \varphi \partial^\alpha g \rangle = \langle g \partial^\alpha f + f \partial^\alpha g, \varphi \rangle \end{aligned}$$

valid for $\varphi \in \mathcal{D}(U)$.

IV.3.7 Schwartz functions

We denote $\mathcal{S}(\mathbf{R}^n)$ the space of infinitely differentiable functions φ on $\mathbf{R}^n$ with complex values, such that, for all α and β in $\mathbf{N}^n$, the function $X^\beta \partial^\alpha \varphi$ is bounded on $\mathbf{R}^n$. We equip $\mathcal{S}(\mathbf{R}^n)$ with the locally convex topology defined by the countable family of semi-norms $(q_{\alpha, \beta})_{(\alpha, \beta) \in \mathbf{N}^n \times \mathbf{N}^n}$, where $q_{\alpha, \beta}$ is defined by

$$q_{\alpha, \beta}(\varphi) = \sup_{x \in \mathbf{R}^n} |x^\beta \partial^\alpha \varphi(x)| = \|X^\beta \partial^\alpha \varphi\|_\infty$$

for $\varphi \in \mathcal{S}(\mathbf{R}^n)$. This topology is Hausdorff. It is also defined by the semi-norms $\tilde{q}_{\alpha,k}$ defined by

$$\tilde{q}_{\alpha,k}(\varphi) = \sup_{x \in \mathbf{R}^n} \|x\|^k |(\partial^\alpha \varphi)(x)|.$$

for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$, where $k \in \mathbf{N}$ and $\alpha \in \mathbf{N}^n$.

We say that $\mathcal{S}(\mathbf{R}^n)$ is the Schwartz space or the space of Schwartz functions on $\mathbf{R}^n$.

Note. — Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. For every $k \in \mathbf{N}$, we have

$$\lim_{\|x\| \rightarrow +\infty} \|x\|^k \varphi(x) = 0,$$

since the function $x \mapsto \|x\|^{k+1} \varphi(x)$ is bounded.

Example. — The function γ_n defined on $\mathbf{R}^n$ by $\gamma_n(x) = \exp(-\|x\|^2)$ belongs to $\mathcal{S}(\mathbf{R}^n)$. Indeed, one proves by induction on k that, for every integer $k \in \mathbf{N}$, there exists a polynomial $P_k \in \mathbf{R}[X]$ such that $\partial_k \gamma_1 = P_k \gamma_1$.

For all $\alpha = (\alpha_i) \in \mathbf{N}^n$ and $\beta = (\beta_i) \in \mathbf{N}^n$, and every $x = (x_i) \in \mathbf{R}^n$, it then follows that

$$|(X^\beta \partial^\alpha \gamma_n)(x)| = \prod_{i=1}^n |x_i|^{\beta_i} |P_{\alpha_i}(x_i)| \gamma_1(x_i),$$

which is a bounded quantity as x varies in $\mathbf{R}^n$.

Let $\alpha \in \mathbf{N}^n$ and $\beta \in \mathbf{N}^n$. If $\varphi \in \mathcal{S}(\mathbf{R}^n)$, then $X^\beta \partial^\alpha (\varphi)$ is still a Schwartz function; the map $\varphi \mapsto X^\beta \partial^\alpha \varphi$ thus defined from $\mathcal{S}(\mathbf{R}^n)$ into itself is continuous.

The space $\mathcal{S}(\mathbf{R}^n)$ is a topological algebra. More precisely, if φ_1 and φ_2 belong to $\mathcal{S}(\mathbf{R}^n)$, then $\varphi_1 \varphi_2$ is a Schwartz function such that

$$q_{\alpha,\beta}(\varphi_1 \varphi_2) \leq \sum_{0 \leq \gamma \leq \alpha} \binom{\alpha}{\gamma} q_{\gamma,\beta}(\varphi_1) q_{\alpha-\gamma,0}(\varphi_2) \quad (2)$$

for all α and $\beta \in \mathbf{N}^n$ (prop. 1 of IV, p. 196).

The canonical inclusion of $\mathcal{S}(\mathbf{R}^n)$ into the space $\mathcal{C}^\infty(\mathbf{R}^n)$, equipped with the topology described in §3 of IV, p. 200, is continuous, since

$$\sup_{x \in K} |\partial^\alpha \varphi(x)| \leq q_{\alpha,0}(\varphi)$$

for every compact subset K of $\mathbf{R}^n$, all $\alpha \in \mathbf{N}^n$ and every Schwartz function φ .

Lemma 7. — Let $k \in \mathbf{N}$ and $\alpha \in \mathbf{N}^n$. For every function $\varphi \in \mathcal{S}(\mathbf{R}^n)$ and every real number $T > 0$, we have

$$\tilde{q}_{\alpha,k}(\varphi) \leq T^k \sup_{\|x\| \leq T} |\partial^\alpha \varphi(x)| + \frac{1}{T} \tilde{q}_{\alpha,k+1}(\varphi). \quad (3)$$

Indeed, one has

$$\begin{aligned} \tilde{q}_{\alpha,k}(\varphi) &\leq \sup_{\|x\| \leq T} \|x\|^k |\partial^\alpha \varphi(x)| + \sup_{\|x\| > T} \|x\|^k |\partial^\alpha \varphi(x)| \\ &\leq T^k \sup_{\|x\| \leq T} |\partial^\alpha \varphi(x)| + \frac{1}{T} \sup_{\|x\| > T} \|x\|^{k+1} |\partial^\alpha \varphi(x)|. \end{aligned}$$

PROPOSITION 10. — Let B be a bounded subset of $\mathcal{S}(\mathbf{R}^n)$. The topology induced on B by $\mathcal{S}(\mathbf{R}^n)$ coincides with the topology induced by $\mathcal{C}^\infty(\mathbf{R}^n)$.

Since the inclusion of $\mathcal{S}(\mathbf{R}^n)$ into $\mathcal{C}^\infty(\mathbf{R}^n)$ is continuous, it suffices to show that, for every open subset V of $\mathcal{S}(\mathbf{R}^n)$, the intersection $V \cap B$ is open in B for the topology induced by $\mathcal{C}^\infty(\mathbf{R}^n)$.

Let V be an open subset of $\mathcal{S}(\mathbf{R}^n)$. Let $\varphi_0 \in V \cap B$. There exist a finite set I , a family $(\alpha_i, k_i)_{i \in I} \in (\mathbf{N}^n \times \mathbf{N})^I$ and a real number $\varepsilon > 0$ such that V contains the set of $\varphi \in \mathcal{S}(\mathbf{R}^n)$ satisfying

$$\sup_{i \in I} \tilde{q}_{\alpha_i, k_i}(\varphi - \varphi_0) \leq \varepsilon.$$

Since B is bounded in $\mathcal{S}(\mathbf{R}^n)$, there exists $M > 0$ such that the seminorms $\tilde{q}_{\alpha_i, k_i+1}$ for $i \in I$ are bounded by M on B . Let $\delta > 0$ and $T > 0$ be real numbers. By inequality (3), as soon as $\varphi \in B$ satisfies the bound

$$\sup_{i \in I} \sup_{\|x\| \leq T} |\partial^{\alpha_i}(\varphi - \varphi_0)| \leq \delta, \quad (4)$$

we have

$$\sup_{i \in I} \tilde{q}_{\alpha_i, k_i}(\varphi - \varphi_0) \leq \delta T^k + \frac{2M}{T}.$$

Set $T = \frac{4M}{\varepsilon}$, then $\delta = \frac{\varepsilon}{2T^k}$. One observes that $V \cap B$ contains the neighborhood of φ_0 in B for the topology induced by $\mathcal{C}^\infty(\mathbf{R}^n)$ which is defined by (4). This concludes the proof.

COROLLARY. — Let $(\varphi_m)_{m \in \mathbf{N}}$ be a bounded sequence in $\mathcal{S}(\mathbf{R}^n)$. For every function $\varphi \in \mathcal{S}(\mathbf{R}^n)$, the following assertions are equivalent:

- a) The sequence (φ_m) converges to φ in $\mathcal{S}(\mathbf{R}^n)$;
- b) The sequence (φ_m) converges to φ in $\mathcal{C}^\infty(\mathbf{R}^n)$.

Remark. — A sequence (φ_m) in $\mathcal{C}^\infty(\mathbf{R}^n)$ converges if and only if, for every $\alpha \in \mathbf{N}^n$ the sequence $(\partial^\alpha \varphi_m)$ converges to a function $\varphi^{(\alpha)}$ in $\mathcal{C}(\mathbf{R}^n)$ endowed with the topology of compact convergence. We then have $\varphi^{(\alpha)} = \partial^\alpha \varphi$ and (φ_m) converges to $\varphi^{(0)}$.

Indeed, the condition is necessary. Conversely, if the sequences $(\partial^\alpha \varphi_m)$ converge to functions $\varphi^{(\alpha)}$ for all $\alpha \in \mathbf{N}^n$, it then follows from FVR, II, p. 2, th. 1, that $\varphi^{(\alpha)} = \partial^\alpha \varphi^{(0)}$, which means that the sequence (φ_m) converges to $\varphi^{(0)}$ in $\mathcal{C}^\infty(\mathbf{R}^n)$.

PROPOSITION 11. — The space $\mathcal{S}(\mathbf{R}^n)$ is a Fréchet space and a Montel space.

As the space $\mathcal{C}^\infty(\mathbf{R}^n)$ is complete (EVT, III, p. 9, example *b*), the corollary of Proposition 10 implies that every Cauchy sequence in $\mathcal{S}(\mathbf{R}^n)$ converges in $\mathcal{S}(\mathbf{R}^n)$ since it is bounded and converges in $\mathcal{C}^\infty(\mathbf{R}^n)$.

The space $\mathcal{S}(\mathbf{R}^n)$ is therefore a Fréchet space; in particular, it is barrelled (EVT, III, p. 25, cor. of prop. 2). Let B be a bounded subset of $\mathcal{S}(\mathbf{R}^n)$ and $(\varphi_m)_{m \in \mathbf{N}}$ a sequence with values in B . Since $\mathcal{C}^\infty(\mathbf{R}^n)$ is a Montel space (EVT, IV, p. 18, example (4)), there exists a subsequence of $(\varphi_m)_{m \in \mathbf{N}}$ which converges in $\mathcal{C}^\infty(\mathbf{R}^n)$, hence in $\mathcal{S}(\mathbf{R}^n)$ (proposition 10). Hence B is relatively compact in $\mathcal{S}(\mathbf{R}^n)$. It follows that $\mathcal{S}(\mathbf{R}^n)$ is a Montel space.

IV.3.8 Inclusions of function spaces in the space of Schwartz functions

PROPOSITION 12. — The space $\mathcal{D}(\mathbf{R}^n)$ is contained in $\mathcal{S}(\mathbf{R}^n)$, and the inclusion of $\mathcal{D}(\mathbf{R}^n)$ into $\mathcal{S}(\mathbf{R}^n)$ is continuous with dense image.

Let $B \subset \mathcal{D}(\mathbf{R}^n)$ be a bounded subset, and let K be a compact subset of $\mathbf{R}^n$ such that $B \subset$

$\mathcal{C}_K^\infty(\mathbf{R}^n)$. Let $\alpha \in \mathbf{N}^n$ and $k \in \mathbf{N}$. For every function $\varphi \in B$, it follows that

$$\tilde{q}_{\alpha,k}(\varphi) \leq \left(\sup_{x \in K} \|x\|^k \right) p_{\alpha,K}(\varphi),$$

so B is bounded in $\mathcal{S}(\mathbf{R}^n)$. The continuity of the inclusion then follows from the fact that the spaces $\mathcal{S}(\mathbf{R}^n)$ and $\mathcal{D}(\mathbf{R}^n)$ are bornological.

Let us prove that $\mathcal{D}(\mathbf{R}^n)$ is dense in $\mathcal{S}(\mathbf{R}^n)$. Let B be the unit ball of $\mathbf{R}^n$ and $\eta \in \mathcal{D}(\mathbf{R}^n)$ a test function with support contained in $2B$ such that $0 \leq \eta \leq 1$ and $\eta(x) = 1$ for all $x \in B$ (lemma 1, a) of IV, p. 196).

Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. For every integer $m \geq 1$ and every $x \in \mathbf{R}^n$, set $\eta_m(x) = \eta(x/m)$. Finally define $\varphi_m = \eta_m \varphi$; we have $\varphi_m \in \mathcal{D}(\mathbf{R}^n)$.

Since $\partial^\alpha \eta_m = m^{-|\alpha|} (\partial^\alpha \eta)(x/m)$ for all $\alpha \in \mathbf{N}^n$ and $x \in \mathbf{R}^n$, from formula (2) of IV, p. 210 we deduce that the sequence (φ_m) is bounded in $\mathcal{S}(\mathbf{R}^n)$.

Let C be a compact subset of $\mathbf{R}^n$. The sequence $(\varphi_m)_{m \geq 1}$ converges to φ in $\mathcal{C}_K^\infty(\mathbf{R}^n)$ since φ_m coincides with φ on C for all sufficiently large m . Thus the sequence (φ_m) converges to φ in $\mathcal{C}^\infty(\mathbf{R}^n)$, and the corollary of Proposition 10 of IV, p. 211 allows us to conclude that the sequence (φ_m) converges to φ in $\mathcal{S}(\mathbf{R}^n)$.

Lemma 8. — Let B be a bounded subset of $\mathcal{D}(\mathbf{R}^n)$. The topology induced on B by the topology of $\mathcal{S}(\mathbf{R}^n)$ coincides with the topology induced by $\mathcal{D}(\mathbf{R}^n)$.

Since the inclusion of $\mathcal{D}(\mathbf{R}^n)$ into $\mathcal{S}(\mathbf{R}^n)$ is continuous, the topology on B induced by $\mathcal{D}(\mathbf{R}^n)$ is finer than the one induced by $\mathcal{S}(\mathbf{R}^n)$. Moreover, there exists a compact subset K of $\mathbf{R}^n$ such that $B \subset \mathcal{C}_K^\infty(\mathbf{R}^n)$. For every $\alpha \in \mathbf{N}^n$, we then have $p_{\alpha,K}(\varphi) \leq \tilde{q}_{\alpha,0}(\varphi)$, which implies that the topology induced by $\mathcal{S}(\mathbf{R}^n)$ is finer than the one induced by $\mathcal{D}(\mathbf{R}^n)$.

PROPOSITION 13. — Let $p \in [1, +\infty]$. The space $\mathcal{S}(\mathbf{R}^n)$ is contained in $\mathcal{L}^p(\mathbf{R}^n)$ and the canonical injection of $\mathcal{S}(\mathbf{R}^n)$ into $\mathcal{L}^p(\mathbf{R}^n)$ is continuous. The image of $\mathcal{S}(\mathbf{R}^n)$ in $L^p(\mathbf{R}^n)$ is dense if $p \neq +\infty$.

The first assertion is immediate for $p = +\infty$. Suppose now that $p \in [1, +\infty[$. Let m be an integer such that $n + 1 < mp$. For every $\varphi \in \mathcal{S}(\mathbf{R}^n)$ and $x \in \mathbf{R}^n$, we have

$$\|x\|^{n+1} |\varphi(x)|^p \leq \tilde{q}_{0,m}(\varphi)^p$$

hence $\varphi \in \mathcal{L}^p(\mathbf{R}^n)$ by prop. 3 of IV, p. 199. Moreover, we obtain

$$N_p(\varphi) \leq a_n^{1/p} \tilde{q}_{0,0}(\varphi) + b_n \tilde{q}_{0,m}(\varphi),$$

where

$$a_n = \int_{\|x\| \leq 1} d\mu(x), \quad b_n = \int_{\|x\| \geq 1} \frac{1}{\|x\|^{n+1}} d\mu(x),$$

therefore the injection of $\mathcal{S}(\mathbf{R}^n)$ into $\mathcal{L}^p(\mathbf{R}^n)$ is continuous.

As the space $\mathcal{D}(\mathbf{R}^n)$ is contained in $\mathcal{S}(\mathbf{R}^n)$, Proposition 4 of IV, p. 202 implies that $\mathcal{S}(\mathbf{R}^n)$ is dense in $L^p(\mathbf{R}^n)$ if $p \neq +\infty$.

IV.3.9 Functions of polynomial growth

Definition 3. — A function $f: \mathbf{R}^n \rightarrow \mathbf{C}$ has polynomial growth if there exists an integer $k \geq 1$ such that the map defined by $x \mapsto (1 + \|x\|)^{-k} f(x)$ is bounded on $\mathbf{R}^n$.

Every function with polynomial growth is locally bounded. Every polynomial function on $\mathbf{R}^n$ has polynomial growth.

PROPOSITION 14. — Let $f \in \mathcal{C}^\infty(\mathbf{R}^n)$. Assume that for every α in $\mathbf{N}^n$, the function $\partial^\alpha f$ is of polynomial growth. The linear map from the space $\mathcal{S}(\mathbf{R}^n)$ into itself defined by $\varphi \mapsto f\varphi$ is continuous.

For every φ in $\mathcal{S}(\mathbf{R}^n)$, the function $f\varphi$ belongs to $\mathcal{C}^\infty(\mathbf{R}^n)$. By hypothesis, for every $\alpha \in \mathbf{N}^n$, there exists an integer $k_\alpha \geq 0$ and a real number C_α such that $|\partial^\alpha f(x)| \leq C_\alpha (1 + \|x\|)^{k_\alpha}$ for all x in $\mathbf{R}^n$. Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. Let $\alpha \in \mathbf{N}^n$ and $k \in \mathbf{N}$. By prop. 1 of IV, p. 196, it follows

$$\begin{aligned} \tilde{q}_{\alpha,k}(f\varphi) &\leq \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} \sup_{x \in \mathbf{R}^n} \|x\|^k |\partial^\beta f(x) \partial^{\alpha-\beta} \varphi(x)| \\ &\leq \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} C_\beta \sup_{x \in \mathbf{R}^n} \|x\|^k (1 + \|x\|)^{k_\beta} |\partial^{\alpha-\beta} \varphi(x)|. \end{aligned}$$

Let $\beta \in \mathbf{N}^n$ such that $0 \leq \beta \leq \alpha$. For every $x \in \mathbf{R}^n$, we have

$$\begin{aligned} \|x\|^k (1 + \|x\|)^{k_\beta} |\partial^{\alpha-\beta} \varphi(x)| &\leq \sup_{\|x\| \leq 1} 2^{k_\beta} |\partial^{\alpha-\beta} \varphi(x)| \\ &\quad + \sup_{x \in \mathbf{R}^n} 2^{k_\beta} \|x\|^{k+k_\beta} |\partial^{\alpha-\beta} \varphi(x)| \end{aligned}$$

whence finally

$$\tilde{q}_{\alpha,k}(f\varphi) \leq \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} 2^{k\beta} C_{\beta} (\tilde{q}_{\alpha-\beta,0}(\varphi) + \tilde{q}_{\alpha-\beta,k+k_{\beta}}(\varphi)),$$

which implies the proposition.

IV.3.10 Tempered distributions

DEFINITION 4. — We call the space of tempered distributions on $\mathbf{R}^n$ the dual space of $\mathcal{S}(\mathbf{R}^n)$ endowed with the topology of bounded convergence. It is denoted $\mathcal{S}'(\mathbf{R}^n)$.

Since $\mathcal{S}(\mathbf{R}^n)$ is bornological, the space $\mathcal{S}'(\mathbf{R}^n)$ is complete (TVS, III, p. 24, cor. 1). Since $\mathcal{S}(\mathbf{R}^n)$ is a Montel space, the same holds for $\mathcal{S}'(\mathbf{R}^n)$ (EVT, IV, p. 19, prop. 9). The space $\mathcal{S}'(\mathbf{R}^n)$ is therefore reflexive (EVT, IV, p. 16, th. 2).

Lemma 9. — A linear map f from $\mathcal{S}(\mathbf{R}^n)$ into $\mathbf{C}$ is a tempered distribution if and only if for every family $(M_{\alpha,k})_{(\alpha,k) \in \mathbf{N}^n \times \mathbf{N}}$ in $\mathbf{R}_+$ the linear form f is bounded on the set of functions $\varphi \in \mathcal{S}(\mathbf{R}^n)$ such that $\tilde{q}_{\alpha,k}(\varphi) \leq M_{\alpha,k}$ for all $(\alpha,k) \in \mathbf{N}^n \times \mathbf{N}$.

Indeed, the space $\mathcal{S}(\mathbf{R}^n)$ is bornological (EVT, III, p. 12, prop. 2) and every bounded subset of $\mathcal{S}(\mathbf{R}^n)$ is contained in one of the bounded sets described in the statement.

Lemma 10. — Let F be a filter on $\mathcal{S}'(\mathbf{R}^n)$ having a countable base or containing a simply bounded set. Then F converges to a tempered distribution if and only if $\langle f, \varphi \rangle$ converges according to F for every Schwartz function φ .

In particular, a sequence $(f_m)_{m \in \mathbf{N}}$ of tempered distributions converges to a tempered distribution f if and only if one has $\langle f_m, \varphi \rangle \rightarrow \langle f, \varphi \rangle$ for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$.

Since $\mathcal{S}'(\mathbf{R}^n)$ is a Fréchet space, hence barrelled, (EVT, III, p. 25, cor. of prop. 2) the lemma follows from the Banach-Steinhaus theorem (EVT, III, p. 26, cor. 3 of th. 1).

The canonical injection j of $\mathcal{D}(\mathbf{R}^n)$ into $\mathcal{S}(\mathbf{R}^n)$ is continuous and its image is dense (Lemma 12 of IV, p. 212), hence the transpose of j , which is the restriction map of tempered distributions to the subspace $\mathcal{D}(\mathbf{R}^n)$ is an injective continuous linear map from $\mathcal{S}'(\mathbf{R}^n)$ into $\mathcal{D}'(\mathbf{R}^n)$. We will identify $\mathcal{S}'(\mathbf{R}^n)$ with a subspace of $\mathcal{D}'(\mathbf{R}^n)$ via this map.

Let $\alpha \in \mathbf{N}^n$. The linear map $\varphi \mapsto \partial^{\alpha} \varphi$ from $\mathcal{S}(\mathbf{R}^n)$ into $\mathcal{S}(\mathbf{R}^n)$ is continuous. Its transpose is therefore a continuous linear map from $\mathcal{S}'(\mathbf{R}^n)$ into $\mathcal{S}'(\mathbf{R}^n)$ (EVT, IV, p. 6, cor., b)). We

denote ∂^α the continuous linear map $(-1)^{|\alpha|} \partial^\alpha$ from $\mathcal{S}'(\mathbf{R}^n)$ into $\mathcal{S}'(\mathbf{R}^n)$. This definition is compatible with definition 2 of IV, p. 208 for distributions.

Let $h \in \mathcal{C}^\infty(\mathbf{R}^n)$ be a function such that $\partial^\alpha h$ is of polynomial growth for every $\alpha \in \mathbf{N}^n$. The transpose of the continuous linear map $\varphi \mapsto h\varphi$ (prop. 14 of IV, p. 214) is a continuous linear map on $\mathcal{S}'(\mathbf{R}^n)$, denoted $f \mapsto hf$.

IV.3.11 Interpretation of functions as tempered distributions

DEFINITION 5. — A measure ν on $\mathbf{R}^n$ is said to be tempered if there exists an integer $r \in \mathbf{N}$ such that the continuous map $x \mapsto (1 + \|x\|)^{-r}$ is ν -integrable on $\mathbf{R}^n$.

In other words, a measure ν is tempered if there exists $r \in \mathbf{N}$ such that the function defined by $x \mapsto \|x\|^{-r}$ is ν -integrable on the complement of the unit ball in $\mathbf{R}^n$. In particular, every bounded measure on $\mathbf{R}^n$ is tempered. More generally, if f is a μ -measurable function of polynomial growth and if ν is tempered, then the measure $f \cdot \nu$ is tempered.

The set $\mathcal{M}^t(\mathbf{R}^n)$ of tempered measures on $\mathbf{R}^n$ is a vector subspace of the space $\mathcal{M}(\mathbf{R}^n; \mathbf{C})$ of complex measures on $\mathbf{R}^n$.

PROPOSITION 15. — Let ν be a tempered measure on $\mathbf{R}^n$. The restriction of ν to $\mathcal{S}(\mathbf{R}^n)$ is a tempered distribution. It is zero if and only if the measure ν is zero.

Since ν is tempered, there exists a positive integer k such that the map $x \mapsto \|x\|^{-k}$ is ν -integrable on the complement of the unit ball in $\mathbf{R}^n$. For every Schwartz function $\varphi \in \mathcal{S}(\mathbf{R}^n)$, we have

$$|\langle \nu, \varphi \rangle| \leq \left(\int_{\|x\| \leq 1} d\nu \right) \tilde{q}_{0,0}(\varphi) + \left(\int_{\|x\| > 1} \|x\|^{-k} d\nu \right) \tilde{q}_{0,k}(\varphi),$$

therefore the restriction of ν to $\mathcal{S}(\mathbf{R}^n)$ is a tempered distribution.

The last assertion follows from prop. 6 of IV, p. 206, since $\mathcal{D}(\mathbf{R}^n)$ is contained in $\mathcal{S}(\mathbf{R}^n)$.

We will identify the space $\mathcal{M}^t(\mathbf{R}^n)$ with a subspace of $\mathcal{S}'(\mathbf{R}^n)$.

PROPOSITION 16. — Let $p \in [1, +\infty]$ and $f \in \mathcal{L}^p(\mathbf{R}^n)$. Then the measure $f \cdot \mu$ with density f with respect to Lebesgue measure is tempered. The map $f \mapsto f \cdot \mu$ from $L^p(\mathbf{R}^n)$ into $\mathcal{S}'(\mathbf{R}^n)$ thus defined is continuous.

Let q be the conjugate exponent of p and let $r \geq 0$ be such that $rq > n$. For every $x \in \mathbf{R}^n$

, write $g(x) = (1 + \|x\|)^{-r}$. The function g belongs to $\mathcal{L}^q(\mathbf{R}^n)$ by prop. 3 of IV, p. 199. By Hölder's inequality, we have

$$\int_{\mathbf{R}^n}^* (1 + \|x\|)^{-r} |f(x)| d\mu(x) \leq N_q(g) N_p(f) < +\infty,$$

therefore the measure $f \cdot \mu$ is tempered.

Let $f \in L^p(\mathbf{R}^n)$ and $\varphi \in \mathcal{S}'(\mathbf{R}^n)$. Hölder's inequality implies

$$|\langle f \cdot \mu, \varphi \rangle| = \left| \int_{\mathbf{R}^n} f(x) \varphi(x) d\mu(x) \right| \leq \|f\|_p \|\varphi\|_q$$

and the continuity of the mapping $f \mapsto f \cdot \mu$ then follows from prop. 13 of IV, p. 213.

For every $p \in [1, +\infty]$, we shall identify $L^p(\mathbf{R}^n)$ with a subspace of $\mathcal{S}'(\mathbf{R}^n)$ by the linear mapping $f \mapsto f \cdot \mu$.

PROPOSITION 17. — The spaces $\mathcal{D}(\mathbf{R}^n)$ and $\mathcal{S}(\mathbf{R}^n)$ are dense in $\mathcal{S}'(\mathbf{R}^n)$.

It suffices to show that $\mathcal{D}(\mathbf{R}^n)$ is dense in $\mathcal{S}'(\mathbf{R}^n)$. Let λ be a continuous linear form on $\mathcal{S}'(\mathbf{R}^n)$ vanishing on $\mathcal{D}(\mathbf{R}^n)$. Since the space $\mathcal{S}(\mathbf{R}^n)$ is reflexive, there exists a function $\varphi \in \mathcal{S}(\mathbf{R}^n)$ such that $\lambda(f) = \langle f, \varphi \rangle$ for all $f \in \mathcal{S}'(\mathbf{R}^n)$. We therefore have

$$0 = \lambda(\psi) = \langle \psi, \varphi \rangle = \int_{\mathbf{R}^n} \psi \varphi d\mu$$

for all $\psi \in \mathcal{D}(\mathbf{R}^n)$. The measure $\varphi \cdot \mu$ on $\mathbf{R}^n$ is therefore zero (prop. 6 of IV, p. 206), whence $\varphi = 0$. The proposition then follows from the Hahn–Banach theorem (EVT, II, p. 46, cor. 1).

IV.3.12 Fourier transform of tempered distributions

Since every Schwartz function φ is integrable on $\mathbf{R}^n$ (prop. 13 of IV, p. 213), it admits a Fourier transform $\mathcal{F}(\varphi)$ (resp. a Fourier cotransform $\overline{\mathcal{F}}(\varphi)$) which is identified with the continuous and bounded function on $\mathbf{R}^n$ defined by

$$y \mapsto \int_{\mathbf{R}^n} \varphi(x) \exp(-2i\pi x \cdot y) d\mu(x)$$

(resp. the function) $y \mapsto \int_{\mathbf{R}^n} \varphi(x) \exp(2i\pi x \cdot y) d\mu(x)$).

Lemma 11. — Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. The function $\mathcal{F}(\varphi)$ is infinitely differentiable on $\mathbf{R}^n$ and we have

$$\mathcal{F}(X^\alpha \varphi) = (-2i\pi)^{-|\alpha|} \partial^\alpha (\mathcal{F}(\varphi)),$$

$$\mathcal{F}(\partial^\alpha \varphi) = (2i\pi)^{|\alpha|} X^\alpha \mathcal{F}(\varphi)$$

for all α in $\mathbf{N}^n$.

We may assume that $n \geq 1$. Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. The function defined by $(x, y) \mapsto \varphi(x) \exp(2i\pi x \cdot y)$ from $\mathbf{R}^n \times \mathbf{R}^n$ into $\mathbf{C}$ satisfies the hypotheses of Corollary 1 of IV, p. 198 for every integer k . The Fourier transform of φ is therefore infinitely differentiable and satisfies

$$\partial^\alpha (\mathcal{F} \varphi)(y) = (-2i\pi)^{|\alpha|} \int_{\mathbf{R}^n} x^\alpha \varphi(x) \exp(-2i\pi x \cdot y) d\mu(x)$$

for all $y \in \mathbf{R}^n$, which implies the first formula.

Let us prove the second formula. By induction on $|\alpha|$ it suffices to do it when $|\alpha| = 1$, and one easily reduces to the case $\alpha = (1, 0, \dots, 0)$. Let us write every $x \in \mathbf{R}^n$ in the form $x = (x_1, x')$ with $x' \in \mathbf{R}^{n-1}$, and denote by μ_1 (resp. μ') the Lebesgue measure on $\mathbf{R}^{n-1}$ (resp. $\mathbf{R}$). By the Lebesgue–Fubini theorem (INT, V, p. 96, § 8, n° 4, th. 1), for every $y = (y_1, y') \in \mathbf{R} \times \mathbf{R}^{n-1}$, it follows that

$$\begin{aligned} \mathcal{F}(\partial_1 \varphi)(y) &= \int_{\mathbf{R}^{n-1}} \exp(-2i\pi x' \cdot y') \\ &\times \left(\int_{\mathbf{R}} (\partial_1 \varphi)(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1) \right) d\mu'(x'). \end{aligned}$$

For every compact interval $[a, b]$ in $\mathbf{R}$ and every $x' \in \mathbf{R}^{n-1}$, we have

$$\begin{aligned} \int_a^b (\partial_1 \varphi)(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1) &= \\ [\varphi(x_1, x') \exp(-2i\pi x_1 y_1)]_a^b &+ \\ + 2i\pi y_1 \int_a^b \varphi(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1) & \end{aligned}$$

by integration by parts (FVR, II, p. 10, formula (10)). When a tends to $-\infty$ and b tends to

$+\infty$, the first term on the right-hand side converges to 0 since $\varphi \in \mathcal{S}(\mathbf{R}^n)$. The second term converges by Lebesgue's theorem (INT, IV, p. 137, § 3, n° 7, th. 6) to

$$2i\pi y_1 \int_{\mathbf{R}} \varphi(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1),$$

since the map $x_1 \mapsto \varphi(x_1, x')$ is integrable on $\mathbf{R}$. Since $x_1 \mapsto \partial_1 \varphi(x_1, x')$ is also integrable on $\mathbf{R}$, it follows that

$$\begin{aligned} & \int_{\mathbf{R}} \partial_1 \varphi(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1) \\ &= 2i\pi y_1 \int_{\mathbf{R}} \varphi(x_1, x') \exp(-2i\pi x_1 y_1) d\mu_1(x_1) \end{aligned}$$

and finally, applying the Lebesgue–Fubini theorem again, we conclude that

$$\mathcal{F}(\partial_1 \varphi)(y) = 2i\pi y_1 \int_{\mathbf{R}^n} \varphi(x) \exp(-2i\pi x \cdot y) d\mu(x),$$

as desired.

PROPOSITION 18. — The restriction to $\mathcal{S}(\mathbf{R}^n)$ of the Fourier transformation is an automorphism of topological vector spaces whose inverse is the restriction of the Fourier cotransformation.

Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. By the preceding lemma, the Fourier transform of φ belongs to $\mathcal{C}^\infty(\mathbf{R}^n)$. Moreover, for $\alpha \in \mathbf{N}^n$ and $\beta \in \mathbf{N}^n$, we have

$$\begin{aligned} X^\beta \partial^\alpha (\mathcal{F}(\varphi)) &= (-2i\pi)^{|\alpha|} X^\beta \mathcal{F}(X^\alpha \varphi) \\ &= (-1)^{|\alpha|} (2i\pi)^{|\alpha|-|\beta|} \mathcal{F}(\partial^\beta (X^\alpha \varphi)). \end{aligned}$$

In particular, the function $X^\beta \partial^\alpha (\mathcal{F}(\varphi))$ is bounded. Since α and β are arbitrary in $\mathbf{N}^n$, this means that $\mathcal{F}(\varphi)$ belongs to $\mathcal{S}(\mathbf{R}^n)$. Moreover, this calculation implies

$$q_{\alpha, \beta}(\mathcal{F}(\varphi)) \leq (2\pi)^{|\alpha|-|\beta|} \|\partial^\beta (X^\alpha \varphi)\|_1$$

for $(\alpha, \beta) \in \mathbf{N}^n \times \mathbf{N}^n$ and $\varphi \in \mathcal{S}(\mathbf{R}^n)$.

Since the inclusion of the space $\mathcal{S}(\mathbf{R}^n)$ in $L^1(\mathbf{R}^n)$ is continuous (prop. 13 of IV, p. 213), the map $q_{\alpha, \beta} \circ \mathcal{F}$ of $\mathcal{S}(\mathbf{R}^n)$ in $\mathbf{R}$ is continuous. It follows that the Fourier transformation is

continuous from the space $\mathcal{S}(\mathbf{R}^n)$ into itself (cf. EVT, II, p. 7, prop. 5, c)). One verifies similarly that the Fourier cotransformation is continuous from the space $\mathcal{S}(\mathbf{R}^n)$ into itself. By the Fourier inversion formula (theorem 3 of II, p. 222), the Fourier transformation and the Fourier cotransformation are mutually inverse isomorphisms.

DEFINITION 6. — The Fourier transformation (resp. Fourier cotransformation) on $\mathcal{S}'(\mathbf{R}^n)$ is called the transpose of the Fourier transformation on $\mathcal{S}(\mathbf{R}^n)$ (resp. of the Fourier cotransformation).

We still denote by $\mathcal{F}$ (resp. $\overline{\mathcal{F}}$) the Fourier transformation (resp. the Fourier cotransformation) on $\mathcal{S}'(\mathbf{R}^n)$. The Fourier transformation on $\mathcal{S}'(\mathbf{R}^n)$ is therefore an automorphism of topological vector spaces whose inverse is the Fourier cotransformation. For every $f \in \mathcal{S}'(\mathbf{R}^n)$, the tempered distribution $\mathcal{F}(f)$ (resp. $\overline{\mathcal{F}}(f)$) is defined by the formula

$$\langle \mathcal{F}(f), \varphi \rangle = \langle f, \mathcal{F}(\varphi) \rangle \quad (\text{resp. } \langle \overline{\mathcal{F}}(f), \varphi \rangle = \langle f, \overline{\mathcal{F}}(\varphi) \rangle)$$

for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$.

PROPOSITION 19. — Let f be a tempered distribution associated with a bounded measure $\nu \in \mathcal{M}^1(\mathbf{R}^n)$ (resp. with $g \in L^2(\mathbf{R}^n)$). The Fourier transformation of f in $\mathcal{S}'(\mathbf{R}^n)$ is the tempered distribution associated with the Fourier transformation of the measure ν (resp. with the Fourier transformation of g).

Let ν be a bounded measure on $\mathbf{R}^n$ and let f be the tempered distribution associated with ν . The Fourier transformation $\mathcal{F}(\nu)$ is a continuous and bounded function on $\mathbf{R}^n$ (prop. 3 of II, p. 207). The tempered distribution associated with this function satisfies

$$\langle \mathcal{F}(\nu), \varphi \rangle = \int_{\mathbf{R}^n} \mathcal{F}(\nu) \varphi d\mu = \int_{\mathbf{R}^n} \mathcal{F}(\varphi) d\nu = \langle f, \mathcal{F}(\varphi) \rangle = \langle \mathcal{F}(f), \varphi \rangle$$

for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$, where the second equality is prop. 13 of II, p. 221, which is applicable here since the measure $\varphi \cdot \mu$ is bounded. The tempered distribution associated with $\mathcal{F}(\nu)$ is therefore $\mathcal{F}(f)$.

When f is the tempered distribution associated with $g \in L^2(\mathbf{R}^n)$, we follow a completely analogous procedure, using formula (29) of II, p. 221.

A similar statement also holds for the Fourier cotransformation.

Remark. — The elementary formulas concerning the Fourier transformation of measures remain valid for the Fourier transformation of tempered distributions. For example, for $f \in \mathcal{S}'(\mathbf{R}^n)$ and $\alpha \in \mathbf{N}^n$, we have

$$\begin{aligned}\mathcal{F}(\partial^\alpha f) &= (2i\pi)^{|\alpha|} X^\alpha \mathcal{F}(f) \\ \mathcal{F}(X^\alpha f) &= (-2i\pi)^{-|\alpha|} \partial^\alpha (\mathcal{F}(f))\end{aligned}$$

by Lemma 11,

IV.3.13 Distributions and tempered distributions on a vector space

Let u be a bijective linear map from $\mathbf{R}^n$ into $\mathbf{R}^n$. The map $\varphi \mapsto \varphi \circ u$ is an automorphism of $\mathcal{S}(\mathbf{R}^n)$ (resp. of $\mathcal{D}(\mathbf{R}^n)$); its transpose is an automorphism of $\mathcal{S}'(\mathbf{R}^n)$ (resp. of $\mathcal{D}'(\mathbf{R}^n)$).

Let E be a real vector space of finite dimension n . Let $v : \mathbf{R}^n \rightarrow E$ be an isomorphism of vector spaces. We denote by $\mathcal{S}(E)$ (resp. $\mathcal{D}(E)$) the set of mappings $\varphi : E \rightarrow \mathbf{C}$ such that $\varphi \circ v \in \mathcal{S}(\mathbf{R}^n)$ (resp. such that $\varphi \circ v \in \mathcal{D}(\mathbf{R}^n)$). By the preceding remark, these spaces do not depend on the choice of v ; they are isomorphic to $\mathcal{S}(\mathbf{R}^n)$ (resp. $\mathcal{D}(\mathbf{R}^n)$). We denote by $\mathcal{S}'(E)$ (resp. $\mathcal{D}'(E)$) the dual of $\mathcal{S}(E)$ (resp. of $\mathcal{D}(E)$) equipped with the topology of bounded convergence. It is a topological vector space isomorphic to $\mathcal{S}'(\mathbf{R}^n)$ (resp. to $\mathcal{D}'(\mathbf{R}^n)$).

Let E and F be real vector spaces of dimension n , in duality with respect to a bilinear form $b : E \times F \rightarrow \mathbf{R}$. The locally compact commutative group E is in duality with F with respect to the mapping

$$(x, y) \mapsto \exp(2i\pi b(x, y))$$

from $E \times F$ into $\mathbf{U}$ (cf. cor. 1 of II, p. 235). We equip E and F with Haar measures that are dual to each other with respect to this mapping.

The space $\mathcal{S}(E)$ is contained in $L^1(E)$; the Fourier transformation of E induces, by passing to subspaces and by duality, an isomorphism of topological vector spaces from $\mathcal{S}(E)$ into $\mathcal{S}(F)$, whose transpose is an isomorphism of topological vector spaces from $\mathcal{S}'(F)$ into $\mathcal{S}'(E)$.

IV.3.14 Sobolev Spaces

Let U be an open subset of $\mathbf{R}^n$. Let p be a real number ≥ 1 and let k be a natural number. We denote by $W^{k,p}(U)$ the space of distributions $f \in \mathcal{D}'(U)$ such that, for every $\alpha \in \mathbf{N}^n$ with $|\alpha| \leq k$, the distribution $\partial^\alpha f$ is associated with an element of $L^p(U)$.

In particular, when $U = \mathbf{R}^n$, the elements of $W^{k,p}(U)$ are tempered distributions.

The map from $W^{k,p}(U)$ into $\mathbf{R}_+$ that assigns to $f \in W^{k,p}(U)$

$$\|f\|_{k,p} = \left(\sum_{|\alpha| \leq k} \|\partial^\alpha f\|_p^p \right)^{1/p}$$

is a norm on $W^{k,p}(U)$. The space $W^{k,p}(U)$ will always be equipped with this norm; this normed space is called the Sobolev space of index k and exponent p .

The space $\mathcal{D}(U)$ is contained in $W^{k,p}(U)$. We denote $W_0^{k,p}(U)$ the closure of $\mathcal{D}(U)$ in $W^{k,p}(U)$. This is a closed subspace of $W^{k,p}(U)$.

One has $W_0^{k,p}(\mathbf{R}^n) = W^{k,p}(\mathbf{R}^n)$, but the spaces $W^{k,p}(U)$ and $W_0^{k,p}(U)$ are distinct in general (cf. exercises 12 of IV, p. 334 and 14 of IV, p. 334).

We also denote $H^k(U) = W^{k,2}(U)$ and $H_0^k(U) = W_0^{k,2}(U)$.

The norm of $H^k(U)$ is a pre-Hilbertian norm, associated with the positive Hermitian form on $H^k(U)$ defined by

$$(f_1, f_2) \mapsto \sum_{|\alpha| \leq k} \int_U \overline{\partial^\alpha f_1} \partial^\alpha f_2 d\mu.$$

The Hilbert space $H^k(U)$ coincides with the space denoted $\mathcal{H}^k$ in EVT, V, p. 6, example (3).

One has $W^{0,p}(U) = L^p(U)$ and $H^0(U) = L^2(U)$ by definition; moreover $W_0^{0,p}(U) = L^p(U)$ by prop. 4, b) of IV, p. 202.

PROPOSITION 20. — The Sobolev spaces $W^{k,p}(U)$ and $W_0^{k,p}(U)$ are Banach spaces of countable type. In particular, the spaces $H^k(U)$ and $H_0^k(U)$ are Hilbert spaces of countable type.

It suffices to prove the assertions concerning $W^{k,p}(U)$.

Let I be the set of $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq k$. The linear map u from $W^{k,p}(U)$ into $L^p(U)^I$ that associates with f the family $(\partial^\alpha f)_{\alpha \in I}$ is injective; it is continuous and strict by definition

of the norm on $W^{k,p}(U)$. To prove that $W^{k,p}(U)$ is complete, it suffices to prove that its image under u is closed. Now, let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $W^{k,p}(U)$ such that $(u(f_n))_{n \in \mathbb{N}}$ converges. Let $(g_\alpha)_{\alpha \in I} \in L^p(U)^I$ be its limit. For $\alpha \in I$, the sequence $(\partial^\alpha f_n)_{n \in \mathbb{N}}$ converges in $L^p(U)$ to g_α . A fortiori, the convergence takes place in $\mathcal{D}'(U)$. Set $f = g_0$. For every $\varphi \in \mathcal{D}(U)$, it follows that

$$\begin{aligned} \langle \partial^\alpha f, \varphi \rangle &= (-1)^{|\alpha|} \langle f, \partial^\alpha \varphi \rangle \\ &= \lim_{n \rightarrow +\infty} (-1)^{|\alpha|} \langle f_n, \partial^\alpha \varphi \rangle = \lim_{n \rightarrow +\infty} \langle \partial^\alpha f_n, \varphi \rangle = \langle g_\alpha, \varphi \rangle. \end{aligned}$$

This proves that $g_\alpha = \partial^\alpha f$ for all $\alpha \in I$, hence $(g_\alpha)_{\alpha \in I} = u(f)$ belongs to the image of u .

The space $W^{k,p}(U)$ is identified by u with a subspace of the space $L^p(U)^I$; the latter is of countable type (prop. 2 of IV, p. 180 and TG, IX, p. 19, cor., (ii))), hence so is $W^{k,p}(U)$ (loc. cit., (i)).

PROPOSITION 21. — Let N be the Euclidean norm on $\mathbf{R}^n$. Let k be an integer ≥ 0 . The Sobolev space $H^k(\mathbf{R}^n)$ is the space of $f \in \mathcal{S}'(\mathbf{R}^n)$ such that $(1 + N^k)\mathcal{F}(f)$ belongs to $L^2(\mathbf{R}^n)$.

We proceed by induction on k . When $k = 0$, the result is a consequence of the Plancherel theorem (II, p. 215, th. 1). Suppose that $k = 1$. For $f \in \mathcal{S}'(\mathbf{R}^n)$, we have $(1 + N)\mathcal{F}f \in L^2(\mathbf{R}^n)$ if and only if $f \in L^2(\mathbf{R}^n)$ and $N\mathcal{F}f \in L^2(\mathbf{R}^n)$. Moreover, $N\mathcal{F} \in L^2(\mathbf{R}^n)$ if and only if, for every $\alpha \in \mathbf{N}^n$ such that $|\alpha| = 1$, we have $X^\alpha \mathcal{F}f \in L^2(\mathbf{R}^n)$. Since one has $\mathcal{F}(\partial^\alpha f) = 2i\pi X^\alpha \mathcal{F}f$, this condition means that $\mathcal{F}(\partial^\alpha f) \in L^2(\mathbf{R}^n)$ for every α with $|\alpha| = 1$, that is, $\partial^\alpha f \in L^2(\mathbf{R}^n)$ for every α with $|\alpha| = 1$. It follows that the assertion is true for $k = 1$.

Suppose now that $k \geq 2$ and that the assertion concerning $H^\ell(\mathbf{R}^n)$ is valid for every positive integer $\ell \leq k-1$. Let $f \in \mathcal{S}'(\mathbf{R}^n)$. By definition, one has $f \in H^k(\mathbf{R}^n)$ if and only if $f \in L^2(\mathbf{R}^n)$ and, for every $\beta \in \mathbf{N}^n$ such that $|\beta| \leq 1$, the distribution $\partial^\beta f$ belongs to $H^{k-1}(\mathbf{R}^n)$. This is equivalent, by the induction hypothesis, to $f \in L^2(\mathbf{R}^n)$ and $(1 + N^{k-1})\mathcal{F}(\partial^\beta f) \in L^2(\mathbf{R}^n)$ for every $\beta \in \mathbf{N}^n$ such that $|\beta| \leq 1$. Since $\mathcal{F}(\partial^\beta f) = (2i\pi X)^\beta \mathcal{F}(f)$, the condition $f \in H^k(\mathbf{R}^n)$ is equivalent to saying that $\mathcal{F}f \in L^2(\mathbf{R}^n)$ and $(1 + N^{k-1})X^\beta \mathcal{F}f \in L^2(\mathbf{R}^n)$ for every $\beta \in \mathbf{N}^n$ such that $|\beta| \leq 1$.

The inequalities

$$1 + N^k \leq 1 + N^{k-1} \sum_{\substack{\beta \in \mathbb{N}^n \\ |\beta| \leq 1}} |X^\beta| \leq 1 + n^{1/2} N^k \leq (1 + n^{1/2})(1 + N^k)$$

then imply that $f \in H^k(\mathbf{R}^n)$ if and only if $(1 + N^k)\mathcal{F} \in L^2(\mathbf{R}^n)$.

IV.4 PARTIAL OPERATORS

IV.4.1 Partial operators

In this section, K is a commutative field.

We recall (E, II, §3, p. 9–10) that a graph⁽²⁾ is a set all of whose elements are ordered pairs. If A and B are sets, a correspondence between A and B is a triple (Γ, A, B) , where Γ is a graph contained in $A \times B$; its domain (also called domain) is $\text{pr}_1(\Gamma)$, and the set of its values is $\text{pr}_2(\Gamma)$. A correspondence is a function (E, II, p. 13, def. 9) if its graph is functional and if its set of departure coincides with its domain of definition. Every subset of a functional graph is a functional graph.

DEFINITION 1. — Let E and F be vector spaces over K . A partial operator u from E to F is a correspondence (Γ, E, F) between E and F satisfying the following conditions:

- (i) The graph Γ is a vector subspace of $E \times F$;
- (ii) The graph Γ is functional.

If $E = F$, we say that u is a partial operator on E .

Let u be a partial operator from E to F . The graph Γ of the correspondence u is called the graph of the partial operator u , and is also denoted Γ_u . We denote $\mathcal{P}(E; F)$ the set of partial operators from E to F ; we simply denote $\mathcal{P}(E) = \mathcal{P}(E; E)$.

Giving a partial operator from E to F amounts to giving a vector subspace D of E and a linear map u from D to F , the associated partial operator being the correspondence (Γ, E, F) where $\Gamma \subset D \times F$ is the graph of u .

The domain of definition of a partial operator u is simply called the domain of u , and denoted $\text{dom}(u)$.

Every linear map u from E to F is a partial operator from E to F .

If $D \subset E$ is a vector subspace, we will denote by 1_D the partial operator with domain D , which is the identity map on D , that is, the correspondence (Δ_D, E, E) where Δ_D is the diagonal of $D \times D$ (E, II, p. 13, def. 8). We shall denote by 0_D the partial operator with domain D that is zero on D , that is, the correspondence $(D \times \{0\}, E, F)$.

Partial operators $u = (\Gamma, E, F)$ and $u' = (\Gamma', E, F)$ from E to F are equal if and only if $\text{dom}(u) = \text{dom}(u')$ and if the linear maps u and u' from $\text{dom}(u)$ into F coincide.

Following E, II, §3, the following notions are defined:

- (i) The image of a subset A of E under u is the subset $u(A \cap D)$ of F ; it is simply denoted $u(A)$. The inverse image under u of a subset B of F is the subset $\bar{u}(B)$ of D .

If A (resp. B) is a vector subspace of E (resp. of F), then its image under u (resp. its inverse image) is a vector subspace of F (resp. of E).

The image of u is the vector subspace $u(D)$ of F , also denoted $\text{Im}(u)$. One says that u is a surjective partial operator if $\text{Im}(u) = F$. The kernel of u is the vector subspace $u^{-1}(\{0\})$ of E , also denoted $\text{Ker}(u)$. The kernel of u is reduced to 0 if and only if the linear map u of $\text{dom}(u)$ in F is injective. We then say that u is injective. If u is injective and surjective, it is said to be bijective.

- (ii) If E, F , and G are vector spaces over K and $u = (\Gamma, E, F)$, $v = (\Gamma', F, G)$ are partial operators from E to F and from F to G , respectively, the composed correspondence $v \circ u = (\Gamma' \circ \Gamma, E, G)$ is a partial operator from E into G . Its domain is $u^{-1}(\text{dom}(v))$. If H is a vector space over K and $w = (\Gamma'', G, H)$ a partial operator from G into H , we have $w \circ (v \circ u) = (w \circ v) \circ u$. One will sometimes write vu instead of $v \circ u$.

- (iii) In particular, for every partial operator u from E into F and every $a \in K$ the partial operators $au = (a1_F) \circ u$ and $ua = u \circ (a1_E)$ are defined. They are equal if $a \neq 0$ or if the domain of u is equal to E ; we have $u0 = 0_E$ and $0u = 0_{\text{dom}(u)}$.

Let E be a vector space. By the preceding, the set $\mathcal{P}(E)$ endowed with the composition law defined by $(u, v) \mapsto u \circ v$ is an associative unital magma (A, I, p. 4, def. 5 and A, I, p. 12, def. 2) with identity element 1_E . For every $n \in \mathbb{N}$, we denote by u^n the composite ${}^n u$ (A, I, p. 13).

Moreover, we define the following notions:

- (i) If $u = (\Gamma, E, F)$ is a partial operator from E to F , and if G is a vector subspace of E , the reduction of u to G is the partial operator $(\Gamma \cap (G \times F), E, F)$ from E to F . Its domain is $\text{dom}(u) \cap G$; it will sometimes be denoted $u|_G$ when no confusion with the restriction of u to the subspace G is to be feared.
- (ii) Let v an injective partial operator from F to E . The inverse correspondence $v^{-1} = (\Gamma^{-1}, E, F)$ of v is then a partial operator such that $\text{dom}(v^{-1}) = \text{Im}(v)$. One says that v^{-1} is the inverse partial operator of v . We have the equalities $v \circ v^{-1} = 1_{\text{dom}(v^{-1})}$ in $\mathcal{P}(E)$ and $v^{-1} \circ v = 1_{\text{dom}(v)}$ in $\mathcal{P}(F)$. The partial operator v^{-1} is injective and we have $(v^{-1})^{-1} = v$.
- (iii) Let E, F , and G be vector spaces. Let u (resp. v) be an injective partial operator from E to F (resp. from F to G). Then the partial operator $v \circ u$ is injective and $(v \circ u)^{-1} = u^{-1} \circ v^{-1}$.
- (iv) If u and v are partial operators from E to F , we say that v is an extension of u , and we write $u \subset v$ if the graph of u is contained in the graph of v . This implies that $\text{dom}(u) \subset \text{dom}(v)$ and that u is the reduction of v to $\text{dom}(u)$. The relation “ $u \subset v$ ” is an order relation on $\mathcal{P}(E; F)$. We have for example $au \subset ua$ for all $a \in K$ and every $u \in \mathcal{P}(E; F)$.
- (v) Let E be a vector space over K and $(F_i)_{i \in I}$ a family of vector spaces over K . For $i \in I$, let u_i a partial operator from E to F_i . The product partial operator of the u_i is the partial operator from E to the product vector space of the spaces F_i whose domain is the intersection D of the spaces $\text{dom}(u_i)$ and which assigns to $x \in D$ the family $(u_i(x))_{i \in I}$. It is denoted $(u_i)_{i \in I}$.
- (vi) Let $A: F \times F \rightarrow F$ be the linear map $(x, y) \mapsto x + y$. Let u and v be partial operators from E to F . The sum $u + v$ is the partial operator $A \circ (u, v)$ from E to F . Its domain is $\text{dom}(u) \cap \text{dom}(v)$. For u, v, w in $\mathcal{P}(E; F)$, we have $(u + v) + w = u + (v + w)$.

Let G be a vector space over K . For all u and v in $\mathcal{P}(E; F)$ and every $w \in \mathcal{P}(F; G)$, we have $w \circ u + w \circ v \subset w \circ (u + v)$. In general, there is no equality in this formula (exercise 1 of IV,

p. 344), but this is the case when the domain of w is equal to F . For $w \in \mathcal{P}(G; E)$, we have $u \circ w + v \circ w = (u + v) \circ w$.

(vii) Let L be an extension of the field K . Let E and F be vector spaces over K and $E_{(L)} = L \otimes_K E$, $F_{(L)} = L \otimes_K F$ the L -vector spaces obtained by extension of scalars from K to L (A, II, p. 82). For every partial operator u from E to F , we denote $u_{(L)}$ the partial operator from $E_{(L)}$ into $F_{(L)}$ whose graph is the vector subspace $L \otimes_K \Gamma_u$ of $E_{(L)} \times F_{(L)}$; its domain is $L \otimes_K \text{dom}(u)$ and it coincides on this domain with the unique linear map that sends $1 \otimes x$ onto $1 \otimes u(x)$ for all $x \in \text{dom}(u)$.

Let v be a partial operator from E to F . We have $u \subset v$ if and only if $u_{(L)} \subset v_{(L)}$.

(viii) Let E_1, F_1, E_2, F_2 of K -vector spaces. Let u (resp. v) be a partial operator from E_1 in F_1 (resp. of E_2 in F_2). We denote by $u \otimes v$ the partial operator from $E_1 \otimes F_1$ into $E_2 \otimes F_2$ with domain $\text{dom}(u) \otimes \text{dom}(v)$ such that $(u \otimes v)(x \otimes y) = u(x) \otimes v(y)$ for all $(x, y) \in E_1 \times E_2$.

IV.4.2 Closed, closable, and densely defined operators

In this section, K denotes a commutative topological field (TG, III, p. 54).

DEFINITION 2. — Let E and F be topological vector spaces over K (EVT, I, p. 1, def. 1).

Let $u \in \mathcal{P}(E; F)$ be a partial operator from E to F .

We say that u is an operator with dense domain if the domain of u is dense in E .

We say that u is closed if the graph of u is closed in the topological vector space $E \times F$. We say that u is closable if it has a closed extension.

Let E, F , and G be topological vector spaces over K . Every extension of an operator $u \in \mathcal{P}(E; F)$ with dense domain has dense domain. Moreover, if $v \in \mathcal{L}(E; F)$, then $u + v$ is an operator with dense domain. If $v : F \rightarrow G$ (resp. $w : G \rightarrow E$) is an isomorphism of topological vector spaces, then $v \circ u$ (resp. $u \circ w$) is an operator with dense domain.

Examples. — Let E and F be topological vector spaces over K .

- 1) Assume the space F is Hausdorff. Let u be a linear map from E into F . If u is continuous, then the partial operator u is closed (TG, I, p. 53, cor. 2). Suppose moreover that K is a

non-discrete valued field and that E and F are metrizable and complete topological vector spaces over K . By the closed graph theorem (EVT, I, p. 19, cor. 5), the partial operator defined by u is then closed if and only if u is continuous.

- 2) Suppose the space E is Hausdorff. Let v be an injective continuous linear map from F into E . The partial operator $v^{-1} \in \mathcal{P}(E; F)$ (cf. IV, p. 226) is then closed, since its graph is the image of the graph of v , which is closed, by the isomorphism of topological vector spaces from $F \times E$ into $E \times F$ defined by $(y, x) \mapsto (x, y)$.

PROPOSITION 1. — Let E and F be topological vector spaces over K . A partial operator u from E into F is closable if and only if the closure of the graph Γ_u of u in $E \times F$ is a functional graph. There then exists a unique partial operator v from E to F whose graph is $\overline{\Gamma}_u$, and it is the smallest closed extension of u .

If the closure of the graph of u in $E \times F$ is a functional graph, it is the graph of a partial operator, and the latter is a closed extension of u , which is therefore closable. Conversely, suppose that $u \subset w$ with w closed. The closure $\overline{\Gamma}_u$ of the graph of u in $E \times F$ is contained in Γ_w , hence $\overline{\Gamma}_u$ is a functional graph.

The last assertion follows from the fact that if w is a closed extension of u , then the graph of w contains $\overline{\Gamma}_u$.

DEFINITION 3. — Let E and F be topological vector spaces over K . Let u be a closable operator from E into F . The closed operator whose graph is $\overline{\Gamma}_u$ is called the closure of u . It is denoted $\overline{u}$.

Remark. — Let E and F be topological vector spaces over K . Let u be a closable operator from E to F . The domain of the closure of u is contained in the closure of the domain of u in E . In general, it is distinct from it (exercise 1 of IV, p. 344, b)).

If $u \in \mathcal{P}(E; F)$ is closable and $\text{dom}(u) = E$, then $u = \overline{u}$ is closed, since then $\text{dom}(\overline{u}) = \text{dom}(u)$.

PROPOSITION 2. — Let $K = \mathbb{R}$ and let E and F be topological vector spaces over $\mathbb{R}$. Let u be a partial operator from E into F . Then u is densely defined (resp. is closed, is closable) if and only if $u_{(\mathbb{C})}$ has dense domain (resp. is closed, is closable).

Suppose that the domain of u is dense in E . Every neighborhood of 0 in $E_{(\mathbb{C})}$ contains a

neighborhood of the form $V + iV$ (EVT, II, p. 65), where V is a neighborhood of 0 in E , hence contains an element of the domain of $u_{(C)}$; this partial operator is therefore densely defined. The converse is also true, since the map from $E_{(C)}$ into E that maps $x + iy$ to x for all $(x, y) \in E \times E$ is continuous and surjective.

The graph of $u_{(C)}$ is identified with the complexified topological vector space of the graph of u . It is therefore closed in $E_{(C)} \times E_{(C)}$ if the graph of u is closed in $E \times E$. Conversely, we have $\Gamma_u = \Gamma_{u_{(C)}} \cap (E \times E)$ in $E_{(C)} \times E_{(C)}$; since $E \times E$ is closed in $E_{(C)} \times E_{(C)}$, the partial operator u is closed when $u_{(C)}$ is.

A partial operator v from E into F is an extension of u if and only if $v_{(C)}$ is an extension of $u_{(C)}$, so the partial operator $u_{(C)}$ is closable if u is closable. Conversely, if $u_{(C)}$ is closable, the partial operator u is also closable, since $\overline{\Gamma_u} = \overline{\Gamma_{u_{(C)}}} \cap (E \times E)$, which is then a functional graph (prop. 1).

Lemma 1. — Let E, F , and G be topological vector spaces over K . Let u be a closed operator from E to F .

- a) For every $v \in \mathcal{L}(E; F)$, the partial operator $u + v$ is closed.
- b) For every $v \in \mathcal{L}(G; E)$, the partial operator $u \circ v$ is closed.

Let us prove *a*). Let γ be the map $(x, y) \mapsto (x, y - v(x))$ from $E \times F$ into itself; it is continuous. For every $(x, y) \in E \times F$, we have $\gamma(x, y) \in \Gamma_u$ if and only if $x \in \text{dom}(u)$ and $y = u(x) + v(x)$, that is, $\gamma^{-1}(\Gamma_u) = \Gamma_{u+v}$. The assertion follows.

Let us prove *b*). The map $\eta = (v, 1_F)$ from $G \times F$ into $E \times F$ is continuous; for every $(z, y) \in G \times F$, we have $\eta(z, y) = (v(z), y)$, hence $\overline{\eta}(\Gamma_u) = \Gamma_{u \circ v}$. The assertion follows.

Let E and F be topological vector spaces over K , with F being Hausdorff. Let $a \in K$. If $u \in \mathcal{P}(E; F)$ is closable, then so is au . If $a \neq 0$, we have $\overline{au} = a\overline{u}$, and u is closed if and only if au is closed. If $a = 0$, the closure of au is $0_{\overline{\text{dom}(u)}}$, and $a\overline{u}$ is equal to $0_{\text{dom}(\overline{u})}$; it may therefore happen that u is closed but that au is not.

PROPOSITION 3. — Let E and F be topological vector spaces over K , with F being Hausdorff. Let u be a closed partial operator from E to F . The kernel of u is a closed subspace of E .

Indeed, the kernel of u is the inverse image of the closed subspace $\Gamma_u \cap (E \times \{0\})$ of $E \times F$ under the continuous linear map $x \mapsto (x, 0)$ from E into $E \times F$.

In the rest of this section, we assume that K is a non-discrete valued field.

DEFINITION 4. — Let E and F be normed spaces over K and let u be a partial operator from E to F . For x in $\text{dom}(u)$, we denote

$$\|x\|_u = (\|x\|_E^2 + \|u(x)\|_F^2)^{1/2}.$$

The map $x \mapsto \|x\|_u$ thus defined is a norm on $\text{dom}(u)$. We denote E_u the normed space thus obtained.

Remarks. — Let E and F be normed spaces over K , and let u be a partial operator from E into F .

- 1) The canonical injection of E_u in E is continuous since we have $\|x\| \leq \|x\|_u$ for all $x \in E$. In particular, every subspace of $\text{dom}(u)$ that is closed in E is closed in E_u .
- 2) If E and F are Hilbert spaces, then the space E_u is a pre-Hilbert space, since the norm on E_u comes from the positive Hermitian form

$$(x, y) \mapsto (x | y)_u = \langle x | y \rangle + \langle u(x) | u(y) \rangle$$

on $\text{dom}(u)$.

PROPOSITION 4. — Let E and F be Banach spaces (resp. Hilbert spaces) and let u be a partial operator from E into F . Then u is closed if and only if the normed space E_u is a Banach space (resp. a Hilbert space).

It suffices to treat the case of Banach spaces. The norm of E_u is obtained, by transport of structure via the bijective linear map $(x, y) \mapsto x$ of Γ_u into $\text{dom}(u)$, from the norm obtained by restriction to the subspace Γ_u with the norm $(x, y) \mapsto (\|x\|_E^2 + \|y\|_F^2)^{1/2}$ on the Banach space $E \oplus F$. The space E_u is therefore a Banach space if and only if the subspace Γ_u of $E \oplus F$ is closed.

If u and v are partial operators from E to F and from F to G , respectively, and if u is closed, then the partial operator $v \circ u$ is not closed in general, even if v is continuous (exercise 1 of IV,

p. 344, c)). Nevertheless, we have the following sufficient condition:

Lemma 2. — Let E , F , and G be normed spaces over K , with F being a Banach space. Let u be a closed partial operator from E to F , and let $v \in \mathcal{L}(F; G)$. If there exists $C \in \mathbf{R}_+$ such that

$$\|u(x)\| \leq C(\|x\| + \|(v \circ u)(x)\|)$$

for all $x \in \text{dom}(v \circ u) = \text{dom}(u)$, that is, if the linear map $x \mapsto u(x)$ from $E_{v \circ u}$ into F is continuous, then $v \circ u$ is closed.

Denote by $w = v \circ u$. Let $(x_n, w(x_n))_{n \in \mathbf{N}}$ be a sequence in the graph of w which converges in $E \times G$. Let x be the limit of the sequence (x_n) . The hypothesis implies that the sequence $(u(x_n))_{n \in \mathbf{N}}$ is then a Cauchy sequence in F ; it converges to an element y of F . Since u is closed, we have $x \in \text{dom}(u)$ and $y = u(x)$. Then $w(x_n) = v(u(x_n))$ tends to $v(y) = v(u(x))$ since v is continuous, hence the graph of w is closed.

Let u be a closed partial operator on a Banach space E , and let F be a subspace of $\text{dom}(u)$. If F is dense in the Banach space E_u , then the reduction of u to F is closable, and its closure is equal to u . One then says that F is a core for u .

IV.4.3 Examples of partial operators

In this section, $K = \mathbf{R}$ or $\mathbf{C}$.

Examples. — 1) Let X be a locally compact topological space and let μ a positive measure on X . We fix elements p_1 and p_2 of $[1, +\infty[$.

Let g a function μ -measurable on X with values in K . Denote by D the subspace of $\mathcal{L}_K^{p_1}(X, \mu)$ consisting of functions f in $\mathcal{L}_K^{p_1}(X, \mu)$ such that $gf \in \mathcal{L}_K^{p_2}(X, \mu)$. The linear mapping from D in $\mathcal{L}_K^{p_2}(X, \mu)$ defined by $f \mapsto gf$ determines a partial operator of $\mathcal{L}_K^{p_1}(X, \mu)$ in $\mathcal{L}_K^{p_2}(X, \mu)$, which we denote m_g .

The vector subspace of functions μ -negligible in $\mathcal{L}_K^{p_1}(X, \mu)$ is contained in D , and the image under m_g of a function μ -negligible is still μ -negligible. We will denote $\tilde{m}_g$ the partial operator from $L_K^{p_1}(X, \mu)$ in $L_K^{p_2}(X, \mu)$ deduced from m_g by passing to quotients. This is called the operator of multiplication by g of $L_K^{p_1}(X, \mu)$ in $L_K^{p_2}(X, \mu)$. Of the functions g_1 and g_2 locally equal μ -almost everywhere define the same multiplication operator.

PROPOSITION 5. — The multiplication operator $\widetilde{m}_g$ of $L_K^{p_1}(X, \mu)$ in $L_K^{p_2}(X, \mu)$ is a closed operator with dense domain.

Let us first prove that the partial operator $\widetilde{m}_g$ is closed. Let $(f_n, h_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{L}_K^{p_1}(X, \mu) \times \mathcal{L}_K^{p_2}(X, \mu)$ such that the sequence $(\tilde{f}_n, \tilde{h}_n)$ of the classes of f_n and h_n belongs to the graph of $\widetilde{m}_g$ and converges in $L_K^{p_1}(X, \mu) \times L_K^{p_2}(X, \mu)$ as n tends to infinity. Let (f, h) be a pair in $\mathcal{L}_K^{p_1}(X, \mu) \times \mathcal{L}_K^{p_2}(X, \mu)$ such that the pair $(\tilde{f}, \tilde{h})$ of their classes is the limit of $(\tilde{f}_n, \tilde{h}_n)$.

There exists a sequence $(f_{n_k})_{k \in \mathbb{N}}$ extracted from the sequence $(f_n)_n$ such that $f_{n_k}(x)$ converges to $f(x)$ for μ -almost every x (INT, IV, p. 131, § 3, no. 4, th. 3). This implies that $h_{n_k}(x) = g(x)f_{n_k}(x)$ converges μ -almost everywhere to $g(x)f(x)$. Moreover, the sequence (h_{n_k}) converges to h in the space $\mathcal{L}_K^{p_2}(X, \mu)$. The functions h and gf are therefore equal μ -almost everywhere (loc. cit.). Thus $\tilde{f}$ belongs to the domain of $\widetilde{m}_g$ and $\tilde{h} = \widetilde{m}_g(\tilde{f})$. This proves that $\widetilde{m}_g$ is closed.

Let us prove that the domain of $\widetilde{m}_g$ is dense in $L_K^{p_1}(X, \mu)$. It suffices to verify that the classes of functions $f \in \mathcal{H}(X; K)$ belong to the closure of the domain of $\widetilde{m}_g$ in $L_K^{p_1}(X, \mu)$. Let $f \in \mathcal{H}(X; K)$ and let $\tilde{f}$ its class in $L_K^{p_1}(X, \mu)$. For every integer $n \in \mathbb{N}$, denote φ_n the characteristic function of the set of elements $x \in X$ such that $|g(x)| \leq n$ and we set $f_n = f\varphi_n$. We then have $|gf_n| \leq n|f|$, which belongs to $\mathcal{L}_K^{p_2}(X, \mu)$, hence f_n belongs to the domain of $\widetilde{m}_g$. For every element x of X , the sequence $(f_n(x))_{n \in \mathbb{N}}$ converges to $f(x)$ as $n \rightarrow +\infty$; moreover, we have $|f_n| \leq |f|$, which belongs to $\mathcal{L}_K^{p_1}(X, \mu)$. By Lebesgue's theorem (INT, IV, p. 137, § 3, n°7, th. 6), the sequence of classes of f_n converges to $\tilde{f}$ in $L_K^{p_1}(X, \mu)$. Thus, the class of f belongs to the closure of the domain of $\widetilde{m}_g$.

In the following proposition, we assume that $p_1 = p_2 = 2$.

PROPOSITION 6. — a) Let g' a function μ -measurable on X such that $|g| \leq |g'|$. We have $\text{dom}(\widetilde{m}_{g'}) \subset \text{dom}(\widetilde{m}_g)$ and $\text{dom}(\widetilde{m}_{g'})$ is a core of the partial operator $\widetilde{m}_g$;

b) Let F be a subspace of $\mathcal{L}_K^2(X, \mu)$ whose intersection with $\mathcal{H}(X; K)$ is dense in $\mathcal{H}(X; K)$ and whose image G in $L_K^2(X, \mu)$ is contained in $\text{dom}(\widetilde{m}_g)$. If $|g|^2$ is locally μ -integrable, then $\mathcal{H}(X; K)$ is contained in the domain of $\widetilde{m}_g$ and G is a core of m_g .

Let us prove a). If $f \in \mathcal{L}_K^2(X, \mu)$ belongs to the domain of $m_{g'}$, so that $fg' \in \mathcal{L}_K^2(X, \mu)$, the hypothesis implies that $fg \in \mathcal{L}_K^2(X, \mu)$, whence the result.

Let us prove that $\text{dom}(\tilde{m}_{g'})$ is a core of $\tilde{m}_g$, that is, the domain of $\tilde{m}_{g'}$ is dense in the Hilbert space $E_{\tilde{m}_g}$. Let $h \in \mathcal{L}_K^2(X, \mu)$ whose class $\tilde{h}$ belongs to $E_{\tilde{m}_g}$ and is orthogonal to $\text{dom}(\tilde{m}_{g'})$. This means that

$$(\tilde{h} | \tilde{h}')_{\tilde{m}_g} = \int_X \bar{h} h' (1 + |g|^2) d\mu = 0$$

for every function $h' \in \mathcal{L}_K^2(X, \mu)$ whose class $\tilde{h}'$ belongs to $\text{dom}(\tilde{m}_{g'})$.

Let C be a compact subset of X and let φ its characteristic function. Let $n \in \mathbb{N}$. Let φ_n be the characteristic function of the set μ -integrable C_n of the $x \in C$ such that $|h(x)| \leq n$ and we set $h'_n = \varphi_n h$. The class of h'_n belongs to the domain of $\tilde{m}_{g'}$ since $|g' h'_n| \leq n \varphi$; it follows that

$$0 = \int_X \bar{h} h'_n (1 + |g|^2) d\mu = \int_X |h|^2 \varphi_n (1 + |g|^2) d\mu.$$

This implies that h is zero for μ -almost all $x \in C_n$ and so, since n is arbitrary, that h is zero for μ -almost all $x \in C$. It finally follows that h is zero μ -almost everywhere, since C is arbitrary and h is moderate (INT, V, p. 9, § 1, n°3, cor.).

Consider now the assertion b). Since $|g|^2$ is locally μ -integrable, the function fg belongs to $\mathcal{L}_K^2(X, \mu)$ if $f \in \mathcal{H}(X; K)$, hence $\mathcal{H}(X; K)$ is contained in the domain of $\tilde{m}_g$.

Let $h \in \mathcal{L}_K^2(X, \mu)$ whose class $\tilde{h}$ belongs to $E_{\tilde{m}_g}$ and is orthogonal to G . We then have

$$0 = (\tilde{h} | \tilde{h}')_{\tilde{m}_g} = \int_X \bar{h} h' (1 + |g|^2) d\mu$$

for all $h' \in F$ of class h' . Given the hypothesis on F , this means that the measure $h(1 + |g|^2) \cdot \mu$ is zero, hence h is zero μ -almost everywhere since h is moderate.

Let p a real number ≥ 1 . Let h be an element of $\mathcal{L}_K^\infty(X, \mu)$. The operator $\tilde{m}_h$ of multiplication by h is an endomorphism of $L_K^p(X, \mu)$. Assume that the set Y of $x \in X$ such that $h(x) = 0$ is locally μ -negligible. The endomorphism $\tilde{m}_h$ is then injective (Lemma 7 of IV, p. 186). Denote h^{-1} the function on X equal to 0 on Y and to $x \mapsto 1/h(x)$ on $X - Y$. The reciprocal partial operator $\tilde{m}_h^{-1}$ is the operator of multiplication by h^{-1} of $L_K^p(X, \mu)$ in $L_K^p(X, \mu)$, that is, $\tilde{m}_h^{-1} = \tilde{m}_{h^{-1}}$. Indeed, the image of $\tilde{m}_h$ is the space of classes of functions $g \in \mathcal{L}_K^p(X, \mu)$ of the form $g = hf$ for $f \in \mathcal{L}_K^p(X, \mu)$. This condition is equivalent to $g(x)/h(x) = f(x)$ for all $x \in X - Y$ and

$g(x) = 0$ if $x \in Y$ This implies that the domain of $\widetilde{m}_h^{-1}$ in $L_K^p(X, \mu)$ is the domain of $\widetilde{m}_{h^{-1}}$, and that the equality $\widetilde{m}_h^{-1} = \widetilde{m}_{h^{-1}}$ is valid.

In what follows, we will sometimes simply denote by m_h the partial multiplication operator by h on $L_K^{p_1}(X, \mu)$ in $L_K^{p_2}(X, \mu)$.

- 2) Let E be a Hilbert space over K and $B = (e_i)_{i \in I}$ an orthonormal basis of E . Let $(\lambda_i)_{i \in I}$ a family of elements of K . Let D be the vector subspace of E consisting of the elements $x \in E$ such that the family $(\lambda_i \langle e_i | x \rangle)_{i \in I}$ is square-summable in K . The space D is dense in E since it contains the vector e_i for all $i \in I$. The partial operator u with domain D given by

$$x \mapsto \sum_{i \in I} \lambda_i \langle e_i | x \rangle e_i$$

is called a partial diagonal operator in the basis B , and $(\lambda_i)_{i \in I}$ is called the family of eigenvalues of u .

The operator u is closed. Indeed, let $(x_n, u(x_n))_{n \in \mathbb{N}}$ a sequence of elements of the graph of u which converges in $E \times E$ and let (x, y) its limit. We then have $\langle e_i | x_n \rangle \rightarrow \langle e_i | x \rangle$ for all $i \in I$ and

$$\langle e_i | u(x_n) \rangle = \lambda_i \langle e_i | x_n \rangle \rightarrow \langle e_i | y \rangle$$

for all $i \in I$. Consequently, $\lambda_i \langle e_i | x \rangle = \langle e_i | y \rangle$ for all $i \in I$, which proves that $x \in D$ and $u(x) = y$ that is, u is closed.

This example is in fact a special case of the preceding one, applied to the topological space $X = I$ equipped with the discrete topology and the counting measure μ , since E is identified with the space $\ell^2(I) = L^2(I, \mu)$ by the mapping $x \mapsto (\langle e_i | x \rangle)_{i \in I}$ (EVT, V, p. 23, cor. 2) and u is then identified with the multiplication operator m_λ , where λ is the function $i \mapsto \lambda_i$.

- 3) The set $N_R = N \cup \{\infty, \omega\}$ is equipped with the total order described in VAR, R2, p. 10, such that $n < \infty < \omega$ for all $n \in N$. Let $r \in N_R$. Let $n \in N$ and U an open subset of R^n . Let $k \in N$ such that $k \leq r$. Let $(n_\alpha)_{|\alpha| \leq k}$ a family of elements of $\mathcal{C}^r(U)$, where the multi-indices considered belong to N^n . The family (n_α) defines a scalar differential operator D of order $\leq k$ on U (cf. VAR, R2, 14.1.6, 14.1.4).

For every integer m such that $k \leq m \leq r$, the differential operator D defines a linear map from $C^m(U)$ in $C^{m-k}(U)$ which sends $f \in C^m(U)$ onto

$$D(f) = \sum_{|\alpha| \leq k} n_\alpha \partial^\alpha (f).$$

The same formula defines a continuous linear map from $\mathcal{D}(U)$ in $\mathcal{D}'(U)$ (def. 2 of IV, p. 208).

DEFINITION 5. — Let E be a vector subspace of $\mathcal{D}'(U)$ containing $\mathcal{D}(U)$. A differential operator associated with D on E is any partial operator on E that is an extension of the partial operator with domain $\mathcal{D}(U)$ defined by $\varphi \mapsto D(\varphi)$.

Suppose for example that the coefficients n_α are bounded functions on U . Let μ Lebesgue measure on U . If p is an element of $[1, +\infty[$ one can then define a differential operator on $L^p(U)$ associated with D whose domain is the Sobolev space $W^{k,p}(U)$ (no. 14 of IV, p. 221), since in this case one has $n_\alpha \partial^\alpha (f) \in L^p(U)$ for all $f \in W^{k,p}(U)$ and every $|\alpha| \leq k$.

IV.4.4 Adjoint

In this section, K is one of the fields $\mathbb{R}$ or $\mathbb{C}$, and E and F denote Hilbert spaces over K .

Let u be a densely defined operator from E to F . Let D denote the domain of u . For $y \in F$, let λ_y the linear form on D given by $\lambda_y(x) = \langle y | u(x) \rangle$ for all $x \in D$. We denote D^* the set of vectors $y \in F$ such that λ_y is continuous on D . It is a vector subspace of F . Let $y \in D^*$; since D is dense in E , the linear form λ_y extends uniquely to a continuous linear form on E , which we still denote by λ_y . According to EVT, V, p. 15, th. 3, there exists a unique element $u^*(y)$ in E such that $\lambda_y(x) = \langle u^*(y) | x \rangle$ for all $x \in E$. The map $y \mapsto u^*(y)$ is linear from D^* in E .

DEFINITION 6. — The partial operator from F to E with domain D^* defined by $y \mapsto u^*(y)$ is called the adjoint of u . It is denoted u^* .

Remark. — We therefore have $y \in D^*$ if and only if there exists $c \in \mathbb{R}_+$ such that $|\langle y | u(x) \rangle| \leq c\|x\|$ for all $x \in D$. The element $u^*(y)$ is then characterized by the relation

$$\langle y | u(x) \rangle = \langle u^*(y) | x \rangle \quad (1)$$

for all $x \in D$. We then have $|\langle y|u(x)\rangle| \leq \|u^*(y)\|\|x\|$ for all $x \in D$.

In the case where u is a continuous linear map from E into F , its adjoint in the sense of the preceding definition coincides with the adjoint defined in EVT, V, p. 38, def. 1, since D^* is equal to F in this case.

Let $v \in \mathcal{P}(E; F)$ such that $u \subset v$. We then have $v^* \subset u^*$.

Let $v \in \mathcal{P}(E; F)$ such that $u + v$ has dense domain. The partial operator v then has dense domain and one has $u^* + v^* \subset (u + v)^*$. In general, there is no equality (exercise 9 of IV, p. 347). If $v \in \mathcal{L}(E; F)$, then $u + v$ is densely defined and $(u + v)^* = u^* + v^*$. This is the case, for example, if $F = E$ and if $v = \lambda 1_E$ where $\lambda \in K$.

Let G be a Hilbert space over K and let $v \in \mathcal{P}(F; G)$ be an operator with dense domain. If $v \circ u$ has dense domain, then $u^* \circ v^* \subset (v \circ u)^*$. In general, there is no equality (loc. cit.). If u (resp. v) is an isomorphism, then $v \circ u$ has dense domain and one has $(v \circ u)^* = u^* \circ v^*$. This is the case, for example, if $E = F$ (resp. $F = G$) and $u = \lambda 1_E$ (resp. $v = \lambda 1_F$) where $\lambda \in K^*$.

We denote s the isometric isomorphism of Hilbert spaces from $E \oplus F$ in $F \oplus E$ defined by $s(x, y) = (-y, x)$ for all $(x, y) \in E \oplus F$.

PROPOSITION 7. — Let u be a partial operator with dense domain from E into F .

- a) The graph of u^* is equal to $s(\Gamma_u^\circ) = s(\Gamma_u)^\circ$;
- b) The partial operator u^* is closed;
- c) The kernel of u^* is the orthogonal complement of the image of u .

Denote by $W = s(\Gamma_u)^\circ$. Since the linear map s is unitary, we have $W = s(\Gamma_u^\circ)$.

One has $(y, x) \in W$ if and only if

$$\langle (y, x) | (-u(x'), x') \rangle = 0$$

for all $x' \in \text{dom}(u)$, that is, if

$$\langle y | u(x') \rangle = \langle x | x' \rangle$$

for all $x' \in \text{dom}(u)$. When $y \in \text{dom}(u^*)$ and $x = u^*(y)$, this property is true (cf. formula

(1), p. 236). Conversely, if this condition holds, we deduce that $|\langle y | u(x') \rangle| \leq \|x\| \|x'\|$ for all $x' \in \text{dom}(u)$, which implies that y belongs to $\text{dom}(u^*)$; we then have $u^*(y) = x$. Hence $W = \Gamma_{u^*}$.

The operator u^* is closed, because the space $s(\Gamma_u)^\circ$ is closed in $F \oplus E$.

Let us prove assertion c). If y is orthogonal to the image of u , the linear form $\lambda_y : x \mapsto \langle y | u(x) \rangle$ on D is zero, hence $y \in \text{dom}(u^*)$ and $u^*(y) = 0$. Conversely, let $y \in \text{dom}(u^*)$. We then have $u^*(y) = 0$ if and only if y is orthogonal to $u(x)$ for all $x \in D$ (formula (1), p. 236).

PROPOSITION 8. — Let u be an operator with dense domain from E to F . Then u^* has dense domain if and only if u is closable. When this is the case, the closure $\bar{u}$ of u is equal to u^{**} , and the adjoint of $\bar{u}$ is equal to u^* .

By prop. 7, the partial operator u^* is closed. Suppose that the domain D^* of u^* is dense in F . Let u^{**} be the adjoint of u^* ; it is a closed partial operator from E to F . Let us prove that $u \subset u^{**}$, which will imply that u is closable. Let $x \in \text{dom}(u)$. By definition of u^* , the linear forms on D^* given by $y \mapsto \langle x | u^*(y) \rangle$ and $y \mapsto \langle u(x) | y \rangle$ are equal; we therefore have $x \in \text{dom}(u^{**})$ and $u^{**}(x) = u(x)$ whence the assertion.

Conversely, suppose that u is closable; we have $\Gamma_{\bar{u}} = \overline{\Gamma_u}$ (prop. 1 of IV, p. 228). Let $y \in F$ be a vector orthogonal to $\text{dom}(u^*)$. The element $(y, 0)$ of $F \oplus E$ then belongs to the orthogonal of the graph of u^* . Now, by prop. 7, a), we have

$$\Gamma_{u^*}^\circ = (s(\Gamma_u)^\circ)^\circ = \overline{s(\Gamma_u)} = s(\overline{\Gamma_u})$$

It follows therefore that $(0, y) \in \Gamma_{\bar{u}}$, whence $y = \bar{u}(0) = 0$. The orthogonal of $\text{dom}(u^*)$ being reduced to 0, the space $\text{dom}(u^*)$ is dense in F .

Finally, prop. 7, applied to u^* implies that

$$\Gamma_{u^{**}} = s^{-1}(\Gamma_{u^*}^\circ) = s^{-1}(s(\Gamma_u^\circ)) = \overline{\Gamma_u},$$

hence $u^{**} = \bar{u}$, then $\bar{u}^* = (u^*)^{**} = \overline{u^*} = u^*$ as u^* is closed.

COROLLARY. — If u is a closed partial operator with dense domain from E into F , then u^* has dense domain and one has $u^{**} = u$.

DEFINITION 7. — Let u be a partial operator on E . One says that u is symmetric if u is densely defined and if u^* is an extension of u . One says that u is self-adjoint if u is densely defined and $u^* = u$.

u is said to be essentially self-adjoint if it is closable and if its closure $\bar{u}$ is self-adjoint.

We say that u is a partial operator bounded below operator if u is symmetric and if there exists a real number c such that $\langle x|u(x) \rangle \geq c\|x\|^2$ for all x belonging to the domain of u . One then says that c is a lower bound of u . If $c = 0$, one also says that u is a positive partial operator.

We denote $\mathcal{A}(E)$ the set of partial self-adjoint operators on E .

Remarks. — 1) For an operator u to be densely defined on E and symmetric, it is necessary and sufficient that one have

$$\langle x | u(y) \rangle = \langle u(x) | y \rangle \quad (2)$$

for all $(x, y) \in \text{dom}(u)^2$. This formula indeed shows that the domain of u is contained in that of u^* then that u^* and u coincide on the domain of u . In particular, it follows that $\langle x|u(x) \rangle \in \mathbf{R}$ for all $x \in \text{dom}(u)$.

As will be seen in various examples, formula (2) can often be verified by a straightforward calculation. On the other hand, the exact determination of the domain of the adjoint, which alone makes it possible to know whether a symmetric operator is self-adjoint or not, can be very delicate.

- 2) A self-adjoint partial operator u is essentially self-adjoint (cf. prop. 8).
- 3) Let u a symmetric partial operator on E . The operator u is closable (prop. 7, b)). It satisfies $\text{dom}(u) \subset \text{dom}(u^*)$, and u is self-adjoint if and only if $\text{dom}(u) = \text{dom}(u^*)$. Moreover, the closure $\bar{u}$ of u is symmetric since $\bar{u} \subset u^* = \bar{u}^*$ (prop. 8).
- 4) Suppose $K = \mathbf{C}$. Let $u \in \mathcal{P}(E; E)$ a densely defined partial operator. The condition $\langle x|u(x) \rangle \in \mathbf{R}$ for all $x \in \text{dom}(u)$ implies that u is symmetric (EVT, V, p. 2, remark); in particular, if $\langle x|u(x) \rangle \in \mathbf{R}_+$ for all $x \in \text{dom}(u)$, then u is positive.
- 5) Let u and v be symmetric partial operators on E . If u is self-adjoint and if $u \subset v$, then

$v \subset v^* \subset u^* = u$ therefore $u = v$.

- 6) An essentially self-adjoint partial operator u is symmetric, since $u \subset \bar{u}$ implies $\bar{u} = \bar{u}^* \subset u^*$, hence $u \subset u^*$.
- 7) Let u and v be symmetric partial operators on E . If $u + v$ has dense domain, for example if u or v belongs to $\mathcal{L}(E)$, then $u + v$ is symmetric. In general, the partial operator $u + v$ is not self-adjoint, even if u and v are (exercise 9 of IV, p. 347).
- 8) Let u a symmetric partial operator on E . A real number c is a lower bound of u if and only if the operator $u - c \cdot 1_E$ is positive.

Lemma 3. — Suppose that $K = \mathbb{R}$. Let u be a partial operator with dense domain from E into F .

- a) The adjoint of $u_{(C)}$ is $(u^*)_{(C)}$;
- b) Suppose that $E = F$; the partial operator u is symmetric (resp. self-adjoint) if and only if the partial operator $u_{(C)}$ is symmetric (resp. self-adjoint).

Let us prove a). Let $y \in F_{(C)}$ and let us write $y = y_1 + iy_2$ with $y_1, y_2 \in F$. For all $(x_1, x_2) \in E \times E$, we have

$$\begin{aligned} \langle u_{(C)}(x_1 + ix_2) | y \rangle &= \langle u(x_1) | y_1 \rangle + i \langle u(x_1) | y_2 \rangle \\ &\quad - i \langle u(x_2) | y_1 \rangle + \langle u(x_2) | y_2 \rangle. \end{aligned}$$

If $y \in \text{dom}(u^*)_{(C)}$, we deduce that $y \in \text{dom}((u_{(C)})^*)$ and that $u_{(C)}^*(y) = (u_{(C)})^*(y)$, hence $u_{(C)}^* \subset (u_{(C)})^*$.

Conversely, suppose that y belongs to $\text{dom}((u_{(C)})^*)$. If one takes $x_2 = 0$ (resp. $x_1 = 0$) in the above formula, one verifies that $y_1 \in \text{dom}(u^*)$ (resp. that $y_2 \in \text{dom}(u^*)$), whence $y \in \text{dom}(u^*)_{(C)}$.

The assertion a) implies that $u_{(C)}$ is symmetric (resp. self-adjoint) if u is.

Conversely, suppose that $u_{(C)}$ is symmetric. The relation $\langle u(x) | y \rangle = \langle x | u(y) \rangle$ for all $(x, y) \in \text{dom}(u_{(C)}) \times \text{dom}(u_{(C)})$ implies that u is symmetric by taking x and y in the subspace $\text{dom}(u)$ of $\text{dom}(u_{(C)})$. If $u_{(C)}$ is self-adjoint, what precedes shows that u is symmetric; since $\text{dom}(u^*) =$

$\text{dom}(u_{(C)}^*) \cap F$, assertion a) implies that $\text{dom}(u^*) = \text{dom}(u_{(C)}) \cap F = \text{dom}(u)$, hence u is self-adjoint.

IV.4.5 Elementary criteria for self-adjoint operators

PROPOSITION 9. — Let $v \in \mathcal{L}(F; E)$ be a continuous injective linear map from F into E whose image is dense in E . The adjoint of v is a continuous injective linear map from E into F and one has $(v^*)^{-1} = (v^{-1})^*$. In particular, if $E = F$, the endomorphism v is Hermitian if and only if the partial operator v^{-1} is self-adjoint.

The partial operator v^{-1} is a closed operator with dense domain from E into F (example 2 of IV, p. 228) and the adjoint v^* of v is a continuous linear map from E into F ; it is injective, since the image of v is dense in E (EVT, V, p. 41, prop. 4). Let s (resp. s') be the isometric isomorphism $(x, y) \mapsto (-y, x)$ from $E \oplus F$ onto $F \oplus E$ (resp. the isometric isomorphism $(y, x) \mapsto (-x, y)$ from $F \oplus E$ onto $E \oplus F$) and let ι (resp. ι') be the isometric isomorphism $(y, x) \mapsto (x, y)$ from $F \oplus E$ onto $E \oplus F$ (resp. the isometric isomorphism $(x, y) \mapsto (y, x)$ from $E \oplus F$ onto $F \oplus E$). One then has $s \circ \iota = -\iota' \circ s'$, whence

$$\Gamma_{(v^{-1})^*} = s(\Gamma_{v^{-1}})^\circ = s(\iota(\Gamma_v))^\circ = -\iota'(s'(\Gamma_v))^\circ = -\iota'(\Gamma_{v^*}) = \Gamma_{(v^*)^{-1}}$$

by prop. 7 of IV, p. 236. The proposition follows.

PROPOSITION 10 (Hellinger–Toeplitz). — Let u be a symmetric partial operator on the Hilbert space E . If the domain of u is equal to E , then $u \in \mathcal{L}(E)$ and u is Hermitian.

Indeed, the partial operator u is closable (prop. 8 of IV, p. 237), and hence closed since its domain is E (IV, p. 228, remark). One then concludes by invoking EVT, I, p. 19, cor. 5.

COROLLARY. — Let u be a symmetric partial operator on E . If the partial operator u induces a bijective linear map from $\text{dom}(u)$ onto E , then u is self-adjoint.

Indeed, the partial operator u^{-1} inverse to u is symmetric with domain E (prop. 9), hence u^{-1} is a self-adjoint element of $\mathcal{L}(E)$ (prop. 10), and u is then Hermitian (prop. 9).

PROPOSITION 11. — Let u be a symmetric partial operator on E and let $\lambda \in \mathbb{C}$. If $u + \lambda 1_E$ and $u + \overline{\lambda} 1_E$ are surjective, then u is self-adjoint.

It suffices to show that $\text{dom}(u^*) \subset \text{dom}(u)$. Let $x \in \text{dom}(u^*)$. By hypothesis there exists

$y \in \text{dom}(u)$ such that $u(y) + \overline{\lambda}y = u^*(x) + \overline{\lambda}x$. Let us show that $y = x$. For all $z \in \text{dom}(u)$, it follows that

$$\begin{aligned}\langle (u + \lambda 1_E)(z) | x \rangle &= \langle z | (u^* + \overline{\lambda} 1_E)(x) \rangle \\ &= \langle z | (u + \overline{\lambda} 1_E)(y) \rangle = \langle (u + \lambda 1_E)(z) | y \rangle,\end{aligned}$$

since u is symmetric. Since the operator $u + \lambda 1_E$ is surjective, one indeed has $y = x$, hence $x \in \text{dom}(u)$.

We shall see later (cf. prop. 17 of IV, p. 248) that if $\lambda \in \mathbf{C} - \mathbf{R}$, then the converse is true.

PROPOSITION 12. — Let u be a closed operator with dense domain from E into F . The partial operator $u^* \circ u$ on E is self-adjoint and positive. Its domain is a core for u .

Denote by v the partial operator the partial operator $1_E + u^* \circ u$. Its domain is $\text{dom}(u^* \circ u)$, which is contained in $\text{dom}(u)$. For all $x \in \text{dom}(u)$ and $y \in \text{dom}(v)$, we have $u(y) \in \text{dom}(u^*)$ and

(3)

$$\langle x | v(y) \rangle = \langle x | y \rangle + \langle x | (u^* \circ u)(y) \rangle = \langle x | y \rangle + \langle u(x) | u(y) \rangle, \quad (4)$$

$$\langle y | v(y) \rangle = \|y\|^2 + \|u(y)\|^2.$$

Formula (4) implies that the partial operator v is injective. Moreover, by prop. 7 of IV, p. 236, a), one has $F \oplus E = \Gamma_{u^*} \oplus s(\Gamma_u)$. Let $x \in E$. There exists $y' \in \text{dom}(u^*)$ and $x' \in \text{dom}(u)$ such that

$$(0, x) = (y', u^*(y')) + (-u(x'), x') = (y' - u(x'), x' + u^*(y')).$$

We therefore have $y' = u(x')$, whence $x' \in \text{dom}(u^* \circ u) = \text{dom}(v)$ and

$$x = x' + u^*(y') = x' + (u^* \circ u)(x') = v(x').$$

The partial operator v on E is therefore surjective, and induces a bijective linear map from

$\text{dom}(v)$ to E .

Let $x \in E$ orthogonal to $\text{dom}(v)$. Let us write $x = v(x')$ where $x' \in \text{dom}(v)$. By formula (4), we obtain

$$0 = \langle x' | x \rangle = \langle x' | v(x') \rangle = \|x'\|^2 + \|u(x')\|^2$$

whence $x' = 0$, then $x = 0$. The domain of v is therefore dense in E .

The partial operator v has dense domain; it is bijective and the formula (3) shows that it is symmetric. We conclude that v is self-adjoint by applying the corollary of proposition 10. Consequently, $u^* \circ u = v - 1_E$ is self-adjoint. Moreover, the formula

$$\langle x | (u^* \circ u)(x) \rangle = \|u(x)\|^2$$

for all $x \in \text{dom}(u^* \circ u)$ implies that $u^* \circ u$ is positive.

Finally, let $y \in E_u$ orthogonal to $\text{dom}(u^* \circ u)$. There exists an element x in the domain of $u^* \circ u$ such that $y = v(x) = x + (u^* \circ u)(x)$. We then have $0 = (x | y)_u = \langle y | y \rangle$, whence $y = 0$.

IV.4.6 Differential operators

Let $n \in \mathbb{N}$ and U an open subset of $\mathbb{R}^n$. We equip $\mathbb{R}^n$ and U with the Lebesgue measure denoted μ .

Let $k \in \mathbb{N}$ and $h \in \mathbb{N}$ such that $h \geq k$. Let D be a scalar differential operator on U of order $\leq k$, with coefficients $(n_\alpha)_{|\alpha| \leq k}$ of class C^h on U . Suppose that for every α such that $|\alpha| \leq k$ and every β such that $0 \leq \beta \leq \alpha$, the function $\partial^\beta n_\alpha$ is bounded on U .

Let tD the scalar differential operator on U transpose of D (VAR, R2, 14.3.2); it is of order $\leq k$ and of class C^{h-k} ; for $\varphi \in \mathcal{D}(U)$, we have

$${}^tD(\varphi) = \sum_{|\alpha| \leq k} (-1)^{|\alpha|} \partial^\alpha (\bar{n}_\alpha \varphi)$$

(loc. cit.); in particular, the coefficients of tD are bounded on U .

We denote by D_- the partial operator on $L^2(U)$ with domain $\mathcal{D}(U)$ defined by

$$\varphi \mapsto D(\varphi) = \sum_{|\alpha| \leq k} n_\alpha \partial^\alpha \varphi. \quad (5)$$

Let H_D the space of $f \in L^2(U)$ such that the distribution

$$D(f) = \sum_{|\alpha| \leq k} n_\alpha \partial^\alpha f$$

belongs to $L^2(U)$; we denote by D_+ the partial operator with domain H_D defined by $f \mapsto D(f)$.

Since we have $\partial^\alpha f \in L^2(U)$ if $f \in H^k(U)$ and $|\alpha| \leq k$, the Sobolev space $H^k(U)$ is contained in H_D ; in general, these spaces are distinct.

One has $D_- \subset D_+$, and these are differential operators associated with D on $L^2(U)$ (def. 5 of IV, p. 235).

PROPOSITION 13. — Let u be a partial operator on $L^2(U)$. If $D_- \subset u \subset D_+$, then u is closable and $({}^tD)_- \subset u^* \subset ({}^tD)_+$.

Let φ and ψ in $\mathcal{D}(U)$. We then have

$$\begin{aligned} \langle \varphi | D(\psi) \rangle &= \sum_{|\alpha| \leq k} \int_U n_\alpha \bar{\varphi} \partial^\alpha \psi d\mu \\ &= \sum_{|\alpha| \leq k} (-1)^{|\alpha|} \langle \partial^\alpha (\bar{n}_\alpha \varphi) | \psi \rangle = \langle {}^tD(\varphi) | \psi \rangle \end{aligned}$$

(cf. VAR, R2, 14.3.8). Since $\mathcal{D}(U)$ is dense in $L^2(U)$ this implies that $\varphi \in \text{dom}(u^*)$ and $u^*(\varphi) = {}^tD(\varphi)$. We therefore have $({}^tD)_- \subset u^*$ In particular, u^* is densely defined and u is closable (prop. 8 of IV, p. 237).

Let $f \in \text{dom}(u^*)$ and $\varphi \in \mathcal{D}(U)$. Since $\mathcal{D}(U) \subset \text{dom}(u)$, the distribution associated with $u^*(f)$ satisfies

$$\langle u^*(f), \varphi \rangle = \langle \bar{\varphi} | u^*(f) \rangle = \langle u(\bar{\varphi}) | f \rangle.$$

Since $D_- \subset u$, one computes $u(\bar{\varphi})$ by the formula (5), whence

$$\begin{aligned} \langle u^*(f), \varphi \rangle &= \sum_\alpha \langle n_\alpha \partial^\alpha \bar{\varphi} | f \rangle = \sum_\alpha \langle \partial^\alpha \bar{\varphi} | \bar{n}_\alpha f \rangle \\ &= \sum_\alpha \langle \bar{n}_\alpha f, \partial^\alpha \varphi \rangle = \sum_\alpha (-1)^{|\alpha|} \langle \partial^\alpha (\bar{n}_\alpha f), \varphi \rangle = \langle {}^tD(f), \varphi \rangle. \end{aligned}$$

The distributions $u^*(f)$ and ${}^tD(f)$ are therefore equal; the distribution f therefore belongs to H_{tD} and $u^*(f) = {}^tD(f)$, whence $u^* \subset ({}^tD)_+$.

Remark. — The proposition means that the adjoint of a partial operator u such that $D_- \subset u \subset D_+$ can always be computed in the sense of distributions: the elements f of the domain of u^* are elements of $L^2(U)$ such that the distribution ${}^tD(f)$ belongs to $L^2(U)$, and we have $u^*(f) = {}^tD(f)$.

We say that D is formally symmetric if ${}^tD = D$ as a scalar differential operator. If that is the case, the partial operator D_- is symmetric.

Consider the particular case of the second-order scalar differential operator defined by

$$\Delta = - \sum_{i=1}^n \partial_i^2.$$

A Laplacian on U is any partial operator u , self-adjoint on $L^2(U)$ such that $\Delta_- \subset u$ (cf. VAR, R2, 14.4.3, p. 83). We shall see later (IV, p. 261, example) that there always exists at least one Laplacian on $L^2(U)$; there may exist more than one (exercise 17 of IV, p. 358).

IV.4.7 Spectrum and resolvent

Lemma 4. — Let E be a complex Banach space and let u be a closed injective partial operator on E such that $u^{-1} \in \mathcal{L}(E)$. Let $v \in \mathcal{L}(E)$

such that $\|v\| < \|u^{-1}\|^{-1}$. Then the partial operator $u + v$ is injective, we have $(u + v)^{-1} \in \mathcal{L}(E)$ and

$$(u + v)^{-1} = u^{-1} \circ \sum_{k=0}^{+\infty} (-vu^{-1})^k, \quad (6)$$

where the series is absolutely convergent in $\mathcal{L}(E)$. Moreover, we have

$$\|(u + v)^{-1}\| \leq \frac{\|u^{-1}\|}{1 - \|v\|\|u^{-1}\|}.$$

Since $\|v\|\|u^{-1}\| < 1$, the series with general term $(-vu^{-1})^k$ is absolutely convergent in $\mathcal{L}(E)$ and its sum is the inverse of the endomorphism $1_E + vu^{-1}$ (prop. 2 of I, p. 22). Consequently, the partial operator $(1_E + vu^{-1}) \circ u = u + v$ (IV, p. 226, rem. vi) is injective and the reciprocal

partial operator $(u + v)^{-1} = u^{-1} \circ (1_E + vu^{-1})^{-1}$ (IV, p. 226, rem. iii) belongs to $\mathcal{L}(E)$. Since

$$\left\| (1_E + vu^{-1})^{-1} \right\| \leq \sum_{k=0}^{+\infty} (\|v\| \|u^{-1}\|)^k = \frac{1}{1 - \|v\| \|u^{-1}\|},$$

the lemma is proved.

DEFINITION 8. — Let u be a closed operator with dense domain on a complex Banach space E . The resolvent set of u is the set of complex numbers λ such that the partial operator $\lambda 1_E - u$ is injective and its inverse $(\lambda 1_E - u)^{-1}$ belongs to $\mathcal{L}(E)$.

The spectrum of u , denoted $\text{Sp}(u)$, is the complement of the resolvent set in $\mathbb{C}$.

If $\lambda \in \mathbb{C} - \text{Sp}(u)$, we denote by $R(u, \lambda) \in \mathcal{L}(E)$ the inverse of $\lambda 1_E - u$. The map from $\mathbb{C} - \text{Sp}(u)$ into $\mathcal{L}(E)$ that sends λ to $R(u, \lambda)$ is called the resolvent of u .

Remarks. — Let E be a complex Banach space and let u be a closed operator with dense domain on E .

- 1) If $u \in \mathcal{L}(E)$, its spectrum coincides with the spectrum of the element u of the algebra $\mathcal{L}(E)$ (I, p. 2, def. 1).
- 2) Let D be the domain of u . For every $\lambda \in \mathbb{C} - \text{Sp}(u)$, we have

$$1_E = (\lambda 1_E - u) \circ R(u, \lambda), \quad 1_D = R(u, \lambda) \circ (\lambda 1_E - u). \quad (7)$$

Moreover, for λ_1 and λ_2 in the resolvent set of u , we have

$$R(u, \lambda_1) - R(u, \lambda_2) = (\lambda_2 - \lambda_1) R(u, \lambda_2) \circ R(u, \lambda_1), \quad (8)$$

$$R(u, \lambda_1) \circ R(u, \lambda_2) = R(u, \lambda_2) \circ R(u, \lambda_1), \quad (9)$$

Indeed, one has

$$R(u, \lambda_1) - R(u, \lambda_2) = R(u, \lambda_1) \circ 1_E - 1_D \circ R(u, \lambda_2).$$

Since $1_E = (\lambda_2 1_E - u) \circ R(u, \lambda_2)$ and $1_D = R(u, \lambda_1) \circ (\lambda_1 1_E - u)$, we obtain

$$\begin{aligned}
R(u, \lambda_1) - R(u, \lambda_2) &= \lambda_2 R(u, \lambda_1) \circ R(u, \lambda_2) \\
&- R(u, \lambda_1) \circ u \circ R(u, \lambda_2) - \lambda_1 R(u, \lambda_1) \circ R(u, \lambda_2) \\
&+ R(u, \lambda_1) \circ u \circ R(u, \lambda_2),
\end{aligned}$$

whence the first formula. Exchanging the roles of λ_1 and λ_2 we deduce the second formula.

3) Let $\lambda \in \mathbf{C}$. By the closed graph theorem (EVT, I, p. 19, cor. 5), if the linear map from $\text{dom}(u)$ to E defined by $x \mapsto (\lambda 1_E - u)(x)$ is bijective, then its inverse, whose graph is closed, is continuous from E to E . Therefore λ belongs to the resolvent set of u if and only if the map $x \mapsto (\lambda 1_E - u)(x)$ from $\text{dom}(u)$ into E is bijective.

4) If λ belongs to the spectrum of u , one of the following properties holds:

- (i) The kernel of $\lambda 1_E - u$ is not reduced to 0; one then says that λ is an eigenvalue of u , and that the dimension of $\text{Ker}(\lambda 1_E - u)$ is its multiplicity;
- (ii) The partial operator $\lambda 1_E - u$ is injective and its image is not dense in E ; one says that λ belongs to the residual spectrum of u ;
- (iii) The partial operator $\lambda 1_E - u$ is injective, its image is dense in E , but $\lambda 1_E - u$ is not surjective; one says that λ belongs to the continuous spectrum of u .

5) Let λ be a complex number belonging to the resolvent set of u . The resolvent $R(u, \lambda)$ is an injective linear map from E into E ; its image is the domain of u and $u = \lambda 1_E - R(u, \lambda)^{-1}$ (cf. IV, p. 226). Conversely, this property characterizes the resolvent set and the resolvent. Precisely, let $\lambda \in \mathbf{C}$; if there exists a continuous injective linear map w from E into E such that $u = \lambda 1_E - w^{-1}$, then λ belongs to the resolvent set of u and $w = R(u, \lambda)$.

In particular, if v is a closed densely defined operator on E , and if $\lambda \in \mathbf{C}$ is a complex number not belonging to $\text{Sp}(u) \cup \text{Sp}(v)$, then the equality $R(u, \lambda) = R(v, \lambda)$ implies that $u = v$.

6) We define the spectrum of a closable partial operator u as the spectrum of its closure.

There exist closed operators whose spectrum is empty, or whose spectrum is equal to $\mathbf{C}$ (Exercise 12 of IV, p. 347).

Let E be a complex Banach space and u a closed densely defined operator on E . If F is a complex Banach space and if $v : E \rightarrow F$ is an isomorphism, then we have $\text{Sp}(v \circ u \circ v^{-1}) = \text{Sp}(u)$ and $R(v \circ u \circ v^{-1}, \lambda) = v \circ R(u, \lambda) \circ v^{-1}$ for all $\lambda \notin \text{Sp}(u)$.

PROPOSITION 14. — Let E be a complex Banach space. Let u be a closed operator with dense domain on E and let U be its resolvent set. a) For every $\lambda \in U$ the open disk in $\mathbb{C}$ with center λ and radius $\|R(u, \lambda)\|^{-1}$ is contained in U ;

b) The set U is open in $\mathbb{C}$;

c) Suppose that $\text{Sp}(u)$ is not empty. Let $\lambda \in U$ and let δ be the distance in $\mathbb{C}$ of λ to the spectrum of u . We have $\delta > 0$ and $\|R(u, \lambda)\| \geq 1/\delta$;

d) The map $\lambda \mapsto R(u, \lambda)$ is a holomorphic map from U to $\mathcal{L}(E)$. For every integer $k \in \mathbb{N}$ and every $\lambda \in U$, we have

$$\frac{\partial^k}{\partial \lambda^k} R(u, \lambda) = (-1)^k k! R(u, \lambda)^{k+1}.$$

Let $\lambda \in U$. For every $\mu \in \mathbb{C}$ such that $\|(\mu - \lambda)1_E\| < \|R(u, \lambda)\|^{-1}$, Lemma 4 applied to the injective partial operator $\lambda 1_E - u$ and with $v = (\mu - \lambda)1_E$ implies that the partial operator $\mu 1_E - u = \lambda 1_E - u + v$ is injective and has a continuous inverse. This implies a). By loc. cit., we also have the formula

$$R(u, \mu) = R(u, \lambda) \circ \sum_{k \in \mathbb{N}} (\lambda - \mu)^k R(u, \lambda)^k,$$

which implies that the resolvent of u is holomorphic in U .

The assertion b) follows immediately from a). If $\text{Sp}(u)$ is nonempty, the distance from λ to $\text{Sp}(u)$ is strictly positive (TG, IX, p. 13, prop. 2), whence c).

The last part of the assertion d) is proved by induction on k , the case $k = 1$ being a consequence of formula (8), p. 245.

PROPOSITION 15. — Let u be a closed operator with dense domain on a complex Banach space E , and let λ be a complex number belonging to the resolvent set of u .

a) The subset $\text{Sp}(R(u, \lambda)) \setminus \{0\}$ of $\mathbb{C}$ is the image of the spectrum of u by the map $\mu \mapsto$

$(\lambda - \mu)^{-1}$ from $\mathbf{C} \setminus \{\lambda\}$ into $\mathbf{C}^*$;

b) For every $\mu \neq \lambda$ in $\mathbf{C}$, one has

$$\text{Ker}(\mu 1_E - u) = \text{Ker}\left((\lambda - \mu)^{-1} 1_E - R(u, \lambda)\right).$$

Let us prove assertion a). For every $\mu \neq \lambda$, we compute

$$\mu 1_E - u = (\lambda - \mu)((\lambda - \mu)^{-1} 1_E - R(u, \lambda))(\lambda 1_E - u).$$

Since $\lambda \notin \text{Sp}(u)$ and $\mu \neq \lambda$, the linear map $(\lambda - \mu)(\lambda 1_E - u)$ is a bijection from $\text{dom}(u)$ onto E . Consequently, this formula implies that $\mu 1_E - u$ is a bijection from $\text{dom}(u)$ onto E if and only if $(\lambda - \mu)^{-1} 1_E - R(u, \lambda)$ is a bijection from E onto E , which implies the assertion.

Let us prove b). If $\mu \neq \lambda$ and $x \in \text{Ker}((\lambda - \mu)^{-1} 1_E - R(u, \lambda))$, we have $x \in \text{dom}(u)$ and the formula $1_E = (\lambda 1_E - u) \circ R(u, \lambda)$ implies that $x \in \text{Ker}(\mu 1_E - u)$. Conversely, if $x \in \text{Ker}(\mu 1_E - u)$ and $\mu \neq \lambda$, the formula $1_{\text{dom}(u)} = R(u, \lambda) \circ (\lambda 1_E - u)$ implies $R(u, \lambda)(x) = (\lambda - \mu)^{-1}x$.

PROPOSITION 16. — Let u be a closed operator with dense domain on a complex Hilbert space E . The spectrum of u^* is the image of the spectrum of u under complex conjugation and, for every element λ of the resolvent set of u , we have $R(u, \lambda)^* = R(u^*, \bar{\lambda})$. In particular, if u is self-adjoint, the endomorphism $R(u, \lambda)$ is normal for every $\lambda \notin \text{Sp}(u)$.

Let $\lambda \in \mathbf{C} - \text{Sp}(u)$ be an element of the resolvent set of u . We have $u = \lambda 1_E - R(u, \lambda)^{-1}$, hence

$$u^* = \bar{\lambda} 1_E - (R(u, \lambda)^{-1})^* = \bar{\lambda} 1_E - (R(u, \lambda)^*)^{-1}$$

(IV, p. 236 and prop. 9 of IV, p. 239). By Remark 5, it follows that $\bar{\lambda} \in \mathbf{C} - \text{Sp}(u^*)$ and that $R(u, \lambda)^* = R(u^*, \bar{\lambda})$. Consequently, the spectrum of u^* is contained in the image of $\text{Sp}(u)$ by complex conjugation. The equality is obtained by applying this property to u^* since $u^{**} = u$ (corollary of Proposition 8 in IV, p. 237). The last assertion then follows from formula (9), p. 245.

COROLLARY. — Let u be a partial self-adjoint operator on a complex Hilbert space E . If

E is not zero, then the spectrum of u is not empty.

Suppose that $\text{Sp}(u)$ is empty. Then u is injective and $u^{-1} = -R(u, 0)$ is an injective endomorphism of E such that $\text{Sp}(u^{-1}) \subset \{0\}$ (prop. 15, a)), hence $\text{Sp}(u^{-1}) = \{0\}$ (I, p. 26, cor. 1). Since u^{-1} is normal (prop. 16), this implies that u^{-1} is zero (I, p. 110, example 1), which is a contradiction.

Lemma 5. — Let E be a complex Hilbert space and let u be a closed densely defined partial operator on E . Let $\lambda \in \mathbb{C}$. Suppose there exists a real number $c > 0$ such that

$$\|u(x) - \lambda x\| \geq c\|x\| \quad \text{for all } x \in \text{dom}(u), \quad (10)$$

$$\|u^*(x) - \bar{\lambda}x\| \geq c\|x\| \quad \text{for all } x \in \text{dom}(u^*), \quad (11)$$

Then λ belongs to the resolvent set of u and $\|R(u, \lambda)\| \leq c^{-1}$.

The hypothesis implies that $u - \lambda 1_E$ and $u^* - \bar{\lambda} 1_E$ are injective. Let F be the image of $u - \lambda 1_E$. The space F is dense in E , since its orthogonal is equal to $\text{Ker}(u^* - \bar{\lambda} 1_E)$ (prop. 7, c) of IV, p. 236), which is zero.

Let us prove that the space F is closed. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\text{dom}(u)$ such that the sequence $(u(x_n) - \lambda x_n)_{n \in \mathbb{N}}$ converges to $y \in F$. Inequality (10) implies that the sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in E . Let $x \in E$ be its limit. The sequence $(x_n, u(x_n))$ converges to $(x, y + \lambda x)$ in $E \times E$; since the graph of u is closed, we therefore have $x \in \text{dom}(u)$ and $u(x) = y + \lambda x$, which proves that $y \in F$.

We conclude that $F = E$. Thus, the partial operator $u - \lambda 1_E$ is bijective, whence $\lambda \notin \text{Sp}(u)$ (Remark 3). Inequality (10) then implies that $\|R(u, \lambda)\| \leq c^{-1}$.

PROPOSITION 17. — Let E be a complex Hilbert space and let u be a self-adjoint partial operator on E .

- a) The spectrum of u is contained in $\mathbf{R}$;
- b) If u is positive, then the spectrum of u is contained in $\mathbf{R}_+$;
- c) Suppose that E is nonzero. Let $\lambda \notin \text{Sp}(u)$ and let $\delta > 0$ be the distance from λ to the

spectrum of u . We have $\|R(u, \lambda)\| = \delta^{-1}$.

Let $(a, b) \in \mathbf{R} \times \mathbf{R}$ and $\lambda = a + ib$. Let $x \in \text{dom}(u)$. Since u is self-adjoint, we have $\langle x|u(x) \rangle \in \mathbf{R}$, whence

$$\begin{aligned}\|u(x) - \lambda x\|^2 &= \|u(x)\|^2 - 2a\langle x|u(x) \rangle + (a^2 + b^2)\|x\|^2 \\ &= \|u(x) - \bar{\lambda}x\|^2.\end{aligned}$$

Suppose that $b \neq 0$. We then obtain

$$\begin{aligned}\|u(x) - \lambda x\|^2 &= \|u(x) - \bar{\lambda}x\|^2 \geq (\|u(x)\| - a\|x\|)^2 + b^2\|x\|^2 \\ &\geq b^2\|x\|^2.\end{aligned}$$

By Lemma 5, we therefore have $\lambda \notin \text{Sp}(u)$, whence assertion a).

Suppose that u is also positive. If $b = 0$ and $a < 0$, we obtain similarly for $x \in \text{dom}(u)$ the inequality

$$\|u(x) - \lambda x\|^2 = \|u(x) - \bar{\lambda}x\|^2 \geq (\|u(x)\| - a\|x\|)^2 \geq a^2\|x\|^2,$$

hence $\lambda \notin \text{Sp}(u)$ (loc. cit.), which proves b).

Let us finally prove c). By prop. 16, the resolvent $R(u, \lambda)$ is a normal endomorphism of E . Its norm is therefore equal to its spectral radius (cor. 1 of I, p. 108), whence

$$\|R(u, \lambda)\| = \sup_{\mu \in \text{Sp}(R(u, \lambda))} |\mu|$$

(theorem 1 of I, p. 24). The spectrum of $R(u, \lambda)$ cannot be reduced to $\{0\}$, for in that case one would have $\|R(u, \lambda)\| = 0$, so $\text{dom}(u)$, the image of $R(u, \lambda)$, would be zero, and E as well. Prop. 15 therefore implies that

$$\|R(u, \lambda)\| = \sup_{\mu \in \text{Sp}(u)} \frac{1}{|\lambda - \mu|} = \frac{1}{\delta}.$$

COROLLARY. — Let u be a self-adjoint operator on E .

a) The residual spectrum of u is empty;

b) For all $\lambda \neq \mu$ in $\mathbb{C}$, the eigenspaces of u relative to λ and μ are orthogonal.

Let us prove a). Let λ belong to the spectrum of u ; it is a real number (prop. 17). We have $\text{Ker}(\lambda 1_E - u) = \text{Im}(\lambda 1_E - u)^\circ$ (prop. 7 of IV, p. 236), hence λ is not an eigenvalue of u if the partial operator $\lambda 1_E - u$ has dense image. This implies by definition that the residual spectrum of u is empty.

Let us prove b). By the proposition, we may assume that λ and μ are real. Let $x \in \text{dom}(u)$ such that $u(x) = \lambda x$ and $y \in \text{dom}(u)$ such that $u(y) = \mu y$. We then have

$$\lambda \langle x | y \rangle = \langle u(x) | y \rangle = \langle x | u(y) \rangle = \mu \langle x | y \rangle,$$

whence $\langle x | y \rangle = 0$.

Remark. — If u is a closed symmetric operator that is not self-adjoint, its spectrum is not contained in $\mathbb{R}$ (cf. Cor. 10 of IV, p. 257 below), and it is possible that the eigenspaces of u corresponding to λ and $\bar{\lambda}$ need not be orthogonal (exercise 11 of IV, p. 347).

IV.4.8 Pseudospectrum

DEFINITION 9. — Let E be a complex Banach space and let u be a closed partial operator on E . Let ε be a strictly positive real number. We call the ε -pseudospectrum of u the union of the spectrum of u and the set of complex numbers λ belonging to the resolvent set of u such that $\|R(u, \lambda)\| > \varepsilon^{-1}$. We denote this set by $\text{PSp}_\varepsilon(u)$.



Some authors define the ε -pseudo-spectrum of u as the set $T_\varepsilon(u)$, which is the union of $\text{Sp}(u)$ and the set of $\lambda \in \mathbb{C} - \text{Sp}(u)$ such that $\|R(u, \lambda)\| \geq \varepsilon^{-1}$. The closure of $\text{PSp}_\varepsilon(u)$ is contained in $T_\varepsilon(u)$ but this inclusion can be strict, even if E is a Hilbert space (cf. exercises 18 of IV, p. 348 and 19 of IV, p. 349).

PROPOSITION 18. — Let E be a complex Banach space and u a closed partial operator on E . Let $\varepsilon \in \mathbb{R}_+^*$. The set $\text{PSp}_\varepsilon(u)$ is an open subset of $\mathbb{C}$. It is nonempty if E is nonzero.

If E is zero, then $\text{PSp}_\varepsilon(u)$ is empty. Suppose that E is nonzero. By prop. 14 of IV, p. 246, the set $\text{PSp}_\varepsilon(u)$ is an open subset of $\mathbb{C}$.

The set $\text{PSp}_\varepsilon(u)$ is not empty if the spectrum of u is not empty. If $\text{Sp}(u)$ is empty, then Liouville's theorem (VAR, R1, p. 29, 3.3.6) implies that the holomorphic function on $\mathbb{C}$ defined by $\lambda \mapsto R(u, \lambda)$ is not bounded, so there exists λ in $\mathbb{C}$ such that $\|R(u, \lambda)\| > \varepsilon^{-1}$.

PROPOSITION 19. — Let E be a complex Banach space and u a closed partial operator on E . Then we have $\text{PSp}_\varepsilon(u) \subset \text{PSp}_\delta(u)$ if $0 < \varepsilon < \delta$ and

$$\bigcap_{\varepsilon \in \mathbf{R}_+^*} \text{PSp}_\varepsilon(u) = \text{Sp}(u).$$

The first assertion follows from the definition. For the second, the spectrum of u is contained in $\text{PSp}_\varepsilon(u)$ for all $\varepsilon > 0$ by definition, and if $\lambda \notin \text{Sp}(u)$, then $\lambda \notin \text{PSp}_\varepsilon(u)$ when $\varepsilon < \|R(u, \lambda)\|^{-1}$.

PROPOSITION 20. — Let E be a complex Banach space and u a closed partial operator on E . Let $\varepsilon \in \mathbf{R}_+^*$. For every $\lambda \in \mathbb{C}$ the following conditions are equivalent:

- (i) One has $\lambda \in \text{PSp}_\varepsilon(u)$;
- (ii) Either $\lambda \in \text{Sp}(u)$, or there exists $x \in \text{dom}(u)$ such that $\|x\| = 1$ and $\|(\lambda 1_E - u)(x)\| < \varepsilon$;
- (iii) There exists $v \in \mathcal{L}(E)$ such that $\|v\| < \varepsilon$ and $\lambda \in \text{Sp}(u + v)$.

We may assume that E is nonzero. Condition (ii) is a reformulation of the definition, and hence of condition (i).

Suppose condition (i) holds, and let us prove (iii). If λ belongs to the spectrum of u one may take $v = 0$ in (iii).

Suppose therefore that $\lambda \notin \text{Sp}(u)$. By definition of $\text{PSp}_\varepsilon(u)$, there exists $y \in E$ such that $\|y\| = 1$ and $\|R(u, \lambda)y\| > \varepsilon^{-1}$. Set $x = R(u, \lambda)y$. We have $x \neq 0$. By the Hahn-Banach theorem (EVT, II, p. 67, cor. 2), the linear form ℓ on $\mathbb{C}x$ such that $\ell(x) = 1$ admits a continuous extension $\ell_1 \in E'$ such that $\|\ell_1\| = \|\ell\|$; we therefore have $\|\ell_1\| = \|x\|^{-1} < \varepsilon$. For every $e \in E$ set $v(e) = \ell_1(e)y$. We have $v \in \mathcal{L}(E)$ and $v(x) = y$. We obtain $(u + v)x = u(x) + y = \lambda x$, hence $\lambda \in \text{Sp}(u + v)$. Since moreover $\|v\| = \|\ell_1\| < \varepsilon$ the condition (iii) is satisfied.

Conversely, let $v \in \mathcal{L}(E)$ such that $\|v\| < \varepsilon$ and $\lambda \in \text{Sp}(u + v)$. The partial operator $\lambda 1_E - (u + v)$ is therefore either not injective or does not have a continuous inverse; by Lemma 4 of IV, p. 243, applied to the injective partial operator $\lambda 1_E - u$ and to $-v$, we therefore have $\|v\| \geq \|R(u, \lambda)\|^{-1}$. It follows that condition (iii) implies (i).

COROLLARY. — Let E be a complex Banach space and u a closed partial operator on E . Let $\varepsilon > 0$.

- a) For every $v \in \mathcal{L}(E)$, we have $\text{PSp}_\varepsilon(u) \subset \text{PSp}_{\varepsilon+\|v\|}(u + v)$;
- b) Let $\delta > 0$ and let D_δ be the open disk with center 0 and radius δ in $\mathbb{C}$. We have $\text{PSp}_\varepsilon(u) + D_\delta \subset \text{PSp}_{\varepsilon+\delta}(u)$.

Let $\lambda \in \text{PSp}_\varepsilon(u)$. There exists an endomorphism $w \in \mathcal{L}(E)$ such that $\|w\| < \varepsilon$ and $\lambda \in \text{Sp}(u + w)$ (prop. 20). Since $u + w = (u + v) + (w - v)$ and $\|w - v\| < \varepsilon + \|v\|$, we have $\lambda \in \text{PSp}_{\varepsilon+\|v\|}(u + v)$ (loc. cit.).

Let $\mu \in D_\delta$; we have $\lambda + \mu \in \text{Sp}(u + (w + \mu 1_E))$ and $\|w + \mu 1_E\| < \varepsilon + \delta$, hence $\lambda + \mu \in \text{PSp}_{\varepsilon+\delta}(u)$ (loc. cit.).

PROPOSITION 21. — Let E be a complex Banach space and u a closed partial operator on E . Let $\varepsilon \in \mathbf{R}_+^*$. Every bounded connected component of $\text{PSp}_\varepsilon(u)$ meets the spectrum of u .

Let U be a connected component of $\text{PSp}_\varepsilon(u)$ which does not meet $\text{Sp}(u)$. The set U is open and closed in $\text{PSp}_\varepsilon(u)$ and its closure $\overline{U}$ in $\mathbb{C}$ therefore satisfies $\overline{U} \cap \text{PSp}_\varepsilon(u) = U$.

Since $\text{Sp}(u)$ is contained in $\text{PSp}_\varepsilon(u)$ and U does not meet $\text{Sp}(u)$, this equality proves that the set $\overline{U}$ is disjoint from $\text{Sp}(u)$, hence contained in the resolvent set of u . Moreover, it implies that the set $\overline{U} - U$ does not meet $\text{PSp}_\varepsilon(u)$.

We therefore have $\|R(u, \lambda)\| \leq \varepsilon^{-1}$ for all λ in $\overline{U} - U$, while $\|R(u, \lambda)\| > \varepsilon^{-1}$ for $\lambda \in U$. If the set U is bounded, its closure $\overline{U}$ is compact and there exists $\lambda_0 \in \overline{U}$ such that $\|R(u, \lambda)\| \leq \|R(u, \lambda_0)\|$ for $\lambda \in \overline{U}$. The foregoing implies that $\lambda_0 \in U$ which contradicts the maximum principle (VAR, R1, p. 29, 3.3.7) since the resolvent of u is holomorphic in the resolvent set of u (prop. 14 of IV, p. 246).

IV.4.9 Multiplication operators

Let X be a locally compact space and μ a positive measure on X . We consider the multiplication operators on $L^2(X, \mu)$; these are closed operators with dense domain (prop. 5 of IV, p. 232). For any μ -measurable function g on X , we will denote by m_g the partial multiplication operator by g on $L^2(X, \mu)$.

PROPOSITION 22. — Let g be a function μ -measurable on X .

- a) The spectrum of m_g is the μ -essential image S of g ;
- b) Let $\lambda \in \mathbb{C} - \text{Sp}(m_g)$. The resolvent $R(m_g, \lambda)$ is the multiplication operator m_h , where h is the function on X defined by $h(x) = 0$ if $g(x) = \lambda$ and $h(x) = (\lambda - g(x))^{-1}$ otherwise.

Let us prove that $\mathbb{C} - S$ is contained in the resolvent set of m_g . Let $\lambda \in \mathbb{C} - S$. There exists an open neighborhood U of λ such that the subset $Y = g^{-1}(U)$ of X is locally μ -negligible. The function k defined on X by $k(x) = (\lambda - g(x))^{-1}$ if $x \notin Y$ and $k(x) = 0$ if $x \in Y$ belongs to $\mathcal{L}^\infty(X, \mu)$ (Lemma 5 of IV, p. 184); the operator of multiplication by k is therefore an endomorphism of $L^2(X, \mu)$.

Since $|gk| \leq 1 + |\lambda k|$, we have

$$|gkf| \leq |f| + |\lambda kf|$$

for $f \in \mathcal{L}^2(X, \mu)$, which implies that the image of m_k is contained in the domain of m_g . Conversely, let $f \in \mathcal{L}^2(X, \mu)$ whose class $\tilde{f}$ belongs to the domain of m_g . Outside the locally μ -negligible set Y , we have $f(x) = k(x)(\lambda - g(x))f(x)$, hence $\tilde{f}$ is in the image of m_k . The same formula proves that λ belongs to the resolvent set of m_g and that $R(m_g, \lambda) = m_k$. Since the set Y is locally μ -negligible, the multiplication operator m_k coincides with the operator m_h described in assertion b).

Let us prove conversely that $\mathbb{C} - \text{Sp}(m_g)$ is contained in $\mathbb{C} - S$. Let $\lambda \in \mathbb{C} - \text{Sp}(m_g)$. Let $M > \|R(m_g, \lambda)\|$ be a real number. Let Y denote the set of $x \in X$ such that $|\lambda - g(x)| < M^{-1}$. Let us prove that Y is locally μ -negligible, which will imply that λ does not belong to S , and will conclude the proof.

Let K be a compact subset of X . Let φ be the characteristic function of $Y \cap K$; this is

an element of $\mathcal{L}^2(X, \mu)$, and we denote its class in $L^2(X, \mu)$ by $\tilde{\varphi}$. Let ψ be a function in $\mathcal{L}^2(X, \mu)$ whose class in $L^2(X, \mu)$ is $R(m_g, \lambda)(\tilde{\varphi})$. We have $R(m_g, \lambda)(\tilde{\varphi}) \in \text{dom}(m_g)$ and $(\lambda - m_g)(R(m_g, \lambda)(\tilde{\varphi})) = \tilde{\varphi}$, hence $(\lambda - g(x))\psi(x) = 1$ for μ -almost all $x \in Y \cap K$. This implies

$$\|R(m_g, \lambda)\|^2 \|\tilde{\varphi}\|^2 \geq \|R(m_g, \lambda)(\tilde{\varphi})\|^2 \geq M^2 \mu(Y \cap K) = M^2 \|\tilde{\varphi}\|^2.$$

In view of the choice of M , this means that φ is zero μ -almost everywhere. Thus, the set $Y \cap K$ is μ -negligible. The set Y is therefore locally μ -negligible (INT, IV, p. 172, § 5, n° 2, prop. 5).

PROPOSITION 23. — Let g be a μ -measurable function on X . The adjoint of the multiplication operator m_g is $m_{\bar{g}}$.

For every integer $n \geq 1$, let φ_n be the characteristic function of the set of elements $x \in X$ such that $|g(x)| \leq n$ and let $\tilde{\varphi}_n$ be its class in $L^2(X, \mu)$. Let $f \in \mathcal{L}^2(X, \mu)$ whose class $\tilde{f}$ belongs to $\text{dom}(m_g^*)$ and let ψ be a function whose class is $m_g^*(\tilde{f})$.

For every $h \in \mathcal{L}^2(X, \mu)$ whose class $\tilde{h}$ belongs to $\text{dom}(m_g)$, we also have $\tilde{h}\tilde{\varphi}_n \in \text{dom}(m_g)$, and hence $\langle \tilde{f} | m_g(\tilde{h}\tilde{\varphi}_n) \rangle = \langle m_g^*(\tilde{f}) | \tilde{h}\tilde{\varphi}_n \rangle$. This entails the equality

$$\int_X \overline{(f\bar{g} - \psi)} \varphi_n h d\mu = 0.$$

Since the domain of m_g is dense in $L^2(X, \mu)$, we deduce that $(f\bar{g} - \psi)\varphi_n$ is zero μ -almost everywhere. Since n is arbitrary, this means that $m_g^*(\tilde{f})$ is the class in $L^2(X, \mu)$ of $f\bar{g}$. In particular, since $m_g^*(\tilde{f}) \in L^2(X, \mu)$ we conclude that f belongs to the domain of $m_{\bar{g}}$ and that $m_g^*(\tilde{f}) = m_{\bar{g}}(\tilde{f})$.

The adjoint of m_g is therefore an extension of $m_{\bar{g}}$. Moreover, we have

$$\langle f | m_g(h) \rangle = \int_X \bar{f} \cdot (gh) d\mu = \int_X f \bar{g} h d\mu$$

for all $f \in L^2(X, \mu)$ and $h \in \text{dom}(m_g)$, which proves that the linear form $h \mapsto \langle f | m_g(h) \rangle$ is continuous when $m_{\bar{g}}(f)g$ belongs to $L^2(X, \mu)$. Consequently, the domain of $m_{\bar{g}}$ is contained in that of m_g^* , which concludes the proof.

COROLLARY. — Let g be a μ -measurable function on X . The multiplication operator m_g on $L^2(X, \mu)$ is self-adjoint (resp. positive) if and only if the function g is locally μ -almost everywhere real-valued (resp. locally μ -almost everywhere positive).

The first assertion follows from the proposition. If g is locally μ -almost everywhere positive, we have $\langle f | m_g(f) \rangle = \int_X g |f|^2 d\mu \geq 0$ for all $f \in L^2(X, \mu)$, so the partial operator m_g is positive.

Conversely, if m_g is positive, then its spectrum is contained in $\mathbf{R}_+$ (prop. 17 of IV, p. 248); since it is the μ -essential image of g (prop. 22, a)), this means that g is locally μ -almost everywhere positive.

Lemma 6. — Let g_1 and g_2 be functions μ -measurable on X .

a) The partial operator $m_{g_1} + m_{g_2}$ is closable and its closure is the multiplication operator

$$m_{g_1+g_2};$$

b) One has $m_{g_1} \circ m_{g_2} \subset m_{g_1 g_2}$;

c) Suppose that g_1 is bounded. We then have $m_{g_2} \circ m_{g_1} = m_{g_1 g_2}$. Moreover, the domain of m_{g_2} is contained in the domain of $m_{g_1 g_2}$, and $m_{g_1} \circ m_{g_2}$ is the reduction of $m_{g_1 g_2}$ to $\text{dom}(m_{g_2})$.

It is elementary that $m_{g_1+g_2}$ is an extension of $m_{g_1} + m_{g_2}$; this latter operator is therefore closable, and $\overline{m_{g_1} + m_{g_2}} \subset m_{g_1+g_2}$.

Let $f \in \mathcal{L}^2(X, \mu)$ such that the function $h = (g_1 + g_2)f$ belongs to $\mathcal{L}^2(X, \mu)$. For every integer $n \geq 1$, denote by X_n the set of $x \in X$ such that $|g_1(x)| + |g_2(x)| \leq n$ and let φ_n be the characteristic function of X_n . We have $\varphi_n f \in \text{dom}(m_{g_1} + m_{g_2})$. Since $(\varphi_n f)(x)$ converges to $f(x)$ for all $x \in X$ and since for every $n \in \mathbf{N}$ we have $|\varphi_n f| \leq |f|$ and $|(g_1 + g_2)\varphi_n f| \leq |(g_1 + g_2)f| = |h|$, with $h \in \mathcal{L}^2(X, \mu)$ by hypothesis, Lebesgue's theorem (INT, IV, p. 137, § 3, n° 7, th. 6) implies that the sequence of pairs of classes in $L^2(X, \mu) \times L^2(X, \mu)$ of $(\varphi_n f, (g_1 + g_2)\varphi_n f)$ which belong to the graph of $m_{g_1} + m_{g_2}$ converges to the pair of classes of (f, h) in $L^2(X, \mu)$. The closure of $m_{g_1} + m_{g_2}$ is therefore indeed equal to $m_{g_1+g_2}$.

It is elementary that $m_{g_1} \circ m_{g_2} \subset m_{g_1 g_2}$. Suppose that g_1 is bounded, so that m_{g_1} is a continuous linear map on $L^2(X, \mu)$. Then the domain of $m_{g_2} \circ m_{g_1}$ is the set of equivalence classes of functions $f \in \mathcal{L}^2(X, \mu)$ such that $g_2(g_1 f) \in \mathcal{L}^2(X, \mu)$, that is, the domain of $m_{g_1 g_2}$.

. We therefore have $m_{g_2} \circ m_{g_1} = m_{g_1 g_2}$.

It is also elementary that $\text{dom}(m_{g_1} \circ m_{g_2}) = \text{dom}(m_{g_2})$ is contained in $\text{dom}(m_{g_1 g_2})$, and that the restriction of $m_{g_1 g_2}$ to this space is equal to $m_{g_1} \circ m_{g_2}$.

One should beware of believing that $m_{g_1} \circ m_{g_2} = m_{g_1 g_2}$ in general (exercise 10 of IV, p. 347). Nevertheless, we have the following partial result:

PROPOSITION 24. — Let g be a μ -measurable function on X .

a) One has $m_{\bar{g}} \circ m_g = m_{|g|^2}$;

b) For all integers $k, \ell \in \mathbb{N}$, we have $m_{g^k \bar{g}^\ell} = m_g^k m_{\bar{g}}^\ell$.

One has $m_{\bar{g}} \circ m_g \subset m_{|g|^2}$ by Lemma 6. Conversely, from the inequality $|g| \leq 1 + |g|^2$ we deduce that $\text{dom}(m_{|g|^2}) = \text{dom}(m_{\bar{g}} \circ m_g)$, whence the first assertion.

Let $k, \ell \in \mathbb{N}$. We have $m_g^k m_{\bar{g}}^\ell \subset m_{g^k \bar{g}^\ell}$ (loc. cit.). The domain of $m_g^k m_{\bar{g}}^\ell$ is the set of classes in $L^2(X, \mu)$ of functions $h \in \mathcal{L}^2(X, \mu)$ such that $|g|^j h$ belongs to $\mathcal{L}^2(X, \mu)$ for every integer j such that $0 \leq j \leq k + \ell$. The inequalities

$$|g^j h| \leq |h| + |g^{k+\ell} h|,$$

valid for $0 \leq j \leq k + \ell$, allow one to observe that $\text{dom}(m_g^k m_{\bar{g}}^\ell)$ is equal to $\text{dom}(m_{g^k \bar{g}^\ell})$, whence assertion b).

IV.4.10 Self-adjoint extensions of a symmetric operator

In this section, we propose to classify the self-adjoint extensions of symmetric operators on a complex Hilbert space, and in particular to find conditions ensuring the existence of a self-adjoint extension of such an operator.

Let E be a complex Hilbert space. For every partial operator u on E , we shall write in this section $u + i$ and $u - i$ instead of $u + i1_E$ and $u - i1_E$.

Let u be a closed operator with dense domain on E . We recall (cf. definition 4 of IV, p. 230 and proposition 4 of IV, p. 230) that we denote by E_u the Hilbert space $\text{dom}(u)$ equipped with the Hermitian form

$$(x | y)_u = \langle x | y \rangle + \langle u(x) | u(y) \rangle.$$

We denote $\|x\|_u$ the norm of an element x of the Hilbert space E_u . Every subspace of $\text{dom}(u)$ that is closed in E is closed in E_u (IV, p. 230, remark 1).

DEFINITION 10. — Let u be a closed symmetric operator on a complex Hilbert space E . Let $E_+ = \text{Ker}(u^* - i)$ and $E_- = \text{Ker}(u^* + i)$. The pair (E_+, E_-) of subspaces of $\text{dom}(u^*)$ is called the deficiency pair of u . The subspaces E_+ and E_- are closed subspaces of E (prop. 3 of IV, p. 229). The pair $(\dim(E_+), \dim(E_-))$ of Hilbertian dimensions of these subspaces is called the deficiency index of u .

PROPOSITION 25. — Let u be a closed symmetric operator on a complex Hilbert space E . We have

$$\text{Ker}(u^* - i)^\circ = \text{Im}(u + i),$$

$$\text{Ker}(u^* + i)^\circ = \text{Im}(u - i).$$

By prop. 7, c) of IV, p. 236, it suffices to show that the image of $u + i$ (resp. of $u - i$) is closed in E . Since u is symmetric, we have $\langle (u + i)(x) | (u + i)(y) \rangle = (x | y)_u$ for all x and y in $\text{dom}(u)$. The map $x \mapsto u(x) + ix$ from E_u into E is therefore isometric. Its image is closed in E (lemma 8 of I, p. 107). Since this is also the image of $u + i$, the latter is closed in E . Likewise, the image of $u - i$ is closed in E .

Lemma 7. — Let u be a closed symmetric operator on a complex Hilbert space E and v a closed symmetric extension of u . The domain of u is a closed subspace of the Hilbert space E_v .

Since v is an extension of u , we have $(x | y)_v = (x | y)_u$ for x and y in $\text{dom}(u)$. The canonical injection of E_u into E_v is therefore isometric, and the conclusion follows (lemma 8 of I, p. 107).

PROPOSITION 26. — Let u be a closed symmetric operator on a complex Hilbert space E . Let (E_+, E_-) be the deficiency pair of u . The spaces E_+ , E_- and $\text{dom}(u)$ are closed and mutually orthogonal subspaces of E_{u^*} whose Hilbertian sum is equal to E_{u^*} .

Since u is symmetric, we have $u \subset u^*$, hence the space $\text{dom}(u)$ is closed in E_{u^*} (lemma 7). The subspaces E_+ and E_- are closed in E and contained in $\text{dom}(u^*)$, hence closed in E_{u^*} .

Let $x \in E_+$. We have $u^*(x) = ix$, whence $\langle x|u(y) \rangle = -i\langle x|y \rangle$ for all $y \in \text{dom}(u)$. Consequently, for every $y \in \text{dom}(u)$, we have

$$(x|y)_{u^*} = \langle x|y \rangle + \langle u^*(x)|u^*(y) \rangle = i(\langle x|u(y) \rangle - \langle x|u^*(y) \rangle) = 0$$

since u is symmetric. The spaces E_+ and $\text{dom}(u)$ are therefore orthogonal in E_{u^*} . Likewise, E_- and $\text{dom}(u)$ are orthogonal in E_{u^*} .

Let $x \in E_+$ and $y \in E_-$. Then

$$(x|y)_{u^*} = \langle x|y \rangle + \langle u^*(x)|u^*(y) \rangle = \langle x|y \rangle - \langle x|y \rangle = 0,$$

hence E_+ and E_- are orthogonal in E_{u^*} .

Let $x \in E_{u^*}$ orthogonal to the closed subspace $\text{dom}(u) \oplus E_+ \oplus E_-$. For every $y \in \text{dom}(u)$, we have $u^*(y) = u(y)$ since $u \subset u^*$, whence

$$0 = (x|y)_{u^*} = \langle x|y \rangle + \langle u^*(x)|u^*(y) \rangle = \langle x|y \rangle + \langle u^*(x)|u(y) \rangle.$$

This implies that $z = u^*(x)$ belongs to the domain of u^* and satisfies $u^*(z) = -x$. Let $x_- = z - ix$. It is an element of $\text{dom}(u^*)$ satisfying $u^*(x_-) = -x - iz = -ix_-$. Hence x_- belongs to E_- . But, for every $w \in E_-$, we have

$$\begin{aligned} -i\langle x_-|w \rangle &= -i\langle z|w \rangle + \langle x|w \rangle \\ &= \langle u^*(x)|u^*(w) \rangle + \langle x|w \rangle = (x|w)_{u^*} = 0. \end{aligned}$$

Taking $w = x_-$, we deduce that $x_- = 0$, that is, $z = ix$; since $z = u^*(x)$, we therefore have $x \in E_+$, whence $x = 0$ since x is orthogonal to E_+ in E_{u^*} . This completes the proof that the direct sum $\text{dom}(u) \oplus E_+ \oplus E_-$ is equal to E_{u^*} .

COROLLARY. — Let u be a closed symmetric operator on E . Then u is self-adjoint if and only if the deficiency index of u is equal to $(0,0)$.

Indeed, the symmetric operator u is self-adjoint if and only if $\text{dom}(u^*) = \text{dom}(u)$; but the proposition shows that $\text{dom}(u^*)$ is the Hilbert sum of $\text{dom}(u)$ and of the deficiency subspaces of u .



Let u be a closed symmetric partial operator. The spectrum of u is contained in $\mathbf{R}$ if and only if u is self-adjoint. Indeed, we know that $\text{Sp}(u) \subset \mathbf{R}$ if u is self-adjoint (prop. 17 of IV, p. 248); if u is not self-adjoint, one of the subspaces $\text{Ker}(u^* + i)$ or $\text{Ker}(u^* - i)$ is nonzero, hence the image of $u + i$ or of $u - i$ is a proper subspace of E (prop. 25 of IV, p. 256), so that i or $-i$ belongs to $\text{Sp}(u)$.

DEFINITION 11. — Let u be a closed symmetric operator on a complex Hilbert space E . A boundary condition for u is a partially isometric linear map (EVT, V, p. 41, def. 3) from $\text{Ker}(u^* - i)$ into $\text{Ker}(u^* + i)$.

Let u be a closed symmetric operator on a complex Hilbert space E and b a boundary condition for u . Let $I = \text{Ker}(b)^\circ$ be the initial subspace of b . We denote by u_b the reduction of u^* to the subspace of $\text{dom}(u^*)$ that is the direct sum of $\text{dom}(u)$ and the graph in $E_+ \oplus E_-$ of the restriction of b to I . Since $E_{u^*} = \text{dom}(u) \oplus E_+ \oplus E_-$ (prop. 26), the partial operator u_b is well defined.

In other words, the domain of u_b is the space of elements $x \in E$ of the form $x = x_0 + y + b(y)$, where $x_0 \in \text{dom}(u)$ and y belongs to the initial subspace of b . We then have $u_b(x_0 + y + b(y)) = u(x_0) + iy - ib(y)$.

THEOREM 1 (von Neumann). — Let u be a closed symmetric operator on a complex Hilbert space E and (E_+, E_-) its deficiency pair.

- a) For every boundary condition b for u , the partial operator u_b is a closed symmetric extension of u ;
- b) The map $b \mapsto u_b$ is a bijection from the set of boundary conditions for u onto the set of closed symmetric extensions of u ;
- c) For every boundary condition b for u , we have

$$\text{Im}(u_b + i) = \text{Im}(u + i) \oplus \text{Im}(b)$$

$$\text{Im}(u_b - i) = \text{Im}(u - i) \oplus \text{Ker}(b)^\circ;$$

d) For every boundary condition b for u , the deficiency pair of u_b is $(\text{Ker}(b), \text{Ker}(b^*))$.

Let b be a boundary condition for u and $I \subset E_+$ its initial subspace. The restriction of u^* to $\Gamma_{b|I} \subset E_+ \oplus E_-$ is the continuous linear map defined by $x + b(x) \mapsto ix - ib(x)$ for $x \in I$. The graph of u_b is the direct sum of the graph of u and the graph of this linear map; it is therefore closed. The closed operator u_b is an extension of u such that $u_b \subset u^*$.

Let γ_1 and η_1 be elements of I . Consider the elements

$$\gamma = \gamma_1 + b(\gamma_1), \quad \eta = \eta_1 + b(\eta_1)$$

of the graph $\Gamma_{b|I}$. We compute

$$\begin{aligned} \langle \gamma | u^*(\eta) \rangle &= i(\langle b(\gamma_1) | \eta_1 \rangle - \langle \gamma_1 | b(\eta_1) \rangle) + i(\langle \gamma_1 | \eta_1 \rangle - \langle b(\gamma_1) | b(\eta_1) \rangle) \\ &= i(\langle b(\gamma_1) | \eta_1 \rangle - \langle \gamma_1 | b(\eta_1) \rangle) \end{aligned}$$

since b is isometric on I . From this we deduce that we have

$$\langle u^*(\gamma) | \eta \rangle = \langle \gamma | u^*(\eta) \rangle \quad (12)$$

for all γ and η in $\Gamma_{b|I}$.

Let x and y be elements of $\text{dom}(u)$, γ and η be elements of $\Gamma_{b|I}$. We obtain

$$\begin{aligned} \langle x + \gamma | u_b(y + \eta) \rangle &= \langle x | u(y) \rangle + \langle x | u^*(\eta) \rangle + \langle \gamma | u(y) \rangle + \langle \gamma | u^*(\eta) \rangle \\ &= \langle u(x) | y \rangle + \langle u(x) | \eta \rangle + \langle u^*(\gamma) | y \rangle + \langle u^*(\gamma) | \eta \rangle \\ &= \langle u_b(x + \gamma) | y + \eta \rangle, \end{aligned}$$

where we used the fact that u is symmetric and that γ and η belong to the domain of u^* as well as formula (12). The operator u_b is therefore a closed symmetric extension of u . Assertion a) is proved.

The map $b \mapsto u_b$ is injective. Indeed, a partially isometric map from E_+ into E_- is uniquely

determined by its initial subspace and by its restriction to that subspace. Now the domain of u_b determines the initial subspace I of b and the graph of the restriction of b to I .

Let us prove that the map $b \mapsto u_b$ is surjective. Let w be a closed symmetric extension of u . We have $w \subset w^* \subset u^*$. The domain of w is therefore a subspace of $\text{dom}(u^*)$ containing $\text{dom}(u)$, and w is the reduction of u^* to this subspace. Let G be the intersection of the domain of w and of $E_+ \oplus E_-$. This is a closed subspace of E_{u^*} (Lemma 7) and we have $\text{dom}(w) = \text{dom}(u) \oplus G$ by proposition 26.

Let $x \in E_+$ and $y \in E_-$ such that $x + y \in G$. Since w is symmetric and is the reduction of u^* to $\text{dom}(w)$, we have

$$\langle x + y | u^*(x + y) \rangle = \langle u^*(x + y) | x + y \rangle.$$

This equality is equivalent to

$$\begin{aligned} & \langle x | ix \rangle + \langle x | -iy \rangle + \langle y | ix \rangle + \langle y | -iy \rangle \\ &= \langle ix | x \rangle + \langle ix | y \rangle + \langle -iy | x \rangle + \langle -iy | y \rangle, \end{aligned}$$

that is, to $\|x\|^2 = \|y\|^2$. In particular, the canonical projection $p_+ : G \rightarrow E_+$ is injective, and if I denotes its image, the space G is the graph of an isometric map b_0 from I into E_- . For x in I and y in E_- such that $x + y \in G$, we have

$$\begin{aligned} \|x + y\|_{u^*}^2 &= \|x\|_{u^*}^2 + \|y\|_{u^*}^2 = 2(\|x\|^2 + \|y\|^2) \\ &= 4\|x\|^2 = 4\|p_+(x + y)\|^2. \end{aligned}$$

Thus, considering G as a closed subspace of the Hilbert space E_{u^*} , the map $x \mapsto \frac{1}{2}p_+(x)$ is an isometry from G onto I . In particular, I is closed in E_+ (lemma 8 of I, p. 107). There then exists a unique boundary condition b for u for which I is the initial space and which coincides with b_0 on I . We have $\text{dom}(w) = \text{dom}(u_b)$, whence $w = u_b$. This establishes assertion b).

Let us prove the assertions c) and d). Let b be a boundary condition for u and I its initial subspace. For $x_0 \in \text{dom}(u)$ and $y \in I$, we have

$$u_b(x_0 + y + b(y)) - i(x_0 + y + b(y)) = (u - i)(x_0) - 2ib(y).$$

One has $\text{Im}(u - i) = E_-^\circ$ (prop. 25). Since the image of b is contained in E_- , this formula shows that $\text{Im}(u_b - i) = \text{Im}(u - i) \oplus \text{Im}(b)$. By loc. cit., we therefore have $\text{Ker}(u_b^* + i) = E_- \cap \text{Im}(b)^\circ = \text{Ker}(b^*)$.

Similarly, one verifies that $\text{Im}(u_b + i) = \text{Im}(u + i) \oplus I$ and hence (loc. cit.) that $\text{Ker}(u_b^* - i) = I^\circ \cap E_+ = \text{Ker}(b)$.

COROLLARY 1. — Let u be a closed symmetric operator on a complex Hilbert space E . The map $b \mapsto u_b$ induces a bijection between the set of isometric isomorphisms from $\text{Ker}(u^* - i)$ onto $\text{Ker}(u^* + i)$ and the set of self-adjoint extensions of u . In particular, there exists a self-adjoint extension of u if and only if the two components of the deficiency index of u are equal.

This follows from the corollary of Proposition 26 and the theorem, as well as from EVT, V, p. 25, cor. 2.

COROLLARY 2. — Let E be a real Hilbert space. Let u be a closed symmetric operator on E . The partial operator $u_{(\mathbb{C})}$ on $E_{(\mathbb{C})}$ is symmetric and closed, and admits at least one self-adjoint extension.

The partial operator $u_{(\mathbb{C})}$ is closed and symmetric by prop. 2 of IV, p. 228 and lemma 3 of IV, p. 239. The $\mathbb{R}$ -linear isomorphism j from $E_{(\mathbb{C})}$ to $E_{(\mathbb{C})}$ such that $j(z \otimes x) = \bar{z} \otimes x$ for all $z \in \mathbb{C}$ and every $x \in E$ induces an isomorphism of Hilbert spaces from $\text{Ker}(u_{(\mathbb{C})}^* - i)$ onto $\text{Ker}(u_{(\mathbb{C})}^* + i)$, and we can apply Corollary 1.

COROLLARY 3. — Let u be a symmetric operator on E . The following conditions are equivalent:

- (i) The operator u is essentially self-adjoint;
- (ii) The spaces $\text{Ker}(u^* + i)$ and $\text{Ker}(u^* - i)$ are zero;
- (iii) The spaces $\text{Im}(u + i)$ and $\text{Im}(u - i)$ are dense in E ;
- (iv) The partial symmetric operator u has a unique self-adjoint extension.

Conditions (ii) and (iii) are equivalent by Proposition 7, c) of IV, p. 236.

The partial operator u is essentially self-adjoint if and only if $\bar{u}$ is self-adjoint, that is, if $\text{Ker}(\bar{u}^* - i) = \text{Ker}(\bar{u}^* + i) = \{0\}$ (corollary of prop. 26). Since $\bar{u}^* = u^*$ (prop. 8 of IV, p. 237),

condition (i) is therefore equivalent to condition (ii). Moreover, the preceding corollary shows that (ii) implies that u has a unique self-adjoint extension, which is condition (iv).

Finally, suppose that u has a unique self-adjoint extension v . Then the same is true of $\bar{u}$, since every self-adjoint extension of $\bar{u}$ is one of u , hence is equal to v . By the preceding corollary, the spaces $\text{Ker}(\bar{u}^* + i)$ and $\text{Ker}(\bar{u}^* - i)$ must be zero, whence condition (ii) using again $\bar{u}^* = u^*$.

Example. — Let U be an open subset of $\mathbf{R}^n$ endowed with Lebesgue measure μ . Let Δ be the scalar differential operator

$$\Delta = - \sum_{i=1}^n \partial_i^2$$

on U . Denote by u the partial operator on the real Hilbert space $L^2_{\mathbf{R}}(U, \mu)$ with domain $\mathcal{D}_{\mathbf{R}}(U)$ (IV, p. 201, remark) defined by $\varphi \mapsto - \sum_{i=1}^n \partial_i^2 \varphi$. We have $u_{(\mathbf{C})} = \Delta_-$, which is a closable partial operator (prop. 13 of IV, p. 242) and symmetric (since Δ is formally symmetric), hence u is closable and symmetric (proposition 2 of IV, p. 228 and lemma 3 of IV, p. 239). By the above corollary, there therefore exists a self-adjoint extension of $u_{(\mathbf{C})}$. This is a Laplacian on U (IV, p. 243).

IV.5 NORMAL PARTIAL OPERATORS AND THE SPECTRAL THEOREM

IV.5.1 Boundedification

Let E be a complex Hilbert space. We denote by s the unitary map $(x, y) \mapsto (-y, x)$ on $E \oplus E$. Let u be a closed partial operator with dense domain on E . The partial operator u^*u is self-adjoint and positive (prop. 12 of IV, p. 241), hence $-1 \notin \text{Sp}(u^*u)$ (prop. 17 of IV, p. 248). We denote by $W(u) = (1_E + u^*u)^{-1} = -R(u^*u, -1)$; this is a positive injective endomorphism of E .

Denote by p_1 and p_2 the two canonical projections of Γ_u into E . These are elements of $\mathcal{L}(\Gamma_u; E)$, and we have the equality of correspondences $p_2 = u \circ p_1$. Let $(j, |p_1^*|)$ be the polar decomposition (def. 4 of I, p. 140) of the endomorphism $p_1^* \in \mathcal{L}(E; \Gamma_u)$, so that $p_1^* = |p_1^*| \circ j$.

The map j is a partial isometry from E into Γ_u .

DEFINITION 1. — The endomorphism $p_2 \circ j$ of E is called the boundedification of u ; we denote by $b(u)$ this endomorphism.

One has $\|b(u)\| \leq 1$.

PROPOSITION 1. — We have the formulas

$$|p_1^*| = W(u)^{1/2}, \quad (1)$$

$$b(u) = u \circ W(u)^{1/2} = u \circ |p_1^*|, \quad (2)$$

$$1_E - b(u)^*b(u) = W(u), \quad (3)$$

$$b(u)W(u) = W(u^*)b(u), \quad b(u)W(u)^{1/2} = W(u^*)^{1/2}b(u), \quad (4)$$

Let $x \in E$. Set $y = W(u)(x) \in \text{dom}(u^*u)$. We have $y \in \text{dom}(u)$ and $u(y) \in \text{dom}(u^*)$. For every element $(y_1, u(y_1))$ of Γ_u , where $y_1 \in \text{dom}(u)$, one computes

$$\begin{aligned} \langle (y, u(y)) | (y_1, u(y_1)) \rangle &= \langle y | y_1 \rangle + \langle u(y) | u(y_1) \rangle \\ &= \langle y + u^*u(y) | y_1 \rangle = \langle x | p_1(y_1, u(y_1)) \rangle. \end{aligned}$$

This means that $p_1^*(x) = (y, u(y))$, whence $p_1 \circ p_1^*(x) = y = W(u)(x)$. Consequently, one has $p_1 \circ p_1^* = W(u)$, hence $|p_1^*| = W(u)^{1/2}$, that is, formula (1). In particular, since $\text{Im}(p_1) = \text{Im}(|p_1^*|)$ by the cor. of prop. 11 of I, p. 140, it follows that $\text{dom}(u) = \text{Im}(p_1) = \text{Im}(W(u)^{1/2})$.

The partial operator $u \circ W(u)^{1/2}$ therefore has domain E . The relation $p_1 \circ j = (p_1^*)^* \circ j = |p_1^*| = W(u)^{1/2}$ (prop. 11, a) of I, p. 140) then implies that

$$b(u) = p_2 \circ j = u \circ p_1 \circ j = u \circ W(u)^{1/2},$$

which is formula (2).

One has $\text{Ker}(j) = \text{Ker}(|p_1^*|)$ (prop. 10, b) of I, p. 139) and formula (1) therefore imply

that the partial isometry $j : E \rightarrow \Gamma_u$ is injective, hence isometric, whence $j^* \circ j = 1_E$. Writing $j = (p_1 \circ j, b(u))$, we find

$$1_E = j^* \circ j = (p_1 \circ j)^* (p_1 \circ j) + b(u)^* b(u) = W(u) + b(u)^* b(u),$$

whence formula (3).

Let $x \in \text{dom}(u)$ and set $y = W(u)(x)$. We have $y \in \text{dom}(u)$ and the formula $y + u^* u(y) = x$ implies that $u^* u(y) \in \text{dom}(u)$. We then have

$$\begin{aligned} W(u^*)^{-1}(u(y)) &= (1_E + uu^*)(u(y)) \\ &= u(y) + uu^* u(y) = u(y + u^* u(y)) = u(x), \end{aligned}$$

which means that the reduction of $u \circ W(u)$ to the domain of u is equal to $W(u^*) \circ u$. Since the image of $|p_1^*|$ is equal to that of p_1 , and hence to the domain of u (cor. of prop. 11 of I, p. 140), it follows

$$u \circ W(u) \circ |p_1^*| = W(u^*) \circ u \circ |p_1^*| = W(u^*) \circ b(u)$$

(formula (2)). Moreover, $|p_1^*| = W(u)^{1/2}$ commutes with $W(u)$, hence we obtain

$$b(u) \circ W(u) = u \circ |p_1^*| \circ W(u) = u \circ W(u) \circ |p_1^*| = W(u^*) \circ b(u).$$

This relation implies that $b(u) \circ f(W(u)) = f(W(u^*)) \circ b(u)$ for every function $f \in \mathcal{C}(\mathbf{R}_+)$ (prop. 11 of I, p. 113), hence in particular that $b(u) \circ W(u)^{1/2} = W(u^*)^{1/2} \circ b(u)$. This proves formula (4) and concludes the proof.

COROLLARY. — We have $b(u)^* = b(u^*)$.

Denote by q_1 and q_2 the projections $\Gamma_{u^*} \rightarrow E$ and let $(k, |q_1^*|)$ be the polar decomposition of q_1^* , so that $b(u^*) = q_2 \circ k$. For all x and y in E , we have $j(x) \in \Gamma_u$ and $k(y) \in \Gamma_{u^*}$, and since Γ_u is orthogonal to $s(\Gamma_{u^*})$ by prop. 7 of IV, p. 236, it follows

$$0 = \langle j(x) | s(k(y)) \rangle = \langle p_1 \circ j(x) | -q_2 \circ k(y) \rangle + \langle p_2 \circ j(x) | q_1 \circ k(y) \rangle.$$

Since $p_1 \circ j = |p_1^*|$ and $q_1 \circ k = |q_1^*|$ (prop. 11, b) of I, p. 140) we obtain

$$\langle b(u)x \mid |q_1^*|y \rangle = \langle |p_1^*|x \mid b(u^*)y \rangle,$$

hence $|q_1^*|b(u) = b(u^*)^*|p_1^*|$. Using formulas (1) and (4), we conclude from this that $b(u)|p_1^*| = |q_1^*|b(u) = b(u^*)^*|p_1^*|$, and since the image of $|p_1^*|$ is the domain of u which is dense in E , we deduce $b(u) = b(u^*)^*$.

We denote $\Omega(E)$ the set of $v \in \mathcal{L}(E)$ such that $1_E - v^*v$ is positive and injective. For $v \in \Omega(E)$, the Hermitian endomorphism $(1_E - v^*v)^{1/2}$ is injective, since its square is, and we denote by $B(v)$ the partial operator $v \circ ((1_E - v^*v)^{1/2})^{-1}$.

Note that $1_E - v^*v$ is positive if and only if $\|v\| \leq 1$, since we have $\langle x \mid (1_E - v^*v)(x) \rangle = \|x\|^2 - \|v(x)\|^2$ for all $x \in E$.

Lemma 1. — The subset $\Omega(E)$ is self-adjoint in $\mathcal{L}(E)$.

Let $v \in \Omega(E)$. We have $\|v^*\| = \|v\| \leq 1$. Therefore the endomorphism $1_E - vv^*$ is positive. It is injective: if $x \in \text{Ker}(1_E - vv^*)$, we have $vv^*(x) = x$, whence $v^*(v(v^*(x))) = v^*(x)$, then $v^*(x) = 0$ since $1_E - v^*v$ is injective, and finally $x = v(v^*(x)) = 0$. The lemma follows from this.

PROPOSITION 2. — The map $u \mapsto b(u)$ defines a bijection from the set of closed densely defined partial operators on E onto the set $\Omega(E)$. The inverse bijection is given by $v \mapsto B(v)$.

The relation $1_E - b(u)^*b(u) = W(u)$ (formula (3)) implies that $b(u)$ belongs to $\Omega(E)$ since $W(u)$ is positive and injective. Moreover, since $b(u) = u \circ W(u)^{1/2}$ (formula (2)), we obtain

$$u = b(u) \circ (W(u)^{1/2})^{-1} = b(u) \circ ((1 - b(u)^*b(u))^{1/2})^{-1} = B(b(u)).$$

Conversely, let $v \in \Omega(E)$. Set $w = (1_E - v^*v)^{1/2}$ and set $u = B(v) = v \circ w^{-1}$. The domain of u is the image of w , which is dense in E since w is Hermitian and injective (EVT, V, p. 41, prop. 4 (i)). For every $x \in \text{dom}(u)$, it follows that

$$\begin{aligned} \|u(x)\|^2 &= \langle (v^*v)(w^{-1}(x)) \mid w^{-1}(x) \rangle \\ &= -\langle (1_E - v^*v)(w^{-1}(x)) \mid w^{-1}(x) \rangle + \|w^{-1}(x)\|^2 \\ &= -\langle w(x) \mid w^{-1}(x) \rangle + \|w^{-1}(x)\|^2 \end{aligned}$$

since $(1_E - v^*v) \circ w^{-1}$ is the reduction of w to the domain of w^{-1} . Since w is self-adjoint, it follows that

$$\|w^{-1}(x)\|^2 = \|x\|^2 + \|v \circ w^{-1}(x)\|^2 = \|x\|^2 + \|u(x)\|^2,$$

and the partial operator $B(v) = u = v \circ w^{-1}$ is therefore closed by Lemma 2 of IV, p. 231.

Let us finally prove that $b(B(v)) = v$. Since $u = v \circ w^{-1}$, we have $v = u \circ w$ and it therefore suffices to check that $w = W(u)^{1/2}$ by formula (2), or even that $1_E - v^*v = W(u)$.

The endomorphism v^* belongs to $\Omega(E)$. By lemma 1, set $w' = (1_E - vv^*)^{1/2}$. It is an injective positive endomorphism of E , and we have $v \circ w = w' \circ v$ (prop. 11 of I, p. 113). The graph of u is the set of elements of the form $(w(x), v(x))$ for $x \in E$. By prop. 7 of IV, p. 236, the graph of u^* is $s(\Gamma_u)^\circ$. Since, for all x and $y \in E$, we have

$$\langle (w'(x), v^*(x)) | (-v(y), w(y)) \rangle = -\langle x | w'v(y) \rangle + \langle x | vw(y) \rangle = 0,$$

the graph of u^* contains the elements of the form $(w'(x), v^*(x))$ for $x \in E$, and we then have $u^*(w'(x)) = v^*(x)$. In particular, the domain of u^* contains the image of w' .

Let $x \in E$ and set $y = w^2(x) = (1_E - v^*v)(x)$, hence $x = y + v^*v(x)$. We have $y \in \text{dom}(u)$ and $u(y) = v(w^{-1}(y)) = v(w(x)) = w'(v(x))$. In particular, $u(y) \in \text{Im}(w) \subset \text{dom}(u^*)$, and $u^*(u(y)) = v^*(v(x))$. It follows therefore $x = y + v^*v(x) = y + u^*u(y)$, whence $y = W(u)(x)$, that is,

$$(1_E - v^*v)(x) = W(u)(x).$$

We have therefore shown that $1_E - v^*v = W(u)$, as desired.

COROLLARY. — The endomorphism $b(u)$ is Hermitian if and only if u is self-adjoint.

This follows from the injectivity of the map $u \mapsto b(u)$ (prop. 2) and from the formula $b(u^*) = b(u)^*$ (cor. of prop. 1).

IV.5.2 Normal partial operators and the spectral theorem

In this section, E is a complex Hilbert space.

DEFINITION 2. — Let u be a partial operator on E . One says that u is normal if u is closed with dense domain and if its bounded transform $b(u)$ is a normal endomorphism of E .

If $u \in \mathcal{L}(E)$ this definition coincides with EVT, V, p. 42, def. 4, by virtue of the formulas $1_E - b(u)^*b(u) = W(u) = (1_E + u^*u)^{-1}$ (prop. 1 of IV, p. 262) and $b(u^*) = b(u)^*$ (cor. of loc. cit.).

If u is a self-adjoint partial operator on E , then $b(u)$ is Hermitian (cor. of prop. 2 of IV, p. 264) hence u is normal.

Let D be the open unit disk in $\mathbb{C}$. We denote by β the function from $\mathbb{C}$ to D defined by $\beta(z) = z/\sqrt{1 + |z|^2}$. It is a homeomorphism, whose inverse satisfies $\beta^{-1}(z) = z/\sqrt{1 - |z|^2}$ for $z \in D$.

Let $u \in \mathcal{L}(E)$. It follows from formulas (2) and (3) of proposition 1 of IV, p. 262 that $u = \beta^{-1}(b(u))$, and hence

$$b(u) = \beta(u). \quad (5)$$

Lemma 2. — Let X be a locally compact topological space and let μ be a positive measure on X .

- a) Let g be a function μ -measurable on X . The multiplication operator m_g on $L^2(X, \mu)$ is normal and $b(m_g) = m_{\beta \circ g}$;
- b) Let $h: X \rightarrow D$ be a function μ -measurable. The endomorphism m_h belongs to $\Omega(L^2(X, \mu))$ and $B(m_h) = m_{\beta^{-1} \circ h}$.

The partial operator m_g is closed with dense domain. Since $m_g^*m_g = m_{\bar{g}}m_g = m_{|g|^2}$ (prop. 23 of IV, p. 253 and prop. 24 of IV, p. 255) we have

$$W(m_g) = -R(m_{|g|^2}, -1) = m_{(1+|g|^2)^{-1}}$$

(prop. 22 of IV, p. 252). This implies

$$b(m_g) = m_g \circ W(m_g)^{1/2} = m_g \circ m_{(1+|g|^2)^{-1/2}} = m_{\beta \circ g}$$

according to formula (2) of IV, p. 262, the cor. of prop. 6 of IV, p. 187 and lemma 6 of IV, p. 254. Finally applying prop. 6 of IV, p. 187, we obtain assertion a).

Let us prove b). Since $1 - |h(x)|^2 > 0$ for all $x \in X$, we have $m_h \in \Omega(L^2(X, \mu))$ (cor. of prop. 6 of IV, p. 187). Since the map $u \mapsto b(u)$ is injective (prop. 2 of IV, p. 264) and the endomorphisms $b(B(m_h))$ and $b(m_{\beta^{-1} \circ h})$ are equal to m_h according to assertion a), we indeed have $B(m_h) = m_{\beta^{-1} \circ h}$.

THEOREM 1 ("Spectral theorem"). — Let u be a normal partial operator on the Hilbert space E . There exist a locally compact topological space X , a positive measure μ on X , an isometric isomorphism θ of $L^2(X, \mu)$ onto E and a continuous function g on X , such that $u = \theta \circ m_g \circ \theta^{-1}$.

The endomorphism $b(u)$ of E is normal by definition. By the spectral theorem for normal endomorphisms of E (corollary 1 of IV, p. 193), there therefore exist a locally compact topological space $\tilde{X}$, a positive measure μ on $\tilde{X}$, an isometric isomorphism $\tilde{\theta}$ of $L^2(\tilde{X}, \mu)$ onto E , and a bounded continuous function h on $\tilde{X}$, so that $b(u)$ coincides with $\tilde{\theta} \circ m_h \circ \tilde{\theta}^{-1}$.

Let N be the closed subset of $x \in \tilde{X}$ such that $|h(x)| \geq 1$. Since the endomorphism $1 - b(u)^*b(u) = \tilde{\theta} \circ m_{1-|h|^2} \circ \tilde{\theta}^{-1}$ is positive and injective, the set N is locally μ -negligible (lemma 7 of IV, p. 186 and cor. of prop. 6 of IV, p. 187). We then set $X = \tilde{X} - N$. This is a locally compact space, and the restriction of functions to X induces an isometric isomorphism of $L^2(\tilde{X}, \mu)$ onto $L^2(X, \mu|_X)$ (prop. 1 of IV, p. 179). The composition of $\tilde{\theta}$ and of the restriction of functions to X therefore induces an isometric isomorphism θ of $L^2(X, \mu|_X)$ onto E . Since $b(u) = \theta \circ m_{h|_X} \circ \theta^{-1}$, it follows that $u = \theta \circ m_{\beta^{-1} \circ (h|_X)} \circ \theta^{-1}$ (lemma 2). The theorem follows from this formula.

Lemma 3. — Let u be a normal partial operator on E . We have

$$\text{Sp}(b(u)) \cap D = \beta(\text{Sp}(u)).$$

One has $b(u) = u \circ W(u)^{1/2}$ (prop. 1 of IV, p. 262). The endomorphism $W(u)^{1/2}$ is injective and its image is the domain of u (formula (1) of IV, p. 262).

Let $\lambda \in \mathbb{C}$. The number λ belongs to the resolvent set of u if and only if $(u - \lambda 1_E) \circ W(u)^{1/2}$ is a bijective linear map from E onto E (remark 3 of IV, p. 245). Now, by formula (3) of IV, p. 262, we have

$$(u - \lambda 1_E) \circ W(u)^{1/2} = b(u) - \lambda(1_E - b(u)^*b(u))^{1/2} = f_\lambda(b(u))$$

where f_λ is the continuous function defined on $\overline{D}$ by $z \mapsto z - \lambda(1 - |z|^2)^{1/2}$. The endomorphism $f_\lambda(b(u))$ is bijective if and only if its spectrum does not contain 0. Since $\text{Sp}(f_\lambda(b(u))) = f_\lambda(\text{Sp}(b(u)))$ (cor. 2 of prop. 7 of I, p. 111), this is the case if and only if 0 does not belong to the set $f_\lambda(\text{Sp}(b(u)))$. Thus, $\lambda \in \text{Sp}(u)$ if and only if there exists $z \in \text{Sp}(b(u)) \cap \overline{D}$ such that

$$z - \lambda(1 - |z|^2)^{1/2} = 0.$$

This equality, if valid, implies that $z \in D$, and means that $\lambda = \beta^{-1}(z)$. We conclude that $\text{Sp}(u) = \beta^{-1}(\text{Sp}(b(u)) \cap D)$, as stated.

Let u be a partial normal operator on E . Let $f \in \mathcal{K}(\text{Sp}(u))$. The map $z \mapsto f(\beta^{-1}(z))$ from $\text{Sp}(b(u)) \cap D$ into $\mathbb{C}$ is continuous and compactly supported; the unique mapping f_β from $\text{Sp}(b(u))$ into $\mathbb{C}$ which extends it by zero is therefore continuous.

DEFINITION 3. — For every function $f \in \mathcal{K}(\text{Sp}(u))$, we define the endomorphism $f(u)$ of E by $f(u) = f_\beta(b(u))$.

The map $f \mapsto f_\beta$ is a morphism of complex algebras from $\mathcal{K}(\text{Sp}(u))$ into $\mathcal{C}(\text{Sp}(b(u)))$, so the map $f \mapsto f(u)$ is a morphism of complex algebras from $\mathcal{K}(\text{Sp}(u))$ into $\mathcal{L}(E)$. We have $\bar{f}(u) = f(u)^*$ since $\bar{f}_\beta = \overline{f_\beta}$. If $f \geq 0$ then $f_\beta \geq 0$, hence $f(u)$ is a positive endomorphism of E .

Remark. — Suppose that u is a normal endomorphism of E , so that $\text{Sp}(u)$ is a compact subset of $\mathbb{C}$. Since the bounded transform $b(u)$ coincides with $\beta(u)$ (formula (5)), and since $\text{Sp}(b(u)) = \beta(\text{Sp}(u))$ (cor. 2 of prop. 7 of I, p. 111), the spectrum of $b(u)$ is a compact subset of D . For every continuous function $f \in \mathcal{C}(\text{Sp}(u))$, the function f_β coincides with the restriction of $f \circ \beta^{-1}$ to $\text{Sp}(u)$. Consequently, $f_\beta(\beta(u))$ coincides with the endomorphism $f(u)$ defined by the continuous functional calculus of u . The above definition is therefore compatible with that of the continuous functional calculus of the star algebra $\mathcal{L}(E)$.

Let $f \in \mathcal{K}(\text{Sp}(u))$. We have

$$\|f(u)\| \leq \|f_\beta\|_\infty = \|f\|_\infty,$$

so that, for all x and y in E , we obtain the majorization $|\langle x|f(u)y\rangle| \leq \|x\|\|y\|\|f\|_\infty$. The map $f \mapsto \langle x|f(u)y\rangle$ is therefore a bounded measure on $\text{Sp}(u)$ of total mass $\leq \|x\|\|y\|$ (INT, IV, p. 154, § 4, no. 7). If $x = y$ it is a positive measure, since $f(u)$ is positive when $f \geq 0$.

DEFINITION 4. — Let u be a normal partial operator on a complex Hilbert space E . Let x and y be in E . The bounded measure on $\text{Sp}(u)$ defined by $f \mapsto \langle x|f(u)y\rangle$ for $f \in \mathcal{K}(\text{Sp}(u))$ is called the spectral measure of (x, y) relative to u . When $x = y$, we say that it is the spectral measure of x relative to u .

Remarks. — 1) When u is continuous, this definition coincides with that of def. 2 of IV, p. 190.

2) Let j be the canonical inclusion of $\text{Sp}(u)$ into $\mathbb{C}$. Since $\text{Sp}(u)$ is closed, the map j is μ -proper for every bounded measure μ on $\text{Sp}(u)$ (INT, V, p. 68, § 6, no. 1, prop. 1). One often identifies the spectral measures of u with the measures on $\mathbb{C}$ which are their images under j (cf. INT, V, p. 84, § 7, no. 2).

The map from $E \times E$ into the Banach space $\mathcal{M}^1(\text{Sp}(u))$ that associates to (x, y) the spectral measure of (x, y) relative to u is sesquilinear and continuous with norm ≤ 1 .

Lemma 4. — Let X be a locally compact topological space, let μ be a positive measure on X and let g be a function μ -measurable on X . Let m_g be the multiplication operator by g on $L^2(X, \mu)$.

- a) The application $f \mapsto f(m_g)$ from $\mathcal{K}(\text{Sp}(m_g))$ into $\mathcal{L}(L^2(X, \mu))$ is given by $f \mapsto m_{f \circ g}$;
- b) For f_1 and f_2 in $\mathcal{L}^2(X, \mu)$ with classes $\tilde{f}_1$ and $\tilde{f}_2$ in $L^2(X, \mu)$ the spectral measure of $(\tilde{f}_1, \tilde{f}_2)$ relative to m_g is the restriction to $\text{Sp}(m_g)$ of the image measure $g(\bar{f}_1 f_2 \cdot \mu)$.

The partial operator m_g is normal and $b(m_g) = m_{\beta \circ g} = m_{g(1+|g|^2)^{-1/2}}$ (lemma 2). Moreover, $\beta(g(x))$ belongs to $\text{Sp}(m_{\beta \circ g})$ for all x outside a locally μ -negligible $Y \subset X$ (proposition 22 of IV, p. 252 and lemma 1 of IV, p. 181). Since $f_\beta(\beta(g(x))) = f(g(x))$ for all $x \in X - Y$, we deduce that $f(m_g) = m_{f \circ g}$ by definition 3 and the corollary of Proposition 6 of IV, p. 187.

Let f_1 and f_2 be in $\mathcal{L}^2(X, \mu)$ with classes $\tilde{f}_1$ and $\tilde{f}_2$ in $L^2(X, \mu)$. Since the measure $\nu = \bar{f}_1 f_2 \cdot \mu$ is bounded, the image measure $g(\nu)$ is defined (INT, V, p. 69, § 6, no. 1, remark 1). Let $f \in \mathcal{K}(\text{Sp}(m_g))$. We have $f(m_g) = m_{f \circ g}$ according to a), whence

$$\langle \tilde{f}_1 | f(m_g) \tilde{f}_2 \rangle = \int_X \bar{f}_1 (f \circ g) f_2 d\mu = \int_X (f \circ g) (\bar{f}_1 f_2 d\mu) = \int_C f d\nu$$

(INT, V, p. 69, § 6, n° 1, formula (1)), which establishes assertion b).

Example. — Let $n \in \mathbf{N}$. We equip $\mathbf{R}^n$ with Lebesgue measure, and we identify $\mathbf{R}^n$ and its dual group as in Corollary 3 of II, p. 236. Let N denote the Euclidean norm on $\mathbf{R}^n$ and let $\mathcal{F}$ be the Fourier transform on $\mathcal{S}(\mathbf{R}^n)$ and on $\mathcal{S}'(\mathbf{R}^n)$ (no. 12 of IV, p. 217).

Let $\Delta_{\mathcal{S}}$ be the partial operator on $L^2(\mathbf{R}^n)$ whose domain is the space $\mathcal{S}(\mathbf{R}^n)$ and which satisfies

$$\Delta_{\mathcal{S}}(\varphi) = - \sum_{i=1}^n \partial_i^2 \varphi$$

for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$. We then have $\mathcal{F}(\Delta_{\mathcal{S}}(\varphi)) = 4\pi^2 N^2 \mathcal{F}(\varphi)$ for every function $\varphi \in \mathcal{S}(\mathbf{R}^n)$ (remark 12 of IV, p. 220). Since the Fourier transform is an automorphism of $\mathcal{S}(\mathbf{R}^n)$ (prop. 18 of IV, p. 219), this means that the partial operator $u = \mathcal{F} \circ \Delta_{\mathcal{S}} \circ \mathcal{F}^{-1}$ is the reduction of the multiplication operator $m_{4\pi^2 N^2}$ on the space $\mathcal{S}(\mathbf{R}^n)$. The partial operator u is closable and symmetric; its closure is the positive self-adjoint partial operator $m_{4\pi^2 N^2}$ (prop. 6 of IV, p. 232, applied to the space $\mathcal{D}(\mathbf{R}^n) \subset \mathcal{S}(\mathbf{R}^n)$ using prop. 4 of IV, p. 202). Consequently, the partial operator $\Delta_{\mathcal{S}}$ is essentially self-adjoint. We denote Δ its closure; it is a positive self-adjoint operator, and it is the unique Laplacian on $\mathbf{R}^n$ (cf. example from No. 6 of IV, p. 243 and corollary of Prop. 26 of IV, p. 257).

By prop. 21 of IV, p. 223, the Sobolev space $H^2(\mathbf{R}^n)$ is the set of $f \in L^2(\mathbf{R}^n)$ such that $(1 + N^2)\mathcal{F}(f)$ belongs to $L^2(\mathbf{R}^n)$, that is, such that $\mathcal{F}(f)$ belongs to the domain of $m_{4\pi^2 N^2}$. Since the Fourier transform is an isometric isomorphism of the space $L^2(\mathbf{R}^n)$ onto itself (th. 1 of II, p. 215), the domain of the Laplacian Δ is $H^2(\mathbf{R}^n)$. We have $\Delta = \mathcal{F}^{-1} \circ m_{4\pi^2 N^2} \circ \mathcal{F}$. In particular, the spectrum of Δ is equal to $\mathbf{R}_+$.

IV.5.3 Universally measurable functional calculus

In this section the Hilbert spaces considered are complex.

Let u be a partial normal operator on a Hilbert space E . We denote by $\mu_{x,y}$ (resp. μ_y) the

spectral measure of $(x, y) \in E \times E$ (resp. of y) relative to u .

Let $y \in E$. The map $\mathcal{H}(\text{Sp}(u)) \rightarrow E$ defined by $f \mapsto f(u)(y)$ satisfies

$$\|f(u)(y)\|^2 = \int_{\text{Sp}(u)} |f|^2 \mu_y = \|f\|_{\mathcal{L}^2(\text{Sp}(u), \mu_y)}^2.$$

There therefore exists a unique isometric linear map ev_y from the space $L^2(\text{Sp}(u), \mu_y)$ into E such that $\text{ev}_y(\tilde{f}) = f(u)(y)$ if $\tilde{f}$ is the class of a function $f \in \mathcal{H}(\text{Sp}(u))$.

Let f be a universally measurable function defined on the spectrum of u . We denote by D_f the set of elements $y \in E$ such that f belongs to $\mathcal{L}^2(\text{Sp}(u), \mu_y)$.

PROPOSITION 3. — Let f be a universally measurable function defined on the spectrum of u . The set D_f is a dense vector subspace of E . The map $y \mapsto \text{ev}_y(f)$ is a partial normal operator on E whose domain is D_f , and which is denoted $f(u)$.

For every $x \in E$ and every $y \in D_f$, we have $f \in \mathcal{L}^1(\text{Sp}(u), \mu_{x,y})$ and

$$\langle x | f(u)y \rangle = \int_{\text{Sp}(u)} f \mu_{x,y}. \quad (6)$$

We shall prove the following more precise statement:

PROPOSITION 4. — Let f be a universally measurable function on the spectrum of u .

- a) The set D_f is a dense vector subspace of E . The map $f(u) : y \mapsto \text{ev}_y(f)$ from D_f into E is linear and coincides with 1_E if $f = 1$;
- b) Let $g \in \mathcal{L}_u(\text{Sp}(u))$. For every $y \in D_f$ and every $x \in D_g$, we have $fg \in \mathcal{L}^1(\text{Sp}(u), \mu_{x,y})$, and

$$\langle g(u)x | f(u)y \rangle = \int_{\text{Sp}(u)} \bar{g}f \mu_{x,y}. \quad (7)$$

- c) Suppose that $E = L^2(X, \mu)$ where X is a locally compact topological space and μ is a positive measure on X , and that $u = m_h$, where $h : X \rightarrow \mathbf{C}$ is μ -measurable. We have $f(m_h) = m_{f \circ h}$.

By Theorem 1 of IV, p. 266, we may assume that we are in the situation of assertion c), namely, $E = L^2(X, \mu)$ and $u = m_h$, where X is a locally compact topological space, μ is a

positive measure on X and $h: X \rightarrow \mathbf{C}$ is μ -measurable. We shall denote by $\tilde{\varphi}$ the class in $L^2(X, \mu)$ of a function $\varphi \in \mathcal{L}^2(X, \mu)$.

Let $S = \text{Sp}(m_h)$. We denote by $\mu_{\tilde{\varphi}_1, \tilde{\varphi}_2}$ (resp. $\mu_{\tilde{\varphi}}$) the spectral measure of $(\tilde{\varphi}_1, \tilde{\varphi}_2)$ (resp. of $\tilde{\varphi}$) relative to m_h for all $(\varphi_1, \varphi_2) \in \mathcal{L}^2(X, \mu)^2$ (resp. every $\varphi \in \mathcal{L}^2(X, \mu)$).

Let $\varphi \in \mathcal{L}^2(X, \mu)$. The spectral measure $\mu_{\tilde{\varphi}}$ is equal to the image measure $h(|\varphi|^2 \cdot \mu)$ on S (Lemma 4 of IV, p. 269). Since the function f is μ_{φ} -measurable, we have $\varphi \in D_f$ if and only if the integrals

$$\int_S^* |f|^2 d\mu_{\tilde{\varphi}} = \int_S^* |f|^2 h(|\varphi|^2) d\mu = \int_X^* |f \circ h|^2 |\varphi|^2 d\mu$$

are finite (INT, V, p. 70, § 6, no. 2, prop. 2). This means that D_f is the domain of the multiplication operator $m_{f \circ h}$, which is a dense subspace of E (prop. 5 of IV, p. 232).

The restriction of $\text{ev}_{\tilde{\varphi}}$ to the function classes $g \in \mathcal{H}(S)$ is the map which associates to a class $\tilde{g}$ the element $g(m_h)(\tilde{\varphi}) = m_{g \circ h}(\tilde{\varphi})$ (Lemma 4 of IV, p. 269). The map $\text{ev}_{\tilde{\varphi}}$ therefore coincides with the isometric map from $L^2(S, \mu_{\tilde{\varphi}})$ into $L^2(X, \mu)$ obtained by passing to quotients from the map $g \mapsto (g \circ h) \cdot \varphi$ from $\mathcal{L}^2(S, \mu_{\tilde{\varphi}})$ into $\mathcal{L}^2(X, \mu)$.

In particular, we therefore have $\text{ev}_{\tilde{\varphi}}(f) = m_{f \circ h}(\tilde{\varphi})$, so the map $f(m_h)$ from D_f into E coincides with the partial operator $m_{f \circ h}$. This proves assertion c); if $f=1$, we find $\text{ev}_{\tilde{\varphi}}(1) = \tilde{\varphi}$, which also completes the proof of a).

Let us prove assertion b). Let φ_1 and φ_2 be functions such that $\tilde{\varphi}_1 \in D_f$ and $\tilde{\varphi}_2 \in D_g$. It follows from (INT, V, loc. cit.)

$$\begin{aligned} \int_S^* |fg| d\mu_{\tilde{\varphi}_1, \tilde{\varphi}_2} &= \int_X^* |(f \circ h)(g \circ h) \varphi_1 \varphi_2| d\mu \\ &\leq \| (f \circ h) \varphi_1 \| \| (g \circ h) \varphi_2 \| \end{aligned}$$

which is finite, hence $fg \in \mathcal{L}^1(S, \mu_{\tilde{\varphi}_1, \tilde{\varphi}_2})$. Moreover, we then have

$$\begin{aligned} \langle g(m_h)(\tilde{\varphi}_2) | f(m_h)(\tilde{\varphi}_1) \rangle &= \int_X \overline{(g \circ h) \varphi_2} (f \circ h) \varphi_1 d\mu \\ &= \int_S \overline{\tilde{g}} \tilde{f} d\mu_{\tilde{\varphi}_1, \tilde{\varphi}_2}, \end{aligned}$$

which concludes the proof.

DEFINITION 5. — The map $f \mapsto f(u)$ from $\mathcal{L}_u(\text{Sp}(u))$ into the set of normal partial

operators on E defined by prop. 3 of IV, p. 271 is called the universally measurable functional calculus mapping associated with u .

More generally, let T be a set and let $g: \text{Sp}(u) \times T \rightarrow \mathbf{C}$ be a map; let $t \in T$ be such that the function $g_t: z \mapsto g(z, t)$ is universally measurable on $\text{Sp}(u)$. We then denote by $g(u, t) = g_t(u)$.

Formula (7) implies in particular that

$$\|f(u)(y)\|^2 = \int_{\text{Sp}(u)} |f|^2 \mu_y \quad (8)$$

for all $f \in \mathcal{L}_u(\text{Sp}(u))$ and every $y \in D_f$. From this, taking $f = 1$ we deduce that μ_y is a positive measure of total mass $\|y\|^2$.

If u is a normal partial operator on a Hilbert space E and $f \in \mathcal{K}(\text{Sp}(u))$, then $D_f = E$ and the partial operator $f(u)$ is then continuous, since

$$\|f(u)y\| \leq \|f\|_\infty \|y\|$$

for all $y \in E$ according to (8). The endomorphism $f(u)$ coincides with that of Definition 3 of IV, p. 268.

COROLLARY 1. — a) For every $f \in \mathcal{L}_u(\text{Sp}(u))$, the partial operator $f(u)$ is normal and $f(u)^* = \bar{f}(u)$. Moreover $f(u)$ is positive if $f \geq 0$, and self-adjoint if f takes real values;

b) Let $k \in \mathbf{N}$ and $f(z) = z^k$ for $z \in \text{Sp}(u)$. We have $f(u) = u^k$;

c) Let $\lambda \in \mathbf{C} - \text{Sp}(u)$ and $f(z) = (\lambda - z)^{-1}$ for $z \in \text{Sp}(u)$. We have $f(u) = R(u, \lambda)$;

d) One has $\beta(u) = b(u)$;

e) Let $f \in \mathcal{L}_u(\text{Sp}(u))$ and $g \in \mathcal{L}_u(\text{Sp}(u))$ such that $|g| \leq 1 + |f|$. The domain of $g(u)$ is a core for $f(u)$.

In view of the spectral theorem (th. 1 of IV, p. 266), this follows from the preceding proposition combined, respectively, with:

a) lemma 2, a) of IV, p. 266, prop. 23 of IV, p. 253, and its corollary;

- b) prop. 24, b) of IV, p. 255;
- c) Prop. 22, b) of IV, p. 252;
- d) Lemma 2, a) of IV, p. 266;
- e) Prop. 6, b) of IV, p. 232.

Note. — For $f = \text{Id}_{\text{Sp}(u)}$, we have $f(u) = u$ (assertion b) for $k = 1$). The domain of u therefore coincides with the set of $x \in E$ such that the identity map of $\text{Sp}(u)$ belongs to $\mathcal{L}^2(\text{Sp}(u), \mu_x)$; it contains in particular the elements $x \in E$ such that the measure μ_x is compactly supported.

The following corollary generalizes the corollary of Prop. 16 of IV, p. 247 and Prop. 17 of IV, p. 248.

COROLLARY 2. — Let u be a partial normal operator on the Hilbert space E . Assume that E is nonzero.

- a) The spectrum of u is nonempty;
- b) For every $\lambda \in \mathbb{C} - \text{Sp}(u)$, the norm of the resolvent $R(u, \lambda)$ is equal to $1/\delta$, where $\delta > 0$ is the distance in $\mathbb{C}$ from λ to the spectrum of u .
- c) For every $\varepsilon > 0$, the ε -pseudo-spectrum $\text{PSp}_\varepsilon(u)$ is the set of $\lambda \in \mathbb{C}$ at distance $< \varepsilon$ from $\text{Sp}(u)$.

Suppose that the spectrum of u is empty. Then u is injective, and the endomorphism $u^{-1} = -R(u, 0)$ is normal (cor. 1, c) and a)). Since $\text{Sp}(u^{-1}) \subset \{0\}$ (prop. 15, a)), we have $\text{Sp}(u^{-1}) = \{0\}$ (I, p. 26, cor. 1). Since u^{-1} is normal, this implies that $u^{-1} = 0$ (I, p. 110, example 1), which is a contradiction since E is not zero.

To prove b), one may assume that u is the multiplication operator m_g on $L^2(X, \mu)$, where g is a continuous function on a locally compact topological space X equipped with a positive measure μ (th. 1 of IV, p. 266). Let $\lambda \in \mathbb{C} - \text{Sp}(u)$ and let $\delta > 0$ be the distance from λ to the spectrum of u . To prove that $\|R(u, \lambda)\| = \delta^{-1}$, we reduce to the case $\lambda = 0$ by replacing u by $u - \lambda 1_E$. The real number δ is then the distance from 0 to the spectrum of m_g . Since the latter

coincides with the μ -essential image of g , the result is a consequence of the lemma 3 of IV, p. 182.

Finally, the assertion c) follows from b) and the definition of $\text{PSp}_\varepsilon(u)$ (IV, p. 250, def. 9).

COROLLARY 3. — Let u be a partial normal operator on a Hilbert space E . Let $f \in \mathcal{L}_u(\text{Sp}(u))$. For all x and y in E , the spectral measure of (x, y) relative to $f(u)$ is the image measure $f(\mu)$, where μ is the spectral measure of (x, y) relative to u .

In view of the spectral theorem (th. 1 of IV, p. 266), one may suppose that u is the multiplication operator m_g on $L^2(X, \mu)$, where X is a locally compact topological space, μ is a positive measure on X and $g \in \mathcal{C}(X)$. Let f_1 and f_2 be in $\mathcal{L}^2(X, \mu)$ with classes $\tilde{f}_1$ and $\tilde{f}_2$ in $L^2(X, \mu)$. Since $f(m_g) = m_{f \circ g}$ (prop. 4, c)), the spectral measure of $(\tilde{f}_1, \tilde{f}_2)$ relative to $f(m_g)$ is the image measure $(f \circ g)(\tilde{f}_1 \tilde{f}_2 \cdot \mu)$ (lemma 4, b) of IV, p. 269). This measure is equal to the image measure $f(g(\tilde{f}_1 \tilde{f}_2 \cdot \mu))$ (INT, V, p. 72, § 6, n° 4, prop. 4, a)), whence the assertion (lemma 4, b) of IV, p. 269).

COROLLARY 4. — Let $g \in \mathcal{L}_u(\text{Sp}(f(u)))$. We have $g(f(u)) = (g \circ f)(u)$.

One has $g \circ f \in \mathcal{L}_u(\text{Sp}(u))$ (lemma 5 of IV, p. 184). For all x and y in E , denote by $\mu'_{x,y}$ the spectral measure of (x, y) relative to $f(u)$. By the preceding corollary and INT, V, p. 71, § 6, n°2, th. 1, one has $g \in \mathcal{L}^2(\text{Sp}(f(u)), \mu'_{x,y})$ if and only if $g \circ f \in \mathcal{L}^2(\text{Sp}(u), \mu_{x,y})$, so the domain of $g(f(u))$ is equal to the domain of $(g \circ f)(u)$. For all $x \in E$ and $y \in \text{dom}(g(f(u)))$, one then has the formula

$$\begin{aligned} \langle x | g(f(u))y \rangle &= \int_{\text{Sp}(f(u))} g \mu'_{x,y} \\ &= \int_{\text{Sp}(f(u))} g f(\mu_{x,y}) = \langle x | (g \circ f)y \rangle \end{aligned}$$

(loc. cit.), whence $g(f(u)) = (g \circ f)(u)$.

PROPOSITION 5. — Let u be a normal partial operator on a Hilbert space E .

- a) If $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$, then $f(u) \in \mathcal{L}(E)$;
- b) If $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$ and $g \in \mathcal{L}_u(\text{Sp}(u))$, then $f(u) \circ g(u) \subset (fg)(u)$;
- c) The map $f \mapsto f(u)$ from $\mathcal{L}_u^\infty(\text{Sp}(u))$ into $\mathcal{L}(E)$ is a continuous unital morphism of star algebras. In particular, one has $\|f(u)\| \leq \|f\|_\infty$ for $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$;

d) If $u \in \mathcal{L}(E)$, then for all $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$, the endomorphism $f(u)$ belongs to the bicommutant of u in $\mathcal{L}(E)$.

Let $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$. We have $D_f = E$ and $f(u)$ is a continuous endomorphism of E by formula (8), p. 272, whence the assertion *a*).

Let $g \in \mathcal{L}_u(\text{Sp}(u))$ Let $y \in D_g$. We have $y \in D_{fg}$ and, for all $x \in E$, it follows that $\langle \bar{f}(u)x | g(u)y \rangle = \langle x | (fg)(u)y \rangle$ by formula (7), p. 271, whence $f(u)(g(u)y) = (fg)(u)(y)$, which proves *b* .

By *a*) and *b*), the map $f \mapsto f(u)$ of $\mathcal{L}_u^\infty(\text{Sp}(u))$ in $\mathcal{L}(E)$ is a unital morphism of involutive algebras; consequently, it is a continuous morphism of norm ≤ 1 (I, p. 104, prop. 2), whence the assertion *c*).

Suppose that u is an endomorphism of E . Let $v \in \mathcal{L}(E)$ commuting with u . We then have $v \circ f(u) = f(u) \circ v$ for $f \in \mathcal{C}(\text{Sp}(u))$ by the properties of the continuous functional calculus (I, p. 110, remark). Let x and y be in E . The formulas

$$\langle x | (v \circ f(u))y \rangle = \langle x | (f(u) \circ v)y \rangle, \quad (9)$$

valid for every function $f \in \mathcal{C}(\text{Sp}(u))$, mean that the spectral measures of $(v^*(x), y)$ and of $(x, v(y))$ relative to u are equal. This equality implies, by formula (6), p. 271, that formula (9) is valid for all $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$. We therefore have $v \circ f(u) = f(u) \circ v$.

COROLLARY. — Let f and g in $\mathcal{L}_u(\text{Sp}(u))$ and $(x, y) \in D_f \times D_g$. The spectral measure of $(f(u)x, g(u)y)$ relative to u is the measure $\bar{f}g \cdot \mu_{x,y}$, where $\mu_{x,y}$ is the spectral measure of (x, y) relative to u .

Denote by ν the spectral measure of $(f(u)x, g(u)y)$ relative to u . For every function $\varphi \in \mathcal{H}(\text{Sp}(u))$, we have

$$\begin{aligned} \int_{\text{Sp}(u)} \varphi \nu &= \langle f(u)x | \varphi(u)(g(u)y) \rangle \\ &= \langle f(u)x | (\varphi g)(u)y \rangle = \int_{\text{Sp}(u)} \bar{f} \varphi g \mu_{x,y}, \end{aligned}$$

(prop. 5, b)) whence $\nu = \bar{f}g \cdot \mu_{x,y}$.

PROPOSITION 6. — Let u be a normal partial operator on a Hilbert space E . Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{L}_u(\text{Sp}(u))$ which converges pointwise to $f \in \mathcal{L}_u(\text{Sp}(u))$ and such that there

exists $g \in \mathcal{L}_u(\text{Sp}(u))$ satisfying $|f_n| \leq g$ for all $n \in \mathbb{N}$. We then have $\text{dom}(g(u)) \subset \text{dom}(f_n(u))$ for all $n \in \mathbb{N}$ and $\text{dom}(g(u)) \subset \text{dom}(f(u))$. Moreover, for every element y of the domain of $g(u)$, we have

$$f(u)(y) = \lim_{n \rightarrow +\infty} f_n(u)(y).$$

In particular, if $f_n \in \mathcal{L}_u^\infty(\text{Sp}(u))$ for all $n \in \mathbb{N}$ and if the functions f_n are uniformly bounded, then $f_n(u)$ converges to $f(u)$ in the space $\mathcal{L}(E)$ endowed with the topology of pointwise convergence.

Denote by $\mu_{x,y}$ (resp. μ_x) the spectral measure of (x, y) (resp. of x) relative to u .

Let $y \in \text{dom}(g(u))$, so that $g \in \mathcal{L}^2(\text{Sp}(u), \mu_y)$. The condition $|f_n| \leq g$ implies that $f_n \in \mathcal{L}^2(\text{Sp}(u), \mu_y)$, hence $y \in \text{dom}(f_n(u))$.

Since (f_n) converges simply to f and that $|f_n| \leq g$, we have $f \in \mathcal{L}^2(\text{Sp}(u), \mu_y)$ by the Lebesgue theorem (INT, IV, p. 137, § 3, n° 7, th. 6), hence $y \in \text{dom}(f(u))$. Moreover, the sequence (f_n) converges to f in $\mathcal{L}^2(\text{Sp}(u), \mu_y)$, so the norm of $f_n(u)y$ converges to the norm of $f(u)y$.

Let $x \in E$. The functions f and g , as well as the functions f_n for all $n \in \mathbb{N}$, belong to $\mathcal{L}^1(\text{Sp}(u), \mu_{x,y})$ (prop. 3). By the Lebesgue theorem (INT, IV, loc. cit.), the sequence (f_n) converges to f in $\mathcal{L}^1(\text{Sp}(u), \mu_{x,y})$, whence

$$\langle x | f_n(u)y \rangle = \int_{\text{Sp}(u)} f_n \mu_{x,y} \rightarrow \int_{\text{Sp}(u)} f \mu_{x,y} = \langle x | f(u)y \rangle.$$

One concludes that $f_n(u)(y)$ converges to $f(u)(y)$ (EVT, V, p. 17, prop. 10).

PROPOSITION 7. — Let X be a locally compact topological space and let ν be a measure on X . Let $g: C \times X \rightarrow C$ be a continuous function with compact support. For $z \in C$, set

$$h(z) = \int_X g(z, x) d\nu(x).$$

- a) The map h from C to C is continuous and bounded;
- b) The map from X to $\mathcal{L}(E)$ defined by $x \mapsto g(u, x)$ is ν -integrable and we have

$$h(u) = \int_X g(u, x) d\nu(x).$$

The function h is bounded because g is continuous with compact support, and it is continuous by INT, IV, p. 144, § 4, n° 3, cor. 1. Since g is continuous and has compact support, the map $x \mapsto g(u, x)$ is continuous from X into $\mathcal{L}(E)$. This map has compact support, hence is bounded and integrable on X with respect to ν . Denote by

$$v = \int_X g(u, x) d\nu(x) \in \mathcal{L}(E).$$

Let y and z be elements of E , and let μ be the spectral measure of (y, z) relative to u . We have

$$\langle y | v(z) \rangle = \int_X \langle y | g(u, x) z \rangle d\nu(x) = \int_X \left(\int_{\text{Sp}(u)} g(\lambda, x) d\mu(\lambda) \right) d\nu(x)$$

(formula (6), p. 271). Since $g \in \mathcal{H}(\mathbb{C} \times X)$, it follows that

$$\langle y | v(z) \rangle = \int_{\text{Sp}(u)} \left(\int_X g(\lambda, x) d\nu(x) \right) d\mu(\lambda) = \langle y | h(u) z \rangle$$

by INT, III, p. 84, § 4, n° 1, th. 2 and formula (6), p. 271. This proves that $v = h(u)$, as desired.

IV.5.4 Spectral projectors

Let u be a normal partial operator on a complex Hilbert space E . Let A be a universally measurable subset of $\text{Sp}(u)$ and φ_A its characteristic function. Since φ_A is bounded and satisfies $\varphi_A^2 = \varphi_A$, the endomorphism $\varphi_A(u)$ of E is an orthogonal projector of E . It is called the spectral projector of u defined by A . We denote $p_A = \varphi_A(u)$. We have $p_\emptyset = 0$ and $p_{\text{Sp}(u)} = 1_E$.

Let A be a universally measurable subset of $\mathbb{C}$. The spectral projector of u defined by A is the spectral projector $p_{\text{Sp}(u) \cap A}$. It is also denoted simply p_A . For every function $f \in \mathcal{L}_u(\text{Sp}(u))$, for all $x \in E$ and every $y \in \text{dom}(f(u))$, we have the formula

$$\langle p_A(x) | f(u)y \rangle = \int_{\text{Sp}(u) \cap A} f d\mu, \quad (10)$$

where μ is the spectral measure of (x, y) relative to u (formula (7), p. 271). Let A and B be universally measurable subsets of $\text{Sp}(u)$. Since $\varphi_A \varphi_B = \varphi_{A \cap B}$, we have $p_A \circ p_B = p_{A \cap B}$ (prop. 5 of IV, p. 275, c)). In particular, if A and B are disjoint, the images of p_A and of p_B are orthogonal.

PROPOSITION 8. — Let $(A_i)_{i \in \mathbb{I}}$ be a countable family of pairwise disjoint universally measurable subsets of $\text{Sp}(u)$ and let p_i be the spectral projector of u defined by A_i . The union A of the sets A_i is a universally measurable subset of $\text{Sp}(u)$ and the series $\sum_i p_i$ converges to p_A in $\mathcal{L}(E)$ endowed with the topology of pointwise convergence.

The set A is universally measurable by INT, IV, p. 177, § 5, n° 4, cor. 2. The series $\sum_i \varphi_{A_i}$ converges simply to φ_A and its partial sums are bounded by 1. The assertion therefore follows from prop. 6 of IV, p. 276.

PROPOSITION 9. — Let A be a closed subset of $\text{Sp}(u)$. Let φ_A be the characteristic function of A and $p_A = \varphi_A(u)$ the spectral projector of u defined by A . Denote by E_A the image of p_A . This is a closed subspace of E .

- a) The subspace E_A is the set of $x \in E$ such that the support of the spectral measure of x relative to u is contained in A ;
- b) If A is bounded in $\mathbb{C}$, then E_A is contained in the domain of u ;
- c) For all x belonging to the domain of u , we have $p_A(x) \in \text{dom}(u)$ and $u(p_A(x)) \in E_A$ in particular $u(x) \in E_A$ if $x \in \text{dom}(u) \cap E_A$.

For x in E , we denote by μ_x the spectral measure of x relative to u .

Let us prove a). Let x be an element of E_A . Let $z \in \mathbb{C} - A$ and let U be a relatively compact open neighborhood of z which does not meet A . For every function $f \in \mathcal{K}(\mathbb{C})$ with support contained in U , we have

$$\int_{\text{Sp}(u)} f d\mu_x = \langle x | f(u)x \rangle = \langle p_A(x) | f(u)x \rangle = \int_{\text{Sp}(u) \cap A} f d\mu_x = 0$$

by formula (10). This means that z does not belong to the support of μ_x . Consequently, the support of μ_x is contained in A .

Conversely, let $x \in E$ such that the support of μ_x is contained in A . It follows that

$$\langle x | p_A(x) \rangle = \int_{\text{Sp}(u) \cap A} \mu_x = \int_{\text{Sp}(u)} \mu_x = \langle x | x \rangle,$$

(loc. cit.) therefore $\|p_A(x) - x\|^2 = \|p_A(x)\|^2 - \|x\|^2 \leq 0$ and consequently $p_A(x) = x$, that is, $x \in E_A$.

Assertion b) follows from a) and the remark in IV, p. 273.

Let us prove c). The domain of u is the set of $x \in E$ such that the identity function of $\text{Sp}(u)$ belongs to $\mathcal{L}^2(\text{Sp}(u), \mu_x)$ (loc. cit.). Since $\mu_{p_A(x)} = \varphi_A \cdot \mu_x$ for $x \in E$ (corollary of Prop. 5 of IV, p. 275), we have $p_A(x) \in \text{dom}(u)$ if $x \in \text{dom}(u)$.

Let $x \in \text{dom}(u)$ and $y = p_A(x)$; we have $y \in \text{dom}(u) \cap E_A$ according to c). The spectral measure of $u(y)$ relative to u is $|\text{Id}_{\text{Sp}(u)}|^2 \cdot \mu_y$ (loc. cit.). Since μ_y is supported in A , the same holds for $\mu_{u(y)}$, hence $u(y) \in E_A$ according to a).

COROLLARY. — Let $\lambda \in \text{Sp}(u)$. The image of $p_{\{\lambda\}}$ is the eigensubspace of u relative to λ .

Let $x \in \text{dom}(u)$. Denote by μ (resp. ν) the spectral measure of the element x relative to u (resp. the spectral measure of $(x, u(x))$ relative to u). We have $\nu = \text{Id}_{\text{Sp}(u)} \cdot \mu$ (corollary of Prop. 5 of IV, p. 275). If $u(x) = \lambda x$, we also have $\nu = \lambda \mu$, whence the equality $\text{Id}_{\text{Sp}(u)} \cdot \mu = \lambda \mu$. This implies that the support of μ is contained in $\{\lambda\}$ (cf. INT, V, p. 46, § 5, no. 3, cor. 2) and therefore x belongs to the image of $p_{\{\lambda\}}$ (prop. 9, a)).

Conversely, suppose that x belongs to the image E_λ of $p_{\{\lambda\}}$. Then x belongs to $\text{dom}(u)$ and $u(x)$ also belongs to E_λ (loc. cit., b)). Since $\varphi_{\{\lambda\}} \cdot (\text{Id}_{\text{Sp}(u)} - \lambda) = 0$, we have the relation $p_{\{\lambda\}} \circ (u - \lambda 1_E) \subset 0$ (prop. 5 of IV, p. 275, c)), whence $0 = p_{\{\lambda\}}(u(x)) - \lambda p_{\{\lambda\}}(x) = u(x) - \lambda x$.

PROPOSITION 10. — Let $\lambda \in \mathbb{C}$. We have $\lambda \in \text{Sp}(u)$ if and only if, for every open neighborhood V of λ in $\mathbb{C}$, the spectral projector p_V of u relative to V is nonzero.

If $\lambda \notin \text{Sp}(u)$, then there exists an open neighborhood V of λ in $\mathbb{C}$ which does not meet $\text{Sp}(u)$, and then $p_V = p_\emptyset = 0$.

Conversely, suppose that there exists an open neighborhood V of λ in $\mathbf{C}$ such that $p_V = 0$. Let $c > 0$ be such that the disk with center λ and radius c is contained in V .

Let $x \in \text{dom}(u)$ and let μ_x the spectral measure of x relative to u . Since $\mu_x(V) = \langle x | p_V(x) \rangle = 0$, one computes

$$\begin{aligned} \int_{\mathbf{C}} |z - \lambda|^2 d\mu_x(z) &= \int_{\mathbf{C}-V} |z - \lambda|^2 d\mu_x(z) \\ &\geq c^2 \int_{\mathbf{C}-V} d\mu_x(z) = c^2 \int_{\mathbf{C}} d\mu_x(z) = c^2 \|x\|^2. \end{aligned}$$

But, on the other hand, we have

$$\|u(x) - \lambda x\|^2 = \int_{\mathbf{C}} |z - \lambda|^2 d\mu_x(z) = \|u^*(x) - \bar{\lambda}x\|^2$$

(formula (8), p. 272), whence $\|u(x) - \lambda x\| \geq c\|x\|$ and $\|u^*(x) - \bar{\lambda}x\| \geq c\|x\|$. It follows that λ belongs to the resolvent set of u (lemma 5 of IV, p. 248). This concludes the proof.

COROLLARY. — Let A be an open set in $\mathbf{C}$ and $n \in \mathbf{N}$. If A contains n elements of $\text{Sp}(u)$, then the dimension of the image of the spectral projector p_A of u relative to A is at least equal to n .

Let $\lambda_1, \dots, \lambda_n$ be distinct elements of the spectrum of u belonging to A . There exists a family $(V_i)_{1 \leq i \leq n}$ of pairwise disjoint open sets of $\mathbf{C}$ such that $\lambda_i \in V_i$ for $1 \leq i \leq n$. Let B be the union of the sets V_i . The image of p_A contains the image of the spectral projector p_B ; moreover, since p_B is the sum of the projectors p_{V_i} , and since the image of p_{V_i} is orthogonal to the image of p_{V_j} for all $i \neq j$, the result follows from Prop. 10.

IV.5.5 The Helffer–Sjöstrand formula

In this issue, E denotes a complex Hilbert space. We shall obtain a formula for certain cases of the functional calculus of a partial self-adjoint operator, expressed directly in terms of the resolvent of the operator in question.

We endow $\mathbf{R}$ (resp. $\mathbf{C}$) with Lebesgue measure, denoted μ and we identify the group $\mathbf{R}$ with its dual group by the map $(x, y) \mapsto \exp(2i\pi xy)$ (cf. Corollary 3 of II, p. 236).

For any function f defined and differentiable on an open set U of $\mathbf{R}^2$, identified with $\mathbf{C}$ with real coordinates x and y , we denote

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right)$$

(cf. VAR, R2, 8.8.10, p. 24).

Lemma 5. — Let $f \in \mathcal{D}(\mathbf{R})$. There exists a function $\tilde{f}$ in $\mathcal{D}(\mathbf{C})$ which coincides with f on $\mathbf{R}$ and satisfies

$$\frac{\partial \tilde{f}}{\partial \bar{z}}(x, 0) = 0 \quad (11)$$

for all $x \in \mathbf{R}$. We then have

$$\frac{\partial \tilde{f}}{\partial y}(x, 0) = if'(x) \quad (12)$$

for all $x \in \mathbf{R}$ and there exists a real number $C \geq 0$ such that

$$\left| \frac{\partial \tilde{f}}{\partial \bar{z}}(x, y) \right| \leq C|y| \quad (13)$$

for all $(x, y) \in \mathbf{R}^2$

There exists $\varphi \in \mathcal{D}(\mathbf{R})$ whose support is contained in $[-2, 2]$ and which is equal to 1 on $[-1, 1]$ (lemma 1 of IV, p. 196). Set

$$\tilde{f}(x, y) = (f(x) + iyf'(x))\varphi(y)$$

for $(x, y) \in \mathbf{R}^2$. We have $\tilde{f} \in \mathcal{D}(\mathbf{C})$ and $\tilde{f}$ coincides with f on $\mathbf{R}$. Moreover, for any $(x, y) \in \mathbf{R}^2$, it follows that

$$\frac{\partial \tilde{f}}{\partial \bar{z}}(x, y) = \frac{1}{2} \left((if(x) - yf'(x))\varphi'(y) + iyf''(x)\varphi(y) \right).$$

As the function φ is equal to 1 in a neighborhood of 0, we have $\varphi'(0) = 0$ whence (11).

Let $\tilde{f}$ be in $\mathcal{D}(\mathbf{C})$ satisfying (11). Formula (12) follows from this, and the bound (13) is obtained using the mean value theorem (FVR, I, p. 23, th. 2).

We say that $\tilde{f}$ is an almost analytic extension of f .

Lemma 6. — Let $\varepsilon > 0$. The function σ_ε defined on $\mathbf{R}$ by

$$\sigma_\varepsilon(x) = \frac{2i\varepsilon x}{x^2 + \varepsilon^2}$$

belongs to $L^2(\mathbf{R})$ Its Fourier transform is the class in $L^2(\mathbf{R})$ of the function η_ε which vanishes at 0 and satisfies

$$\eta_\varepsilon(y) = \frac{2\pi\varepsilon y}{|y|} e^{-2\pi\varepsilon|y|}$$

for all $y \neq 0$.

One has $\sigma_\varepsilon \in L^2(\mathbf{R})$ by prop. 3 of IV, p. 199, and the class of the function η_ε belongs to $L^2(\mathbf{R}) \cap L^1(\mathbf{R})$. For every $x \in \mathbf{R}$, we have

$$\begin{aligned} \overline{\mathcal{F}}(\eta_\varepsilon)(x) &= 2\pi\varepsilon \int_{\mathbf{R}_+} e^{2\pi(ix-\varepsilon)y} dy - 2\pi\varepsilon \int_{\mathbf{R}_-} e^{2\pi(ix+\varepsilon)y} dy \\ &= \frac{\varepsilon}{\varepsilon-ix} - \frac{\varepsilon}{\varepsilon+ix} = \frac{2i\varepsilon x}{x^2 + \varepsilon^2}, \end{aligned}$$

whence $\overline{\mathcal{F}}(\eta_\varepsilon) = \sigma_\varepsilon$ The result then follows from the Fourier inversion formula in $L^2(\mathbf{R})$ (corollary of Theorem 2 of II, p. 220).

Lemma 7. — Let $f \in \mathcal{D}(\mathbf{R})$ and let $\tilde{f} \in \mathcal{D}(\mathbf{C})$ be an almost analytic extension of f . For $\varepsilon > 0$, define f_ε on $\mathbf{R}$ by

$$f_\varepsilon(x) = -\frac{1}{2i\pi} \int_{\mathbf{R}} \left(\frac{\tilde{f}(y+i\varepsilon)}{y-x+i\varepsilon} - \frac{\tilde{f}(y-i\varepsilon)}{y-x-i\varepsilon} \right) dy.$$

Then f_ε is continuous and bounded, and f_ε converges to f in $\mathcal{C}_b(\mathbf{R})$ as ε tends to 0.

The continuity of f_ε is a consequence of INT, IV, p. 144, § 4, n° 3, cor. 1, since $\tilde{f}$ is compactly supported. Moreover, if r is such that the support of $\tilde{f}$ is contained in $[-r, r] \times \mathbf{R}$, we have

$$|f_\varepsilon(x)| \leq 2\|\tilde{f}\|_\infty \int_{-r}^r \frac{1}{\sqrt{(x-y)^2 + \varepsilon^2}} dy \leq \frac{4r}{\varepsilon} \|\tilde{f}\|_\infty,$$

hence f_ε is bounded.

The first-order Taylor expansion (FVR, I, p. 30, prop. 3) and formula (13) show that there exist $M \geq 0$ and a function ϱ_1 with values in R^2 such that

$$\tilde{f}(y + i\gamma) = f(y) + i\gamma f'(y) + \gamma^2 \varrho_1(y; \gamma), \text{ and } |\varrho_1(y; \gamma)| \leq M,$$

for all $y \in \mathbf{R}$ and $\gamma \in \mathbf{R}$. Since the map $y \mapsto \tilde{f}(y + i\gamma)$ has support contained in $[-r, r]$ for all $\gamma \in \mathbf{R}$, the map $y \mapsto \varrho_1(y; \gamma)$ has support contained in $[-r, r]$, for all $\gamma \in \mathbf{R}$.

Let g_ε the function defined on $\mathbf{R}^2$ by

$$g_\varepsilon(x, y) = \frac{\tilde{f}(y + i\varepsilon)}{y - x + i\varepsilon} - \frac{\tilde{f}(y - i\varepsilon)}{y - x - i\varepsilon}.$$

For every $(x, y) \in \mathbf{R}^2$ and $\varepsilon > 0$, we obtain

$$g_\varepsilon(x, y) = -\frac{2i\varepsilon}{(x-y)^2 + \varepsilon^2} f(y) + \frac{2i\varepsilon(y-x)}{(x-y)^2 + \varepsilon^2} f'(y) + \varepsilon^2 \left(\frac{\varrho_1(y; \varepsilon)}{y-x+i\varepsilon} - \frac{\varrho_1(y; -\varepsilon)}{y-x-i\varepsilon} \right).$$

Let $x \in \mathbf{R}$ and $\varepsilon > 0$. We have

$$\begin{aligned} \left| \varepsilon^2 \int_{\mathbf{R}} \left(\frac{\varrho_1(y; \varepsilon)}{y-x+i\varepsilon} - \frac{\varrho_1(y; -\varepsilon)}{y-x-i\varepsilon} \right) dy \right| &\leq 2M\varepsilon^2 \int_{-r}^r \frac{dy}{\sqrt{(x-y)^2 + \varepsilon^2}} \\ &\leq 4M\varepsilon. \end{aligned}$$

Consequently, for every $x \in \mathbf{R}$, it follows that

$$f_\varepsilon(x) = -\frac{1}{2i\pi} \int_{\mathbf{R}} g_\varepsilon(x, y) dy = (f * \delta_\varepsilon)(x) - \frac{1}{2i\pi} (f' * \sigma_\varepsilon)(x) + k_\varepsilon(x)$$

where δ_ε and σ_ε are the functions on $\mathbf{R}$ defined by

$$\delta_\varepsilon(x) = \frac{1}{\pi} \frac{\varepsilon}{x^2 + \varepsilon^2}, \quad \sigma_\varepsilon(x) = \frac{2i\varepsilon x}{x^2 + \varepsilon^2},$$

and $\|k_\varepsilon\|_\infty \leq 2M\varepsilon$.

The function $\mathcal{F}(f') \in \mathcal{S}(\mathbf{R})$ is integrable (prop. 18 of IV, p. 219 and prop. 13 of IV, p. 213). By lemma 6, it follows that

$$\begin{aligned} \int_{\mathbf{R}} |\mathcal{F}(f')(y) \mathcal{F}(\sigma_\varepsilon)(y)| dy &= 2\pi\varepsilon \int_{\mathbf{R}} |\mathcal{F}(f')(y)| e^{-2\pi\varepsilon|y|} dy \\ &\leq 2\pi\varepsilon \int_{\mathbf{R}} |\mathcal{F}(f')(y)| dy, \end{aligned}$$

hence $\mathcal{F}(f') \mathcal{F}(\sigma_\varepsilon)$ converges to 0 in $L^1(\mathbf{R})$ as ε tends to 0. Since f' and σ_ε belong to

$L^2(\mathbf{R})$, prop. 14 of II, p. 223 implies that $f' * \sigma_\varepsilon = \overline{\mathcal{F}}(\mathcal{F}(f')\mathcal{F}(\sigma_\varepsilon))$ converges to 0 in $\mathcal{C}_b(\mathbf{R})$.

For every $\varepsilon > 0$, the positive measure $\delta_\varepsilon \cdot dx$ on $\mathbf{R}$ has total mass 1 (cf. FVR, III, p. 7). The set of measures $\delta_\varepsilon \cdot dx$ for $\varepsilon > 0$ and the filter induced on this set by the filter of neighborhoods of 0 in $\mathbf{R}_+^*$ satisfy the hypotheses of lemma 4 of INT, VIII, p. 137, § 2, no. 7. The measures $\delta_\varepsilon \cdot dx$ therefore converge in $\mathcal{M}^1(\mathbf{R})$ toward the point measure ε_0 as ε tends to 0. It follows that $f * \delta_\varepsilon \rightarrow f$ in $\mathcal{C}_b(\mathbf{R})$ (INT, VIII, p. 163, § 4, no. 4). The lemma is proved.

We denote by μ the Lebesgue measure on $\mathbf{C}$.

THEOREM 2 (Helffer–Sjöstrand). — Let u be a partial self-adjoint operator on E . Let $f \in \mathcal{D}(\mathbf{R})$ and $\tilde{f} \in \mathcal{D}(\mathbf{C})$ be an almost analytic extension of f . Let h be the map from $\mathbf{C}$ into $\mathcal{L}(E)$ defined by

$$h(\lambda) = \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) R(u, \lambda)$$

if $\lambda \in \mathbf{C} - \mathbf{R}$ and $h(\lambda) = 0$ if $\lambda \in \mathbf{R}$. Then h is μ -integrable on $\mathbf{C}$ and

$$f(u) = -\frac{1}{\pi} \int_{\mathbf{C}} h(\lambda) d\mu(\lambda).$$

The map h is measurable and its support is compact. Since u is self-adjoint, we have $\|R(u, \lambda)\| \leq |\mathcal{I}(\lambda)|^{-1}$ for all $\lambda \in \mathbf{C} - \mathbf{R}$ (prop. 17 of IV, p. 248). The map h is therefore bounded according to formula (13), and consequently it is integrable on $\mathbf{C}$.

Let $\varepsilon \in \mathbf{R}_+^*$. We denote F_ε^+ (resp. F_ε^-) the closed set in $\mathbf{C}$ of the $\lambda \in \mathbf{C}$ such that $\mathcal{I}(\lambda) \geq \varepsilon$ (resp. $\mathcal{I}(\lambda) \leq -\varepsilon$). We have

$$\int_{\mathbf{C}} h(\lambda) d\mu(\lambda) = \lim_{\varepsilon \rightarrow 0} \left(\int_{F_\varepsilon^+} h(\lambda) d\mu(\lambda) + \int_{F_\varepsilon^-} h(\lambda) d\mu(\lambda) \right).$$

Let $r > 0$ such that the support of h is contained in $\mathbf{C} = [-r, r]^2$. Denote by R_ε^+ the box $[-2r, 2r] \times [\varepsilon, 2r]$ in $\mathbf{C}$. It is a locally polyhedral subset of $\mathbf{C}$ (VAR, R2, 11.3, p. 48). It satisfies the following conditions:

- (i) One has $R_\varepsilon^+ \subset 2C \cap F_\varepsilon^+$;

- (ii) The set R_ε^+ contains the intersection of F_ε^+ and the support of h ;
- (iii) The regular boundary ∂R_ε^+ (VAR, R2, 11.3.2, p. 49) contains the segment $S_\varepsilon = [-r, r] + i\varepsilon \subset C$;
- (iv) One has $h(\lambda) = 0$ if $\lambda \in \partial R_\varepsilon^+ - S_\varepsilon$.

Denote by $d\lambda$ (resp. $d\bar{\lambda}$) the differential 1-form on $\mathbf{C}$ of the identity map of $\mathbf{C}$ (resp. of complex conjugation). Denote by g the function on $\mathbf{C} - \mathbf{R}$ with values in $\mathcal{L}(E)$ such that $g(\lambda) = \tilde{f}(\lambda)R(u, \lambda)$ for $\lambda \in \mathbf{C} - \mathbf{R}$. Let $\omega = gd\lambda$; it is a differential form of degree 1 on $\mathbf{C} - \mathbf{R}$ with compact support and values in $\mathcal{L}(E)$. Since the resolvent of u is holomorphic (prop. 14 of IV, p. 246), we have

$$d\omega = \left(\frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda)R(u, \lambda) + \tilde{f}(\lambda) \frac{\partial}{\partial \bar{z}}R(u, \lambda) \right) d\bar{\lambda} \wedge d\lambda = -h(\lambda)d\lambda \wedge d\bar{\lambda}.$$

The vector measure associated with $d\omega$ (VAR, R2, 10.4.3, p. 43) is the density measure $-2ih$ with respect to Lebesgue measure. Applying Stokes' formula to the locally polyhedral set R_ε^+ and to the differential form ω (VAR, R2, 11.3.4, p. 49), we thus obtain

$$\frac{i}{2} \int_{R_\varepsilon^+} d\omega = \frac{i}{2} \int_{\partial R_\varepsilon^+} \omega = \frac{i}{2} \int_{S_\varepsilon} \omega = \frac{i}{2} \int_{-r}^r \tilde{f}(y + i\varepsilon)R(u, y + i\varepsilon)dy,$$

whence

$$\int_{F_\varepsilon^+} h(\lambda)d\mu(\lambda) = -\frac{i}{2} \int_{\mathbf{R}} \tilde{f}(y + i\varepsilon)R(u, y + i\varepsilon)dy.$$

Reasoning in the same way for F_ε^- , we obtain

$$\int_{F_\varepsilon^-} h(\lambda)d\mu(\lambda) = -\frac{i}{2} \int_{\mathbf{R}} \tilde{f}(y - i\varepsilon)R(u, y - i\varepsilon)dy,$$

and we conclude that the integral of h over $\mathbf{C}$ is the limit when $\varepsilon \rightarrow 0$ of

$$v_\varepsilon = \frac{i}{2} \int_{\mathbf{R}} (\tilde{f}(y + i\varepsilon)R(u, y + i\varepsilon) - \tilde{f}(y - i\varepsilon)R(u, y - i\varepsilon))dy.$$

By prop. 7, we have $v_\varepsilon = \pi f_\varepsilon(u)$, where f_ε is the function defined on $\mathbf{R}$ by

$$f_\varepsilon(x) = -\frac{1}{2i\pi} \int_{\mathbf{R}} \left(\frac{\tilde{f}(y+i\varepsilon)}{y-x+i\varepsilon} - \frac{\tilde{f}(y-i\varepsilon)}{y-x-i\varepsilon} \right) dy.$$

Since $f_\varepsilon \rightarrow f$ as $\varepsilon \rightarrow 0$ in $\mathcal{C}_b(\mathbf{R})$ (Lemma 7), the endomorphism $v_\varepsilon = \pi f_\varepsilon(u)$ converges to $\pi f(u)$ in $\mathcal{L}(E)$ (prop. 5 of IV, p. 275). The theorem is proved.

IV.5.6 Resolvent topologies and continuity of the functional calculus

In this issue, E denotes a complex Hilbert space. Recall that we denote by $\mathcal{A}(E)$ the set of partial self-adjoint operators on E . We will extend to $\mathcal{A}(E)$ the continuity properties of item 8 of IV, p. 194.

DEFINITION 6. — Let T be a locally convex topology on $\mathcal{L}(E)$. The T -resolvent topology on $\mathcal{A}(E)$ is the coarsest topology such that the maps $u \mapsto R(u, \lambda)$ from $\mathcal{A}(E)$ into $\mathcal{L}(E)$ endowed with the topology T are continuous for every $\lambda \in \mathbf{C} - \mathbf{R}$.

PROPOSITION 11. — Let T be a locally convex topology on $\mathcal{L}(E)$ which is coarser than the Banach space topology of $\mathcal{L}(E)$.

- a) Let $f \in \mathcal{C}_0(\mathbf{R})$. For every sequence $(u_n)_{n \in \mathbf{N}}$ in $\mathcal{A}(E)$ which converges to u for the topology $\mathcal{T}$ -resolvent, the sequence $(f(u_n))_{n \in \mathbf{N}}$ converges to $f(u)$ in the space $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}$;
- b) Suppose that everything $u \in \mathcal{A}(E)$ admits a countable fundamental system of neighborhoods for the topology $\mathcal{T}$ -resolvent. The map of $\mathcal{C}_0(\mathbf{R}) \times \mathcal{A}(E)$ in the space $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}$ defined by $(f, u) \mapsto f(u)$ is continuous.

Let us prove a). The topology $\mathcal{T}$ on $\mathcal{L}(E)$ is the upper bound of the topologies defined by the semi-norms continuous for the topology $\mathcal{T}$ (EVT, II, p. 26, cor. and following remark). It therefore suffices to prove that for every seminorm p on $\mathcal{L}(E)$ which is continuous for the topology $\mathcal{T}$ the sequence $p(f(u_n) - f(u))$ converges to 0 (TG, I, p. 51, prop. 10).

Let p be such a seminorm. Denote by $\mathcal{L}(E)_p$ the separated Banach space completion of the space $\mathcal{L}(E)$ equipped with the seminorm p , and denote by ϖ the canonical map from $\mathcal{L}(E)$ into $\mathcal{L}(E)_p$; it is continuous, and one has $p(u) = \|\varpi(u)\|_p$ for all $u \in \mathcal{L}(E)$, where the norm is that of the space $\mathcal{L}(E)_p$.

Since $\mathcal{T}$ is coarser than the Banach space topology of $\mathcal{L}(E)$, there exists $c \geq 0$ such that $p(u) \leq c\|u\|$ for all $u \in \mathcal{L}(E)$.

Suppose first that $f \in \mathcal{D}(\mathbf{R})$. Let $\tilde{f}$ an almost analytic extension of f . Denote by μ the Lebesgue measure on $\mathbf{C}$. By the Helffer–Sjöstrand formula (th. 2 of IV, p. 284), one has

$$\varpi(f(u_n)) = -\frac{1}{\pi} \int_{\mathbf{C}} \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) \varpi(R(u_n, \lambda)) d\mu(\lambda) \quad (14)$$

for all $n \in \mathbf{N}$.

Let $\lambda \in \mathbf{C} - \mathbf{R}$. By hypothesis, the sequence of resolvents $R(u_n, \lambda)$ converges to $R(u, \lambda)$ in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}$, hence the sequence $(\varpi(R(u_n, \lambda)))_n$ converges to $\varpi(R(u, \lambda))$ to the Banach space $\mathcal{L}(E)_p$.

For every $n \in \mathbf{N}$, we have

$$\left\| \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) R(u_n, \lambda) \right\| \leq \left| \frac{1}{\mathcal{T}(\lambda)} \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) \right|$$

(prop. 17 of IV, p. 248) therefore

$$\left\| \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) \varpi(R(u_n, \lambda)) \right\|_p \leq c \left| \frac{1}{\mathcal{T}(\lambda)} \frac{\partial \tilde{f}}{\partial \bar{z}}(\lambda) \right|.$$

The estimate (13) of IV, p. 281 shows that the right-hand side of this inequality is a bounded function for $\lambda \in \mathbf{C} - \mathbf{R}$; it is integrable on $\mathbf{C}$ since $\tilde{f}$ has compact support. One deduces from Lebesgue's theorem (INT, IV, p. 137, § 3, n° 7, th. 6) and from the Helffer–Sjöstrand formula applied to f we deduce that $\varpi(f(u_n))$ converges to $\varpi(f(u))$, hence $p(f(u_n) - f(u))$ tends to 0.

Suppose now that $f \in \mathcal{C}_0(\mathbf{R})$. Let $\varepsilon > 0$. There exists a function f_ε in $\mathcal{D}(\mathbf{R})$ such that $\|f_\varepsilon - f\|_\infty \leq \varepsilon$ (cf. prop. 4, a) of IV, p. 202 and INT, III, p. 45, § 1, n° 2, prop. 3). According to prop. 5, c) of IV, p. 275, one then has $\|f(u) - f_\varepsilon(u)\| \leq \varepsilon$ and $\|f(u_n) - f_\varepsilon(u_n)\| \leq \varepsilon$ for all $n \in \mathbf{N}$. Consequently, it follows that

$$p(f(u_n) - f(u)) \leq 2c\varepsilon + p(f_\varepsilon(u_n) - f_\varepsilon(u))$$

for all $n \in \mathbf{N}$. Since $f_\varepsilon \in \mathcal{D}(\mathbf{R})$, the sequence $(p(f_\varepsilon(u_n) - f_\varepsilon(u)))_{n \in \mathbf{N}}$ converges to 0 by

the preceding case, and therefore $p(f(u_n) - f(u))$ converges to 0. This concludes the proof of a).

Let us prove assertion b). Under the hypothesis concerning $\mathcal{T}$, it follows from TG, IX, p. 17, prop. 10 and the following remark, and from assertion a), that the map $u \mapsto f(u)$ of $\mathcal{A}(E)$ in $\mathcal{L}(E)$ is continuous when $f \in \mathcal{C}_0(\mathbf{R})$. Since the maps $f \mapsto f(u)$ of $\mathcal{C}_0(\mathbf{R})$ in $\mathcal{L}(E)$ are continuous with norm ≤ 1 for all $u \in \mathcal{A}(E)$ (prop. 5, c) of IV, p. 275), assertion b) follows from TG, X, p. 13, cor. 3.

Examples. — 1) Let $\mathcal{S}$ be a set of bounded subsets of E . The $\mathcal{S}$ -topology on $\mathcal{L}(E)$ (EVT, III, p. 13) is a locally convex topology less fine than the Banach space topology of $\mathcal{L}(E)$.

2) Let T_b the Banach space topology of $\mathcal{L}(E)$. For the topology T_b -resolvent, every $u \in \mathcal{A}(E)$ admits a countable fundamental system of neighborhoods.

Indeed, it suffices to show that for every $\lambda \in \mathbf{C} - \mathbf{R}$ and every $\varepsilon > 0$, there exist $\lambda' \in \mathbf{Q} + i\mathbf{Q}^*$ and an integer $n \geq 1$ such that every partial operator $v \in \mathcal{A}(E)$ satisfying $\|R(v, \lambda') - R(u, \lambda')\| < 1/n$ also satisfies condition $\|R(v, \lambda) - R(u, \lambda)\| < \varepsilon$; by writing

$$\begin{aligned} \|R(v, \lambda) - R(u, \lambda)\| &\leq \|R(v, \lambda) - R(v, \lambda')\| \\ &+ \|R(v, \lambda') - R(u, \lambda')\| + \|R(u, \lambda') - R(u, \lambda)\|, \end{aligned}$$

this property follows from formula (8) of IV, p. 245, from the estimates of proposition 17 of IV, p. 248, and from the fact that $\mathbf{Q} + i\mathbf{Q}^*$ is everywhere dense in $\mathbf{C} - \mathbf{R}$.

The conclusion of the proposition is not valid in general if $\mathcal{C}_0(\mathbf{R})$ is replaced by $\mathcal{C}_b(\mathbf{R})$ (exercise 29 of IV, p. 366, e)). Nevertheless, one has the following result.

COROLLARY. — Let $\mathcal{T}_s$ the topology of simple convergence on $\mathcal{L}(E)$ Let $f \in \mathcal{C}_b(\mathbf{R})$. For every sequence $(u_n)_{n \in \mathbf{N}}$ in $\mathcal{A}(E)$ which converges to u for the topology $\mathcal{T}_s$ -resolvent, the sequence $(f(u_n))_{n \in \mathbf{N}}$ converges to $f(u)$ in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_s$.

Let (u_n) be a sequence in $\mathcal{A}(E)$ which converges to u for the topology T_s -resolvent. Let $x \in E$ and let $\varepsilon > 0$.

For every integer $N \in \mathbf{N}$, let $\varphi_N \in \mathcal{K}(\mathbf{R})$ be a function with support contained in $[-(N+1), N+1]$ such that $0 \leq \varphi_N \leq 1$ and $\varphi_N(t) = 1$ for all $t \in [-N, N]$. The functions φ_N converge pointwise to 1 as N tends to infinity, and satisfy $|\varphi_N| \leq 1$; hence prop. 6 of IV, p. 276 implies

that there exists $N \in \mathbf{N}$ such that $\|\varphi_N(u)x - x\| \leq \varepsilon$. Set $f_N = f\varphi_N$.

For every $n \in \mathbf{N}$, we have

$$\begin{aligned} \|f(u_n)x - f(u)x\| &\leq \|f(u_n)x - f_N(u_n)x\| + \\ &\|f_N(u_n)x - f_N(u)x\| + \|f_N(u)x - f(u)x\|. \end{aligned} \quad (15)$$

For every $v \in \mathcal{A}(E)$, we have

$$\begin{aligned} \|f(v)x - f_N(v)x\| &= \|(f(1 - \varphi_N))(v)x\| \\ &\leq \|f(v)\| \|x - \varphi_N(v)x\| \leq \|f\|_\infty \|x - \varphi_N(v)x\| \end{aligned}$$

(prop. 5, c) of IV, p. 275). Since $\varphi_N \in \mathcal{C}_0(\mathbf{R})$, assertion a) of prop. 11 applied to the topology $\mathcal{T}_s$ -resolvent implies that $\varphi_N(u_n)x$ converges to $\varphi_N(u)x$ as $n \rightarrow +\infty$. Consequently, for every n sufficiently large, one has

$$\|x - \varphi_N(u_n)x\| \leq \|x - \varphi_N(u)x\| + \varepsilon \|x\| \leq (1 + \|x\|)\varepsilon.$$

Inequality (15) then becomes

$$\|f(u_n)x - f(u)x\| \leq \|f\|_\infty (2 + \|x\|)\varepsilon + \|f_N(u_n)x - f_N(u)x\|$$

for all sufficiently large n . One concludes the proof by means of assertion a) of loc. cit., applied to the function $f_N \in \mathcal{C}_0(\mathbf{R})$ and to the topology T_s -resolvent.

IV.5.7 Polar decomposition

Lemma 8. — Let E be a separated topological vector space. Let $(E_1, \|\cdot\|_1)$ and $(E_2, \|\cdot\|_2)$ be dense subspaces of E endowed with Hilbert structures such that the canonical injections of E_1 and E_2 into E are continuous. Let F be a subspace of $E_1 \cap E_2$, dense in E_1 and E_2 for these Hilbert structures. If $\|x\|_1 = \|x\|_2$ for all $x \in F$, then $E_1 = E_2$ and $\|\cdot\|_1 = \|\cdot\|_2$.

Let $x \in E_1$. There exists a sequence $(x_n)_{n \in \mathbf{N}}$ in F such that x_n converges to x in the Hilbert space E_1 . Since $\|x_n - x_m\|_2 = \|x_n - x_m\|_1$ for all integers n and m , the sequence (x_n) is a Cauchy sequence in E_2 . Let y be its limit. We have $\|y\|_2 = \lim \|x_n\|_2 = \|x\|_1$. Since the canonical

injections of E_1 and E_2 into E are continuous, we have $x_n \rightarrow x$ in E and likewise $x_n \rightarrow y$ in E . Thus, it follows that $x = y$ and $\|x\|_1 = \|x\|_2$. In particular, $E_1 \subset E_2$. We conclude by symmetry.

Let E be a complex Hilbert space. Let u be a positive self-adjoint partial operator on E . For every $\alpha \in \mathbf{R}_+$ we set $u^\alpha = f(u)$, where f is the continuous map $x \mapsto x^\alpha$ of $\mathbf{R}_+$ into $\mathbf{R}$. This is a positive self-adjoint partial operator (cor. 1, a) of IV, p. 273). When $u \in \mathcal{L}(E)$, this notation is compatible with the notation relative to the C^* -algebra $\mathcal{L}(E)$ (cf. prop. 16 of I, p. 118). If α is a positive integer, the partial operator u^α coincides with the partial operator defined by composition $u \circ \dots \circ u$ (cor. 1, b) of IV, p. 273).

Let $\beta \in \mathbf{R}_+$. We have $u^{\alpha\beta} = (u^\alpha)^\beta$ (cor. 4 of IV, p. 274). In particular, if $\alpha > 0$, then the partial operator $u^{1/\alpha}$ is the unique positive self-adjoint partial operator v on E such that $v^\alpha = u$.

Suppose that $0 \leq \alpha \leq \beta$. We then have $\text{dom}(u^\beta) \subset \text{dom}(u^\alpha)$: indeed, for every real number $x \geq 0$, we have $x^\alpha \leq 1 + x^\beta$, and the result then follows from the definition of the domain of u^α (cf. prop. 3 of IV, p. 271).

Let u be a closed densely defined operator from E into a complex Hilbert space F . The partial operator u^*u is self-adjoint and positive (prop. 12 of IV, p. 241). We denote $|u| = (u^*u)^{1/2}$.

PROPOSITION 12. — Let u be a closed densely defined operator from E into a complex Hilbert space F .

- a) The domain of the positive self-adjoint partial operator $|u|$ coincides with the domain of u ;
- b) There exists a unique partially isometric linear map j from E into F such that $u = j|u|$ and $\text{Ker}(j) = \text{Ker}(u)$;
- c) Let u_1 be a positive self-adjoint operator on E and j_1 a partially isometric linear map from E into F such that $u = j_1 u_1$ and $\text{Ker}(j_1) = \text{Ker}(u_1)$. We then have $u_1 = |u|$ and $j_1 = j$.

The domain of u^*u is contained in $\text{dom}(u)$ and in $\text{dom}(|u|)$. It is dense in the Hilbert spaces E_u (loc. cit.) and $E_{|u|}$ (cor. 1, e) of IV, p. 273) and, for every $x \in \text{dom}(u^*u)$, we have

$$\begin{aligned} \|x\|_u^2 &= \|x\|^2 + \langle u(x)|u(x) \rangle = \|x\|^2 + \langle x|(u^*u)(x) \rangle \\ &= \|x\|^2 + \langle |u|(x)||u|(x) \rangle = \|x\|_{|u|}^2. \end{aligned} \tag{16}$$

Consequently, the Hilbert spaces E_u and $E_{|u|}$ are equal (lemma 8), hence $\text{dom}(u) = \text{dom}(|u|)$ and $\|u(x)\| = \||u|(x)\|$ for all $x \in \text{dom}(u)$.

Formula (16) implies that $\text{Ker}(u) = \text{Ker}(|u|)$ and that there exists a unique isometric linear map v from $\overline{\text{Im}(|u|)}$ onto $\overline{\text{Im}(u)}$ satisfying $v(|u|(x)) = u(x)$ for all $x \in \text{dom}(|u|)$. Since $|u|$ is self-adjoint, we have $\text{Im}(|u|)^\circ = \text{Ker}(|u|)$ (prop. 7, c) of IV, p. 236). There therefore exists a unique partial isometry j from E into F which extends v and vanishes on $\text{Ker}(|u|) = \text{Ker}(u)$. Since $E = \text{Ker}(u) \oplus \overline{\text{Im}(|u|)}$, this map is the unique partially isometric map such that $u = j|u|$ and $\text{Ker}(j) = \text{Ker}(u)$.

Let us prove c). We have $u_1 j_1^* j_1 u_1 \subset (j_1 u_1)^* j_1 u_1 = u^* u$. The linear map $j_1^* j_1$ is the orthogonal projector with kernel $\text{Ker}(j_1) = \text{Ker}(u_1)$ (EVT, V, p. 41, prop. 5 (ii)) and therefore with image $\text{Ker}(u_1)^\circ = \overline{\text{Im}(u_1)}$ (prop. 7, c) of IV, p. 236). Consequently, $u_1^2 \subset u^* u$, whence $u_1^2 = u^* u$ since these two operators are self-adjoint. Thus, it follows that $u_1 = (u^* u)^{1/2}$ and the uniqueness assertion of b) ultimately shows that $j_1 = j$.

DEFINITION 7. — Let u be a closed operator with dense domain from E into a complex Hilbert space F . The pair $(j, |u|)$ determined by prop. 12 is called the polar decomposition of u .

Suppose that $u \in \mathcal{L}(E; F)$. Its polar decomposition in the sense of this definition coincides with that of Definition 4 in I, p. 140.

IV.5.8 Self-adjoint operators defined by a positive partial Hermitian form

In this issue, E denotes a complex Hilbert space.

DEFINITION 8. — Let D be a vector subspace of E . A partial Hermitian form on E with domain D is a correspondence $q = (\Gamma, E \times E, \mathbb{C})$ whose domain of definition is $D \times D$, whose graph Γ is functional, and such that the map from $D \times D$ into $\mathbb{C}$ defined by Γ is a Hermitian form. We denote $\text{dom}(q) = D$.

We say that q is a positive partial form if the Hermitian form it defines is positive.

Let u be a partial self-adjoint operator on E . For all elements x and y of E , we denote $\mu_{x,y}$ (resp. μ_x) as the spectral measure of (x, y) (resp. of x) relative to u .

Let $(j, |u|)$ be the polar decomposition of u (definition 7 of IV, p. 290). Let D' denote the domain of $|u|^{1/2}$. By definition, it is the space of $x \in E$ such that the function $z \mapsto |z|$ is integrable

on $\text{Sp}(u)$ with respect to the measure μ_x . It contains $\text{dom}(u)$.

Let $(x, y) \in D' \times D'$. The identity function of $\text{Sp}(u)$ is integrable with respect to the measure $\mu_{x,y}$ (prop. 4, b) of IV, p. 271); we set

$$q_u(x, y) = \int_{\text{Sp}(u)} t d\mu_{x,y}(t).$$

If $y \in \text{dom}(u)$, we have $q_u(x, y) = \langle x | u(y) \rangle$ by formula (6) of IV, p. 271. The map q_u is a Hermitian form on D' . It defines a partial Hermitian form on E , said to be associated with u . If the operator u is positive, then the form q_u is a positive partial Hermitian form.

DEFINITION 9. — Let q be a positive partial Hermitian form on E . We denote by E_q the separated pre-Hilbert space $\text{dom}(q)$ equipped with the scalar product

$$(x | y)_q = \langle x | y \rangle + q(x, y).$$

We denote by $\|x\|_q$ the norm of $x \in E_q$. The form q is said to be closed if E_q is a Hilbert space.

PROPOSITION 13. — Let u be a positive self-adjoint partial operator on E . Let q_u be the positive partial form associated with u .

a) The domain of q_u is the domain of $u^{1/2}$, and for all x and y in $\text{dom}(u^{1/2})$, we have

$$q_u(x, y) = \langle u^{1/2}(x) | u^{1/2}(y) \rangle;$$

b) The positive partial form q_u is closed. The domain of u is a core for q_u .

Since u is positive, we have $|u| = u$. The domain $\text{dom}(|u|^{1/2})$ of q_u therefore coincides with $\text{dom}(u^{1/2})$ and we have

$$\begin{aligned} q_u(x, y) &= \int_{\text{Sp}(u)} t d\mu_{x,y}(t) = \int_{\text{Sp}(u)} t^{1/2} \cdot t^{1/2} d\mu_{x,y}(t) \\ &= \langle u^{1/2}(x) | u^{1/2}(y) \rangle \end{aligned}$$

for x and y in $\text{dom}(q_u)$ (prop. 4, b) of IV, p. 271). This proves the assertion a).

Moreover, the pre-Hilbert space E_{q_u} then coincides with the pre-Hilbert space $E_{u^{1/2}}$ associated with $u^{1/2}$ (def. 4 of IV, p. 230). Since $u^{1/2}$ is a closed partial operator, this space is a Hilbert space (prop. 4 of IV, p. 230) and $\text{dom}(u)$ is dense in $E_{u^{1/2}}$ by assertion e) of Cor. 1 of

IV, p. 273.

Let q be a partial Hermitian form on E with domain D dense in E . For $y \in D$, denote by λ_y the linear form $x \mapsto q(y, x)$ on D . We denote by $\tilde{D}$ the set of $y \in D$ such that the linear form λ_y is continuous on D .

Let $y \in \tilde{D}$. Since D is dense in E , there exists a unique continuous linear form on E extending λ_y . We still denote it by λ_y . There exists a unique element $u(y)$ in E such that $\lambda_y(x) = \langle u(y) | x \rangle$ for all $x \in E$ (EVT, V, p. 15, th. 3). The map $y \mapsto u(y)$ is linear from $\tilde{D}$ into E ; we say that the partial operator u on E with domain $\tilde{D}$ is the partial operator representing q . We therefore have

$$q(x, y) = \langle x | u(y) \rangle$$

for $y \in \text{dom}(u)$ and $x \in D$.

Remark. — Let q be a closed positive partial form and q' a continuous positive Hermitian form on E . The positive Hermitian form defined on $\text{dom}(q)$ by $(x, y) \mapsto q(x, y) + q'(x, y)$ is a closed positive partial Hermitian form with the same domain as q . We denote it by $q + q'$.

According to EVT, V, p. 16, cor. 2, there exists a unique linear map $u' \in \mathcal{L}(E)$ such that $q'(x, y) = \langle x | u'(y) \rangle$ for all $(x, y) \in E \times E$. The endomorphism u' is positive and the partial operator representing the closed positive partial Hermitian form $q + q'$ is $u + u'$.

PROPOSITION 14. — Let q be a closed positive partial form with dense domain on E and let u be the partial operator representing q .

- a) The domain of u is dense in the Hilbert space E_q ;
- b) The partial operator u is self-adjoint and positive.

Since q is closed, the pre-Hilbert space E_q of Definition 9 is a Hilbert space.

Let us show that the partial operator $u + 1_E$ with domain $\text{dom}(u)$ is bijective. Since

$$\langle x | (u + 1_E)(x) \rangle = q(x, x) + \|x\|^2 \geq \|x\|^2$$

for all $x \in \text{dom}(u)$, this partial operator is injective.

Let $y \in E$. The linear form on E_q defined by $x \mapsto \langle y | x \rangle$ is continuous since $\|x\| \leq \|x\|_q$.

There therefore exists $y' \in E_q$ such that

$$\langle y | x \rangle = (y' | x)_q = \langle y' | x \rangle + q(y', x)$$

for all $x \in E_q$ (EVT, V, p. 15, th. 3). By definition, this means that y' belongs to the domain of the partial operator $\tilde{u}$ representing the partial form $(x, y) \mapsto (x | y)_q$ with domain $\text{dom}(q)$, and that $\tilde{u}(y') = y$. Since $\tilde{u} = u + 1_E$ by the remark above, we deduce from this that the partial operator $u + 1_E$ is also surjective, hence bijective.

Let us prove that the domain of u is dense in E_q . Let $y \in E_q$ be orthogonal to $\text{dom}(u)$ in E_q . There exists $y' \in \text{dom}(u)$ such that $u(y') + y' = y$. We then have

$$0 = (y | y')_q = \langle y | y' \rangle + q(y, y') = \langle y | y' + u(y') \rangle = \|y\|^2$$

hence $y = 0$. The assertion a) is therefore proved.

Since $\text{dom}(q)$ is dense in E and $\|x\| \leq \|x\|_q$ for all $x \in \text{dom}(q)$, assertion a) implies that $\text{dom}(u)$ is dense in E .

Since the form q is Hermitian (resp. positive), for all x and y in $\text{dom}(u)$, we have

$$\langle y | u(x) \rangle = q(y, x) = \overline{q(x, y)} = \overline{\langle x | u(y) \rangle} = \langle u(y) | x \rangle,$$

(resp. $\langle x | u(x) \rangle = q(x, x) \geq 0$) so that u is symmetric (resp. positive). Finally, the partial operator $u + 1_E$ is self-adjoint by the corollary of Proposition 10 of IV, p. 240, and the same is true of u .

THEOREM 3. — The map that associates the positive partial form q_u with a positive self-adjoint operator u on E is a bijection between the set of positive self-adjoint partial operators on E and the set of closed positive partial forms with dense domain on E . The inverse bijection associates with a positive partial form q the partial operator representing q .

By Propositions 13, b) and 14, b), the maps described in the statement are well defined. Let us show that they are inverse bijections of one another.

Let u be a positive self-adjoint partial operator on E and q the positive partial form associated with u . Denote by v the positive self-adjoint operator representing q . Let $y \in \text{dom}(u) \subset$

$\text{dom}(u^{1/2}) = \text{dom}(q)$. For every $x \in \text{dom}(q) = \text{dom}(u^{1/2})$, we have

$$q(y, x) = \langle u^{1/2}(y) | u^{1/2}(x) \rangle = \langle u(y) | x \rangle,$$

hence y belongs to the domain of v and satisfies $v(y) = u(y)$. The partial operator v is therefore an extension of u ; since u and v are self-adjoint, they are equal.

Conversely, let q be a closed positive partial form with dense domain and let u be the positive self-adjoint partial operator representing q . We have $\text{dom}(u) \subset \text{dom}(u^{1/2})$. For x and y in $\text{dom}(u)$, we have

$$q_u(x, y) = \langle u^{1/2}(x) | u^{1/2}(y) \rangle = \langle x | u(y) \rangle = q(x, y)$$

by Proposition 13, a). Thus, the Hilbert spaces E_q and E_{q_u} both contain $\text{dom}(u)$ as a dense subspace (Prop. 14, a) and Prop. 13, b), respectively) and $\|x\|_q = \|x\|_{q_u}$ for all $x \in \text{dom}(u)$. It follows that $E_q = E_{q_u}$ and that $q = q_u$ (lemma 8).

COROLLARY. — Let q be a closed positive partial form on E and u the positive self-adjoint operator representing q . The domain of q is equal to the domain of $(1_E + u)^{1/2}$, and we have

$$\|x\|_q = \|(1_E + u)^{1/2}x\|$$

for all $x \in \text{dom}(q)$.

The domain of q is equal to the domain of $u^{1/2}$ (prop. 13, a), which coincides with that of $(1_E + u)^{1/2}$ (cf. prop. 3 of IV, p. 271). For every $x \in \text{dom}(u) \subset \text{dom}(u^{1/2})$, we have

$$\|(1_E + u)^{1/2}x\|^2 = \langle x | (1_E + u)(x) \rangle = \|x\|^2 + \langle x | u(x) \rangle = \|x\|_q^2.$$

As the domain of u is dense in the Hilbert space E_q (prop. 14, a)), this formula extends by continuity to all $x \in \text{dom}(u^{1/2})$.

Example. — Let u be a positive partial operator on E which is not necessarily closed. One defines a positive partial form q with domain $\text{dom}(u)$ by

$$q(x, y) = \langle x | u(y) \rangle$$

for x and y in $\text{dom}(u)$. Let E_q be the separated pre-Hilbert space of Definition 9 of IV, p. 292. Let $\tilde{E}_q$ be the completed Hilbert space of E_q , still denoted $(x, y) \mapsto (x | y)_q$ for the scalar product. Since the canonical inclusion ι of E_q in E is continuous, it admits a unique continuous extension, denoted $\tilde{\iota}$, of $\tilde{E}_q$ in E . The Hermitian form q is continuous on E_q , hence also extends to a unique continuous positive Hermitian form $\tilde{q}$ on $\tilde{E}_q$.

Let us prove that the linear map $\tilde{\iota}$ is injective. Let $x \in \text{Ker}(\tilde{\iota})$. Consider a sequence $(x_n)_{n \in \mathbb{N}}$ in E_q which converges to x in $\tilde{E}_q$. We then have

$$\lim_{n \rightarrow +\infty} \iota(x_n) = \tilde{\iota}(x) = 0,$$

therefore the sequence $(x_n)_{n \in \mathbb{N}}$ converges to 0 in E . Let $y \in E_q$. Since $\tilde{q}$ is continuous on $\tilde{E}_q$, it follows that

$$\begin{aligned} (x | y)_q &= \lim_{n \rightarrow +\infty} (x_n | y)_q = \lim_{n \rightarrow +\infty} (\langle x_n | y \rangle + q(x_n, y)) \\ &= \lim_{n \rightarrow +\infty} (\langle x_n | y \rangle + \langle x_n | u(y) \rangle) = 0. \end{aligned}$$

As the space E_q is dense in $\tilde{E}_q$, we deduce that $x = 0$, as desired.

Identifying $\tilde{E}_q$ with its image under $\tilde{\iota}$ in E , we interpret $\tilde{q}$ as a closed positive partial form with dense domain extending q . The positive self-adjoint operator associated with $\tilde{q}$ is a self-adjoint extension of u . It is called the Friedrichs extension of u .

Note. — Let $c \in \mathbf{R}_+$. Let q be a closed, densely defined partial Hermitian form such that $q(x, x) \geq -c\|x\|^2$ for all x belonging to the domain of q . This means that the closed partial Hermitian form with domain $\text{dom}(q)$ defined by $\tilde{q}(x, y) = q(x, y) + c\langle x | y \rangle$ is a positive partial form. By Theorem 3, such forms therefore correspond to self-adjoint partial operators u on E such that $u + c1_E$ is positive, that is, such that u is bounded below (def. 7 of IV, p. 237) by $-c$.

Conversely, let u be a symmetric partial operator on E such that

$$\langle x | u(x) \rangle \geq -c\|x\|^2 \tag{17}$$

for all $x \in \text{dom}(u)$. Then $v = u + c1_E$ is a positive partial operator on E . The Friedrichs extension of u is the self-adjoint operator $\tilde{v} - c1_E$, where $\tilde{v}$ is the Friedrichs extension of v ; it is a self-adjoint extension of u that does not depend on the choice of the real number c satisfying (17).

IV.5.9 Variational principles for the spectrum of positive operators

In this number, E is a nonzero complex Hilbert space.

PROPOSITION 15. — Let u be a self-adjoint partial operator on E . Suppose that u is bounded below by $c \in \mathbf{R}$ (def. 7 of IV, p. 237). We have

$$\inf(\text{Sp}(u)) = \inf_{x \in \text{dom}(u) - \{0\}} \frac{\langle x | u(x) \rangle}{\|x\|^2} \in [c, +\infty[.$$

Let m be the right-hand side of the equality to be proved. Let $x \in E$ and μ_x be the spectral measure of x relative to u . We have

$$\begin{aligned} \langle x | u(x) \rangle &= \int_{\text{Sp}(u)} t d\mu_x(t) \\ &\geq \inf(\text{Sp}(u)) \int_{\text{Sp}(u)} d\mu_x(t) = \inf(\text{Sp}(u)) \|x\|^2, \end{aligned}$$

(formula (6), p. 271), hence $\inf(\text{Sp}(u)) \leq m$. Conversely, the partial operator $u - m$ is positive, therefore $\inf(\text{Sp}(u)) - m = \inf(\text{Sp}(u - m \cdot 1_E)) \geq 0$ (prop. 17 of IV, p. 248).

Proposition 9 of I, p. 139 corresponds to the particular case of this proposition when u is a Hermitian element of $\mathcal{L}(E)$, which is then necessarily bounded below. In this case, we also have

$$\sup(\text{Sp}(u)) = \sup_{x \in \text{dom}(u) - \{0\}} \frac{\langle x | u(x) \rangle}{\|x\|^2}$$

(loc. cit.); if u does not belong to $\mathcal{L}(E)$ then this supremum is $+\infty$.

DEFINITION 10. — Let u be a normal partial operator on E . A complex number $\lambda \in \text{Sp}(u)$ belongs to the sensitive spectrum of u if λ is isolated in $\text{Sp}(u)$ and if λ is an eigenvalue of finite multiplicity.

We denote by $\text{Sp}_s(u)$ the sensitive spectrum of u . Its complement in $\text{Sp}(u)$ is called the

essential spectrum of u and denoted $\text{Sp}_e(u)$.

The essential spectrum of a normal partial operator u on E is closed in $\mathbb{C}$, since $\text{Sp}(u)$ is closed in $\mathbb{C}$ and the complement of the essential spectrum is open in $\text{Sp}(u)$. The sensitive spectrum of u is not necessarily closed in $\mathbb{C}$ (Exercise 36 of IV, p. 369).

Lemma 9. — Let E be a complex Hilbert space and let u be a normal partial operator on E . Let $\lambda \in \text{Sp}(u)$. We have $\lambda \in \text{Sp}_s(u)$ if and only if there exists an open neighborhood V of λ in $\mathbb{C}$ such that the spectral projector p_V of u defined by V has finite rank.

Let $\lambda \in \text{Sp}_s(u)$. There exists an open neighborhood V of λ in $\mathbb{C}$ such that $\text{Sp}(u) \cap V = \{\lambda\}$; the spectral projector $p_V = p_{\{\lambda\}}$ is then of finite rank (cor. of prop. 9 of IV, p. 278).

Conversely, suppose that there exists an open neighborhood V of λ such that the spectral projector p_V has finite rank $n \in \mathbb{N}$. By the corollary of Proposition 10 of IV, p. 279, the intersection $V \cap \text{Sp}(u)$ contains at most n elements, therefore λ is isolated in $\text{Sp}(u)$. This is therefore an eigenvalue of u of spectral multiplicity at most n , by the corollary of prop. 9 of IV, p. 278.

In the rest of this number, we assume that E is infinite-dimensional; the analogues of the results below when E is finite-dimensional are deduced from No. 4 of IV, p. 153.

Let u be a self-adjoint operator bounded below on E . Let c be a lower bound of u ; the spectrum of u is contained in $[c, +\infty[$ (prop. 15). Suppose that the essential spectrum of u is not empty. We then denote by ϱ_e the lower bound of the essential spectrum of u , so that $\varrho_e \geq c$ and ϱ_e is an element of the spectrum of u . We denote $\text{Sp}_h(u) = \text{Sp}(u) \cap [\varrho_e, +\infty[$; it is a closed subset of $\text{Sp}(u)$, hence also of $\mathbb{C}$, such that $\inf(\text{Sp}_h(u)) = \varrho_e$; it is called the upper part of the spectrum of u . If the essential spectrum of u is empty, we denote $\text{Sp}_h(u) = \emptyset$ and $\varrho_e = +\infty$.

The intersection $\text{Sp}_b(u) = \text{Sp}(u) \cap [0, \varrho_e[$ is contained in the sensitive spectrum of u and its elements are therefore isolated eigenvalues of finite multiplicity; this is called the lower part of the spectrum of u . Let E_b be the closed subspace of E generated by the eigenspaces corresponding to the eigenvalues $\lambda \in \text{Sp}_b(u)$. By the cor. of prop. 9 of IV, p. 278 and prop. 8 of IV, p. 278, the space E_b is the image of the spectral projector $p_{\text{Sp}_b(u)}$ defined by $\text{Sp}_b(u)$. Since $\text{Sp}(u)$ is the disjoint union of $\text{Sp}_b(u)$ and $\text{Sp}_h(u)$, the orthogonal complement E_h of E_b is the image of the spectral projector $p_{\text{Sp}_h(u)}$ defined by $\text{Sp}_h(u)$. If $\text{Sp}_b(u)$ is finite, then the space E_b is

finite-dimensional; the space E_h is then nonzero, since E is assumed to be infinite-dimensional.

Lemma 10. — One has

$$\langle x | u(x) \rangle \geq \varrho_e \|x\|^2. \quad (18)$$

for all $x \in E_h \cap \text{dom}(u)$.

If $x \in E_h \cap \text{dom}(u)$, then the support of the spectral measure μ_x of x relative to u is contained in the interval $[\varrho_e, +\infty[$ (prop. 9, a) of IV, p. 278), hence

$$\langle x | u(x) \rangle = \int_{\mathbf{R}} t d\mu_x(t) \geq \varrho_e \|x\|^2.$$

Lemma 11. — Suppose that E_b is finite-dimensional. Then, for every real number $\varepsilon > 0$, the image of the spectral projector of u defined by $[\varrho_e, \varrho_e + \varepsilon]$ is infinite-dimensional.

If E_b is finite-dimensional, then the essential spectrum of u is nonempty, hence ϱ_e is finite and belongs to $\text{Sp}_e(u)$. Moreover, the lower spectrum $\text{Sp}_b(u)$ is finite, so there exists $\delta > 0$ such that $[\varrho_e - \delta, \varrho_e + \delta] \cap \text{Sp}(u) \subset [\varrho_e, \varrho_e + \delta]$. The assertion then follows from Lemma 9.

PROPOSITION 16. — The set $\text{Sp}_b(u) \subset [c, \varrho_e[$ is countable, discrete, and closed in $[0, \varrho_e[$. It is the set of values of a strictly increasing sequence $(\nu_k)_{0 \leq k < \text{Card}(\text{Sp}_b(u))}$ of positive real numbers; if $\text{Sp}_b(u)$ is infinite, then the sequence (ν_k) converges to ϱ_e .

Let $T = \text{Sp}_b(u) \cap [0, \varrho_e[$. It is a closed and discrete subset of $[c, \varrho_e[$ by definition. For every integer $i \geq 1$, the set $\text{Sp}_b(u) \cap [c, \varrho_e - 1/i]$ is compact and discrete, hence finite. Since T is the union of these sets for $i \geq 1$, the set T is countable.

This concludes the proof if $\text{Sp}_b(u)$ is finite. If $\text{Sp}_b(u)$ is infinite, one concludes by applying Lemma 2 of III, p. 90 to the image S of T under the map $\lambda \mapsto \varrho_e - \lambda$.

For every integer k such that $0 \leq k < \text{Card}(\text{Sp}_b(u))$, we denote by $n_k \geq 1$ the multiplicity of the eigenvalue ν_k of u . We denote $\overline{\mathbf{N}} = \mathbf{N} \cup \{+\infty\} \subset \overline{\mathbf{R}}$. Let $\mathbf{M} \in \overline{\mathbf{N}}$ be the sum of the multiplicities n_k . This is the Hilbertian dimension of E_b , if one agrees to say that a space of infinite Hilbertian dimension has dimension $+\infty \in \overline{\mathbf{N}}$.

For every integer n such that $0 \leq n < M$, one defines $\lambda_n(u) = \nu_k$, where $k \geq 0$ is the unique integer such that

$$n_0 + \dots + n_{k-1} \leq n < n_0 + \dots + n_k.$$

If $n \in \mathbb{N}$ satisfies $n \geq M$, we set $\lambda_n(u) = \varrho_e$. This case can occur only if $\text{Sp}_b(u)$ is finite, in which case ϱ_e is finite, since E is assumed to be of infinite dimension.

By construction, the sequence $(\lambda_n(u))_{n \in \mathbb{N}}$ is increasing and, for every element λ of $\text{Sp}_b(u)$, the number of integers n such that $\lambda_n(u) = \lambda$ is equal to the multiplicity of λ as an eigenvalue of u . The sequence $(\lambda_n(u))_{n \in \mathbb{N}}$ tends to $+\infty$ if and only if the essential spectrum of u is empty.

For every $n \in \mathbb{N}$, we denote by F_n (resp. F_n^u) the set of vector subspaces $F \subset E$ of dimension n (resp. the set of vector subspaces $F \subset \text{dom}(u)$ of dimension n).

Let $n \in \mathbb{N}$ such that $n < M$ and $F \in \mathcal{F}_n^u$. One says that F is adapted to u if F admits an orthonormal basis $(f_i)_{0 \leq i \leq n-1}$ such that $u(f_i) = \lambda_i(u)f_i$ for $0 \leq i \leq n-1$.

Let $F \in F_n^u$ adapted to u . The space F is contained in E_b ; it is generated by eigenvectors of u for eigenvalues $\lambda \leq \lambda_{n-1}(u)$, and it contains the eigenspaces corresponding to the eigenvalues $\lambda < \lambda_{n-1}(u)$. Moreover, for every universally measurable subset A of $\text{Sp}(u)$, the space F is stable under the spectral projector p_A defined by A (prop. 9, c) of IV, p. 278).

Lemma 12. — Let $F \in F_n^u$ be a subspace adapted to u . Every eigenvector of u belonging to the space $F^\circ \cap E_b$ has an eigenvalue $\geq \lambda_n(u)$.

Let ℓ be such that $\lambda_{n-1}(u) = \nu_\ell$. Let $x \in F^\circ \cap E_b$ be an eigenvector of u for an eigenvalue λ and let $k < \text{Card}(\text{Sp}_b(u))$ be such that $\lambda = \nu_k$.

The condition $k < \ell$ is impossible, because $\nu_k < \nu_\ell$, and the space F would then contain the eigenspace for the eigenvalue ν_k , contradicting the fact that x is orthogonal to F . If $k = \ell$, then x is an eigenvector for the eigenvalue $\lambda_{n-1}(u)$; since $x \in F^\circ$, the multiplicity n_k is strictly greater than the number of integers $i < n$ such that $\lambda_i(u) = \nu_k$, which implies that $\lambda_n(u) = \lambda_{n-1}(u)$. Finally, if $k > \ell$, we have $\lambda = \nu_k > \nu_\ell = \lambda_{n-1}(u)$, hence $\lambda \geq \lambda_n(u)$.

For every subspace F of E , we denote by

$$\tilde{r}_F(u) = \inf_{\substack{x \in \text{dom}(u) \cap F^\circ \\ x \neq 0}} \frac{\langle x | u(x) \rangle}{\|x\|^2},$$

$$\tilde{R}_F(u) = \sup_{\substack{x \in \text{dom}(u) \cap F \\ x \neq 0}} \frac{\langle x | u(x) \rangle}{\|x\|^2}.$$

PROPOSITION 17. — a) For every integer $n \in \mathbb{N}$, one has

$$\lambda_n(u) = \sup_{F \in \mathcal{F}_n} \tilde{r}_F(u) = \inf_{F \in \mathcal{F}_{n+1}^u} \tilde{R}_F(u);$$

b) For every integer $n < M$ and for every subspace $F \in \mathcal{F}_n^u$ adapted to u , we have $\lambda_n(u) = \tilde{r}_F(u)$;

c) For every integer $n < M$ and for every subspace $F \in \mathcal{F}_{n+1}^u$ adapted to u , we have $\lambda_n(u) = \tilde{R}_F(u)$.

There exists an orthonormal basis $(e_n)_{0 \leq n < M}$ of the space E_b such that $e_n \in \text{dom}(u)$ and $u(e_n) = \lambda_n(u)e_n$ for every n such that $0 \leq n < M$. For every $x \in E_b \cap \text{dom}(u)$, we therefore have

$$\langle x | u(x) \rangle = \sum_{0 \leq n < M} \lambda_n(u) |\langle e_n | x \rangle|^2.$$

For every integer n such that $1 \leq n < M + 1$, let F_n be the subspace of dimension n of E_b generated by $(e_0, \dots, e_{n-1})$. We have $F_n \subset \text{dom}(u)$ and F_n is adapted to u .

Let $n \in \mathbb{N}$ and $F \in \mathcal{F}_n$. Let us prove that $\tilde{r}_F(u) \leq \lambda_n(u)$ and, consequently, that

$$\sup_{F \in \mathcal{F}_n} \tilde{r}_F(u) \leq \lambda_n(u). \quad (19)$$

If $0 \leq n < M$, in particular if M is infinite, then the restriction to F_{n+1} of the orthogonal projection of E onto F is not injective, hence there exists $x \neq 0$ in F_{n+1} orthogonal to F . Since

$$\langle x | u(x) \rangle = \sum_{0 \leq i \leq n} \lambda_i(u) |\langle e_i | x \rangle|^2 = \lambda_n(u) |\langle e_n | x \rangle|^2 \leq \lambda_n(u) \|x\|^2,$$

we have $\tilde{r}_F(u) \leq \lambda_n(u)$.

If $M \in \mathbb{N}$ and $n \geq M$, then for every real number $\varepsilon > 0$, there exists x of norm 1 in $\text{dom}(u)$ which is orthogonal to F and whose spectral measure μ_x has support contained in $[\varrho_e, \varrho_e + \varepsilon]$

(lemma 11 and prop. 9, a) of IV, p. 278), whence

$$\tilde{r}_F(u) \leq \langle x | u(x) \rangle = \int_{\text{Sp}(u)} t d\mu_x(t) \leq \varrho_e + \varepsilon = \lambda_n(u) + \varepsilon$$

by definition. Since $\varepsilon > 0$ is arbitrary, we therefore have $\tilde{r}_F(u) \leq \lambda_n(u)$. Inequality (19) is therefore established.

Let $n \in \mathbf{N}$ and $F \in \mathcal{F}_{n+1}^u$. Let us prove that $\tilde{R}_F(u) \geq \lambda_n(u)$ and, consequently, that

$$\inf_{F \in \mathcal{F}_{n+1}^u} \tilde{R}_F(u) \geq \lambda_n(u). \quad (20)$$

If $0 \leq n < M$, observe that the restriction to F of the orthogonal projector onto F_n is not injective, hence there exists a vector $x \neq 0$ in F orthogonal to F_n . Set $x_b = p_{\text{Sp}_b(u)}(x)$ and $x_h = p_{\text{Sp}_h(u)}(x)$. We therefore have $x = x_b + x_h$. The elements x_b and x_h belong to the domain of u (prop. 9, c) of IV, p. 278) and are orthogonal to F_n . We have the lower bound

$$\langle x_b | u(x_b) \rangle = \sum_{i \geq n} \lambda_i(u) |\langle e_i | x_b \rangle|^2 \geq \lambda_n(u) \|x_b\|^2,$$

and $\langle x_h | u(x_h) \rangle \geq \varrho_e \|x_h\|^2$ (formula (18)), whence

$$\begin{aligned} \langle x | u(x) \rangle &= \langle x_b | u(x_b) \rangle + \langle x_h | u(x_h) \rangle \\ &\geq \lambda_n(u) \|x_b\|^2 + \varrho_e \|x_h\|^2 \geq \lambda_n(u) \|x\|^2. \end{aligned}$$

If M is finite and $n \geq M$, there exists $x \neq 0$ in F orthogonal to E_b , hence $x \in E_h$, and

$$\langle x | u(x) \rangle \geq \varrho_e \|x\|^2 = \lambda_n(u) \|x\|^2$$

by (18). Inequality (20) is therefore proved.

We shall now prove the assertions b) and c), which imply that inequalities (19) and (20) are equalities when $n < M$.

Let us prove assertion b). Let $F \in \mathcal{F}_n^u$ be a space adapted to u . We have the Hilbert sum

$$F^\circ = E_h \oplus \bigoplus_{\substack{\lambda \in \text{Sp}_b(u) \\ \lambda > \lambda_{n-1}(u)}} \text{Ker}(u - \lambda \cdot 1_E) \oplus (F^\circ \cap \text{Ker}(u - \lambda_{n-1}(u) \cdot 1_E)).$$

Let $x \in F^\circ - \{0\}$. Let us write $x = x_h + y + z$, where $x_h \in E_h$ and y (resp. z) belongs to the second (resp. third) space of the above decomposition. Using (18) again and the fact that every eigenvalue $\lambda > \lambda_{n-1}(u)$ of u is $\geq \lambda_n(u)$, we obtain

$$\begin{aligned}\langle x | u(x) \rangle &= \langle x_h | u(x_h) \rangle + \langle y | u(y) \rangle + \langle z | u(z) \rangle \\ &\geq \varrho_e \|x_h\|^2 + \lambda_n(u) \|y\|^2 + \lambda_{n-1}(u) \|z\|^2.\end{aligned}$$

If $z \neq 0$, then by Lemma 12, we have $\lambda_n(u) = \lambda_{n-1}(u)$. We therefore deduce that $\langle x | u(x) \rangle \geq \lambda_n(u) \|x\|^2$, whence $\tilde{r}_F(u) \geq \lambda_n(u)$. Combined with (19), this implies the assertion *b*).

Let us prove assertion *c*). Let $F \in \mathcal{F}_{n+1}^u$ be a subspace adapted to u . Let $(f_i)_{0 \leq i \leq n}$ be an orthonormal basis of F such that $u(f_i) = \lambda_i(u) f_i$ for $0 \leq i \leq n$. For every $x \in F$, we have

$$\langle x | u(x) \rangle = \sum_{0 \leq i \leq n} \lambda_i(u) |\langle f_i | x \rangle|^2 \leq \lambda_n(u) \|x\|^2,$$

with equality if $x = f_n$, hence $\tilde{R}_F(u) = \lambda_n(u)$.

Let us finally prove that (19) and (20) are equalities when $n \geq M$. In this case, M is finite, hence E_b is contained in the domain of u ; moreover, we have $\lambda_n(u) = \varrho_e$ by definition.

There exists $F \in \mathcal{F}_n$ containing E_b . Every element $x \neq 0$ of $\text{dom}(u)$ orthogonal to F is therefore orthogonal to E_b , and satisfies

$$\frac{\langle x | u(x) \rangle}{\|x\|^2} \geq \varrho_e$$

(formula (18)), hence $\tilde{r}_F(u) \geq \varrho_e = \lambda_n(u)$, and consequently

$$\sup_{F \in \mathcal{F}_n} \tilde{r}_F(u) \geq \lambda_n(u).$$

Let $\varepsilon > 0$. Since E_b is finite-dimensional, there exists by Lemma 11 an orthonormal family $(x_i)_{i \in I}$ of elements of E such that the subspace F generated by E_b and $(x_i)_{i \in I}$ has dimension $n+1$ and is contained in $\text{dom}(u)$, and such that $\langle x_i | u(x_i) \rangle \leq \varrho_e + \varepsilon$ for all $i \in I$. We then have $\tilde{R}_F(u) \leq \varrho_e + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, we conclude that

$$\inf_{F \in \mathcal{F}_{n+1}^u} \tilde{R}_F(u) \leq \varrho_e = \lambda_n(u).$$

The proposition is proved.

IV.5.10 Compact perturbation and essential spectrum

In this number, E is an infinite-dimensional complex Hilbert space.

Lemma 13. — Let I be an orthonormal family in E . The family I converges weakly to 0 along the filter of complements of finite subsets of I .

Let $x \in E$. By Bessel's inequality (EVT, V, p. 21, prop. 4) and TG, IV, p. 37, th. 1, the family $(|\langle e_i | x \rangle|^2)_{i \in I}$ is summable, hence $\langle e_i | x \rangle$ tends to 0 along the filter of complements of finite subsets of I (TG, III, p. 38, prop. 1).

PROPOSITION 18. — Let u be a self-adjoint partial operator on E and let λ be a real number. We have $\lambda \in \text{Sp}_e(u)$ if and only if there exists an orthonormal sequence $(x_n)_{n \in \mathbb{N}}$ in E such that $u(x_n) - \lambda x_n$ tends to 0 in E .

Suppose first that $\lambda \in \text{Sp}_e(u)$. If the spectral projector of u relative to $\{\lambda\}$ is of infinite rank, every orthonormal sequence (x_n) in its image answers the question (cf. cor. of prop. 9 of IV, p. 278). In what follows we shall assume that this is not the case.

For every $k \in \mathbb{N}$, let J_k be the set of $t \in [\lambda - 1, \lambda + 1]$ such that

$$\frac{1}{k+2} < |t - \lambda| \leq \frac{1}{k+1}.$$

The sets J_k are pairwise disjoint. Moreover, for every integer $K \in \mathbb{N}$, the sets $(J_k)_{k \geq K}$ form a partition of the set

$$I_K = [\lambda - 1/(K+1), \lambda + 1/(K+1)] - \{0\}.$$

Since the spectral projector of u relative to $I_K \cup \{0\}$ has infinite rank (lemma 9 of IV, p. 297) and the one relative to $\{\lambda\}$ is assumed to have finite rank, it follows from prop. 8 of IV, p. 278 that there exists a sequence $(k_n)_{n \in \mathbb{N}}$ strictly increasing in $\mathbb{N}$ such that the spectral projector p_n of u relative to J_{k_n} is nonzero. Let x_n be a vector of norm 1 in the image of p_n . The sequence (x_n) is orthonormal, since the image of p_n is orthogonal to that of p_m for all $n \neq m$ in $\mathbb{N}$.

Let $n \in \mathbb{N}$. Denote by μ_n the spectral measure of x_n relative to u ; its support is contained

in J_{k_n} (prop. 9 of IV, p. 278). It follows

$$\|u(x_n) - \lambda x_n\|^2 = \int_{\mathbf{C}} |t - \lambda|^2 d\mu_n(t) \leq \frac{1}{k_n^2},$$

therefore the sequence $(x_n)_{n \in \mathbf{N}}$ has the required property.

Suppose conversely that there exists an orthonormal sequence $(x_n)_{n \in \mathbf{N}}$ in E such that $u(x_n) - \lambda x_n \rightarrow 0$. Denote by μ_n the spectral measure of x_n relative to u .

Let $\varepsilon > 0$. Denote by p_ε be the spectral projector of u relative to the open interval $I_\varepsilon =]\lambda - \varepsilon, \lambda + \varepsilon[$. For every $n \in \mathbf{N}$, it follows that

$$\begin{aligned} 1 = \|x_n\|^2 &= \mu_n(I_\varepsilon) + \mu_n(\mathbf{R} - I_\varepsilon) \\ &\leq \mu_n(I_\varepsilon) + \frac{1}{\varepsilon^2} \int_{\mathbf{R} - I_\varepsilon} |t - \lambda|^2 d\mu_n(t) \\ &= \|p_\varepsilon(x_n)\|^2 + \frac{1}{\varepsilon^2} \|u(x_n) - \lambda x_n\|^2. \end{aligned}$$

The hypothesis on the sequence (x_n) therefore implies that the norm of $p_\varepsilon(x_n)$ cannot tend to 0. Since the orthonormal sequence (x_n) converges weakly to 0 in E (Lemma 13), the projector p_ε cannot be compact (cor. of prop. 6 of III, p. 6) and is consequently of infinite rank. Since this holds for every $\varepsilon > 0$, lemma 9 of IV, p. 297 allows us to conclude that $\lambda \in \text{Sp}_e(u)$.

The following theorem is the analogue for self-adjoint partial operators of Theorem 3 in III, p. 93.

THEOREM 4. — Let u be a self-adjoint partial operator on E . The essential spectrum of u is the intersection of the sets $\text{Sp}(u + v)$ where v ranges over the set of compact Hermitian endomorphisms of E .

The partial operator $u + v$ is self-adjoint for every Hermitian endomorphism v of E since $(u + v)^* = u^* + v^*$ (cf. IV, p. 236).

Let us prove that if v is compact, then $\text{Sp}_e(u) \subset \text{Sp}_e(u + v)$. Let $\lambda \in \text{Sp}_e(u)$ and let $(x_n)_{n \in \mathbf{N}}$ be an orthonormal sequence in E such that $u(x_n) - \lambda x_n$ converges to 0 (prop. 18). The sequence (x_n) converges weakly to 0 in E (lemma 13), and since the endomorphism v is compact, the sequence $(v(x_n))_{n \in \mathbf{N}}$ converges to 0 in E (cor. of prop. 6 of III, p. 6). Consequently, the sequence $(u + v)(x_n) - \lambda x_n$ converges to 0 in E , and Prop. 18 implies that $\lambda \in \text{Sp}_e(u + v)$.

The essential spectrum of u is therefore contained in the intersection of the sets $\text{Sp}(u + v)$

for Hermitian $v \in \mathcal{L}^c(E)$.

Conversely, let $\lambda \in \text{Sp}_s(u)$. Denote by E_λ the eigenspace of u relative to λ , and p_λ the orthogonal projector with image E_λ ; it is the spectral projector of u relative to $\{\lambda\}$ (corollary of Prop. 9 of IV, p. 278). By definition of the sensitive spectrum, the projector p_λ is of finite rank, hence compact. Set $w = u + p_\lambda$; it is a self-adjoint partial operator. To conclude the proof, let us verify that λ belongs to the resolvent set of w .

One has $E_\lambda \subset \text{dom}(u)$ and the spaces E_λ and $\text{dom}(u) \cap E_\lambda^\circ$ are stable under u (prop. 9 of IV, p. 278).

Let $x \in \text{dom}(u)$. Let us write $x = p_\lambda(x) + y$ where $y \in \text{dom}(u) \cap E_\lambda^\circ$. From what precedes, we have

$$\|w(x) - \lambda x\|^2 = \|p_\lambda(x)\|^2 + \|w(y) - \lambda y\|^2. \quad (21)$$

Let V be an open neighborhood of λ not meeting the spectrum of u , and let $c > 0$ be such that the disk with center λ and of radius c is contained in V . Let μ_y be the spectral measure of y relative to u . Since y is orthogonal to E_λ , the support of μ_y does not meet V (loc. cit.). We then have

$$\|w(y) - \lambda y\|^2 = \|u(y) - \lambda y\|^2 = \int_{\mathbb{C}} |t - \lambda|^2 d\mu_y(t) \geq c^2 \|y\|^2. \quad (22)$$

We conclude from (21) and (22) that $\|w(x) - \lambda x\|^2 \geq \inf(c^2, 1) \|x\|^2$, since w is self-adjoint and $\lambda \in R$. The conclusion then follows from Lemma 5 of IV, p. 248.

COROLLARY. — Let u be a self-adjoint partial operator on E and v a compact Hermitian endomorphism of E . We have $\text{Sp}_e(u + v) = \text{Sp}_e(u)$.

This follows immediately from the theorem.

IV.5.11 Perturbation

In this number, E is a complex Hilbert space.

If u is a self-adjoint partial operator on E and $v \in \mathcal{L}(E)$ is a Hermitian endomorphism, then $u + v$ is self-adjoint (cf. IV, p. 236). This is not the case in general when v is a symmetric partial

operator (exercise 9 of IV, p. 347). We shall nevertheless obtain positive results of this type in this number.

DEFINITION 11. — Let u be a partial operator on E . A partial operator v on E is said to be bounded relative to u if $\text{dom}(u) \subset \text{dom}(v)$ and if v defines, by restriction to the subspace, a continuous linear map from E_u into E .

Let u be a partial operator on E . Let v be a partial operator bounded relative to u . There exists a real number m such that

$$\|v(x)\| \leq m(\|x\| + \|u(x)\|)$$

for all $x \in \text{dom}(u)$. The relative norm of v with respect to u , denoted by $\|v\|_u$, is the lower bound of the real numbers $a \geq 0$ such that there exists a real number b such that

$$\|v(x)\| \leq a\|u(x)\| + b\|x\|$$

for all $x \in \text{dom}(u)$. The relative norm of v is therefore less than the norm of the restriction of v to the space E_u .

Note. — Let u be a partial operator on E . Every endomorphism $v \in \mathcal{L}(E)$ is bounded relative to u and its relative norm is zero since $\|v(x)\| \leq \|v\|\|x\|$ for all $x \in \text{dom}(u)$, which allows us to take $a = 0$ in the inequality above.

THEOREM 5 (Kato–Rellich). — Let u be a self-adjoint partial operator on E and v a symmetric partial operator bounded relative to u . If the relative norm $\|v\|_u$ is < 1 , then the partial operator $u + v$ whose domain is $\text{dom}(u)$ is self-adjoint.

Since $\|v\|_u < 1$ there exist, by definition, positive real numbers a and b such that $a < 1$ and $\|v(x)\| \leq a\|u(x)\| + b\|x\|$ for all $x \in \text{dom}(u)$.

Let $t \in \mathbb{R}^*$. Set $w_t = v \circ R(u, it)$. We have $\text{dom}(w_t) = E$ since the image of $R(u, it)$ is contained in the domain of u , which is contained in the domain of v by hypothesis. Let $x \in \text{dom}(u)$. Since u is self-adjoint, we have

$$\begin{aligned} \|(it - u)x\|^2 &= \|itx\|^2 + \|u(x)\|^2 - it\langle x|u(x)\rangle + it\langle u(x)|x\rangle \\ &= |t|^2\|x\|^2 + \|u(x)\|^2, \end{aligned}$$

whence the inequalities $\|u(x)\| \leq \|(it - u)x\|$ and $\|x\| \leq |t|^{-1} \|(it - u)x\|$. But then it follows that

$$\|v(x)\| \leq (a + b|t|^{-1}) \|(it - u)x\|.$$

In particular, set $x = R(u, it)y$ for $y \in E$; we obtain

$$\|w_t(y)\| \leq (a + b|t|^{-1}) \|y\|.$$

We deduce that $w_t \in \mathcal{L}(E)$ and that $\|w_t\| \leq a + |t|^{-1}b$.

Since $a < 1$, there exists $t \in \mathbf{R}_+^*$ such that $a + b|t|^{-1} < 1$. We then have $\|w_t\| < 1$ and $\|w_{-t}\| < 1$; the endomorphisms $1_E - w_t$ and $1_E - w_{-t}$ of E are therefore invertible (prop. 2 of I, p. 22). Since we have the formula $(1_E - w_t) \circ (it - u) = it - (u + v)$, this implies that $u + v - it$ is surjective; likewise, the partial operator $u + v + it$ is surjective. It follows that $u + v$ is self-adjoint (prop. 11 of IV, p. 240).

IV.5.12 Operators with compact resolvent

In this number, E denotes a complex Hilbert space of infinite dimension.

PROPOSITION 19. — Let u be a self-adjoint partial operator on E . The following conditions are equivalent:

- (i) There exists an orthonormal basis $B = (e_j)_{j \in J}$ of E such that u is diagonal in the basis B and the absolute value of the family of eigenvalues of u tends to infinity along the filter of complements of finite subsets of J ;
- (ii) For every λ belonging to the resolvent set of u , the resolvent $R(u, \lambda)$ is compact;
- (iii) There exists a complex number λ belonging to the resolvent set of u such that the resolvent $R(u, \lambda)$ is compact;
- (iv) The spectrum of u coincides with the sensitive spectrum of u .

Suppose that (i) is satisfied and let $(\lambda_j)_{j \in J}$ be the family of eigenvalues of u . Let μ be the

counting measure on J . The spectrum of u is the image μ -essential of the family (λ_j) (prop. 22 of IV, p. 252); it is the set of values of this family. For every $\lambda \notin \text{Sp}(u)$, the resolvent $R(u, \lambda)$ is diagonal in the basis B and the family of its eigenvalues is $((\lambda - \lambda_j)^{-1})_{j \in J}$ (loc. cit.). Since this family converges to 0, the endomorphism $R(u, \lambda)$ is compact (prop. 2, (iii) of IV, p. 148). This proves that (i) implies (ii).

Since the resolvent set of u is nonempty (cf. prop. 17 of IV, p. 248), properties (ii) and (iii) are equivalent by formula (8) of IV, p. 245 and proposition 3 of III, p. 5.

Since $R(u, \lambda)$ is normal for $\lambda \notin \text{Sp}(u)$ (prop. 16 of IV, p. 247), condition (iii) implies (iv) by prop. 15 of IV, p. 247 and prop. 5 of III, p. 90.

Finally, let us prove that (iv) implies (i). Let $\mathcal{O}$ be the set of orthonormal subsets of E consisting of eigenvectors for u . The set $\mathcal{O}$, ordered by inclusion, is of finite character (E, III, p. 34, def. 2) since O belongs to $\mathcal{O}$ if and only if the sets formed of at most two elements of O belong to $\mathcal{O}$. By E, III, p. 35, th. 1, there exists a maximal element O of $\mathcal{O}$. Let F be the closed subspace of E generated by O . For $e \in O$, there exists a unique $\lambda(e) \in \mathbf{R}$ such that $u(e) = \lambda(e)e$ (prop. 17 of IV, p. 248).

The set of values of the map λ from O to $\mathbf{R}$ coincides with the spectrum of u . Indeed, on the one hand this set is contained in the spectrum of u and on the other hand, if there exists $\lambda_0 \in \text{Sp}(u)$ which is not a value of λ , then λ_0 is an eigenvalue of u by hypothesis. There exists a vector $e \in \text{dom}(u)$ of norm 1 such that $u(e) = \lambda_0 e$ and $e \notin O$; the subset $O \cup \{e\}$ is orthonormal (since eigenspaces of u corresponding to distinct eigenvalues are orthogonal by assertion *b*) of the cor. of prop. 17 of IV, p. 248); it belongs to $\mathcal{O}$, contradicting the maximality of O .

Suppose that $F \neq E$. Then F° is nonzero. The endomorphism $R(u, i)$ of E is normal (prop. 16 of IV, p. 247). It leaves the subspace F° of E stable (lemma 4 of I, p. 135). Let v be the endomorphism of F° deduced from $R(u, i)$ by passing to subspaces. It is a continuous and normal endomorphism of F° (loc. cit.) whose spectrum is contained in that of $R(u, i)$. Since F° is nonzero, the spectrum of v is nonempty (cor. 1 of I, p. 26). Moreover, the spectrum of v is not reduced to 0, since the normal endomorphism v is nonzero (example 1 of I, p. 110). Let $s \in \text{Sp}(v) - \{0\}$. Since s belongs to the spectrum of $R(u, i)$, there exists $e \in O$ such that

$s = (i - \lambda(e))^{-1}$, and s is an eigenvalue of $R(u, i)$ (prop. 15, a) of IV, p. 247). Since s is nonzero, it is an isolated point of $\text{Sp}(R(u, i))$ by hypothesis, hence also of $\text{Sp}(v)$. Thus, s is an eigenvalue of v (prop. 5, c) of I, p. 134). Let $e \in F^\circ$ be an eigenvector of norm 1 of v ; it is also an eigenvector of u (prop. 15, b) of IV, p. 247), and the set $O \cup \{e\}$ contradicts the fact that O is maximal in O . Hence $F = E$.

The family of elements of O is therefore an orthonormal basis of E consisting of eigenvectors of u . The spectrum of u is closed in $\mathbb{R}$, and the sensitive spectrum is discrete; since these sets coincide, the set of elements $\lambda \in \text{Sp}(u)$ such that $|\lambda| \leq R$ is compact, hence finite, for every $R > 0$. Thus the family of absolute values of the eigenvalues of u tends to infinity along the filter of complements of finite subsets of O . Hence (iv) implies (i).

DEFINITION 12. — Let u be a self-adjoint partial operator on E . One says that u has compact resolvent if the equivalent conditions of Prop. 19 are satisfied.

PROPOSITION 20. — Let u be a self-adjoint partial operator on E . Then u has compact resolvent if and only if the canonical injection j of the Hilbert space E_u into E is compact.

Suppose that u has compact resolvent. There exists $\lambda \notin \text{Sp}(u)$ such that $R(u, \lambda)$ is compact (prop. 19, (iii)). Let B be the unit ball of the Hilbert space E_u . Since u is a continuous linear map from E_u into E , the subset $B' = (\lambda 1_E - u)(B)$ of E is bounded. Since $R(u, \lambda)$ is compact, the subset $B = R(u, \lambda)(B')$ is relatively compact in E (Remark 1 of III, p. 2). This proves that j is compact.

Conversely, suppose that the linear map j is compact. The complex number i belongs to the resolvent set of u (prop. 17 of IV, p. 248). We have $u \circ R(u, i) = -1_E + iR(u, i)$. Let B be the unit ball of E . For every $x \in B$, we have

$$\|u \circ R(u, i)(x)\| = \|-x + iR(u, i)(x)\| \leq 1 + \|R(u, i)\|,$$

and consequently

$$\begin{aligned} \|R(u, i)x\|_u^2 &= \|R(u, i)x\|^2 + \|u \circ R(u, i)x\|^2 \\ &\leq \|R(u, i)\|^2 + (1 + \|R(u, i)\|)^2. \end{aligned}$$

The image C of B under $R(u, i)$ is therefore bounded in E_u . Since j is compact by hypothesis,

the set $C = j(C)$ is relatively compact in E (III, loc. cit.). Consequently, the resolvent $R(u, i)$ is compact and u has compact resolvent (prop. 19, (iii)).

COROLLARY. — Let q be a closed positive partial form on E and u the positive self-adjoint operator representing q . The following conditions are equivalent:

- (i) The partial operator u has compact resolvent;
- (ii) The positive endomorphism $\sqrt{(1_E + u)^{-1}} = (1_E + u)^{-1/2}$ of E is compact;
- (iii) The canonical injection j of the Hilbertian space E_q (def. 9 of IV, p. 292) in E is compact.

Since u is positive, the real number -1 belongs to the resolvent set of u (prop. 17 of IV, p. 248), hence the positive endomorphism $v = \sqrt{(1_E + u)^{-1}}$ is defined, and we have $v = (1_E + u)^{-1/2}$ by the functional calculus.

The endomorphism v is compact if and only if $v^2 = (1_E + u)^{-1}$ is compact (prop. 6 of III, p. 91), that is, if and only if u has compact resolvent (prop. 19, (iii)). This proves that conditions (i) and (ii) are equivalent.

Suppose that the endomorphism v is compact. Let B be the unit ball of the Hilbert space E_q . Since $(1_E + u)^{1/2}$ is a continuous linear map from E_q into E (cor. of th. 3 of IV, p. 294), the subset $B' = (1_E + u)^{1/2}(B)$ of E is bounded, hence the subset $j(B) = B = v(B')$ is relatively compact in E (Remark 1 of III, p. 2). This proves that j is compact. Thus (ii) implies (iii).

The linear map $\tilde{v}: x \mapsto (1_E + u)^{-1/2}(x)$ from E into E_q is well defined and isometric (corollary of Th. 3 of IV, p. 294). Since $v = j \circ \tilde{v}$, condition (iii) implies (ii) (Prop. 3 of III, p. 5).

Example. — Let $n \in \mathbb{N}$. Let U be an open subset of $\mathbb{R}^n$. We equip U with Lebesgue measure, denoted μ . Let Δ be the scalar differential operator $-\sum_{i=1}^n \partial_i^2$ on U . The partial operator Δ_- with domain $\mathcal{D}(U)$ defined by $\varphi \mapsto \Delta(\varphi)$ is closable (prop. 13 of IV, p. 242) and symmetric (IV, p. 243). It is positive, since for every $\varphi \in \mathcal{D}(U)$, we have

$$\begin{aligned} \langle \varphi | \Delta_-(\varphi) \rangle &= \int_U \overline{\varphi} \Delta(\varphi) d\mu = - \sum_{i=1}^n \int_U \overline{\varphi} \partial_i^2 \varphi d\mu \\ &= \sum_{i=1}^n \int_U \overline{\partial_i \varphi} \partial_i \varphi d\mu = \int_U \sum_{i=1}^n |\partial_i \varphi|^2 d\mu \geq 0. \end{aligned}$$

We denote by Δ_D the Friedrichs extension of the positive symmetric partial operator Δ_- (IV, p. 295, example); it is a Laplacian on U , called the Dirichlet Laplacian on U .

Let q be the positive partial form associated with Δ_D . The domain of q is the Hilbertian space completion of $\mathcal{D}(U)$ for the positive-definite Hermitian form defined by

$$(\varphi_1, \varphi_2) \mapsto \int_U \overline{\varphi_1} \varphi_2 + \sum_{i=1}^n \int_U \overline{\partial_i \varphi_1} \partial_i \varphi_2$$

for all $(\varphi_1, \varphi_2) \in \mathcal{D}(U) \times \mathcal{D}(U)$. In other words, the domain of q is the Sobolev space $H_0^1(U)$ (No. 14 of IV, p. 221).

*Suppose that U is bounded. The canonical injection of $H_0^1(U)$ into $L^2(U)$ is compact; the Dirichlet Laplacian on U is therefore an operator with compact resolvent (corollary of Prop. 20). Since the Hilbert space $H_0^1(U)$ is of countable type (prop. 20 of IV, p. 222), and since the image of Δ_D is of infinite dimension, there exists an increasing sequence $(\lambda_n)_{n \geq 0}$ of real numbers tending to $+\infty$ and an orthonormal basis $(f_n)_{n \in \mathbb{N}}$ of $L^2(U)$ whose elements belong to the domain of Δ_D , such that $\Delta_D(f_n) = \lambda_n f_n$ for all $n \in \mathbb{N}$. One can prove (“Weyl’s law”) that when T tends to $+\infty$, we have

$$\sum_{\substack{n \in \mathbb{N} \\ \lambda_n \leq T}} 1 \sim \frac{c_n}{(2\pi)^n} m T^{n/2}$$

where $c_n = \pi^{n/2} / \Gamma(1 + n/2)$ is the volume of the unit ball in R^n (INT, V, p. 101, § 8, no. 7) and $m > 0$ is the Lebesgue measure of U .*

Exercises

Exercises for Section 1

- 1) Let $E = \ell_C^2(\mathbf{N})$ and let $u \in \mathcal{L}(E)$ be the endomorphism such that $u(x_0, \dots, x_n, \dots) = (0, x_0, \dots, x_n, \dots)$ for all $(x_n)_{n \in \mathbf{N}}$ in E . Let (e_n) be the canonical orthonormal basis of E .
 - a) For every $v \in \mathcal{L}(E)$ such that $\|v\| < 1/2$, the vector e_0 is not adherent to the image of $u + v$.
 - b) The open ball in $\mathcal{L}(E)$ of radius $1/2$ centered at u contains no diagonalizable endomorphism of E .
- 2) Let E be a complex Hilbert space. Prove directly that an endomorphism u of E is a Fredholm endomorphism if and only if $\pi(u)$ is invertible in $\mathcal{Calk}(E)$ (Atkinson's theorem). (Use the spectral properties of compact operators, cf. § 1 of IV, p. 146.)
- 3) Let E be a complex Hilbert space and let $B = (e_i)_{i \in I}$ be an orthonormal basis of E . For every $i \in I$, we denote by p_i the orthogonal projector with image Ce_i .
 - a) Let K be a precompact subset of E . For every $\varepsilon > 0$ there exists a finite subset J of I such that, for every $x \in K$, $\|x - \sum_{i \in J} p_i(x)\| \leq \varepsilon$.
 - b) Let $u \in \mathcal{D}_B(E)$ and let us denote by $\lambda = (\lambda_i)_{i \in I}$ its family of eigenvalues. The family $(\lambda_i p_i)$ is summable in $\mathcal{L}(E)$ endowed with the topology of precompact convergence and whose sum is equal to u (cf. prop. 1 of IV, p. 147)
- 4) Let E be a complex Hilbert space of infinite dimension. There exists a compact endomorphism of E that is not of finite trace.

- 5) a) Let $E = L^2([0, 1], \mu)$ where μ is the Lebesgue measure. The image A of the algebra $L^\infty([0, 1], \mu)$ by the map that associates to f the endomorphism of multiplication by f is a maximal commutative subalgebra of $\mathcal{L}(E)$ which is not of the form $\mathcal{D}_B(E)$ for an orthonormal basis B of E . (Verify that the algebra A contains no minimal projector.)
- b) Let E be an infinite-dimensional Hilbert space. There exists a maximal commutative subalgebra of $\mathcal{L}(E)$ which is not of the form $\mathcal{D}_B(E)$ for B an orthonormal basis of E .
- 6) Let E be a complex Hilbert space.
- a) Let u be a compact normal endomorphism of E . Every closed subspace of E invariant under u is invariant under u^* .
- b) If E is infinite-dimensional, there exist a normal endomorphism u of E and a closed subspace $F \subset E$ such that:
- (i) One has $u(F) \subset F$;
 - (ii) The space F is not invariant under u^* ;
 - (iii) The space F° is not invariant under u ;
 - (iv) The endomorphism of F induced by u by passing to subspaces is not normal.
- 7) The goal of this exercise is to give a more direct proof of Theorem 1 of IV, p. 149. Let E be an infinite-dimensional complex Hilbert space and let u be a compact and normal endomorphism of E .
- a) Show that, for every nonzero complex number λ , the eigenspace $\text{Ker}(u - \lambda 1_E)$ is of finite dimension.
- b) Suppose that u is Hermitian. Show that there exists $x_0 \neq 0$ in E such that

$$\frac{\|u(x_0)\|^2}{\|x_0\|^2} = \sup_{x \in E - \{0\}} \frac{\|u(x)\|^2}{\|x\|^2}.$$

- c) Show that x_0 is an eigenvector of u for an eigenvalue λ such that $|\lambda| = \|u\|$.
- d) Show that there exists a countable set I , an orthonormal family $(e_i)_{i \in I}$ in E and a family $(\lambda_i)_{i \in I}$ of real numbers such that

$$u(x) = \sum_{i \in I} \lambda_i \langle e_i | x \rangle e_i$$

for all $x \in E$.

- e) Extend the result to the case where u is compact and normal. (Write $u = u_1 + iu_2$ where u_1 and u_2 are compact, Hermitian, and mutually commuting.)
- 8) We endow the group $\mathbf{R}/\mathbf{Z}$ with its normalized Haar measure μ .
- a) Let $h \in \mathcal{L}^2(\mathbf{R}/\mathbf{Z})$. For $f \in \mathcal{L}^2(\mathbf{R}/\mathbf{Z})$ we define a function $C_h(f)$ on $\mathbf{R}/\mathbf{Z}$ by the formula

$$C_h(f)(x) = \int_{\mathbf{R}/\mathbf{Z}} f(t) h(x-t) d\mu(t).$$

The map $f \mapsto C_h(f)$ is a continuous endomorphism of $\mathcal{L}^2(\mathbf{R}/\mathbf{Z})$, and by passing to quotients defines a compact normal endomorphism of $L^2(\mathbf{R}/\mathbf{Z})$.

- b) Determine the spectrum of C_h and the spectral multiplicity of the elements of this spectrum. (Expand h in a Fourier series.)
- c) There exists $h \in \mathcal{C}(\mathbf{R}/\mathbf{Z})$ such that C_h is not of finite trace.
- 9) We denote by E the complex Hilbert space $L^2([0, 1])$ of functions that are square-integrable on $[0, 1]$ with respect to Lebesgue measure μ .

Let $u_0 \in \mathcal{L}(E)$ be the Volterra endomorphism on E , such that

$$u_0(f)(x) = \int_{[0, x]} f(t) d\mu(t)$$

for all $f \in L^2([0, 1])$ and almost all $x \in [0, 1]$.

- a) The endomorphism u_0 is compact and $\text{Sp}(u_0) = \{0\}$.

- b) The endomorphism $u_0^* u_0$ has finite trace; compute its trace and deduce the value of the series

$$\sum_{n=1}^{+\infty} \frac{1}{n^2}.$$

- c) Let v_0 be the rank-1 endomorphism of E defined by

$$v_0(f) = \left(\int_0^1 f(t) d\mu(t) \right) \cdot 1,$$

where 1 denotes the class of the constant function 1 . Let $u = u_0 - v_0$. Show that u is a Hilbert–Schmidt operator, and compute the kernel N such that

$$u(f)(x) = \int_0^1 N(x, y) f(y) d\mu(y)$$

for $f \in E$ and almost all $x \in [0, 1]$.

- d) Determine the eigenvalues of u and their spectral multiplicity. (Compute $u(e_n)$, where $e_n(t) = e^{2i\pi nt}$ for all $n \in \mathbb{Z}$.)
- e) Compute the trace of $u^* u$ and deduce anew the value of the series whose general term is $1/n^2$ for $n \geq 1$.
- f) Generalize these arguments to show that, if $k \geq 1$ is a strictly positive integer, then

$$\sum_{n=1}^{+\infty} \frac{1}{n^{2k}} \in \pi^{2k} \mathbf{Q}^*.$$

- 10) Let E be a complex Hilbert space. Let A be a star-subalgebra of $\mathcal{L}(E)$. Suppose that the representation of A in E is irreducible and that A contains a nonzero compact endomorphism.

- a) The star-subalgebra A contains a nonzero finite-rank projector. (Show that A contains a nonzero compact Hermitian endomorphism and apply the functional calculus.)
- b) The star-subalgebra A contains a rank-1 projector. (If $p \in A$ is a nonzero projector of

minimal rank, show that the algebra pAp is contained in $\mathbb{C}p$.)

c) It follows that A contains $\mathcal{L}^c(E)$.

d) Conversely, every star-subalgebra of $\mathcal{L}(E)$ containing $\mathcal{L}^c(E)$ is irreducible.

11) Let E be a complex Hilbert space. Let $A \subset \mathcal{L}(E)$ be a set of compact endomorphisms such that $u \circ v = v \circ u$ for every pair (u, v) of elements of A and $u^* \in A$ for all $u \in A$.

Show that there exists a countable orthonormal family $(e_i)_{i \in I}$ and a family $(\Lambda_i)_{i \in I}$ of mappings $A \rightarrow \mathbb{C}$ such that

$$u(x) = \sum_{i \in I} \Lambda_i(u) \langle e_i | x \rangle e_i$$

for every element u of A and every x in E . If A is a unital subalgebra of $\mathcal{L}(E)$, then Λ_i is a character of A for every $i \in I$.

12) If E is a complex Hilbert space and u is a compact normal endomorphism of E , show that u has a cyclic vector (cf. def. 3 of IV, p. 191) if and only if the following conditions are satisfied:

(i) The kernel of u is reduced to 0;

(ii) The spectral multiplicity of every nonzero eigenvalue of u is equal to 1.

13) Let E be a complex Hilbert space of countable type, let $(e_n)_{n \in \mathbb{N}}$ be an orthonormal basis of E , and let u be an endomorphism of E . Show that u is compact if and only if one has

$$\lim_{n \rightarrow +\infty} \sup_{x \in \{e_0, \dots, e_{n-1}\}^\circ - \{0\}} \frac{\|u(x)\|^2}{\|x\|^2} = 0.$$

14) Let $X = [0, 1]$ be equipped with the Lebesgue measure, and let $N: X \times X \rightarrow \mathbf{R}_+$ be such that $N(x, y) = |x - y|$. The endomorphism u_N of $L^2(X)$ is not positive.

15) Let X be a locally compact topological space, let μ be a positive bounded measure on X with support X , and let $N \in \mathcal{C}_b(X \times X)$.

a) There exists an endomorphism u of $\mathcal{C}_b(X)$ such that

$$u(f)(y) = \int_X N(x, y) f(x) d\mu(x)$$

for every function $f \in \mathcal{C}_b(X)$ and every $y \in X$. In what follows, we assume that u is compact; this is the case, for example, if the space X is compact.

b) The image of u_N is contained in $\mathcal{C}_b(X)$ and u_N defines a continuous linear map from $L^2_{\mathbb{C}}(X)$ into $\mathcal{C}_b(X)$.

c) Let $f \in L^2_{\mathbb{C}}(X)$ and $\lambda \in \mathbb{C}^*$ such that $u_N(f) = \lambda f$. Then $f \in \mathcal{C}_b(X)$.

d) There exists a countable family $(f_i)_{i \in I}$ in $\mathcal{C}_b(X)$ and a family $(\lambda_i)_{i \in I}$ of nonzero real numbers such that for all $f \in L^2_{\mathbb{C}}(X)$, we have

$$u(f) = \sum_{i \in I} \lambda_i \langle f_i | f \rangle f_i$$

in $\mathcal{C}_b(X)$.

e) Suppose that u_N is a positive endomorphism of $L^2_{\mathbb{C}}(X)$. For every $i \in I$, we denote by $h_i \in \mathcal{C}_b(X \times X)$ the function defined by $h_i(x, y) = \overline{f_i(x)} f_i(y)$. Then

$$N = \sum_{i \in I} \lambda_i h_i$$

in $\mathcal{C}(X \times X)$ endowed with the topology of compact convergence. (Prove that the series converges using Dini's theorem (TG, X, p. 34, th. 1), then prove the equality.)

16) Let $X = [0, 1]$ endowed with the Lebesgue measure.

a) For any sequence $(f_n)_{n \in \mathbb{N}}$ of continuous functions on X that is orthonormal in $L^2_{\mathbb{C}}(X)$, there exists a kernel $k \in \mathcal{C}(X \times X)$ such that

b) The endomorphism u_k of $L^2_{\mathbb{C}}(X)$ is positive;

c) For every $n \in \mathbb{N}$, the function f_n is an eigenfunction of u_k whose eigenvalue is nonzero;

d) The family (f_n) is a basis of the image of the endomorphism u_k .

- e) There exist kernels $k \in \mathcal{C}(X \times X)$ and orthonormal families of eigenfunctions (f_n) of the endomorphism u_k of $L^2_{\mathcal{C}}(X)$ consisting of bounded functions, such that the sequence $(\|f_n\|_{\infty})$ is not bounded.

17) Let E and F be Hilbert spaces. The bilinear map $(u, v) \mapsto \text{Tr}(uv)$ defines an isometric isomorphism from the Banach space $\mathcal{L}_1(E; F)'$ into the space $\mathcal{L}(F; E)$.

18) Let E be a complex Hilbert space and let I_E be the set defined in no. 3 of IV, p. 151.

Let $n \in I_E$. Let F be a complex Hilbert space. The map $u \mapsto \lambda_n(|u|)$ from $\mathcal{L}^c(E; F)$ into R_+ is continuous.

19) We keep the notations of Lemma 5 of IV, p. 161. Show that if the image of f is contained in $\mathbf{R}$, if f is strictly convex, and if $\varrho_n > 0$, then there is equality in inequality (6) if and only if $a_i = b_i$ for $0 \leq i \leq n$.

20) Let E be a pre-Hilbert space, let F be a Banach space, and let $u : E \rightarrow F$ be a linear map.

a) Prove that u is a compact linear map if and only if $\|u(e_n)\| \rightarrow 0$ for every orthonormal sequence (e_n) of E . (To prove that this condition is sufficient, use TG, IX, p. 20, prop. 14.)

b) Suppose that E is a Hilbert space of countable type and that $F = E$. For there to exist an orthonormal basis $(e_n)_{n \in \mathbf{N}}$ of E such that $\|u(e_n)\| \rightarrow 0$, it is necessary and sufficient that u not be a Fredholm endomorphism.

c) Suppose that E is a Hilbert space and that $F = E$. If one has $\langle u(e_n) | e_n \rangle \rightarrow 0$ for every orthonormal sequence (e_n) of E , then u is a compact linear map.

21) Let E be a Hilbert space of countable type.

a) Let A be an infinite set of unit vectors of E which converges weakly to 0 along the filter of complements of finite sets. Prove that there exists a sequence $(x_n)_{n \geq 1}$ with values in A and an orthonormal sequence $(e_n)_{n \geq 1}$ in E such that $\|x_n - e_n\| \leq 1/n$ for all $n \geq 1$.

b) Let u be a linear map from E into E , not necessarily continuous. Suppose that for every

orthonormal sequence (e_n) in E , the sequence $(u(e_n))$ converges weakly to 0. Prove that u is continuous and compact.

- c) Let u be an endomorphism of E . There exists an orthonormal basis $(e_n)_{n \in \mathbb{N}}$ of E such that the sequence $(u(e_n))_{n \in \mathbb{N}}$ converges weakly to 0 if and only if there does not exist an endomorphism v of E such that $v \circ u - 1_E$ is compact.

22) Let E be a complex Hilbert space. Let u and v be positive endomorphisms of E .

- a) For every real number p such that $0 < p < 1$ and every $x \in R_+$, we have

$$x^p = \frac{\sin(p\pi)}{\pi} \int_0^{+\infty} t^p \left(\frac{1}{t} - \frac{1}{t+x} \right) dt.$$

- b) Suppose that $u \leq v$. For every real number p such that $0 < p < 1$, we have $u^p \leq v^p$.

(Use Lemma 14 of I, p. 119, b) and the preceding question.)

- c) Let $n \in \mathbb{N}$. We set $a = v^{n/2} \circ u^n \circ v^{n/2}$ and $b = (v^{1/2} \circ u \circ v^{1/2})^n$. If $a \leq 1_E$, then $b \leq 1_E$.

- d) Suppose that u and v have finite trace. For every integer $n \in \mathbb{N}$, we have

$$\text{Tr} \left((v^{1/2} \circ u \circ v^{1/2})^n \right) \leq \text{Tr} (v^{n/2} \circ u^n \circ v^{n/2}).$$

(Let a and b be as in question c); by using Lemma 5 of IV, p. 161, reduce to proving that the largest eigenvalue of $\widehat{\Lambda}^n b$ is less than that of $\widehat{\Lambda}^n a$; replace E by $\widehat{\Lambda}^n E$ and u, v by $\widehat{\Lambda}^n u$ and $\widehat{\Lambda}^n v$, respectively, then apply the preceding question.)

Exercises for Section 2

Unless stated otherwise, in the exercises of this section, E denotes a complex Hilbert space.

- 1) Let u be an endomorphism of E whose image is closed in E . Let q be an orthogonal projector of E and let v be the endomorphism $x \mapsto q(u(x))$ of the image of E .

- a) The adjoint of v is the endomorphism $x \mapsto q(u^*(x))$ of the image of E .
- b) If u is Hermitian, then v is Hermitian.
- c) It is possible that u is normal and that v is not.
- 2) Prove Theorem 1 of IV, p. 149 by using directly the spectral theorem (cor. of th. 1 of IV, p. 193).
- 3) Let $S \subset \mathbb{C}$ be a compact subset of $\mathbb{C}$. Let $\varphi: \mathcal{C}(S) \rightarrow \mathcal{L}(E)$ be a continuous unital morphism of involutive algebras. There exists a unique normal endomorphism u of E whose spectrum is S and which satisfies $\varphi(f) = f(u)$ for all $f \in \mathcal{L}^\infty(S)$.
- 4) Let $u \in \mathcal{L}(E)$ be normal. The convex hull of the spectrum of u is the closure in $\mathbb{C}$ of the numerical range $\iota(u)$ of u (cf. I, p. 135, def. 2).
- 5) Let $u \in \mathcal{L}(E)$ be such that $\|u\| \leq 1$. For every $\lambda \in \mathbb{C}$, let p_λ be the orthogonal projector of E onto the eigenspace of u corresponding to λ . For all x and y in E , we have

$$\lim_{N \rightarrow +\infty} \frac{1}{N} \sum_{n=1}^N |\langle x | u^n(y) \rangle|^2 = \sum_{\substack{\lambda \in \mathbb{C} \\ |\lambda| = 1}} |\langle p_\lambda(y) | p_\lambda(x) \rangle|^2 = \sum_{\substack{\lambda \in \mathbb{C} \\ |\lambda| = 1}} |\langle y | p_\lambda(x) \rangle|^2$$

(one may use Exercise 64 of II, p. 300).

- 6) Let E be a pre-Hilbert space, let F be a Banach space, and let $u: E \rightarrow F$ be a linear map.
- a) Prove that u is a compact linear map if and only if $\|u(e_n)\| \rightarrow 0$ for every orthonormal sequence (e_n) of E . (To prove that this condition is sufficient, use TG, IX, p. 20, prop. 14.)
- b) Assume that E is a separable Hilbert space and that $F = E$. For there to exist an orthonormal basis (e_n) of E such that $\|u(e_n)\| \rightarrow 0$ it is necessary and sufficient that u not be a Fredholm operator.
- c) We assume that E is a Hilbert space and that we have $\langle u(e_n), e_n \rangle \rightarrow 0$ for every orthonormal sequence (e_n) of H . Prove that u is a compact linear map.

7) Let u and v be continuous endomorphisms of E .

a) Prove that $\text{Im}(u) + \text{Im}(v) = \text{Im}((u \circ u^* + v \circ v^*)^{1/2})$.

b) If u is positive and has closed image; prove that $\text{Im}(u) = \text{Im}(u^{1/2})$.

c) If u, v are positive with closed image, and if $u+v$ is also closed-image, we have $\text{Im}(u+v) = \text{Im}(u) + \text{Im}(v)$.

8) Let F be a complex Hilbert space and let $u: E \rightarrow F$ be a continuous linear map with closed image.

a) Prove that there exists a unique continuous linear map $v \in \mathcal{L}(F; E)$ satisfying the following conditions:

(i) $u \circ v \circ u = u$ and $v \circ u \circ v = v$;

(ii) $u \circ v$ and $v \circ u$ are self-adjoint.

We denote this map by $u^\dagger$ (Moore–Penrose pseudoinverse).

b) Prove that $\text{Im}(u) = \text{Im}(u \circ u^\dagger)$. Show that for $x \in E$ and $y \in \text{Im}(u)$, the conditions $u(x) = y$ and $x - u^\dagger(y) \in \text{Im}(1_E - u^\dagger \circ u)$ are equivalent.

9) Let u and v be endomorphisms of E with closed images. If $u + v$ also has closed image, we set $u : v = u \circ (u + v)^\dagger \circ v$, where $(u + v)^\dagger$ is the Moore–Penrose pseudoinverse of $u + v$.

a) Prove that $u : v$ is self-adjoint.

b) One has $u : v = v : u$ and $\text{Im}(u : v) = \text{Im}(u) \cap \text{Im}(v)$.

c) Assume that u and v are orthogonal projectors. Prove that $2u : v$ is the orthogonal projector with image $\text{Im}(u) \cap \text{Im}(v)$.

10) A continuous endomorphism u of E is said to be hyponormal if one has $u^* \circ u \geq u \circ u^*$, and is said to be paranormal if $\|u^2(x)\| \geq \|u(x)\|^2$ for all $x \in E$.

a) If u is paranormal, then u^n is paranormal for every $n \in \mathbb{N}$.

- b) If u is paranormal and invertible, then u^{-1} is paranormal.
- c) A hyponormal endomorphism is paranormal.
- d) Let u be a paranormal endomorphism whose spectrum is contained in the unit circle; then u is unitary.
- e) If u is a paranormal endomorphism, its norm $\|u\|$ is equal to its spectral radius $\varrho(u)$.
- f) We assume that u is paranormal.

11) Let u be a continuous endomorphism of E .

- a) For u to be paranormal, it is necessary and sufficient that

$$(u^*)^2 \circ u^2 - 2t(u^* \circ u) + t^2 1_E \geq 0$$

for every real number $t > 0$.

- b) For u to be paranormal, it is necessary and sufficient that

$$p \circ q^2 \circ p - 2tp^2 + t1_E \geq 0$$

for all $t > 0$, where $p = (u \circ u^*)^{1/2}$ and $q = (u^* \circ u)^{1/2}$.

- 12) a) Let u and v be continuous positive endomorphisms of E . Suppose that for every $t > 0$, we have

$$2tu^2 \circ (u^2 + t^2 1_E)^{-1} \leq v \leq \frac{1}{2t}(u^2 + t^2 1_E).$$

Prove that $u = v$.

- b) For u to be a normal endomorphism, it is necessary and sufficient that u and u^* be paranormal and have the same kernel.
- 13) a) Let A be an infinite set of unit vectors of E which converges weakly to 0 along the filter of complements of finite sets. Prove that there exists a sequence $(x_n)_{n \geq 1}$ with

values in A and an orthonormal sequence $(e_n)_{n \geq 1}$ in E such that $\|x_n - e_n\| \leq 1/n$ for all $n \geq 1$.

b) Let u be an endomorphism of E , not necessarily continuous. Suppose that for every orthonormal sequence (e_n) in E , the sequence $(u(e_n))$ converges weakly to 0. Prove that u is continuous and compact.

c) Let u be a continuous endomorphism of E . For there to exist an orthonormal basis (e_n) of E such that the sequence $(u(e_n))$ converges weakly to 0, it is necessary and sufficient that for every endomorphism v of E , $v \circ u - 1_E$ is not compact.

14) Let X be a locally compact space, μ a positive bounded measure on X such that $\mu(X) = 1$, and let $f : X \rightarrow X$ be a μ -measurable map such that $f(\mu) = \mu$. We denote by u_f the unitary endomorphism of $L^2_{\mathbb{C}}(X, \mu)$ defined by $\varphi \mapsto f \circ \varphi$ (cf. exercise 16 of I, p. 190). For every $n \in \mathbb{N}$, we denote by $f^n = f \circ \dots \circ f$ (the n -fold composition), and for every subset A of X , we denote $f^{-n}(A) = g^{-1}(A)$ with $g = f^n$.

We say that f is weakly mixing relative to μ if, for all μ -measurable subsets A and B of X , we have

$$\lim_{N \rightarrow +\infty} \frac{1}{N} \sum_{n=0}^N |\mu(A \cap f^{-n}(B)) - \mu(A)\mu(B)| = 0.$$

a) The following conditions are equivalent:

(i) The map f is weakly mixing relative to μ ;

(ii) For all $\varphi_1, \varphi_2 \in L^2_{\mathbb{C}}(X, \mu)$, we have

$$\lim_{N \rightarrow +\infty} \frac{1}{N} \sum_{n=0}^N |\langle \varphi_1 | u_f^n(\varphi_2) \rangle - \langle 1 | \varphi_2 \rangle \langle \varphi_1 | 1 \rangle| = 0;$$

(iii) The map $f \times f : X \times X \rightarrow X \times X$ is $(\mu \otimes \mu)$ -ergodic (cf. exercise 16 of I, p. 190);

(iv) The eigenspace associated with the eigenvalue 1 has dimension 1, and for every $\lambda \in \mathbb{C} - \{1\}$, the eigenspace of u_f relative to λ is reduced to 0.

(To prove that (iii) implies (i), use Exercise 64 of II, p. 300.)

b) Consider the example from part (e) of Exercise 16 in I, p. 190. The map f is weakly mixing relative to μ .

c) Consider the example from question f) of exercise 16 in I, p. 190. The map f is not weakly mixing relative to μ .

15) Let $G = (S, F, o, t, \iota)$ be a graph (TA, II, p. 155, def. 1). We assume that G is connected, that S is countable, and that S has at least two elements. For every $x \in S$, we denote by $v(x)$ the number of edges of G of which x is an endpoint; we have $v(x) \geq 1$ for all $x \in S$, and we assume that $v(x)$ is finite for all $x \in S$. For all x and y in S , we denote by $a(x, y)$ the number of arrows in G with endpoints x and y . We assume that $a(x, y)$ is finite for all $(x, y) \in S^2$.

We define a measure μ on S by $\mu(\{x\}) = v(x)$ for all $x \in S$. The measure μ is positive. We denote by $\mathcal{F}(G)$ the vector space of functions on S with complex values. We denote by $\widetilde{M}_G$ the endomorphism of $\mathcal{F}(G)$ such that

$$\widetilde{M}_G(\varphi)(x) = \frac{1}{v(x)} \sum_y a(x, y) \varphi(y),$$

for all $\varphi \in \mathcal{F}(G)$ and every $x \in S$ ("Markov operator").

a) The endomorphism $\widetilde{M}_G$ induces, by passage to subspaces, an endomorphism M_G of $L^2_C(S, \mu)$.

b) The endomorphism M_G is self-adjoint. It satisfies

$$\langle \varphi | 1 - M_G(\varphi) \rangle = \frac{1}{2} \sum_{(x,y) \in S^2} a(x, y) |\varphi(x) - \varphi(y)|^2$$

$$\langle \varphi | 1 + M_G(\varphi) \rangle = \frac{1}{2} \sum_{(x,y) \in S^2} a(x, y) |\varphi(x) + \varphi(y)|^2$$

for all $\varphi \in L^2(S, \mu)$. Moreover, we have $-1 \leq M_G \leq 1$.

c) Let $d \geq 1$ be an integer and suppose that G is the Cayley graph of Z^d relative to the set $\{e_i\}$, where (e_i) is the canonical basis of Z^d (TA, II, p. 221, exercise 7). The spectral radius $\varrho(M_G)$ of M_G is equal to 1.

d) Let $d \geq 3$ be an integer. Suppose that G is an infinite tree (TA, II, p. 157) d -regular, that is, an infinite tree in which every vertex has d neighbors. Let x_0 be a fixed vertex of G . For every $\lambda \in \mathbb{C}$ there exists a unique function $\varphi_\lambda : S \rightarrow \mathbb{C}$ such that $\varphi_\lambda(x_0) = 1$ and $\widetilde{M}_G(\varphi_\lambda) = \lambda \varphi_\lambda$. Moreover, for every $x \in S$ the value of φ_λ at x depends only on the distance from x to x_0 (cf. TA, II, p. 222, exercise 10).

e) The spectrum of M_G is the interval $[-2\sqrt{d-1}/d, 2\sqrt{d-1}/d]$ ("Kesten's theorem"). In particular, we have $\varrho(M_G) = 2\sqrt{d-1}/d$.

16) We keep the notation from the previous exercise, and we assume that S and F are finite. We denote by h_G the Cheeger constant of the graph G , defined by

$$h_G = \inf_{\emptyset \neq V \subset S} \frac{\text{Card}(E(V, S - V))}{\inf(\text{Card}(V), \text{Card}(S - V))},$$

where $E(V, W)$ denotes the set of arrows of G having one endpoint in V and one in W .

a) One has $\varrho(M_G) = 1$ and $\lambda = 1$ is an eigenvalue of M_G of multiplicity 1. We have $-1 \in \text{Sp}(M_G)$ if and only if G is bipartite (cf. TA, II, p. 219, exercise 2); in this case, the spectrum of M_G is invariant under $\lambda \mapsto -\lambda$ and the spectral multiplicity of λ is equal to that of $-\lambda$ for all $\lambda \in \mathbb{C}$.

We denote by $L_0^2(S, \nu)$ the orthogonal complement in $L_C^2(S, \nu)$ of the union of the eigenspaces of M_G corresponding to the eigenvalues 1 and -1. The endomorphism M_G induced by passage to subspaces is an endomorphism of $L_0^2(S, \nu)$, denoted $M_{G,0}$. We have $\varrho(M_{G,0}) < 1$. We denote by $\lambda_1(G)$ the smallest nonzero eigenvalue of $1 - M_G$.

b) Let $\nu_+ = \sup \nu(x)$ and $\nu_- = \inf \nu(x)$. We have

$$\lambda_1(G) \leq \frac{2\nu_+}{\nu_-^2} h_G.$$

- c) Let $\varphi: S \rightarrow \mathbf{R}$ be a nonconstant function on S with real values; let $m = \inf \varphi(x)$ and $M = \sup \varphi(x)$. Let $t_0 \in \mathbf{R}$ be such that, for every $t \in \mathbf{R}$, the set $W_t = \overline{\varphi}^{-1}(]-\infty, t_0[)$ satisfies $\text{Card}(W_t) \leq \text{Card}(S)/2$ if and only if $t < t_0$. Let ν be a diffuse measure (INT, V, p. 67, def. 5) supported on $[m, M]$. Set

$$A = \frac{1}{2} \sum_{(x,y) \in S^2} a(x,y) \nu([\varphi(x), \varphi(y)])$$

$$B = \sum_{x \in S} \nu([t_0, \varphi(x)]),$$

where, if $a < b$ are real numbers, we denote $\nu([b, a]) = \nu([a, b])$. There exists $t \in \mathbf{R}$ such that

$$\frac{\text{Card}(E(W_t, S - W_t))}{\inf(\text{Card}(W_t), \text{Card}(S - W_t))} \leq \frac{A}{B}.$$

- d) One has

$$h_G \leq \nu_+ \sqrt{2\lambda_1(G)}.$$

(Apply the previous question with φ an eigenfunction of M_G for the eigenvalue $\lambda_1(G)$ and with a suitably chosen measure ν .)

- e) One says that a family $(G_j)_{j \in J}$ of finite graphs $G_j = (S_j, F_j, o_j, t_j, i_j)$ is a family of expander graphs if the following conditions hold:

- (i) One has $\text{Card}(S_j) \rightarrow +\infty$ according to the filter of complements of finite subsets of J ;
- (ii) There exists $C \geq 0$ such that for every j and every $x \in S_j$, the valence of G_j at x is $\leq C$;
- (iii) There exists $\delta > 0$ such that $h_{G_j} \geq \delta$ for all $j \in J$.

Show that (G_j) is a family of expander graphs if and only if (i), (ii) are satisfied as well as the condition

- (iii') There exists $\delta > 0$ such that $\lambda_1(G_j) \geq \delta$ for all $j \in J$.

- 17) a) Let $n \in \mathbf{N}$. Let E be a Hilbert space of dimension n , let $u \in \mathcal{L}(E)$, and let $x \in E$ be of norm 1 such that $u(x) = 0$. For every vector $y \in E$ of norm 1, we denote by $\Delta_u(y)$ the discriminant of the quadratic form defined on y° by $z \mapsto \langle z | u(z) \rangle$.

For every $y \in E$ of norm 1, we have $|\langle x | y \rangle|^2 \Delta_u(x) = \Delta_u(y)$. (Note that $\Delta_u(y) = \langle a | \bigwedge u^{n-1}(a) \rangle / \langle a | a \rangle$ for every nonzero element a of the space $\bigwedge^{n-1} y^\circ \subset \bigwedge^{n-1} E$, the space $\bigwedge^{n-1} E$ being endowed with its canonical Hilbertian structure, cf. EVT, V, p.33; then verify the formula separately when x is orthogonal to y and when $x = y$.)

- b) Let A be a matrix of type (n, n) with complex coefficients such that ${}^t \bar{A} = A$. For every integer i such that $1 \leq i \leq n$, let A_i be the matrix of type $(n-1, n-1)$ obtained by deleting the i -th row and the i -th column of A . Let $(e_1, \dots, e_n)$ be an orthonormal basis of C^n , endowed with its canonical Hilbertian structure, consisting of eigenvectors for A , and let λ_i be the eigenvalue of u relative to e_i . Let us write $e_i = (e_{i,j})_{1 \leq j \leq n}$.

For all (i, j) , we have

$$|e_{i,j}|^2 \prod_{k \neq i} (\lambda_i - \lambda_k) = \det(\lambda_i - A_j).$$

(Reduce to the case where $\lambda_i = 0$, then apply the preceding question to the space C^n , to the endomorphism $u: y \mapsto Ay$, and to the vector $x = e_i$.)

- 18) Let X be a compact metrizable space and let μ be a positive measure on X . Denote by $E = L^2_C(X, \mu)$. Let $(\mathcal{X}_n)_{n \in \mathbf{N}}$ be a sequence of finite partitions of X into measurable sets such that X_{n+1} is finer than X_n for all $n \in \mathbf{N}$, and such that the largest of the diameters of a part of X_n tends to 0 as $n \rightarrow +\infty$.

Let $E_n \subset E$ be the space of $f \in L^2(X, \mu)$ such that f is μ -almost everywhere constant on Y for every part Y of X_n . We set $E_{-1} = \{0\}$. For every $n \geq -1$, let p_n be the orthogonal projector onto E with image E_n .

- a) The sequence $(p_n)_{n \in \mathbf{N}}$ converges to the identity in the space $\mathcal{L}_s(E)$ of endomorphisms of E endowed with the topology of simple convergence. (First consider the limit of $p_n(f)$ as $f \in \mathcal{C}(X)$.)

b) For every $n \in N$, the endomorphism $q_n = p_n - p_{n-1}$ is the orthogonal projector onto $F_n = E_n \cap E_{n-1}^\circ$, and the spaces F_n are pairwise orthogonal.

c) One has $\sum_n q_n = 1_E$ in $\mathcal{L}_s(E)$, and the space E is the Hilbertian sum of the spaces F_n .

19) Let $X \subset \mathbf{R}$ be a compact subset of $\mathbf{R}$ and let μ be a positive measure on X and let u be the endomorphism of multiplication by the identity function of X .

We denote $E = L^2(X, \mu)$ and we denote by $\mathcal{L}_s(E)$ the space of endomorphisms of E endowed with the topology of simple convergence.

Let $\varepsilon > 0$. Let $r > 0$ such that $X \subset]-r, r[$ and let $\ell \geq 1$ such that $r2^{-\ell} < \varepsilon$. For every integer $n \in \mathbf{N}$ and every integer j such that $|j| \leq 2^{n+\ell}$, let

$$X_{n,j} = X \cap \left] \frac{rj}{2^{n+\ell}}, \frac{r(j+1)}{2^{n+\ell}} \right]$$

a) For every integer $n \in N$, the family $\mathcal{X}_n = (X_{n,j})_j$ is a finite partition of X ; the sets $X_{n,j}$ are measurable and of diameter $\leq r2^{-n-\ell}$. For every $n \in N$, the partition X_{n+1} is finer than the partition X_n . We can therefore apply to $(\mathcal{X}_n)$ the results of the previous exercise, whose notation we reuse, in particular the spaces E_n and F_n and the orthogonal projectors p_n and q_n . The spaces E_n are finite-dimensional.

b) The series

$$v = \sum_{n \in \mathbf{N}} q_n u q_n$$

converges in the space $\mathcal{L}_s(E)$; there exists an orthonormal basis B of E such that $v \in \mathcal{D}_B(E)$.

c) The endomorphism $w = v - u$ is compact. (Consider the endomorphisms u_n of multiplication by the functions $g_n \in E_n$ such that, for every integer j such that $|j| \leq 2^{n+\ell}$, and for all $x \in X_{n,j}$, we have $g_n(x) = rj2^{-n-\ell}$; then verify that $\|u - u_n\| \leq r2^{-n-\ell}$, and such that $u_n p_n = p_n u_n$.)

20) Let E be a complex Hilbert space of countable type and $u \in \mathcal{L}(E)$ be a Hermitian en-

domorphism. Let ε be a strictly positive real number. There exists an orthonormal basis B of E , a Hermitian diagonal endomorphism $v \in \mathcal{D}_B(E)$ and a compact endomorphism $w \in \mathcal{L}^c(E)$ such that $u = v + w$ and $\|w\| < \varepsilon$ (First consider the case where u admits a cyclic vector, and then apply the preceding exercise.)

21) Let E be a complex Hilbert space of countable type and $u \in \mathcal{L}(E)$ a normal endomorphism. Let ε be a strictly positive real number. There exists an orthonormal basis B of E , a diagonal endomorphism $v \in \mathcal{D}_B(E)$ and a compact endomorphism $w \in \mathcal{L}^c(E)$ such that $u = v + w$ and $\|w\| < \varepsilon$.

22) Let G be a locally compact commutative topological group and μ a Haar measure on G .

Let $u \in \mathcal{L}(L^2(G, \mu))$ be such that u commutes with the multiplication operator m_χ for every character $\chi \in \widehat{G}$. Prove that there exists $g \in L^\infty(G, \mu)$ such that $u = m_g$.

23) Let u be an endomorphism of E and let A be the closed unital subalgebra of $\mathcal{L}(E)$ generated by u .

If v is an endomorphism of E such that $u \circ v - v \circ u$ belongs to A , then $u \circ v - v \circ u$ is quasi-nilpotent. In particular, we have $u \circ v - v \circ u \neq 1_E$ if E is nonzero. (Use exercise 10 of I, p. 167; note that the result is valid for any Banach space E .)

24) Let $A \subset \mathcal{L}(E)$ be a unital self-adjoint subalgebra of $\mathcal{L}(E)$. We denote by A'' the bicommutant of A . We denote by $\mathcal{L}(E)_s$ the space $\mathcal{L}(E)$ endowed with the topology of simple convergence.

a) The space A'' is a unital self-adjoint subalgebra of $\mathcal{L}(E)$ which is closed in $\mathcal{L}(E)_s$.

b) Let $v \in A''$ and $x \in E$. The vector $v(x)$ belongs to the closure in E of the set of $u(x)$ for $u \in A$. (Let F be this closure; observe that the orthogonal projector of E with image F belongs to A' .)

c) The algebra A'' coincides with the closure of A in $\mathcal{L}(E)_s$. (For every finite family $(x_i)_{i \in I}$ of elements of E , apply the preceding question to the space E^I , to the image $\widetilde{A}$ of A by the unital involutive morphism $f : \mathcal{L}(E) \rightarrow \mathcal{L}(E^I)$ such that $f(u)(y_i) = (u(y_i))_{i \in I}$ and to

the element (x_i) of E^I .)

The equality of A'' and the closure of A in $\mathcal{L}(E)_s$ is von Neumann's bicommutant theorem.

25) Let U be an open subset of $\mathbf{R}$. A universally measurable function $f : U \rightarrow \mathbf{R}$ is called a Loewner function if, for every complex Hilbert space E and all Hermitian endomorphisms u and v of E such that $\text{Sp}(u) \subset U$ and $\text{Sp}(v) \subset U$, the condition $u \leq v$ in $\mathcal{L}(E)$ implies $f(u) \leq f(v)$. We denote by $\text{Loe}(U)$ the set of Loewner functions on U .

a) Let $f \in \text{Loe}(U)$. The function f is increasing on U . The constant function 1 and the identity function of U belong to $\text{Loe}(U)$.

b) Let $f \in \text{Loe}(\mathbf{R}_+^*)$. Let E be a complex Hilbert space and let u be a positive endomorphism of E . For every orthogonal projector p of E and for every $\varepsilon > 0$, we have $f(u)_p \leq f((1 + \varepsilon)u_p)$, where we denote by v_p the endomorphism of $\text{Im}(p)$ defined by $x \mapsto p(v(x))$ for all $v \in \mathcal{L}(E)$. (Identify E with the Hilbert direct sum $\text{Im}(p) \oplus \text{Im}(1_E - p)$ and represent u as a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Note that $u \leq \begin{pmatrix} (1 + \varepsilon)a & 0 \\ 0 & (1 + \varepsilon^{-1})d \end{pmatrix}$ for all $\varepsilon > 0$.)

c) Let $f \in \text{Loe}(\mathbf{R}_+^*)$. The function f is concave on $\mathbf{R}_+^*$ (Let v be an endomorphism of C^2 diagonal in the canonical basis and let $g \in \text{SO}(\mathbf{R}^2)$ viewed as an endomorphism of C^2 . Apply the previous question to the orthogonal projector with image Ce_1 and with $u = g^*vg$.)

d) The set $\text{Loe}(\mathbf{R})$ is the set of functions of the form $f(x) = ax + b$ where $(a, b) \in \mathbf{R}_+ \times \mathbf{R}$. (Apply the previous question to f as well as to the function $x \mapsto -f(-x)$.)

e) Let $t \in \mathbf{R}_+^*$. The function $f : x \mapsto -(t + x)^{-1}$ belongs to $\text{Loe}(\mathbf{R}_+^*)$.

f) For any bounded positive measure ν on $\mathbf{R}_+^*$ such that the function $x \mapsto x^{-1}$ is ν -integrable, the function $f : \mathbf{R}_+^* \rightarrow \mathbf{R}$ defined by

$$f(x) = \int_{\mathbf{R}_+^*} \frac{x}{x+t} d\nu(t)$$

belongs to $\text{Loe}(\mathbf{R}_+^*)$; it is differentiable on $\mathbf{R}_+^*$ and moreover satisfies the following condi-

tions:

- (i) $\lim_{t \rightarrow 0} f(t) = 0$;
- (ii) $\lim_{t \rightarrow 0} f'(t)$ exists in $\mathbf{R}$;
- (iii) f has a finite limit at $+\infty$.

26) Let E be a complex Hilbert space and let a be a positive endomorphism of E . We denote by $\|x\|_a = \langle x|a(x) \rangle$; it is a seminorm on E . We likewise denote by $\|u\|_a$ the seminorm of an endomorphism u of the space E equipped with this seminorm.

We call a function $f: \mathbf{R}_+^* \rightarrow \mathbf{R}_+^*$ an interpolation function if, for every complex Hilbert space E , every positive endomorphism a of E and every $u \in \mathcal{L}(E)$, the conditions $\|u\| \leq 1$ and $\|u\|_a \leq 1$ imply that $\|f(u)\|_a \leq 1$.

- a) A function $f: \mathbf{R}_+^* \rightarrow \mathbf{R}_+^*$ is an interpolation function if and only if, for every complex Hilbert space E , every positive endomorphism a of E and every $u \in \mathcal{L}(E)$, the conditions $u^*u \leq 1_E$ and $u^*au \leq a$ imply $u^*f(a)u \leq f(a)$.
- b) Any Loewner function on $\mathbf{R}_+^*$ is an interpolation function. (Given E and $u \in \mathcal{L}(E)$ of norm ≤ 1 , set $w = \sqrt{1_E - u^*u}$ and $w' = \sqrt{1_E - uu^*}$; consider $F = E \oplus E$ and note that the endomorphism v of F defined by $v(x, y) = (u(x) + w'(y), -w(x) + u(y))$ is an isometry; then consider $a': (x, y) \mapsto (a(x), 0)$ and apply part b) of the previous exercise to the orthogonal projector with image $E \oplus \{0\}$.)
- c) Every interpolation function is a Loewner function on $\mathbf{R}_+^*$. (Let $0 \leq u \leq v$ be endomorphisms of a complex Hilbert space E with v invertible; set $F = E \oplus E$ and consider the endomorphism $a = u \oplus v$ of F as well as the endomorphism w represented by the matrix
$$\begin{pmatrix} 0 & 1_E \\ v^{-1/2}u^{1/2} & 0 \end{pmatrix}$$
; show that $w^*w \leq 1_F$ and $w^*aw \leq a$.)
- d) Let f be an interpolation function. Let E be a complex Hilbert space; we denote $\mathcal{R}(u) = \frac{1}{2}(u + u^*)$ for all $u \in \mathcal{L}(E)$. For u and v in $\mathcal{L}(E)$, prove that $e^{-tv^*} \circ u \circ e^{-tv} \leq u$ for all $t \in \mathbf{R}_+$ if and only if $\mathcal{R}(u \circ v) \geq 0$. (For a given $x \in E$, compute the derivative of the

function $t \mapsto \langle e^{-tv}x | (ue^{-tv})x \rangle$.)

Hence deduce that for all endomorphisms u and v of E such that the conditions

$$u \geq 0, \quad \mathcal{R}(v) \geq 0, \quad \mathcal{R}(uv) \geq 0$$

are satisfied, we have $\mathcal{R}(f(u)v) \geq 0$.

27) We propose to prove the converse of the assertion in question *b* of Exercise 25. For this, one will use the results and notations of Exercise 23 of V, p. 503 below. In particular, G denotes the group $\mathbf{R}_+^*$ equipped with the Haar measure $\mu = x^{-1}dx$ in duality with the group $i\mathbf{R}$ by the mapping $(x, it) \mapsto x^{it}$. We denote by X the identity map of G .

We denote by co , si , and ta the functions $it \mapsto \cos(i\pi t)$, $it \mapsto \sin(i\pi t)$, and $it \mapsto \tan(i\pi t)$ on $i\mathbf{R}$, respectively.

Let $f \in \text{Loe}(G)$ satisfy conditions (i), (ii), and (iii) of question *f* of Exercise 25.

- a) Let α be the function $x \mapsto x/(1+x)$ on G . We have $I_\alpha =]-1, 0[$ and $\widehat{\alpha}(s) = -\pi / \sin(\pi s)$ for $s \in B_f$.
- b) The function ta is bounded on $i\mathbf{R}$. We set $v = F \circ m_{ta} \circ \overline{F}$; it is an endomorphism of $L^2(G)$ such that $\|v\| \leq 1$ and $\mathcal{R}(v) = 0$.
- c) Let $\varepsilon > 0$ be a real number. We denote by u_ε the multiplication endomorphism by $h_\varepsilon = X/(1 + \varepsilon X)$ on $L^2(G)$. We have $u_\varepsilon \geq 0$ and for every $\varphi \in L^2(G)$ and every bounded continuous function f on G , we have

$$\langle \varphi | \mathcal{R}(f(u_\varepsilon)v) \varphi \rangle = \frac{1}{2} \langle f \circ h_\varepsilon | Q(\varphi) \rangle$$

$$Q(\varphi) = \overline{\varphi}v(\varphi) + \overline{v(\varphi)}\varphi.$$

- d) Let F be the vector subspace of $L^2(i\mathbf{R})$ generated by the functions $t \mapsto e^{\alpha(it-s_0)^2}$ where $\alpha \in \mathbf{R}_+^*$ and $s_0 \in \mathbf{C}$; it is dense in $L^2(i\mathbf{R})$.
- e) Let $\psi \in F$. There exists $\varphi \in L^2(G)$ such that $\widehat{\varphi} = \text{co} \cdot \psi$. Let $g = Q(\varphi)$. We have $g \in L^1(G)$, and there exists a holomorphic function on $\mathbf{C}$ whose restriction to $i\mathbf{R}$

coincides with $\hat{g}$. Moreover

$$\hat{g} = \frac{1}{2i\pi}(\psi^* * \psi) \text{ si.}$$

f) For every $\varepsilon > 0$, we have $\mathcal{R}(u_\varepsilon \circ v) \geq 0$. (Compute $\langle \varphi | \mathcal{R}(u_\varepsilon u) \varphi \rangle$ when φ is a function as in the preceding question, and use convolution on G as in question *d* of Exercise 23 of V, p. 503.)

g) Let $f \in \text{Loc}(\mathbf{R}_+^*)$. Let g be a function associated with $\psi \in F$ as in question *e*). We then have

$$\int_G f g d\mu \geq 0$$

and

$$\int_{\mathbf{R}} \hat{f}(-\sigma - it) \sin(\pi(\sigma + it))(\psi^* * \psi)(\sigma + it) dt \geq 0$$

for all $\sigma \in]0, 1[$.

h) Let $\sigma \in]0, 1[$. The function q defined by $q(s) = \hat{f}(-s) \sin(\pi s)$ for s such that $\mathcal{R}(s) = \sigma$ satisfies

$$\sum_{i=0}^n \sum_{j=0}^n \bar{t}_i t_j q\left(\frac{1}{2}(\bar{z}_i + z_j)\right) \geq 0$$

for all $n \in \mathbf{N}$ and all families $(t_i)_{0 \leq i \leq n}$ and $(z_i)_{0 \leq i \leq n}$ of complex numbers such that $\mathcal{R}(z_i) = \sigma$ for every i .

i) There exists a positive measure ν_0 on G such that $\hat{\nu}_0(it) = \hat{f}(-it) \sin(i\pi t)/\pi$ for all $t \in \mathbf{R}$; we have $X^\sigma \in L^1(G, \nu_0)$ if $0 < \sigma < 1$.

j) Let $\nu = \check{\nu}_0$. The measure ν is bounded. (Verify that $] -1, 0[\subset I_\nu$ and then that $\hat{\nu}(-t)$ is bounded when $t \rightarrow 0$ to show that $\nu([1, +\infty[)$ is finite.) For every x , we have

$$f(x) = \int_G \frac{x}{x+t} d\nu(t).$$

- 28) a) Let $D \subset \mathbb{C}$ be the open unit disk. Let f be a germ of a holomorphic function in a neighborhood of $\overline{D}$. For every $z \in D$, we have

$$f(z) = i\mathcal{I}(f(0)) + \frac{1}{2\pi} \int_0^{2\pi} \mathcal{R}(f(e^{i\theta}))k(e^{i\theta}, z)d\theta$$

where $k(z, w) = (z + w)/(z - w)$ for $(z, w) \in D \times D$. (Note that

$$k(e^{i\theta}, z) = 1 + 2 \sum_{n=1}^{+\infty} z^n e^{-in\theta}$$

and use the Taylor expansion of f around 0.)

- b) Let f be a holomorphic function on D such that $\mathcal{R}(f(z)) > 0$ for all $z \in D$. The measures $\mu_r = \frac{1}{2\pi} \mathcal{R}(f(re^{i\theta}))d\theta$ on $[0, 2\pi]$ converge vaguely as $r \rightarrow 1$ to a bounded positive measure ν , and we have

$$f(z) = i\mathcal{I}(f(0)) + \frac{1}{2\pi} \int_0^{2\pi} \mathcal{R}(f(e^{i\theta}))k(e^{i\theta}, z)d\nu(\theta)$$

for all $z \in D$. (Reduce to the case where $f(0) = 1$ and observe that the measures μ_r are then positive measures of total mass 1.)

- c) Let H be the open subset of $\mathbb{C}$ consisting of $z \in \mathbb{C}$ such that $\mathcal{I}(z) > 0$. Let $f : H \rightarrow \mathbb{C}$ be a holomorphic function. The image of f is contained in H if and only if there exists a bounded positive measure ν on $\mathbb{R}$ and a real number $a \geq 0$ such that

$$f(z) = \mathcal{R}(f(i)) + az + \int_{\mathbb{R}} \frac{1+xz}{x-z} d\nu(x)$$

for all $z \in H$.

- d) Let I be a nonempty open interval in $\mathbb{R}$ and let $f : I \rightarrow \mathbb{R}$. Denote by $J = \mathbb{R} - I$. The following conditions are equivalent («Loewner's theorem»):

- (i) The function f belongs to $\text{Loe}(I)$;
- (ii) There exists an analytic function $\tilde{f}$ on $(\mathbb{C} - \mathbb{R}) \cup I$ whose restriction to I coincides with f

and which satisfies $\mathcal{T}(f(z)) > 0$ if $\mathcal{T}(z) > 0$;

(iii) There exists a bounded measure ν on J and a pair $(a, b) \in \mathbf{R}_+ \times \mathbf{R}$ such that

$$f(x) = ax + b + \int_J \frac{1+xy}{y-x} d\nu(y)$$

for all $x \in I$.

(First verify directly that (iii) implies (i) and (ii); to show that (i) implies (iii), treat separately the case where $I = \mathbf{R}$ and reduce the other cases to the case where $I = \mathbf{R}_+^*$, for which one applies the preceding exercise; finally, to show that (ii) implies (iii), apply part c) and verify that the measure ν on $\mathbf{R}$ given by it is supported on J .)

Exercises for Section 3

1) The function φ_0 on $\mathbf{R}$ defined by $\varphi_0(x) = 0$ if $x \leq 0$ and $\varphi_0(x) = e^{-1/x}$ if $x > 0$ is of class C^∞ and satisfies $\varphi_0^{(k)}(0) = 0$ for every integer $k \in \mathbf{N}$.

2) Let $x_0 \in \mathbf{R}^n$, $r > 0$ and $t > 1$. There exists $\varphi \in \mathcal{D}(\mathbf{R}^n)$ such that

(i) One has $0 \leq \varphi \leq 1$;

(ii) One has $\varphi(x) = 1$ for $\|x - x_0\| \leq r$ and $\varphi(x) = 0$ for $\|x - x_0\| \geq tr$;

(iii) For every $\alpha \in \mathbf{N}^n$, we have

$$\sup_{x \in \mathbf{R}^n} |\partial^\alpha \varphi(x)| \leq \alpha!((t-1)r)^{|\alpha|}.$$

3) Let $n \in \mathbf{N}$. We denote by μ_n the Lebesgue measure on $\mathbf{R}^n$.

a) Let $\mathcal{H}_1$ be the image of the map $\partial_1 : \mathcal{D}(\mathbf{R}^n) \rightarrow \mathcal{D}(\mathbf{R}^n)$; it is the space of $\varphi \in \mathcal{D}(\mathbf{R}^n)$ such that

$$\int_{\mathbf{R}} \varphi(x, y) d\mu_1(x) = 0$$

for all $y \in \mathbf{R}^{n-1}$.

- b) Let $\varphi_0 \in \mathcal{D}(\mathbf{R})$ be such that its integral over $\mathbf{R}$ is equal to 1. For every $\varphi \in \mathcal{D}(\mathbf{R}^n)$, there exist $\varphi_1 \in \mathcal{H}_1$ such that

$$\varphi(x, y) = \varphi_1(x, y) + \varphi_0(x) \int_{\mathbf{R}} \varphi(t, y) d\mu_1(t)$$

for all $(x, y) \in \mathbf{R} \times \mathbf{R}^{n-1}$.

- c) For every distribution $f \in \mathcal{D}'(\mathbf{R}^n)$, there exists $g \in \mathcal{D}'(\mathbf{R}^n)$ such that $\partial_1 g = f$. If g_1 and g_2 are in $\mathcal{D}'(\mathbf{R}^n)$ satisfy $\partial_1 g_1 = \partial_1 g_2 = f$, then there exists a distribution $h \in \mathcal{D}'(\mathbf{R}^{n-1})$ such that

$$\langle g_1 - g_2, \varphi \rangle = \langle h, \psi \rangle$$

for all $\varphi \in \mathcal{D}(\mathbf{R}^n)$, where $\psi \in \mathcal{D}(\mathbf{R}^{n-1})$ is defined by

$$\psi(y) = \int_{\mathbf{R}} \varphi(t, y) d\mu_1(t).$$

- d) Let U be a connected open subset of $\mathbf{R}^n$. Let $f \in \mathcal{D}'(U)$ be a distribution such that $\partial_i f = 0$ for $1 \leq i \leq n$. The distribution f is the distribution associated with a constant function.

- 4) We denote by μ the Lebesgue measure on $\mathbf{R}$.

- a) Let ν be a measure on $\mathbf{R}$ and $a \in \mathbf{R}$ such that $\nu(\{a\}) = 0$. For $x \in \mathbf{R}$ we set

$$f(x) = \begin{cases} \nu([a, x]) & \text{if } x \geq a, \\ \nu([x, a]) & \text{if } x \leq a. \end{cases}$$

The function f is continuous and the associated distribution satisfies $f' = \nu$.

- b) Let f be a distribution on $\mathbf{R}$ such that f' is a measure ν . The distribution f is associated with the continuous function on $\mathbf{R}$ defined as in the preceding question.

- c) Let f be a distribution on $\mathbf{R}$ such that $f^{(k)}$ is a measure for every integer $k \geq 0$. Then f is

the distribution associated with a function in $\mathcal{C}^\infty(\mathbf{R})$.

- 5) The family of distributions associated with point measures ε_n for $n \in \mathbf{Z}$ is summable in $\mathcal{S}'(\mathbf{R})$. Its sum ω satisfies $\mathcal{F}(\omega) = \omega$. (This formula is equivalent to the Poisson formula for the locally compact group $G = \mathbf{R}^n$ equipped with Lebesgue measure and the subgroup $H = \mathbf{Z}^n$, cf. II, § 1, no. 7.)
- 6) Let A be a commutative unital algebra over $\mathbf{C}$ such that the following conditions are satisfied:
- (i) The algebra $\mathcal{C}(\mathbf{R})$ is a unital subalgebra of A ;
 - (ii) There exists a derivation $d: A \rightarrow A$ whose restriction to $\mathcal{C}^1(\mathbf{R})$ coincides with the usual derivation of functions.
- a) The identity map X of $\mathbf{R}$ is invertible in A . (Prove that the map $g: \mathbf{R} \rightarrow \mathbf{R}$ such that $g(0) = 0$ and $g(x) = x(\log(|x|) - 1)$ for $x \neq 0$ is continuous and satisfies $gX = 1$.)
- b) One has $d \circ d(|X|) = 0$.
- c) There does not exist a bilinear map $m: \mathcal{D}'(\mathbf{R}) \times \mathcal{D}'(\mathbf{R}) \rightarrow \mathcal{D}'(\mathbf{R})$ such that $m(f, g) = fg$ for every pair $(f, g) \in \mathcal{D}(\mathbf{R})^2$, and such that

$$m(f, g)' = m(f', g) + m(f, g')$$

for all $(f, g) \in \mathcal{D}'(\mathbf{R})^2$ ("impossibility of multiplication of distributions").

- d) Give examples of sequences (f_n) and (g_n) of test functions on $\mathbf{R}$ such that $f_n \rightarrow \varepsilon_0$ and $g_n \rightarrow \varepsilon_0$ in $\mathcal{D}'(\mathbf{R})$, and such that f_n^2 and g_n^2 converge in $\mathcal{D}'(\mathbf{R})$ to different limits.
- 7) Let $n \in \mathbf{N}$. We denote by $\tilde{\gamma}$ the linear representation of $\mathbf{R}^n$ on the space of functions from $\mathbf{R}^n$ to $\mathbf{C}$ defined by

$$\tilde{\gamma}(h)f(x) = f(x + h)$$

for all $h \in \mathbf{R}^n$ and every function $f: \mathbf{R}^n \rightarrow \mathbf{C}$.

- a) For every $h \in \mathbf{R}^n$, the map $\tilde{\gamma}(h)$ defines, by passing to subspaces, an automorphism of the locally convex space of $\mathcal{D}(\mathbf{R}^n)$, which is still denoted $\tilde{\gamma}(h)$.
- b) We denote γ the automorphism of $\mathcal{D}'(\mathbf{R}^n)$ transposed to $\tilde{\gamma}(-h)$. If $f \in \mathcal{D}(\mathbf{R}^n)$, then $\gamma(h)f = \tilde{\gamma}(h)f$ for all $h \in \mathbf{R}^n$.
- c) The map $h \mapsto \gamma(h)$ is a linear representation of $\mathbf{R}^n$ in $\mathcal{D}'(\mathbf{R}^n)$.
- d) For every $h \in \mathbf{R}^n$ and every $\alpha \in \mathbf{N}^n$, we have $\partial^\alpha \circ \gamma(h) = \gamma(h) \circ \partial^\alpha$.
- e) Let i be an integer such that $1 \leq i \leq n$ and e_i the i -th vector of the canonical basis of $\mathbf{R}^n$. For every distribution $f \in \mathcal{D}'(\mathbf{R}^n)$, we have

$$\lim_{\substack{h \rightarrow 0 \\ h \neq 0}} \frac{1}{h} (\gamma(h e_i) f - f) = \partial_i f$$

in $\mathcal{D}'(\mathbf{R}^n)$ endowed with the weak topology.

- 8) Let $n \in \mathbf{N}$ and let U be an open subset of $\mathbf{R}^n$. A distribution $f \in \mathcal{D}'(U)$ is said to be positive if $\langle f, \varphi \rangle \geq 0$ for every $\varphi \in \mathcal{D}(U)$ such that $\varphi \geq 0$. Prove that f is positive if and only if f is the distribution associated with a positive measure on U .
- 9) Let $n \in \mathbf{N}$ and U be an open subset of $\mathbf{R}^n$. Let $k \in \mathbf{N}$ and $p \in]1, +\infty[$. The space $W^{k,p}(U)$ is reflexive.
- 10) Let $n \in \mathbf{N}$ and U be an open subset of $\mathbf{R}^n$. Let $k \in \mathbf{N}$ and $p \in [1, +\infty[$. Let $V \subset U$ be a relatively compact open subset.

Let $\varphi \in \mathcal{D}(\mathbf{R}^n)$ be a positive function with support contained in the unit ball of $\mathbf{R}^n$ and integral 1. For every $\varepsilon > 0$ and $x \in \mathbf{R}^n$ we set $\varphi_\varepsilon(x) = \varepsilon^{-n} \varphi(x/\varepsilon)$. Let $f \in W^{k,p}(U)$.

- a) The restriction f_V of f to V belongs to $W^{k,p}(V)$.
- b) Let $\tilde{f}$ be the function on $\mathbf{R}^n$ which coincides with f on U and vanishes outside U . When $\varepsilon \rightarrow 0$, the restriction of $\tilde{f} * \varphi_\varepsilon$ to V belongs to $W^{k,p}(V)$ and converges to f_V in $W^{k,p}(V)$.

11) Let $n \in \mathbf{N}$ and let U be an open subset of $\mathbf{R}^n$. Let $k \in \mathbf{N}$ and $p \in [1, +\infty[$. We denote by $H^{k,p}(U)$ the closure of $\mathcal{C}^\infty(U)$ in $W^{k,p}(U)$.

For $x \in U$, let $\delta(x)$ be the distance from x to the boundary of U in $\mathbf{R}^n$; let $U_{-1} = U_0 = \emptyset$ and

$$U_k = \{x \in U \mid \|x\| < k, \delta(x) > k^{-1}\}$$

for every integer $k \geq 1$. Set $V_k = U_{k+1} \cap (\mathbf{R}^n - \overline{U}_{k-1})$ for $k \geq 1$. The open sets V_k cover U ; let $(\psi_k)_{k \geq 1}$ be a partition of unity subordinate to the covering $(V_k)_{k \geq 1}$ consisting of functions in $\mathcal{D}(U)$.

a) For $k \geq 1$, the function

$$\varphi_k = \sum_{\substack{j \geq 1 \\ \text{Supp}(\psi_j) \subset V_k}} \psi_j$$

belongs to $\mathcal{D}(U)$ and has support contained in V_k ; we have

$$\sum_{k \geq 1} \varphi_k = 1$$

on U .

b) Let $f \in W^{k,p}(U)$ and $\varepsilon > 0$. Let φ and φ_ε for $\varepsilon > 0$ functions as in Exercise 10. For every $k \geq 1$, there exists $\varepsilon_k > 0$ such that

$$\|\varphi_{\varepsilon_k} * (\psi_k f) - \psi_k f\|_{k,p} < \frac{\varepsilon}{2^k}.$$

c) The function

$$f_\varepsilon = \sum_{k \geq 1} \varphi_{\varepsilon_k} * (\psi_k f)$$

belongs to $\mathcal{C}^\infty(U)$ and $\|f - f_\varepsilon\|_{k,p} < \varepsilon$.

d). The space $H^{k,p}(U)$ is equal to $W^{k,p}(U)$.

e) The space $H^{k,p}(U)$ is identified with the completion of the space $\mathcal{C}^k(U)$ for the norm $f \mapsto \|f\|_{k,p}$.

f) Let V be an open subset of $\mathbf{R}^n$ and let $\varphi: U \rightarrow V$ be a diffeomorphism of class C^k . Assume that the partial derivatives of order $\leq k$ of φ (resp. of φ^{-1}) are bounded and uniformly continuous on U (resp. on V). The map $f \mapsto f \circ \varphi$ is an isomorphism of topological vector spaces from $W^{k,p}(V)$ to $W^{k,p}(U)$. (Use the result of the previous question.)

12) a) For all integers $n \in \mathbf{N}$ and $k \in \mathbf{N}$, and for every real number $p \geq 1$, we have

$$W^{k,p}(\mathbf{R}^n) = W_0^{k,p}(\mathbf{R}^n).$$

b) Let $p \geq 1$ be a real number. Let $U \subset \mathbf{R}^2$ be the set of $(x, y) \in \mathbf{R}^2$ such that $0 < |x| < 1$ and $0 < y < 1$. The characteristic function φ of $U \cap (\mathbf{R}_+^* \times \mathbf{R})$ belongs to $W^{1,p}(U)$; if $\varepsilon > 0$ is sufficiently small, there does not exist a function f of class C^1 in the neighborhood of $\overline{U}$ such that $\|f - \varphi\|_{1,p} < \varepsilon$.

13) Let U be an open subset of $\mathbf{R}^n$ and $j: U \rightarrow \mathbf{R}^n$ be the canonical inclusion. For every function $f: U \rightarrow \mathbf{C}$, we denote by $j_!f$ the zero extension of f to $\mathbf{R}^n$, that is, the function from $\mathbf{R}^n$ to $\mathbf{C}$ which coincides with f in U and is zero outside U .

Let $k \geq 0$ be an integer and $p \geq 1$ a real number.

a) Let $f \in W^{k,p}(U)$. For every $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq k$, we have $\partial^\alpha(j_!f) = j_!(\partial^\alpha(f))$ in $\mathcal{D}'(\mathbf{R}^n)$.

b) The map $f \mapsto j_!f$ induced by passage to subspaces, is an isometric linear map from $W_0^{k,p}(U)$ into $W^{k,p}(\mathbf{R}^n)$.

14) Let U be an open subset of $\mathbf{R}^n$ and let $k \geq 1$ be an integer. Let $p \geq 1$ be a real number. Assume that the complement $\mathbf{R}^n - U$ is not negligible for Lebesgue measure. Let j denote the canonical inclusion of U into $\mathbf{R}^n$.

a) There exists a relatively compact open rectangle B in $\mathbf{R}^n$ such that $B \cap U$ and $B \cap (\mathbf{R}^n - U)$ are not negligible.

b) Let $\tilde{f} \in \mathcal{D}(\mathbf{R}^n)$ which equals 1 in $B \cap U$. Let f be the restriction of $\tilde{f}$ to U . Then f does

not belong to $W_0^{k,p}(U)$ (Otherwise, the extension by zero $j_1 f$ would belong to $W^{k,p}(\mathbf{R}^n)$; the restriction to B of $j_1 f$ would be a distribution such that $\partial_i(j_1 f) = 0$ for $1 \leq i \leq n$, hence would be the distribution associated with a constant.)

15) Let $n \geq 1$ be an integer and let p be a real number such that $1 \leq p < n$.

- a) Let α be a real number such that $0 < \alpha < n/p - 1$. Let g be a function on $\mathbf{R}_+^*$, infinitely differentiable, positive, such that $g(x) = x^{-\alpha}$ for x sufficiently small and $g(x) = 0$ for x sufficiently large; the function from $\mathbf{R}^n$ to $\mathbf{C}$ defined by $x \mapsto g(\|x\|)$ belongs to $W^{1,p}(\mathbf{R}^n)$.
- b) Let $i \mapsto q_i$ be a bijection from $\mathbf{N}$ to $\mathbf{Q}^n$ and let f be the map from $\mathbf{R}^n$ to $\mathbf{C}$ defined by

$$f(x) = \sum_{i \in \mathbf{N}} 2^{-i} g(\|x - q_i\|)$$

for $x \in \mathbf{R}^n$. The series defining f converges in $W^{1,p}(\mathbf{R}^n)$; for every open set U of $\mathbf{R}^n$ the restriction of f to U is not almost everywhere differentiable. c) There exists a function $f \in W^{1,n}(\mathbf{R}^n)$ which is continuous at no point. (Similar method with the function $x \mapsto \log(|\log(|x|)|)$.)

16) Let $n \geq 1$ and $k \geq 1$ be integers. Let $p \geq 1$ be a real number. Let U be a bounded open subset of $\mathbf{R}^n$.

- a) Suppose that $n \neq kp$. Let q be a real number such that $1 \leq q \leq np/(n - kp)$. The inclusion of $\mathcal{D}(U)$ in $L^q(U)$ admits a continuous extension i from $W_0^{k,p}(U)$ into $L^q(U)$.
- b) Suppose that $1 \leq q \leq np/(n - kp)$. The linear map i is compact ("Rellich-Kondrachov theorem").
- c) Suppose that $k \geq 1$ and that $q = np/(n - p)$. The linear map i is not compact.

17) Let $n \geq 1$ be an integer. We denote by μ the Lebesgue measure on $\mathbf{R}^n$.

- a) A bounded subset B of $L^2(\mathbf{R}^n, \mu)$ is compact if and only if, for every $\varepsilon > 0$, there exists a compact subset K of $\mathbf{R}^n$ such that

$$\int_{\mathbf{R}^n - K} |f|^2 + \int_{\mathbf{R}^n - K} |\hat{f}|^2 < \varepsilon$$

for all $f \in B$

- b) Let $k \in \mathbf{N}$. Let K be a compact subset of $\mathbf{R}^n$. We denote by $H_k^2(K)$ the closed subspace of the Sobolev space $H_k^2(\mathbf{R}^n)$ consisting of functions vanishing outside the compact set K . The canonical injection $H_k^2(K) \rightarrow H_k^2(\mathbf{R}^n)$ is compact.

- 18) Let $n \in \mathbf{N}$ and U be an open subset of $\mathbf{R}^n$. The goal of this exercise is to prove that $\mathcal{D}'(U)$ is a bornological space.

If K is a compact subset of U , we equip $\mathcal{C}_K^\infty(U)'$ with the topology of bounded convergence. We denote by $\pi_K: \mathcal{D}'(U) \rightarrow \mathcal{C}_K^\infty(U)'$ the linear map which assigns to a distribution its restriction to $\mathcal{C}_K^\infty(U)$; it is continuous.

Let $S \subset \mathcal{D}'(U)$ be a convex balanced bornivorous set.

- a) There exists a compact subset K of U such that $\text{Ker}(\pi_K) \subset S$. (Otherwise, consider an increasing sequence $(K_m)_{m \in \mathbf{N}}$ of compact subsets of U whose interiors cover U and choose f_m in $\text{Ker}(\pi_{K_m}) - S$; then the set of elements mf_m for $m \in \mathbf{N}$ is a bounded subset not absorbed by S .)
- b) The set T of distributions $f \in \mathcal{D}'(U)$ such that $f + \text{Ker}(\pi_K) \subset S$ is a convex balanced bornivorous set.
- c) The set $\pi_K(T)$ is a convex balanced bornivorous subset of $\mathcal{C}_K^\infty(U)'$. (Let A be a bounded subset of $\mathcal{C}_K^\infty(U)'$; show that there exists a neighborhood W of 0 in $\mathcal{D}(U)$ such that $A \subset (W \cap \mathcal{C}_K^\infty(U))^\circ$ and deduce that A is contained in $\pi_K(W^\circ)$, hence is absorbed by $\pi_K(T)$.)
- d) Deduce that S is a neighborhood of 0, and conclude that the space $\mathcal{D}'(U)$ is bornological. (Note that $\pi_K(T)$ is a neighborhood of 0, and observe that S contains $\pi_K^{-1}(\pi_K(T))$.)

- 19) Let $n \in \mathbf{N}$ and let $U \subset \mathbf{R}^n$ be an open set.

- a) Let $f \in \mathcal{D}'(U)$. The union V of the open sets $W \subset U$ such that the restriction of f to W

is zero is open in U .

The support of f is called the complement of V ; it is a closed subset of U .

b) Let $f \in \mathcal{D}'(U)$ and let $\varphi \in \mathcal{D}(U)$ whose support does not meet the support of f . We have

$$\langle f, \varphi \rangle = 0.$$

c) Let $f \in \mathcal{D}'(U)$ and $\alpha \in \mathbb{N}^n$. The support of $\partial^\alpha f$ is contained in the support of f .

d) If $f \in \mathcal{D}'(U)$ is the distribution associated with a measure ν , then the support of f coincides with the support of ν .

e) Let f be a distribution with compact support in U . The linear form f extends to a continuous linear form on the space $\mathcal{C}^\infty(U)$.

f) The space of distributions with compact support in U is identified with the dual of the space $\mathcal{C}^\infty(U)$.

g) Let $f \in \mathcal{D}'(\mathbb{R}^n)$ be a distribution with compact support. We have $f \in \mathcal{S}'(\mathbb{R}^n)$. h)

Let $f \in \mathcal{D}'(\mathbb{R}^n)$ be a distribution with compact support. The Fourier transform of f belongs to $\mathcal{C}^\infty(\mathbb{R}^n)$; it is a function of polynomial growth together with all its derivatives.

20) Let $n \in \mathbb{N}$. We denote by V_n the open set in $\mathbb{C}$ consisting of the complex numbers λ such that $\Re(\lambda) > -n - 1$ and $\lambda \notin \{-1, -2, \dots, -n\}$.

Let $\lambda \in V_0$. We denote by x_+^λ the distribution on $\mathbb{R}$ defined by the locally integrable function

$$x \mapsto \begin{cases} 0 & \text{if } x \leq 0, \\ x^\lambda & \text{if } x > 0. \end{cases}$$

a) The map $\lambda \mapsto x_+^\lambda$ of V_0 in $\mathcal{D}'(\mathbb{R})$ is analytic.

b) Let $n \in \mathbb{N}$. For every $\lambda \in V_n$, the map defined on $\mathcal{D}(\mathbb{R})$ by

$$\begin{aligned} \varphi \mapsto & \int_0^1 x^\lambda \left(\varphi(x) - \varphi(0) - x\varphi'(0) - \dots - \frac{x^{n-1}}{(n-1)!} \varphi^{(n-1)}(0) \right) dx \\ & + \int_1^{+\infty} x^\lambda \varphi(x) dx + \sum_{k=1}^n \frac{\varphi^{(k-1)}(0)}{(k-1)!(\lambda+k)} \end{aligned}$$

is a tempered distribution; it does not depend on the integer n such that $\lambda \in V_n$, and coincides with x_+^λ when $\lambda \in V_0$.

We shall denote by x_+^λ the distribution thus defined for every λ belonging to the union of the sets V_n for $n \in \mathbb{N}$.

- c) Let $n \in \mathbb{N}$ and let $\lambda \in \mathbb{C}$ such that $-n - 1 < \Re(\lambda) < -n$. For every Schwartz function φ on $\mathbb{R}$, we have

$$\langle x_+^\lambda, \varphi \rangle = \int_0^{+\infty} x^\lambda \left(\varphi(x) - \varphi(0) - x\varphi'(0) - \dots - \frac{x^{n-1}}{(n-1)!} \varphi^{(n-1)}(0) \right) dx.$$

The map from V_n in $\mathcal{D}'(\mathbb{U})$ defined by $\lambda \mapsto x_+^\lambda$ is analytic.

- d) For every λ belonging to the union of the sets V_m , we have the formula

$$(x_+^\lambda)' = \lambda x_+^{\lambda-1}.$$

- 21) a) Let $f \in \mathcal{S}(\mathbb{R})$. The function g on $\mathbb{R}$ such that $g(0) = 2f'(0)$ and $g(x) = (f(x) - f(-x))/x$ for all $x \neq 0$ is integrable on $\mathbb{R}$ with respect to Lebesgue measure. The map that assigns to f the integral of g over $\mathbb{R}$ is a tempered distribution, denoted vp , and called the Cauchy principal value of $1/x$.

- b) For every $f \in \mathcal{S}(\mathbb{R})$, we have

$$\langle \text{vp}, f \rangle = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R} - [-\varepsilon, \varepsilon]} \frac{f(x)}{x} dx.$$

- c) The restriction of vp to $\mathbb{R} - \{0\}$ is the distribution associated with the locally integrable function $x \mapsto 1/x$ on $\mathbb{R} - \{0\}$.
- d) The distribution vp coincides with the derivative of the distribution associated with the locally integrable function on $\mathbb{R}$ taking value 0 at 0 and defined for $x \neq 0$ by $x \mapsto \log(|x|)$.
- e) Let $\varepsilon > 0$. We denote by p_ε the function $x \mapsto 1/(x + i\varepsilon)$ on $\mathbb{R}$. We have

$$\lim_{\varepsilon \rightarrow 0} p_\varepsilon = \text{vp} - i\pi \varepsilon_0$$

in the space $\mathcal{S}'(\mathbf{R})$ endowed with the weak topology, where ε_0 is the tempered distribution associated with the point mass at 0.

22) The distribution on $\mathbf{R}$ associated with the exponential function is not tempered.

23) Let $n \in \mathbf{N}$ and let U be an open set of $\mathbf{R}^n$. We denote by μ the Lebesgue measure on U .

For $m \in \mathbf{N}$, we denote $m = (m_1, \dots, m_n) \in \mathbf{N}^n$.

a) Let $f \in \mathcal{D}(U)$. Suppose that there exists an integer $m \geq 1$ and a real number $c \geq 0$ such that

$$|\langle f, \varphi \rangle| \leq c \int_U |\partial^m \varphi| d\mu$$

for all $\varphi \in \mathcal{D}(U)$. There then exists a function $g \in L^\infty(U)$ such that $f = \partial^m g$. (The linear map $\varphi \mapsto \partial^m \varphi$ is injective on $\mathcal{D}(U)$; let E be its image; consider the linear form on E into $\mathbf{C}$ that maps $\partial^m \varphi$ to $\langle f, \varphi \rangle$ and apply the Hahn–Banach theorem.)

b) Let V be a relatively compact open set contained in U . There exists an integer $m \geq 1$ and $c \geq 0$ such that

$$|\langle f, \varphi \rangle| \leq c \int_U |\partial^m \varphi| d\mu$$

for all $\varphi \in \mathcal{D}(U)$ with support contained in V .

c) For every relatively compact open set $V \subset U$, there exists a continuous function $g \in \mathcal{C}(V)$ and $\alpha \in \mathbf{N}^n$ such that the restriction of f to V is equal to $\partial^\alpha g$. d) Give an example of a distribution $f \in \mathcal{D}'(\mathbf{R}^n)$ such that f is not of the form $f = \partial^\alpha g$ for a continuous function $g \in \mathcal{C}(\mathbf{R}^n)$ and $\alpha \in \mathbf{N}^n$.

24) Let $n \in \mathbf{N}$ and let U be an open set of $\mathbf{R}^n$. Let $f \in \mathcal{D}'(U)$ be a distribution whose support C is compact in U .

- a) There exists an integer $m \in \mathbb{N}$ such that the distribution f extends to a continuous linear form on the space $\mathcal{C}^m(U)$ endowed with the topology of uniform convergence on compact subsets of functions and their partial derivatives of order $\leq m$.

The smallest integer m satisfying this property is called the order of f .

- b) Let $\varphi \in \mathcal{D}(U)$ such that $\partial^\alpha \varphi$ vanishes on C for every $\alpha \in \mathbb{N}^n$ such that $|\alpha| \leq m$. We then have $\langle f, \varphi \rangle = 0$. (Consider, for $\delta > 0$ sufficiently small, the neighborhood U_δ of C consisting of the $x \in U$ such that $d(x, C) \leq \delta$, and show that there exists $\eta \geq 0$, tending to 0 as δ tends to 0, such that, for some $C_0 \geq 0$, one has $|\partial^\alpha \varphi(x)| \leq C_0 \delta^{m-|\alpha|} \eta$ for $x \in U_\delta$ and $|\alpha| \leq m$; show that there exists a function $\beta_\delta \in \mathcal{D}(U)$ equal to 1 on $U_{\delta/4}$ and with support contained in U_δ satisfying $|\partial^\alpha \beta_\delta| \leq C_1 \delta^{-|\alpha|}$ for some $C_1 \geq 0$; finally observe that $\langle f, \varphi \rangle = \langle f, \varphi \beta_\delta \rangle$ and that $\varphi \beta_\delta$ tends to 0 in $\mathcal{C}^m(U)$.)
- c) Let $x \in U$. The space of distributions f whose support of f is reduced to x has as an algebraic basis the distributions $\partial^\alpha \varepsilon_x$, where $\alpha \in \mathbb{N}^n$.
- d) Let $n = 1$. The formula

$$\langle f, \varphi \rangle = \lim_{m \rightarrow +\infty} \left(\sum_{1 \leq k \leq m} \varphi(k^{-1}) - m\varphi(0) - \log(m)\varphi'(0) \right)$$

defines a distribution on $\mathbf{R}$, of order 1, whose support is compact, equal to the set $C = \{0\} \cup \{1/(n+1) \mid n \in \mathbb{N}\}$. There exist sequences (φ_n) in $\mathcal{D}(\mathbf{R})$ such that $\partial^\alpha \varphi_n$ converges uniformly to 0 on C for every $\alpha \in \mathbb{N}^n$ such that $|\alpha| \leq 1$, but $\langle f, \varphi_n \rangle \rightarrow +\infty$.

- 25) Let $n \in \mathbb{N}$. We denote by N the Euclidean norm on $\mathbf{R}^n$ and let μ be the Lebesgue measure.

We denote by $\mathcal{D}_{L^1}(\mathbf{R}^n)$ the vector subspace of $\mathcal{C}^\infty(\mathbf{R}^n)$ whose elements are the functions φ such that $\partial^\alpha \varphi \in L^1(\mathbf{R}^n)$ and $\partial^\alpha \varphi \in \mathcal{C}_0(\mathbf{R}^n)$ for all $\alpha \in \mathbb{N}^n$. a) The space $\mathcal{D}_{L^1}(\mathbf{R}^n)$, endowed with the topology defined by the seminorms

$$q_\alpha(\varphi) = \int_{\mathbf{R}^n} |\partial^\alpha \varphi| d\mu, \quad \alpha \in \mathbb{N}^n,$$

is a Fréchet space.

b) The space $\mathcal{D}(\mathbf{R}^n)$ is contained in and is dense in $\mathcal{D}_{L^1}(\mathbf{R}^n)$. One can identify the dual space $\mathcal{D}'_{L^1}(\mathbf{R}^n)$ of $\mathcal{D}_{L^1}(\mathbf{R}^n)$ with a subspace of $\mathcal{D}'(\mathbf{R}^n)$.

c) Let $f \in \mathcal{D}'(\mathbf{R}^n)$. We have $f \in \mathcal{D}'_{L^1}(\mathbf{R}^n)$ if and only if there exists a finite set I , a family $(\alpha_i)_{i \in I}$ in $\mathbf{N}^n$ and a family $(g_i)_{i \in I}$ in $L^\infty(\mathbf{R}^n)$ such that

$$f = \sum_{i \in I} \partial^{\alpha_i} g_i.$$

(Determine I and the family (α_i) so that $|\langle f, \varphi \rangle|$ is bounded when φ is such that $q_{\alpha_i}(\varphi) \leq 1$ for all $i \in I$; identify $\mathcal{D}_{L^1}(\mathbf{R}^n)$ with a vector subspace of $L^1(\mathbf{R}^n)^m$ by the mapping $\varphi \mapsto (\partial^{\alpha_i} \varphi)_{i \in I}$, and apply the Hahn-Banach theorem.)

d) Let $f \in \mathcal{D}'(\mathbf{R}^n)$. We have $f \in \mathcal{D}'_{L^1}(\mathbf{R}^n)$ if and only if there exists a continuous function g with moderate growth on $\mathbf{R}^n$ and $\alpha \in \mathbf{N}^n$ such that $f = \partial^\alpha g$. e) Let $f \in \mathcal{D}'(\mathbf{R}^n)$. There exists an integer $m \in \mathbf{N}$ such that $(1 + N^2)^{-m/2} f$ belongs to $\mathcal{D}'_{L^1}(\mathbf{R}^n)$.

f) Let $f \in \mathcal{D}'(\mathbf{R}^n)$. We have $f \in \mathcal{S}'(\mathbf{R}^n)$ if and only if there exists a bounded continuous function $g: \mathbf{R}^n \rightarrow \mathbf{C}$, an integer $m \in \mathbf{N}$ and $\alpha \in \mathbf{N}^n$ such that $f = \partial^\alpha ((1 + N^2)^{m/2} g)$.

26) Let $n \in \mathbf{N}$. We equip $\mathbf{R}^n$ with Lebesgue measure μ and identify $\mathbf{R}^n$ with its dual by the mapping $(x, y) \mapsto \exp(2i\pi x \cdot y)$.

a) For any function $f \in \mathcal{S}(\mathbf{R}^n)$, the map $g \mapsto f * g$ is an automorphism of the locally convex space of $\mathcal{S}(\mathbf{R}^n)$.

b) Let $f \in \mathcal{S}'(\mathbf{R}^n)$ and $g \in \mathcal{S}(\mathbf{R}^n)$. The map $\varphi \mapsto \langle f, \check{g} * \varphi \rangle$ is a tempered distribution, called the convolution of f and g . If $f \in \mathcal{S}(\mathbf{R}^n)$, it coincides with the distribution associated with $f * g$. We shall still denote by $f * g$ the tempered distribution thus defined.

In the following questions, we fix $f \in \mathcal{S}'(\mathbf{R}^n)$ and $g \in \mathcal{S}(\mathbf{R}^n)$. c) One has $f * g \in \mathcal{C}^\infty(\mathbf{R})$ and

$$\partial^\alpha (f * g) = \partial^\alpha (f) * g = f * \partial^\alpha (g)$$

for all $\alpha \in \mathbf{N}^n$.

d) For every $y \in \mathbf{R}^n$, let g_y be the function $x \mapsto g(y - x)$ on $\mathbf{R}^n$. The function $f * g$ satisfies

$$(f * g)(y) = \langle f, g_y \rangle$$

for all $y \in \mathbf{R}^n$.

e) Let $h \in \mathcal{S}(\mathbf{R}^n)$. We have $(f * g) * h = f * (g * h)$.

f) The function $f * g$ has polynomial growth, as do all its derivatives. (Use Exercise 25.)

g) One has $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$.

h) The Fourier transform of fg is an infinitely differentiable function on $\mathbf{R}^n$ and has polynomial growth, as do all its derivatives. Moreover, for every $y \in \mathbf{R}^n$, we have

$$\mathcal{F}(fg)(y) = \langle f, g e_{-y} \rangle$$

where e_y denotes the character $x \mapsto \exp(2i\pi x \cdot y)$ of $\mathbf{R}^n$.

27) Let $n \in \mathbf{N}$. We denote by N the Euclidean norm on $\mathbf{R}^n$ and let μ be the Lebesgue measure on every open subset of $\mathbf{R}^n$. Let Δ be the differential operator

$$\Delta = - \sum_{i=1}^n \partial_i^2$$

viewed as an endomorphism of $\mathcal{D}'(\mathbf{R}^n)$.

For $k \in \mathbf{R}$, we denote by $S^k(\mathbf{R}^n)$ the space of tempered distributions f in $\mathcal{S}'(\mathbf{R}^n)$ such that $(1 + N^k)\mathcal{F}(f)$ belongs to $L^2(\mathbf{R}^n)$.

Let $U \subset \mathbf{R}^n$ be an open subset of $\mathbf{R}^n$. We denote by $S_U^k(\mathbf{R}^n)$ the space of distributions $f \in \mathcal{D}'(\mathbf{R}^n)$ such that for every test function $\varphi \in \mathcal{D}(\mathbf{R}^n)$ with support contained in U , we have $\varphi f \in S^k(\mathbf{R}^n)$.

a) The space $S^k(\mathbf{R}^n)$ is a Hilbert space with the inner product

$$\langle f_1 | f_2 \rangle = \int_{\mathbf{R}^n} (1 + N^k) \overline{\mathcal{F}(f_1)} \mathcal{F}(f_2) d\mu.$$

When $k \in \mathbf{N}$, we have $W^k(\mathbf{R}^n) = S^k(\mathbf{R}^n)$ and the norms of these two Banach spaces are equivalent; moreover, we then have $W^k(\mathbf{R}^n) \subset S_{\mathbf{U}}^k(\mathbf{R}^n)$.

b) The dual of $S^k(\mathbf{R}^n)$ is identified with $S^{-k}(\mathbf{R}^n)$ by the bilinear form

$$(f, g) \mapsto \int_{\mathbf{R}^n} \mathcal{F}(f)(-x) \mathcal{F}(g)(x) d\mu(x)$$

for $(f, g) \in S^{-k}(\mathbf{R}^n) \times S^k(\mathbf{R}^n)$.

c) Suppose that U is relatively compact. We have

$$\bigcup_{k \in \mathbf{R}} S_{\mathbf{U}}^k(\mathbf{R}^n) = \mathcal{D}'(\mathbf{R}^n).$$

(Let $f \in \mathcal{D}'(\mathbf{R}^n)$; apply the results of Exercise 19 to φf , where $\varphi \in \mathcal{D}(\mathbf{R}^n)$ is equal to 1 on U .)

d) Let $k \in \mathbf{R}$ and $f \in S^k(\mathbf{R}^n)$. For every integer i such that $1 \leq i \leq n$, we have $\partial_i f \in S^{k-1}(\mathbf{R}^n)$. Moreover, if $\Delta(f) \in S^k(\mathbf{R}^n)$, then $f \in S^{k+2}(\mathbf{R}^n)$.

e) Let $k \in \mathbf{R}$ and $f \in S_{\mathbf{U}}^k(\mathbf{R}^n)$. If $\Delta(f) \in S_{\mathbf{U}}^k(\mathbf{R}^n)$, then $f \in S_{\mathbf{U}}^{k+2}(\mathbf{R}^n)$.

f) Let $k > n/2$ be a real number and $m \in \mathbf{N}$ be an integer such that $m < k - n/2$. The space $S^k(\mathbf{R}^n)$ is contained in $\mathcal{C}^m(\mathbf{R}^n)$ (Using the Cauchy-Schwarz inequality, verify that $x \mapsto x^\alpha \mathcal{F}f(x)$ belongs to $L^1(\mathbf{R}^n)$ for all $\alpha \in \mathbf{N}^n$ such that $|\alpha| \leq m$. Then show that $\mathcal{F}(f)$ is differentiable on $\mathbf{R}$ using the Fourier inversion formula.)

g) Let $k > n/2$ be a real number and $m \in \mathbf{N}$ be an integer such that $m < k - n/2$. The restriction to U of an element of $S_{\mathbf{U}}^k(\mathbf{R}^n)$ belongs to $\mathcal{C}^m(U)$ (For every $x \in U$, verify that one obtains a function $g \in \mathcal{C}^m(U)$ well defined by setting $g(x) = (\varphi_x f)(x)$, where $\varphi_x \in \mathcal{D}(\mathbf{R}^n)$ is a function with support in U equal to 1 in a neighborhood of x ; finally verify that the restriction of f to U is equal to g .)

h) Let $g \in \mathcal{D}'(\mathbf{R}^n)$. If $f \in \mathcal{D}'(\mathbf{R}^n)$ satisfies $\Delta(f) = g$ and if the restriction of g to U is in $\mathcal{C}^\infty(U)$, then the restriction of f to U is in $\mathcal{C}^\infty(U)$ ("elliptic regularity theorem").

28) Let n and m be integers, and let $U \subset \mathbf{R}^n$ and $V \subset \mathbf{R}^m$ be open sets. a) There exists a

unique injective linear map u from $\mathcal{D}(U) \otimes \mathcal{D}(V)$ into $\mathcal{D}(U \times V)$ such that $u(\varphi \otimes \psi)$ is the function $(x, y) \mapsto \varphi(x)\psi(y)$ for all $(\varphi, \psi) \in \mathcal{D}(U) \times \mathcal{D}(V)$. Moreover, the image of u is dense in $\mathcal{D}(U \times V)$. One identifies the space $\mathcal{D}(U) \otimes \mathcal{D}(V)$ with a subspace of $\mathcal{D}(U \times V)$ by means of the map u .

- b) For all distributions f on U and g on V , there exists a unique distribution $f \otimes g$ on $U \times V$ such that $\langle f \otimes g, \varphi \otimes \psi \rangle = \langle f, \varphi \rangle \langle g, \psi \rangle$ for all $(\varphi, \psi) \in \mathcal{D}(U) \times \mathcal{D}(V)$.
- c) Let $f \in \mathcal{D}(U)$ and $g \in \mathcal{D}(V)$. For every $\eta \in \mathcal{D}(U \times V)$ and every $x \in U$, let η_x be the map $y \mapsto \eta(x, y)$. We have $\eta_x \in \mathcal{D}(V)$ and the map φ which assigns to each $x \in U$ $\langle f, \eta_x \rangle$ belongs to $\mathcal{D}(U)$ and satisfies

$$\langle f \otimes g, \eta \rangle = \langle f, \varphi \rangle.$$

29) Let n and m be integers and let $U \subset \mathbf{R}^n$ and $V \subset \mathbf{R}^m$ be open sets. We resume the notation of the previous exercise. Let u be a continuous linear map from $\mathcal{D}(U)$ into $\mathcal{D}'(V)$.

- a) There exists a unique continuous linear form $\tilde{\kappa}$ defined on the subspace $\mathcal{D}(U) \otimes \mathcal{D}(V)$ of $\mathcal{D}(U \times V)$ such that $\tilde{\kappa}(\varphi \otimes \psi) = \langle u(\varphi), \psi \rangle$ for all $(\varphi, \psi) \in \mathcal{D}(U) \times \mathcal{D}(V)$.
- b) There exists a distribution $\kappa(u)$ on $U \times V$ which extends $\tilde{\kappa}$, and the map $u \mapsto \kappa(u)$ from $\mathcal{L}(\mathcal{D}(U), \mathcal{D}'(V))$ into $\mathcal{D}'(U \times V)$ is continuous.
- c) The map κ is bijective ("Schwartz kernel theorem"; to prove the surjectivity of κ , given a distribution η on $U \times V$, define $u(\varphi)$ as the distribution $\psi \mapsto \langle \eta, \varphi \otimes \psi \rangle$ on V .)

30) Let $n \in \mathbf{N}$. We denote by $(x_1, \dots, x_n)$ the dual basis of the canonical basis of $\mathbf{R}^n$.

- a) Let $c \geq 0$ and $r > 0$ be real numbers. There exists a function u defined and analytic on an open neighborhood U of 0 in $\mathbf{R}^n$ such that

$$\left(r - \sum_{j=1}^{n-1} x_j - u \right) \partial_n u - cr \sum_{j=1}^{n-1} \partial_j u = cr$$

on U and $u(x, 0) = 0$ for $x \in \mathbf{R}^{n-1}$ such that $(x, 0) \in U$. (Consider first the case $n=2$.)

- b) Let U be an open neighborhood of 0 in $\mathbf{R}^n$. Let $(a_1, \dots, a_{n-1}, b)$ be functions from $U \times \mathbf{R}$ to $\mathbf{C}$ that are analytic in a neighborhood of $(0, 0) \in U \times \mathbf{R}$. We denote by Q the map from the vector space of analytic functions on U to itself such that

$$Q(u) = \partial_n u - \sum_{j=1}^{n-1} a_j(x_1, \dots, x_{n-1}, u) \partial_j u + b(x_1, \dots, x_{n-1}, u).$$

There exists at most one function v defined and analytic in an open neighborhood V of 0 contained in U such that $Q(v) = 0$ and such that $v(x, 0) = 0$ for all $x \in \mathbf{R}^{n-1}$ such that $(x, 0) \in V$. (Verify that the coefficients of the Taylor expansion at 0 of such a solution v are uniquely determined.)

- c) Let f and g be formal power series in $\mathbf{C}[[z_1, \dots, z_n]]$. One says that f majorizes g , and one writes $f \ll g$, if

$$f = \sum_{\alpha \in \mathbf{N}^n} a_\alpha z^\alpha, \quad g = \sum_{\alpha \in \mathbf{N}^n} b_\alpha z^\alpha$$

with $|a_\alpha| \leq b_\alpha$ for all $\alpha \in \mathbf{N}^n$. If the formal series g converges in a neighborhood of 0 in $\mathbf{R}^n$, then so does f .

- d) If there exist functions $(\tilde{a}_1, \dots, \tilde{a}_{n-1}, \tilde{b})$ analytic in a neighborhood of 0 such that $a_j \ll \tilde{a}_j$ for $1 \leq j \leq n-1$ and $b \ll \tilde{b}$ (in the sense that the corresponding Taylor expansions at 0 satisfy these conditions), and if the equation

$$\partial_n v - \sum_{j=1}^{n-1} \tilde{a}_j(x_1, \dots, x_{n-1}, v) \partial_j v + \tilde{b}(x_1, \dots, x_{n-1}, v) = 0$$

in the unknown v has an analytic solution in a neighborhood of 0, then the equation $Q(v) = 0$ has a unique analytic solution in a neighborhood of 0.

- e) There exists a unique solution of the equation $Q(v) = 0$ that is analytic in a neighborhood of 0 (“Cauchy–Kowalewski theorem”).

31) We denote by L the differential operator

$$L = -i\partial_1 + \partial_2 - 2(x_1 + ix_2)\partial_3$$

of order 1 on $\mathbf{R}^3$, where (x_1, x_2, x_3) is the dual basis of the canonical basis of $\mathbf{R}^3$, the x_i being viewed as polynomials on $\mathbf{R}^3$ ("Lewy operator").

- a) Let U be an open neighborhood of 0 in $\mathbf{R}^3$. There exists a function $\varphi \in \mathcal{D}(U)$ such that $\varphi(0) = 1$ and such that $L(\varphi)$ is zero in a neighborhood V of 0 contained in U . (Use the Cauchy-Kowalewski theorem.)
- b) Let $u \in \mathcal{C}^\infty(U)$ be a function such that $L(u) = 0$ and $u(0) = 0$. Assume that there exists $\delta > 0$ such that $\Re(u(x)) < -2\delta$ for $x \in U - V$. We set $v_t = \varphi \exp(t(u + \delta))$. We have $v_t \in \mathcal{C}^\infty(U)$ and $L(v_t)$ tends to 0 in $\mathcal{D}(U)$ as $t \rightarrow +\infty$.
- c) Let F be the space of functions $f \in \mathcal{C}^\infty(U)$ such that $\partial^\alpha f$ is bounded on U for every $\alpha \in \mathbf{N}^3$. The space F , equipped with the seminorms $q_\alpha(f) = \sup_{x \in U} |\partial^\alpha f|$ is a Fréchet space.
- d) Let $\psi \in \mathcal{D}(\mathbf{R}^3)$. Denote by f_t the restriction to U of the function $x \mapsto t^3 \psi(tx)$. We have $f_t \in F$ for $t \geq 1$, and there exists a real number $c \geq 0$ such that $q_\alpha(f_t) \leq ct^{|\alpha|+3}$ for all $\alpha \in \mathbf{N}^3$ and every $t \geq 1$.
- e) For every $t \geq 1$, we denote by λ_t the linear form $f \mapsto \int_U f v_t$ on F . It is continuous and for t sufficiently large, one has

$$\lambda_t(f_t) = e^{\delta t} \left(\int_U \psi(x) \exp \left(\sum_{j=1}^3 \xi_j x_j \right) dx_1 dx_2 dx_3 + \varepsilon(t) \right)$$

where $\xi_j = \partial_j u(0)$ and $\varepsilon(t) \rightarrow 0$ as $t \rightarrow +\infty$.

f) Suppose that $(\xi_j) \neq 0$; there exists $f \in F$ such that the integrals $\int_U f v_t$ are not bounded when $t \rightarrow +\infty$. (If the assertion were false, obtain a contradiction by using the previous question and the Banach–Steinhaus theorem.)

g) The functions v_t do not converge to 0 in $\mathcal{D}'(U)$ when $t \rightarrow +\infty$. h) Let $\xi = (0, 0, \xi_3) \in \mathbf{R}^3$ such that $\xi_3 < 0$. There exists a neighborhood U of 0 in $\mathbf{R}^3$ and a function $u \in \mathcal{C}^\infty(U)$ such

that $L(u) = 0$ and $u(0) = 0$, which satisfies the conditions of questions *b)* and *f)*. (Use the Cauchy–Kowalewski theorem.)

- i) Let (v_t) be functions in $\mathcal{D}(U)$ such that $L(v_t) \rightarrow 0$, and $f \in \mathcal{D}(U)$ such that $\int_U f v_t$ is not bounded when $t \rightarrow +\infty$ (question *g)*); the equation $Lv=f$ has no solution $v \in \mathcal{D}'(U)$ (“Lewy’s theorem”). (Note that the formal adjoint of L coincides with L .)

32) Let $n \geq 1$ be an integer. We denote by Δ the Laplacian on $\mathcal{D}'(\mathbf{R}^n)$, defined by

$$\Delta(f) = - \sum_{i=1}^n \partial_i^2 f.$$

Let u be a locally integrable function on $\mathbf{R}^n$ such that $\Delta(u)$ is a locally integrable function.

Let $v \in \mathcal{L}^\infty(\mathbf{R}^n)$ be the function defined by

$$v(x) = \begin{cases} 0 & \text{if } u(x) = 0, \\ \overline{u(x)} / |u(x)| & \text{if } u(x) \neq 0. \end{cases}$$

- a) For $\varepsilon > 0$, set $u_\varepsilon = (|u|^2 + \varepsilon^2)^{1/2}$. Show that the distribution $-\Delta(u_\varepsilon) - \mathcal{R}(\overline{u}/u_\varepsilon \Delta(u))$ is positive. (First consider the case where $u \in \mathcal{C}^\infty(\mathbf{R}^n)$, then use convolution to treat the general case.)
- b) The distribution $-\Delta(|u|) - \mathcal{R}(v \Delta(u))$ is positive (“Kato inequality”).

33) Let $m \geq 1$ be an integer.

- a) There exists a unique linear map A^+ (resp. A^-) from $H^m(\mathbf{R})$ to $H^{m-1}(\mathbf{R})$ such that

$$A^+ f(x) = f'(x) - x f(x) \quad (\text{resp. } A^- f(x) = f'(x) + x f(x))$$

for all $f \in \mathcal{D}(\mathbf{R})$ and every $x \in \mathbf{R}$.

- b) The map A^+ is a Fredholm operator of index 1 and A^- is a Fredholm operator of index -1. (Consider the images of the Hermite functions under A^+ and A^- , cf. Exercise 3 of II, p. 263.)

Exercises for Section 4

Unless stated otherwise, the Banach spaces and Hilbert spaces below are assumed to be complex.

- 1)
 - a) There exist vector spaces E, F, G and partial operators u, v from E to F , and w from F to G , such that $w \circ (u + v) \neq w \circ u + w \circ v$.
 - b) Give an example of a closable operator u on a topological vector space such that the domain of the closure of u is strictly contained in the closure of the domain of u .
 - c) Give an example of closed operators u and v on a Hilbert space such that u is continuous but $u \circ v$ is not closed.
- 2) Let E and F be Banach spaces, and let G be a subspace of E . Let u be a finite-rank linear map from G into F . Suppose that u is not continuous on G .
 - a) The partial operator from E to F with domain G defined by u is not closable.
 - b) Suppose that E and G are Hilbert spaces. Compute the adjoint of the partial operator defined by u .
- 3) Let $K = \mathbb{R}$ or $\mathbb{C}$. Let E, F be topological vector spaces over K and u an operator with dense domain from E into F . Let D' be the set of continuous linear forms $\lambda \in F'$ such that there exists a continuous linear form $\mu \in E'$ coinciding with $\lambda \circ u$ on the domain of u .
 - a) The set D' is a vector subspace of F' , the linear form μ is determined in a unique way by λ , and the map $\lambda \mapsto \mu$ from D' to E' is linear.

One calls the transpose of u , and denotes ${}^t u$ the partial operator from F' into E' whose domain is D' and which is given by $\lambda \mapsto \mu$ for $\lambda \in D'$.

- b) Suppose that E and F are locally convex. The operator u is closable if and only if the duality $F \times \text{dom}({}^t u) \rightarrow K$ is separating.
- c) Suppose that E and F are locally convex and that F is semi-reflexive. The operator u is closable if and only if ${}^t u$ has dense domain.

4) Let E be a complex Hilbert space and u an injective self-adjoint partial operator on E . The partial operator u^{-1} is self-adjoint.

5) Let E be a complex Banach space. A continuous semigroup on E is called a map $s: \mathbf{R}_+ \rightarrow \mathcal{L}(E)$ such that

(i) $s(0) = 1_E$;

(ii) The map from $\mathbf{R}_+ \times E \rightarrow E$ defined by $(t, x) \mapsto s(t)x$ is continuous;

(iii) For all t_1 and t_2 in $\mathbf{R}_+$, we have $s(t_1 + t_2) = s(t_1)s(t_2)$.

We consider a continuous semigroup s on E .

a) There exist real numbers $C \geq 0$ and $M \geq 0$ such that

$$\|s(t)\| \leq Ce^{Mt}$$

for all $t \in \mathbf{R}_+$. (Let $a > 0$; show that $\|s(t)\|$ is bounded for $t \in [0, a]$.) We fix such numbers in the rest of the exercise.

b) Let D be the set of $x \in E$ such that the mapping ψ_x of $\mathbf{R}_+$ to E defined by $t \mapsto s(t)x$ is differentiable at 0. The space D is dense in E ; the map u from D into E defined by $u(x) = -\psi'_x(0)$ is a closed partial operator on E , which is called the infinitesimal generator of the semigroup s .

c) Let λ a complex number such that $\Re(\lambda) < -M$. The formula

$$v_\lambda = - \int_{\mathbf{R}_+} e^{\lambda t} s(t) dt$$

defines an endomorphism of E .

d) The image of v_λ is contained in $\text{dom}(u)$ and we have $u \circ v_\lambda = -1_E - \lambda v_\lambda$.

e) One has $u \circ v_\lambda = v_\lambda \circ u$.

f) The complex number λ belongs to the resolvent set of u and we have $v_\lambda = -R(u, \lambda)$.

g) One has

$$\|v_\lambda^n\| \leq \frac{C}{|\mathcal{R}(\lambda) + M|^n}$$

for all $n \in \mathbb{N}$.

6) Let E be a complex Banach space and u a closed partial operator with dense domain on E .

Suppose there exist real numbers $C \geq 0$ and $M \geq 0$ such that

(i) The set $]-\infty, -M[$ is contained in the resolvent set of u ;

(ii) For every integer $n \in \mathbb{N}$ and every real number $\lambda < -M$, we have

$$\|R(u, \lambda)^n\| \leq C|\lambda + M|^{-n}.$$

a) For every real number $y < -M$, we denote

$$u_y = -y1_E + y^2R(u, y) \in \mathcal{L}(E).$$

The endomorphisms u_y commute pairwise; for every $x \in \text{dom}(u)$, we have $u_y(x) \rightarrow u(x)$ as $y \rightarrow -\infty$.

b) The map from R_+ in $\mathcal{L}(E)$ defined by $s_y(t) = e^{-tu_y}$ is a continuous semigroup on E such that $\|s_y(t)\| \leq Ce^{Mt}$ for all $t \in R_+$.

c) For all y_1, y_2 and t in $\mathbf{R}_+^*$, and every x in $\text{dom}(u)$, we have

$$\|s_{y_1}(t)x - s_{y_2}(t)x\| \leq tC^2 \|u_{y_1}(x) - u_{y_2}(x)\|.$$

d) Let $t \in \mathbf{R}_+$. For every x in E , the function defined on $\mathbf{R}_+^*$ by $y \mapsto s_y(t)x$ converges as $y \rightarrow -\infty$ to an element $s(t)x$ of E . The map $s(t)$ is linear and continuous with norm $\leq Ce^{Mt}$.

e) The map $t \mapsto s(t)$ is a continuous semigroup on E .

- f) The partial operator u is the infinitesimal generator of the continuous semigroup s (“Hille–Yosida theorem”). (First verify that the infinitesimal generator of s is an extension of u .)
- g) Let $\tilde{s}$ a continuous semigroup whose u is the infinitesimal generator. We have $\tilde{s} = s$.
- 7) Let E be a complex Banach space and u a partial operator on E . We say that u is accretive if, for every $x \in \text{dom}(u)$ there exists a continuous linear form $\ell \in E'$ such that
- (i) $\Re(\ell(u(x))) \geq 0$;
- (ii) One has $\|\ell\| = \|x\|$ and $\ell(x) = \|x\|^2$.

We say that a continuous semigroup s on E is contractive if $\|s(t)\| \leq 1$ for all $t \in \mathbf{R}_+$.

- a) The partial operator u is the infinitesimal generator of a continuous contractive semigroup on E if and only if u is accretive and if there exists $\lambda > 0$ such that the partial operator $\lambda 1_E + u$ is surjective. (To show that the condition is necessary, let s be the contractive semigroup whose infinitesimal generator is u ; for $x \in \text{dom}(u)$, consider the function defined by $t \mapsto \ell(s(t)x)$; it is differentiable at 0, and its derivative at 0 is $-\Re(\ell(u(x)))$.)
- b) Suppose that u is closed and that u and ${}^t u$ are accretive. Then u is the infinitesimal generator of a continuous contractive semigroup on E .
- 8) Let E be a complex Banach space. Let s be a continuous contractive semigroup on E , and let u be its infinitesimal generator. We have

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{t}{n} u\right)^{-n} x = s(t)x$$

for all $x \in E$.

- 9) a) Find Hilbert spaces E and F and partial operators u and v from E into F such that u and $u + v$ are densely defined and $u^* + v^*$ is different from $(u + v)^*$.
- b) Find Hilbert spaces E , F , and G and densely defined partial operators u and v from E into F and from F into G , respectively, such that $v \circ u$ is densely defined and $u^* \circ v^*$ is different from $(v \circ u)^*$.

- c) Find a Hilbert space E and partial self-adjoint operators u and v on E such that $u + v$ is densely defined but is not self-adjoint.
- 10) Give an example of multiplication operators. m_{g_1} and m_{g_2} such that $m_{g_1} \circ m_{g_2} \neq m_{g_1 g_2}$.
- 11) We identify the space $\mathcal{D}(]0, 1[)$ with a subspace of $L^2([0, 1])$. Let u the partial operator on $L^2([0, 1])$ with domain $\mathcal{D}(]0, 1[)$ such that $u(f) = i^{-1}f'$ for every function $f \in \mathcal{D}(]0, 1[)$.
- a) The partial operator u is closable and $\bar{u}$ is symmetric.
- b) The eigenspaces of $\bar{u}$ relative to i and to $-i$ are not orthogonal.
- 12) Let $S \subset \mathbb{C}$ a closed subset. Give an example of a closed operator with dense domain u on a Hilbert space E such that $\text{Sp}(u) = S$.
- 13) Let u a non-closable partial operator on a Hilbert space E . There does not exist a complex number λ such that the linear map from $\text{dom}(u)$ to E defined by $x \mapsto \lambda x - u(x)$ is bijective and has a continuous inverse.
- 14) Let u a symmetric partial operator on a complex Hilbert space E . The map $\lambda \mapsto \dim \text{Ker}(\lambda 1_E - u^*)$, of $\mathbb{C}$ in $\bar{\mathbb{N}}$ is locally constant on the resolvent set of u .
- 15) Let E be the complex Hilbert space $\mathbb{C}^2$ and let $u \in \mathcal{L}(E)$ the endomorphism such that $u(ae_1 + be_2) = be_1$.
- a) Show that the spectrum of u is reduced to 0 and that, for all $\lambda \in \mathbb{C}^*$, we have the formula

$$\|R(u, \lambda)\|^2 = \frac{2}{1 + 2|\lambda|^2 - \sqrt{1 + 4|\lambda|^2}}.$$

Deduce the form of the pseudo-spectrum from this. $\text{PSp}_\varepsilon(u)$ for all $\varepsilon > 0$.

- 16) Find an example showing that the conclusion of Prop. 21 of IV, p. 251, is not valid in general for bounded components of the spectrum.

17) a) Let U be a connected open subset of $\mathbb{C}$ and let $V \subset U$ be an open set. Let E be a complex Banach space. Let $f: U \rightarrow E$ be a holomorphic map, $z_0 \in U$ and $M > 0$ such that $\|f(z)\| \leq M$ for all $z \in V$ and such that $\|f(z_0)\| < M$. We then have $\|f(z)\| < M$ for all $z \in U$.

b) Let E be a complex Banach space and $u \in \mathcal{L}(E)$. Let $U \subset \mathbb{C}$ an open subset of the unbounded connected component of $\mathbb{C} - \text{Sp}(u)$. If $M > 0$ is such that $\|R(u, \lambda)\| \leq M$ for all $\lambda \in U$, then $\|R(u, \lambda)\| < M$ for all $\lambda \in U$. In particular, if $\mathbb{C} - \text{Sp}(u)$ is connected, then the norm of the resolvent of u cannot be constant on a nonempty open subset of $\mathbb{C} - \text{Sp}(u)$.

18) Let E be the complex vector space of bounded sequences $(x_k)_{k \in \mathbb{Z}}$ of complex numbers. For every $x = (x_k)$ in E , we denote

$$\|x\|_E = |x_0| + \sup_{k \in \mathbb{Z} - \{0\}} |x_k|.$$

a) The map $x \mapsto \|x\|_E$ is a norm on E ; the space E equipped with this norm is a Banach space.

b) Let $M > 2$ be a real number. Let $\lambda = (\lambda_k)_{k \in \mathbb{Z}}$ the sequence defined by $\lambda_0 = M^{-1}$ and $\lambda_k = 1$ if $k \neq 0$. We denote by u the linear map from E to E such that $u((x_k)_{k \in \mathbb{Z}}) = (\lambda_k x_{k+1})_{k \in \mathbb{Z}}$. This is a continuous endomorphism of E .

c) The open disk with center 0 and radius $1/M$ is contained in the resolvent set of u .

d) For every $\lambda \in \mathbb{C}$ such that $|\lambda| < \inf(M^{-1}, \frac{1}{2} - M^{-1})$, we have $\|R(u, \lambda)\| \leq M$. (Use formula (6) of IV, p. 244).

e) Let $e_0 \in E$ the sequence $(x_k)_{k \in \mathbb{N}}$ defined by $x_0 = 1$ and $x_k = 0$ if $k \neq 0$. Let $\lambda \in \mathbb{C}$ such that $|\lambda| < \inf(M^{-1}, \frac{1}{2} - M^{-1})$. We have $\|R(u, \lambda)e_0\|_E = M$, hence $\|R(u, \lambda)\| = M$.

(This example, like that of the following exercise, is due to E. Shargorodsky, "On the level sets of the resolvent norm of a linear operator", Bull. London Math. Soc. 40 (2008), 493–504.)

19) Let E be the Hilbert space $\ell_2(\mathbb{N}^*)$. Let $(\alpha_n)_{n \in \mathbb{N}^*}$ a sequence of real numbers ≥ 2 such that α_n tends to $+\infty$ as $n \rightarrow +\infty$. Set $\beta_n = 1 + \alpha_n^{-1}$ for all $n \in \mathbb{N}^*$.

a) Let $F \subset E$ the space of sequences $(x_k)_{k \in \mathbb{N}^*} \in E$ such that the series

$$\sum_{n \in \mathbb{N}^*} \alpha_n^2 |x_{2n}|^2$$

converges. It is a dense subspace of E .

b) There exists a closed partial operator. u on E with domain F such that for every $x = (x_k)_{k \in \mathbb{N}^*}$ in F , we have $u(x) = (y_k)$ where

$$\begin{pmatrix} y_{2k-1} \\ y_{2k} \end{pmatrix} = \begin{pmatrix} 0 & \alpha_k \\ \beta_k & 0 \end{pmatrix} \begin{pmatrix} x_{2k-1} \\ x_{2k} \end{pmatrix}$$

for all $k \in \mathbb{N}^*$.

c) Let u_n the endomorphism

$$x \mapsto \begin{pmatrix} 0 & \alpha_n \\ \beta_n & 0 \end{pmatrix} x$$

of the Hilbertian space C^2 equipped with the canonical inner product. For every $\lambda \in C$ such that $|\lambda| < 1$, the resolvent of u_n at λ is defined and we have

$$\lim_{n \rightarrow +\infty} \|R(u_n, \lambda)\| = 1.$$

d) For every $n \in \mathbb{N}^*$ and every $\lambda \in C$ such that $|\lambda| < \frac{1}{2}$, we have $\|R(u_n, \lambda)\| < 1$.

e) The open disk with center 0 and radius $\frac{1}{2}$ is contained in the resolvent set of u , and we have $\|R(u, \lambda)\| = 1$ for all $\lambda \in C$ such that $|\lambda| < \frac{1}{2}$.

20) If a and b are real numbers with $a < b$, we denote $\mathcal{C}^2([a, b])$ the vector space of continuous functions on $[a, b]$ whose restriction to $]a, b[$ is of class C^2 . We denote μ Lebesgue measure.

a) Let q_1 and q_2 of real-valued functions in $\mathcal{C}^2([a, b])$ such that $q_1 \geq q_2$. For $i \in \{1, 2\}$, let $f_i \in \mathcal{C}^2([a, b])$ a solution of the equation

$$f_i'' + q_i f_i = 0.$$

Suppose that $f_2(a) = f_2(b) = 0$ and that $f_2 \neq 0$. If the function f_1 does not vanish in $]a, b[$, then $q_1 = q_2$ and there exists $c \in \mathbb{C}$ such that $f_1 = c f_2$. (Reduce to the case where f_1 and f_2 are strictly positive on $]a, b[$; consider the Wronskian $w = f_1 f_2' - f_1' f_2$ and prove that $w(a) = w(b)$ then that $w = 0$.)

b) Let $q \in \mathcal{C}^2([0, 1])$ satisfying $q \geq 0$. We denote $\widetilde{SL}_q$ the partial differential operator on $L^2([0, 1], \mu)$ with domain

$$D = \{f \in \mathcal{C}^2([0, 1]) \mid f(0) = f(1) = 0\}$$

such that

$$\widetilde{SL}_q(f) = -f'' + qf$$

Sturm–Liouville operator. The operator $\widetilde{SL}_q$ is symmetric and essentially self-adjoint. We denote SL_q the closure of $\widetilde{SL}_q$.

c) Let $\lambda \in \text{Sp}(SL_q)$ an eigenvalue of SL_q . The eigenspace associated with λ has dimension 1 and is generated by a real-valued function belonging to D .

d) Every eigenvalue λ of SL_q satisfies $\lambda \geq \pi^2$. (First prove that λ is positive using question a) with $q_1 = q$ and $q_2 = 0$.)

e) For $s > 0$, there exists a unique function $f_s \in \mathcal{C}^2([0, 1])$ which is a solution of the linear differential equation

$$-f_s'' + qf - s^2 f = 0$$

and satisfies $f_s(0) = 0$ and $f_s'(0) = s$. Set $f(x, s) = f_s(x)$; the map f is a function of class C^2 onto $]0, 1[\times \mathbb{R}_+^*$. The eigenvalues of SL_q are of the form $\lambda = s^2$ where s satisfies $f(1, s) = 0$.

f) For $t \in [0, 1]$ and $s > 0$, let $N(t, s) \in \overline{\mathbf{N}}$ the number of $x \in [0, t]$ such that $f(x, s) = 0$. We have $N(t, s) \in \mathbf{N}$ and the function $s \mapsto N(t, s)$ is increasing for every t fixed. (Use question a).)

g) Let $t \in [0, 1]$ and $s_0 > 0$ Let $m = N(t, s_0) \in \mathbf{N}$. Denote by

$$0 = x_1(s_0) < x_2(s_0) < \cdots < x_m(s_0)$$

the zeros of f_{s_0} in $[0, t]$. For $s \geq 0$ let

$$0 = x_1(s) < \cdots < x_m(s)$$

the m first zeros of f_s in $[0, t]$. The functions $s \mapsto x_j(s)$ are of class $\mathbf{C}^1$ and we have

$$\partial_1(x_j(s), s)x_j'(s) = -\partial_2 f(x_j(s), s).$$

h) The set of eigenvalues of SL_q is the set of values of an increasing sequence of strictly positive real numbers

$$0 < \lambda_0 < \lambda_1 < \cdots < \lambda_n < \dots$$

tending toward $+\infty$. Moreover, the eigenspace corresponding to λ_n has dimension 1; if φ_n of which is a basis, we have $\varphi_n \in \mathcal{C}^2([0, 1])$ and the function φ_n vanishes at exactly n distinct points of $[0, 1]$. (To prove that there exists an infinite number of eigenvalues of SL_q to use the question a) with $q_1 = M - s^2$, where $M \in \mathbf{R}$ is well chosen, to verify that $N(1, s)$ tends to $+\infty$ as $s \rightarrow +\infty$).

21) We resume the notation of the preceding exercise. We denote by φ_n the unique eigenfunction of SL_q for the eigenvalue λ_n such that $\varphi_n'(0) = \lambda_n$.

a) Let $M = \sup\{q(x)\}$. For every $n \in \mathbf{N}$, we have

$$\pi^2(n+1)^2 \leq \lambda_n \leq \pi^2(n+1)^2 + M,$$

and moreover

$$\lambda_n = \pi^2(n+1)^2 \left(1 + O\left(\frac{1}{n^2}\right)\right)$$

as $n \rightarrow +\infty$.

b) Let $s > 0$ Let $f \in \mathcal{C}^2([0, 1])$ such that

$$-f'' + s^2 f = -qf, \quad f(0) = 0, \quad f'(0) = s.$$

We then have $(1 - s^{-1}u_s)f = g_s$, where $g_s(x) = \sin(sx)$ and u_s is the integral operator on $\mathcal{C}([0, 1])$ given by

$$u_s(f)(x) = \int_0^x \sin(s(x-y))q(y)f(y)d\mu(y).$$

c) For s sufficiently large, we have $s \notin \text{Sp}(u_s)$.

d) For every fixed x , we have

$$\varphi_n(x) = \sin((n+1)x) + O(n^{-1})$$

as $n \rightarrow +\infty$.

22) We resume the notation of the preceding exercise.

a) There exist functions f_1 and f_2 in $\mathcal{C}^2([a, b])$ such that $-f_i'' + qf_i = 0$ and

$$f_1(0) = 0, \quad f_1'(0) = 1$$

$$f_2(1) = 0, \quad f_2'(1) = -1.$$

These functions are the unique solutions of these equations, and the function $w = f_1 f_2' - f_1' f_2$ is constant.

b) Let $g \in \mathcal{C}([0, 1])$. There exists a unique function $\varphi \in \mathcal{C}^2([0, 1])$ such that

$$-\varphi'' + q\varphi = g,$$

and we have $\varphi = c_1 f_1 + c_2 f_2$, where

$$c_1(x) = -\frac{1}{w} \int_x^1 g(y) f_2(y) d\mu(y), \quad c_2(x) = -\frac{1}{w} \int_0^x g(y) f_1(y) d\mu(y)$$

for all $x \in [0, 1]$.

- c) The partial operator SL_q is surjective; its inverse is the map $g \mapsto \varphi$ and there exists a function $G \in \mathcal{C}([0, 1]^2)$ such that

$$SL_q^{-1}(g)(x) = \int_0^1 G(x, y) g(y) d\mu(y)$$

for all $g \in \mathcal{C}([0, 1])$ and almost all $x \in [0, 1]$. We have $G(x, y) = G(y, x)$ for all $(x, y) \in [0, 1]^2$.

- d) The operator SL_q^{-1} is compact and injective on $L^2([0, 1], \mu)$.

- e) The functions (φ_n) form an orthonormal basis of $L^2([0, 1])$, and the spectrum of SL_q coincides with the set of its eigenvalues.

- 23) Let $E = \ell_{\mathbb{C}}^2(\mathbb{N})$ and let u be the endomorphism of E defined by

$$u((x_i)_{i \in \mathbb{N}}) = (0, x_1, \dots, x_n, \dots)$$

for all $(x_i) \in E$

- a) The spectrum of u is the closed unit disk in $\mathbb{C}$.
- b) The residual spectrum of u is the open unit disk in $\mathbb{C}$.
- c) The essential spectrum of u is the unit circle in $\mathbb{C}$.
- 24) Let E be a complex Hilbert space and u a partial symmetric operator on E . Let $F = E \oplus E$ and let v be the partial operator on F with domain $\text{dom}(u^*) \oplus \text{dom}(\bar{u})$ such that $v(x, y) = (\bar{u}(y), u^*(x))$.

- a) The partial operator v is self-adjoint.
- b) Let $\tilde{E}$ the image of the linear map $\delta: x \mapsto (x, x)$. The reduction of v to $\tilde{E}$ is the partial operator with domain $\delta(\text{dom}(u))$ such that $(x, x) \mapsto (u(x), u(x))$.

Exercises for Section 5

Unless stated otherwise, the Hilbert spaces below are assumed to be complex.

- 1) Let E be a Hilbert space and $\mathcal{P}(E) \subset \mathcal{L}(E)$ the set of orthogonal projections in E . Let S be a universally measurable subset of $\mathbb{C}$. We denote by $\mathcal{T}(S)$ the set of universally measurable subsets T of $\mathbb{C}$ such that $T \subset S$. A measure on S with values in the projections of E is a map ϖ of $\mathcal{T}(S)$ in $\mathcal{P}(E)$ such that

- (i) One has $\varpi(\emptyset) = 0$ and $\varpi(S) = 1_E$.
- (ii) If (T_n) is a sequence of pairwise disjoint sets in $\mathcal{T}(S)$ and if T is the union of the sets T_n , then

$$\varpi(T) = \sum_{n \in \mathbb{N}} \varpi(T_n)$$

in the space $\mathcal{L}(E)$ endowed with the topology of simple convergence.

For every normal partial operator u on E with spectrum equal to S , and for $T \in \mathcal{T}(S)$, we denote $p_{T,u}$ the spectral projector of u defined by T .

The map which to a normal partial operator u on E with spectrum S associates the map $T \mapsto p_{T,u}$ is a bijection between the set of normal partial operators with spectrum S and the set of measures on S with values in the projections of E .

- 2) Let E be a complex Hilbert space and let u be a normal partial operator on E . We denote by p_A the spectral projection of u on A . Let $a < b$ be real numbers. We have

$$\frac{1}{2}(p_{[a,b]} + p_{]a,b[}) = \lim_{\varepsilon \rightarrow 0} \frac{1}{2i\pi} \int_a^b (R(u, t - i\varepsilon) - R(u, t + i\varepsilon)) dt,$$

in the space $\mathcal{L}(E)$ endowed with the topology of simple convergence (“Stone formula”).

- 3) Let E be a Hilbert space of countable type. The only closed two-sided ideals of $\mathcal{L}(E)$ are $\{0\}$, $\mathcal{L}^c(E)$ and $\mathcal{L}(E)$. (Use exerc. IV, p. 316, exerc. 10; if I is such an ideal, suppose that there exists in I a nonzero self-adjoint element u which is not compact, and show that for every open interval J in $\mathbb{R}$ such that $0 \notin J$, the spectral projection of u on J belongs to I .)
- 4) Let E be a Hilbert space and u a self-adjoint partial operator on E . The map $\sigma : f \mapsto f(u)$ of $\mathcal{L}_u^\infty(\text{Sp}(u))$ in $\mathcal{L}(E)$ is the unique morphism of involutive algebras satisfying the following three conditions:
 - (i) One has $\|\sigma(f)\| \leq \|f\|_\infty$ for all $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$
 - (ii) If $(f_n)_{n \in \mathbb{N}}$ is a bounded sequence in $\mathcal{L}_u^\infty(\text{Sp}(u))$ and $f_n(t) \rightarrow f(t)$ for all $t \in \text{Sp}(u)$, then $\sigma(f_n) \rightarrow \sigma(f)$ in $\mathcal{L}(E)$ endowed with the topology of simple convergence.
 - (iii) If $(f_n)_{n \in \mathbb{N}}$ is a sequence in $\mathcal{L}_u^\infty(\text{Sp}(u))$ such that $|f_n(t)| \leq |t|$ and $f_n(t) \rightarrow t$ for all $t \in \text{Sp}(u)$, then we have $u(x) = \lim \sigma(f_n)(x)$ for all $x \in \text{dom}(u)$.
- 5) Let u be a normal partial operator on a Hilbert space E . The residual spectrum of u is empty.
- 6) Let E and F be Hilbert spaces. Let u be a closed partial operator with dense domain from E to F . Let $\lambda \in C^*$ belonging to the resolvent set of the self-adjoint operator $u^* \circ u$ on E (cf. prop. 12 of IV, p. 241).
 - a) The partial operator $u \circ u^*$ from F to F is self-adjoint.
 - b) For every vector $x \in \text{dom}(u^*)$, the vector

$$y = (1_F + u \circ R(u^* \circ u, \lambda) \circ u^*)x$$

belongs to $\text{dom}(u \circ u^*)$, and the formula

$$(u \circ u^* - \lambda 1_F)y = x$$

is valid.

c) The partial operator $u \circ R(u^* \circ u, \lambda) \circ u^*$ has dense domain, and is continuous on its domain.

d) The partial operator $u \circ u^* - \lambda 1_F$ is surjective.

e) One has $\lambda \notin \text{Sp}(u \circ u^*)$.

f) We have the identity

$$\lambda R(u \circ u^*, \lambda) - u \circ R(u^* \circ u, \lambda) \circ u^* = 1_F$$

in $\mathcal{L}(F)$ (where $u \circ R(u^* \circ u, \lambda) \circ u^*$ denotes, by abuse of notation, the endomorphism of F which coincides with this partial operator on its domain, cf. question c)).

g) We have the relation

$$u^* \circ R(u \circ u^*, \lambda) \subset R(u^* \circ u, \lambda) \circ u^*.$$

h) The partial operator u induces, by passage to subspaces, an isomorphism of $\text{Ker}(u^* \circ u - \lambda 1_E)$ in $\text{Ker}(u \circ u^* - \lambda 1_F)$.

7) Let A be a set of closed operators with dense domain on a Hilbert space E . Let $\lambda \in \mathbb{C}$.

We say that A is λ -commutative if λ belongs to the resolvent set of u for all $u \in A$, and if the endomorphisms $R(u, \lambda)$ of E , for $u \in A$, are pairwise permutable.

We say that A is commutative in the resolvent sense if there exists $\lambda \in \mathbb{C}$ such that A is λ -commutative.

A set consisting of a single closed densely defined operator u is commutative if and only if the resolvent set of u is nonempty.

a) If $A \subset L(E)$, then A is commutative in the resolvent sense if and only if the union of the spectra of the elements of A is not equal to $\mathbb{C}$, and A is commutative.

b) Let A be a set of closed densely defined operators on E that is λ -commutative. Then A is μ -commutative for every complex number μ belonging to the resolvent set of every $u \in A$.

c) Let A be a set of closed densely defined operators on E , commutative in the resolvent sense, and such that $u^* \in A$ for all $u \in A$. There exists a locally compact topological space X , a positive measure μ on X , an isometric isomorphism v of $L^2(X, \mu)$ on E , and a mapping $u \mapsto g_u$ from A into the set of functions μ -measurable such that $u = v \circ m_{g_u} \circ v^{-1}$ for all $u \in A$.

8) Let u and v be partial normal operators on a Hilbert space E . The following assertions are equivalent:

- a) The endomorphisms $b(u)$ and $b(v)$ are permutable;
- b) For every universally measurable subset A of $\mathbb{C}$, the spectral projections of u and v on A commute;

If the resolvent set of u and v is nonempty, these assertions are equivalent to

- c) For every $\lambda \notin \text{Sp}(u)$ and $\lambda' \notin \text{Sp}(v)$, the resolvents $R(u, \lambda)$ and $R(v, \lambda')$ commute.

If u and v are self-adjoint, these assertions are equivalent to

- d) For every $t \in \mathbb{R}$, the unitary endomorphisms e^{itu} and e^{itv} are permutable.

9) Let u be a closed densely defined partial operator on a Hilbert space E . The following assertions are equivalent:

- a) The operator u is normal;
- b) There exist self-adjoint operators u_1 and u_2 on E , which commute in the resolvent sense, and such that $u = u_1 + iu_2$;
- c) One has $\text{dom}(u) = \text{dom}(u^*)$ and $\|u(x)\| = \|u^*(x)\|$ for all $x \in \text{dom}(u)$;
- d) One has $uu^* = u^*u$.

If the resolvent set of u is nonempty, these assertions are equivalent to

- e) For every $\lambda \notin \text{Sp}(u)$, the resolvent $R(u, \lambda)$ is normal.
- 10) Let E be a complex Hilbert space and A a unital self-adjoint subalgebra of $\mathcal{L}(E)$ which is closed under the topology of simple convergence. (Such an algebra is called a von Neumann algebra.)
 - a) There exists a subset X of $\mathcal{L}(E)$ such that A coincides with the commutant of X in $\mathcal{L}(E)$. (Use Exercise 24 of IV, p. 326.) Conversely, for every self-adjoint subset of X , the commutant of X in $\mathcal{L}(E)$ is a unital self-adjoint subalgebra closed under the topology of pointwise convergence.
 - b) Let $u \in A$ and let $(j, |u|)$ the polar decomposition of u (definition 4 of I, p. 140). The elements j and u belong to A .
 - c) Let $u \in A$ such that u is normal. For every function $f \in \mathcal{L}_u^\infty(\text{Sp}(u))$, the endomorphism $f(u)$ belongs to A .
 - d) Let $u \in A$. The element u is a right (resp. left) zero divisor in A (cf. A, I, p. 93) if and only if the image of u is not dense in E (resp. if u is not injective).
 - e) Let F be the set of partial operators on E whose elements are the closed partial operators with dense domain u such that, for every x in X , the partial operator $x \circ u$ is a reduction of the partial operator $u \circ x$. For every u in F and v in A , the partial operators u^* , $u + v$, $v \circ u$ and $u \circ v$ belong to F .
 - f) Let $u \in F$ and let $(j, |u|)$ the polar decomposition of u (definition 7 of IV, p. 290). The partial operator $|u|$ belongs to F and j belongs to A .
 - g) Let $u \in F$ such that u is normal. For every function $f \in \mathcal{L}_u(\text{Sp}(u))$, the partial operator $f(u)$ belongs to F ; if f is bounded, then $f(u)$ belongs to A .
 - h) Let $u \in A$ injective whose image is dense in E . The partial operator u^{-1} belongs to F .
 - i) For every u in F , there exist elements u_1 and u_2 of A such that u_2 is injective with dense image and $u = u_1 u_2^{-1}$. (First reduce to the case where u is self-adjoint and positive, then

apply the universally measurable functional calculus to the functions $f_1 : t \mapsto t\varphi(t) + \psi(t)$ and $f_2 : t \mapsto \varphi(t) + t^{-1}\psi(t)$, where φ and ψ are the characteristic functions of the intervals $[0, 1[$ and $[1, +\infty[$ respectively.)

j) Assume that every injective element of A with dense image is surjective. The algebra A then satisfies the Ore condition: for any u and s in A , if s is not a zero divisor (neither on the right nor on the left), then there exists u' and s' in A such that s' is not a zero divisor and $us' = su'$.

k) The conclusion of the previous question is not always valid. (Consider $A = \mathcal{L}(E)$.)

11) a) Let q a positive partial form on a complex Hilbert space E . Then q is closed if and only if for every sequence (x_n) of elements of $\text{dom}(q)$ which converges to x in E and satisfies

$$\lim_{n,m \rightarrow +\infty} q(x_n - x_m, x_n - x_m) = 0,$$

we have $x \in \text{dom}(q)$ and $q(x_n - x, x_n - x) \rightarrow 0$.

b) Let $E = L^2(\mathbf{R}, \mu)$, where μ is the Lebesgue measure. The positive Hermitian form q with domain $\mathcal{D}(\mathbf{R})$ defined by $q(\varphi_1, \varphi_2) = \overline{\varphi_1(0)}\varphi_2(0)$ is not closed, and there does not exist a closed partial positive form extending q .

12) We equip $\mathbf{R}$ of Lebesgue measure. Let E be the closed subspace of $L^2(\mathbf{R})$ consisting of the $f \in L^2(\mathbf{R})$ such that $f(-x) = f(x)$ for almost every $x \in \mathbf{R}$. Let $\mathcal{D}_+(\mathbf{R})$ the space of test functions on $\mathbf{R}$ such that $f(x) = f(-x)$ for all $x \in \mathbf{R}$. We identify $\mathcal{D}_+(\mathbf{R})$ with a subspace of E . We denote by $\mathcal{F} \in \mathcal{L}(E)$ the endomorphism of E deduced from the Fourier transform of $L^2(\mathbf{R})$ by passage to the subspaces.

a) For every $f \in \mathcal{D}_+(\mathbf{R})$, we have $\mathcal{F}(f) + \mathcal{F}^*(f) \in \mathcal{D}_+(\mathbf{R})$.

b) Let u be the partial operator on E whose domain is $\mathcal{D}_+(\mathbf{R})$ and which is defined by $u(f) = \mathcal{F}(f) + \mathcal{F}^*(f)$ for $f \in \mathcal{D}_+(\mathbf{R})$. The operator u is essentially self-adjoint.

c) One has $\text{dom}(u^2) = \{0\}$ (Use Exercise 28 of II, p. 273.)

- 13) Let E be a complex Hilbert space and let u be a partial symmetric operator on E . Suppose there exists a real number $c > 0$ such that $\langle x | u(x) \rangle \geq c\|x\|^2$ for all $x \in \text{dom}(u)$.

Show that the following conditions are equivalent:

- (i) The partial operator u is essentially self-adjoint;
- (ii) The image of u is dense in E ;
- (iii) The kernel of u^* is reduced to 0.

- 14) Let $n \in \{1, 2, 3\}$. We denote Δ the Laplace operator on $L^2(\mathbf{R}^n)$, and D its domain. It is a positive partial operator.

- a) There exists a real number $c \geq 0$ such that for every $f \in D$, we have

$$\|f\|_\infty \leq c(\|\Delta f\| + \|f\|).$$

- b) For every real number $a > 0$, there exists a real number $b \geq 0$ such that for every $f \in D$, we have

$$\|f\|_\infty \leq a\|\Delta f\| + b\|f\|.$$

(Apply the previous question to $x \mapsto f(\alpha x)$, where $\alpha \in \mathbf{R}_+^*$.)

- c) Let $g \in \mathcal{L}^\infty(\mathbf{R}^n) + \mathcal{L}^2(\mathbf{R}^n)$. If g is real-valued, then the partial operator $\Delta + m_g$ is self-adjoint.
- d) Let α be a real number such that $0 \leq \alpha < n/2$. Let N be the Euclidean norm on $\mathbf{R}^n$ and g the function on $\mathbf{R}^n$ defined by $g(x) = 1/N(x)^\alpha$ if $x \neq 0$, and $g(0) = 0$. The partial operator $\Delta + m_g$ is self-adjoint.

- 15) Let $n \in \mathbf{N}$ and let Δ the unique Laplacian on $L^2(\mathbf{R}^n)$. Denote by B the unit ball in $\mathbf{R}^n$.

If $g \in \mathcal{L}_u(\mathbf{R}^n)$ is such that the domain of Δ is contained in the domain of m_g , then

$$\sup_{y \in \mathbf{R}^n} \int_B |g(x+y)|^2 dx$$

is finite. (First verify that m_g is Δ -bounded.)

16) a) Let $I =]0, 1[$. The space $H^1(I)$ is contained in $C(\bar{I})$.

b) The differential operator $u: f \mapsto -if'$ onto $L^2(I)$ with domain $\mathcal{D}(I)$ is symmetric and closable. For all $f \in \text{dom}(u^*)$ the derived distribution f' of f belongs to $L^2(I)$ and we have $u^*(f) = if'$.

c) The deficiency spaces of $\bar{u}$ are of dimension 1; deduce from this an explicit description of all self-adjoint extensions of $\bar{u}$.

17) Classify the Laplacians on $I =]0, 1[$.

¶ 18) Let n be an integer ≥ 1 . We denote by D the partial operator on $L^2(T^n)$ with domain $C^\infty(T^n)$ such that

$$D(\varphi) = - \sum_{i=1}^n \partial_i^2 \varphi$$

for all $\varphi \in \mathcal{C}^\infty(T^n)$,

a) The partial operator D is essentially self-adjoint. We denote by Δ its closure.

b) The partial operator Δ is self-adjoint and positive.

c) The partial operator Δ has compact resolvent.

d) Let $\lambda \in \mathbf{R}$. We have $\lambda \in \text{Sp}(\Delta)$ if and only if there exist natural integers $k_1, \dots, k_n$ such that $\lambda = 4\pi^2(k_1^2 + \dots + k_n^2)$.

e) For every eigenvalue λ of Δ , let $N(\lambda)$ the dimension of the corresponding eigenspace. We have

$$\sum_{\lambda \leq T} N(\lambda) = c_n T^{n/2} + O(T^{(n-1)/2})$$

when $T \rightarrow +\infty$.

f) If $n \geq 2$, then $\limsup_{\lambda} N(\lambda) = +\infty$. (In the case $n = 2$, consider $\lambda = 4\pi^2 m$, where m is a product of prime numbers of the form $4k + 1$.) g) If $n = 1$ or $n = 2$, we have

$$\text{Card}(\text{Sp}(u) \cap [0, T]) = o(T),$$

and if $n \geq 3$, we have

$$\liminf_{T \rightarrow +\infty} \frac{1}{T} \text{Card}(\text{Sp}(u) \cap [0, T]) > 0.$$

19) Let u be a partial self-adjoint operator on a Hilbert space E .

a) Let $t \in \mathbb{R}$ and $x \in E$. If the spectral measure of x with respect to u has compact support, then we have

$$e^{iu}(x) = \sum_{n \in \mathbb{N}} \frac{u^n(x)}{n!},$$

where the series converges in E .

b) The set of vectors $x \in E$ such that the spectral measure of x with respect to u has compact support is a dense subspace of E .

20) Let u a partial operator on a Hilbert space E . We denote by $\text{dom}(u)^{\text{an}}$ the space of vectors $x \in E$ such that $x \in \text{dom}(u^n)$ for every integer $n \in \mathbb{N}$, and such that the power series

$$\sum_{n \in \mathbb{N}} \frac{\|u^n(x)\|}{n!} z^n \tag{23}$$

has a radius of convergence > 0 . The elements of $\text{dom}(u)^{\text{an}}$ are said to be the analytic vectors of u .

Let x an element of E such that x belongs to $\text{dom}(u^n)$ for all $n \in \mathbb{N}$. Let E_x the closure of the subspace of E generated by the vectors $u^n(x)$ for $n \in \mathbb{N}$ and let u_x the partial operator on $\overline{E_x}$ with domain E_x deduced from u by passing to subspaces. We say that x is a uniqueness vector of u if u_x is essentially self-adjoint.

- a) Suppose that u is symmetric and that E contains a total subset of uniqueness vectors for u . Then u is essentially self-adjoint. (Show that the image of $u + i1_E$ and that of $u - i1_E$ are dense in E .)

Suppose henceforth that u is symmetric and that E contains a total subset of analytic vectors.

- b) For every analytic vector x of u , the partial operator u_x onto $\overline{E_x}$ admits a self-adjoint extension.
- c) Let x an analytic vector for u and $r > 0$ the radius of convergence of the power series (23). Let ν be a self-adjoint extension of u_x . For every $t \in \mathbf{R}$ such that $|t| < r$, we have

$$\langle e^{it\nu}(x) | x \rangle = \sum_{n \in \mathbf{N}} \frac{(it)^n}{n!} \langle u^n(x) | x \rangle.$$

(Using the spectral measure of x relative to ν , show that the left-hand side is the restriction of an analytic function on the set of $t \in \mathbf{C}$ such that $|t| < r$, and compute the power series expansion of this function at 0.)

- d) For every $t \in \mathbf{R}$, the unitary operator $e^{it\nu}$ depends only on u_x and the operator u_x is essentially self-adjoint. (Use Theorem V, p. 428.) e) Deduce that u is essentially self-adjoint.
- e) A closed symmetric partial operator u on E is self-adjoint if and only if $\text{dom}(u)^{\text{an}}$ is dense in E .
- 21) We denote x the identity function of $\mathbf{R}$. We denote $\mathcal{M}_m(\mathbf{R})$ the set of positive measures ν onto $\mathbf{R}$ such that $x^n \in \mathcal{L}^1(\mathbf{R}, \nu)$ for every integer $n \in \mathbf{N}$. a) Let $\nu \in \mathcal{M}_m(\mathbf{R})$ and set $\mu_n = \nu(x^n)$ for all $n \in \mathbf{N}$ ("moments of the measure ν "). For every integer $n \geq 0$ and for all complex numbers $a_0, \dots, a_n$, we have

$$\sum_{i=0}^n \sum_{j=0}^n \bar{a}_i a_j \mu_{i+j} \geq 0. \quad (24)$$

In what follows, we propose to study the converse assertion of question a) ("Hamburger moment problem"). In the rest of this exercise, we therefore assume given a sequence $\mu =$

$(\mu_n)_{n \geq 0}$ of real numbers satisfying condition (24).

- b) There exists a unique positive sesquilinear form b_μ on $C[X]$ such that $b_\mu(X^i, X^j) = \mu_{i+j}$ for all integers i and j .

We denote E_μ the separated completion Hilbert space of $C[X]$ for the form b_μ . c) The linear map u_μ from $C[X]$ to $C[X]$ defined by $u_\mu(P) = XP$ induces, by passing to quotients, a symmetric partial operator u_μ on the Hilbert space E_μ .

- d) The partial operator u_μ admits a self-adjoint extension.

- e) There exists a measure $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\mu_n = \nu(x^n)$ for all $n \in \mathbf{N}$. f) The sesquilinear form b_μ is separating if and only if, for every measure $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\mu_n = \nu(x^n)$ for all $n \in \mathbf{N}$, the support of ν is infinite. (Observe that the left-hand side of inequality (24) is equal to $b_\mu(p, p)$ for $p = a_0 + \dots + a_n X^n$.)

22) We resume the notation of the preceding exercise.

Let $\mu = (\mu_n)_{n \in \mathbf{N}}$ a sequence of real numbers satisfying conditions (24). We consider the question of uniqueness of a measure $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\mu_n = \nu(x^n)$ for all $n \in \mathbf{N}$. We assume that the sesquilinear form b_μ is separating (cf. question f) of the preceding exercise).

- a) If there exists a measure $\nu \in \mathcal{M}_m(\mathbf{R})$ with compact support such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$, then ν is the unique measure in $\mathcal{M}_m(\mathbf{R})$ such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$.
- b) Let $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$. For every $p \in C[X]$, we have $p \in \mathcal{L}^2(\mathbf{R}, \nu)$ and $\|p\| = b_\mu(p, p)$.
- c) The canonical injection of $C[X]$ in $\mathcal{L}^2(\mathbf{R}, \nu)$ defines by continuity and by passing to the quotient an isometry of E_μ in $L^2(\mathbf{R}, \nu)$. In what follows, we identify E_μ with its image, which is a closed subspace of $L^2(\mathbf{R}, \nu)$.
- d) One has $u_\mu \subset m_x$, where m_x denotes the operator of multiplication by x in $L^2(\mathbf{R}, \nu)$.
- e) The measure ν is the spectral measure of the constant function 1 relative to m_x .

- f) Let $\nu \in \mathcal{M}_m(\mathbf{R})$. We have $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$ if and only if there exists a Hilbert space F and a partial self-adjoint operator ν on F satisfying the following conditions:
- (i) The space E_μ is a closed subspace of F , and the canonical injection from E_μ in F is an isometry;
 - (ii) One has $u_\mu \subset \nu$;
 - (iii) The measure ν is the spectral measure of the constant function $1 \in E_\mu$ relative to ν .
- g) There exists a unique measure $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$ if and only if the partial operator u_μ is essentially self-adjoint.
- h) Let ν_{lg} the positive measure on $\mathbf{R}$ with support in R_+ whose restriction to R_+^* to the density

$$t \mapsto \frac{1}{t} \exp(-\log(t)^2/2)$$

with respect to Lebesgue measure. Let $\tilde{\nu}_{lg}$ the positive measure on $\mathbf{R}$ with support in R_+ whose restriction to R_+^* to the density $t \mapsto 1 + \sin(2\pi \log(t))$ with respect to ν_{lg} .

One has $\nu_{lg} \in \mathcal{M}_m(\mathbf{R})$ and $\tilde{\nu}_{lg} \in \mathcal{M}_m(\mathbf{R})$.

- i) One has $\nu_{lg}(x^n) = \tilde{\nu}_{lg}(x^n)$ for all $n \in \mathbf{N}$.
- 23) We resume the notations of the previous exercise; in particular, we consider a sequence $\mu = (\mu_n)_{n \in \mathbf{N}}$ of real numbers satisfying conditions (24) and such that the sesquilinear form b_μ is separating. We identify $\mathbf{C}[X]$ with a dense subspace of E_μ .
- a) There exists a unique sequence $(p_n)_{n \in \mathbf{N}}$ of polynomials in $\mathbf{C}[X]$ such that the following conditions are satisfied (“orthogonal polynomials of the first kind associated with μ »):
 - (i) The family (p_n) is an orthonormal basis of E_μ ;
 - (ii) The polynomial p_n is of degree n for all $n \in \mathbf{N}$;
 - (iii) The coefficient of X^n of p_n belongs to $\mathbf{R}_+^*$.

Moreover, we have $p_n \in \mathbf{R}[X]$ for all $n \in \mathbf{N}$ and $p_0 = 1/\sqrt{\mu_0}$. We shall also denote $p_{-1} = 0$

b) There exist sequences $(a_n)_{n \geq -1}$ in $\mathbf{R}_+^*$ and $(b_n)_{n \in \mathbf{N}}$ in $\mathbf{R}$ such that $a_{-1} = 1$ and

$$u_\mu(p_n) = a_n p_{n+1} + b_n p_n + a_{n-1} p_{n-1}$$

for all $n \in \mathbf{N}$. These sequences are unique.

c) For every $n \geq 1$, the coefficient of X^n of the orthogonal polynomial p_n is equal to $(a_0 \cdots a_{n-1})^{-1}$.

d) Let $\mathcal{F}$ the complex vector space of sequences $(z_n)_{n \geq -1}$ of complex numbers such that $z_{-1} = 0$. We denote $\tilde{v}_\mu$ the linear map from $\mathcal{F}$ into itself such that $\tilde{v}_\mu(z_n) = (w_n)_{n \in \mathbf{N}}$ where $w_{-1} = 0$ and

$$w_n = a_n z_{n+1} + b_n z_n + a_{n-1} z_{n-1}$$

for all $n \in \mathbf{N}$.

For every $s \in \mathbf{C}$ the sequence $z_s = (p_n(s))_{n \geq -1}$ is the unique sequence in $\mathcal{F}$ such that $(z_s)_0 = p_0(z)$ and $\tilde{v}_\mu(z_s) = s z_s$.

e) Let z and w be elements of $\mathcal{F}$. One defines the sequence $W(z, w) = (W_n(z, w))_{n \in \mathbf{N}}$ by setting

$$W_n(z, w) = a_n(z_{n+1}w_n - z_n w_{n+1})$$

for $n \in \mathbf{N}$. We have

$$W_n(z, w) = \sum_{j=0}^n (\tilde{v}_\mu(z)_j w_j - z_j \tilde{v}_\mu(w)_j)$$

for all $n \in \mathbf{N}$.

f) Let $\mathbf{P} \subset \ell^2(\mathbf{N})$ the space of sequences with finite support. For every $z \in \mathbf{P}$ the sequence

$\tilde{v}_\mu(z)$ belongs to $\ell^2(\mathbf{N})$.

We denote v_μ the partial operator on $\ell^2(\mathbf{N})$ of domain P given by $z \mapsto \tilde{v}_\mu(z)$. It is symmetric.

g) The adjoint of v_μ is the partial operator with domain

$$\text{dom}(v_\mu^*) = \{z \in \ell^2(\mathbf{N}) | \tilde{v}_\mu(z) \in \ell^2(\mathbf{N})\}$$

such that $v_\mu^*(z) = \tilde{v}_\mu(z)$ for $z \in \text{dom}(v_\mu^*)$.

h) For all z and w in $\text{dom}(v_\mu^*)$, the limit of $W_n(\bar{z}, w)$ as $n \rightarrow +\infty$ exists and is equal to $\langle w | v_\mu^*(z) \rangle - \langle v_\mu^*(w) | z \rangle$.

i) Let $s \in \mathbf{C}$. The kernel of $v_\mu^* - s \cdot 1_{\ell^2(\mathbf{N})}$ has dimension ≤ 1 ; it has dimension 1 if and only if $z_s \in \ell^2(\mathbf{N})$, and it is then generated by z_s .

j) The closure of the partial operator v_μ (resp. of u_μ) has deficiency index $(0, 0)$ or $(1, 1)$; the first case holds if and only if there exists $s \in \mathbf{C} - \mathbf{R}$ such that $z_s \notin \ell^2(\mathbf{N})$. (For the case of u_μ , observe that the map $X^n \mapsto e_n$, where (e_n) is the canonical basis of $\ell^2(\mathbf{N})$, induces an isometric isomorphism of E_μ in $\ell^2(\mathbf{N})$ which makes it possible to identify u_μ with v_μ .)

k) Let $s \in \mathbf{C}$ and $f \in \text{Ker}(u_\mu^* - s \cdot 1_{E_\mu})$. If $\langle f | 1 \rangle = 0$, then $f = 0$.

l) Let u_1 and u_2 of the self-adjoint extensions of u_μ onto E_μ . We have $u_1 = u_2$ if and only if there exists $s \in \mathbf{C} - \mathbf{R}$ such that $\langle R(u_1, s)1 | 1 \rangle = \langle R(u_2, s)1 | 1 \rangle$. (Suppose that such an equality holds; set $f_1 = R(u_1, s)1$, and show that f_1 does not belong to the domain of $\bar{u}_\mu$; deduce that u_1 coincides with the closure of the restriction of u_1 to $\text{dom}(\bar{u}_\mu) \oplus \mathbf{C}f_1$. Reason similarly with u_2 and $f_2 = R(u_2, s)1$, then prove that $f_1 = f_2$ to conclude that $u_1 = u_2$.) m) Suppose that u_μ is not essentially self-adjoint. There then exist at least two measures $\nu \in \mathcal{M}_m(\mathbf{R})$ satisfying $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$. n) In what follows, assume that u_μ is essentially self-adjoint. For every $s \in \mathbf{C} - \mathbf{R}$, there exists a sequence $(r_{s,m})_{m \in \mathbf{N}}$ of polynomials such that $(x - s)r_{s,m}$ converges in E_μ to the constant function 1 as $m \rightarrow +\infty$.

m) Let $\nu \in \mathcal{M}_m(\mathbf{R})$ a measure such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$. For every $m \in \mathbf{N}$, the

inequality

$$\left| \int_{\mathbf{R}} \frac{d\nu(t)}{t-s} - \int_{\mathbf{R}} r_{s,m}(t) d\nu(t) \right| \leq \frac{\mu_0^2}{|\mathcal{I}(s)^2|} \|1 - (x-s)r_{s,m}\|_{E_\mu}$$

is valid.

p) The measure ν is unique.

24) We resume the notations of the preceding exercises; in particular, we consider a sequence $\mu = (\mu_n)_{n \in \mathbf{N}}$ of real numbers satisfying conditions (24) and such that the sesquilinear form b_μ is separating.

a) There exists a unique sequence of polynomials $(q_n)_{n \in \mathbf{N}}$ with real coefficients such that for every $s \in C$, the formula

$$q_n(s) = b_\mu \left(1, \frac{p_n - p_n(s)}{X - s} \right)$$

is valid (“orthogonal polynomials of the second kind”).

b) Let $\nu \in \mathcal{M}_m(\mathbf{R})$ a measure on $\mathbf{R}$ such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$. For every $s \in C - \mathbf{R}$, we have

$$\int_{\mathbf{R}} \frac{p_n(t)}{t-s} d\nu(t) = q_n(s) + p_n(s) \int_{\mathbf{R}} \frac{d\nu(t)}{t-s}.$$

c) Let $s \in C - \mathbf{R}$. The sequence $(p_n(s))_{n \in \mathbf{N}}$ belongs to $\ell^2(\mathbf{N})$ if and only if the sequence $(q_n(s))_{n \in \mathbf{N}}$ belongs to $\ell^2(\mathbf{N})$. (Let $\sigma(s) = \int (t-s)^{-1} d\nu(t)$; using the preceding question, prove that the sequence $(q_n(s) + \sigma(s)p_n(s))_{n \in \mathbf{N}}$ belongs to $\ell^2(\mathbf{N})$.)

d) For every $n \in \mathbf{N}$, the formula

$$a_n(p_{n+1}(X)p_n(Y) - p_n(X)p_{n+1}(Y)) = (X - Y) \sum_{j=0}^n p_j(X)p_j(Y)$$

is valid in $C[X, Y]$.

e) For every $n \in \mathbf{N}$, we have the relation

$$p_n q_{n+1} - p_{n+1} q_n = \frac{1}{a_n}.$$

f) If the series with general term a_n^{-1} is divergent, then the measure ν is the unique measure such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$.

25) We resume the notations of the preceding exercises; in particular, we consider a sequence $\mu = (\mu_n)_{n \in \mathbf{N}}$ of real numbers satisfying conditions (24) and such that the sesquilinear form b_μ is separating.

Suppose that $\mu_0 = 1$ and such that the Carleman condition is satisfied, that is, the series with general term $\mu_{2n}^{-1/(2n)}$ for $n \geq 1$ is divergent. a) We have

$$\mu_{2n}^{-1/(2n)} \leq (a_0 \dots a_{n-1})^{-1/n} \leq \frac{e}{n^2} \sum_{j=1}^n \frac{j}{a_{j-1}}$$

for every integer $n \geq 1$. (To establish the second inequality, one may prove and use the inequality $(n/e)^n \leq n!$ for $n \geq 1$.)

b) The series with general term a_n^{-1} for $n \geq 1$ is divergent.

c) There exists a unique measure $\nu \in \mathcal{M}_m(\mathbf{R})$ such that $\nu(x^n) = \mu_n$ for all $n \in \mathbf{N}$.

26) We denote x the identity function of $\mathbf{R}_+$.

a) Let ν a positive measure on R_+ such that x^n is ν -integrable for every integer $n \geq 0$.

Denote by $\mu_n = \nu(x^n)$. For every integer $n \geq 0$ and for all complex numbers $a_0, \dots, a_n$, we have

$$\sum_{i=0}^n \sum_{j=0}^n \bar{a}_i a_j \mu_{i+j} \geq 0$$

$$\sum_{i=0}^n \sum_{j=0}^n \bar{a}_i a_j \mu_{i+j+1} \geq 0.$$

b) If $(\mu_n)_{n \geq 0}$ is a sequence of real numbers satisfying the condition of the preceding question,

then there exists a positive measure ν onto R_+ such that $\mu_n = \nu(x^n)$ for all $n \in \mathbb{N}$. (“Stieltjes moment problem”; use the method of Exercise 21, appealing to the Friedrichs extension.)

27) We denote $M = X^2Y^2(X^2 + Y^2 - 3) \in \mathbf{R}[X, Y]$ (“Motzkin polynomial”).

- a) We have $M(x,y) \geq 0$ for all x and y in $\mathbf{R}$.
- b) There exists no finite family $(p_i)_{i \in I}$ in $\mathbf{R}[X,Y]$ such that $M = \sum p_i^2$. (If such a family existed, show that each p_i must be of the form $a + bXY$ with $a \in \mathbf{R}$ and $b \in \mathbf{R}[X, Y]$ of degree at most 1; then compute the coefficient of $(XY)^2$.)
- c) Let $\tau: \mathbb{N}^2 \rightarrow \mathbb{N} - \{0\}$ the map defined by

$$\tau(0,0) = 1, \quad \tau(1,2) = 2, \quad \tau(2,1) = 3, \quad \tau(1,1) = 4, \quad \tau(1,0) = 5,$$

$$\tau(0,1) = 6, \quad \tau(2,0) = 7, \quad \tau(0,2) = 8, \quad \tau(3,0) = 9, \quad \tau(0,3) = 10$$

and

$$\tau(i,j) = j + 1 + (i + j)(i + j + 1)/2 \text{ if } i + j \geq 4.$$

The map τ is a bijection.

- d) Let $(\nu_n)_{n \geq 1}$ the sequence defined by

$$\nu_i = 1 \text{ if } i \in \{1, 2, 3\}, \quad \nu_4 = 4, \quad \nu_n = (n!)^{(n+1)!} \text{ if } n \geq 5.$$

For n and m in $\mathbb{N}$ we set $\mu_{n,m} = 0$ if n or m is odd and $\mu_{n,m} = \nu_{\tau(n/2, m/2)}$ otherwise. The linear map α of $\mathbf{C}[X, Y]$ in $\mathbf{C}$ such that $\alpha(X^n Y^m) = \mu_{n,m}$ satisfies $\alpha(p^2) \geq 0$ for all $p \in \mathbf{R}[X, Y]$. (It suffices to verify that, for every integer $k \in \mathbb{N}$ the determinant of the matrix $(a_{m,m})_{1 \leq n, m \leq k}$ is positive, where $a_{n,m}$ is defined by $a_{\tau(i,j), \tau(i',j')} = \alpha(X^{i+i'} Y^{j+j'})$ for all (i,j, i', j') ; then proceed by induction on k .)

- e) There exists no positive measure ν onto $\mathbb{R}^2$ such that $(x, y) \mapsto x^n y^m$ is μ -integrable for all $(n, m) \in \mathbb{N}^2$, and such that

$$\int_{\mathbb{R}^2} x^n y^m d\nu(x, y) = \mu_{n,m}$$

for all $(n, m) \in \mathbb{N}^2$.

- f) There exists a Hilbert space E and a densely defined partial operator u on E satisfying the following conditions:
- (i) One has $\text{dom}(u) \subset \text{dom}(u^*)$;
 - (ii) One has $\|u(x)\| = \|u^*(x)\|$ for all $x \in \text{dom}(u)$;
 - (iii) There does not exist a Hilbert space F containing E as a subspace, such that the canonical injection of E into F is isometric, and a partial normal operator v on F such that $u \subset v$.

Compare this result with exercise 24 of IV, p. 352.

- 28) For every integer $k \geq 0$, we denote b_k the number of equivalence relations on the set $\{1, \dots, k\}$.

For every $\lambda > 0$, we denote ϖ_λ the Poisson measure with parameter λ onto $\mathbf{R}$ (cf. exercise 41 of II, p. 282), that is, the measure of total mass 1 supported on $\mathbf{N}$ such that $\varpi_\lambda(j) = e^{-\lambda} \lambda^j / j!$ for all $j \in \mathbf{N}$.

- a) The relation

$$b_{k+1} = \sum_{j=0}^k \binom{k}{j} b_j$$

is valid for every integer $k \in \mathbf{N}$.

- b) For any function $f : \mathbf{N} \rightarrow \mathbb{C}$ such that the function $x \mapsto xf(x)$ is ϖ -integrable, we have

$$\int_{\mathbf{R}} xf(x) d\varpi_\lambda(x) = \lambda \int_{\mathbf{R}} f(x+1) d\varpi_\lambda(x)$$

(“Chen’s formula”).

Additionally, if μ is a bounded measure on $\mathbb{R}$ satisfying this property, then there exists $c \in \mathbb{R}$ such that $\mu = c\varpi_\lambda$.

c) For every $k \in \mathbb{N}$, the function $x \mapsto x^k$ is ϖ_1 -integrable and we have

$$\varpi_1(x \mapsto x^k) = b_k.$$

Additionally, the measure ϖ_1 is the unique positive measure on $\mathbb{R}$ satisfying these equations.

d) For every integer $k \geq 0$, we have

$$b_k = \frac{1}{e} \sum_{j \geq 0} \frac{j^k}{j!}$$

(« Dobiński’s formula ») and

$$b_k = \frac{2k!}{e\pi} \mathcal{I} \left(\int_0^\pi e^{e^{it}} \sin(kt) dt \right)$$

(“Cesàro formula”).

e) For every integer $n \geq 1$, we denote μ_n the positive measure on $\mathbb{R}$ that is the image of the normalized Haar measure of the symmetric group S_n by the mapping f_n of S_n in $\mathbb{R}$ which to $\sigma \in S_n$ associates the number of fixed points of σ . Let $k \in \mathbb{N}$. For every integer $n \geq k$, we have

$$\mu_n(x \mapsto x^k) = b_k.$$

f) The sequence (μ_n) converges weakly when $n \rightarrow +\infty$ toward the Poisson measure ϖ_1 .

29) Let E be a complex Hilbert space. We denote by $\mathcal{T}_b$ (resp. $\mathcal{T}_s$, $\mathcal{T}_f$) the Banach space topology of $\mathcal{L}(E)$ (resp. the topology of pointwise convergence, the locally convex topology defined by the seminorms $u \mapsto \langle x | u(y) \rangle$ for $(x, y) \in E \times E$).

a) Let $\lambda_0 \in \mathbb{C} - \mathbb{R}$. The topology $\mathcal{T}_b$ -resolvent (resp. $\mathcal{T}_s$ -resolvent) is the coarsest topology

on $\mathcal{A}(E)$ such that the map $u \mapsto R(u, \lambda_0)$ of $\mathcal{A}(E)$ in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_b$ (resp. $\mathcal{T}_s$) is continuous.

- b) The topology T_f -resolvent coincides with the topology T_s -resolvent.
- c) Let $X \subset \mathcal{L}(E)$ a bounded set of Hermitian endomorphisms of E . The topology induced on X by the topology T_b -resolvent (resp. the topology T_s -resolvent) coincides with the topology induced by the topology T_b (resp. the topology T_s).
- d) The property from part (a) does not hold for the topology T_f -resolvent.
- e) Give an example of a sequence (u_n) of partial self-adjoint operators that converges for the topology T_b -resolvent and function $f \in \mathcal{C}_b(\mathbf{R})$ such that $f(u_n)$ does not converge to $f(u)$ in $\mathcal{L}(E)$.

f Let (u_n) a sequence of partial self-adjoint operators on E and u a self-adjoint partial operator on E . If there exists a dense subspace F of E_u contained in $\text{dom}(u_n)$ for all n such that $u_n(x)$ converges to $u(x)$ for all $x \in F$, then (u_n) converges to u for the topology $\mathcal{T}_s$ -resolvent.

30) Let E be a complex Hilbert space. Denote by T_b the topology of bounded convergence on $\mathcal{L}(E)$.

- a) Let u a self-adjoint partial operator on E . For every $\lambda \in \mathbf{C} - \mathbf{R}$ such that the imaginary part of λ is negative, we have

$$R(u, \lambda) = i \int_0^\infty e^{-it\lambda} e^{itu} dt.$$

- b) The topology T_b -resolvent is the coarsest topology on $\mathcal{A}(E)$ such that the maps $u \mapsto e^{itu}$ of $\mathcal{A}(E)$ in $\mathcal{L}(E)$ are continuous for every $t \in \mathbf{R}$.

31) Let E be a complex Hilbert space. Denote by T_s the topology of pointwise convergence $\mathcal{L}(E)$. Let H denote the set of complex numbers with strictly positive imaginary part.

Let (u_n) a sequence in $\mathcal{A}(E)$. One assumes that

- (i) There exists $\lambda_+ \in \mathbf{H}$ such that $R(u_n, \lambda_+)$ converges in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_s$; we denote by v_+ its limit;
- (ii) There exists $\lambda_- \in \overline{\mathbf{H}}$ such that $R(u_n, \lambda_-)$ converges in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_s$;
- (iii) The image F of the endomorphism v_+ is dense in E .

Let D be the open disk in $\mathbf{C}$ consisting of $\lambda \in H$ such that

$$|\lambda - \lambda_+| < \frac{1}{|\mathcal{T}(\lambda_+)|}.$$

- a) For every $\lambda \in D$, the series

$$r_\lambda = \sum_{n=0}^{\infty} (\lambda_+ - \lambda) v_+^{n+1}$$

converges in $\mathcal{L}(E)$. Moreover, for every $\lambda \in D$ the sequence $(R(u_n, \lambda))_n$ converges to r_λ in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_s$.

- b) There exists a unique holomorphic map from H to $\mathcal{L}(E)$ which coincides on D with the map $\lambda \mapsto r_\lambda$. We still denote it by $\lambda \mapsto r_\lambda$.
- c) There exists a unique holomorphic map from $\mathbf{C} - \mathbf{R}$ in $\mathcal{L}(E)$ which coincides on $\mathbf{H}$ with the map $\lambda \mapsto r_\lambda$ and which satisfies $r_\lambda^* = r_{\bar{\lambda}}$ for all $\lambda \in \mathbf{C} - \mathbf{R}$. It satisfies $r_{\lambda_-} = v_-$, and for all $\lambda \in \mathbf{C} - \mathbf{R}$ the sequence $(R(u_n, \lambda))$ converges to r_λ in $\mathcal{L}(E)$ endowed with the topology $\mathcal{T}_s$.
- d) For all λ and μ in $\mathbf{C} - \mathbf{R}$, we have $r_\lambda - r_\mu = (\mu - \lambda) = r_\mu r_\lambda$ and $r_\lambda r_\mu = r_\mu r_\lambda$.
- e) The image of r_λ is equal to F for all $\lambda \in \mathbf{C} - \mathbf{R}$.

f) For every $\lambda \in \mathbf{C} - \mathbf{R}$, the endomorphism r_λ is injective; the partial operator $\lambda 1_E - r_\lambda^{-1}$ of domain F does not depend on the choice of $\lambda \in \mathbf{C} - \mathbf{R}$; we denote it u . g) The partial operator u is self-adjoint; for all $\lambda \in \mathbf{C} - \mathbf{R}$, we have $R(u, \lambda) = r_\lambda$.

h) The sequence (u_n) converges to u for the topology T_s -resolvent. (“Kato–Trotter theorem”)

32) Let E be a complex Hilbert space. Let u and v be partial self-adjoint operators on E such that $u + v$ is self-adjoint.

a) For $t \in \mathbf{R}^*$, let $v(t) = t^{-1}(e^{itu}e^{itv} - e^{it(u+v)}) \in \mathcal{L}(E)$, and $w(t)$ the restriction of $v(t)$ to the Hilbert space E_{u+v} . Let B be the set of continuous linear maps $w(t)$. The set B is bounded in the Banach space $\mathcal{L}(E_{u+v}, E)$. (Use the Banach-Steinhaus theorem.)

b) One has $w(t) \rightarrow 0$ as $t \rightarrow 0$ in $\mathbf{R}^*$, into the space $\mathcal{L}(E_{u+v}, E)$ endowed with the topology of compact convergence.

c) Let $x \in E_{u+v}$. We have

$$\lim_{n \rightarrow +\infty} (e^{itu/n} e^{itv/n})^n x = e^{it(u+v)} x.$$

For $n \in \mathbf{N}$, write

$$(e^{itu/n} e^{itv/n})^n - e^{it(u+v)} = \sum_{i=0}^{n-1} (e^{itu/n} e^{itv/n})^i (e^{itu/n} e^{itv/n} - e^{it(u+v)/n}) (e^{it(u+v)/n})^{n-1-i}$$

and apply the result of the previous question to the compact image of the map $t \mapsto e^{it(u+v)} x$ of $[-1, 1]$ in E_{u+v} .

33) Let $n \in \mathbf{N}$. We denote Δ the Laplacian on $L^2(\mathbf{R}^n)$.

a) For every $t \in \mathbf{R}^*$, the endomorphism $e^{it\Delta}$ of $L^2(\mathbf{R}^n)$ is given by

$$e^{it\Delta} f(x) = \frac{1}{(2it)^{n/2}} \int_{\mathbf{R}^n} f(y) e^{i|x-y|^2/(4t)} dy$$

if $f \in \mathcal{L}^1(\mathbf{R}^n) \cap \mathcal{L}^2(\mathbf{R}^n)$ (Use the Fourier transform.)

b) Let C be the unitary operator on $L^2(\mathbf{R}^n)$ obtained by passing to the quotient from the map $f \mapsto C(f)$ defined for $f \in \mathcal{L}^2(\mathbf{R}^n)$ by

$$C(f)(x) = \frac{1}{(2it)^{n/2}} e^{\frac{ix^2}{4t}} \mathcal{F}f\left(\frac{x}{2t}\right)$$

for $x \in \mathbf{R}^n$

One has $e^{it\Delta} \rightarrow C$ as $t \rightarrow +\infty$ in the space $\mathcal{L}(L^2(\mathbf{R}^n))$ endowed with the topology of simple convergence.

- c) Suppose that $1 \leq n \leq 3$. Let $g \in \mathcal{L}^\infty(\mathbf{R}^n) + \mathcal{L}^2(\mathbf{R}^n)$ with real values. Let u denote the partial operator $u = \Delta + m_g$; it is self-adjoint on $L^2(\mathbf{R}^n)$ (Exercise 14). For $f \in \mathcal{L}^1(\mathbf{R}^n) \cap \mathcal{L}^2(\mathbf{R}^n)$ and $t \in \mathbf{R}^*$, we have

$$e^{ituf}(x) = \lim_{k \rightarrow +\infty} \frac{1}{(2it/k)^{kn/2}} \int_{\mathbf{R}^{nk}} f(y_k) \exp\left(\frac{it}{k} \sum_{i=1}^k \left(\frac{|y_i - y_{i-1}|^2}{4(t/k)^2} - g(y_i)\right)\right) dy$$

where we write $y = (y_1, \dots, y_k)$ with $y_i \in \mathbf{R}^n$.

- d) Interpretation of the preceding formula.

- 34) Let E be a complex Hilbert space. For all endomorphisms u and v of E , we denote $[u, v] = uv - vu$ and we define $E_{u,v} = \overline{\text{Im}(u)} \cap \text{Ker}(v)$.

Let p and q orthogonal projectors of E . We set $a = p - q$ and $b = 1_E - p - q$. a) If there exists a unitary endomorphism u of E such that $upu^{-1} = q$ and $uqu^{-1} = p$, then the Hilbert dimensions of $E_{p,q}$ and $E_{1-p,1-q}$ are equal.

- b) We have the relations

$$a^2 + b^2 = 1_E, \quad ab + ba = 0, \quad [p, a] = [q, a] = [p, b] = [q, b] = 0.$$

- c) Up to the question f), included, we will assume that $E_{p,q} = E_{1-p,1-q} = \{0\}$. The endomorphism b is injective; let s the function on $\mathbf{R}$ in $\mathbf{R}$ such that

$$s(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0. \end{cases}$$

The endomorphism $u = s(b)$ is unitary.

d) For every $\varepsilon > 0$, we have $(|b| + \varepsilon)a = a(|b| + \varepsilon)$.

e) One has $uau^{-1} = -a$ (check that $s(b) = \lim_{\varepsilon \rightarrow 0} b(|b| + \varepsilon)^{-1}$) and $ubu^{-1} = b$.

f) One has $upu^{-1} = q$ and $uqu^{-1} = p$.

g) We assume that the Hilbertian dimension of $E_{p,q}$ is equal to that of $E_{1-p,1-q}$. Then there exists a unitary endomorphism u of E such that $upu^{-1} = q$ and $uqu^{-1} = p$. (The spaces $E_{p,q}$ and $E_{1-p,1-q}$ are orthogonal; consider the Hilbert space $F = E_{p,q} \oplus E_{1-p,1-q}$ and apply the preceding case to the orthogonal projectors of F° deduced from p and q by passing to subspaces.)

h) Suppose that $\|p - q\| < 1$. There exists a unitary endomorphism u of E such that $upu^{-1} = q$ and $uqu^{-1} = p$.

i) Suppose that $a = p - q$ is compact. The endomorphism $k = qp$ induces, by passing to subspaces, a Fredholm linear map $\tilde{k}$ of the image of p in the image of q . There exists a unitary endomorphism u of E such that $upu^{-1} = q$ and $uqu^{-1} = p$ if and only if the index of $\tilde{k}$ is zero.

35) Let E be a Hilbert space and let u be a positive, self-adjoint partial operator on E .

a) For every $t \in \mathbf{R}_+^*$, we have $u \circ R(u, -t) \in \mathcal{L}(E)$.

b) The map f of $\mathbf{R}_+^*$ in $\mathcal{L}(E)$ defined by $f(t) = -t^{-1/2}u \circ R(u, -t)$ is integrable with respect to Lebesgue measure on $\mathbf{R}_+^*$.

c) One has

$$\sqrt{u} = \frac{1}{\pi} \int_{\mathbf{R}_+^*} f(t) dt.$$

36) Give an example of a normal partial operator on a complex Hilbert space whose essential spectrum is not closed in $\mathbf{C}$.

37) Let E and F be complex Hilbert spaces, and let u be a closed densely defined partial operator from E into F . If u is surjective, then there exists $v \in \mathcal{L}(F; E)$ such that $u \circ v = 1_F$.

38) Let u be a partial operator in a complex Hilbert space E .

- a) Assume that u is a positive symmetric operator. $F(u)$ the Friedrichs extension of u and let q be the positive partial form associated with $F(u)$. Show that $F(u)$ is the unique self-adjoint extension of u whose domain is contained in $\text{dom}(q)$.
- b) Suppose that u is a closed partial operator with dense domain. Let q the positive partial form of domain $\text{dom}(u)$ defined by $q(x, y) = \langle u(x) | u(y) \rangle$. Prove that the partial self-adjoint operator representing q is $v = u^*u$.
- c) We assume that u is symmetric and that u^2 has dense domain. The partial operator u^2 is symmetric and positive; the Friedrichs extension of u^2 is equal to u^*u .

39) Let $n \geq 1$ an integer. We equip $\mathbf{R}^n$ of Lebesgue measure.

Let v be a positive function that is locally square-integrable on $\mathbf{R}^n$. We denote Δ the Laplacian on $\mathcal{D}(\mathbf{R}^n)$ or on $\mathcal{D}'(\mathbf{R}^n)$, defined by

$$\Delta(f) = - \sum_{i=1}^n \partial_i^2 f.$$

Let D be the partial operator on $L^2(\mathbf{R}^n)$ with domain $\mathcal{D}(\mathbf{R}^n)$ such that

$$D(\varphi) = \Delta(\varphi + v\varphi).$$

- a) The partial operator D is symmetric and positive.
- b) Let $u \in \text{dom}(D^*)$ such that $D^*u + u = 0$. We have $\Delta(u) + u = 0$ where the Laplacian of $u \in L^2(\mathbf{R}^n)$ is computed in the sense of distributions.
- c) The distributions $\Delta(u)$ and $(v + 1)u$ are locally integrable functions, and the distribution $-\Delta(u)$ is positive. (Use Exercise 32 of IV, p. 344.)

- d) We have $u \neq 0$. $|u|$ by infinitely differentiable functions w_ε by convolution, so that $\Delta(w_\varepsilon) \geq 0$ and apply the previous question.) e) The partial operator D is essentially self-adjoint.

40) Let E be a complex Hilbert space.

- a) Let u be a partial normal operator on E and let $(j, |u|)$ be its polar decomposition. We have $|u^*| = |u|$ and the polar decomposition of u^* is $(j^*, |u|)$. We have $j|u| = |u|j$.
- b) Let u be a partial self-adjoint operator on E and let $(j, |u|)$ be its polar decomposition. We have $j = s(u)$, where the function s is defined by

$$s(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0. \end{cases}$$

- c) Let u be a partial self-adjoint operator on E and let $(j, |u|)$ be its polar decomposition. We denote by D the domain of $|u|^{1/2}$, and we define a partial Hermitian form q with domain D by setting

$$q(x, y) = \langle |u|^{1/2}(x) | j|u|^{1/2}(y) \rangle.$$

The partial operator representing q coincides with u ; if u is positive, then q is the positive partial form associated with u .

- d) Let u and v be partial self-adjoint operators on E . Denote by q_u and q_v the partial Hermitian forms defined as in question c). Suppose that $\text{dom}(q_u) \cap \text{dom}(q_v)$ is dense in E . The sum of u and v in the sense of Hermitian forms is the partial operator $u + v$ on E representing the partial form

$$q(x, y) = q_u(x, y) + q_v(x, y)$$

with domain $\text{dom}(q_u) \cap \text{dom}(q_v)$. Show that the partial operator $u + v$ is an extension of

$u + v$; if u and v are positive, then $u + v$ is a positive partial self-adjoint operator.

- e) Give an example where the domain of $u + v$ is not dense, but $u + v$ is defined.
 - f) Let $n \geq 1$ be an integer and $S \subset \mathbb{R}^n$ a closed set of Lebesgue measure zero. Let v be a positive measurable function on $\mathbb{R}^n$ with respect to Lebesgue measure such that v is locally integrable on $\mathbb{R}^n - S$. The partial self-adjoint operator $\Delta + m_v$ is defined.
- 41) Let $n \geq 1$ an integer. We denote by Δ the Laplacian on $\mathbb{R}^n$, and we denote by u the reduction of Δ to the subspace $\mathcal{D}(\mathbb{R}^n - \{0\})$.
- a) The domain of u is dense in the Hilbert space E_q associated with the partial form q associated with Δ if and only if $n \geq 2$. (Identify the dual of E_q with a space of tempered distributions; if $g \in E'_q$ vanishes on $\text{dom}(u)$, show that g is a distribution with support contained in $\{0\}$; conclude by using exercise 24 of IV, p. 338.)
 - b) The domain of u is dense in the Hilbert space E_Δ if and only if $n \geq 4$.
- 42) Let E be a complex Hilbert space and let u be a closed partial operator with dense domain on E . We say that a partial operator v on E is compact relative to u if the domain of v contains the domain of u and if the restriction of v to $\text{dom}(u)$ is a compact linear map from E_u in E .
- a) Every partial operator v that is compact relative to u is bounded relative to u and its relative norm $\|v\|_u$ is zero. (If the relative norm is not zero, construct a sequence (x_n) in $\text{dom}(u)$ converging to 0, such that $\|v(x_n)\| = 1$ for all n and such that $(u(x_n))$ converges weakly to an element y of E ; prove that $y = 0$ and deduce that v is not relatively compact.)
 - b) Let $\lambda \in \mathbb{C} - \text{Sp}(u)$ Let v a partial operator on E whose domain contains $\text{dom}(u)$. Then v is compact relative to u if and only if $v \circ R(u, \lambda)$ is a compact endomorphism of E .
- 43) Let E be a complex Hilbert space.
- a) Let u be a self-adjoint partial operator on E . For every $\lambda \in \mathbb{R}$, we have $\lambda \in \text{Sp}_c(u)$ if and only if there exists a sequence (x_n) in E that converges weakly to 0 in E and satisfies the conditions

$$\liminf_{n \rightarrow +\infty} \|x_n\| > 0, \quad \lim_{n \rightarrow +\infty} (u(x_n) - \lambda x_n) = 0.$$

- b) Let u and v of self-adjoint partial operators on E . If there exists $\lambda \in \mathbb{C} - (\text{Sp}(u) \cup \text{Sp}(v))$ such that $R(u, \lambda) - R(v, \lambda)$ is compact, then $\text{Sp}_e(u) = \text{Sp}_e(v)$.
- c) Let u be a closed symmetric partial operator on E . Let u_1 and u_2 be self-adjoint extensions of u . If the space $\text{Ker}(u^* - i)$ is finite-dimensional, then $\text{Sp}_e(u_1) = \text{Sp}_e(u_2)$.
- d) Let u be a self-adjoint partial operator on E and v a partial operator compact relative to u (see the preceding exercise). The partial operator $u + v$ is self-adjoint and $\text{Sp}_e(u + v) = \text{Sp}_e(u)$.

Chapter V

Unitary representations

For every vector space E , we denote by 1_E the identity map of E . If E , F and G are vector spaces and if $u: F \rightarrow E$ and $v: G \rightarrow F$ are linear maps, one sometimes denotes by uv the linear map $u \circ v$ from G to E .

For a Hilbert space E , one denotes by $\langle x|y \rangle_E$, or simply $\langle x|y \rangle$, the scalar product on E .

If X is a locally compact topological space and μ a measure on X , we denote by $\mathcal{L}^p(X, \mu) = \mathcal{L}_C^p(X, \mu)$ and $L^p(X, \mu) = L_C^p(X, \mu)$ for $p \in [1, +\infty]$.

One denotes by e the identity element of a group G .

Let A be an associative algebra, E a topological vector space and $\pi: A \rightarrow \mathcal{L}(E)$ a representation of A in E . One says that π is non-degenerate if the set of vectors $\pi(a)x$ for $a \in A$ and $x \in E$ is total in E .

V.1 UNITARY REPRESENTATIONS

In this section, the vector spaces are over $K = \mathbb{R}$ or $\mathbb{C}$.

Recall that a linear representation of a group G in a K -vector space E is a homomorphism ϱ from G into the group $\mathbf{GL}(E)$ of automorphisms of E (A, VIII, p. 387, def. 1). One says that E is the space of the representation ϱ , and that the dimension of E is the dimension of ϱ , also denoted $\dim(\varrho)$.

One can identify a representation of G in a K -vector space with a $K[G]$ -module (loc. cit.), and one will use the corresponding terminology, for example concerning direct sums of represen-

tations, morphisms of representations or the representation on C obtained from a representation in an R -vector space by extension of scalars.

A representation $\varrho: G \rightarrow GL(E)$ is said to be faithful if it is an injective homomorphism.



Note that this does not mean that the $K[G]$ -module associated with ϱ is a faithful K -module (A, II, p. 28).

The character of a representation of G in a finite-dimensional K -vector space is the map χ_ϱ from G to K such that $\chi_\varrho(g) = \text{Tr}(\varrho(g))$ for all $g \in G$, cf. A, VIII, p. 388.

V.1.1 Reminders concerning continuous linear representations

Let G be a topological group. One recalls that a continuous linear representation of G in a topological K -vector space E is a linear representation ϱ of G in E such that the map from $G \times E$ to E defined by $(g, x) \mapsto \varrho(g)x$ is continuous (INT, VIII, p. 128, § 2, n° 1, def. 1). This map defines an action of G on E and the image of ϱ is contained in $\mathcal{L}(E)$. One says that a continuous representation ϱ is bounded if its image is bounded in the space $\mathcal{L}(E)$ endowed with the topology of bounded convergence.

Remark. — If $K = \mathbb{R}$ and if ϱ is a continuous linear representation of G in an $\mathbb{R}$ -vector space E , then the map $\varrho_{(\mathbb{C})}: g \mapsto 1 \otimes \varrho(g)$ is a continuous linear representation of G in $E_{(\mathbb{C})}$; if ϱ is bounded, then $\varrho_{(\mathbb{C})}$ is bounded.

For every topological K -vector space E , the homomorphism which to every $g \in G$ associates 1_E is a continuous linear representation of G in E , called the trivial representation of G in E .

Let H be a subgroup of G . The restriction of ϱ to H is a continuous linear representation of H in E , denoted $\text{Res}_H^G(\varrho)$.

Let ϱ_1 and ϱ_2 of continuous linear representations of G in K -vector topological spaces E_1 and E_2 . A G -morphism u of ϱ_1 in ϱ_2 is a morphism of linear representations that is continuous, that is, a continuous linear map $u: E_1 \rightarrow E_2$ such that $u \circ \varrho_1(g) = \varrho_2(g) \circ u$ for all $g \in G$. We denote by $\text{Hom}_G(\varrho_1, \varrho_2)$ the vector space of G -morphisms from ϱ_1 in ϱ_2 . It is a closed

subspace of the space $\mathcal{L}(E_1; E_2)$ equipped with the topology of pointwise convergence.

Let ϱ a continuous linear representation of a group G in a K -vector topological space E . The identity map 1_E of E is a G -morphism of ϱ in ϱ , which is denoted 1_ϱ . If ϱ_1, ϱ_2 and ϱ_3 are continuous linear representations of G into K -topological vector spaces E_1, E_2, E_3 respectively, and if $u : E_1 \rightarrow E_2$ and $v : E_2 \rightarrow E_3$ are G -morphisms, then $v \circ u$ is a G -morphism.

A G -morphism u of ϱ_1 in ϱ_2 is a G -isomorphism if there exists a G -morphism v of ϱ_2 in ϱ_1 such that $v \circ u = 1_{\varrho_1}$ and $u \circ v = 1_{\varrho_2}$. It is necessary and sufficient for this that u be a G -morphism and an isomorphism of topological vector spaces of the space ϱ_1 in the space of ϱ_2 ; its inverse $v = u^{-1}$ is indeed then a G -morphism. If there exists a G -isomorphism from ϱ_1 in ϱ_2 we say that these representations are isomorphic.

DEFINITION 1. — Let ϱ a continuous linear representation of a topological group G in a topological vector space E . π a subrepresentation of ϱ . If F is a closed subspace of E and if, for every $g \in G$ the space F is stable under $\varrho(g)$ and $\pi(g)$ is the endomorphism of F induced by $\varrho(g)$ by passage to the subspaces.

Let ϱ a continuous linear representation of a topological group G in a topological vector space E . The closure of $\{0\}$ in E is a subrepresentation of ϱ . More generally, the closure of a subspace stable under the endomorphisms $\varrho(g)$ is a subrepresentation of ϱ .

A subrepresentation π of ϱ is uniquely determined by a closed subspace F , stable under all endomorphisms $\varrho(g)$. This last condition will often be stated in the form “ F is a G -invariant subspace of E .” One will then also say that F defines a subrepresentation of ϱ or sometimes, by abuse of language, that F is a subrepresentation of ϱ , or of E .

Let F be a subspace of E defining a subrepresentation π of ϱ . For every $g \in G$, the linear map $\varrho(g)$ defines, by passing to the quotient, an endomorphism $\tilde{\varrho}(g)$ of E/F . The map $g \mapsto \tilde{\varrho}(g)$ is a continuous linear representation of G in E/F , and the canonical projection of E onto E/F is a G -morphism. We say that $\tilde{\varrho}$ is the quotient representation of G on E/F ; it is also denoted ϱ/π .

The subspace of elements of E invariant under the action of G is a trivial subrepresentation of ϱ . We denote it by ϱ^G or E^G .

Let A be a subset of E . The closed subspace F generated by the elements $\varrho(g)x$, where $g \in G$ and $x \in A$ is a subrepresentation of ϱ called the subrepresentation of ϱ generated by A ;

if $F = E$, one says that A generates ϱ . If the subrepresentation F is finite-dimensional, one says that the subset A is G -finite.

Suppose that A is reduced to a single element $x \in E$. If A generates ϱ , one says that x is a cyclic vector of ϱ and that ϱ is a cyclic representation; one says that x is a G -finite vector if A is G -finite.

The set of G -finite vectors of ϱ is a vector subspace of E stable under ϱ ; it is not necessarily closed.

Let $(\varrho_i)_{i \in I}$ a family of continuous linear representations of G in locally convex topological K -vector spaces $(E_i)_{i \in I}$. Let E be the direct sum space of the spaces E_i . The map $g \mapsto (\varrho_i(g))$ is a linear representation of G in E , called the sum, or direct sum, of the representations ϱ_i . If I is finite, it is continuous.

If the representations ϱ_i are equal to a representation ϱ for all $i \in I$, we say that the direct sum of the representations ϱ_i is the sum of $\text{Card}(I)$ copies of the representation ϱ , and it is also denoted $\varrho^{\text{Card}(I)}$, or $\text{Card}(I) \varrho$.

Let $(\varrho_i)_{i \in I}$ and $(\pi_j)_{j \in J}$ of finite families of continuous linear representations of G in topological K -vector spaces $(E_i)_{i \in I}$ and $(F_j)_{j \in J}$, respectively. The canonical isomorphism of K -vector spaces

$$\text{Hom}_K \left(\bigoplus_{i \in I} E_i, \bigoplus_{j \in J} F_j \right) \rightarrow \bigoplus_{(i,j) \in I \times J} \text{Hom}_K(E_i, F_j)$$

(A, II, p. 13, cor. 1) induces, by passage to subspaces, an isomorphism

$$\text{Hom}_G \left(\bigoplus_{i \in I} \varrho_i, \bigoplus_{j \in J} \pi_j \right) \rightarrow \bigoplus_{(i,j) \in I \times J} \text{Hom}_G(\varrho_i, \pi_j) \quad (1)$$

which is likewise said to be canonical.

Let ϱ_1 and ϱ_2 of linear representations of G in K -vector spaces E_1 and E_2 . For $u \in \mathcal{L}(E_1; E_2)$ and $g \in G$ we set $\varrho(g)u = \varrho_2(g) \circ u \circ \varrho_1(g^{-1})$. The map $g \mapsto \varrho(g)$ is a linear representation of G in $\mathcal{L}(E_1; E_2)$. The space of invariant elements of this representation coincides with $\text{Hom}_G(\varrho_1, \varrho_2)$.

Let ϱ a continuous linear representation of G in a locally convex space E . Recall (INT, VIII,

p. 131, § 2, no. 2) that the contragredient representation $\check{\varrho}$ of ϱ is the linear representation of G in the dual E' of E defined by $\check{\varrho}(g) = {}^t\varrho(g^{-1})$.

V.1.2 A criterion for continuity

The following result makes it possible to verify that certain linear representations of a direct product of topological groups are continuous.

Lemma 1. — Let G and H be topological groups and E a Banach space. Let ϱ be a bounded representation of G in E and π a bounded representation of H in E . Suppose that $\varrho(g)$ commutes with $\pi(h)$ for all $(g, h) \in G \times H$. The map ϖ of $G \times H$ in $\mathbf{GL}(E)$ defined by $(g, h) \mapsto \varrho(g) \circ \pi(h)$ is a continuous and bounded linear representation of $G \times H$ in E .

The map ϖ is a linear representation of $G \times H$ in E ; let us verify that it is continuous. Since $\|\varpi(g, h)\| \leq \|\varrho(g)\| \|\pi(h)\|$ for all $(g, h) \in G \times H$, the representation ϖ is bounded, hence its image is equicontinuous in $\mathcal{L}(E)$. It then suffices to show that, for every $x \in E$, the map $(g, h) \mapsto \varpi(g, h)x$ is continuous (remark 2 of INT, VIII, p. 129, § 2, no. 1). Let $(g_0, h_0) \in G \times H$. Set $y = \pi(h_0)x$. For every $(g, h) \in G \times H$, it follows that

$$\begin{aligned} \|\varpi(g, h)x - \varpi(g_0, h_0)x\| &\leq \|\varrho(g)(\pi(h)x - y)\| + \|\varrho(g)y - \varrho(g_0)y\| \\ &\leq \|\varrho(g)\| \|\pi(h)x - y\| + \|\varrho(g)y - \varrho(g_0)y\|. \end{aligned}$$

Since ϱ is bounded and that ϱ and π are continuous, this implies the assertion.

V.1.3 Continuous finite-dimensional representations

Let ϱ be a continuous linear representation of a topological group G in a finite-dimensional separated topological K -vector space E . We equip $\mathcal{L}(E)$ with its unique separated topological vector space structure over K ; the morphism $\varrho: G \rightarrow \mathbf{GL}(E)$ is then continuous since the topology of $\mathbf{GL}(E)$ is induced by the topology of $\mathcal{L}(E)$, which coincides with the topology of simple convergence (EVT, I, p. 14, th. 2). Consequently, the representation ϱ is continuous. If G is a real Lie group, then the character of ϱ is an analytic function on G (LIE, III, p. 225, § 8, no. 1, th. 1). The contragredient representation is also continuous when E' is equipped with its unique topology as a separated topological vector space over K . Moreover, for every integer $n \geq 0$ the

representations $T^n(\varrho)$, $S^n(\varrho)$ and $\wedge^n(\varrho)$ (loc. cit.) are continuous, when the corresponding spaces are equipped with their topologies as separated topological vector spaces over K .

Let ϱ_1 and ϱ_2 be continuous representations of a topological group G in separated topological vector spaces E_1 and E_2 of finite dimension. The linear representation $\varrho_1 \otimes \varrho_2$ of G in $E_1 \otimes E_2$ (LIE, III, p. 256, Appendix) is continuous, the space $E_1 \otimes E_2$ being equipped with its topology as a separated topological vector space over K .

V.1.4 Irreducible Representations

In this issue, G is a topological group.

DEFINITION 2. — A representation ϱ of G in a topological K -vector space E is said to be irreducible if $\{0\}$ is not dense in E and if the only subrepresentations of ϱ are ϱ and the representation in the closure of $\{0\}$.

If ϱ is an irreducible representation of G in a separated topological vector space E , then every nonzero element of E is a cyclic vector for ϱ .

Lemma 2. — Let π and ϱ be continuous linear representations of G in separated topological K -vector spaces. We assume that π is irreducible. Every nonzero G -morphism from π to ϱ is injective and every nonzero G -morphism from ϱ to π has dense image.

In particular, if π and ϱ are irreducible and of finite dimension, then every nonzero G -morphism from π to ϱ is an isomorphism.

Since the space of ϱ is separated, the kernel of a G -morphism u from π to ϱ is closed, and hence defines a subrepresentation of π . If the morphism u is not zero, its kernel must therefore be reduced to $\{0\}$, since π is an irreducible representation in a separated topological vector space. Similarly, the closure of the image of a nonzero G -morphism from ϱ to π is a nonzero subrepresentation of π , hence is equal to the space of π .

The last assertion follows from what precedes.

Lemma 3. — Let ϱ be a continuous linear representation of G in a separated topological K -vector space E of finite dimension. If E is nonzero, then there exists an irreducible subrepresentation of ϱ .

Since E is of finite dimension and nonzero, there exists a nonzero G -invariant subspace F

of E of minimal dimension. This subspace is closed in E and defines a subrepresentation of E ; every subrepresentation of F is also a subrepresentation of E , and the representation F is therefore irreducible by minimality.

Remark. — A nonzero representation E of G does not always contain an irreducible subrepresentation (cf. V, p. 426, remark). One will see, however, that this is the case if G is compact, if $K = \mathbb{C}$, and if E is a separated quasi-complete locally convex space over K and nonzero (prop. 7 of V, p. 464).

V.1.5 Unitary representations

DEFINITION 3. — Let G be a topological group and E a Hilbert space over K . A unitary representation of G in E is a continuous linear representation ϱ of G in E such that, for every g in G , the endomorphism $\varrho(g)$ of E is a unitary endomorphism (EVT, V, p. 40) of E .

In other words, a unitary representation is an isometric representation in a Hilbert space. In particular, a unitary representation is bounded.

The trivial representation of a topological group in a Hilbert space is unitary. Every subrepresentation of a unitary representation is unitary. The restriction to a subgroup of a unitary representation is unitary.

DEFINITION 4. — Let G be a topological group and let ϱ and π be unitary representations of G in Hilbert spaces E and F respectively. A G -invariant sesquilinear form on $E \times F$ is a continuous sesquilinear form q on $E \times F$ such that

$$q(x, y) = q(\varrho(g)x, \pi(g)y)$$

for all $g \in G$ and every $(x, y) \in E \times F$.

The vector space of G -invariant sesquilinear forms on $E \times F$ is denoted $\text{Sesq}_G(\varrho, \pi)$ or $\text{Sesq}_G(E, F)$.

Example. — Let G be a topological group and let ϱ be a unitary representation of G in E . The scalar product q on E is a G -invariant sesquilinear form on $E \times E$.

Lemma 4. — Let G be a topological group and let ϱ be a homomorphism of G into the unitary group $U(E)$ of a Hilbert space E . Then ϱ is a unitary representation of G if and only if ϱ

is continuous at the element e of G for the topology of simple convergence on $U(E)$. It suffices that this property hold for every x in a total subset of E .

The condition is evidently necessary. It is sufficient by remark 2 of INT, VIII, p. 129, § 2, no. 1, since the image of ϱ is equicontinuous in $\mathcal{L}(E)$ and since the continuity of the map $g \mapsto \varrho(g)x$ at e implies its continuity on G .

Let G be a topological group. If ϱ_1 and ϱ_2 are unitary representations of G and if u belongs to $\text{Hom}_G(\varrho_1, \varrho_2)$, then u^* belongs to $\text{Hom}_G(\varrho_2, \varrho_1)$. Indeed, since ϱ_1 and ϱ_2 are unitary representations, one has for every $g \in G$

$$\begin{aligned} u^* \circ \varrho_2(g) &= u^* \circ \varrho_2(g^{-1})^* = (\varrho_2(g^{-1}) \circ u)^* \\ &= (u \circ \varrho_1(g^{-1}))^* = \varrho_1(g) \circ u^*. \end{aligned}$$

Lemma 5. — Let G be a topological group and let ϱ be a unitary representation of G in a complex Hilbert space E . The space $\text{Hom}_G(\varrho, \varrho)$ is a unital stellar subalgebra of $\mathcal{L}(E)$.

The preceding shows that $\text{Hom}_G(\varrho, \varrho)$ is a unital and self-adjoint subalgebra of $\mathcal{L}(E)$. Since it is closed in $\mathcal{L}(E)$, it is a stellar subalgebra of $\mathcal{L}(E)$.

Lemma 6. — Let π and ϱ be unitary representations of a topological group G in Hilbert spaces E and F respectively. Let D be a dense subspace of E which is stable under π . Let u be a closed partial operator from E to F with domain D such that $u \circ \pi(g) = \varrho(g) \circ u$ for all $g \in G$. Then the domain of u^* is stable under ϱ and one has the relation $u^* \circ \varrho(g) = \pi(g) \circ u^*$ for all $g \in G$.

Let $g \in G$ and $x \in \text{dom}(u^*)$. For every $y \in \text{dom}(u)$, we have

$$\begin{aligned} \langle \varrho(g)x \mid u(y) \rangle &= \langle x \mid \varrho(g^{-1})u(y) \rangle = \langle x \mid u(\pi(g^{-1})y) \rangle \\ &= \langle u^*(x) \mid \pi(g^{-1})y \rangle = \langle \pi(g)(u^*(x)) \mid y \rangle, \end{aligned}$$

since $\varrho(g)^* = \varrho(g)^{-1} = \varrho(g^{-1})$. This proves that $\varrho(g)x \in \text{dom}(u^*)$ and that $u^*(\varrho(g)x) = \pi(g)(u^*(x))$. In particular, the domain of $u^* \circ \varrho(g)$ contains the domain of u^* , and $\pi(g) \circ u^* \subset u^* \circ \varrho(g)$.

But moreover, if $x \in \text{dom}(u^* \circ \varrho(g))$, then $x = \varrho(g^{-1})(\varrho(g)x)$ belongs to $\text{dom}(u^*)$ by virtue of the preceding applied to g^{-1} . We conclude that $u^* \circ \varrho(g) = \pi(g) \circ u^*$.

PROPOSITION 1. — Let ϱ_1 and ϱ_2 be unitary representations of a topological group G in Hilbert spaces E_1 and E_2 . The map from $\text{Hom}_G(\varrho_1, \varrho_2)$ to $\text{Sesq}_G(\varrho_2, \varrho_1)$ which associates to u the sesquilinear form q_u defined by $q_u(x, y) = \langle x | u(y) \rangle$ is an isomorphism of vector spaces.

If u is a G -morphism, then one has

$$\begin{aligned} q_u(\varrho_2(g)x, \varrho_1(g)y) &= \langle \varrho_2(g)x | u(\varrho_1(g)y) \rangle \\ &= \langle \varrho_2(g)x | \varrho_2(g)u(y) \rangle = \langle x | u(y) \rangle = q_u(x, y) \end{aligned}$$

for all $g \in G$ and every $(x, y) \in E_2 \times E_1$, hence the indicated map is a linear map from $\text{Hom}_G(\varrho_1, \varrho_2)$ to $\text{Sesq}_G(\varrho_2, \varrho_1)$. By EVT, V, p. 16, cor. 2, it is injective.

Conversely, let q be a G -invariant sesquilinear form and u the unique linear map from E_1 to E_2 such that $q(x, y) = \langle x | u(y) \rangle$ for all $(x, y) \in E_2 \times E_1$ (loc. cit.). For every g in G and every (x, y)

in $E_2 \times E_1$, we have

$$\begin{aligned} \langle x | (\varrho_2(g) \circ u \circ \varrho_1(g^{-1}))(y) \rangle &= \langle \varrho_2(g^{-1})x | u(\varrho_1(g^{-1})y) \rangle \\ &= q(\varrho_2(g^{-1})x, \varrho_1(g^{-1})y) = q(x, y) = \langle x | u(y) \rangle, \end{aligned}$$

hence $u = \varrho_2(g) \circ u \circ \varrho_1(g^{-1})$. Consequently, $u \in \text{Hom}_G(\varrho_1, \varrho_2)$. The proposition follows.

Let ϱ be a unitary representation of a topological group G in a Hilbert space E . If $K = \mathbb{C}$, denote by $\bar{E}$ the conjugate space of E (EVT, V, p. 6). If $K = \mathbb{R}$, set $\bar{E} = E$. The conjugate representation $\bar{\varrho}$ is the representation of G in $\bar{E}$ defined by $\bar{\varrho}(g) = \varrho(g)$ for all $g \in G$. It is a unitary representation of G ; by definition, a subspace of $\bar{E}$ is a subrepresentation of $\bar{E}$ if and only if it is a subrepresentation of E .

PROPOSITION 2. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . Denote by u the isometric isomorphism of $\bar{E}$ into E' which associates to x the linear form $y \mapsto \langle x | y \rangle$.

- a) The map u is an isomorphism of the conjugate representation $\bar{\varrho}$ into the contragredient representation $\check{\varrho}$;
- b) Endow E' with the Hilbert space structure obtained by transporting structure via u ; the contragredient representation $\check{\varrho}$ is a unitary representation of G in E' .

By EVT, V, p. 15, th. 3 and the following remark, the map u is an isometric isomorphism.

For every g in G , every x in $\bar{E}$ and every y in E , we have

$$\begin{aligned}\langle y, (\check{\varrho}(g) \circ u)(x) \rangle &= \langle \varrho(g^{-1})y, u(x) \rangle \\ &= \langle x | \varrho(g^{-1})y \rangle = \langle \varrho(g)x | y \rangle = \langle y, u(\bar{\varrho}(g)x) \rangle,\end{aligned}$$

hence $\check{\varrho}(g) \circ u = u \circ \bar{\varrho}(g)$, which proves a ; the assertion b) follows immediately.

COROLLARY. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . The following conditions are equivalent:

- (i) The representation ϱ is irreducible;
- (ii) The contragredient representation $\check{\varrho}$ is irreducible;
- (iii) The conjugate representation $\bar{\varrho}$ is irreducible.

This follows from Proposition 2, and from the remark preceding it concerning subrepresentations of $\bar{E}$.

PROPOSITION 3. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . Let π be a subrepresentation of ϱ , with space F . The orthogonal F° of F in E is a subrepresentation of E such that $E = F \oplus F^\circ$ and F° is isomorphic to E/F .

The space F° is closed. For every $x \in F^\circ$, all $g \in G$ and every $y \in F$, we have $\langle \varrho(g)x | y \rangle = \langle x | \varrho(g^{-1})y \rangle = 0$ since ϱ is unitary and F is a subrepresentation. Therefore $\varrho(g)x \in F^\circ$.

The canonical projection of E onto E/F is a G -morphism, so its restriction to F° is a G -morphism from F° to E/F ; by EVT, V, p. 13, it is an isometric isomorphism.

We shall also denote by π° the representation of G in F° .

PROPOSITION 4. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . A closed subspace F of E is a subrepresentation of ϱ if and only if the orthogonal projector p of E with image F is a G -morphism from ϱ to ϱ .

Suppose that F is a subrepresentation of ϱ . Let $x \in E$ and let us write $x = p(x) + y$ where $y \in F^\circ$. For every g in G , we have

$$\varrho(g)x = \varrho(g)(p(x)) + \varrho(g)y,$$

and since $\varrho(g)(p(x))$ belongs to F and $\varrho(g)y$ to F° (prop. 3), we have $p(\varrho(g)x) = \varrho(g)(p(x))$. Hence p belongs to $\text{Hom}_G(\varrho, \varrho)$.

Conversely, if $p \in \text{Hom}_G(\varrho, \varrho)$ then $1_E - p$ is a G -morphism, hence $F = \text{Ker}(1_E - p)$ is a subrepresentation of ϱ .

V.1.6 Hilbertian direct sum and tensor product of unitary representations

Let G be a topological group. Let $(\varrho_i)_{i \in I}$ be a family of unitary representations of G in Hilbert spaces E_i . Let E be the space of the external Hilbert sum of the E_i . For every g in G and for every $x = (x_i)_{i \in I}$ in E , we have

$$\sum_i \|\varrho_i(g)x_i\|^2 = \sum_i \|x_i\|^2 = \|x\|^2,$$

which proves that $(\varrho_i(g)x_i)_{i \in I}$ is in E and has the same norm as x . Thus the element $\varrho(g): (x_i)_{i \in I} \mapsto (\varrho_i(g)x_i)_{i \in I}$ is a unitary element of $\mathcal{L}(E)$.

Lemma 7. — The map $g \mapsto \varrho(g)$ is a unitary representation of G in E .

By lemma 4 of V, p. 380, it suffices to show that for every i in I and every x in E_i , the map $g \mapsto \varrho_i(g)x$ is continuous at the identity element e of G . This map is the composition of the continuous map $g \mapsto \varrho_i(g)x$ and of the canonical injection of E_i into E . It is therefore continuous.

The representation ϱ is called the Hilbert sum of the unitary representations $(\varrho_i)_{i \in I}$; it is denoted by $\varrho = \bigoplus \varrho_i$.

Let G and H be topological groups. Let ϱ_1 (resp. ϱ_2) be a unitary representation of G (resp. H) in a Hilbert space E_1 (resp. E_2). Let $E = E_1 \widehat{\otimes}_2 E_2$ be the Hilbert tensor product space of E_1 and E_2 (EVT, V, p. 28, def. 1). For $(g, h) \in G \times H$, let $\varrho(g, h)$ be the continuous endomorphism $\varrho_1(g) \widehat{\otimes}_2 \varrho_2(h)$ of E (TVS, V, p. 28); we will denote it simply $\varrho_1(g) \otimes \varrho_2(h)$ whenever no ambiguity is to be feared.

Lemma 8. — a) The map $\varrho: (g, h) \mapsto \varrho(g, h)$ is a unitary representation of $G \times H$ in E ;

b) For any orthonormal basis $(e_i)_{i \in I}$ of E_1 the mapping of the Hilbert sum $\bigoplus_{i \in I} E_2$ to E defined by $(y_i)_{i \in I} \mapsto \sum_{i \in I} e_i \otimes y_i$ is an isometric H -isomorphism of the Hilbert sum $\bigoplus_{i \in I} \varrho_2$ in $\text{Res}_{\{e\} \times H}^{G \times H}(\varrho)$;

c) For any orthonormal basis $(f_j)_{j \in J}$ of E_2 the mapping of the Hilbert sum $\bigoplus_{j \in J} E_1$ to E defined by $(x_j)_{j \in J} \mapsto \sum_{j \in J} x_j \otimes f_j$ is an isometric G -isomorphism of the Hilbert sum $\bigoplus_{j \in J} \varrho_1$ in $\text{Res}_{G \times \{e\}}^{G \times H}(\varrho)$.

The map $(g, h) \mapsto \varrho(g, h)$ is a homomorphism from $G \times H$ to $\mathbf{GL}(E)$ (cf. EVT, V, p. 28, no. 2). Let $(e_i)_{i \in I}$ be an orthonormal basis of E_1 . By EVT, V, p. 29, prop. 3 and cor. 2, the map

$$u: (y_i) \mapsto \sum_{i \in I} e_i \otimes y_i$$

is an isometry from the Hilbert sum $F_2 = \bigoplus_{i \in I} E_2$ onto E . Via this isometry, the representation $\varrho_H: h \mapsto \varrho(e, h)$ of H in E is identified with the direct sum representation of the representations $(\varrho_2)_{i \in I}$ in F_2 . In particular, it is a unitary representation of H in E (lemma 7). Likewise, the homomorphism $\varrho_G: g \mapsto \varrho(g, e)$ is a unitary representation of G in E .

Let $(g, h) \in G \times H$. We have $\varrho(g, h) = \varrho_G(g) \circ \varrho_H(h)$, hence $\varrho(g, h)$ is unitary; moreover, $\varrho_G(g)$ and $\varrho_H(h)$ commute, so lemma 1 of V, p. 377 implies that ϱ is a unitary representation of $G \times H$.

Finally, the assertions concerning the restriction of ϱ to the subgroups $\{e\} \times H$ and $G \times \{e\}$ were obtained in the course of the preceding argument.

The representation $(g, h) \mapsto \varrho_1(g) \otimes \varrho_2(h)$ of $G \times H$ is called the external tensor product of the unitary representations ϱ_1 and ϱ_2 , and is denoted $\varrho_1 \boxtimes \varrho_2$.

Let $n \in \mathbf{N}$ and let $(G_i)_{1 \leq i \leq n}$ be a finite family of topological groups. Let ϱ_i be a unitary representation of G_i in a Hilbert space E_i for $1 \leq i \leq n$. One defines similarly a representation

$$\varrho_1 \boxtimes \cdots \boxtimes \varrho_n$$

of $G_1 \times \cdots \times G_n$ in the Hilbert space $E = E_1 \widehat{\otimes}_2 \cdots \widehat{\otimes}_2 E_n$ (EVT, V, p. 27).

Suppose that $G_i = G$ for $1 \leq i \leq n$. Let $\Delta_n: G \rightarrow G^n$ be the homomorphism defined by $g \mapsto (g, \dots, g)$ for all $g \in G$. We denote by $\varrho_1 \otimes \cdots \otimes \varrho_n$ the unitary representation $(\varrho_1 \boxtimes \cdots \boxtimes \varrho_n) \circ \Delta_n$ of G . This is called the tensor product of the unitary representations ϱ_i .

For every permutation σ of $\{1, \dots, n\}$, the canonical isomorphism

$$E_1 \widehat{\otimes}_2 \dots \widehat{\otimes}_2 E_n \rightarrow E_{\sigma(1)} \widehat{\otimes}_2 \dots \widehat{\otimes}_2 E_{\sigma(n)}$$

(EVT, V, p. 28) is an isometric isomorphism

$$\varrho_1 \otimes \dots \otimes \varrho_n \rightarrow \varrho_{\sigma(1)} \otimes \dots \otimes \varrho_{\sigma(n)}$$

of representations of G .

If $\varrho_i = \varrho$ for $1 \leq i \leq n$, where ϱ is a unitary representation of G , we also denote by $\varrho^{\otimes n}$ the tensor product representation of the ϱ_i , and we say that it is the n th tensor power of ϱ .

V.1.7 Matrix coefficients

DEFINITION 5. — Let G be a topological group and let ϱ be a unitary representation of G in a Hilbert space E . Let x and y be elements of E . The function from G to $\mathbb{K}$ given by $g \mapsto \langle x | \varrho(g)y \rangle$ is called a matrix coefficient of ϱ , or a representative function. If $x = y$, one says that it is a diagonal matrix coefficient. If ϱ is of finite dimension, we say that it is a finite-dimensional matrix coefficient.

The matrix coefficients of ϱ are continuous and bounded functions on G . We denote by $\Upsilon(G)$ (resp. $\Theta(G)$) the set of matrix coefficients of complex unitary representations (resp. complex finite-dimensional representations) of G .

PROPOSITION 5. — Let G be a topological group. The sets $\Theta(G)$ and $\Upsilon(G)$ are unital involutive subalgebras of $\mathcal{C}_b(G)$.

The constant function 1 is a matrix coefficient of the trivial representation of G on $\mathbb{C}$. Let ϱ be a unitary representation of G and let x and y be vectors in the space of ϱ . For every $\lambda \in \mathbb{C}$ and every $g \in G$, we have $\lambda \langle x | \varrho(g)y \rangle = \langle x | \varrho(g)(\lambda y) \rangle$. Moreover, we have

$$\langle x | \varrho(g)y \rangle = \langle \varrho(g^{-1})x | y \rangle = \overline{\langle y | \varrho(g^{-1})x \rangle}$$

for all $g \in G$. Consequently, the sets $\Theta(G)$ and $\Upsilon(G)$ are stable under multiplication by scalars and under conjugation.

Let ϱ_1 and ϱ_2 be unitary representations of G ; let (x_1, y_1) be vectors in the space of ϱ_1 and

(x_2, y_2) be vectors in the space of ϱ_2 . For every $g \in G$, we then have

$$\langle x_1 | \varrho_1(g) y_1 \rangle + \langle x_2 | \varrho_2(g) y_2 \rangle = \langle (x_1, x_2) | (\varrho_1 \oplus \varrho_2)(g) (y_1, y_2) \rangle,$$

$$\langle x_1 | \varrho_1(g) y_1 \rangle \langle x_2 | \varrho_2(g) y_2 \rangle = \langle x_1 \otimes x_2 | (\varrho_1 \otimes \varrho_2)(g) (y_1 \otimes y_2) \rangle,$$

which proves that $\Theta(G)$ and $\Upsilon(G)$ are stable under addition and multiplication. The proposition follows.

V.1.8 Schur's lemma

In this issue, Hilbert spaces are complex.

PROPOSITION 6 (Schur's Lemma). — Let ϱ be a unitary representation of a topological group G in a nonzero Hilbert space E . Then ϱ is irreducible if and only if $\text{Hom}_G(\varrho, \varrho)$ is equal to $\mathbb{C} \cdot 1_E$.

The space $\text{Hom}_G(\varrho, \varrho)$ is a unital stellar subalgebra of $\mathcal{L}(E)$ (lemma 5 of V, p. 380). For every subrepresentation F of E , the orthogonal projector p with image F is an idempotent element of the stellar algebra $\text{Hom}_G(\varrho, \varrho)$ (prop. 4 of V, p. 383). If it is equal to $\mathbb{C} \cdot 1_E$, this means that $p = 0$ or $p = 1_E$, which means that $F = 0$ or $F = E$. Thus ϱ is irreducible.

Conversely, suppose ϱ is irreducible. Let u and v be commuting elements of the stellar algebra $\text{Hom}_G(\varrho, \varrho)$ such that $uv = 0$. Suppose u is nonzero. The kernel F of u then defines a subrepresentation of ϱ different from E , hence F is reduced to $\{0\}$; since F contains the image of v , we deduce that $v = 0$. By Proposition 10 of I, p. 113, we therefore have $\text{Hom}_G(\varrho, \varrho) = \mathbb{C} \cdot 1_E$.

COROLLARY 1. — Let π be an irreducible unitary representation of a topological group G in a Hilbert space E . Let u be a closed partial operator on E . Suppose that u has dense domain, that $\text{dom}(u)$ is stable under π , and that $u \circ \pi(g) = \pi(g) \circ u$ for all $g \in G$. Then $\text{dom}(u) = E$ and u is a scalar multiple of the identity.

The partial operator $u^* \circ u$ is a positive self-adjoint partial operator on E (prop. 12 of IV, p. 241), the same holds for $v = 1_E + u^* \circ u$ and the latter is injective since $-1 \notin \text{Sp}(u^* \circ u)$ (prop. 17 of IV, p. 248). We have $u^* \circ \pi(g) = \pi(g) \circ u^*$ for all $g \in G$ (lemma 6 of V, p. 381), whence $v \circ \pi(g) = \pi(g) \circ v$ for all $g \in G$. Since v is injective, it follows that $v^{-1} \circ \pi(g) = \pi(g) \circ v^{-1}$.

for all $g \in G$. But v^{-1} belongs to $\mathcal{L}(E)$, hence by prop. 6, there exists $\lambda \in \mathbb{C}$ such that $v^{-1} = \lambda 1_E$. Necessarily, $\lambda \neq 0$, which implies that $E = \text{Im}(v^{-1}) = \text{dom}(v) \subset \text{dom}(u)$, whence $\text{dom}(u) = E$. Since u is closed, we have $u \in \mathcal{L}(E)$, therefore $u \in \text{Hom}_G(\pi, \pi)$ and u is a homothety by Prop. 6.

COROLLARY 2. — Let ϱ and π be irreducible unitary representations of a topological group G in Hilbert spaces E and F , respectively. The space $\text{Hom}_G(\varrho, \pi)$ is of dimension 1 if ϱ is isomorphic to π and is zero otherwise. In particular, if ϱ is isomorphic to π , every nonzero G -morphism from ϱ to π is an isomorphism.

Suppose there exists a nonzero G -morphism u from ϱ to π . The linear map $u^* \circ u$ is an element of $\text{Hom}_G(\varrho, \varrho)$; therefore there exists a complex number λ such that $u^* \circ u = \lambda \cdot 1_E$ (prop. 6).

We then have

$$\langle u(x) | u(y) \rangle = \langle x | u^* u(y) \rangle = \lambda \langle x | y \rangle$$

for all x and y in E . In particular, $\lambda \neq 0$ since u is nonzero. Since $\|u(x)\| = |\lambda|^{1/2} \|x\|$ for all $x \in E$, the linear map u is injective, and the image of u is closed in F (Lemma 8 of I, p. 107); it is then a nonzero subrepresentation of the irreducible representation π , whence we deduce that u is surjective. Thus, u is an isomorphism from ϱ to π . The map $v \mapsto u^* \circ v$ is then an isomorphism from the space $\text{Hom}_G(\varrho, \pi)$ to the space $\text{Hom}_G(\varrho, \varrho)$, which is of dimension 1 (prop. 6).

COROLLARY 3. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . The representation ϱ is irreducible if and only if the space $\text{Sesq}_G(\varrho, \varrho)$ has dimension 1. This space is then generated by the scalar product on E .

In view of prop. 1 of V, p. 381, this follows from prop. 6.

COROLLARY 4. — Let ϱ_1 and ϱ_2 be non-isomorphic irreducible unitary representations of a topological group G in Hilbert spaces E_1 and E_2 . The space $\text{Sesq}_G(\varrho_1, \varrho_2)$ is zero.

By prop. 1 of V, p. 381, this follows from cor. 2.

COROLLARY 5. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . Let π be an irreducible unitary representation of G in a Hilbert space F .

- a) For every nonzero G -morphism u from π to ϱ , there exists $\lambda \in \mathbf{R}_+^*$ such that λu is isometric;
- b) Every G -morphism u from π to ρ has closed image;
- c) Every nonzero G -morphism v from ϱ to π is surjective.

Let us prove assertion a), which implies the assertion b) (lemma 8 of I, p. 107). Since the morphism u is nonzero, it is injective (lemma 2 of V, p. 378). The formula $q(x, y) = \langle u(x)|u(y) \rangle$ for x and y in F then defines a continuous sesquilinear form on F . Since u is a G -morphism, we have $q(\pi(g)x, \pi(g)y) = q(x, y)$ for all g in G and $(x, y) \in F \times F$, hence q is G -invariant. By Cor. 3, there exists $\alpha \in \mathbf{R}_+^*$ such that

$$q(x, y) = \langle u(x)|u(y) \rangle = \alpha \langle x|y \rangle$$

for all $(x, y) \in F \times F$, and hence $\alpha^{-1/2}u$ is isometric.

Let us finally prove assertion c). Since π is irreducible, the image of v is dense in F (Lemma 2 of V, p. 378), hence the adjoint v^* is injective (TVS, V, p. 41, prop. 4). By b), the image H of v^* is a closed subspace of E ; it is therefore a subrepresentation of ϱ , and v^* , induced by passage to the subspaces, is a G -isomorphism from F to H . The restriction w of v to H defines a G -morphism from H to F , which is injective since its kernel is the intersection of H and $\text{Ker}(v) = H^\circ$ (loc. cit.). The map w is therefore an isomorphism (Corollary 2); in particular, v is surjective.

COROLLARY 6. — Let G_1 and G_2 be topological groups. Let ϱ_1 be an irreducible unitary representation of G_1 in a Hilbert space E_1 and ϱ_2 an irreducible unitary representation of G_2 in a Hilbert space E_2 . The external tensor product $\varrho_1 \boxtimes \varrho_2$ is an irreducible unitary representation of $G_1 \times G_2$ in $E_1 \widehat{\otimes}_2 E_2$.

The external tensor product $\varrho = \varrho_1 \otimes \varrho_2$ is a unitary representation of $G = G_1 \times G_2$ in the space $E = E_1 \widehat{\otimes}_2 E_2$ by lemma 8 of V, p. 384.

Let q be a sesquilinear form $(G_1 \times G_2)$ -invariant on E . For every pair $(x_1, y_1) \in E_1^2$, the map $(x_2, y_2) \mapsto q(x_1 \otimes x_2, y_1 \otimes y_2)$ belongs to $\text{Sesq}_{G_2}(\varrho_2, \varrho_2)$. Denote by $b(x_1, y_1)$ the unique complex number such that

$$q(x_1 \otimes x_2, y_1 \otimes y_2) = b(x_1, y_1) \langle x_2 | y_2 \rangle$$

for all $(x_2, y_2) \in E_2^2$ (cor. 3). Let $\varepsilon \in E_2$ be of norm 1, so that $b(x_1, y_1) = q(x_1 \otimes \varepsilon, y_1 \otimes \varepsilon)$ for all $(x_1, y_1) \in E_1^2$. This formula implies that the map b is sesquilinear on E_1 . Moreover

$$\|b(x_1, y_1)\| \leq \|q\| \|x_1 \otimes \varepsilon\| \|y_1 \otimes \varepsilon\| = \|q\| \|x_1\| \|y_1\|$$

for all $(x_1, y_1) \in E_1^2$ (EVT, V, p. 26, formula (5)), hence b is continuous.

Let $g \in G$ and $(x_1, y_1) \in E_1^2$. Since ϱ_2 is unitary and q is invariant, it follows that

$$b(\varrho_1(g)x_1, \varrho_1(g)y_1) = q(x_1 \otimes \varrho_2(g^{-1})\varepsilon, y_1 \otimes \varrho_2(g^{-1})\varepsilon) = b(x_1, y_1),$$

so that $b \in \text{Sesq}_{G_1}(\varrho_1, \varrho_1)$. There therefore exists a unique $\lambda \in \mathbf{C}$ such that $b(x_1, y_1) = \lambda \langle x_1 | y_1 \rangle$ for all $(x_1, y_1) \in E_1^2$ (cor. 3), that is,

$$q(x_1 \otimes x_2, y_1 \otimes y_2) = \lambda \langle x_1 | y_1 \rangle \langle x_2 | y_2 \rangle$$

for all $(x_1, y_1, x_2, y_2) \in E_1^2 \times E_2^2$. From this one deduces that the space $\text{Sesq}_{G_1 \times G_2}(\varrho, \varrho)$ has dimension 1, which implies that the representation $\varrho = \varrho_1 \boxtimes \varrho_2$ is irreducible (loc. cit.).

Recall that U denotes the group of complex numbers of modulus 1.

COROLLARY 7. — Let π be an irreducible unitary representation of a topological group G in a Hilbert space E . There exists a continuous homomorphism χ of the center C of G in U such that $\pi(z) = \chi(z) \cdot 1_E$ for all $z \in C$. In particular, if G is commutative, we have $\dim(E) = 1$.

For $z \in C$, the map $\pi(z)$ belongs to $\text{Hom}_G(\pi, \pi)$; it is therefore of the form $\chi(z) \cdot 1_E$ for some complex number $\chi(z)$ (prop. 6). Since $\pi(z)$ is a unitary map, we have $|\chi(z)| = 1$. Moreover, since π is a homomorphism, the map $z \mapsto \chi(z)$ is a homomorphism. Fix $v \neq 0$ in E ; the map $z \mapsto \chi(z)v = \pi(z)v$ from C to U is continuous, and therefore the homomorphism χ is continuous.

Finally, if G is commutative, then $C = G$, so $\pi(g) = \chi(g) \cdot 1_E$ for all g in G ; every subspace of dimension 1 of E is then a subrepresentation of π , and since π is irreducible, the space E

must be of dimension 1.

DEFINITION 6. — Let π be an irreducible unitary representation of a topological group G in a Hilbert space E . The homomorphism χ of the center C of G into $\mathbb{C}$ such that $\pi(z) = \chi(z) \cdot 1_E$ for all z in C is called the central character of π .

Note. — Let π (resp. ϱ) be an irreducible unitary representation of a topological group G (resp. H) in a Hilbert space E (resp. F). Denote by χ (resp. η) the central character of π (resp. ϱ). The central character of the representation $\overline{\pi}$ is $\overline{\chi}$ and the central character of the irreducible unitary representation $\pi \boxtimes \varrho$ of $G \times H$ (cor. 6) is the character $\chi \boxtimes \eta : (g, h) \mapsto \chi(g)\eta(h)$ of $G \times H$.

V.1.9 Semisimplicity

DEFINITION 7. — Let ϱ be a unitary representation of a topological group G in a Hilbert space E . We say that ϱ is semi-simple if there exists a family $(F_i)_{i \in I}$ of irreducible subrepresentations of ϱ such that E is the Hilbert sum of the subspaces F_i .

One sometimes also says that a semisimple unitary representation admits a discrete decomposition or is discretely decomposable.

If $(\varrho_i)_{i \in I}$ is a family of semisimple unitary representations of a topological group G , then the Hilbert sum of the representations ϱ_i is semisimple.

PROPOSITION 7. — Let G be a topological group and let ϱ be a unitary representation of G in a complex Hilbert space E . The representation ϱ is semi-simple if and only if every nonzero subrepresentation of ϱ contains an irreducible subrepresentation.

Suppose that ϱ is semisimple. Let $(F_i)_{i \in I}$ be a family of irreducible subrepresentations of ϱ such that E is the Hilbert sum of the subspaces F_i . Let F be a nonzero subrepresentation of E . There exists $i \in I$ such that the restriction to F of the orthogonal projector with image F_i is not zero. This orthogonal projector then defines, by passing to subspaces, a nonzero G -morphism p_i from F into F_i . By Schur's lemma, the G -morphism p_i is surjective (cor. 5 of V, p. 388). The orthogonal complement in F of the kernel of p_i is a subrepresentation of F isomorphic to F_i (prop. 3 of V, p. 383), hence irreducible.

Let us prove the converse assertion. Let $\mathcal{F}$ be the set of closed subspaces F of E that are

stable under G such that the subrepresentation of ϱ in F is irreducible. Let $\mathcal{O}$ be the set of subsets $\mathcal{G}$ of $\mathcal{F}$ such that the subspaces belonging to $\mathcal{G}$ are pairwise orthogonal.

The set $\mathcal{O}$ is of finite character (E, III, p. 34, def. 2). Let then $\mathcal{G}$ be a maximal element of $\mathcal{O}$ (E, III, p. 35, th. 1). Let E_1 be the subspace of E that is the Hilbert sum of the irreducible subrepresentations F of $\mathcal{G}$. If one had $E_1 \neq E$, the orthogonal complement of E_1 would not be zero, and the subrepresentation of G in E_1° would by hypothesis contain an irreducible subrepresentation. Its space F would be orthogonal to the elements of $\mathcal{G}$, so that $\mathcal{G} \cup \{F\} \in \mathcal{O}$; this contradicts the maximality of $\mathcal{G}$, so $E = E_1$, which concludes the proof.

COROLLARY 1. — Let G be a topological group and let ϱ be a semisimple unitary representation of G in a complex Hilbert space E . Every subrepresentation of ϱ (resp. every quotient representation of ϱ) is semisimple.

If ϱ_1 is a subrepresentation of ϱ , then every nonzero subrepresentation of ϱ_1 contains an irreducible subrepresentation (prop. 7), hence ϱ_1 is semi-simple (loc. cit.).

By prop. 3 of V, p. 383, every quotient representation of ϱ is isomorphic to a subrepresentation of ϱ ; therefore it is semisimple.

COROLLARY 2. — Every finite-dimensional unitary representation ϱ of a topological group G is semisimple.

This follows from prop. 7 and lemma 3 of V, p. 379.

COROLLARY 3. — Let ϱ_1 and ϱ_2 be finite-dimensional unitary representations of a topological group G . The representations ϱ_1 and ϱ_2 are isomorphic if and only if $\chi_{\varrho_1} = \chi_{\varrho_2}$.

This follows from Corollary 2 and from A, VIII, p. 389, prop. 1, b).

V.1.10 Classes of unitary representations

Lemma 9. — Let E be a Hilbert space over K . Let $F \subset E$ be a dense vector subspace of E . Then the Hilbert dimension of E is less than or equal to the dimension of F .

If E has finite Hilbert dimension, this is equal to the dimension of E . Suppose that E has infinite Hilbert dimension. The subspace F is then of infinite dimension. Let B be an orthonormal basis of E and let B' be a basis of F . For every $x \in B$, there exists an element $f(x) \in B'$ such that $\langle x | f(x) \rangle \neq 0$, since otherwise we would have $x \in F^\circ = \{0\}$. One thus defines a map

$f : B \rightarrow B'$.

For every $y \in B'$, the set $f^{-1}(y)$ is contained in the set of $x \in B$ such that $\langle x|y \rangle \neq 0$ and is therefore countable (EVT, V, p. 21, prop. 4). By E, III, p. 50, prop. 4, one obtains $\text{Card}(B) = \text{Card}(f(B)) \leq \text{Card}(B')$.

Denote by $Is_G(\pi_1, \pi_2)$ the relation

“ G is a topological group and π_1 and π_2 are

unitary representations of G in Hilbert spaces over K that are

isomorphic.” With respect to π_1 and π_2 , this is an equivalence relation. For any unitary representation π of G in a Hilbert space over K , we shall denote by $\text{cl}(\pi)$ the equivalence class of π (cf. E, II, p. 47); it is therefore a unitary representation of G isomorphic to π ; one says that $\text{cl}(\pi)$ is the class of π . Unitary representations π_1 and π_2 in Hilbert spaces over K are isomorphic if and only if $\text{cl}(\pi_1) = \text{cl}(\pi_2)$.

Let G be a topological group. Let $\mathfrak{c}$ be a cardinal. The relation

“ λ is a class of unitary representation of G in

a complex Hilbert space of Hilbert dimension $\leq \mathfrak{c}$ ”

is collectivizing in λ (E, II, p. 3). Indeed, every Hilbert space over K of dimension $\leq \mathfrak{c}$ is isometrically isomorphic to a Hilbert subspace of $\ell^2(\mathfrak{c})$ (EVT, V, p. 23, cor. 2), and the assertion then follows from E, II, p. 47.

Let π be an irreducible unitary representation of G in a Hilbert space E . Let x be a nonzero element of E . Since π is irreducible, the vector x is a cyclic vector of π , which implies that the Hilbert dimension of E is $\leq \text{Card}(G)$ (Lemma 9, applied to the dense subspace generated by the elements $\pi(g)x$ for $g \in G$). The classes of irreducible unitary representations of G in a Hilbert space over K therefore belong to the set of classes of unitary representations of G in a Hilbert space over K of Hilbert dimension $\leq \text{Card}(G)$; consequently, they form a set.

DEFINITION 8. — We denote by $\widehat{G}$ the set of classes of irreducible unitary representations of G in a complex Hilbert space. We say that $\widehat{G}$ is the unitary dual of G .

For every irreducible unitary representation π of G in a complex Hilbert space, we therefore have $\text{cl}(\pi) \in \widehat{G}$.

Remarks. — 1) Suppose that G is a commutative group. By Corollary 7 of V, p. 390, every

irreducible unitary representation of G in a complex Hilbert space has dimension 1. If G is locally compact, the set $\widehat{G}$ is identified with the set of unitary characters of G (def. 1 of II, p. 201), and the notation $\widehat{G}$ is therefore compatible with that introduced in Def. 2 of II, p. 201.

- 2) If G is finite, the set $\widehat{G}$ is in bijection with the set of classes of $\mathbb{C}[G]$ -simple modules (A, VIII, p. 47), which is also denoted $\widehat{G}$ in A, VIII, p. 396.
- 3) For $\pi \in \widehat{G}$, we shall identify $\overline{\pi}$ with the class in $\widehat{G}$ of the conjugate representation of π .
- 4) If π is an irreducible unitary representation of G in a finite-dimensional complex Hilbert space, its character χ_π depends only on the class of π . One can therefore speak of the set of characters of finite-dimensional complex irreducible unitary representations of G .

V.1.11 Isotypic components

DEFINITION 9. — Let G be a topological group and let π be a continuous irreducible representation of G . Let ϱ be a continuous representation of G in a separated locally convex space E . The π -isotypic component of ϱ is called the closure in E of the sum of the spaces of all subrepresentations of ϱ isomorphic to π . This subspace is denoted $M_\pi(\varrho)$.

The space $M_\pi(\varrho)$ is a closed subspace of E , which defines a subrepresentation of ϱ . This space depends only on the class of π in $\widehat{G}$.

PROPOSITION 8. — Let G be a topological group. Let ϱ be a unitary representation of G in a complex Hilbert space E .

- a) Let π_1 and π_2 be non-isomorphic irreducible unitary representations of G . The subspaces $M_{\pi_1}(\varrho)$ and $M_{\pi_2}(\varrho)$ are orthogonal;
- b) Let ϱ' be a unitary representation of G and u a G -morphism from ϱ to ϱ' . For every irreducible unitary representation π of G , we have $u(M_\pi(\varrho)) \subset M_\pi(\varrho')$.

Let E_1 and E_2 be subspaces of E defining subrepresentations isomorphic to π_1 and π_2 respectively. The orthogonal projector of E with image E_2 defines by passing to subspaces an element of $\text{Hom}_G(\pi_1, \pi_2)$, which is zero (cor. 2 of V, p. 387), which proves that E_2 is orthogonal to E_1 . This proves assertion a).

For assertion *b*), note that $M_\pi(\varrho)$ is, by definition, the closure in E of the space generated by the elements $x \in E$ belonging to a closed subspace $F \subset E$ stable under ϱ such that the subrepresentation ϱ_F of ϱ in F is isomorphic to π . It therefore suffices to prove that, in this case, one has $u(x) \in M_\pi(\varrho')$. Let $H = u(F)$; it is a closed subspace of the space of ϱ' (cor. 5 of V, p. 388) stable under ϱ' , and u , induced by passage to the subspaces, is a surjective G -morphism from F to H . If H is not zero, then this G -morphism is an isomorphism by Corollary 2 of V, p. 387. The representation of G on H is therefore isomorphic to π , whence it follows that $u(x)$ belongs to $M_\pi(\varrho')$.

For every vector space H and every family $(H_i)_{i \in I}$ of subspaces of H , we say that the spaces $(H_i)_{i \in I}$ are in direct sum if the family $(H_i)_{i \in I}$ satisfies the equivalent conditions of Prop. 11 of A, II, p. 18.

PROPOSITION 9. — Let G be a topological group and let ϱ be a unitary representation of G in a complex Hilbert space E . Let π be an irreducible unitary representation of G in a complex Hilbert space E_π . Denote by v the linear map from $\text{Hom}_G(\pi, \varrho) \otimes E_\pi$ into E such that $v(u \otimes x) = u(x)$ for all $(u, x) \in \text{Hom}_G(\pi, \varrho) \times E_\pi$.

The linear map v is injective and its image is the sum of the spaces of all subrepresentations of ϱ which are isomorphic to π . In particular, the image of v is dense in $M_\pi(\varrho)$.

Corollary 5 of V, p. 388 implies that the image of v is the sum of the spaces of all subrepresentations of ϱ which are isomorphic to π .

Let us prove that v is injective. Let $(u_i)_{i \in I}$ be a basis of the vector space $\text{Hom}_G(\pi, \varrho)$. For $i \in I$, denote by F_i the image of u_i and $\tilde{u}_i: E_\pi \rightarrow F_i$ the G -isomorphism deduced from u_i by passage to the subspaces.

Let us first prove, by induction on the cardinality of a finite subset J of I , that the subspaces $(F_i)_{i \in J}$ are in direct sum.

The case where J is empty is immediate. Suppose that J is nonempty and that the required property holds for subsets of I of cardinality at most $\text{Card}(J) - 1$.

Let j be a fixed element of J . The induction hypothesis implies that the subspaces F_i for $i \in J - \{j\}$ are in direct sum. Let F' be their sum; it now suffices to prove that $F_j \cap F' = \{0\}$.

Suppose that the intersection of F_j and of F' is not reduced to 0; this intersection is a sub-

representation of F_j , and since the latter is irreducible, we therefore have $F_j \cap F' = F_j$, whence $F_j \subset F'$.

For $i \in J - \{j\}$, denote by $\text{pr}_i : F' \rightarrow F_i$ the projection and by $\iota_i : F_i \rightarrow F'$ the inclusion. We have a canonical isomorphism

$$\text{Hom}_G(F_j, F') \rightarrow \bigoplus_{i \in J - \{j\}} \text{Hom}_G(F_j, F_i)$$

(formula (1), p. 377) such that $u : F_j \rightarrow F'$ has as image the family $(\text{pr}_i \circ u)_{i \in J - \{j\}}$ (A, II, p. 13, cor. 1). Corollary 2 of V, p. 387 implies that $\tilde{u}_i \circ \tilde{u}_j^{-1}$, which is nonzero, is a basis of the space $\text{Hom}_G(F_j, F_i)$.

Therefore, the family of G-morphisms $(\iota_i \circ \tilde{u}_i \circ \tilde{u}_j^{-1})_{i \in J - \{j\}}$ is a basis of $\text{Hom}_G(F_j, F')$.

Denote by ι the inclusion of F_j in F' ; it is an element of $\text{Hom}_G(F_j, F')$, hence it is a linear combination of the G-morphisms $\iota_i \circ \tilde{u}_i \circ \tilde{u}_j^{-1}$ for $i \in J - \{j\}$. It follows that u_j is a linear combination of the maps u_i for $i \neq j$, which contradicts the linear independence of the family $(u_i)_{i \in I}$. The assertion is therefore proved by induction.

Let us now prove that v is injective. Let w be an element of $\text{Hom}_G(\pi, \varrho) \otimes E_\pi$. There exists a unique family $(x_i)_{i \in I}$ in E_π with finite support such that

$$w = \sum_{i \in I} u_i \otimes x_i$$

(A, II, p. 62, cor. 1) and we then have

$$v(w) = \sum_{i \in I} u_i(x_i).$$

From what precedes, the condition $v(w) = 0$ therefore implies that $u_i(x_i) = 0$ for all $i \in I$, whence $x_i = 0$ for all i since u_i is injective, and therefore $w = 0$.

PROPOSITION 10. — Let G be a topological group. Let π be an irreducible unitary representation of G in a complex Hilbert space E_π and let ϱ be a unitary representation of G in a complex Hilbert space E .

There exist families $(E_i)_{i \in I}$ of closed invariant subspaces of E such that the subrepresentation of ϱ in E_i is isomorphic to π for all $i \in I$ and such that $M_\pi(\varrho)$ is the Hilbert sum of the spaces E_i

. Moreover, the cardinality of I is independent of the family $(E_i)_{i \in I}$ satisfying these properties.

Let us prove the existence of families satisfying the stated conditions. Let $\mathcal{O}$ be the set of subsets C of $\text{Hom}_G(\pi, \varrho) \setminus \{0\}$ such that the subspaces $u(E_\pi)$ for $u \in C$ are pairwise orthogonal. The set $\mathcal{O}$ is ordered by inclusion. It is nonempty because the empty set is a member, and it is of finite character (E, III, p. 34, def. 2) since C belongs to $\mathcal{O}$ if and only if every two-element subset of C belongs to $\mathcal{O}$. By E, III, p. 35, th. 1, there exists a maximal element C of $\mathcal{O}$. Let F be the closure of the subspace generated by the spaces $u(E_\pi)$ for $u \in C$; it is the Hilbert sum of the spaces $u(E_\pi)$ for $u \in C$. We shall prove that $F = M_\pi(\varrho)$, which will prove that the family $(u(E_\pi))_{u \in C}$ has the required properties.

By definition, $u(E_\pi) \subset M_\pi(\varrho)$ for all $u \in C$, hence F is contained in $M_\pi(\varrho)$. To prove the reverse inclusion, it suffices to prove that if v is a G -morphism from π to ϱ , then its image is contained in F . Let p be the orthogonal projector of E with image F° ; it is a G -morphism, since F is a subrepresentation of E (prop. 4 of V, p. 383). The image of the G -morphism $p \circ v$ is orthogonal to F ; it is therefore zero (otherwise $C \cup \{p \circ v\} \in \mathcal{O}$, which contradicts the maximality of C), and consequently the image of v is contained in F .

Now let $(E_i)_{i \in I}$ and $(F_j)_{j \in J}$ be families of pairwise orthogonal closed invariant subspaces of E , such that the subrepresentation of G in E_i (resp. in F_j) is isomorphic to π for all $i \in I$ (resp. for every $j \in J$), and such that $M_\pi(\varrho)$ is the Hilbert sum of the family $(E_i)_{i \in I}$ and of the family $(F_j)_{j \in J}$.

If I is finite, then

$$\dim \text{Hom}_G(\pi, M_\pi(\varrho)) = \dim \text{Hom}_G\left(E_\pi, \bigoplus_{i \in I} E_i\right) = \text{Card}(I)$$

(formula (1) of V, p. 377 and cor. 2 of V, p. 387). For every finite subset L of J , we then have

$$\begin{aligned} \text{Card}(L) &= \dim \text{Hom}_G\left(E_\pi, \bigoplus_{j \in L} F_j\right) \\ &\leq \dim \text{Hom}_G(\pi, M_\pi(\varrho)) = \text{Card}(I) \end{aligned}$$

(loc. cit.). This proves that J is then finite and that $\text{Card}(I) = \text{Card}(J)$, as desired. Likewise, if J is finite, then I is finite and has the same cardinal.

Now suppose that I and J are infinite. For $j \in J$, denote by p_j the orthogonal projector of E with image F_j . For $i \in I$, let x_i be a nonzero element of E_i ; it is a cyclic vector of E_i . Observe that since p_j , induced by passage to the subspaces, is a G -morphism from E_i to F_j , the space $p_j(E_i)$ is zero if and only if $p_j(x_i) = 0$ (indeed, the space $E_i \cap \text{Ker}(p_j)$ is either zero or equal to E_i).

For every $j \in J$, there exists an element $f(j) \in I$ such that $p_j(E_{f(j)})$ is not reduced to 0 (otherwise, the orthogonal projector p_j would be zero on $M_\pi(\varrho)$). We have thus defined a map f from J to I . For every $i \in I$, the set $f^{-1}(i)$ is countable. Indeed, this set is contained in the set of $j \in J$ such that $p_j(E_i)$ is nonzero, that is, those such that $p_j(x_i) \neq 0$. Now, by (EVT, V, p. 20, cor. 2), we have

$$\sum_{j \in J} \|p_j(x_i)\|^2 = \|x_i\|^2,$$

hence the set of $j \in J$ such that $p_j(x_i) \neq 0$ is countable. By E, III, p. 50, prop. 4, we conclude that $\text{Card}(J) = \text{Card}(f(J)) \leq \text{Card}(I)$. By interchanging the roles of I and J , we conclude that $\text{Card}(I) = \text{Card}(J)$.

COROLLARY. — Let G be a topological group and ϱ be a unitary representation of G in a complex Hilbert space E . The Hilbert sum of the π -isotypic components of G for $\pi \in \widehat{G}$ is the largest semisimple subrepresentation of ϱ .

Indeed, the π -isotypic components of G for $\pi \in \widehat{G}$ are pairwise orthogonal (prop. 8, a), and each is semisimple (prop. 10, a), hence the Hilbert sum F of the spaces $M_\pi(\varrho)$ for $\pi \in \widehat{G}$ defines a semisimple subrepresentation of ϱ . Since moreover every irreducible subrepresentation of ϱ is a subrepresentation of an isotypic component of ϱ , the corollary follows.

DEFINITION 10. — Let G be a topological group. Let ϱ be a unitary representation of G in a complex Hilbert space E and π be an irreducible unitary representation of G in a complex Hilbert space.

One calls the multiplicity of π in ϱ the cardinal of the set I for any family $(E_i)_{i \in I}$ of closed subspaces of E stable under G such that the subrepresentation of ϱ induced in E_i is isomorphic to π for every $i \in I$ and such that $M_\pi(\varrho)$ is the Hilbert sum of the subspaces E_i .

If the multiplicity of π in ϱ is finite, then it is equal to the dimension of the space $\text{Hom}_G(\pi, \varrho)$

(resp. to the dimension of $\text{Hom}_G(\varrho, \pi)$) by formula (1) of V, p. 377 and corollary 2 of V, p. 387. This is not always the case in general.

Remark. — It is possible that a unitary representation ϱ is nonzero but that all isotypic components of ϱ relative to all irreducible representations of G are zero (cf. V, p. 426, remark).

V.2 REPRESENTATIONS OF LOCALLY COMPACT GROUPS

In this paragraph, vector spaces are over the field $K = \mathbb{R}$ or $\mathbb{C}$. We denote by G a locally compact group equipped with a left Haar measure μ . We denote $\mathcal{L}^p(G) = \mathcal{L}_C^p(G, \mu)$ and $L^p(G) = L_C^p(G, \mu)$ for all $p \in [1, +\infty]$.

V.2.1 Continuity of certain representations

PROPOSITION 1. — Let H be a locally compact group. Let ϱ_1 (resp. ϱ_2) be a continuous representation of G (resp. of H) in a locally convex separated K -vector space E_1 (resp. E_2). Let F denote the space $\mathcal{L}(E_1; E_2)$ endowed with the topology of compact convergence. The representation ϱ of $G \times H$ into F defined by $\varrho(g, h)u = \varrho_2(h) \circ u \circ \varrho_1(g^{-1})$ for $(g, h) \in G \times H$ is continuous.

Denote by $\mathcal{L}_c(E_1; E_2)$ the space $\mathcal{L}(E_1; E_2)$ equipped with the topology of compact convergence. Let $\mathfrak{F}_1$ (resp. $\mathfrak{F}_2, \mathfrak{F}_3$) be a filter in G converging to e (resp. a filter in $\mathcal{L}_c(E_1; E_2)$ converging to 0, a filter in H converging to e). Since G and H are locally compact, there exist elements $C \in \mathfrak{F}_1$ and $D \in \mathfrak{F}_3$ which are relatively compact. The set $\varrho_1(C^{-1})$ is equicontinuous in $\mathcal{L}(E_1)$ (cf. INT, VIII, p. 129, § 2, no. 1, remark 2, a'). By prop. 9 of EVT, III, p. 33 and prop. 4 of EVT, III, p. 31, the map defined by $(u, v) \mapsto u \circ v$ of $\mathcal{L}_c(E_1; E_2) \times \varrho_1(C^{-1})$ in $\mathcal{L}_c(E_1; E_2)$ is continuous. The filter base $\mathfrak{F}_2 \circ \varrho_1(\mathfrak{F}_1^{-1})$ therefore converges to 0 in $\mathcal{L}_c(E_1; E_2)$. The set $\varrho_2(D)$ is equicontinuous in $\mathcal{L}(E_2)$ (cf. INT, VIII, p. 129, § 2, n° 1, rem. 2), and for every $x \in E_2$, the set $\varrho_2(D)x \subset E_2$ is relatively compact. Consequently, $\varrho_2(D)$ is relatively compact in $\mathcal{L}(E_2)$ endowed with the topology of compact convergence (TG, X, p. 18, cor. 1). The map defined by $(u, v) \mapsto u \circ v$ of $\overline{\varrho_2(D)} \times \mathcal{L}_c(E_1; E_2)$ in $\mathcal{L}_c(E_1; E_2)$ is continuous by prop.

9 of EVT, III, p. 33 and prop. 4 of EVT, III, p. 31. Hence the filter base $\varrho(\mathfrak{F}_1 \times \mathfrak{F}_3)(\mathfrak{F}_2)$ converges to 0 in $\mathcal{L}_c(E_1; E_2)$. This implies the assertion.

COROLLARY. — Let ϱ be a continuous representation of G on a separated locally convex K -vector space E . The contragredient representation $\check{\varrho}$ is continuous when E' is endowed with the topology of compact convergence.

This follows from the proposition by taking $H = \{e\}$, $\varrho_1 = \varrho$ and ϱ_2 the trivial representation on K .

Remark. — The contragredient representation is not necessarily continuous when E' is endowed with the strong topology (cf. INT, VIII, p. 191, § 2, exercise 3, d)). One can show that it is continuous if the space E is semi-reflexive (loc. cit., c)).

V.2.2 Extension of representations to spaces of measures

In this section, we assume that $K = \mathbb{C}$.

Let ϱ be a continuous linear representation of G in a separated locally convex quasi-complete space E . Let us denote by $\mathcal{M}_c(G)$ the space of measures with compact support on G equipped with the topology of compact convergence; it is the dual of the space $\mathcal{C}(G)$. For every measure $\nu \in \mathcal{M}_c(G)$ we set

$$\varrho(\nu) = \int_G \varrho(g) d\nu(g).$$

This is an element of $\mathcal{L}(E)$. In particular, we have $\varrho(\varepsilon_g) = \varrho(g)$ for all $g \in G$.

According to INT, VIII, p. 136, § 2, no. 6, the map $\nu \mapsto \varrho(\nu)$ is a continuous linear map from $\mathcal{M}_c(G)$ to the space $\mathcal{L}(E)$ equipped with the topology of compact convergence. According to INT, VIII, p. 145, § 3, no. 3, prop. 11, it is a morphism of unital algebras.

Let $x \in E$. The map from $\mathcal{M}_c(G)$ to E defined by $\nu \mapsto \varrho(\nu)x$ is continuous when $\mathcal{M}_c(G)$ is equipped with the topology of compact convergence (INT, VI, p. 27, § 1, no. 7, prop. 16 and EVT, III, p. 31, prop. 4).

For $f \in \mathcal{L}^1(G)$ vanishing outside a compact subset of G , the measure $f \cdot \mu$ has compact support and we shall denote by $\varrho^\mu(f)$, or $\varrho(f)$ when no ambiguity is possible, the endomorphism $\varrho(f \cdot \mu)$ of E . This endomorphism depends only on the class $\tilde{f}$ of f in $L^1(G)$, and we shall also

denote it by $\varrho^\mu(\tilde{f})$ or $\varrho(\tilde{f})$.

Similarly, let ϱ be a continuous bounded linear representation of G in a Banach space E . For every bounded measure $\nu \in \mathcal{M}^1(G)$ we set

$$\varrho(\nu) = \int_G \varrho(g) d\nu(g).$$

According to INT, VIII, loc. cit., the map $\nu \mapsto \varrho(\nu)$ is a continuous unital morphism of Banach algebras from the algebra $\mathcal{M}^1(G)$ of bounded complex measures on G into the Banach algebra $\mathcal{L}(E)$.

Let ρ be the function $g \mapsto \|\varrho(g)\|$ on G ; the algebra denoted $\mathcal{M}^\rho$ in INT, VIII, p. 145, prop. 11 (whose elements are the measures ν such that $\rho \in \mathcal{L}^1(\nu)$) coincides with the Banach algebra $\mathcal{M}^1(G)$. Indeed, set $M = \sup \rho$. We have $M > 0$ since $\rho(e) = 1$, and $M^{-1} \leq \rho \leq M$ since $\|\varrho(e)\| \leq \|\varrho(g)\| \|\varrho(g^{-1})\|$ for all $g \in G$; thus $\rho \in \mathcal{L}^1(\nu)$ if and only if ν is a bounded measure.

If $f \in \mathcal{L}^1(G)$, one also denotes $\varrho^\mu(f)$ or simply $\varrho(f)$ the endomorphism $\varrho(f \cdot \mu)$, and likewise for the class $\tilde{f}$ of f in $L^1(G)$.

Lemma 1. — Let ϱ be a unitary representation of G in a Hilbert space E . The map $\nu \mapsto \varrho(\nu)$ of $\mathcal{M}^1(G)$ into $\mathcal{L}(E)$ is a unital morphism of involutive Banach algebras.

By what precedes, it suffices to show that the morphism $\nu \mapsto \varrho(\nu)$ is involutive. Let $\nu \in \mathcal{M}^1(G)$. The measure ν^* is the conjugate measure of the measure $\tilde{\nu}$ (I, p. 99, example 4). According to the definition of the conjugate measure (INT, III, p. 52, § 1, no. 5), one computes

$$\begin{aligned} \langle x | \varrho(\nu)y \rangle &= \int_G \langle x | \varrho(g)y \rangle d\nu(g) \\ &= \int_G \langle \varrho(g^{-1})x | y \rangle d\nu(g) = \int_G \langle \varrho(g)x | y \rangle d\tilde{\nu}(g) \\ &= \overline{\int_G \langle y | \varrho(g)x \rangle d\nu^*(g)} = \langle \varrho(\nu^*)x | y \rangle \end{aligned}$$

for all x and y in E , whence $\varrho(\nu)^* = \varrho(\nu^*)$.

Let ϱ be a continuous linear representation (resp. continuous and bounded) of G in a locally convex quasi-complete space E (resp. a Banach space E). If F is a closed subspace of E defining a subrepresentation ϱ_F of ϱ , then for every measure $\nu \in \mathcal{M}_c(G)$ (resp. $\nu \in \mathcal{M}^1(G)$) the linear map $\varrho(\nu)$ leaves the subspace F stable and coincides with $\varrho_F(\nu)$ on F .

Conversely, let $F \subset E$ be a closed subspace, stable under the linear maps $\varrho(f)$ for every function $f \in \mathcal{B}(G)$. Then F defines a subrepresentation of ϱ (INT, VIII, p. 139, § 2, no. 7, prop. 10).

Recall (A, VIII, p. 388) that a function $f: G \rightarrow \mathbb{C}$ is said to be central if, for all g and h in G , we have $f(gh) = f(hg)$. This amounts to saying that f is invariant under conjugation.

DEFINITION 1. — A bounded measure $\nu \in \mathcal{M}^1(G)$ is said to be central if one has $\varepsilon_g * \nu = \nu * \varepsilon_g$ for all $g \in G$.

If G is unimodular and $f \in \mathcal{C}(G)$ is μ -integrable, the measure $f \cdot \mu$ is central if and only if f is central.

Let ϱ be a continuous bounded representation of G in a Banach space E (resp. a continuous representation of G in a quasi-complete separated locally convex space E). For every bounded central measure ν on G (resp. any compactly supported central measure ν on G), the linear map $\varrho(\nu)$ is a G -morphism from ϱ to ϱ . Indeed, for every $g \in G$, we have

$$\varrho(\nu)\varrho(g) = \varrho(\nu * \varepsilon_g) = \varrho(\varepsilon_g * \nu) = \varrho(g)\varrho(\nu).$$

V.2.3 Criterion for semisimplicity

PROPOSITION 2. — Let ϱ be a unitary representation of G in a complex Hilbert space E . Let $\mathfrak{F}$ be a filter base on $\mathcal{M}_c(G)$ converging to the measure ε_e for the topology of compact convergence. Suppose that there exists $M \in \mathfrak{F}$ such that $\varrho(\nu)$ is a compact endomorphism of E for every $\nu \in M$.

The unitary representation ϱ is then semisimple and every irreducible unitary representation of G has finite multiplicity in ϱ .

Let us first prove a lemma.

Lemma 2. — Let ϱ_1 be a nonzero unitary representation of G isomorphic to a subrepresentation of ϱ . There exists a measure $\nu \in M$ such that $\varrho_1(\tilde{\nu} * \nu)$ is a nonzero compact Hermitian endomorphism of the space of ϱ_1 . In particular, there exists a nonzero real number λ such that the eigenspace of $\varrho_1(\tilde{\nu} * \nu)$ relative to λ is nonzero.

We may assume that ϱ_1 is a subrepresentation of ϱ . Let E_1 be its space. For every measure

$\nu \in M$, the endomorphism $\varrho_1(\nu)$ is deduced from $\varrho(\nu)$ by passing to subspaces. The hypothesis and prop. 3 of III, p. 5 therefore imply that $\varrho_1(\nu)$ is compact.

Since $\mathfrak{F}$ converges to the measure ε_e in $\mathcal{M}_c(G)$ endowed with the topology of compact convergence and the space E_1 is nonzero, there exists $\nu \in M$ such that $\nu = \varrho_1(\nu)$ is a nonzero endomorphism of E_1 (cf. n°2 of V, p. 400). The endomorphism $u = \varrho_1(\tilde{\nu} * \nu)$ is equal to $\nu^* \circ \nu$ (lemma 1 of V, p. 401); it is therefore nonzero (EVT, V, p. 39, prop. 2), Hermitian and compact, since ν is compact.

Since u is Hermitian and nonzero, its spectrum is not reduced to zero (example 1 of I, p. 110). Finally, since u is compact, every $\lambda \in \text{Sp}(u) \setminus \{0\}$ is an eigenvalue of u (prop. 2 of III, p. 83).

Let us now prove the proposition.

We will use prop. 7 of V, p. 391 to establish that ϱ is semisimple. Let ϱ_1 be a nonzero subrepresentation of ϱ and $E_1 \subset E$ its space; we must show that ϱ_1 contains an irreducible subrepresentation.

Let $\nu \in M$ such that $u = \varrho_1(\tilde{\nu} * \nu)$ is a nonzero compact Hermitian endomorphism of the space of ϱ_1 and let λ be nonzero such that the eigenspace F of u relative to λ is nonzero (lemma 2). The space F is finite-dimensional (prop. 5 of III, p. 90). There exists a subrepresentation ϱ_2 of ϱ_1 on a subspace E_2 of E_1 such that $E_2 \cap F$ is nonzero and of minimal dimension. Let x be a nonzero element of $E_2 \cap F$ and let E_3 be the subrepresentation of ϱ_1 generated by x . We then have $E_3 \subset E_2$, whence $E_3 \cap F = E_2 \cap F$ by minimality of the dimension of $E_2 \cap F$.

Let us show that the representation E_3 is irreducible. Let $E_4 \subset E_3$ be a closed subspace stable under G . We have $E_2 \cap F = (E_4 \cap F) \oplus (E_4^\circ \cap F)$, where E_4° denotes the orthogonal complement of E_4 in E_3 (indeed, if $y \in E_2 \cap F$, let $y_4 \in E_4$ be its orthogonal projection onto E_4 ; since E_4 and E_4° are stable under u , the vector $u(y_4)$ is the orthogonal projection of $u(y) = \lambda y$ onto E_4 , whence $u(y_4) = \lambda y_4$, which means that $y_4 \in E_4 \cap F$, and the assertion follows). The minimality of the dimension of $E_2 \cap F$ then implies that either $E_4 \cap F = E_2 \cap F$, or $E_4^\circ \cap F = E_2 \cap F$. In the first case, we have $x \in E_4$, whence $E_4 = E_3$, and in the second, we have $x \in E_4^\circ$, whence $E_4^\circ = E_3$ and $E_4 = \{0\}$. The representation E_3 is therefore irreducible.

We conclude by prop. 7 of V, p. 391 that the representation ϱ is semisimple.

Let π be an irreducible unitary representation of G whose multiplicity in ϱ is nonzero; let E_π be its space. There exists a measure $\nu \in M$ and a nonzero real number λ such that $u = \pi(\check{\nu} * \nu)$ is a nonzero compact Hermitian endomorphism of E_π and the eigenspace of u relative to λ is nonzero (lemma 2). Let F be the eigenspace of $\nu = \varrho(\check{\nu} * \nu)$ relative to λ . For every subrepresentation E_1 of E isomorphic to π , the endomorphism of E_1 deduced from ν by passing to subspaces is identified with u , and the intersection of E_1 and F is nonzero. Thus, the multiplicity of π in ϱ is less than the dimension of the space F , which is finite (prop. 5 of III, p. 90).

Example. — Suppose that G is unimodular, for example that G is a real semisimple Lie group. Let $\Gamma \subset G$ be a discrete subgroup such that the quotient $X = \Gamma \backslash G$ is compact. The group G acts on X on the right by multiplication. We denote by β the counting measure on Γ and we set $\tilde{\mu} = \mu / \beta$ (INT, VII, p. 44, § 2, n° 2, def. 1); this is a bounded G -invariant measure on X .

For every $f \in \mathcal{L}^2(X, \tilde{\mu})$ and every $g \in G$, we define the function $\varrho(g)f \in \mathcal{L}^2(X, \tilde{\mu})$ by $\varrho(g)f(x) = f(x \cdot g)$. The map $\varrho(g)$ is a continuous linear map, which induces by passing to quotients a unitary map on $L^2(X, \tilde{\mu})$, still denoted $\varrho(g)$. The map ϱ is a unitary representation of G in $L^2(X, \tilde{\mu})$ (INT, VII, p. 135, § 2, n° 5, prop. 8).

Let $\varphi \in \mathcal{K}(G)$. For all compact subsets L_1 and L_2 of G , the intersection T of Γ and of the compact set $L_1 \text{Supp}(\varphi)L_2^{-1}$ is finite. For every $(g, h) \in L_1 \times L_2$, the series $\sum_{\gamma \in \Gamma} \varphi(g^{-1}\gamma h)$ coincides with the finite sum $\sum_{\gamma \in T} \varphi(g^{-1}\gamma h)$. Since G is locally compact, the sum of this series, denoted $k_\varphi(g, h)$, is a continuous function on $G \times G$.

Let $(g, h) \in G \times G$ and let $(\gamma, \eta) \in \Gamma \times \Gamma$. We have $k_\varphi(\gamma g, \eta h) = k_\varphi(g, h)$, hence k_φ defines by passing to the quotient a continuous function on $X \times X$, which we denote $\tilde{k}_\varphi$. We also have $\tilde{k}_\varphi \in \mathcal{L}^2(X \times X, \tilde{\mu} \otimes \tilde{\mu})$, since the space X is assumed compact.

We denote by $\dot{x}$ the image in X of an element x of G under the canonical projection. Let $\varphi \in \mathcal{K}(G)$. For $f \in \mathcal{L}^2(X, \tilde{\mu})$ and $x \in G$, we have

$$\begin{aligned} \varrho(\varphi)f(\dot{x}) &= \int_G \varphi(g)(\varrho(g)f)(\dot{x})d\mu(g) = \int_G \varphi(g)f(\dot{x} \cdot g)d\mu(g) \\ &= \int_G \varphi(x^{-1}y)f(\dot{y})d\mu(y) = \int_{\Gamma \backslash G} \left(\sum_{\gamma \in \Gamma} \varphi(x^{-1}\gamma y) \right) f(\dot{y})d\tilde{\mu}(\dot{y}) \end{aligned}$$

(INT, VII, p. 46, § 2, no. 3, prop. 5). Since $\tilde{k}_\varphi$ belongs to the space $\mathcal{L}^2(X \times X, \tilde{\mu} \otimes$

$\tilde{\mu}$), it follows that the endomorphism $\varrho(\varphi)$ of $L^2(X, \tilde{\mu})$ coincides with the Hilbert-Schmidt endomorphism with kernel $\tilde{k}_\varphi$; this endomorphism is therefore compact (cor. 1 of III, p. 33).

There exists a sequence $(\varphi_n)_{n \in \mathbb{N}}$ of functions in $\mathcal{H}_+(G)$ with integral 1 such that $\varphi_n \cdot \mu$ converges to ε_e in $\mathcal{M}_c(G)$ endowed with the topology of compact convergence (INT, VIII, p. 139, § 2, no. 7, cor. 2). Prop. 2 consequently implies that the representation ϱ is semisimple and that the multiplicities of the irreducible unitary representations of G in ϱ are finite.

The irreducible unitary representations of G whose multiplicity in ϱ is nonzero are called Γ -automorphic representations of the group G .

V.2.4 Regular representations

Let p be a real number ≥ 1 and μ' a right Haar measure on G . For every $g \in G$ and every function $f \in \mathcal{L}^p(G, \mu)$, we denote by $\gamma_G^{(p)}(g)f$ the function $x \mapsto f(g^{-1}x)$ on G . The map $g \mapsto \gamma_G^{(p)}(g)$ is a continuous linear representation of G in $\mathcal{L}^p(G)$. By passing to quotients, it induces a continuous isometric linear representation of G in $L^p(G)$ (INT, VIII, p. 135, § 2, no. 5, prop. 8), denoted $\gamma_G^{(p)}$.

Similarly, for every $g \in G$ and every function $f \in \mathcal{L}^p(G, \mu')$, we denote by $\delta_G^{(p)}(g)f$ the function $x \mapsto f(xg)$ on G . The map $g \mapsto \delta_G^{(p)}(g)$ is a continuous linear representation of G in $\mathcal{L}^p(G, \mu')$. By passing to quotients, it induces a continuous isometric linear representation of G in $L^p(G, \mu')$ (cf. INT, VIII, p. 136, § 2, no. 5).

We say that $\gamma_G^{(p)}$ is the left regular representation of G in $\mathcal{L}^p(G)$ or $L^p(G)$ and that $\delta_G^{(p)}$ is the right regular representation of G in $\mathcal{L}^p(G, \mu')$ or $L^p(G, \mu')$.

Lemma 3. — Let p be a real number ≥ 1 . The left (resp. right) regular representation of G in $L^p(G, \mu)$ (resp. in $L^p(G, \mu')$) is faithful.

More precisely, let q be the conjugate exponent of p and let g be an element of G such that $g \neq e$.

a) There exists a function $\varphi \in \mathcal{H}(G)$, positive and nonzero in $L^q(G, \mu)$, such that

$$\langle \varphi, \gamma_G^{(p)}(g)\varphi \rangle = 0;$$

b) There exists a function $\varphi \in \mathcal{H}(G)$, positive and nonzero in $L^q(G, \mu')$, such that

$$\langle \varphi, \delta_G^{(p)}(g)\varphi \rangle = 0.$$

Consider the case of the left regular representation $\gamma_G^{(p)}$, that of the right regular representation being similar. The assertion *a*) implies that $\gamma_G^{(p)}$ is faithful, for if $g \neq e$ is an element of G , and if φ is as in *a*), one has $\varphi \neq \gamma_G^{(p)}(g)\varphi$ since $\langle \varphi, \varphi \rangle = \int_G \varphi^2 > 0$.

Let us therefore prove *a*). Let $g \neq e$ in G . Let C be a compact symmetric neighborhood of e such that $g \notin C^2$ and let $\varphi \in \mathcal{K}(G)$ be a continuous positive function with integral 1 and support contained in C ; the function φ is nonzero in $L^p(G, \mu)$. Since $C \cap g^{-1}C = \emptyset$, we have

$$\langle \varphi, \gamma_G^{(p)}(g)\varphi \rangle = \int_G \varphi(x)\varphi(g^{-1}x)d\mu(x) = 0.$$

Let ϱ be a continuous and bounded representation of G in a Banach space E . For every $f \in L^1(G)$ (resp. $f' \in L^1(G, \mu')$) and every $g \in G$, we have

$$\varrho(g)\varrho(f \cdot \mu) = \varrho(\varepsilon_g * (f \cdot \mu)) = \varrho(\gamma_G^{(1)}(g)f \cdot \mu), \quad (1)$$

$$\varrho(f' \cdot \mu')\varrho(g) = \varrho((f' \cdot \mu') * \varepsilon_g) = \varrho(\delta_G^{(1)}(g^{-1})f' \cdot \mu'), \quad (2)$$

(INT, VIII, p. 144, § 3, no. 2, formula (5)).

Suppose G is unimodular, and set $\mu' = \mu$. The biregular representation of G in $\mathcal{L}^p(G)$ (resp. in $L^p(G)$) is the representation $\varrho_G^{(p)}$ of $G \times G$ in $\mathcal{L}^p(G)$ (resp. in $L^p(G)$) such that

$$\varrho_G^{(p)}(g_1, g_2) = \gamma_G^{(p)}(g_1)\delta_G^{(p)}(g_2) = \delta_G^{(p)}(g_2)\gamma_G^{(p)}(g_1).$$

It is a continuous linear representation (lemma 1 of V, p. 377). It satisfies

$$(\varrho_G^{(p)}(g_1, g_2)f)(x) = f(g_1^{-1}xg_2)$$

for all $f \in \mathcal{L}^p(G)$, all $(g_1, g_2) \in G \times G$ and every $x \in G$.

Note. — The biregular representation of G in $L^p(G, \mu)$ is not necessarily faithful; its kernel is the image of the center of G under the map $g \mapsto (g, g)$ (exercise 4 of V, p. 487).

When $p = 2$ the left regular representation $\gamma_G^{(2)}$ of G in the complex Hilbert space $L^2(G, \mu)$

is unitary, since it is isometric. Likewise, the right regular representation $\delta_G^{(2)}$ in $L^2(G, \mu')$ is unitary.

We will simply denote $\gamma_G = \gamma_G^{(2)}$ and $\delta_G = \delta_G^{(2)}$ and these representations will be called the left and right regular representations of G .

If G is unimodular, the biregular representation $\varrho_G^{(2)}$ of $G \times G$ in $L^2(G, \mu)$ is unitary. We will simply denote it by ϱ_G .

Lemma 4. — Let p be a real number ≥ 1 . For $f \in \mathcal{L}^1(G)$, the linear map $\gamma_G^{(p)}(f)$ coincides with the endomorphism of $L^p(G)$ defined by $\varphi \mapsto f *^\mu \varphi$.

This follows from INT, VIII, p. 157, § 4, n°2, prop. 6, taking into account formula (14) of INT, VIII, p. 165.

Note. — Recall that, for any function f on G , one defines the function $\check{f}$ on G by $\check{f}(g) = f(g^{-1})$ for all $g \in G$ (INT, VII, p. 12, § 1, n° 1, formula (12)).

One checks that if $f \in \mathcal{L}^1(G)$, the linear map $u = \delta_G^{(p)}(f)$ satisfies the relation

$$\widetilde{u(\varphi)} = f * \check{\varphi}$$

for all $\varphi \in L^p(G, \mu')$.

V.2.5 Equivariant functions

In this subsection, we fix a closed subgroup H of G . We denote by ϖ the canonical projection of G onto G/H .

In addition to the Haar measure μ on G , we fix a left Haar measure β on H .

By INT, VII, p. 56, § 2, no. 5, th. 2, there exists a continuous function $\kappa : \mathbf{G} \rightarrow \mathbf{R}_+^*$ such that $\kappa(xh) = \Delta_H(h)\Delta_G(h)^{-1}\kappa(x)$ for every $(x, h) \in \mathbf{G} \times \mathbf{H}$. We fix such a function κ . We denote ν the measure $(\kappa \cdot \mu)/\beta$ on G/H ; by loc. cit., it is a non-zero positive measure quasi-invariant under G . Its support is equal to G/H (INT,

VII, p. 10, § 1, no. 1). By INT, VII, p. 43, § 2, no. 2, prop. 4, the measure ν is the unique measure on G/H such that the measure $\nu^\#$ on G is equal to $\kappa \cdot \mu$. We equip the space G/H with the measure ν (so that we write, for example, $\mathcal{L}^p(G/H) = \mathcal{L}^p(G/H, \nu)$).

We say that a set $S \subset G$ is negligible modulo H if $\varpi(S)$ is ν -negligible. This condition

does not depend on the choice of Haar measures on G and on H . It implies that S is locally μ -negligible (INT, VII, p. 47, § 2, no. 3, prop. 6, a)). We say that a property of an element of G is true almost everywhere modulo H if the set of elements of G for which this property does not hold is negligible modulo H .

Let π a unitary representation of H in a Hilbert space E . We denote by $\mathcal{F}_\pi(G)$ the vector space of functions f on G with values in E such that $f(xh) = \pi(h)f(x)$ for all $(x, h) \in G \times H$. We say that the elements of $\mathcal{F}_\pi(G)$ are the π -equivariant functions on G .

The space $\mathcal{F}_1(G)$ associated with the trivial representation of G on $\mathbb{C}$ is identified with the space $\mathcal{F}(G/H)$ of complex-valued functions on G/H by the map $f \mapsto f \circ \varpi$ of $\mathcal{F}(G/H)$ in $\mathcal{F}_1(G)$.

For any function f in $\mathcal{F}_\pi(G)$, the function $\|f\|$ belongs to $\mathcal{F}_1(G)$ since π is unitary. This allows us to identify $\|f\|$ with a function on G/H , and we shall write, for example, $\|f(xH)\|$ for the value of this function at an element xH of G/H .

A function f of $\mathcal{F}_\pi(G)$ will be said to vanish outside a compact set modulo H if $\|f\|$ vanishes outside a compact subset of G/H . This is equivalent to saying that there exists a compact subset K of G such that f vanishes outside $K \cdot H$ (TG, III, p. 33, prop. 10).

We denote $\mathcal{K}_\pi(G)$ the space of continuous functions on G belonging to $\mathcal{F}_\pi(G)$ which have compact support modulo H .

A space analogous to $\mathcal{H}_\pi(G)$ was defined in INT, VII, p. 39, § 2, no. 1, when π is a continuous homomorphism of H into $\mathbf{R}_+^*$.

Let $p \in [1, +\infty[$. For every $f \in \mathcal{F}_\pi(G)$, we denote

$$N_p(f) = \left(\int_{G/H}^* \|f\|^p d\nu \right)^{1/p}.$$

It is a real number or $+\infty$. We denote $\mathcal{F}_\pi^p(G, \nu)$, or simply $\mathcal{F}_\pi^p(G)$, the subspace of functions $f \in \mathcal{F}_\pi(G)$ such that $N_p(f)$

is finite. The space $\mathcal{F}_\pi^p(G)$ equipped with the map N_p is a semi-normed space.

The space $\mathcal{K}_\pi(G)$ is contained in $\mathcal{F}_\pi^p(G)$. Its closure in $\mathcal{F}_\pi^p(G)$ is denoted by $\mathcal{L}_\pi^p(G, \nu)$, or simply $\mathcal{L}_\pi^p(G)$; we say that it is the space of π -equivariant functions on G of power p^e integrable modulo H .

The assertions of the following proposition, when π is the trivial representation of dimension 1, are consequences of INT, IV, p. 128, § 3, no. 3, prop. 6 and p. 131, § 3, no. 4, th. 3.

PROPOSITION 3. — a) Let $(f_n)_{n \in \mathbb{N}}$ a sequence in $\mathcal{F}_\pi^p(G)$ such that the series

$$\sum_{n=0}^{+\infty} N_p(f_n)$$

converges. Then the series with general term $f_n(g)$ is absolutely convergent for g outside a set T which is negligible modulo H . Let f be the function on G with values in E which is equal to the sum of this series on $G - T$ and equal to 0 on T . We then have $f \in \mathcal{F}_\pi^p(G)$ and the series with general term f_n converges to f in the space $\mathcal{F}_\pi^p(G)$;

b) Let (f_n) a sequence converging in $\mathcal{L}_\pi^p(G)$ to a function f . There exists a sequence (f_{n_k}) extracted from (f_n) such that $f_{n_k}(g)$ converges to $f(g)$ for all g outside a set negligible modulo H ;

c) The semi-normed spaces $\mathcal{F}_\pi^p(G)$ and $\mathcal{L}_\pi^p(G)$ are complete.

In the assertion a), saying that the series with general term f_n converges to f in the space $\mathcal{F}_\pi^p(G)$ means that the sequence of partial sums $f_0 + \dots + f_n$ converges to f in $\mathcal{F}_\pi^p(G)$. One then also says that f is a sum of this series.

Let us prove a). By prop. 6 of INT, IV, p. 128, § 3, no. 3, there exists a set ν -negligible $S \subset G/H$ such that the series with general term $\|f_n(gH)\|$ converges absolutely for $gH \notin S$. Moreover, the function h which is equal to the sum of this series for $gH \notin S$ and which is zero for $gH \in S$ satisfies $N_p(h) < +\infty$.

The set $T = \overline{\omega}(S)$ is negligible modulo H . For every $g \notin T$, the series with general term $f_n(g)$ is absolutely convergent in E . Let us define $f(g) = \sum f_n(g)$ for $g \notin T$ and $f(g) = 0$ otherwise. We have $f \in \mathcal{F}_\pi^p(G)$. Note that $\|f(gH)\| \leq h(gH)$ for all $g \in G$, whence $N_p(f) \leq N_p(h) < +\infty$. We therefore have $f \in \mathcal{F}_\pi^p(G)$. Similarly, it follows that

$$N_p\left(f - \sum_{n=0}^k f_n\right) \leq \sum_{n=k+1}^{+\infty} N_p(f_n)$$

for all $k \in \mathbb{N}$, hence the series $\sum f_n$ converges to f in $\mathcal{F}_\pi^p(G)$. The assertion a) is proved.

Let us prove *b*). The sequence $(\|f_n - f\|)_{n \in \mathbb{N}}$ converges to 0 in the space $\mathcal{L}^p(G/H)$. By INT, IV, p. 131, § 4, no. 3, th. 3, there exists a subsequence $(\|f_{n_k} - f\|)_{k \in \mathbb{N}}$ which converges ν -almost everywhere to 0. The sequence $(f_{n_k}(g))_{k \in \mathbb{N}}$ then converges to $f(g)$ for all g outside a set negligible modulo H .

Finally let us prove *c*). Let $(f_n)_{n \in \mathbb{N}}$ a Cauchy sequence in $\mathcal{F}_\pi^p(G)$. There exists a strictly increasing sequence of integers $(n_k)_{k \in \mathbb{N}}$ such that $N_p(f_{n_{k+1}} - f_{n_k}) \leq 2^{-k}$ for all $k \in \mathbb{N}$. For every $k \in \mathbb{N}$ set $h_k = f_{n_{k+1}} - f_{n_k} \in \mathcal{F}_\pi^p(G)$. By *a*), the series with general term h_k converges in $\mathcal{F}_\pi^p(G)$; denote it h its sum. For every $\ell \in \mathbb{N}$, it follows that

$$f_{n_0} + \sum_{k=0}^{\ell} h_k = f_{n_{\ell+1}},$$

hence $f_{n_0} + h$ is a cluster point of the sequence (f_n) . The latter is therefore convergent (TG, II, p. 14, cor. 2 of prop. 5). Thus the space $\mathcal{F}_\pi^p(G)$ is complete; since the space $\mathcal{L}_\pi^p(G)$ is closed in $\mathcal{F}_\pi^p(G)$, it is also complete.

COROLLARY. — Let $(f_n)_{n \in \mathbb{N}}$ a Cauchy sequence in $\mathcal{L}_\pi^p(G)$ and let $f \in \mathcal{F}_\pi(G)$ such that $f_n(g)$ converges to $f(g)$ almost everywhere modulo H . Then $f \in \mathcal{L}_\pi^p(G)$ and $(f_n)_{n \in \mathbb{N}}$ converges to f in $\mathcal{L}_\pi^p(G)$.

Indeed, the function f coincides almost everywhere modulo H with the limit of the sequence (f_n) in $\mathcal{L}_\pi^p(G)$.

We denote $L_\pi^p(G)$ the separated normed topological vector space associated with the seminormed space $\mathcal{L}_\pi^p(G)$; it is a Banach space.

Let f a function on G with values in E . We denote S_f the set of $x \in G$ such that the function $h \mapsto f(xh)$ onto H does not belong to $\mathcal{L}_E^1(H)$. We have $S_f \cdot h = S_f$ for all $h \in H$.

Let $x \in G - S_f$. Since π is a unitary representation, the function $h \mapsto \pi(h)^* f(xh)$ is measurable (as a composition of the map $h \mapsto (h, f(xh))$ from H to $H \times E$ which is then measurable and of the continuous map $(g, x) \mapsto \pi(g)^* x$ from $H \times E$ to E , cf. INT, IV, p. 174, § 5, no. 3, th. 1), and it is integrable on H .

One defines a function f^π on G by setting

$$f^\pi(x) = \int_H \pi(h)^* f(xh) d\beta(h) \quad (3)$$

for $x \in G - S_f$ and $f^\pi(x) = 0$ if $x \in S_f$.

PROPOSITION 4. — Let $f \in \mathcal{L}_E^1(G)$.

- a) The set S_f is negligible modulo H and $f^\pi \in \mathcal{F}_\pi(G)$;
- b) Let C be a compact subset of G . If f is continuous and supported in C , then S_f is empty, the function f^π belongs to $\mathcal{H}_\pi(G)$ and its support is contained in $C \cdot H$.

The first part of a) follows from INT, VII, p. 57, § 2, no. 5, c). Let $w \in H$. If $x \in G - S_f$, then $xw \in G - S_f$ and

$$\begin{aligned} f^\pi(xw) &= \int_H \pi(h) * f(xwh) d\beta(h) \\ &= \int_H \pi(w^{-1}y) * f(xy) d\beta(y) = \pi(w) f^\pi(x), \end{aligned}$$

whereas if $x \in S_f$, then $f^\pi(xw) = 0 = \pi(w) f^\pi(x)$. The function f^π therefore belongs to $\mathcal{F}_\pi(G)$.

Suppose now that f is continuous and supported in C . For every $x \in G$, the map $h \mapsto f(xh)$ then belongs to $\mathcal{K}(H)$, hence is integrable, which proves that $S_f = \emptyset$.

Let us prove that f^π is continuous. Let $x \in G$. Let U be a relatively compact open neighborhood of x in G .

For every $y \in U$, we have

$$\begin{aligned} \|f^\pi(y) - f^\pi(x)\| &\leq \int_H \|f(yh) - f(xh)\| d\beta(h) \\ &= \int_{H \cap (y^{-1}C \cup x^{-1}C)} \|f(yh) - f(xh)\| d\beta(h) \\ &\leq \beta(H \cap U^{-1}C) \sup_{h \in U^{-1}C} \|f(yh) - f(xh)\|, \end{aligned}$$

and the continuity of f^π then follows from the uniform continuity of f on G .

Finally, if $x \in G$ satisfies $f^\pi(x) \neq 0$, there exist $h \in H$ such that $f(xh) \neq 0$ thus xh belongs to C and x belongs to $C \cdot H$. Since $C \cdot H$ is closed in G , we conclude that the support of f^π is contained in $C \cdot H$.

Let C be a compact subset of G . Let $u \in \mathcal{K}_+(G)$ a function such that $u(x) > 0$ for all $x \in C$. Let v be the function on G defined by

$$v(x) = \int_H u(xh) d\beta(h) \tag{4}$$

for all $x \in G$; with the previous notation, we have $v = u^1$, corresponding to the trivial 1-dimensional representation of H . The function v is continuous and positive; it belongs to $\mathcal{F}_1(G)$, its support is contained in $\text{Supp}(u) \cdot H$ and we have

$$\inf_{x \in C \cdot H} v(x) > 0 \quad (5)$$

(INT, VII, p. 39–40, § 2, no. 1, prop. 1 and lemma 1, a)).

Lemma 5. — Let $f \in \mathcal{F}_\pi(G)$ a function μ -measurable and zero outside of $C \cdot H$ such that the function $\|f\|$ is v -integrable on G/H . Let s be the function on G with values in E such that

$$s(x) = \begin{cases} v(x)^{-1}f(x) & \text{if } x \in C \cdot H \\ 0 & \text{if } x \in G - C \cdot H. \end{cases}$$

- a) The function s is μ -measurable; it belongs to $\mathcal{F}_\pi(G)$ and is zero outside of $C \cdot H$;
- b) The function us belongs to $\mathcal{L}^1(G)$ and $(us)^\pi = f$ almost everywhere modulo H .

The function s vanishes outside $C \cdot H$ by definition; it is μ -measurable since the function f is μ -measurable and $v(x) > 0$ for all $x \in C \cdot H$ (INT, IV, p. 193, § 5, no. 10, prop. 16). We have $s \in \mathcal{F}_\pi(G)$ because $v \in \mathcal{F}_1(G)$.

The function f is locally μ -integrable, since $\|f\|$ is a function v -integrable (INT, VII, p. 47, § 2, no. 3, prop. 6, c), noting that the measure $v^\# = \kappa \cdot \mu$ is equivalent to μ), hence the function s is likewise by formula (5). The function us is measurable and zero outside a compact subset of G ; since u is bounded, it follows that us is μ -integrable.

For every $x \in G - S_{us}$, we have

$$\begin{aligned} (us)^\pi(x) &= \int_H \pi(h)^*(u(xh)s(xh))d\beta(h) \\ &= \int_H u(xh)s(x) d\beta(h) = v(x)s(x), \end{aligned}$$

since $s(xh) = \pi(h)s(x)$; the last assertion follows from prop. 4, a).

When $H = \{e\}$ and π is of dimension 1, the following proposition is none other than INT, IV, p. 184, § 5, no. 6, th. 5.

PROPOSITION 5. — Let $p \in [1, +\infty[$. The space $\mathcal{L}_\pi^p(G)$ is the space of functions

$f \in \mathcal{F}_\pi(G)$ such that f is μ -measurable and that the function $\|f\|$ belongs to $\mathcal{L}^p(G/H)$.

Let $f \in \mathcal{L}^p_\pi(G)$. It is the limit in $\mathcal{L}^p_\pi(G)$ of a sequence of elements of $\mathcal{H}_\pi(G)$. By prop. 3, b) and Egoroff's theorem (INT, IV, p. 175, § 5, no. 4, th. 2), the function f is therefore μ -measurable; consequently, the function $\|f\|$ on G/H is ν -measurable (INT, VII, p. 47, § 2, no. 3, prop. 6, b)). Since $N_p(f)$ is finite, the function $\|f\|$ belongs to $\mathcal{L}^p(G/H)$ (INT, IV, p. 184, § 5, no. 6, th. 5).

Let us prove the converse assertion. Let f a function μ -measurable belonging to $\mathcal{F}_\pi(G)$ such that $\|f\| \in \mathcal{L}^p(G/H)$. Let $\varepsilon > 0$. Let us prove that there exists $\tilde{f} \in \mathcal{H}_\pi(G)$ such that

$$N_p(f - \tilde{f})^p = \int_{G/H}^* \|f - \tilde{f}\|^p d\nu \leq \varepsilon,$$

which will conclude the proof.

Suppose first that there exists a compact subset C of G such that f vanishes outside $C \cdot H$. Let $u \in \mathcal{H}_+(G)$ a function such that $u(x) > 0$ for all $x \in C$ and let us define $v = u^1$ as above. Denote φ the characteristic function of the support of u .

Let q be the conjugate exponent of p . If $p = 1$, let w be the constant function on G equal to $\sup_{x \in G} u(x)$. If $p > 1$, we define

$$w(x) = \left(\int_H u(xh)^q d\beta(h) \right)^{1/q}$$

for all $x \in G$. In all cases, the function w is continuous and positive; it belongs to $\mathcal{H}_1(G)$ (Prop. 4 applied to the trivial representation of dimension 1 and to the function u^q) hence it is bounded on G . We set $W = \sup_{x \in G} w(x)$.

Let s the function on G with values in $\mathbf{E}$ defined by Lemma 5 applied to f . Set $g = \varphi s$. The function g is μ -measurable, and satisfies $\|g\| \leq \|f\| / \inf_{x \in C \cdot H} v(x)$. Since g is zero outside the support of u and that κ is continuous, we have $g \in \mathcal{L}^p_E(G, \kappa \cdot \mu)$. Let $\tilde{g}$ a function

in $\mathcal{H}_E(G)$ such that

$$\int_G^* \|g - \tilde{g}\|^p \kappa d\mu \leq \frac{\varepsilon}{W^p}.$$

We have $us = ug$, hence $f = (us)^\pi = (ug)^\pi$ almost everywhere modulo H (lemma 5, b)).

Let $\tilde{f} = (u\tilde{g})^\pi$; we have $\tilde{f} \in \mathcal{K}_\pi(G)$ (prop. 4, b)). For every $x \in G - S_{ug}$, we obtain

$$\|(ug)^\pi(x) - (u\tilde{g})^\pi(x)\| \leq \int_H^* u(xh) \|g(xh) - \tilde{g}(xh)\| d\beta(h)$$

whence

$$\|(ug)^\pi(x) - (u\tilde{g})^\pi(x)\|^p \leq w(x)^p \int_H^* \|g(xh) - \tilde{g}(xh)\|^p d\beta(h),$$

by Hölder's inequality in the case $p > 1$. Since S_{ug} is negligible modulo H, it then follows that

$$\begin{aligned} \int_{G/H}^* \|f - \tilde{f}\|^p d\nu &\leq W^p \int_{G/H}^* \left(\int_H^* \|g(xh) - \tilde{g}(xh)\| d\beta(h) \right)^p d\nu(xH) \\ &= W^p \int_G^* \|g - \tilde{g}\|^p \kappa d\mu \leq \varepsilon \end{aligned}$$

according to INT, VII, p. 46, § 2, no. 3, prop. 5, b), which is applicable because $g - \tilde{g}$ vanishes outside a compact subset of G. This implies the required property when f is zero outside a compact subset modulo H.

Now consider the general case. Since $\|f\| \in \mathcal{L}^p(G/H)$ by hypothesis, there exists a compact subset L of G/H such that

$$\int_{(G/H)-L}^* \|f\|^p d\nu \leq \frac{\varepsilon}{2}$$

(cf. INT, IV, p. 152, § 4, no. 6, th. 4). Let φ_L the characteristic function of L and set $f_L = (\varphi_L \circ \varpi)f$. We have

$$\int_{G/H}^* \|f - f_L\|^p d\nu = \int_{(G/H)-L}^* \|f\|^p d\nu \leq \frac{\varepsilon}{2}.$$

The function f_L is μ -measurable and zero outside a compact set modulo H. It belongs to $\mathcal{L}_\pi^p(G)$ therefore, by the preceding case, there exists $\tilde{f} \in \mathcal{K}_\pi(G)$ such that $N_p(f_L - \tilde{f}) \leq (\frac{\varepsilon}{2})^{1/p}$, whence

$$\int_{G/H}^* \|f - \tilde{f}\|^p d\nu \leq \varepsilon,$$

as desired.

Let us consider the case $p = 2$. For f_1 and f_2 in $\mathcal{F}_\pi(G)$, the function $x \mapsto \langle f_1(x)|f_2(x) \rangle$ belongs to $\mathcal{F}_1(G)$ since the representation π is unitary, and therefore defines by passing to the quotient a function on G/H which we identify as before with $\langle f_1|f_2 \rangle$. We have the bound $|\langle f_1|f_2 \rangle| \leq \|f_1\| \|f_2\|$ in $\mathcal{F}(G/H)$.

If f_1 and f_2 belong to $\mathcal{L}_\pi^2(G)$, then the function $\langle f_1|f_2 \rangle$ belongs to $\mathcal{L}_1^1(G)$. In particular, it is integrable on G/H . The map

$$(f_1, f_2) \mapsto \int_{G/H} \langle f_1|f_2 \rangle d\nu$$

is a positive hermitian form on $\mathcal{L}_\pi^2(G)$; the associated semi-norm is the semi-norm N_2 . In particular, the space $\mathcal{L}_\pi^2(G)$ is a pre-Hilbert space and $L_\pi^2(G)$ is the Hilbert space associated with $\mathcal{L}_\pi^2(G)$.

V.2.6 Induced representations

We keep the notations and conventions of the preceding number concerning the measures β on H and ν on G/H , as well as the function $\kappa: G \rightarrow \mathbf{R}_+^*$.

There exists a continuous function η of $G \times G/H$ in $\mathbf{R}_+^*$ such that

$$\eta(x, yH) = \frac{\kappa(xy)}{\kappa(x)}$$

for all $(x, y) \in G \times G$, and $\gamma_{G/H}(x)\nu = (y \mapsto \eta(x^{-1}, y)) \cdot \nu$ for $x \in G$ (INT, VII, p. 56, § 2, no. 5, th. 2, c)).

Let $p \in [1, +\infty[$ and let π a unitary representation of H in a complex Hilbert space E .

Lemma 6. — Let $f \in \mathcal{H}_{\overline{\pi}}(G)$. For every $g \in G$, the function

$$\tilde{f}: x \mapsto \eta(g^{-1}, xH)^{1/p} f(g^{-1}x)$$

of G in E belongs to $\mathcal{H}_{\overline{\pi}}(G)$ and satisfies $N_p(\tilde{f}) = N_p(f)$.

One verifies without difficulty that $\tilde{f} \in \mathcal{H}_{\overline{\pi}}(G)$. Since

$$N_p(\tilde{f})^p = \int_{G/H}^* \|\tilde{f}\|^p d\nu = \int_{G/H}^* \gamma_{G/H}(g)(\|f\|^p)(y) \eta(g^{-1}, y) d\nu(y)$$

and that $(y \mapsto \eta(g^{-1}, y)) \cdot \nu = \gamma_{G/H}(g) \nu$, we obtain $N_p(\tilde{f}) = N_p(f)$.

It follows from this lemma that there exists a continuous and isometric representation $\tilde{\pi}$ from G into $L^p_{\tilde{\pi}}(G)$ such that for $f \in \mathcal{K}_{\tilde{\pi}}(G)$ and $g \in G$, the element $\tilde{\pi}(g)f$ is the class of the function $\tilde{f}$ defined above. If $p = 2$, then this representation is unitary.

DEFINITION 2. — One says that the unitary representation of G in the space $L^2_{\tilde{\pi}}(G)$ thus defined is the unitary representation of G induced by the representation π of H with respect to κ . We denote it by $\text{Ind}_H^G(\pi, \kappa)$, or simply $\text{Ind}_H^G(\pi)$.

Remarks. — 1) Let ϱ a unitary representation of H in a complex Hilbert space F , and let $u: \pi \rightarrow \varrho$ an H -morphism. For every function $f \in \mathcal{K}_{\tilde{\pi}}(G)$, denote $v(f)$ the function $g \mapsto u(f(g))$ from G to F ; it belongs to $\mathcal{K}_{\tilde{\varrho}}(G)$ and satisfies $N_p(v(f)) \leq \|u\| N_p(f)$. The linear mapping from $\mathcal{K}_{\tilde{\pi}}(G)$ in $\mathcal{K}_{\tilde{\varrho}}(G)$ which to f associates $v(f)$ extends therefore to a continuous H -morphism from $\text{Ind}_H^G(\pi)$ in $\text{Ind}_H^G(\varrho)$ which is denoted $\text{Ind}_H^G(u)$. We have $\text{Ind}_H^G(1_{\pi}) = 1_{\text{Ind}_H^G(\pi)}$. Let σ a unitary representation of H , and let $v: \varrho \rightarrow \sigma$ an H -morphism; we have $\text{Ind}_H^G(v \circ u) = \text{Ind}_H^G(v) \circ \text{Ind}_H^G(u)$.

In other words, the construction which to π associates $\text{Ind}_H^G(\pi)$ and with u associates $\text{Ind}_H^G(u)$ is a functor from the category of unitary representations of H to that of unitary representations of G (cf. CAT, I, § 2, in preparation).

2) Let $\kappa': G \rightarrow \mathbf{R}_+^*$ such that

$$\frac{\kappa'(xh)}{\kappa'(x)} = \frac{\Delta_H(h)}{\Delta_G(h)} = \frac{\kappa(xh)}{\kappa(x)}$$

for all $(x, h) \in G \times H$. Let ν' the quasi-invariant measure $(\kappa' \cdot \mu)/\beta$ on G/H . The function $\kappa' \kappa^{-1}$ defines, by passing to the quotient, a continuous function $\xi: G/H \rightarrow \mathbf{R}_+^*$ such that $\nu' = \xi \cdot \nu$. The endomorphism α of $\mathcal{K}_{\tilde{\pi}}(G)$ defined by $f \mapsto (\kappa' \kappa^{-1})^{1/p} f$ satisfies

$$\int_{G/H}^* \|f\|^p d\nu' = \int_{G/H}^* \|\alpha(f)\|^p d\nu$$

and makes it possible to identify the spaces $\mathcal{L}_{\pi}^p(G, \nu')$ and $\mathcal{L}_{\pi}^p(G, \nu)$, as well as the spaces $L_{\pi}^p(G, \nu')$ and $L_{\pi}^p(G, \nu)$. Moreover, α induces an isometric isomorphism of representations $\text{Ind}_H^G(\pi, \kappa')$ and $\text{Ind}_H^G(\pi, \kappa)$.

V.2.7 Case of a closed central subgroup

In this issue, we assume that the group G is unimodular.

Consider a closed subgroup Z of the center of G , and let dz denote a Haar measure on Z . We endow the quotient group G/Z with the Haar measure $\nu = \mu/dz$. Let χ a unitary character of Z . The quotient group G/Z is unimodular by INT, VII, p. 61, § 2, no. 7, cor. of prop. 11.

Lemma 7. — One has

$$N_1(\check{f}) = N_1(f), \quad \langle f_1 | f_2 \rangle = \langle \check{f}_1 | \check{f}_2 \rangle. \quad (6)$$

for all $f \in \mathcal{F}_{\chi}(G)$ and for all f_1 and f_2 in $\mathcal{L}_{\chi}^2(G)$.

These formulas are consequences of the definitions.

For every $g \in G$ and every $f \in \mathcal{F}_{\chi}(G)$, the functions $x \mapsto f(g^{-1}x)$ and $x \mapsto f(xg)$ belong to $\mathcal{F}_{\chi}(G)$. They are denoted respectively $\gamma_{G,\chi}(g)f$ and $\delta_{G,\chi}(g)f$. The maps $\gamma_{G,\chi}$ and $\delta_{G,\chi}$ are linear representations of G in $\mathcal{F}_{\chi}(G)$. For every $z \in Z$, we have

$$\gamma_{G,\chi}(gz) = \overline{\chi(z)} \gamma_{G,\chi}(g), \quad \delta_{G,\chi}(gz) = \chi(z) \delta_{G,\chi}(g). \quad (7)$$

Let $f \in \mathcal{F}_{\chi}(G)$ and $g \in G$; we have

$$|\gamma_{G,\chi}(g)f| = \gamma_{G/Z}(gZ)|f|, \quad |\delta_{G,\chi}(g)f| = \delta_{G/Z}(gZ)|f|, \quad (8)$$

where all functions appearing in these equalities are identified with functions on G/Z .

The subspace $\mathcal{K}_{\chi}(G)$ of $\mathcal{F}_{\chi}(G)$ is stable under $\gamma_{G,\chi}$ and $\delta_{G,\chi}$. Let p a real number ≥ 1 . The formulas (8) imply that the representations $\gamma_{G,\chi}$ and $\delta_{G,\chi}$ restricted to $\mathcal{K}_{\chi}(G)$, extend to continuous, isometric linear representations of G into $\mathcal{L}_{\chi}^p(G)$ which will be denoted $\gamma_{G,\chi}^{(p)}$ and $\delta_{G,\chi}^{(p)}$. By passing to quotients, these representations also define isometric representations of G in $L_{\chi}^p(G)$ denoted in the same way.

The representations $\gamma_{G,\chi}^{(2)}$ and $\delta_{G,\chi}^{(2)}$ in $L_\chi^2(G)$ are unitary, and will be denoted simply $\gamma_{G,\chi}$ and $\delta_{G,\chi}$, respectively, when there is no confusion with the representations in $\mathcal{F}_\chi(G)$. We also denote $\varrho_{G,\chi}$ the continuous representation of $G \times G$ in $\mathcal{L}_\chi^2(G)$ or $L_\chi^2(G)$ defined by

$$\varrho_{G,\chi}(g, h) = \gamma_{G,\chi}(g) \circ \delta_{G,\chi}(h) = \delta_{G,\chi}(h) \circ \gamma_{G,\chi}(g)$$

for all $(g, h) \in G \times G$ (cf. Lemma 1 of V, p. 377).

The representation $\gamma_{G,\chi}$ onto $L_\chi^2(G)$ is none other than the induced representation $\text{Ind}_Z^G(\overline{\chi})$.

When $Z = \{e\}$ the following lemma results from INT, VIII, p. 166, § 4, no. 5, prop. 12, since G is unimodular.

Lemma 8. — Let $f_1 \in \mathcal{H}_\chi(G)$ and $f_2 \in \mathcal{L}_\chi^2(G)$.

- a) The function f onto G defined by $f(g) = \langle f_1 | \gamma_{G,\chi}(g) f_2 \rangle$ for all $g \in G$ belongs to $\mathcal{L}_\chi^2(G)$ and satisfies $N_2(f) \leq N_1(f_1) N_2(f_2)$;
- b) The function f onto G defined by $f(g) = \langle f_1 | \delta_{G,\chi}(g) f_2 \rangle$ for all $g \in G$ belongs to $\mathcal{L}_\chi^2(G)$ and satisfies $N_2(f) \leq N_1(f_1) N_2(f_2)$.

Let us prove a), the proof of assertion b) being similar. The function f is continuous, hence μ -measurable. For every $z \in Z$ and every $g \in G$, we have

$$f(gz) = \langle f_1 | \gamma_{G,\chi}(gz) f_2 \rangle = \overline{\chi(z)} \langle f_1 | \gamma_{G,\chi}(g) f_2 \rangle$$

(formula (7), p. 417), hence $f \in \mathcal{F}_{\overline{\chi}}(G)$. Prop. 5 of V, p. 413 implies that it now suffices to prove that $N_2(f) \leq N_1(f_1) N_2(f_2)$.

Suppose first that f_2 belongs to $\mathcal{H}_\chi(G)$. For every $g \in G$, one has by definition

$$f(g) = \int_{G/Z} \overline{f_1} \gamma_{G,\chi}(g) f_2 d\nu,$$

where the function $\overline{f_1} \gamma_{G,\chi}(g) f_2$ is identified with a function on G/Z .

Let us define a function f_3 onto G by setting $f_3(g) = 0$ if $f_1(g) = 0$ and $f_3(g) = f_1(g) |f_1(g)|^{-1/2}$ otherwise. The function f_3 belongs to $\mathcal{F}_\chi(G)$ and satisfies $f_1 = |f_1|^{1/2} f_3$; it

is μ -measurable and zero outside a compact modulo Z since f_1 is. Since $|f_1|^{1/2} \in \mathcal{H}_1(G)$, it follows that

$$\overline{f_1} \gamma_{G,\chi}(g) f_2 = |f_1|^{1/2} \overline{f_3} \gamma_{G,\chi}(g) f_2,$$

$$|\overline{f_3} \gamma_{G,\chi}(g) f_2| = |f_3| |\gamma_{G,\chi}(g) f_2|$$

for all $g \in G$.

Let $g \in G$. By the Cauchy–Schwarz inequality, we have

$$\begin{aligned} |f(g)|^2 &\leq \left(\int_{G/Z} |f_1|^{1/2} |f_3| |\gamma_{G,\chi}(g) f_2| d\nu \right)^2 \\ &\leq \left(\int_{G/Z} |f_1| d\nu \right) \left(\int_{G/Z} |f_3|^2 |\gamma_{G,\chi}(g) f_2|^2 d\nu \right). \end{aligned}$$

Since we have $|f_3|^2 = |f_1|$ in $\mathcal{H}(G/Z)$, we deduce by integrating over G/Z that

$$\int_{G/Z} |f(g)|^2 d\nu(g) \leq N_1(f_1) \int_{G/Z} \left(\int_{G/Z} |f_1(x)| |f_2(g^{-1}x)|^2 d\nu(x) \right) d\nu(g).$$

The function on $G/Z \times G/Z$ deduced from the function

$$(g, x) \mapsto |f_1(x)| |f_2(g^{-1}x)|^2$$

by passing to quotients is $(\nu \otimes \nu)$ measurable and compactly supported, hence it is $(\nu \otimes \nu)$ -moderate (INT, V, p. 4, § 5, no. 2, def. 2). By prop. 7 of INT, V, p. 93, § 8, no. 3, it follows that

$$\begin{aligned} &\int_{G/Z} \left(\int_{G/Z} |f_1(x)| |f_2(g^{-1}x)|^2 d\nu(x) \right) d\nu(g) \\ &= \int_{G/Z} |f_1(x)| \left(\int_{G/Z} |f_2(g^{-1}x)|^2 d\nu(g) \right) d\nu(x) = N_1(f_1) N_2(f_2)^2. \end{aligned}$$

Consequently, we have $N_2(f)^2 \leq N_1(f_1)^2 N_2(f_2)^2$ which establishes the desired property in this case.

Let us consider the general case. Denote by u the linear map from $\mathcal{H}_\chi(G)$ in $\mathcal{L}_\chi^2(G)$ which to f_2 associates f . Let $f_2 \in \mathcal{L}_\chi^2(G)$ and let $(f_{2,n})_{n \in \mathbb{N}}$ a sequence in $\mathcal{H}_\chi(G)$ which converges to f_2 in $\mathcal{L}_\chi^2(G)$. Let $f_n = u(f_{2,n})$; the sequence $(f_n)_{n \in \mathbb{N}}$ is Cauchy in $\mathcal{L}_\chi^2(G)$ since the previous

case implies that $N_2(f_n - f_m) \leq N_1(f_1)N_2(f_{2,n} - f_{2,m})$ for all n and m in $\mathbf{N}$. Let $f \in \mathcal{L}_{\chi}^2(G)$ such that (f_n) converges to f (prop. 3, c) of V, p. 409). Since $N_2(f_n) \leq N_1(f_1)N_2(f_{2,n})$ for all $n \in \mathbf{N}$, it follows that $N_2(f) \leq N_1(f_1)N_2(f_2)$.

There exists a subsequence $(f_{n_k})_{k \in \mathbf{N}}$ such that $f_{n_k}(g)$ converges to $f(g)$ for all $g \in G$ outside of a negligible subset of G modulo Z (loc. cit., b)). But on the other hand, for every $g \in G$, we have

$$f_{n_k}(g) = \langle f_1 | \gamma_{G,\chi}(g)f_{2,n_k} \rangle \rightarrow \langle f_1 | \gamma_{G,\chi}(g)f_2 \rangle.$$

Consequently, we have $f(g) = \langle f_1 | \gamma_{G,\chi}(g)f_2 \rangle$ for all g outside a negligible subset of G modulo Z . Since $f \in \mathcal{L}_{\chi}^2(G)$ and that $N_2(f) \leq N_1(f_1)N_2(f_2)$, the lemma is proved.

V.2.8 Square-integrable representations

In this issue, the Hilbert spaces considered are complex. It is assumed that G is unimodular, and its center is denoted by C . For every closed subgroup Z of C , we denote by β_Z a Haar measure on Z , and we equip G/Z with the Haar measure $\nu_Z = \mu/\beta_Z$ (INT, VII, p. 44, § 2, no. 2, def. 1).

Let π an irreducible unitary representation of G in a Hilbert space E and $\chi \in \widehat{C}$ its central character (def. 6 of V, p. 390). For all x and y in E , we denote $f_{x,y}$ the matrix coefficient $g \mapsto \langle x | \pi(g)y \rangle$; it is a continuous function on G with complex values.

Let x and y in E . For $z \in C$ and $g \in G$, we have

$$f_{x,y}(zg) = \langle x | \pi(zg)y \rangle = \langle x | \chi(z)\pi(g)y \rangle = \chi(z)f_{x,y}(g), \quad (9)$$

hence $f_{x,y}$ belongs to the space $\mathcal{F}_{\chi}(G)$ (V, p. 408). Moreover, for every $(g_1, h_1) \in G \times G$ and every $g \in G$, we have

$$f_{\pi(g_1)x, \pi(h_1)y}(g) = \langle \pi(g_1)x | \pi(g)\pi(h_1)y \rangle = f_{x,y}(g_1^{-1}gh_1). \quad (10)$$

Relation (9) justifies the following definition.

DEFINITION 3. — Let π an irreducible unitary representation of G in a Hilbert space E . We say that π is square-integrable modulo the center if the function on G/C deduced from the

function $|f_{x,y}|$ by passage to the quotient belongs to $\mathcal{L}^2(G/C)$ for all $(x,y) \in E \times E$.

This condition does not depend on the choice of a Haar measure on G/C .

If the matrix coefficients of π belong to $\mathcal{L}^2(G)$, one says that π is square-integrable; the existence of an irreducible unitary representation of G that is square-integrable implies that the center of G is compact (exercise 5 of V, p. 487).

There exist groups G which admit no square-integrable representation, even modulo the center (cf. exercise 32 of V, p. 516).

PROPOSITION 6. — Let π an irreducible unitary representation of G in a Hilbert space E . Let χ the central character of π . Then π is square-integrable modulo the center if and only if there exist nonzero elements x_0 and y_0 of E such that the function on G/C induced by $|f_{x_0,y_0}|$ by passage to the quotient belongs to $\mathcal{L}^2(G/C)$.

Suppose there exist elements x_0 and y_0 nonzero elements of E such that $|f_{x_0,y_0}| \in \mathcal{L}^2(G/C)$. It suffices to show that π is then square-integrable modulo the center.

Let F be the set of elements $x \in E$ such that $|f_{x,y_0}|$ belongs to $\mathcal{L}^2(G/C)$. This is a vector subspace of E ; it contains x_0 , hence is nonzero. Relation (10) above implies that F is stable under π ; since the representation π is irreducible, the space F is therefore dense in E .

Let $x \in F$. Since f_{x,y_0} belongs to $\mathcal{F}_\chi(G)$ and is μ -measurable, and $|f_{x,y_0}| \in \mathcal{L}^2(G/C)$, we have $f_{x,y_0} \in \mathcal{L}^2_\chi(G)$ (prop. 5 of V, p. 413 applied to $Z = C$). Let u denote the partial operator from E to $L^2_\chi(G)$ whose domain is F and which associates to $x \in F$ the class of f_{x,y_0} in $L^2_\chi(G)$.

Let us show that the partial operator u is closed. Let $(x_n, u(x_n))_{n \in \mathbb{N}}$ a sequence of elements of the graph of u which converges in $E \times L^2_\chi(G)$. Let x the limit of (x_n) and let $f \in L^2_\chi(G)$ a function whose class is the limit of the sequence $(u(x_n))$.

The function $u(x_n)$ is the class of the matrix coefficient f_{x_n,y_0} . For every $g \in G$, we have

$$f_{x_n,y_0}(g) = \langle x_n | \pi(g) y_0 \rangle \rightarrow \langle x | \pi(g) y_0 \rangle = f_{x,y_0}(g)$$

as $n \rightarrow +\infty$. We therefore have $f_{x,y_0} \in \mathcal{L}^2_\chi(G)$ and $f = f_{x,y_0}$ almost everywhere modulo C (cor. of prop. 3 of V, p. 409); this means that $u(x)$ is the class of f in $L^2_\chi(G)$. We have therefore proved that u is closed.

The domain F of u is stable under π , and relation (10) shows that u satisfies $u \circ \pi(g) =$

$\gamma_{G,\chi}(g) \circ u$ for all $g \in G$. Consequently we also have the equality $u^* \circ \gamma_{G,\chi}(g) = \pi(g) \circ u^*$ for all $g \in G$ (lemma 6 of V, p. 381), whence $(u^* \circ u) \circ \pi(g) = \pi(g) \circ (u^* \circ u)$. Now the partial operator $u^* \circ u$ is self-adjoint (prop. 12 of IV, p. 241), and in particular closed (prop. 7 of IV, p. 236), so cor. 1 of V, p. 387 implies that the domain of $u^* \circ u$ is equal to E . A fortiori, we have $F = E$, that is, the function $|f_{x,y_0}|$ belongs to $\mathcal{L}^2(G/C)$ for all $x \in E$.

Let $(x, y) \in E \times E$. We have $f_{x,y_0} \in \mathcal{L}^2_\chi(G)$. One proves mutatis mutandis, using the representation $\delta_{G,\chi}$ instead of $\gamma_{G,\chi}$, that the set of $y \in E$ such that the function $|f_{x,y}|$ belongs to $\mathcal{L}^2(G/C)$ is equal to E . The proposition follows.

In the rest of this section, we fix a closed subgroup Z of C such that C/Z is compact.

Lemma 9. — Let π an irreducible unitary representation of G in a Hilbert space E which is square-integrable modulo the center. Let χ the restriction to Z of the central character of π . For all x and y in E , we have $f_{x,y} \in \mathcal{L}^2_\chi(G)$.

The function $f_{x,y}$ is continuous hence μ -measurable. We have $f_{x,y} \in \mathcal{F}_\chi(G)$ by formula (9), p. 420. Moreover, by INT, VII, p. 64, § 2, no. 8, cor. 1, c), we have $N_2(f_{x,y}) < +\infty$ since C/Z is compact and $|f_{x,y}| \in \mathcal{L}^2(G/C)$. The assertion therefore follows from Proposition 5 of V, p. 413.

PROPOSITION 7. — Let π be an irreducible unitary representation of G in a Hilbert space E which is square-integrable modulo the center. Let χ be the restriction to Z of the central character of π .

There exists a real number $c > 0$ and a unique $(\mathbf{G} \times \mathbf{G})$ -isometric morphism w of the unitary representation $\overline{\pi} \boxtimes \pi$ in $L^2_\chi(G)$ such that, for all $(x, y) \in \overline{E} \times E$, the element $w(x \otimes y)$ is the class in $L^2_\chi(G)$ of the function $c^{1/2}f_{x,y}$.

For every $(x, y) \in E \times E$, we have $f_{x,y} \in \mathcal{L}^2_\chi(G)$ (Lemma 9). Let v denote the unique linear map from $\overline{E} \otimes E$ in $L^2_\chi(G)$ such that $v(x \otimes y)$ is the class of $f_{x,y}$ for all $(x, y) \in \overline{E} \times E$.

We prove below the following lemma.

Lemma 10. — There exists a real number $c > 0$ such that the linear map $w = c^{1/2}v$ is isometric.

This lemma being assumed valid, note that formula (10), p. 420, can be written as

$$v(\overline{\pi}(g_1)x \otimes \pi(h_1)y) = \varrho_{G,\chi}(g_1, h_1)v(x \otimes y) \quad (11)$$

for all $(g_1, h_1) \in \mathbf{G} \times \mathbf{G}$ and every $(x, y) \in \overline{\mathbf{E}} \times \mathbf{E}$. The isometric linear map w from $\overline{\mathbf{E}} \otimes \mathbf{E}$ in $L^2_\chi(G)$ admits a continuous extension, still denoted w , to $\overline{\mathbf{E}} \widehat{\otimes}_2 \mathbf{E}$. By continuity and linearity, formula (11) implies that w is a $(G \times G)$ -morphism of $\overline{\pi} \boxtimes \pi$ in $L^2_\chi(G)$, which concludes the proof of the proposition.

Let us prove the lemma. For all $x \in E$, we denote u_x the linear map $y \mapsto v(x \otimes y) = f_{x,y}$ from E into $L^2_\chi(G)$. We have $u_x \in \text{Hom}_G(\pi, \gamma_{G,\chi})$. By formula (10), according to Corollary 5 of V, p. 388, there exists a real number $\lambda_x \geq 0$ such that $\lambda_x u_x$ is isometric.

Let x and y in E . We have

$$\|f_{x,y}\|^2 = \int_{G/Z} \bar{f}_{x,y} f_{x,y} d\nu_Z = \int_{G/Z} \check{\bar{f}}_{x,y} \check{f}_{x,y} d\nu_Z$$

since G/Z is unimodular (Lemma 7 of V, p. 417). Noting that $\check{f}_{x,y} = \bar{f}_{y,x}$, we obtain

$$\lambda_x \|y\|^2 = \|f_{x,y}\|^2 = \int_{G/Z} f_{y,x} \bar{f}_{y,x} d\nu_Z = \|f_{y,x}\|^2 = \lambda_y \|x\|^2.$$

This means that the positive real number $\lambda_x / \|x\|^2$ is independent of the choice of the nonzero element x of E . It is strictly positive since, for every nonzero x , the function $f_{x,x}$ is continuous and takes the value $\|x\|^2 > 0$ at e , whence $\|u_x(x)\| = \|f_{x,x}\| > 0$. Denote by c^{-1} this real number.

For every $(x, y) \in E \times E$, it follows that

$$\|v(x \otimes y)\|^2 = \|f_{x,y}\|^2 = \lambda_x \|y\|^2 = c^{-1} \|x\|^2 \|y\|^2 = c^{-1} \|x \otimes y\|^2.$$

Using EVT, V, p. 29, cor. 1, we deduce that the linear map $w = c^{1/2}v$ of $\overline{\mathbf{E}} \widehat{\otimes}_2 \mathbf{E}$ in $L^2_\chi(G)$ is isometric, as desired.

COROLLARY. — Let π an irreducible unitary representation of G and let χ the restriction to Z of its central character. The representation π is square-integrable modulo the center if and only if it is isomorphic to a subrepresentation of the representation $\gamma_{G,\overline{\chi}}$ from G into $L^2_{\overline{\chi}}(G)$ (resp. of the representation $\delta_{G,\chi}$ from G into $L^2_\chi(G)$).

Let us prove the assertion concerning $\gamma_{G,\overline{\chi}}$, the second being proved in a similar manner.

Suppose first that there exists a subrepresentation E of the representation $\gamma_{G,\overline{\chi}}$ isomorphic to π . Since the space E is not zero, there exists a function $f_1 \in \mathcal{H}_{\overline{\chi}}(G)$ whose class $\tilde{f}_1$ that is not orthogonal to E . We have $f_1 \in \mathcal{L}_{\overline{\chi}}^1(G)$. Denote by $\tilde{f}_{1,E}$ the orthogonal projection of $\tilde{f}_1$ on E ; it is a nonzero element of E . Let moreover $\tilde{f}_2 \in E$ non-zero. The map $h : g \mapsto \langle \tilde{f}_{1,E} | \gamma_{G,\overline{\chi}}(g) \tilde{f}_2 \rangle$ is a matrix coefficient of π . For every $g \in G$, we have $h(g) = \langle \tilde{f}_1 | \gamma_{G,\overline{\chi}}(g) \tilde{f}_2 \rangle$ since $\tilde{f}_1 - \tilde{f}_{1,E}$ is orthogonal to E . By Lemma 8 of V, p. 418, the function h belongs to $\mathcal{L}_{\chi}^2(G)$, so Proposition 6 implies that π is square-integrable modulo the center.

Conversely, suppose that π is square-integrable modulo the center. Let $x_0 \in E$ a nonzero vector. By prop. 7 and formula (10), the map $y \mapsto \check{f}_{x_0,y}$ is an injective G -morphism from π in $\gamma_{G,\overline{\chi}}$.

DEFINITION 4. — Let Z be a closed subgroup of C such that C/Z is compact. Let π be an irreducible unitary representation of G which is square-integrable modulo the center. The unique real number $c > 0$ satisfying the property of Proposition 7 is called the formal degree of π relative to Z . It is denoted by $c_Z(\pi)$.

The formal degree depends on the choice of Haar measure on Z . If the Haar measure β_Z on Z is multiplied by a real number $t > 0$, then the measure $\nu_Z = \mu / \beta_Z$ on G/Z is multiplied by t^{-1} , and the formal degree of π is multiplied by t .

The formal degree is characterized by the following property:

PROPOSITION 8 (Orthogonality relations). — Let π be an irreducible unitary representation of G in a Hilbert space E that is square-integrable modulo the center. Then one has

$$c_Z(\pi) \int_{G/Z} \overline{\langle x | \pi(g) x' \rangle} \langle y | \pi(g) y' \rangle d\nu_Z(g) = \overline{\langle x | y \rangle} \langle x' | y' \rangle$$

for all $(x, y, x', y') \in E^4$

Denote by w the morphism of Prop. 7. We have

$$\int_{G/Z} \overline{\langle x | \pi(g) x' \rangle} \langle y | \pi(g) y' \rangle d\nu_Z(g) = \langle f_{x,x'} | f_{y,y'} \rangle$$

and according to loc. cit., it follows that

$$\begin{aligned}\langle f_{x,x'} | f_{y,y'} \rangle &= \frac{1}{c_Z(\pi)} \langle w(x \otimes x') | w(y \otimes y') \rangle \\ &= \frac{1}{c_Z(\pi)} \langle x \otimes x' | y \otimes y' \rangle = \frac{1}{c_Z(\pi)} \langle x | y \rangle_{\bar{E}} \langle x' | y' \rangle_E,\end{aligned}$$

whence the result.

In addition to the previous proposition, we also have the following relations for non-isomorphic square-integrable irreducible representations.

PROPOSITION 9. — Let π_1 and π_2 non-isomorphic irreducible unitary representations of G in Hilbert spaces E_1 and E_2 . One assumes that π_1 and π_2 are square-integrable modulo the center and the restrictions to Z of their central characters coincide. Then

$$\int_{G/Z} \overline{\langle x | \pi_1(g)x' \rangle} \langle y | \pi_2(g)y' \rangle d\nu_Z(g) = 0$$

for all $(x, x', y, y') \in E_1^2 \times E_2^2$

For $i = 1, 2$, denote w_i the morphism from Proposition 7 for the representation π_i . By Lemma 8, b) of V, p. 384 and assertion b) of Proposition 8 of V, p. 394, the image of w_i is contained in the component π_i -isotypic of $\delta_{G,\chi}$. By assertion a) of loc. cit., the image of w_1 is therefore orthogonal to the image of w_2 . Consequently, it follows that $\langle w_1(x \otimes x') | w_2(y \otimes y') \rangle = 0$ for all $(x, x', y, y') \in E_1^2 \times E_2^2$, which is the desired formula.

Remark. — The orthogonality relations of Props. 8 and 9 generalize those of A, VIII, p. 399 (see also the case where G is compact in § 2 of V, p. 457).

V.2.9 Subrepresentations of the regular representation of a commutative group

In this section, G is a locally compact commutative group and μ a Haar measure on G . We denote by $\hat{G}$ the dual group of G (def. 2 of II, p. 201) and by $\hat{\mu}$ the dual Haar measure of μ onto $\hat{G}$ (def. 4 of II, p. 214). The notions of measurability will always be relative to μ and $\hat{\mu}$.

We propose to determine all subrepresentations of the left regular representation γ_G of G in $L^2(G, \mu)$. Since G is commutative, we also have $\delta_G(g) = \gamma_G(g^{-1})$, so these subrepresentations are also the subrepresentations of the right regular representation.

For every measurable subset M of $\widehat{G}$, we denote E_M the set of $f \in L^2(G, \mu)$ such that the Fourier transform $\mathcal{F}_G(f)$ is zero almost everywhere on M (cf. no. 3 of II, p. 210). This is the kernel of the continuous linear map $f \mapsto \varphi_M \mathcal{F}_G(f)$ of $L^2(G, \mu)$ in $L^2(\widehat{G}, \widehat{\mu})$, where φ_M denotes the characteristic function of M (cf. th. 1 of II, p. 215), and it is therefore a closed subspace of $L^2(G, \mu)$.

We say that measurable subsets M and N of $\widehat{G}$ are equal up to a locally negligible set if $(M \cup N) - (M \cap N)$ is locally negligible. Equivalently, measurable subsets M and N are equal up to a locally negligible set if and only if the characteristic functions of M and N are equal in $L^\infty(\widehat{G}, \widehat{\mu})$.

PROPOSITION 10. — a) Let M be a measurable subset of $\widehat{G}$. The space E_M is a subrepresentation of the representation γ_G ;

b) Let M and N be measurable subsets of $\widehat{G}$. We have $E_M = E_N$ if and only if M and N are equal up to a locally negligible set;

c) Any subrepresentation of γ_G is of the form E_M for a measurable subset M of $\widehat{G}$.

Let η the canonical map from G to $\widehat{G}$ (cf. II, p. 216, remark 1). Since E_M is closed, the assertion a) follows from the formulas

$$\gamma_G(x)(f) = \varepsilon_x * f, \quad \mathcal{F}_G(\varepsilon_x * f) = \eta(x) \mathcal{F}_G(f)$$

for $x \in G$ and $f \in L^2(G, \mu)$.

Let M and N be measurable subsets equal up to a locally negligible set. Since φ_M is then equal to φ_N in $L^\infty(\widehat{G}, \widehat{\mu})$, this condition implies that $E_M = E_N$.

Conversely, suppose that M and N are not equal up to a locally negligible set. Possibly after interchanging the roles of M and N , there then exists a compact subset K such that the set $L = K \cap (M - (M \cap N))$ is not negligible. Let $\varphi_L \in L^2(\widehat{G}, \widehat{\mu})$ the class of the characteristic function of L , and set $f = \overline{\mathcal{F}_G}(\varphi_L) \in L^2(G, \mu)$; we have $\mathcal{F}_G(f) = \varphi_L$ (corollary of Theorem 2 in II, p. 220). Then f belongs to E_N since $L \cap N$ is empty, but not in E_M , since $\varphi_M \mathcal{F}_G(f) = \varphi_M \varphi_L = \varphi_L$. We therefore have $E_M \neq E_N$, which proves b).

Let now E be a subrepresentation of γ_G . Let p_E the orthogonal projector from $L^2(G, \mu)$

with image E , and let $q_E = \mathcal{F}_G \circ p_E \circ \mathcal{F}_G^{-1}$. The projector p_E belongs to $\text{Hom}_G(\gamma_G, \gamma_G)$ (prop. 4 of V, p. 383), hence it commutes with $\gamma_G(f)$ for all $f \in L^1(G, \mu)$ (cf. V, p. 401). This means that it commutes with the endomorphisms $\varphi \mapsto f * \varphi$ for $f \in L^1(G, \mu)$. By lemma 4 of V, p. 407, consequently, the endomorphism q_E of $L^2(\widehat{G}, \widehat{\mu})$ commutes with the endomorphism of multiplication by g for every function $g \in \mathcal{C}_0(\widehat{G})$ belonging to the image of the Fourier transform of $L^1(G, \mu)$ in $\mathcal{C}_0(\widehat{G})$ (prop. 14 of II, p. 223). Since the image of the Fourier transform is dense in $\mathcal{C}_0(\widehat{G})$ (corollary of Prop. 5 of II, p. 209), the continuity of the morphism $g \mapsto m_g$ implies that q_E commutes with m_g for every function $g \in \mathcal{C}_0(\widehat{G})$.

By the corollary of Proposition 7 of IV, p. 188, there therefore exists a function $\varphi \in \mathcal{L}^\infty(\widehat{G}, \widehat{\mu})$ such that $q_E = m_\varphi$. Since $q_E = q_E^2$, we have $m_\varphi = m_\varphi^2 = m_{\varphi^2}$, whence $\varphi = \varphi^2$ in $L^\infty(\widehat{G}, \widehat{\mu})$ (prop. 5 of IV, p. 186); this means that φ is equal in $L^\infty(\widehat{G}, \widehat{\mu})$ to the class of the characteristic function of a measurable subset N of $\widehat{G}$. Set $M = \widehat{G} - N$. Let $f \in L^2(G, \mu)$. We have $f \in E$ if and only if $p_E(f) = f$ if and only if $\varphi \mathcal{F}_G(f) = \mathcal{F}_G(f)$, which is equivalent to $f \in E_M$.

Note. — Let χ a character of G . If G is not compact, the χ -isotypic component of the regular representation of G in $L^2(G)$

is trivial. If G is compact, the χ -isotypic component of the regular representation of G has dimension 1, and the function $\chi \in L^2(G)$ is a basis for it.

V.2.10 Unitary representations of the group $\mathbf{R}$

In this section, E denotes a complex Hilbert space. The group $\mathbf{R}$ is equipped with Lebesgue measure.

Lemma 11. — Let u be a partial self-adjoint operator on E . For $t \in \mathbf{R}$ set $\varrho(t) = e^{itu} \in \mathcal{L}(E)$. Then the map ϱ is a unitary representation of $\mathbf{R}$.

The operator $\varrho(t)$ is defined by the universally measurable functional calculus (def. 5 of IV, p. 272); it is an endomorphism of E since the function $x \mapsto e^{itx}$ is bounded on $\text{Sp}(u)$ (prop. 5, a) of IV, p. 275).

According to prop. 5 of IV, p. 275, we have $\varrho(0) = 1_E$, $\varrho(t)^* = \varrho(-t)$ for all $t \in \mathbf{R}$ and $\varrho(t_1 + t_2) = \varrho(t_1)\varrho(t_2)$ for all t_1 and t_2 in $\mathbf{R}$. In particular, the endomorphism $\varrho(t)$ is unitary

for every $t \in \mathbf{R}$. For every $x \in E$, the map from $\mathbf{R}$ to E defined by $x \mapsto \varrho(t)x$ is continuous at $t = 0$ by prop. 6 of IV, p. 276, hence ϱ is a unitary representation of $\mathbf{R}$ in E (V, p. 380, lemma 4).

Lemma 12. — Let ϱ a unitary representation of $\mathbf{R}$ in E . Let D be the set of elements x of E such that the map $\psi_x : t \mapsto \varrho(t)x$ is differentiable at 0. The map u of D into E given by $x \mapsto i^{-1}\psi'_x(0)$ defines a symmetric partial operator on E .

Let $f \in \mathcal{D}(\mathbf{R})$ and $x \in E$. Set $y = \varrho(f)x$. For every $h \in \mathbf{R}$, we have

$$\begin{aligned}\psi_y(h) &= \varrho(h)\varrho(f)x = \varrho(f)\varrho(h)x \\ &= \int_{\mathbf{R}} f(t)\varrho(t+h)x dt = \int_{\mathbf{R}} f(t-h)\psi_x(t)dt.\end{aligned}$$

The function of $\mathbf{R}^2$ to $\mathbf{C}$ defined by $(t, h) \mapsto f(t-h)$ is infinitely differentiable; its derivative with respect to h is the function defined by $(t, h) \mapsto -f'(t-h)$, which is bounded by $\|f'\|_{\infty}$. When $h=0$, this derivative is zero for t outside a compact set. Since $\|\psi_x(t)\| = \|x\|$ for all t , it follows from prop. 2 of IV, p. 197 that the function ψ_y is differentiable at 0 and that its derivative is given by

$$\psi'_y(0) = - \int_{\mathbf{R}} f'(t)\psi_x(t)dt = -\varrho(f')x.$$

We therefore have $y \in D$. Since the space $\mathcal{D}(\mathbf{R})$ is dense in $L^1(\mathbf{R})$, it follows from INT, VIII, p. 139, § 2, no. 7, prop. 10, together with (prop. 4 of IV, p. 202), that the space D is dense in E .

Let x_1 and x_2 in D . We compute

$$\begin{aligned}\langle u(x_1) | x_2 \rangle &= \lim_{h \rightarrow 0} \frac{i}{h} \langle (\varrho(h) - 1_E)x_1 | x_2 \rangle \\ &= \lim_{h \rightarrow 0} \frac{i}{h} \langle x_1 | (\varrho(-h) - 1_E)x_2 \rangle = \langle x_1 | u(x_2) \rangle.\end{aligned}$$

Therefore, the partial operator u is symmetric.

DEFINITION 5. — Let ϱ a unitary representation of $\mathbf{R}$ in E . The symmetric partial operator defined in Lemma 12 is called the infinitesimal generator of ϱ .

THEOREM 1 (Stone). — The mapping σ which to a unitary representation ρ of $\mathbf{R}$ in E associates its infinitesimal generator u defines a bijection from the set of unitary representations

of $\mathbf{R}$ in E onto the set of partial self-adjoint operators on E . The inverse bijection τ associates to a partial self-adjoint operator the unitary representation $t \mapsto e^{itu}$.

Let us first prove a few lemmas.

Lemma 13. — Let ϱ a unitary representation of $\mathbf{R}$ in E and let u its infinitesimal generator.

- a) The domain of u is the set of $x \in E$ such that the mapping $\psi_x: t \mapsto \varrho(t)x$ is differentiable on $\mathbf{R}$;
- b) For every $t \in \mathbf{R}$ and every $x \in \text{dom}(u)$, we have $\varrho(t)x \in \text{dom}(u)$ and $\psi'_x(t) = iu(\varrho(t)x)$;
- c) The partial operator u is essentially self-adjoint.

For every $x \in E$, denote ψ_x the map from $\mathbf{R}$ in E defined by $\psi_x(t) = \varrho(t)x$. For every $t \in \mathbf{R}$ and every $h \in \mathbf{R}$, we have

$$\psi_x(t+h) - \psi_x(t) = \varrho(t)(\psi_x(h) - \psi_x(0)),$$

which proves that $\text{dom}(u)$ is the space of elements $x \in E$ such that ψ_x is differentiable on $\mathbf{R}$ and establishes that $\psi'_x(t) = \varrho(t)\psi'_x(0) = i\varrho(t)(u(x))$ for all $t \in \mathbf{R}$.

Let $x \in E$ and $t \in \mathbf{R}$. We have $\psi_{\varrho(t)x}(s) = \psi_x(s+t)$ for all $s \in \mathbf{R}$. Consequently, one has $\varrho(t)x \in \text{dom}(u)$ if $x \in \text{dom}(u)$, and moreover $u(\varrho(t)x) = i^{-1}\psi'_x(t) = \varrho(t)u(x)$ by a). We then obtain assertion b).

Let us prove c). Let $x \in \text{Ker}(u^* - i1_E)$. Let us prove that $x = 0$. Let $y \in \text{dom}(u)$, and let f be the function on $\mathbf{R}$ defined by $f(t) = \langle \psi_y(t) | x \rangle$ for $t \in \mathbf{R}$. The function f is bounded since $\|\psi_y(t)\| = \|y\|$ for all $t \in \mathbf{R}$; it is differentiable on $\mathbf{R}$ and, for every $t \in \mathbf{R}$, assertion b) implies

$$\begin{aligned} f'(t) &= \langle \psi'_y(t) | x \rangle = \langle iu(\varrho(t)y) | x \rangle \\ &= -i\langle \varrho(t)y | u^*(x) \rangle = -i\langle \psi_y(t) | ix \rangle = f(t) \end{aligned}$$

since $u^*(x) = ix$. We therefore have $f(t) = f(0)e^t$ for all $t \in \mathbf{R}$ (FVR, IV, p. 27). Since f is bounded, the function f is zero and, in particular, we have $\langle y | x \rangle = f(0) = 0$. Since the space $\text{dom}(u)$ is dense in E , we conclude that $x=0$. Similarly, we show that $\text{Ker}(u^* + i1_E)$ is reduced to 0. By Cor. 3 of IV, p. 261, the partial operator u is therefore essentially self-adjoint.

Lemma 14. — Let u be a partial self-adjoint operator on E , and let $\varrho(t) = e^{itu}$ the unitary representation of $\mathbf{R}$ defined by u . The infinitesimal generator of ϱ is equal to u .

Let $x \in \text{dom}(u)$. Set $\psi_x(t) = \varrho(t)x$ for all $t \in \mathbf{R}$. For every nonzero real number h , we have

$$\frac{1}{h}(\psi_x(t+h) - \psi_x(t)) = \left(\frac{1}{h}(e^{ihu} - 1_E) \right) e^{itu}x = \left(\frac{1}{h}(e^{ihu} - 1_E) \right) \varrho(t)x.$$

As h tends to 0, we have

$$\frac{1}{h}(e^{iht} - 1) \rightarrow it$$

for all $t \in \mathbf{R}$. Moreover,

$$\left| \frac{1}{h}(e^{iht} - 1) \right| = |t| \left| \frac{1}{h} \int_0^h e^{its} ds \right| \leq |t|.$$

It then follows from prop. 6 of IV, p. 276 that the function ψ_x is differentiable on $\mathbf{R}$ and satisfies $\psi'_x(t) = iu(\varrho(t)x)$ for all $t \in \mathbf{R}$. Consequently, the domain of u is contained in the set of $x \in E$ such that ψ_x is differentiable at 0 and that we then have $\psi'_x(0) = iu(x)$. This means by definition that the infinitesimal generator of ϱ is an extension of u . These two operators are therefore equal since they are symmetric and u is self-adjoint (IV, p. 238, remark 5).

Lemma 15. — Let ϱ a unitary representation of $\mathbf{R}$ in E . Then the infinitesimal generator u of ϱ is self-adjoint and $\varrho(t) = e^{itu}$ for all $t \in \mathbf{R}$.

By Lemma 13(c), the partial operator u is essentially self-adjoint. Its closure $\bar{u}$ is therefore a self-adjoint operator. Denote π the unitary representation of $\mathbf{R}$ defined by $\pi(t) = e^{it\bar{u}}$ (lemma 11).

For every $x \in E$, denote ψ_x (resp. $\tilde{\psi}_x$) the map from $\mathbf{R}$ in E defined by $\psi_x(t) = \varrho(t)x$ (respectively by $\tilde{\psi}_x(t) = \pi(t)x$).

By Lemma 13, parts (a) and (b), the space $\text{dom}(u)$ is the subspace of E consisting of the elements $x \in E$ such that the mapping ψ_x is differentiable on $\mathbf{R}$, and for all $x \in \text{dom}(u)$ and every $t \in \mathbf{R}$, we have $\psi'_x(t) = iu(\psi_x(t))$. Likewise, the space $\text{dom}(\bar{u})$ is the subspace of E consisting of the elements $x \in E$ such that the mapping $\tilde{\psi}_x$ is differentiable on $\mathbf{R}$, and for all

$x \in \text{dom}(\bar{u})$ and every $t \in \mathbf{R}$, we have $\widetilde{\psi}'_x(t) = i\bar{u}(\widetilde{\psi}_x(t))$.

Let $x \in \text{dom}(u) \subset \text{dom}(\bar{u})$. Set $f = \psi_x - \widetilde{\psi}_x$. This is a differentiable function from $\mathbf{R}$ into E . For every $t \in \mathbf{R}$, it follows that

$$f'(t) = iu(\psi_x(t)) - i\bar{u}(\widetilde{\psi}_x(t)) = i\bar{u}(f(t))$$

since $\psi_x(t) \in \text{dom}(u)$ and $u \subset \bar{u}$. Set $g = \|f\|^2$; it is a differentiable map from $\mathbf{R}$ in $\mathbf{R}$ such that $g(0) = 0$. For $t \in \mathbf{R}$, we obtain by FVR, I, p. 28, prop. 2

$$\begin{aligned} g'(t) &= \langle f'(t)|f(t) \rangle + \langle f(t)|f'(t) \rangle \\ &= \langle i\bar{u}(f(t))|f(t) \rangle + \langle f(t)|i\bar{u}(f(t)) \rangle = 0 \end{aligned}$$

since $\bar{u}$ is self-adjoint. Hence $f = 0$, so $\varrho(t)x = \pi(t)x$ for all $t \in R$. The continuous endomorphisms $\varrho(t)$ and $\pi(t)$ of E coincide on $\text{dom}(u)$, and are therefore equal for all $t \in R$. Thus, we have $\pi = \varrho$; since $\bar{u}$ is the infinitesimal generator of π (Lemma 14), we have $\bar{u} = u$, which proves that u is self-adjoint.

We can now prove Theorem 1.

The maps σ and τ are well-defined (Lemma 15 and Lemma 11, respectively).

Let ϱ a unitary representation of $\mathbf{R}$. Denote by u its infinitesimal generator. The relation $\varrho(t) = e^{itu}$ for all $t \in \mathbf{R}$ (lemma 15) proves that $\tau \circ \sigma$ is the identity map.

Let u be a partial self-adjoint operator on E . Lemma 14 shows that the infinitesimal generator of the unitary representation $t \mapsto e^{itu}$ is equal to u , hence $\sigma \circ \tau$ is the identity map.

Let u be a partial self-adjoint operator on E and let $\varrho(t) = e^{itu}$ for $t \in R$ the unitary representation of R in E associated with it.

Let $x \in \text{dom}(u)$. The equation $\partial_t \varrho(t)x = iu(\varrho(t)x)$ which is then satisfied (Lemma 13, b)) is called the Schrödinger equation.

COROLLARY. — Let ϱ a unitary representation of $\mathbf{R}$ in a Hilbert space E . There exists a locally compact space X , a positive measure μ on X and a continuous function g on X with real values such that ϱ is isomorphic to the representation π of $\mathbf{R}$ in $L^2(X, \mu)$ defined by $\pi(t)f = e^{itg}f$ for all $t \in \mathbf{R}$ and every $f \in L^2(X, \mu)$.

Let u the self-adjoint operator on E such that $\varrho(t) = e^{itu}$ for all $t \in \mathbf{R}$ (Theorem 1). There

exists a locally compact space X , a positive measure μ on X , an isometric isomorphism θ of $L^2(X, \mu)$ on E and a continuous function g on X with real values such that $u = \theta \circ m_g \circ \theta^{-1}$ (theorem 1 of IV, p. 266). The assertion follows from the formula $e^{itu} = \theta \circ e^{itm_g} \circ \theta^{-1} = \theta \circ m_{e^{itg}} \circ \theta^{-1}$ (lemma 4 of IV, p. 269).

Remark. — Suppose that u is an endomorphism of E . The unitary representation of $\mathbf{R}$ in E defined by $\varrho(t) = e^{itu}$ then verifies the inequality $\|\varrho(t) - 1_E\| \leq |t|\|u\|$ for all $t \in \mathbf{R}$, and the map ϱ from $\mathbf{R}$ into the Banach space $\mathcal{L}(E)$ is the unique solution of the linear differential equation

$$\frac{1}{i} \frac{d\varrho}{dt} = u \circ \varrho$$

(cf. FVR, IV, p. 26, §6).

Example. — Let ϱ the regular representation of $\mathbf{R}$ in $L^2(\mathbf{R})$. The infinitesimal generator of ϱ is the closure of the differential operator with domain $\mathcal{D}(\mathbf{R})$ defined by $f \mapsto -if'$.

V.3 FUNCTIONS OF POSITIVE TYPE

In this section, all vector spaces, as well as all Hilbert spaces and algebras considered, are over $\mathbf{C}$, unless otherwise stated.

V.3.1 Universally positive kernels

In this issue, X is a separated topological space.

Theorem 1. — Let $f \in \mathcal{C}(X \times X)$. The following conditions are equivalent:

- (i) For every compact subset Y of X and every positive measure μ on Y , the endomorphism of $L^2(Y, \mu)$ defined by the kernel $f|_{(Y \times Y)}$ (def. 1 of III, p. 29) is positive, in other words, we have

$$\int_{Y \times Y} \overline{h(x)} h(y) f(x, y) d(\mu \otimes \mu)(x, y) \geq 0$$

for all $h \in \mathcal{L}^2(Y, \mu)$;

(ii) For every integer $n \in \mathbb{N}$, every family $(x_i)_{0 \leq i \leq n}$ in X and any family $(t_i)_{0 \leq i \leq n}$ of complex numbers, we have

$$\sum_{i=0}^n \sum_{j=0}^n \bar{t}_i t_j f(x_i, x_j) \geq 0;$$

(iii) There exists a complex Hilbert space E and a continuous map $g: X \rightarrow E$ of total image such that $f(x, y) = \langle g(x) | g(y) \rangle$ for all x and y in X ;

(iv) There exists a complex Hilbert space E and a continuous map $g: X \rightarrow E$ such that $f(x, y) = \langle g(x) | g(y) \rangle$ for all x and y in X .

It is clear that (iii) implies (iv), and one sees that (i) implies (ii) by considering the discrete measure μ which is the image of the counting measure on $\{0, \dots, n\}$ by the mapping $i \mapsto x_i$ and the function h such that

$$h(x) = \sum_{\substack{0 \leq i \leq n \\ x_i = x}} t_i.$$

Let us prove that (iv) implies (i). Suppose there exists a complex Hilbert space E and a continuous map $g: X \rightarrow E$ such that $f(x, y) = \langle g(x) | g(y) \rangle$ for all $(x, y) \in X \times X$. Let Y be a compact subset of X , μ a positive measure on Y and $h \in \mathcal{L}^2(Y, \mu)$. We have

$$\begin{aligned} & \int_{Y \times Y} \overline{h(x)} h(y) f(x, y) d(\mu \otimes \mu)(x, y) \\ &= \int_{Y \times Y} \overline{h(x)} h(y) \langle g(x) | g(y) \rangle d(\mu \otimes \mu)(x, y) \\ &= \left\langle \int_Y h(x) g(x) d\mu(x) \mid \int_Y h(y) g(y) d\mu(y) \right\rangle \geq 0 \end{aligned}$$

by INT, V, p. 97, § 8, no. 4, prop. 9.

Let us finally prove that (ii) implies (iii). Let $\tilde{E}$ be the space of complex measures with finite support on X . For μ_1 and μ_2 in $\tilde{E}$ we set

$$\langle \mu_1 | \mu_2 \rangle = \int_{X \times X} f(x, y) (\bar{\mu}_1 \otimes \mu_2)(x, y).$$

The sesquilinear form thus defined on $\tilde{E}$ is a positive Hermitian form. Indeed, let $\mu \in \tilde{E}$; there exists a finite family $(x_i)_{0 \leq i \leq n}$ in X and complex numbers $(t_i)_{0 \leq i \leq n}$ such that $\mu =$

$\sum_{i=0}^n t_i \varepsilon_{x_i}$. We then have

$$\langle \mu | \mu \rangle = \sum_{i=0}^n \sum_{j=0}^n \bar{t}_i t_j f(x_i, x_j) \geq 0$$

by hypothesis. One defines $\tilde{g}: \mathbf{X} \rightarrow \tilde{\mathbf{E}}$ by $\tilde{g}(x) = \varepsilon_x$. The image of the map $\tilde{g}$ generates $\tilde{\mathbf{E}}$. Moreover, for every $(x, y) \in \mathbf{X} \times \mathbf{X}$, one has on the one hand $f(x, y) = \langle \tilde{g}(x) | \tilde{g}(y) \rangle$ and on the other hand

$$\|\tilde{g}(x) - \tilde{g}(y)\|^2 = f(x, x) + f(y, y) - f(x, y) - f(y, x),$$

which implies that $\tilde{g}$ is continuous since f is continuous.

Let E be the separated completion Hilbert space of $\tilde{E}$ (EVT, V, p. 8, cor. of prop. 4) and $g: X \rightarrow E$ the composition of $\tilde{g}$ and of the canonical map $\tilde{E} \rightarrow E$. Then the map g is continuous, its image is total in E , and one has $f(x, y) = \langle g(x) | g(y) \rangle$ for all $(x, y) \in X \times X$.

The method used to prove that (ii) implies (iii) is called the Gelfand-Naimark-Segal construction.

DEFINITION 1. — One says that a function $f \in \mathcal{C}(X \times X)$ is a universally positive kernel on X if it satisfies the equivalent conditions of Theorem 1.

If f is a universally positive kernel on X , a pair (E, g) satisfying condition (iv) of loc. cit. is called a Hilbertian realization of f ; if condition (iii) is satisfied, one says that it is a cyclic Hilbertian realization.

We denote $\text{Noy}_+(X)$ the set of universally positive kernels on X .

Let X' be a separated topological space and $h: X \rightarrow X'$ a continuous map. The map $f \mapsto f \circ (h, h)$ of $\mathcal{C}(X' \times X')$ in $\mathcal{C}(X \times X)$ induces by passage to subspaces a map $\text{Noy}_+(X') \rightarrow \text{Noy}_+(X)$.

PROPOSITION 1. — Let f be a universally positive kernel on X . Let (E_1, g_1) and (E_2, g_2) be Hilbertian realizations of f . One supposes that (E_1, g_1) is a cyclic Hilbertian realization. There then exists a unique continuous linear map $u: E_1 \rightarrow E_2$ such that $g_2 = u \circ g_1$. This map is isometric. If (E_2, g_2) is also cyclic, then u is an isomorphism.

The uniqueness of u follows from the fact that the image of g_1 is total in E_1 . Let $F = \mathbf{C}^{(X)}$

and let $(e_x)_{x \in X}$ the canonical basis of F . For $j = 1$ and $j = 2$, denote u_j the linear map from F in E_j determined by $u_j(e_x) = g_j(x)$, and denote by F_j its image; the space F_1 is dense in E_1 . Let $t = \sum t_x e_x$ be an element of F . We have

$$\begin{aligned} \|u_1(t)\|^2 &= \sum_{x,y} \bar{t}_x t_y \langle g_1(x) | g_1(y) \rangle = \sum_{x,y} \bar{t}_x t_y f(x,y) \\ &= \sum_{x,y} \bar{t}_x t_y \langle g_2(x) | g_2(y) \rangle = \|u_2(t)\|^2. \end{aligned}$$

Consequently, there exists a linear isometric map v from F_1 in F_2 such that $u_2 = v \circ u_1$, and in particular $g_2(x) = v(g_1(x))$ for all $x \in X$. Since the image of g_1 is total in E_1 , this map extends to an isometric linear map u from E_1 in E_2 such that $g_2 = u \circ g_1$.

By Lemma 8 of I, p. 107, the image of u is closed in E_2 . If (E_2, g_2) is a cyclic Hilbert realization, the image of u is also dense in E_2 and u is therefore an isomorphism.

PROPOSITION 2. — The set $\text{Noy}_+(X)$ is a self-adjoint cone in the space $\mathcal{C}(X \times X)$; it is stable under product and it is closed when one equips $\mathcal{C}(X \times X)$ with the topology of simple convergence.

It is elementary that if $f \in \text{Noy}_+(X)$, then $tf \in \text{Noy}_+(X)$ for every real number $t \geq 0$, and $\bar{f} \in \text{Noy}_+(X)$.

If (E_1, g_1) and (E_2, g_2) are Hilbertian realizations of universally positive kernels f_1 and f_2 on X , then the pair $(E_1 \oplus E_2, g_1 + g_2)$ (resp. the pair $(E_1 \widehat{\otimes} E_2, g_1 \otimes g_2)$) is a Hilbertian realization of $f_1 + f_2$ (resp. of $f_1 f_2$); they are therefore universally positive kernels.

The characterization (ii) of $\text{Noy}_+(X)$ (theorem 1) implies that this set is closed in $\mathcal{C}(X \times X)$ endowed with the topology of simple convergence.

V.3.2 Further remarks on the holomorphic functional calculus

For every subset X of $\mathbb{C}$, we denote by X^* the image of X under complex conjugation. Let U be an open subset of $\mathbb{C}$ and $g : U \rightarrow \mathbb{C}$ a holomorphic function. The function $f^* : z \mapsto \overline{g(\bar{z})}$ is then defined and holomorphic on U^* . The map $f \mapsto f^*$ is a continuous bijection from $\mathcal{O}(U)$ to $\mathcal{O}(U^*)$ such that $(f_1 f_2)^* = f_1^* f_2^*$ and $(f_1 + f_2)^* = f_1^* + f_2^*$ for f_1 and f_2 in $\mathcal{O}(U)$.

Let C be a compact subset of $\mathbb{C}$. The maps $f \mapsto f^*$ of $\mathcal{O}(U)$ in $\mathcal{O}(U^*)$ as U ranges over the open subsets of $\mathbb{C}$ containing C , induce a continuous bijection of the space $\mathcal{O}(C)$ to $\mathcal{O}(C^*)$ (I, p.

49, no. 1), which is also denoted $f \mapsto f^*$ and which satisfies $(f_1 f_2)^* = f_1^* f_2^*$ and $(f_1 + f_2)^* = f_1^* + f_2^*$ for f_1 and f_2 in $\mathcal{O}(C)$.

PROPOSITION 3. — Let A be a unital involutive Banach algebra. Let $a \in A$. The spectrum of a^* is the image $\text{Sp}_A(a)^*$ of $\text{Sp}_A(a)$ by complex conjugation. For every $f \in \mathcal{O}(\text{Sp}_A(a))$, we have $f(a)^* = f^*(a^*)$.

The first assertion follows from I, p. 97. By the preceding, the map φ of $\mathcal{O}(\text{Sp}_A(a))$ into A defined by $f \mapsto (f^*(a^*))^*$ is a continuous unital morphism of $\mathcal{O}(\text{Sp}_A(a))$ in A such that the image of the germ in a neighborhood of $\text{Sp}_A(a)$ of the identity function of $\mathbb{C}$ is equal to a . Consequently, φ is the map $f \mapsto f(a)$ of the holomorphic functional calculus (I, p. 74, th. 5).

Lemma 1. — Let D be the open unit disk in $\mathbb{C}$. There exists a unique holomorphic function f defined on D such that $f(z)^2 = 1 - z$ for all $z \in D$ and $f(0) = 1$. We have $f^* = f$.

The radius of convergence of the power series

$$\sum_{n=0}^{+\infty} \binom{1/2}{n} (-z)^n$$

is 1 and its sum f defines a holomorphic function on D (VAR, R1, p. 27, 3.2.9) taking the value 1 at 0. It satisfies $f(x) = \sqrt{1-x}$ for all $x \in D \cap \mathbb{R}$ (FVR, III, p. 19), hence $f(z)^2 = 1 - z$ for $z \in D$ since the difference $f(z)^2 - (1 - z)$ is a holomorphic function all of whose successive derivatives vanish at 0 (VAR, R1, p. 27, 3.2.5). By definition, one verifies that $f^* = f$.

Let g a holomorphic function on D such that $g(z)^2 = 1 - z$ for all $z \in D$ and such that $g(0) = 1$. The function g does not vanish and the continuous function f/g on D takes values in $\{-1, 1\}$; since D is connected and $f(0) = g(0)$, we have $f = g$.

V.3.3 Positive linear forms

DEFINITION 2. — Let A be an involutive algebra. A linear form λ on A is said to be positive if $\lambda(a^*a) \in \mathbf{R}_+$ for all $a \in A$.

If A is an involutive Banach algebra, we denote by A'_+ the set of continuous positive linear forms on A .

Let A be an involutive Banach algebra. The set A'_+ is a pointed convex cone in the real vector space of $\mathbb{C}$ -linear forms on A .

Lemma 2. — Let A be an involutive Banach algebra and let λ be a positive linear form on A .

a) For all a and b in A , we have $\lambda(a^*b) = \overline{\lambda(b^*a)}$ and

$$|\lambda(a^*b)|^2 \leq \lambda(a^*a) \lambda(b^*b);$$

b) If A is unital, then the linear form λ is continuous and its norm is equal to $\lambda(1)$.

The map $(a, b) \mapsto \lambda(a^*b)$ is a positive Hermitian form on A ; it therefore satisfies $\lambda(a^*b) = \overline{\lambda(b^*a)}$ and $|\lambda(a^*b)|^2 \leq \lambda(a^*a) \lambda(b^*b)$ for all a and b in A (EVT, V, p. 2, remark and p. 3, prop. 2).

Let us prove b). Let $a \in A$ be a Hermitian element of norm < 1 . The spectral radius of a is less than $\|a\|$, hence the spectrum of a is contained in the open unit disk D of $\mathbb{C}$ (th. 1 of I, p. 24). Let f be a holomorphic function on D such that $f(z)^2 = 1 - z$ for all $z \in D$ (Lemma 1 of V, p. 435). Apply the holomorphic functional calculus to the element a and to the function f . The element $b = f(a)$ satisfies $b^2 = 1 - a$ (I, p. 74, Thm. 5). By Prop. 3 of V, p. 435, we also have $f(a)^* = f^*(a^*) = f(a)$, hence b is Hermitian. It then follows that $\lambda(1 - a) = \lambda(b^*b) \geq 0$, whence $\lambda(a) \leq \lambda(1)$.

Now let $b \in A$ be of norm < 1 . The element b^*b is Hermitian and $\|b^*b\| < 1$, so applying a) with $a = 1$, we find

$$|\lambda(b)|^2 \leq \lambda(1) \lambda(b^*b) \leq \lambda(1)^2,$$

with equality if $b = 1$. The linear form λ is therefore continuous, and its norm is equal to $\lambda(1)$. The assertion b) is proved.

Example. — Let X be a compact topological space and let A be the star algebra $\mathcal{C}(X)$. Positive linear forms on A are identified with positive measures on X (INT, III, p. 52, § 1, n° 6, th. 1).

Lemma 3. — Let A be a unital involutive Banach algebra admitting an approximate unit (I, p. 120, def. 7). Let λ be a continuous positive linear form on A .

- a) For all a in A , we have $\lambda(a^*) = \overline{\lambda(a)}$ and $|\lambda(a)|^2 \leq \|\lambda\| \lambda(a^*a)$;
- b) Let $\tilde{A}$ be the involutive algebra obtained from A by adjoining a unit element, and let e be its unit element. There exists a continuous positive linear form $\tilde{\lambda}$ on $\tilde{A}$ which extends λ and such that $\tilde{\lambda}(e) = \|\lambda\|$;
- c) For all a and b in A , we have $|\lambda(b^*ab)| \leq \|a\| \lambda(b^*b)$.

Let us prove a). Let $\mathfrak{F}$ be an approximate unit of A . Let $a \in A$. Using Lemma 2, a) and the definition of an approximate unit, one finds

$$\lambda(a^*) = \lim_{f, \mathfrak{F}} \lambda(fa^*) = \lim_{f, \mathfrak{F}} \overline{\lambda(af^*)} = \lim_{f, \mathfrak{F}} \overline{\lambda((fa^*)^*)} = \overline{\lambda(a)},$$

whence (loc. cit.)

$$|\lambda(a)|^2 = \lim_{f, \mathfrak{F}} |\lambda(fa)|^2 \leq \lambda(a^*a) \limsup_{f, \mathfrak{F}} \lambda(ff^*) \leq \|\lambda\| \lambda(a^*a),$$

since $\mathfrak{F}$ is a filter on the unit ball of A .

To prove b), one may assume that λ is not zero, and then that $\|\lambda\| = 1$. For $a \in A$ and $z \in \mathbb{C}$, set $\tilde{\lambda}(a + z \cdot e) = \lambda(a) + z$. The map $\tilde{\lambda}$ is a continuous linear form on $\tilde{A}$ which extends λ and satisfies $\tilde{\lambda}(e) = 1$. It is positive: for all $a \in A$ and $z \in \mathbb{C}$, according to a), one computes

$$\begin{aligned} \tilde{\lambda}((a + z \cdot e)^*(a + z \cdot e)) &= \lambda(a^*a) + \bar{z}\lambda(a) + z\lambda(a^*) + |z|^2 \\ &= |z + \lambda(a)|^2 + \lambda(a^*a) - |\lambda(a)|^2 \geq 0. \end{aligned}$$

Finally, let b be an element of A . The map $a \mapsto \tilde{\lambda}(b^*ab)$ is a positive linear form on the unital involutive Banach algebra $\tilde{A}$. It is therefore continuous with norm equal to its value at e (Lemma 2, b), which is equal to $\lambda(b^*b)$, whence assertion c).

PROPOSITION 4. — Let A be an involutive Banach algebra.

- a) For every Hilbert space E , every morphism of involutive algebras $\varphi: A \rightarrow \mathcal{L}(E)$ and every vector $x \in E$, the linear form defined on A by $\lambda(a) = \langle x | \varphi(a)x \rangle$ is a continuous positive linear form;
- b) Let $\lambda \in A'_+$. The map f from $A \times A$ to $\mathbb{C}$ defined by $f(a, b) = \lambda(a^*b)$ for all $(a, b) \in A \times A$

is a universally positive kernel on A .

Let us prove a). For every $a \in A$, we have

$$\lambda(a^*a) = \langle x | \varphi(a^*a)x \rangle = \|\varphi(a)x\|^2 \geq 0$$

hence λ is a positive linear form on A ; it is continuous since φ is continuous (Prop. 2 of I, p. 104).

Let us prove b). The function f is continuous. For every $n \in \mathbb{N}$, every family $(a_i)_{0 \leq i \leq n}$ in A and every family $(t_i)_{0 \leq i \leq n}$ of complex numbers, we have

$$\sum_{i=0}^n \sum_{j=0}^n \bar{t}_i t_j f(a_i, a_j) = \lambda \left(\left(\sum_{i=0}^n t_i a_i \right)^* \left(\sum_{j=0}^n t_j a_j \right) \right) \geq 0,$$

whence the result (Def. 1 of V, p. 433).

DEFINITION 3. — Let A be an involutive Banach algebra. Let $\lambda \in A'_+$ be a continuous positive linear form on A . A Hilbertian realization of λ is a triplet (E, x, φ) consisting of a Hilbert space E , an element x of E , and a morphism of involutive algebras φ from A into $\mathcal{L}(E)$ such that $\lambda(b^*a) = \langle \varphi(b)x | \varphi(a)x \rangle$ for all $(a, b) \in A^2$.

If the set of elements $\varphi(a)x$ for $a \in A$ is total in E , we say that (E, x, φ) is a cyclic Hilbert-space realization of λ .

PROPOSITION 5. — Let A be an involutive Banach algebra admitting an approximate unit. Every continuous positive linear form on A admits a cyclic Hilbert realization (E, x, φ) . If A is unital, then there exists such a realization where the morphism φ is unital.

We may assume that A is a unital algebra (Lemma 3, b)).

Let λ be a positive continuous linear form on A . Let (E, g) be a cyclic Hilbertian realization of the universally positive kernel f on A defined by $f(a, b) = \lambda(a^*b)$ (prop. 4 and th. 1). The map g is continuous, its image is total in E , and we have $\lambda(a^*b) = \langle g(a) | g(b) \rangle$ for all $(a, b) \in A^2$. Set $x = g(1) \in E$.

The map g from A into E is linear: indeed, for $(a, b, c) \in A^3$ and $(s, t) \in \mathbb{C}^2$, it follows that

$$\begin{aligned}\langle g(c) | g(sa + tb) \rangle &= \lambda(c^*(sa + tb)) \\ &= s\lambda(c^*a) + t\lambda(c^*b) = \langle g(c) | sg(a) + tg(b) \rangle,\end{aligned}$$

whence the assertion since the image of g is total in E .

In particular, the image F of g is a dense vector subspace of E . The kernel of g is a left ideal of A : if $g(b) = 0$, then for all a and c in A , it follows that

$$\langle g(ab) | g(c) \rangle = \lambda((ab)^*c) = \lambda(b^*(a^*c)) = \langle g(b) | g(a^*c) \rangle = 0,$$

hence $g(ab) = 0$ since F is dense in E .

Let $a \in A$. Since the kernel of g is a left ideal of A , there exists a linear map $\tilde{\varphi}(a) : F \rightarrow F$ such that $\tilde{\varphi}(a)(g(b)) = g(ab)$ for all $b \in A$. Moreover, for every $b \in A$, it follows that

$$\|\tilde{\varphi}(a)(g(b))\|^2 = \|g(ab)\|^2 = \lambda(b^*a^*ab) \leq \|a^*a\|\lambda(b^*b) \leq \|a\|^2\|g(b)\|^2$$

(lemma 3, c)), hence $\tilde{\varphi}(a)$ is continuous and of norm $\leq \|a\|$. Consequently, there exists a unique endomorphism $\varphi(a) \in \mathcal{L}(E)$ which induces $\tilde{\varphi}(a)$ by passage to the subspaces.

Let a and b be in A . We have

$$(\varphi(a) \circ \varphi(b))(g(c)) = g(abc) = \varphi(ab)(g(c))$$

for all c in A , hence $\varphi(ab) = \varphi(a) \circ \varphi(b)$ since F is dense in E .

The map φ from A into $\mathcal{L}(E)$ is linear: indeed, for every $(a, b) \in A^2$ and every $(s, t) \in \mathbb{C}^2$, we have

$$\varphi(sa + tb)(g(c)) = g((sa + tb)c) = (s\varphi(a) + t\varphi(b))g(c)$$

for all $c \in A$, whence $\varphi(sa + tb) = s\varphi(a) + t\varphi(b)$ since F is dense in E . Since $\|\varphi(b)\| \leq \|b\|$ for all $b \in A$, the map φ is continuous. We also have $\varphi(1) = 1_E$ since $\varphi(1)(g(a)) = g(a)$ for all $a \in A$ and since F is dense in E .

Finally, let a, b, c be in A . We have

$$\begin{aligned}\langle g(c) \mid \varphi(a^*)(g(b)) \rangle &= \langle g(c) \mid g(a^*b) \rangle = \lambda(c^*a^*b) \\ &= \langle g(ac) \mid g(b) \rangle = \langle \varphi(a)(g(c)) \mid g(b) \rangle\end{aligned}$$

whence $\varphi(a^*) = \varphi(a)^*$ since F is dense in E .

In conclusion, the map φ is a continuous morphism of involutive algebras from A into $\mathcal{L}(E)$ and $g(a) = \varphi(a)x$ for all $a \in A$; consequently, the triple (E, x, φ) is a cyclic Hilbert realization of λ .

PROPOSITION 6. — Let A be an involutive Banach algebra and let λ be a continuous positive linear form on A . Let (E_1, x_1, φ_1) and (E_2, x_2, φ_2) be Hilbertian realizations of λ . Suppose that (E_1, x_1, φ_1) is a cyclic Hilbertian realization.

- a) There exists a unique continuous linear map u from E_1 to E_2 which is a morphism of representations from φ_1 into φ_2 (I, p. 11) and which satisfies $u(\varphi_1(a)x_1) = \varphi_2(a)x_2$ for all $a \in A$;
- b) The linear map u is isometric;
- c) If (E_2, x_2, φ_2) is cyclic, then u is an isomorphism;
- d) If (E_2, x_2, φ_2) is cyclic and if A is unital, then $u(x_1) = x_2$.

For $j \in \{1, 2\}$, define $\gamma_j: A \rightarrow E_j$ by $\gamma_j(a) = \varphi_j(a)x_j$ for all $a \in A$. By definition, the pairs (E_j, γ_j) are Hilbert realizations of the universally positive kernel f on A defined by $(a, b) \mapsto \lambda(a^*b)$. The Hilbert realization (E_1, γ_1) is cyclic; by prop. 1 of V, p. 434, there therefore exists a unique continuous linear map $u: E_1 \rightarrow E_2$ such that $\gamma_2 = u \circ \gamma_1$, and this map is isometric. Moreover, if (E_2, x_2, φ_2) is also cyclic, then u is an isomorphism. To prove a) , b) and c) , it suffices to prove that u is a morphism of representations from φ_1 into φ_2 .

Let $a \in A$. For every $b \in A$, we have

$$\begin{aligned}(u \circ \varphi_1(a))(\gamma_1(b)) &= (u \circ \varphi_1(ab))x_1 = (u \circ \gamma_1)(ab) \\ &= \gamma_2(ab) = \varphi_2(a)(\gamma_2(b)) = \varphi_2(a)(u(\gamma_1(b))),\end{aligned}$$

therefore the continuous linear maps $u \circ \varphi_1(a)$ and $\varphi_2(a) \circ u$ coincide on the subspace of E_1 generated by the image of γ_1 ; this subspace is dense in E_1 by hypothesis, hence $u \circ \varphi_1(a) =$

$\varphi_2(a) \circ u$.

Let us finally prove d). Suppose therefore that A is unital and that (E_2, x_2, φ_2) is cyclic. There exists a cyclic Hilbert-space realization (E_3, x_3, φ_3) of λ such that φ_3 is a unital morphism (prop. 5). Let $j \in \{1, 2\}$. Applying the preceding to (E_j, x_j, φ_j) and (E_3, x_3, φ_3) , one sees that there exists an isometric isomorphism $\tilde{u}_j$ from E_j into E_3 such that $\tilde{u}_j \circ \varphi_j(a) = \varphi_3(a) \circ \tilde{u}_j$ for all $a \in A$. Taking $a = 1_A$, one sees that φ_j is unital. One then has $x_2 = \varphi_2(1_A)x_2 = u(\varphi_1(1_A)x_1) = u(x_1)$.

Remark. — Keep the notations of the proposition. It is possible that $u(x_1)$ is different from x_2 (exercise 3, b) of V, p. 493). However, the triple $(E_2, u(x_1), \varphi_2)$ is also a Hilbertian realization of λ .

V.3.4 Representations of stellar algebras

Let A be a stellar algebra. Denote by A_+ the closed convex cone of positive elements of A (def. 6 of I, p. 115). A linear form λ on A is positive if and only if $\lambda(A_+) \subset \mathbf{R}_+$ (I, p. 118, th. 2).

PROPOSITION 7. — Let A be a stellar algebra. Every positive linear form λ on A is continuous.

Let us first show that λ is bounded on the intersection of A_+ and the unit ball of A . Suppose this is not the case. Then there exists a sequence $(x_n)_{n \geq 1}$ in A_+ such that $\|x_n\| \leq 1$ and $\lambda(x_n) \geq n$ for every integer $n \geq 1$. The series with general term $n^{-2}x_n$ converges to an element x of A (cf. TG, IV, p. 33, example 3). For every integer $N \geq 1$, since

$$\sum_{n=N+1}^{+\infty} \frac{1}{n^2} x_n \in A_+,$$

(I, p. 116, prop. 14) it follows that

$$\sum_{n=1}^N \frac{1}{n} \leq \sum_{n=1}^N \frac{1}{n^2} \lambda(x_n) = \lambda(x) - \lambda\left(\sum_{n=N+1}^{+\infty} \frac{1}{n^2} x_n\right) \leq \lambda(x),$$

which is absurd (TG, IV, p. 33, example 4).

Since every element of the unit ball of A is a linear combination with coefficients bounded by 1 of at most four positive elements of A of norm ≤ 1 (I, p. 96, lemma 2 and formula (4) of I,

p. 117), we conclude that λ is continuous.

There exist involutive Banach algebras which admit positive linear forms that are not continuous (exercise 3, a) of V, p. 493).

PROPOSITION 8. — Let A be a stellar algebra and let a be a nonzero element of A . There exists a positive linear form $\lambda \in A'_+$ such that $\lambda(a^*a) > 0$.

Consider the real Banach space A_h of Hermitian elements of A . The set A_+ is a closed convex pointed salient cone in A_h (prop. 14 of I, p. 116). The element a^*a is positive (th. 2 of I, p. 118) and nonzero, hence the Hermitian element $-a^*a$ is not positive. By EVT, II, p. 42, cor. 5, there exists a continuous real linear form $\lambda \in A'_h$ such that $\lambda(-a^*a) < 0$ and $\lambda(A_+) \subset \mathbf{R}_+$. The linear form λ extends to a Hermitian $\mathbf{C}$ -linear form on A (cf. I, p. 98), which is positive and possesses the required property.

THEOREM 2 (Gelfand–Naimark). — Let A be a stellar algebra. There exists a Hilbert space E and an isometric morphism of involutive algebras φ from A into $\mathcal{L}(E)$. If A is unital, there exists such a unital morphism.

For every $b \in A - \{0\}$, let λ_b be a continuous positive linear form on A such that $\lambda_b(b^*b) > 0$ (prop. 8). Since A admits an approximate unit (I, p. 121, prop. 18), there exists a Hilbertian realization (E_b, x_b, φ_b) of λ_b (prop. 5). If A is unital, one may suppose that φ_b is unital (loc. cit.). We have $\|\varphi_b(b)\|^2 = \lambda_b(b^*b) \neq 0$.

Let E be the external Hilbert sum of the spaces E_b for b belonging to $A - \{0\}$. For every $a \in A$, denote by $\varphi(a) \in \mathcal{L}(E)$ the unique continuous linear map whose restriction to E_b coincides with $\varphi_b(a)$ for all $b \in A - \{0\}$. The map $a \mapsto \varphi(a)$ is a morphism of involutive algebras; it is injective, hence isometric (prop. 9 of I, p. 112), and satisfies $\varphi(1) = 1_E$ if A is unital. This concludes the proof.

The category whose objects are the stellar algebras and whose morphisms are the morphisms of involutive algebras is therefore equivalent to the category whose objects are the closed involutive subalgebras of the endomorphism algebras of Hilbert spaces, and whose morphisms are the morphisms of involutive algebras.

V.3.5 Functions of positive type on a topological group

In this number, G is a topological group whose identity element is denoted by e . The Hilbert spaces considered are complex.

DEFINITION 4. — A continuous function $\varphi \in \mathcal{C}(G)$ is said to be of positive type on G if the function f defined by $f(g, h) = \varphi(g^{-1}h)$ on $G \times G$ is a universally positive kernel on G .

We denote by $\text{Pos}(G)$ the set of functions of positive type on G and we denote by $\text{Pos}_1(G)$ (resp. $\text{Pos}_{\leq 1}(G)$) the subset of $\varphi \in \text{Pos}(G)$ such that $\varphi(e) = 1$ (resp. such that $\varphi(e) \leq 1$).

Example. — Let ϱ be a unitary representation of G in a Hilbert space E and $x \in E$. Let φ be the diagonal matrix coefficient defined by $\varphi(g) = \langle x | \varrho(g)x \rangle$ for all $g \in G$; the function φ is continuous. It follows that

$$\varphi(g^{-1}h) = \langle x | \varrho(g)^* \varrho(h)x \rangle = \langle \varrho(g)x | \varrho(h)x \rangle$$

for all $(g, h) \in G \times G$. Consequently $\varphi \in \text{Pos}(G)$ (theorem 1 of V, p. 432, (iv)). We have $\varphi \in \text{Pos}_1(G)$ if and only if $\|x\| = 1$.

DEFINITION 5. — Let φ be a function of positive type on G . A Hilbert realization of φ is a pair (ϱ, x) where ϱ is a unitary representation of G in a Hilbert space E and $x \in E$ such that $\varphi(g) = \langle x | \varrho(g)x \rangle$ for all $g \in G$.

If x is a cyclic vector of ϱ , one says that it is a cyclic Hilbert realization of φ .

PROPOSITION 9. — Let $\varphi \in \mathcal{C}(G)$. The following conditions are equivalent:

- (i) The function φ is of positive type on G ;
- (ii) There exists a cyclic Hilbert realization of φ ;
- (iii) There exists a Hilbert realization of φ .

Condition (ii) implies condition (iii), and condition (iii) implies (i) by the example above.

Let us finally prove that (i) implies (ii). Let (E, γ) be a cyclic Hilbert realization of the universally positive kernel defined by $f(g, h) = \varphi(g^{-1}h)$ for $(g, h) \in G \times G$. Let $k \in G$. The continuous function $\gamma_k : g \mapsto \gamma(kg)$ on G satisfies

$$\langle \gamma_k(g) | \gamma_k(h) \rangle = \langle \gamma(kg) | \gamma(kh) \rangle = f(kg, kh) = \varphi((kh)^{-1}kg) = f(g, h)$$

for all $(g, h) \in G \times G$, hence (E, γ_k) is a Hilbert realization of f . By prop. 1 of V, p. 434, there exists a unique unitary operator $\varrho(k)$ in $\mathcal{B}(E)$ such that $\gamma_k = \varrho(k) \circ \gamma$. For every $g \in G$ and every $(k_1, k_2) \in G \times G$, it follows that

$$\varrho(k_1 k_2)(\gamma(g)) = \gamma(k_1 k_2 g) = \varrho(k_1)(\varrho(k_2)(\gamma(g))),$$

whence $\varrho(k_1 k_2) = \varrho(k_1)\varrho(k_2)$ since the image of γ is a total subset of E .

Let $k \in G$. For every $g \in G$, we have $\varrho(g)(\gamma(k)) = \gamma(gk)$, and the map from G to E defined by $g \mapsto \varrho(g)(\gamma(k))$ is therefore continuous. Since the endomorphism $\varrho(g)$ is unitary, one deduces that ϱ is a unitary representation of G in E (lemma 4 of V, p. 380). Then set $x = \gamma(e)$. The set of vectors $\varrho(g)x = \varrho(g)(\gamma(e)) = \gamma(g)$ for g ranging over G is total in E , hence x is a cyclic vector of ϱ . Since

$$\langle x | \varrho(g)x \rangle = f(e, g) = \varphi(g)$$

for all $g \in G$, the pair (ϱ, x) is a cyclic Hilbert realization of φ .

PROPOSITION 10. — Let φ be a function of positive type on G and let (ϱ_1, x_1) and (ϱ_2, x_2) be Hilbert realizations of φ , the Hilbert representation (ϱ_1, x_1) being cyclic. There exists a unique isometric G -morphism u from ϱ_1 into ϱ_2 such that $u(x_1) = x_2$. If (ϱ_2, x_2) is also cyclic, then u is an isomorphism.

For $1 \leq j \leq 2$, let E_j be the space of ϱ_j and γ_j the function on G defined by $\gamma_j(g) = \varrho_j(g)x_j$ for all $g \in G$. The pairs (E_1, γ_1) and (E_2, γ_2) are Hilbertian realizations of the function of positive type $(g, h) \mapsto \varphi(g^{-1}h)$, and (E_1, γ_1) is a cyclic Hilbertian realization. By prop. 1 of V, p. 434, there exists a unique isometric linear map $u : E_1 \rightarrow E_2$ such that $\gamma_2 = u \circ \gamma_1$. In particular, we have $x_2 = \gamma_2(e) = u(\gamma_1(e)) = u(x_1)$. Moreover, for all g and h in G , we have

$$\varrho_2(g)u(\gamma_1(h)) = \varrho_2(g)\gamma_2(h) = \gamma_2(gh) = u(\gamma_1(gh)) = u(\varrho_1(g)\gamma_1(h)),$$

and since the set of elements $\gamma_1(h)$ for $h \in G$ is total in E_1 , this means that u is a G -morphism. By loc. cit., it is an isometric isomorphism if (ϱ_2, x_2) is also cyclic.

PROPOSITION 11. — Let $\varphi \in \text{Pos}_1(G)$ and (ϱ, x) be a cyclic Hilbertian realization of φ . Then ϱ is an irreducible representation of G if and only if φ is an extreme point (EVT, II, p. 57, def. 1) of $\text{Pos}_1(G)$.

Let E be the space of ϱ . Suppose first that ϱ is not irreducible. Let F be a closed subspace of E stable under ϱ , nonzero and different from E . Let us write $x = x_1 + x_2$, where $x_1 \in F$ and $x_2 \in F^\circ$. We then have $1 = \|x\|^2 = \|x_1\|^2 + \|x_2\|^2$. The subrepresentation of E generated by x_1 is contained in F , hence $x_1 \neq x$ since x is a cyclic vector of ϱ , whence $x_2 \neq 0$. Similarly, one verifies that $x_1 \neq 0$.

For $j = 1$ and $j = 2$, let φ_j be the continuous function on G such that

$$\varphi_j(g) = \frac{1}{\|x_j\|^2} \langle x_j | \varrho(g)x_j \rangle$$

for all $g \in G$. We have $\varphi_j \in \text{Pos}_1(G)$ (V, p. 443, example). Since $\varphi = \|x_1\|^2 \varphi_1 + \|x_2\|^2 \varphi_2$, it suffices to check that $\varphi_1 \neq \varphi_2$ to show that φ is not an extreme point of $\text{Pos}_1(G)$; it suffices for that to prove that $\varphi \neq \varphi_1$.

Let us reason by contradiction and suppose that $\varphi = \varphi_1$. Since $\langle x_1 | \varrho(g)x_2 \rangle = 0$ for all $g \in G$, we would have

$$\frac{1}{\|x_1\|^2} \langle x_1 | \varrho(g)x \rangle = \frac{1}{\|x_1\|^2} \langle x_1 | \varrho(g)x_1 \rangle = \varphi_1(g) = \varphi(g) = \langle x | \varrho(g)x \rangle$$

for all $g \in G$, whence $\langle x_1 | y \rangle = \langle \|x_1\|^2 x | y \rangle$ for every element y of the vector subspace of E generated by the elements $\varrho(g)x$ for $g \in G$, therefore, for all $y \in E$, since x is a cyclic vector of ϱ . This would imply that $x_1 = \|x_1\|^2 x$ is also a cyclic vector of ϱ , which is a contradiction, whence the assertion.

Suppose now that ϱ is irreducible and let us prove that φ is an extreme point of $\text{Pos}_1(G)$. Let $\varphi_1 \neq \varphi_2$ be elements of $\text{Pos}_1(G)$ and $t_1, t_2 \in [0, 1]$ such that $t_1 + t_2 = 1$ and $\varphi = t_1 \varphi_1 + t_2 \varphi_2$. For $j \in \{1, 2\}$, let (ϱ_j, x_j) be a cyclic Hilbertian realization of φ_j , and let E_j be the space of ϱ_j . Let $x_3 = t_1^{1/2} x_1 + t_2^{1/2} x_2$. Then $(\varrho_1 \oplus \varrho_2, x_3)$ is a Hilbert realization of φ . Since (ϱ, x) is

cyclic, there exists an isometric G -morphism $u : E \rightarrow E_1 \oplus E_2$ such that $u(x) = x_3$ (prop. 10).

Let $j = 1$ or $j = 2$. Since ϱ is irreducible, there exists $\lambda_j \geq 0$ such that the G -morphism $u_j = \text{pr}_j \circ u$ from E into E_j satisfies

$$\langle u_j(y) | u_j(y') \rangle = \lambda_j \langle y | y' \rangle$$

for all y and y' in E (cor. 5 of V, p. 388, if $u_j \neq 0$, and one may take $\lambda_j = 0$ otherwise). For every $g \in G$, it follows that

$$\begin{aligned} t_j \varphi_j(g) &= \langle t_j^{1/2} x_j | \varrho_j(g)(t_j^{1/2} x_j) \rangle = \langle u_j(x) | \varrho_j(g)(u_j(x)) \rangle \\ &= \langle u_j(x) | u_j(\varrho(g)x) \rangle = \lambda_j \varphi(g). \end{aligned}$$

Since $\varphi_j(e) = \varphi(e) = 1$, we deduce that $\varphi_j = \varphi$ if $t_j \neq 0$. The hypothesis $\varphi_1 \neq \varphi_2$ therefore implies that t_1 or t_2 must be zero, which proves that φ is an extreme point of $\text{Pos}_1(G)$.

Lemma 4. — Let $\varphi \in \text{Pos}(G)$. The function φ is bounded on G and we have $\|\varphi\|_\infty = \varphi(e)$. Moreover, we have

$$|\varphi(g^{-1}h) - \varphi(h)| \leq \sqrt{2\varphi(e)(\varphi(e) - \mathcal{R}(\varphi(g)))} \quad (1)$$

for all $(g, h) \in G \times G$.

Let (ϱ, x) be a Hilbert realization of φ . We have $\varphi(e) = \|x\|^2$. For every $g \in G$, it follows therefore that $|\varphi(g)| = |\langle x | \varrho(g)x \rangle| \leq \varphi(e)$ by the Cauchy-Schwarz inequality. This proves the first assertion.

For every $(g, h) \in G \times G$, we have

$$\varphi(g^{-1}h) - \varphi(h) = \langle x | \varrho(g^{-1}h)x \rangle - \langle x | \varrho(h)x \rangle = \langle \varrho(g)x - x | \varrho(h)x \rangle$$

since ϱ is unitary, whence

$$|\varphi(g^{-1}h) - \varphi(h)| \leq \|\varrho(g)x - x\| \|\varrho(h)x\| \leq \varphi(e)^{1/2} \|\varrho(g)x - x\|.$$

One concludes by observing that

$$\|\varrho(g)x - x\|^2 = 2\|x\|^2 - 2\Re(\langle x|\varrho(g)x\rangle) = 2(\varphi(e) - \Re(\varphi(g))).$$

Remark. — The sets $\text{Pos}(G)$ (resp. $\text{Pos}_1(G)$ and $\text{Pos}_{\leq 1}(G)$) are convex self-adjoint subsets of the stellar algebra $\mathcal{C}_b(G)$. They are closed in the space $\mathcal{C}_b(G)$ equipped with the topology of simple convergence. The set $\text{Pos}(G)$ is a convex cone with vertex 0 in the real Banach space $\mathcal{C}_b(G)$.

V.3.6 Unitary dual of a locally compact group

Let G be a locally compact topological group. One equips G with a left Haar measure μ . For $p \in [1, +\infty]$, we denote by $\mathcal{L}^p(G)$ (resp. $L^p(G)$) the space $\mathcal{L}_C^p(G, \mu)$ (resp. the space $L_C^p(G, \mu)$). We identify the space $\mathcal{C}_b(G)$ with its image in $L^\infty(G)$.

Denote by Δ the modular function of G . Recall that $L^1(G)$ is an involutive Banach algebra whose involution is induced, by passage to quotients, by the map $f \mapsto f^*$ where $f^*(y) = \Delta^{-1}(y)\overline{f(y^{-1})}$ for all $f \in \mathcal{L}^1(G)$ and $y \in G$ (cf. I, p. 99, example 4). The Banach algebra $L^1(G)$ admits an approximate unit according to INT, VIII, p. 172, §4, no. 7, prop. 20.

Let ϱ be a unitary representation of G in a complex Hilbert space E . The map $f \mapsto \varrho(f)$ is a continuous morphism of involutive algebras from $L^1(G)$ to $\mathcal{L}(E)$ (lemma 1 of V, p. 401); it is a non-degenerate representation of $L^1(G)$ in E (INT, VIII, p. 139, § 2, no. 7, prop. 10, (i)).

PROPOSITION 12. — Let E be a complex Hilbert space and $\tilde{\pi}$ a morphism of involutive algebras from $L^1(G)$ to $\mathcal{L}(E)$. If the representation $\tilde{\pi}$ is nondegenerate, then there exists a unique unitary representation π of G in E such that $\tilde{\pi}(f) = \pi(f)$ for all $f \in L^1(G)$.

Uniqueness follows from INT, VIII, p. 139, § 2, n° 7, cor. 3 of lemma 4.

Let us prove the existence of π . We have $\|\tilde{\pi}\| \leq 1$ by prop. 2 of I, p. 104. Let F be the subspace of E generated by the vectors of the form $\tilde{\pi}(f)x$ for f in $L^1(G)$ and x in E ; it is dense in E since the representation $\tilde{\pi}$ is nondegenerate.

For every compact neighborhood V of e , let φ_V be a continuous positive function with support contained in V such that $\int \varphi_V \mu = 1$. Let B be a base of the filter of compact neighborhoods of e .

Let g be in G and f be in $L^1(G)$. We have $(\varphi_V * \varepsilon_g) * f \rightarrow \varepsilon_g * f$ in $L^1(G)$ along the filter

of sections of $\mathfrak{B}$ (INT, VIII, p. 172, § 4, no. 7, prop. 20), hence $\widetilde{\pi}(\varphi_V * \varepsilon_g)\widetilde{\pi}(f)$ converges to $\widetilde{\pi}(\varepsilon_g * f)$ in $\mathcal{L}(E)$. This implies that $\widetilde{\pi}(\varphi_V * \varepsilon_g)$ converges simply along the filter of sections of $\mathfrak{B}$ to a linear map $\pi(g)$ from F to F ; this map is continuous by EVT, III, p. 26, cor. 3, since $\|\widetilde{\pi}(\varphi_V * \varepsilon_g)\| \leq 1$ for every compact neighborhood V of e . There therefore exists a unique endomorphism of E which induces $\pi(g)$ by passage to the subspaces. We still denote by $\pi(g)$ this endomorphism; we have $\|\pi(g)\| \leq 1$.

Let $f \in L^1(G)$. Since $\widetilde{\pi}(\varphi_V * \varepsilon_g)\widetilde{\pi}(f)$ converges to $\widetilde{\pi}(\varepsilon_g * f)$, we have $\pi(g)\widetilde{\pi}(f) = \widetilde{\pi}(\varepsilon_g * f)$.

For $g = e$, this relation shows that $\pi(e)$ is the identity on F , hence $\pi(e) = 1_E$. Let g_1 and g_2 be in G . We have

$$\pi(g_1)\pi(g_2)\widetilde{\pi}(f) = \pi(g_1)\widetilde{\pi}(\varepsilon_{g_2} * f) = \widetilde{\pi}(\varepsilon_{g_1 g_2} * f) = \pi(g_1 g_2)\widetilde{\pi}(f),$$

whence $\pi(g_1)\pi(g_2) = \pi(g_1 g_2)$ on F , and hence on E . This proves that the map $g \mapsto \pi(g)$ is a representation of G in E . Since $\|\pi(g)\| \leq 1$ and $\|\pi(g)^{-1}\| = \|\pi(g^{-1})\| \leq 1$, the endomorphisms $\pi(g)$ of E are isometric, hence unitary (EVT, V, p. 40, prop. 3).

Let $f \in L^1(G)$ and $x \in E$. The map $g \mapsto \varepsilon_g * f$ from G into $L^1(G)$ is continuous (INT, VIII, p. 136, § 2, no. 5 and p. 144, § 3, no. 2, formula (5)), and $\widetilde{\pi}$ is continuous, hence the map $g \mapsto \widetilde{\pi}(\varepsilon_g * f)x = \pi(g)\widetilde{\pi}(f)x$ is continuous at $g = e$. Since F is dense in E , the representation π is unitary (lemma 4 of V, p. 380).

Let f_1 and f_2 be in $L^1(G)$. According to INT, VIII, p. 127, §1, no. 5, prop. 7, we have the relation

$$f_1 * f_2 = \int_G f_1(g)(\varepsilon_g * f_2)d\mu(g)$$

in $L^1(G)$, whence

$$\begin{aligned} \widetilde{\pi}(f_1)\widetilde{\pi}(f_2) &= \widetilde{\pi}(f_1 * f_2) = \int_G f_1(g)\widetilde{\pi}(\varepsilon_g * f_2)d\mu(g) \\ &= \int_G f_1(g)\pi(g)\widetilde{\pi}(f_2)d\mu(g) = \left(\int_G f_1(g)\pi(g)d\mu(g) \right) \widetilde{\pi}(f_2), \end{aligned}$$

using INT, VI, p. 9, § 1, no. 1, prop. 1. It follows that

$$\widetilde{\pi}(f) = \int_G f(g) \pi(g) d\mu(g) = \pi(f)$$

for all $f \in L^1(G)$. The proposition is proved.

PROPOSITION 13. — Let $\varphi \in L^\infty(G)$. Denote by λ_φ the continuous linear form $f \mapsto \langle f, \varphi \rangle$ on $L^1(G)$. Then φ is the class of a function of positive type on G if and only if λ_φ is a continuous positive linear form on $L^1(G)$.

Suppose that λ_φ is a continuous positive linear form on $L^1(G)$. Let $(E, x, \widetilde{\pi})$ be a cyclic Hilbertian realization of λ (prop. 5 of V, p. 438).

Let π be a unitary representation of G in E such that $\pi(f) = \widetilde{\pi}(f)$ for all $f \in L^1(G)$ (prop. 12). We then have $\lambda_\varphi(f) = \langle x | \pi(f)x \rangle$ for all $f \in L^1(G)$.

Since

$$\pi(f) = \int_G f(g) \pi(g) d\mu(g)$$

for all $f \in L^1(G)$, it follows that

$$\int_G f(g) \varphi(g) d\mu(g) = \lambda_\varphi(f) = \langle x | \pi(f)x \rangle = \int_G f(g) \langle x | \pi(g)x \rangle d\mu(g)$$

for all $f \in L^1(G)$. Thus, φ is the class in $L^\infty(G)$ of the function on G defined by $g \mapsto \langle x | \pi(g)x \rangle$, which is a function of positive type on G (Prop. 9 of V, p. 443).

Conversely, let $\varphi \in \text{Pos}(G)$. Let $f \in \mathcal{K}(G)$. We then have

$$\begin{aligned} \lambda_\varphi(f^* * f) &= \int_G \varphi(y) \left(\int_G \Delta(z)^{-1} \overline{f(z^{-1})} f(z^{-1}y) d\mu(z) \right) d\mu(y) \\ &= \int_G \varphi(y) \left(\int_G \overline{f(z)} f(z) d\mu(z) \right) d\mu(y) \end{aligned}$$

The continuous function on $G \times G$ defined by $(z, y) \mapsto \varphi(y) \overline{f(z)} f(zy)$ is bounded and

$$\begin{aligned} &\int_G^* |\varphi(y)| \left(\int_G^* \overline{f(z)} f(zy) d\mu(z) \right) d\mu(y) \\ &\leq \varphi(e) \int_G^* \int_G^* |f(z)| |f(zy)| d\mu(z) d\mu(y) \\ &= \varphi(e) \left(\int_G^* |f(z)| d\mu(z) \right)^2 \end{aligned}$$

By INT, V, p. 94, § 8, no. 3, prop. 8, we therefore deduce from the Lebesgue–Fubini theorem

(INT, V, p. 96, § 8, no. 4, th. 1) that

$$\begin{aligned}\lambda_\varphi(f^* * f) &= \int_G \overline{f(z)} \left(\int_G \varphi(y) f(zy) d\mu(y) \right) d\mu(z) \\ &= \int_{G \times G} \overline{f(z)} f(y) \varphi(z^{-1}y) d(\mu \otimes \mu)(z, y).\end{aligned}$$

This implies that $\lambda_\varphi(f^* * f) \geq 0$ by Theorem 1, (i) of V, p. 432 applied to the measure induced by μ on the support of f (INT, IV, p. 186, § 5, n° 7, def. 4). From this one deduces by continuity that $\lambda_\varphi(f^* * f) \geq 0$ for all $f \in L^1(G)$.

The spaces $L^\infty(G)$ and $\mathcal{C}_b(G)$ are equipped with their Banach space topologies, and we denote the norm by $f \mapsto \|f\|_\infty$. We shall call here the weak topology on $L^\infty(G)$, $\mathcal{C}_b(G)$ or $\text{Pos}(G)$ the topology induced by the weak topology $\sigma(L^\infty(G), L^1(G))$.

Since $L^\infty(G)$ is identified with the dual of $L^1(G)$ (INT, V, p. 61, §5, no. 8, th. 4), every closed ball of $L^\infty(G)$ is compact for the weak topology (EVT, III, p. 17, cor. 3).

COROLLARY. — In $L^\infty(G)$ equipped with the weak topology, the set $\text{Pos}(G)$ is closed, and the set $\text{Pos}_{\leq 1}(G)$ is compact.

The first assertion follows from the proposition, and the second then follows from EVT, III, loc. cit.

In general, $\text{Pos}_1(G)$ is not compact for the weak topology, as shown by the example of the group $\mathbb{R}$ (exercise 10 of V, p. 497).

PROPOSITION 14. — The set of extreme points of $\text{Pos}_{\leq 1}(G)$ is equal to the union of the set of extreme points of $\text{Pos}_1(G)$ and of the zero function. Moreover, the closed convex hull of the set of extreme points of $\text{Pos}_1(G)$ contains $\text{Pos}_1(G)$.

By the corollary of Proposition 13, the set $\text{Pos}_{\leq 1}(G)$ is a cap of the pointed convex cone $\text{Pos}(G)$, which is defined by the gauge $\varphi \mapsto \varphi(e)$ (EVT, II, p. 61, def. 3 and prop. 4). Its extreme points are therefore the zero function and the elements φ of $\text{Pos}_1(G)$ belonging to the extreme generators of $\text{Pos}(G)$ (EVT, II, p. 62, cor. 1). These are the extreme points of $\text{Pos}_1(G)$ (EVT, II, p. 61, prop. 3). The first assertion follows from this.

Let us prove the second assertion. Let $\varphi \in \text{Pos}_1(G)$. Since the set $\text{Pos}_{\leq 1}(G)$ is compact for the weak topology (corollary of Prop. 13), there exists a filter $\mathfrak{F}$ on the convex hull of the extreme points of $\text{Pos}_{\leq 1}(G)$ which converges to φ (EVT, II, p. 59, th. 1). As the closed balls

of the Banach space $L^\infty(G)$ are closed for the weak topology and since $\|\varphi\|_\infty = 1$, we have $\lim_{\psi, \mathfrak{F}} \|\psi\|_\infty = 1$ (indeed, otherwise there would exist a real number $c < 1$ and a filter $\mathfrak{G}$ on the closed ball of radius c in $L^\infty(G)$ which would be finer than $\mathfrak{F}$, which would imply that φ belongs to this closed ball).

According to the description of the extreme points of $\text{Pos}_{\leq 1}(G)$, every element ψ of the convex hull of the extreme points of $\text{Pos}_{\leq 1}(G)$ is a function of positive type on G of the form

$$\psi = \sum_{i \in I} t_i \psi_i,$$

where I is a finite set, ψ_i is an extreme point of $\text{Pos}_1(G)$ for all $i \in I$, and $t_i \in [0, 1]$. We have $\sum_i t_i = \psi(e) = \|\psi\|_\infty \leq 1$ (lemma 4 of V, p. 446). If $\psi \neq 0$, the function

$$\frac{\psi}{\|\psi\|_\infty} = \sum_{i \in I} \frac{t_i}{\psi(e)} \psi_i$$

therefore belongs to the convex hull of the extreme points of $\text{Pos}_1(G)$. Since $\lim_{\psi, \mathfrak{F}} \|\psi\|_\infty^{-1} \psi = \varphi$, we conclude that φ belongs to the closed convex hull of the extreme points of $\text{Pos}_1(G)$. The assertion follows.

V.3.7 Existence of irreducible representations

We keep the notation from the previous number.

THEOREM 3 (Raikov). — The weak topology on $\text{Pos}_1(G)$ coincides with the topology of compact convergence.

We will first prove a few lemmas.

Lemma 5. — Let X be a locally compact topological space and let ν be a positive measure on X . On every bounded subset of $\mathcal{C}_b(X)$ the topology induced by the weak topology $\sigma(L^\infty(X), L^1(X))$ is coarser than the topology of compact convergence.

Let B be a bounded subset of $\mathcal{C}_b(X)$ and let $M \in \mathbb{R}_+$ such that $\|\varphi\|_\infty \leq M$ for all $\varphi \in B$. Let $\psi \in L^1(X, \nu)$ and $\varepsilon > 0$. Fix a compact subset K of X such that

$$\int_{X-K} |\psi| d\nu < \varepsilon.$$

For all φ_1 and φ_2 in B, we then have

$$\begin{aligned}\langle \varphi_1 - \varphi_2, \psi \rangle &= \int_X \psi(\varphi_1 - \varphi_2) d\nu \\ &= \int_K \psi(\varphi_1 - \varphi_2) d\nu + \int_{X-K} \psi(\varphi_1 - \varphi_2) d\nu,\end{aligned}$$

whence

$$|\langle \varphi_1 - \varphi_2, \psi \rangle| \leq \sup_{x \in K} |\varphi_1(x) - \varphi_2(x)| \|\psi\|_1 + 2M\varepsilon,$$

and the lemma follows.

Lemma 6. — Let $\psi \in L^1(G)$. Let B be a bounded subset of the Banach space $L^\infty(G)$. The map $\varphi \mapsto \psi * \varphi$ from B to $\mathcal{C}_b(G)$ is continuous when B is equipped with the weak topology and $\mathcal{C}_b(G)$ with the topology of compact convergence.

Let $\varphi \in L^\infty(G)$ and $\eta \in L^1(G)$. The function $\Delta^{-1}\check{\eta}$ belongs to $L^1(G)$ and $\langle \eta, \check{\psi} \rangle = \langle \Delta^{-1}\check{\eta}, \psi \rangle$ (cf. INT, VII, p. 19, § 1, n°3, formula (22)). The map $\varphi \mapsto \check{\varphi}$ is therefore an automorphism of the space $L^\infty(G)$ endowed with the weak topology.

Let $\varphi \in L^\infty(G)$. By INT, VIII, p. 167, § 4, no. 5, prop. 14, the function $\psi * \varphi$ belongs to $\mathcal{C}_b(G)$ and satisfies

$$\begin{aligned}(\psi * \varphi)(g) &= \int_G \psi(h) \varphi(h^{-1}g) d\mu(h) \\ &= \int_G \psi(gy) \check{\varphi}(y) d\mu(y) = \langle \check{\varphi}, \gamma_G^{(1)}(g^{-1})\psi \rangle\end{aligned}$$

for all $g \in G$. It follows that the linear map $u : \varphi \mapsto \psi * \varphi$ is a continuous map from the space $L^\infty(G)$ equipped with the weak topology to $\mathcal{C}_b(G)$ endowed with the topology of simple convergence.

Let $\varphi \in B$ and $(g, h) \in G \times G$. By the formula above, we have

$$|u(\varphi)(g) - u(\varphi)(h)| \leq \|\check{\varphi}\|_\infty \|(\gamma_G^{(1)}(g^{-1}) - \gamma_G^{(1)}(h^{-1}))\psi\|_1.$$

As B is bounded and the left regular representation of G in $L^1(G)$ is continuous (no. 4 of V, p. 405), this implies that $u(B)$ is an equicontinuous subset of $\mathcal{C}_b(G)$. The assertion then follows from what precedes and from TG, X, p. 16, th. 1.

Lemma 7. — Let $\psi \in L^1(G)$ such that $\psi \geq 0$ and $\int \psi = 1$. Let p denote the semi-norm

on $\mathcal{C}_b(\mathbf{G})$ defined by $p(\varphi) = |\langle \varphi, \psi \rangle|$ for all $\varphi \in \mathcal{C}_b(\mathbf{G})$. For every $\varphi \in \text{Pos}_1(\mathbf{G})$, we have

$$\|\psi * \varphi - \varphi\|_\infty \leq \sqrt{2p(1 - \varphi)}.$$

Let $\varphi \in \text{Pos}_1(\mathbf{G})$ and $x \in \mathbf{G}$. According to INT, VIII, p. 167, § 4, no. 5, prop. 14, we obtain

$$\begin{aligned} |\psi * \varphi(x) - \varphi(x)| &= \left| \int_{\mathbf{G}} (\varphi(y^{-1}x) - \varphi(x)) \psi(y) d\mu(y) \right| \\ &\leq \int_{\mathbf{G}} |\varphi(y^{-1}x) - \varphi(x)| \psi(y) d\mu(y) \\ &\leq \int_{\mathbf{G}} \sqrt{2(1 - \mathcal{R}\varphi(y))} \psi(y) d\mu(y) \end{aligned}$$

by applying the bound (1) from V, p. 446. The Cauchy-Schwarz inequality then implies

$$\begin{aligned} \|\psi * \varphi - \varphi\|_\infty &\leq \sqrt{2} \left(\int_{\mathbf{G}} (1 - \mathcal{R}(\varphi)) \psi d\mu \right)^{1/2} \left(\int_{\mathbf{G}} \psi d\mu \right)^{1/2} \\ &\leq \sqrt{2} \sqrt{p(1 - \varphi)}, \end{aligned}$$

whence the result.

Lemma 8. — Let K be a topological field, and let E and F be topological vector spaces over K . Let f be a map from E into F and X a subset of E . Let $x \in X$. The map $f|_X$ is continuous at x if, for every neighborhood W of 0 in F , there exist a neighborhood U of x in E and a continuous map g from X to F such that $(f - g)(U \cap X) \subset W$.

Let W_0 be a neighborhood of 0 in F and let V_0 be a symmetric neighborhood of 0 in F such that $V_0 + V_0 + V_0 \subset W_0$. Let U_0 be a neighborhood of x in E and let g be a continuous map from X into F such that $(f - g)(U_0 \cap X) \subset V_0$.

There exists a neighborhood U_1 of x in E such that $g(U_1 \cap X) \subset g(x) + V_0$. For every $y \in U_0 \cap U_1 \cap X$, we have

$$f(y) - f(x) = (f(y) - g(y)) + (g(y) - g(x)) + (g(x) - f(x)) \in V_0 + V_0 + V_0 \subset W_0,$$

hence $f(y) \in f(x) + W_0$, whence the result.

Let us now prove Theorem 3. Denote by $\text{Pos}_1(\mathbf{G})_f$ (resp. $\text{Pos}_1(\mathbf{G})_c$) the set $\text{Pos}_1(\mathbf{G})$ en-

dowed with the weak topology (resp. the topology of compact convergence); likewise, denote by $\mathcal{C}_b(G)_f$ (resp. $\mathcal{C}_b(G)_c$) the space $\mathcal{C}_b(G)$ endowed with the weak topology (resp. the topology of compact convergence).

The identity map from $\text{Pos}_1(G)_c$ to $\text{Pos}_1(G)_f$ is continuous (Lemma 5). Conversely, denote by ι the inclusion of $\text{Pos}_1(G)_f$ into $\mathcal{C}_b(G)_c$. Let us prove that ι is continuous to conclude the proof.

Let $\varphi_1 \in \text{Pos}_1(G)$. To verify that ι is continuous at φ_1 , we apply Lemma 8. Let W be a neighborhood of 0 in $\mathcal{C}_b(G)_c$. It suffices to find a linear map u from $\mathcal{C}_b(G)$ to $\mathcal{C}_b(G)$ which induces, by passing to subspaces, a continuous map from $\text{Pos}_1(G)_f$ to $\mathcal{C}_b(G)_c$, as well as a neighborhood U of 0 in $\text{Pos}_1(G)_f$ so that one has $(\iota - u)(U \cap \text{Pos}_1(G)) \subset W$.

There exists $\varepsilon > 0$ such that W contains the set of $\varphi \in \mathcal{C}_b(G)$ which satisfy $\|\varphi\|_\infty \leq \varepsilon$. Since $\varphi_1(e) = 1$, there exists a compact neighborhood V of e in G such that

$$\sup_{x \in V} |1 - \varphi_1(x)| \leq \frac{\varepsilon^2}{4}.$$

Let φ_V be the characteristic function of V , and set $\psi_V = \mu(V)^{-1} \varphi_V$. The linear map $u: \varphi \mapsto \psi_V * \varphi$ from $\text{Pos}_1(G)_f$ to $\mathcal{C}_b(G)_c$ is continuous (Lemma 6).

Denote by q_V the seminorm $\varphi \mapsto |\langle \varphi, \psi_V \rangle|$ on $\mathcal{C}_b(G)$; it is continuous for the weak topology. Since ψ_V is zero outside V , it follows that $q_V(1 - \varphi_1) \leq \varepsilon^2/4$; there therefore exists a neighborhood U of φ_1 in $\mathcal{C}_b(G)_f$ such that $q_V(1 - \varphi) \leq \varepsilon^2/2$ for all $\varphi \in U$.

Let $\varphi \in U \cap \text{Pos}_1(G)$. By Lemma 7, we have

$$\|(\iota - u)(\varphi)\|_\infty \leq \sqrt{2q_V(1 - \varphi)} \leq \varepsilon$$

hence $(\iota - u)(U) \subset W$. The theorem is proved.

In general, the weak topology on $\text{Pos}_{\leq 1}(G)$ does not coincide with the topology of compact convergence, as shown by the example of the group R .

We recall that we denote by $\widehat{G}$ the set of classes of irreducible unitary representations of G (V, p. 393, def. 8).

THEOREM 4 (Gelfand–Raikov). — For every $x \neq e$ in G , there exists an irreducible unitary

representation π of G such that $\pi(x)$ is not the identity endomorphism of the space π .

Let $x \in G$ be such that $\pi(x)$ is the identity for every $\pi \in \widehat{G}$. It follows that $\varphi(x) = \varphi(e) = 1$ for every extreme point φ of $\text{Pos}_1(G)$ (prop. 11 of V, p. 444), and therefore for every $\varphi \in \text{Pos}_1(G)$ by prop. 14, since the linear form $\varphi \mapsto \varphi(e)$ is continuous on $\text{Pos}_1(G)$ endowed with the weak topology (th. 3). But if $x \neq e$, there exists $f \in L^2(G)$ of norm 1 such that $\langle f | \gamma_G(x) f \rangle = 0$ (V, p. 406, lemma 3). Since the left-hand side of this equality is of the form $\varphi(x)$, where $\varphi \in \text{Pos}_1(G)$, this is a contradiction.

If G is not locally compact, it is not always true that the irreducible unitary representations of G separate the points of G .

COROLLARY. — For all elements $g \neq h$ in G , there exists an irreducible representation $\pi \in \widehat{G}$ and a matrix coefficient f of π such that $f(g) \neq f(h)$. In particular, the subalgebra $\Upsilon(G)$ of $\mathcal{C}_b(G)$ separates the points of G .

Let $g \neq h$ be in G . There exists an irreducible representation $\pi \in \widehat{G}$ such that $\pi(g) \neq \pi(h)$ (th. 4), so there exist x and y in the space of π such that $\langle x | \pi(g) y \rangle \neq \langle x | \pi(h) y \rangle$.

V.3.8 Functions of positive type on a locally compact commutative group

In this number, G is a locally compact commutative group and μ denotes a Haar measure on G . We denote by $\widehat{G}$ the dual group of G and by $\widehat{\mu}$ the dual Haar measure of μ (def. 4 of II, p. 214). We denote by F the Fourier transform on the Banach space $\mathcal{M}^1(G)$ of bounded measures on G .

Since $\mathcal{C}_0(G)$ is the closure of $\mathcal{K}(G)$ in $\mathcal{C}_b(G)$, the Banach space $\mathcal{M}^1(G)$, dual of $\mathcal{K}(G)$ endowed with the topology of $\mathcal{C}_b(G)$, is identified with the dual of $\mathcal{C}_0(G)$ (cf. INT, III, p. 56, § 1, no. 8). We endow $\mathcal{M}^1(G)$ with the weak topology associated with this duality.

Lemma 9. — The Fourier transform is a continuous mapping from $\mathcal{M}^1(G)$ equipped with the weak topology to $\mathcal{C}_b(\widehat{G})$ endowed with the topology induced by the weak topology $\sigma(L^\infty(\widehat{G}), L^1(\widehat{G}))$.

Let $\mathfrak{F}$ be a filter on $\mathcal{M}^1(G)$ weakly converging to a bounded measure ν on G . For every $\varphi \in L^1(\widehat{G})$, we have $\mathcal{F}(\varphi) \in \mathcal{C}_0(G)$ (prop. 4 of II, p. 209), hence

$$\begin{aligned}\int_{\widehat{G}} \varphi \mathcal{F}(\nu) d\widehat{\mu} &= \int_G \mathcal{F}(\varphi) d\nu \\ &= \lim_{\eta, \mathfrak{s}} \int_G \mathcal{F}(\varphi) d\eta = \lim_{\eta, \mathfrak{s}} \int_{\widehat{G}} \varphi \mathcal{F}(\eta) d\widehat{\mu},\end{aligned}$$

by the transposition property of the Fourier transform (prop. 13 of II, p. 221). The lemma follows.

THEOREM 5 (Bochner). — A continuous function φ on $\widehat{G}$ belongs to $\text{Pos}(\widehat{G})$ if and only if there exists a bounded positive measure ν on G such that $\varphi = \mathcal{F}(\nu)$.

Let $\nu \in \mathcal{M}^1(G)$ be a positive measure. Its Fourier transform is continuous. For every finite family $(x_i)_{i \in I}$ in $\widehat{G}$ and every finite family $(t_i)_{i \in I}$ of complex numbers, it follows that

$$\begin{aligned}\sum_{i \in I} \sum_{j \in I} \bar{t}_i t_j \mathcal{F}(\nu)(x_i^{-1} x_j) &= \sum_{i \in I} \sum_{j \in I} \bar{t}_i t_j \int_G x_i \bar{x}_j d\nu \\ &= \int_G \left| \sum_{i \in I} \bar{t}_i x_i \right|^2 d\nu \geq 0,\end{aligned}$$

since ν is a positive measure. The Fourier transform of ν is therefore a function of positive type on $\widehat{G}$ (th. 1 of V, p. 432, (ii)).

Let us prove the converse. Endow the set $\text{Pos}_{\leq 1}(\widehat{G})$ with the weak topology, induced as before by the weak topology $\sigma(L^\infty(\widehat{G}), L^1(\widehat{G}))$. The set $\text{Pos}_{\leq 1}(\widehat{G})$ is compact and convex (cor. of prop. 13 of V, p. 448). By prop. 14 of V, p. 450, prop. 11 of V, p. 444, and cor. 7 of V, p. 390, the extreme points of $\text{Pos}_{\leq 1}(\widehat{G})$ are the zero function and the elements of $\widehat{G}$.

Let $\mathcal{N}$ be the set of positive bounded measures of mass ≤ 1 on G ; it is compact in $\mathcal{M}^1(G)$ endowed with the weak topology (EVT, III, p. 17, cor. 3). By Lemma 9 and the first part of the proof, the Fourier transform on G defines, by passing to subspaces, a continuous map from $\mathcal{N}$ to $\text{Pos}_{\leq 1}(\widehat{G})$; by homogeneity, it suffices to show that this map is surjective. The image $\mathcal{F}(\mathcal{N})$ of $\mathcal{N}$ by the Fourier transform is convex and compact; it contains the zero function and the elements of $\widehat{G}$ (indeed, these are of the form $\text{ev}_x: \chi \mapsto \chi(x)$ for an element x of G by th. 2 of II, p. 220, and we have $\text{ev}_x = \mathcal{F}(\varepsilon_{x^{-1}})$). The set $\mathcal{F}(\mathcal{N})$ therefore contains the extreme points of $\text{Pos}_{\leq 1}(\widehat{G})$, whence $\mathcal{F}(\mathcal{N}) = \text{Pos}_{\leq 1}(\widehat{G})$ by the Krein-Milman theorem (EVT, II, p. 59, th. 1). This concludes the proof.

When $G = R^k$ for an integer $k \in \mathbb{N}$, this theorem corresponds to prop. 11 of INT, IX, p. 94, § 6, no. 12.

V.4 REPRESENTATIONS OF COMPACT GROUPS

In this paragraph, all topological vector spaces considered are complex, unless explicitly stated otherwise.

We fix a compact topological group G , whose identity element we denote by e . The group G is unimodular (INT, VII, p. 20, § 1, n° 3, cor. of prop. 3). We denote by μ the normalized Haar measure on G (that is, the one such that $\mu(G) = 1$), and for $1 \leq p \leq +\infty$, we denote by $\mathcal{L}^p(G)$ (resp. $L^p(G)$) the space $\mathcal{L}_C^p(G, \mu)$ (resp. the space $L_C^p(G, \mu)$). Convolutions will always be taken with respect to the measure μ . Let $p \in [1, +\infty]$. We identify $\mathcal{C}(G)$ with a subspace of $L^p(G)$, which is permissible since the support of μ is equal to G .

For every irreducible unitary representation $\pi \in \widehat{G}$, let E_π be the space of π .

V.4.1 Semi-simplicity of finite-dimensional representations

We recall (INT, VII, p. 71, § 3, n° 1, lemma 1) that for every continuous representation ϱ of G in a Hilbert space E , there exists a nondegenerate positive Hermitian form q on E such that the topological vector space structure of E defined by q is identical to the initial structure of E , and such that ϱ is a unitary representation of G in the Hilbert space E equipped with the inner product q .

PROPOSITION 1. — Let ϱ be a finite-dimensional linear representation of G in a separated topological vector space E . There exists an inner product q on E such that ϱ is a unitary representation in the Hilbert space E equipped with q . In particular, ϱ is semisimple.

Since E is finite-dimensional, there exists a Hilbert space structure on E . The result follows by applying the preceding remark and Corollary 2 of V, p. 392.

Remark. — We shall see later (Cor. 4 of V, p. 466) that every unitary representation of G is semi-simple.

COROLLARY. — Let ϱ_1 and ϱ_2 be finite-dimensional representations of G . The representations ϱ_1 and ϱ_2 are isomorphic if and only if their characters are equal.

This follows from the proposition and Corollary 3 of V, p. 392.

V.4.2 Irreducible representations

Lemma 1. — Every irreducible unitary representation of G is square-integrable.

Indeed, the matrix coefficients of an irreducible unitary representation are continuous and bounded, and are therefore square-integrable on G , since G is compact.

PROPOSITION 2. — Let π be an irreducible unitary representation of G in a Hilbert space E . The dimension of E is finite and equal to the formal degree of π .

The representation π being square-integrable (Lemma 1), its formal degree $c_\mu(\pi)$ relative to μ is defined (def. 4 of V, p. 423); it is a strictly positive real number. Let $(e_i)_{i \in I}$ be a finite orthonormal family in E . Let x be an element of E with norm 1 (such exists since E is nonzero). For $i \in I$, we have

$$c_\mu(\pi) \int_G |\langle e_i | \pi(g)x \rangle|^2 d\mu(g) = 1$$

(prop. 8 of V, p. 424). Summing this formula over $i \in I$, we obtain

$$\begin{aligned} \text{Card}(I) &= c_\mu(\pi) \int_G \sum_{i \in I} |\langle e_i | \pi(g)x \rangle|^2 d\mu(g) \\ &\leq c_\mu(\pi) \int_G \|\pi(g)x\|^2 d\mu(g) = c_\mu(\pi) \end{aligned}$$

by Bessel's inequality (EVT, V, p. 21, prop. 4). Hence the cardinality of I is bounded above by $c_\mu(\pi)$. This implies that the dimension of E is finite.

One can then apply the preceding result to an orthonormal basis $(e_i)_{i \in I}$ of E . We obtain, by EVT, V, p. 22, prop. 5,

$$\begin{aligned} \dim(E) &= c_\mu(\pi) \int_G \sum_{i \in I} |\langle e_i | \pi(g)x \rangle|^2 d\mu(g) \\ &= c_\mu(\pi) \int_G \|\pi(g)x\|^2 d\mu(g) = c_\mu(\pi), \end{aligned}$$

which concludes the proof.

In particular, it is therefore permissible to speak of the character χ_π of an irreducible unitary representation π of G . It is a continuous function on G .

COROLLARY 1. — Let π be an irreducible unitary representation of G in a Hilbert space E . The character of π is a Hermitian element of the involutive algebra $L^1(G)$.

Let $x \in G$. Since G is unimodular, we have $\chi_\pi^*(x) = \overline{\text{Tr}(\pi(x^{-1}))}$, whence $\chi_\pi^*(x) =$

$\text{Tr}(\pi(x))$ since π is a unitary representation.

COROLLARY 2. — a) Let π be an irreducible unitary representation of G in a Hilbert space E . We have

$$\int_G \overline{\langle x | \pi(g)x' \rangle} \langle y | \pi(g)y' \rangle d\mu(g) = \frac{1}{\dim(E)} \overline{\langle x | y \rangle} \langle x' | y' \rangle$$

for all $(x, y, x', y') \in E^4$.

b) Let π_1 (resp. π_2) be an irreducible unitary representation of G in a Hilbert space E_1 (resp. E_2). If π_1 and π_2 are not isomorphic, one has

$$\int_G \overline{\langle x | \pi_1(g)x' \rangle} \langle y | \pi_2(g)y' \rangle d\mu(g) = 0$$

for all $(x, x', y, y') \in E_1^2 \times E_2^2$

This follows immediately from prop. 8 of V, p. 424 and prop. 9 of V, p. 424, taking into account lemma 1 and the formula $c_\mu(\pi) = \dim(E)$ (prop. 2).

COROLLARY 3. — Let π_1 and π_2 be irreducible representations of G . In the Hilbert space $L^2(G)$, one has $\langle \chi_{\pi_1} | \chi_{\pi_2} \rangle = 1$ if π_1 and π_2 are isomorphic and $\langle \chi_{\pi_1} | \chi_{\pi_2} \rangle = 0$ otherwise. In other words, the family of characters of the classes $\pi \in \widehat{G}$ is an orthonormal family in $L^2(G)$.

This follows from proposition 1 of V, p. 457 and corollary 2, noting that for every orthonormal basis $(e_i)_{i \in I}$ of the space of a finite-dimensional unitary representation π of G , and for every $g \in G$, we have the formula

$$\chi_\pi(g) = \sum_{i \in I} \langle e_i | \pi(g)e_i \rangle.$$

COROLLARY 4. — a) Let ϱ be a continuous finite-dimensional representation of G . One has

$$\chi_\varrho = \sum_{\pi \in \widehat{G}} \dim(\text{Hom}_G(\pi, \varrho)) \chi_\pi = \sum_{\pi \in \widehat{G}} \dim(\text{Hom}_G(\varrho, \pi)) \chi_\pi.$$

b) Let ϱ_1 and ϱ_2 be continuous finite-dimensional representations of G . One has

$$\langle \chi_{\varrho_1} \mid \chi_{\varrho_2} \rangle = \dim(\text{Hom}_G(\varrho_1, \varrho_2)) = \dim(\text{Hom}_G(\varrho_2, \varrho_1)).$$

c) A continuous finite-dimensional representation ϱ of G is irreducible if and only if $\|\chi_\varrho\|^2 = 1$.

Since ϱ is semi-simple, it is isomorphic to the direct sum of its π -isotypic components $M_\pi(\varrho)$ for $\pi \in \widehat{G}$. Let $m_\pi(\varrho)$ denote the multiplicity of π in ϱ (def. 10 of V, p. 398); we then have

$$\chi_\varrho = \sum_{\pi \in \widehat{G}} m_\pi(\varrho) \chi_\pi.$$

The assertion a) therefore follows from the fact that

$$m_\pi(\varrho) = \dim \text{Hom}_G(\pi, \varrho) = \dim \text{Hom}_G(\varrho, \pi)$$

(formula (1) of V, p. 377 and cor. 2 of V, p. 387).

By bilinearity and orthonormality of characters, the assertion a) implies that

$$\langle \chi_{\varrho_1} \mid \chi_{\varrho_2} \rangle = \sum_{\pi \in \widehat{G}} \dim(\text{Hom}_G(\pi, \varrho_1)) \dim(\text{Hom}_G(\pi, \varrho_2)),$$

and on the other hand we have canonical isomorphisms

$$\text{Hom}_G(\varrho_1, \varrho_2) \rightarrow \bigoplus_{(\pi_1, \pi_2) \in \widehat{G} \times \widehat{G}}$$

(formula (1) of V, p. 377), whence it follows that

$$\dim(\text{Hom}_G(\varrho_1, \varrho_2)) = \sum_{\pi \in \widehat{G}} m_\pi(\varrho_1) m_\pi(\varrho_2),$$

whence the assertion b). The assertion c) also follows from a) and Schur's lemma (prop. 6 of V, p. 386).

COROLLARY 5. — Let π_1 and π_2 be irreducible unitary representations of G . Then

$\pi_1(\overline{\chi}_{\pi_2}) = 0$ if π_1 is not isomorphic to π_2 , and $\pi_1(\overline{\chi}_{\pi_1})$ is multiplication by $1/\dim(\pi_1)$.

The character of π_2 is a continuous central function on G , hence the linear map $u = \pi_1(\overline{\chi}_{\pi_2})$ is defined and belongs to the space $\text{Hom}_G(\pi_1, \pi_1)$. It is a homothety by Schur's lemma (prop. 6 of V, p. 386) whose trace is

$$\begin{aligned}\text{Tr}(u) &= \text{Tr}\left(\int_G \overline{\chi}_{\pi_2}(g) \pi_1(g) d\mu(g)\right) \\ &= \int_G \overline{\chi}_{\pi_2}(g) \chi_{\pi_1}(g) d\mu(g).\end{aligned}$$

By the orthogonality relations, the trace of u is therefore zero if π_1 is not isomorphic to π_2 and equals 1 otherwise. The assertion follows.

Remark. — When G is finite, the Schur orthogonality relation (resp. the character orthogonality formula) coincides with that of A, VIII, p. 399 (resp. that of A, VIII, p. 400, prop. 4). On the other hand, the "second orthogonality relation" of characters of finite groups (A, VIII, p. 402, formula (32)) has no exact analogue when G is compact.

For G finite, the particular case of the second orthogonality formula corresponding to the conjugacy class of e is the formula

$$\frac{1}{\text{Card}G} \sum_{\pi \in \widehat{G}} \dim(\pi) \chi_{\pi}(g) = \begin{cases} 1 & \text{if } g = e \\ 0 & \text{otherwise,} \end{cases}$$

which is also interpreted as the computation of the character of the regular representation γ_G of G , and which is equivalent to the formula $\text{Tr}(\gamma_G(f)) = f(e)$ for every function f from G to $\mathbb{C}$.

Let G again be an arbitrary compact group. For every function $f \in \mathcal{C}(G)$, the endomorphism $\gamma_G(f)$ of $L^2(G)$ coincides with the endomorphism $\varphi \mapsto f * \varphi$ and it has finite trace (lemma 4 of V, p. 407 and corollary 2 of III, p. 33). By loc. cit. and th. 2 of IV, p. 177, we also have

$$\text{Tr}(\gamma_G(f)) = \int_G f(x^{-1}x) d\mu(x) = f(e).$$

PROPOSITION 3. — Suppose that $G = G_1 \times G_2$ where G_1 and G_2 are compact groups. The map b from $\widehat{G}_1 \times \widehat{G}_2$ to $\widehat{G}$ that sends (π_1, π_2) to the class of $\pi_1 \boxtimes \pi_2$ is a bijection.

By cor. 6 of V, p. 389, the map b is well defined.

Let π_1 and π_2 be elements of $\widehat{G}$. Let $\pi \in \widehat{G}$. The π -isotypic component of the restriction ϖ of $\pi_1 \boxtimes \pi_2$ to $G_1 \times \{e\}$ is equal to ϖ if $\pi_1 = \pi$ and is zero otherwise (lemma 8 of V, p. 384). Thus, the unitary representation $\pi_1 \boxtimes \pi_2$ determines π_1 up to isomorphism; likewise, it determines π_2 , which proves that b is injective.

Let us prove that b is surjective. Let π be an irreducible unitary representation of $G_1 \times G_2$ in a Hilbert space E . It is finite-dimensional (prop. 2). Let π_1 be an irreducible unitary representation of G_1 in a Hilbert space E_1 such that the restriction of π to $G_1 \times \{e\}$ contains a subrepresentation isomorphic to π_1 (lemma 3 of V, p. 379). Let $F = \text{Hom}_{G_1}(\pi_1, \pi)$. This is a nonzero finite-dimensional vector space. For $h \in G_2$ and $u \in F$, let $\varrho(h)(u)$ be the linear map from E_1 to E defined by $x \mapsto \pi(e, h)(u(x))$. Since $G_1 \times \{e\}$ and $\{e\} \times G_2$ commute in G , the map $\varrho(h)(u)$ belongs to the space F . The map ϱ is a linear representation of G_2 in the space F ; it is continuous. Since F is nonzero, there exists an irreducible subrepresentation π_2 of ϱ (lemma 3 of V, p. 379). Let E_2 be the space of π_2 . The linear map $v : E_1 \otimes E_2 \rightarrow E$ such that $x \otimes u \mapsto u(x)$ for $x \in E_1$ and $u \in E_2$ is then a nonzero G -morphism from $\pi_1 \boxtimes \pi_2$ to π ; since the representations π and $\pi_1 \boxtimes \pi_2$ are irreducible and finite-dimensional, the morphism v is an isomorphism (Lemma 2 of V, p. 378).

This proposition is to be compared with Theorem 1 of A, VIII, p. 208.

V.4.3 The Peter-Weyl theorem

We recall that we denote by $\Theta(G)$ the space of matrix coefficients of finite-dimensional unitary representations of G . The space $\Theta(G)$ is a unital subalgebra of $\mathcal{C}(G)$ stable under complex conjugation (prop. 5 of V, p. 386).

For $\pi \in \widehat{G}$, let $\varrho_G(\pi)$ be the subspace of $\mathcal{C}(G)$ generated by the matrix coefficients of π . We identify it with a subspace of $L^2(G)$.

The space $\Theta(G)$ coincides with the sum of the spaces $\varrho_G(\pi)$ for $\pi \in \widehat{G}$. Indeed, every element of $\Theta(G)$ is a sum of matrix coefficients of irreducible representations of G , since the finite-dimensional representations of G are semisimple by Prop. 1 of V, p. 457). Moreover, this sum is direct since, by Cor. 2 of V, p. 458, the spaces $\varrho_G(\pi)$ are pairwise orthogonal.

Proposition 4. — The space $\Theta(G)$ is dense in $\mathcal{C}(G)$ and in $L^2(G)$.

The unital subalgebra $\Theta(G)$ of $\mathcal{C}(G)$ is stable under conjugation. It coincides with the subalgebra $\Upsilon(G)$ by prop. 2 of V, p. 457, hence it separates the points of G (cor. of th. 4 of V, p. 454). Consequently, it is dense in $\mathcal{C}(G)$ by TG, X, p. 40, cor. 2, and a fortiori, it is dense in the space $L^2(G)$ (cf. INT, IV, p. 155, §4, no. 7, prop. 13).

Recall that the biregular unitary representation of G is the representation ϱ_G of $G \times G$ into $L^2(G)$ such that $(g_1, g_2) \mapsto \gamma_G(g_1) \delta_G(g_2)$.

PROPOSITION 5. — Let $\pi \in \widehat{G}$. The space $\varrho_G(\pi)$ is a subrepresentation of ϱ_G which is isomorphic to $\overline{\pi} \boxtimes \pi$. In particular, it is an irreducible representation of $G \times G$.

By prop. 7 of V, p. 422 (applied with $Z = \{e\}$), the space $\varrho_G(\pi)$ is a subrepresentation of ϱ_G and the linear map from $\overline{E}_\pi \otimes E_\pi$ to $L^2(G)$ that sends $x \otimes y$ to the matrix coefficient $g \mapsto \langle x | \pi(g)y \rangle$ is a $(G \times G)$ -isomorphism from $\overline{\pi} \boxtimes \pi$ onto $\varrho_G(\pi)$. The subrepresentation $\varrho_G(\pi)$ is therefore irreducible (prop. 3 of V, p. 461).

THEOREM 1 (Peter–Weyl). — Let G be a compact topological group. The biregular representation ϱ_G of G is the Hilbert sum of the subrepresentations $\varrho_G(\pi)$ for $\pi \in \widehat{G}$.

The spaces $\varrho_G(\pi)$ are pairwise orthogonal; these are irreducible subrepresentations of the biregular representation of G (prop. 5) which are pairwise non-isomorphic (prop. 3 of V, p. 461).

The theorem then follows from the fact that the sum $\Theta(G)$ of the spaces $\varrho_G(\pi)$ for π running through $\widehat{G}$ is dense in $L^2(G)$ (prop. 4).

COROLLARY 1. — For every $\pi \in \widehat{G}$, the subrepresentation $\varrho_G(\pi)$ is the $(\overline{\pi} \boxtimes \pi)$ -isotypic component of ϱ_G .

Let F be the $(\overline{\pi} \boxtimes \pi)$ -isotypic component of ϱ_G . The space F contains $\varrho_G(\pi)$ (prop. 5). Moreover, for every $\tau \in \widehat{G}$ different from π , the intersection of F and $\varrho_G(\tau)$ is zero (prop. 3 of V, p. 461). From this we deduce that $F = \varrho_G(\pi)$ by applying the theorem.

COROLLARY 2. — Let π_1 and π_2 be non-isomorphic irreducible representations of G . The $(\overline{\pi}_1 \boxtimes \pi_2)$ -isotypic component of ϱ_G is zero.

COROLLARY 3. — The right (resp. left) regular representation of G is isomorphic to the Hilbert sum

$$\bigoplus_{\pi \in \widehat{G}} \pi^{\dim(\pi)}.$$

By the theorem and proposition 5, the restriction of the biregular representation of G to the subgroup $H = \{e\} \times G$ is isomorphic to the Hilbert sum of the restrictions to H of the representations $\overline{\pi} \boxtimes \pi$ for $\pi \in \widehat{G}$. These are isomorphic to the direct sum of $\dim(\pi)$ copies of π (lemma 8 of V, p. 384). This implies the result for the right regular representation, and the case of the left regular representation is similar.

COROLLARY 4. — Let $\pi \in \widehat{G}$. The π -isotypic component of δ_G (resp. the component $\overline{\pi}$ -isotypic of γ_G) is equal to $\varrho_G(\pi)$.

The argument is similar to that of the preceding corollary, by considering $\varrho_G(\pi)$ as a subrepresentation of δ_G (resp. of γ_G).

Remarks. — 1) If G is finite, these statements correspond to the results of A, VIII, p. 398, remark.

2) Suppose that G is commutative. The set $\widehat{G}$ identifies with the dual group of G (V, p. 393, remark). For every $\chi \in \widehat{G}$, the space $\varrho_G(\chi)$ is the 1-dimensional subspace of $L^2(G)$ generated by χ . Theorem 1 for G is therefore equivalent to the corollary of Theorem 1 of II, p. 215.

3) If there exists an integer $n \geq 0$ such that G is a compact subgroup of $\mathbf{GL}(C^n)$, the identity representation of G in $\mathbf{GL}(C^n)$ is enough to separate the points of G , and one can then prove Proposition 4 directly, followed by the Peter–Weyl theorem, without invoking the Gelfand–Raikov theorem.

4) The Peter–Weyl theorem implies in particular that the natural continuous homomorphism from G into $\prod_{\pi \in \widehat{G}} \mathbf{U}(E_\pi)$ is injective.

V.4.4 Matrix coefficients and G -finite functions

PROPOSITION 6. — The following subspaces of $L^2(G)$ are equal:

a) The space $\Theta(G)$

b) The algebraic direct sum of the subspaces $\varrho_G(\pi)$ of $L^2(G)$.

c) The space of G -finite vectors (cf. V, p. 376) of γ_G ;

d) The space of G -finite vectors of δ_G ;

e) The space of $(G \times G)$ -finite vectors of ϱ_G .

In particular, every G -finite vector of γ_G , δ_G or ϱ_G belongs to $\mathcal{C}(G)$.

Denote by F_a (resp. F_b , F_c , F_d , F_e) the space defined by condition a) (resp. b), c), d), e)).

We have already noted that $F_a = F_b$.

One has $F_b \subset F_c$ since $\varrho_G(\pi)$ is a finite-dimensional subrepresentation of γ_G for all $\pi \in \widehat{G}$. Conversely, let f be a G -finite vector of γ_G . The subspace E_f generated by the functions $\gamma_G(g)f$ for $g \in G$ is a finite-dimensional subrepresentation of γ_G . It is equal to the direct sum of its π -isotypic components for $\pi \in \widehat{G}$ (prop. 1 of V, p. 457). According to cor. 4 of V, p. 463, this implies that E_f is contained in the sum of the spaces $\varrho_G(\pi)$, hence $F_c \subset F_b$.

By an analogous argument, we obtain $F_b = F_d$, and since $\varrho_G(\pi)$ is a subrepresentation of ϱ_G , one establishes in the same way that $F_b = F_e$.

COROLLARY. — Let $E \subset L^2(G)$ be a finite-dimensional vector subspace defining a subrepresentation of γ_G (resp. of δ_G , of ϱ_G). Then E is contained in $\mathcal{C}(G)$.

Indeed, every element of E is a G -finite vector of the representation γ_G .

V.4.5 Representations in a separated quasi-complete space

PROPOSITION 7. — Let ϱ be a continuous representation of G in a separated, locally convex, and quasi-complete space E . The sum of the finite-dimensional subrepresentations of E is dense in E .

Let $x \in E$ and let U be an open neighborhood of x . The set of measures $\nu \in \mathcal{M}(G)$ such that $\varrho(\nu)x \in U$ is open in $\mathcal{M}(G)$ for the topology of compact convergence (cf. no. 2 of V, p. 400). It contains ε_e ; hence it contains a measure of the form $\nu = f_1 \cdot \mu$ where $f_1 \in \mathcal{C}(G)$ (INT, VIII, p. 171, § 4, n° 7, prop. 19) and, consequently, it contains a measure of the form $f_2 \cdot \mu$ where f_2 is a G -finite function (prop. 6 of V, p. 464 and prop. 4 of V, p. 462).

The subrepresentation F of γ_G generated by f_2 is finite-dimensional. Let $\tilde{F}$ be the image of the linear map $f \mapsto \varrho(f)x$ from F into E . The space $\tilde{F}$ is finite-dimensional, and it contains the element $\varrho(f_2)x$ belonging to U . Since $\varrho(g)\varrho(f) = \varrho(\gamma_G(g)f)$ for all $g \in G$ and every $f \in F$ (formula (1) of V, p. 406), the space $\tilde{F}$ is a subrepresentation of ϱ which intersects U . The proposition is proved.

COROLLARY 1. — Let π be a continuous irreducible representation of G in a locally convex separated quasi-complete space E . The space E is finite-dimensional.

COROLLARY 2. — Let ϱ be a continuous representation of G in a locally convex separated quasi-complete space E , and let π be a continuous irreducible representation of G .

- a) The G -morphism $p_\pi = \dim(\pi)\varrho(\overline{\chi}_\pi)$ from E into E is a continuous projector of E whose image is the π -isotypic component of ϱ ;
- b) If ϱ is a unitary representation, then the projector p_π is the orthogonal projector of E with image $M_\pi(\varrho)$.

Let us prove a). The linear map $p_\pi = \dim(\pi)\varrho(\overline{\chi}_\pi)$ is well defined, since E is quasi-complete and G is compact (V, p. 401). It is an element of $\text{Hom}_G(\varrho, \varrho)$ since the character of π is a continuous central function on G .

Let ϖ be a finite-dimensional subrepresentation of E and F its space. The map p_π induces, by passing to subspaces, the endomorphism $\dim(\pi)\varpi(\overline{\chi}_\pi)$ of F . This endomorphism is therefore zero if ϖ is not isomorphic to π and is the identity map of F otherwise (indeed, this is the case if ϖ is irreducible by cor. 5 of V, p. 460, and the general case follows from this since ϖ is semisimple by prop. 1 of V, p. 457).

Finally, since the sum of the finite-dimensional subrepresentations of E is dense in E (prop. 7) and since p_π is continuous, we conclude that p_π is a projector whose image is the π -isotypic component of ϱ . If E is a Hilbert space, the projector p_π is Hermitian by Cor. 1 of V, p. 458, hence it is an orthogonal projector (Lemma 3, (ii) of I, p. 133).

COROLLARY 3. — Let ϱ be a continuous representation of G in a locally convex separated quasi-complete space E . The endomorphism

$$x \mapsto \int_G \varrho(g)x d\mu(g)$$

of E is a projector of E whose image is the space E^G of invariant vectors in E . In particular, if E is finite-dimensional, then

$$\dim(E^G) = \int_G \chi_\varrho(g) d\mu(g).$$

The first assertion is the special case of the preceding corollary when π is the trivial representation of G , of dimension 1. The second follows from it since the dimension of E^G is the trace of the orthogonal projector onto E^G , that is,

$$\dim(E^G) = \text{Tr}\left(\int_G \varrho(g) d\mu(g)\right) = \int_G \chi_\varrho(g) d\mu(g).$$

In particular, when E is finite-dimensional, there exists a vector $x \neq 0$ in E such that $\varrho(g)x = x$ for all $g \in G$ if and only if

$$\int_G \chi_\varrho(g) d\mu(g) \neq 0.$$

COROLLARY 4. — Let ϱ be a unitary representation of G in a Hilbert space E . The space E is the Hilbert sum of the π -isotypic components $M_\pi(\varrho)$ for $\pi \in \widehat{G}$. In particular, the representation ϱ is semisimple.

The isotypic components of ϱ corresponding to non-isomorphic irreducible representations of G are orthogonal (prop. 8 of V, p. 394). Since every finite-dimensional unitary representation of G is semisimple (prop. 1 of V, p. 457), the sum of the isotypic components $M_\pi(\varrho)$ for $\pi \in \widehat{G}$ is dense in E (prop. 7). The corollary follows from this.

V.4.6 Characters and conjugacy classes

Let ϱ be a unitary representation of G in a finite-dimensional Hilbert space E . The character of ϱ satisfies $|\chi_\varrho| \leq \dim(E)$. Let $(e_i)_{i \in I}$ be an orthonormal basis of E ; the character χ_ϱ is the sum of the diagonal matrix coefficients $g \mapsto \langle e_i | \varrho(g)e_i \rangle$ of ϱ . In particular, the character of ϱ is

a G -finite function, and if ϱ is irreducible, then $\chi_\varrho \in \varrho_G(\varrho)$.

We have the following formulas

$$\chi_{\varrho_1 \oplus \varrho_2} = \chi_{\varrho_1} + \chi_{\varrho_2}, \quad \chi_{\varrho_1 \otimes \varrho_2} = \chi_{\varrho_1} \chi_{\varrho_2}, \quad \chi_{\check{\varrho}} = \chi_{\overline{\varrho}} = \overline{\chi_\varrho}$$

(cf. A, VIII, p. 388–389).

PROPOSITION 8. — We have

(1) $\chi_\pi * \chi_\sigma = 0$ for all π, σ belonging to $\widehat{G}$, $\pi \neq \sigma$,

$$\chi_\pi * \chi_\pi = \frac{1}{\dim(\pi)} \chi_\pi \quad \text{for every } \pi \in \widehat{G}. \quad (2)$$

Let π and σ be irreducible representations of G . We have

$$\dim(\pi) (\chi_\pi * \chi_\sigma) = \dim(\pi) \gamma_G(\chi_\pi) \chi_\sigma$$

(lemma 4 of V, p. 407). The function $\dim(\pi) \gamma_G(\chi_\pi) \chi_\sigma$ is the orthogonal projection of χ_σ onto the $\overline{\pi}$ -isotypic component of γ_G (cor. 2 of V, p. 465), that is, onto $\varrho_G(\pi)$ (cor. 4 of V, p. 463). Since χ_σ belongs to $\varrho_G(\sigma)$, the result follows from theorem 1 of V, p. 462.

Lemma 2. — The graph of the equivalence relation « $x \in G$ and $y \in G$ and x is conjugate to y » in G is closed.

Indeed, this graph is the image of the continuous map from the compact space $G \times G$ into itself defined by $(x, y) \mapsto (x, yxy^{-1})$.

We denote by $G^\#$ the space of conjugacy classes of G equipped with the quotient topology; it is a compact space by lemma 2 and TG, I, p. 78, prop. 8. Let $\varpi : G \rightarrow G^\#$ be the canonical projection. The map from $\mathcal{C}(G^\#)$ into $\mathcal{C}(G)$ defined by $f \mapsto f \circ \varpi$ identifies the stellar algebra $\mathcal{C}(G^\#)$ with the stellar subalgebra of $\mathcal{C}(G)$ formed by continuous central functions.

The measures on $G^\#$ are identified with the central measures on G (V, p. 402, def. 1).

The linear form on $\mathcal{C}(G^\#)$ defined by $f \mapsto \int_G f$ is then a positive measure of mass 1 on $G^\#$, which is denoted $\mu^\#$. For every $p \in [1, +\infty]$, we denote by $\mathcal{L}^p(G^\#)$ (resp. $L^p(G^\#)$) the space $\mathcal{L}_C^p(G^\#, \mu^\#)$ (resp. $L_C^p(G^\#, \mu^\#)$). We identify $L^p(G^\#)$ with the closure of $\mathcal{C}(G^\#)$ in $L^p(G)$; in

particular, it is a closed subspace of $L^p(G)$.

We also denote by $\Theta(G^\#) = \Theta(G) \cap \mathcal{C}(G^\#)$ the space of central matrix coefficients of G . It is a unital involutive subalgebra of $\mathcal{C}(G^\#)$.

When G is a Lie group, the space $\Theta(G^\#)$ is also denoted $Z\Theta(G)$ in LIE, IX, p. 71.

Lemma 3. — Let π be an irreducible unitary representation of G . The vector space $L^2(G^\#) \cap \varrho_G(\pi)$ is one-dimensional and generated by the character of π .

Consider the unitary representation σ of G in the space $\varrho_G(\pi)$ defined by $\sigma(g) = \varrho_G(g, g)$. Since the unitary representation $\varrho_G(\pi)$ of $G \times G$ is isomorphic to $\overline{\pi} \boxtimes \pi$ (prop. 5 of V, p. 462), its character is given by

$$\chi_\sigma(g) = \chi_{\overline{\pi} \boxtimes \pi}(g, g) = |\chi_\pi(g)|^2$$

for all $g \in G$. The space $L^2(G^\#) \cap \varrho_G(\pi)$ is the subspace of invariant elements of this representation, and its dimension is equal to

$$\int_G \chi_\sigma(g) d\mu(g) = \int_G |\chi_\pi(g)|^2 d\mu(g) = 1$$

(cor. 3 of V, p. 466 and cor. 3 of V, p. 459). Since the character of π is a nonzero element of $L^2(G^\#) \cap \varrho_G(\pi)$, it is a basis of this space.

PROPOSITION 9. — The family $(\chi_\pi)_{\pi \in \widehat{G}}$ of characters of irreducible unitary representations of G is an orthonormal basis of $L^2(G^\#)$.

By the Peter–Weyl theorem (th. 1 of V, p. 462), the closed subspace $L^2(G^\#)$ of $L^2(G)$ formed by the invariant elements of the unitary representation $g \mapsto \varrho_G(g, g)$ of G is the Hilbert sum of the subspaces $L^2(G^\#) \cap \varrho_G(\pi)$ for $\pi \in \widehat{G}$. The proposition therefore follows from the preceding lemma and cor. 3 of V, p. 459.

COROLLARY 1. — The vector space $\Theta(G^\#)$ is dense in $\mathcal{C}(G^\#)$ and in $L^2(G^\#)$.

The second assertion follows from prop. 9. As for the first assertion, consider the linear representation ϱ of G on the Banach space $\mathcal{C}(G)$ defined by

$$\varrho(g)f(x) = f(g^{-1}xg)$$

for all $f \in \mathcal{C}(G)$ and every $x \in G$. It is a continuous isometric representation and $\mathcal{C}(G^\#)$ is the space of invariant elements of this representation. The continuous projector $p = \varrho(1)$ of $\mathcal{C}(G)$ therefore has as its image $\mathcal{C}(G^\#)$ (cor. 3 of V, p. 466); it has norm ≤ 1 .

Let $f \in \mathcal{C}(G^\#)$ and let $\varepsilon > 0$. There exists $\tilde{f} \in \Theta(G)$ such that $\|f - \tilde{f}\| \leq \varepsilon$ (prop. 4 of V, p. 462), and we then have $\|f - p(\tilde{f})\| = \|p(f - \tilde{f})\| \leq \varepsilon$; since $p(\tilde{f}) \in \mathcal{C}(G^\#)$, we conclude that $\Theta(G^\#)$ is dense in $\mathcal{C}(G^\#)$.

We denote by $R(G)$ the Grothendieck ring of continuous representations of G in separated finite-dimensional complex vector spaces (cf. LIE, IX, p. 70 and A, VIII, p. 182, applied to the additive class of continuous representations of G in separated finite-dimensional complex vector spaces, viewed as $\mathbf{C}[G]$ -modules). Since every continuous linear representation of G in a finite-dimensional separated topological vector space is semisimple (prop. 1 of V, p. 457), the abelian group $R(G)$ is free, and the classes of irreducible unitary representations $\pi \in \widehat{G}$ form a basis (A, VIII, p. 186, prop. 7).

COROLLARY 2. — a) The family of characters of the irreducible representations of G is a basis of $\Theta(G^\#)$;

b) The map $u : R(G) \otimes_{\mathbf{Z}} \mathbf{C} \rightarrow \Theta(G^\#)$ such that $u(\pi \otimes 1) = \chi_\pi$ for every $\pi \in \widehat{G}$ is an isomorphism of algebras over $\mathbf{C}$.

The first assertion follows from Proposition 9 and Corollary 1.

Since the classes of the representations $\pi \in \widehat{G}$ form a basis of the free $\mathbf{Z}$ -module $R(G)$, the map u is well defined. It is a morphism of $\mathbf{C}$ -algebras, and it is an isomorphism by a).

COROLLARY 3. — Let H be a locally compact topological group such that G is a compact subgroup of H . Let ϱ be a unitary representation of H in a Hilbert space E . If the π -isotypic component of the restriction of ϱ to G is finite-dimensional for all $\pi \in \widehat{G}$, then the representation ϱ of H is semisimple and every irreducible unitary representation of H has finite multiplicity in ϱ .

Denote by σ the restriction of ϱ to the compact subgroup G of H . Let π be an irreducible representation of G . The endomorphism $\sigma(\chi_\pi) \in \mathcal{L}(E)$ has finite rank, since its image is the $\overline{\pi}$ -isotypic component of σ (cor. 2 of V, p. 465) and it is of finite dimension by hypothesis. Consequently, the endomorphism $\sigma(f)$ has finite rank for every $f \in \Theta(G^\#)$ and is compact for

every $f \in \mathcal{C}(G^\#)$ (cor. 1 of V, p. 468 and cor. of prop. 2 of III, p. 4).

Denote by j the canonical injection from G into H . This is a continuous map that is ν -proper for every measure ν on G (INT, V, p. 69, § 6, no. 1, remark 1). Let $\mathfrak{B}$ be the filter of compact neighborhoods of the element e in G . For every $V \in \mathfrak{B}$ there exists a continuous positive central function f_V on G with support contained in V whose integral over G equals 1. Let β_V be the image measure $j(f_V \cdot \mu)$; it is a positive measure with compact support on H such that $\varrho(\beta_V) = \sigma(f_V)$ (INT, V, p. 71, § 6, no. 2, th. 1). According to what precedes, the filter base on $\mathcal{M}_c(H)$ obtained as the image of $\mathfrak{B}$ under the map $V \mapsto \beta_V$ satisfies the conditions of Proposition 2 of V, p. 402, and the assertion follows.

COROLLARY 4 (Weyl's equidistribution criterion)

Let M be a set of central positive measures on G such that $\nu(G) = 1$ for all $\nu \in M$. Let $\mathfrak{F}$ be a filter on M . For the filter $\mathfrak{F}$ to converge vaguely to the measure $\mu^\#$ on $G^\#$, it is necessary and sufficient that, for every nontrivial irreducible unitary representation π , one has

$$\lim_{\nu, \mathfrak{F}} \int_G \chi_\pi(x) d\nu(x) = 0.$$

Since $\nu(G^\#) = 1 = \mu^\#(G^\#)$ for $\nu \in M$, the hypothesis means that

$$\lim_{\nu, \mathfrak{F}} \int_{G^\#} \chi_\pi(x) d\nu(x) = \int_{G^\#} \chi_\pi(x) d\mu^\#(x)$$

for every representation $\pi \in \widehat{G}$, hence it is equivalent to the condition

$$\lim_{\nu, \mathfrak{F}} \int_{G^\#} f(x) d\nu(x) = \int_{G^\#} f(x) d\mu^\#(x)$$

for every function $f \in \Theta(G^\#)$ by linearity. Since the space $\Theta(G^\#)$ is dense in $\mathcal{C}(G^\#)$ (corollary 1), the hypothesis is therefore equivalent to the convergence of the filter $\mathfrak{F}$ to $\mu^\#$ in $\mathcal{M}(G^\#)$ endowed with the topology of pointwise convergence in $\mathcal{C}(G)$, which coincides with the topology of vague convergence since G is compact (cf. INT, III, p. 59, § 1, no. 9).

V.4.7 The Fourier cotransformation

We equip the set $\widehat{G}$ with the discrete topology. We denote by $F(\widehat{G})$ the product algebra of the $\text{End}(E_\pi)$ for π belonging to $\widehat{G}$ and $F_b(\widehat{G})$ the product stellar algebra of the $\text{End}(E_\pi)$ (Example 5 of I, p. 103); this is the set of families $(u_\pi)_{\pi \in \widehat{G}}$ such that $\sup \|u_\pi\| < +\infty$.

$$\pi \in \widehat{G}$$

We denote by $F_0(\widehat{G})$ the closed stellar subalgebra of $F_b(\widehat{G})$ consisting of families $(u_\pi)_{\pi \in \widehat{G}}$ such that $\|u_\pi\|$ tends to 0 at infinity.

Let $\nu \in \mathcal{M}^1(G)$. For every $\pi \in \widehat{G}$, we have $\|\pi(\nu)\| \leq \|\nu\|$, so the family $(\pi(\nu))_{\pi \in \widehat{G}}$ belongs to $F_b(\widehat{G})$.

DEFINITION 1. — For every measure $\nu \in \mathcal{M}^1(G)$, the element $(\pi(\nu))_{\pi \in \widehat{G}}$ of $F_b(\widehat{G})$ is denoted $\overline{\mathcal{F}}_G(\nu)$. The map from $\mathcal{M}^1(G)$ into $F_b(\widehat{G})$ thus defined is called the Fourier cotransformation of G , and $\overline{\mathcal{F}}_G(\nu)$ is called the Fourier cotransform of the measure ν .

For $f \in L^1(G)$, we denote by $\overline{\mathcal{F}}_G(f) = \overline{\mathcal{F}}_G(f \cdot \mu)$.

For any representation $\pi \in \widehat{G}$, the map $\nu \mapsto \pi(\nu)$ is a unital morphism of involutive Banach algebras from $\mathcal{M}^1(G)$ into $\text{End}(E_\pi)$ (lemma 1 of V, p. 401). The Fourier cotransformation is therefore a unital morphism of involutive Banach algebras from $\mathcal{M}^1(G)$ into $F_b(\widehat{G})$.

V.4.8 The Fourier transform

We keep the notation of the previous number.

Let $\pi \in \widehat{G}$. The vector space $\text{End}(E_\pi)$ is equipped with the structure of a Hilbert space whose scalar product is given by

$$\langle u_1 | u_2 \rangle = \dim(\pi) \text{Tr}(u_1^* u_2) = \dim(\pi) \text{Tr}(u_2 u_1^*)$$

for u_1, u_2 in $\text{End}(E_\pi)$ (cf. EVT, V, p. 52, th. 1).

We denote by $\|u\|_2 = \sqrt{\langle u | u \rangle}$ the norm of an element u of $\text{End}(E_\pi)$ for $\pi \in \widehat{G}$. For every $g \in G$, we have $\|\pi(g)\|_2 = \dim(\pi)$ since $\pi(g)$ is unitary.

The norm denoted here $\|u\|_2$ differs by a factor $\dim(\pi)$ from the norm defined in EVT, V, p.

52 on the space of Hilbert–Schmidt operators on E_π .

Lemma 4. — Let π be an irreducible representation of G . The map $f \mapsto \pi(f)$ defines, by passing to subspaces, an isometric isomorphism from $\varrho_G(\overline{\pi})$ onto $\text{End}(E_\pi)$.

Set $\varepsilon_\pi(g, h)u = \pi(g) \circ u \circ \pi(h^{-1})$ for all $(g, h) \in G \times G$ and for every $u \in \text{End}(E_\pi)$. The map ε_π is a continuous representation of $G \times G$ in $\text{End}(E_\pi)$. It is unitary. Indeed, since π itself is unitary, it follows that

$$\begin{aligned} \|\varepsilon_\pi(g, h)u\|_2^2 &= \dim(\pi) \text{Tr} \left((\pi(g)u\pi(h^{-1}))^* \pi(g)u\pi(h^{-1}) \right) \\ &= \dim(\pi) \text{Tr}(\pi(h)u^*u\pi(h)^{-1}) = \dim(\pi) \text{Tr}(u^*u) = \|u\|_2^2 \end{aligned}$$

for all $(g, h) \in G \times G$ and every $u \in \text{End}(E_\pi)$.

The map Ψ defined by $f \mapsto \pi(f)$ is a $(G \times G)$ -morphism from $\varrho_G(\overline{\pi})$ into $\text{End}(E_\pi)$ since

$$\pi(\varrho_G(g, h)f) = \int_G f(g^{-1}xh)\pi(x)d\mu(x) = \pi(g)\pi(f)\pi(h^{-1})$$

for all $(g, h) \in G \times G$ and every $f \in \varrho_G(\overline{\pi})$. Since $\varrho_G(\overline{\pi})$ is an irreducible representation (th. 1, a) of V , p. 462), there exists $\lambda \in \mathbf{C}^*$ such that the map $\lambda\Psi$ is zero or isometric (cor. 5, a) of V , p. 388). Let $f = \dim(\pi)\overline{\chi}_\pi \in \varrho_G(\overline{\pi})$. We obtain $\pi(f) = 1_{E_\pi}$ (cor. 2 of V , p. 465), whence $\|\pi(f)\|_2 = \dim(\pi) = \|f\|$ (cor. 3 of V , p. 459). Consequently, the map Ψ is an isometry. Since $\varrho_G(\overline{\pi})$ and $\text{End}(E_\pi)$ have the same dimension, Ψ is an isometric isomorphism.

We denote by $F^2(\widehat{G})$ the Hilbert sum of Hilbert spaces $\text{End}(E_\pi)$. The norm of an element $x \in F^2(\widehat{G})$ is still denoted $\|x\|_2$.

In LIE, IX, p. 79, this space is denoted $L^2(\widehat{G})$, a notation we prefer to avoid here so as not to create confusion with the space $\ell^2(\widehat{G})$. Let $(u_\pi)_{\pi \in \widehat{G}}$ be an element of $F^2(\widehat{G})$. Since

$$\sum_{\pi \in \widehat{G}} \|u_\pi\|_2^2 = \sum_{\pi \in \widehat{G}} \dim(\pi) \text{Tr}(u_\pi^* u_\pi) < +\infty$$

and since $\|u_\pi\|^2 \leq \text{Tr}(u_\pi^* u_\pi)$ (cf. EVT, V , p. 52, formula (33)), the norm in $\mathcal{L}(E_\pi)$ of the endomorphism u_π tends to 0 at infinity. One can therefore identify $F^2(\widehat{G})$ with a subspace of $F_0(\widehat{G})$.

Let $\pi \in \widehat{G}$ and $u \in \text{End}(E_\pi)$. We denote by $\mathcal{F}_\pi(u)$ the function on G defined by

$\mathcal{F}_\pi(u)(g) = \langle \pi(g)|u \rangle = \dim(\pi)\text{Tr}(\pi(g)^*u)$ for all $g \in \mathbf{G}$. It is a continuous function on $\mathbf{G}$. If $\mathbf{G}$ is a compact real Lie group, then the function $\mathcal{F}_\pi(u)$ is analytic on $\mathbf{G}$ (LIE, III, § 8, no. 1, th. 1).

Since $\mathbf{G}$ is compact, one may identify $L^2(\mathbf{G})$ with a subspace of $L^1(\mathbf{G})$.

THEOREM 2. — The Fourier cotransformation of $\mathbf{G}$ induces, by passing to subspaces, an isometric isomorphism of $L^2(\mathbf{G})$ onto $F^2(\widehat{\mathbf{G}})$. Its inverse $\mathcal{F}_\mathbf{G}$ associates to an element $(u_\pi)_{\pi \in \widehat{\mathbf{G}}}$ of $F^2(\widehat{\mathbf{G}})$ the sum of the series

$$\sum_{\pi \in \widehat{\mathbf{G}}} \mathcal{F}_\pi(u_\pi),$$

which converges in $L^2(\mathbf{G})$

By the Peter–Weyl theorem (Th. 1 of V, p. 462), the Hilbert space $L^2(\mathbf{G})$ is the Hilbert sum of the spaces $\varrho_\mathbf{G}(\overline{\pi})$ for $\pi \in \widehat{\mathbf{G}}$. For every $\pi \in \widehat{\mathbf{G}}$, the linear map $f \mapsto \pi(f)$ from $\varrho_\mathbf{G}(\overline{\pi})$ into $\text{End}(E_\pi)$ is an isometric isomorphism (Lemma 4). Consequently, the restriction to $L^2(\mathbf{G})$ of the Fourier cotransformation defines an isometric isomorphism of $L^2(\mathbf{G})$ onto $F^2(\widehat{\mathbf{G}})$.

Let $f \in L^2(\mathbf{G})$. Let $\pi \in \widehat{\mathbf{G}}$ and let $f_{\overline{\pi}} \in \mathcal{C}(\mathbf{G})$ be the orthogonal projection of f onto $\varrho_\mathbf{G}(\overline{\pi})$ (th. 1 of V, p. 462). Since this space is the $\overline{\pi}$ -isotypic component of $\boldsymbol{\delta}_\mathbf{G}$ (cor. 4 of V, p. 463), we have $f_{\overline{\pi}} = \dim(\pi)\boldsymbol{\delta}_\mathbf{G}(\chi_\pi)(f)$ by cor. 2 of V, p. 465. For every $x \in \mathbf{G}$, it follows that

$$\begin{aligned} f_{\overline{\pi}}(x) &= \dim(\pi) \int_{\mathbf{G}} \chi_\pi(g) (\boldsymbol{\delta}_\mathbf{G}(g)f)(x) d\mu(g) \\ &= \dim(\pi) \int_{\mathbf{G}} \chi_\pi(g) f(xg) d\mu(g) \\ &= \dim(\pi) \int_{\mathbf{G}} \chi_\pi(x^{-1}y) f(y) d\mu(y) \\ &= \dim(\pi) \text{Tr}(\pi(x^{-1})\pi(f)) = \langle \pi(x) | \pi(f) \rangle, \end{aligned}$$

that is $f_{\overline{\pi}} = \mathcal{F}_\pi(\pi(f))$.

Since f is the sum in $L^2(\mathbf{G})$ of the family $(f_{\overline{\pi}})$, by the Peter–Weyl theorem, we obtain

$$f = \sum_{\pi \in \widehat{\mathbf{G}}} \mathcal{F}_\pi(\pi(f))$$

where the series converges in $L^2(\mathbf{G})$. This proves the theorem.

DEFINITION 2. — The isometric isomorphism $\mathcal{F}_\mathbf{G}$ of $F^2(\widehat{\mathbf{G}})$ onto $L^2(\mathbf{G})$, which is the

inverse of the isomorphism induced by the Fourier cotransformation, is called the Fourier transform of G . The image of an element of $F^2(\widehat{G})$ is called its Fourier transform.

Remarks. — 1) The Fourier transform of $(u_\pi)_{\pi \in \widehat{G}} \in F^2(\widehat{G})$ is therefore the class in $L^2(G)$ of the series

$$g \mapsto \sum_{\pi \in \widehat{G}} \langle \pi(g) | u_\pi \rangle = \sum_{\pi \in \widehat{G}} \dim(\pi) \operatorname{Tr}(u_\pi \circ \pi(g)^{-1}).$$

2) Let $f \in L^2(G)$. We then have the Plancherel formula

$$\|f\|^2 = \|\overline{\mathcal{F}}_G(f)\|^2 = \sum_{\pi \in \widehat{G}} \|\pi(f)\|^2.$$

Moreover, by Theorem 2, f is the sum in $L^2(G)$ of the family $(f_\pi)_{\pi \in \widehat{G}}$ where

$$\begin{aligned} f_\pi(x) &= \mathcal{F}_\pi(\pi(f))(x) \\ &= \langle \pi(x) | \pi(f) \rangle = \dim(\pi) \int_G f(g) \chi_\pi(x^{-1}g) d\mu(g) \end{aligned}$$

for all $x \in G$ and every $\pi \in \widehat{G}$.

Suppose that G is commutative. Since the irreducible representations of G are of dimension 1 (cor. 7 of V, p. 390) and the dual group $\widehat{G}$ is discrete (prop. 18 of II, p. 233), the algebras $F_b(\widehat{G})$ and $F_0(\widehat{G})$ are identified, respectively, with the algebra $\mathcal{C}_b(\widehat{G})$ of continuous bounded functions on $\widehat{G}$ and the algebra $\mathcal{C}_0(\widehat{G})$ of continuous functions tending to 0 at infinity on $\widehat{G}$. Since G is compact, the Haar measure on $\widehat{G}$ dual to μ is the counting measure $\widehat{\mu}$ (prop. 18 of II, p. 233). The Hilbert sum $F^2(\widehat{G})$ of the spaces $\operatorname{End}(E_\pi)$ for $\pi \in \widehat{G}$ is identified with the Hilbert space $L^2(\widehat{G}, \widehat{\mu})$.

Let $\nu \in \mathcal{M}^1(G)$. Then, for every unitary character $\chi \in \widehat{G}$, we have

$$\chi(\nu) = \int_G \chi \nu$$

which proves that the Fourier cotransformation defined above coincides with the Fourier cotransformation of G defined in II, p. 206.

Every $f \in \mathcal{L}^2(G)$ is integrable and the formula $\|\overline{\mathcal{F}}_G(f)\|_2 = \|f\|$ is the Plancherel formula

(II, p. 215, theorem 1).

PROPOSITION 10. — The Fourier cotransformation is an injective morphism of involutive Banach algebras from $\mathcal{M}^1(G)$ into $F_b(\widehat{G})$ which sends $L^1(G)$ into $F_0(\widehat{G})$.

Since G is compact, the space $\mathcal{C}(G)$ is contained and dense in $\mathcal{L}^1(G)$ and $\mathcal{L}^2(G)$.

Let $\nu \in \mathcal{M}^1(G)$ such that $\overline{\mathcal{F}}_G(\nu) = 0$. For every function f continuous on G , we then have $\overline{\mathcal{F}}_G(\nu * f) = 0$. Now $\nu * f$ belongs to $\mathcal{C}(G)$ (INT, VIII, p. 152, § 4, n°2, prop. 3). Since, by theorem 2, the Fourier cotransformation is injective on $L^2(G)$, and a fortiori, on the space $\mathcal{C}(G)$, we have $\nu * f = 0$ for $f \in \mathcal{C}(G)$. In particular, it follows that

$$\int_G f(x) d\nu(x) = (\nu * \check{f})(e) = 0$$

for every function $f \in \mathcal{C}(G)$ (INT, VIII, loc. cit.), therefore the measure ν is zero.

The image of $\mathcal{C}(G)$ by the Fourier cotransformation is contained in $F^2(\widehat{G})$, and a fortiori in $F_0(\widehat{G})$. Since $F_0(\widehat{G})$ is closed in $F_b(\widehat{G})$ and since $\mathcal{C}(G)$ is dense in $L^1(G)$, the image of $L^1(G)$ by the Fourier cotransformation is also contained in $F_0(\widehat{G})$.

PROPOSITION 11. — a) Let $u = (u_\pi)_{\pi \in \widehat{G}}$ be an element of $F(\widehat{G})$. If the family $(\mathcal{F}_\pi(u_\pi))_{\pi \in \widehat{G}}$ is uniformly summable in $\mathcal{C}(G)$, then its sum f is a continuous function on G whose Fourier cotransform is u ;

b) Let $f \in L^1(G)$. If the family $(\mathcal{F}_\pi(\pi(f)))_{\pi \in \widehat{G}}$ is uniformly summable in $\mathcal{C}(G)$, then its sum is a continuous function on G whose class in $L^1(G)$ is equal to f .

Let us prove assertion a). Since G is compact, the sum f of the family $(\mathcal{F}_\pi(u_\pi))$ belongs to $L^2(G)$ and the series

$$\sum_{\pi \in \widehat{G}} \mathcal{F}_\pi(u_\pi)$$

converges in $L^2(G)$ to f (INT, IV, p. 127, § 3, n° 3, prop. 4). We therefore have $\mathcal{F}_G((u_\pi)_\pi) = f$ in $L^2(G)$, whence $(u_\pi)_{\pi \in \widehat{G}} = \overline{\mathcal{F}}_G(f)$ (theorem 2).

Let us prove assertion b). We may apply assertion a) to the family $(\pi(f))_{\pi \in \widehat{G}}$. The sum g of the family $(\mathcal{F}_\pi(\pi(f)))_{\pi \in \widehat{G}}$ is a continuous function, and hence belongs to $L^2(G)$, such that $\overline{\mathcal{F}}_G(g) = (\pi(f))_{\pi \in \widehat{G}}$. Since $\overline{\mathcal{F}}_G$ is injective on $L^2(G)$, we have $f = g$ in $L^2(G)$, hence in

$L^1(G)$.

The algebra $\mathcal{C}(\widehat{G})$ of complex-valued functions on $\widehat{G}$ is identified with the center of the algebra $F(\widehat{G})$ by the map that associates to $f: \widehat{G} \rightarrow \mathbb{C}$ the central element $(f(\pi)1_{E_\pi})_{\pi \in \widehat{G}}$ of $F(\widehat{G})$. Denote by $\mathcal{M}^1(G^\#)$ the space of central bounded measures on G . It is a closed subspace of the Banach algebra $\mathcal{M}^1(G)$. For every measure $\nu \in \mathcal{M}^1(G^\#)$ and every representation $\pi \in \widehat{G}$, we have $\pi(\nu) \in \text{End}_G(E_\pi)$ hence $\pi(\nu)$ is a scalar multiple of the identity endomorphism of E_π by Schur's lemma (prop. 6 of V, p. 386). The restriction of the Fourier cotransformation to $\mathcal{M}^1(G^\#)$ is therefore identified with a unital morphism of involutive Banach algebras from $\mathcal{M}^1(G^\#)$ into $\mathcal{C}(\widehat{G})$. This morphism is injective; the image of $L^1(G^\#)$ is contained in $\mathcal{C}_b(\widehat{G})$ and its restriction to $L^2(G^\#)$ is an isometric isomorphism onto $L^2(\widehat{G}, \widehat{\mu})$.

V.4.9 Frobenius–Schur indicator and Larsen's alternative

Recall (LIE, VIII, §7, n° 6, def. 2) that an irreducible representation of G in a finite-dimensional complex vector space E is said to be of orthogonal type (resp. of symplectic type) if there exists a nonzero symmetric bilinear form on E invariant under G (resp. if there exists a nonzero alternating bilinear form on E invariant under G). It is said to be of complex type if there exists no nonzero bilinear form on E invariant under G .

When an irreducible representation in E is of orthogonal type (resp. symplectic type), the space of symmetric (resp. alternating) G -invariant bilinear forms on E has dimension 1 and every nonzero G -invariant bilinear form on E is separating.

A representation of orthogonal type is sometimes said to be of real type, and a representation of symplectic type is sometimes said to be of quaternionic type (cf. LIE, IX, App. II).

Let π be an irreducible representation of G in a complex vector space E . The three possibilities above are distinguished by the value of the quantity

$$\text{FS}(\pi) = \int_G \chi_\pi(g^2) d\mu(g),$$

which is called the Frobenius–Schur indicator of π . Indeed we have

$$\text{FS}(\pi) = \begin{cases} 1 & \text{if } \pi \text{ is of orthogonal type,} \\ -1 & \text{if } \pi \text{ is of symplectic type,} \\ 0 & \text{if } \pi \text{ is of complex type.} \end{cases}$$

(LIE, IX, App. 2, p. 103, prop. 3 and p. 105, prop. 4).

When G is a Lie group, one can compute $\text{FS}(\pi)$ using prop. 1 of LIE, IX, p. 69.

DEFINITION 3. — Let ϱ be a finite-dimensional unitary representation of G in a Hilbert space E . Let k be a positive integer. We call the dimension of the subspace of G -invariant elements in the representation $\overline{\varrho}^{\otimes k} \otimes \varrho^{\otimes k}$ the absolute moment of order $2k$ of ϱ , and denote it by $M_{2k}(\varrho)$.

THEOREM 3 (Larsen's alternative). — Suppose that G is infinite. Let π be a faithful unitary representation of G in a Hilbert space E of finite dimension ≥ 2 .

- a) Suppose that the derived group of G is infinite. We have $M_4(\pi) \geq 2$ with equality if and only if G contains $\text{SU}(E)$;
- b) Suppose that $\dim(E) \geq 3$, that π is of orthogonal or symplectic type, and that $\dim(E) \neq 4$ if π is of orthogonal type. Then $M_4(\pi) \geq 3$ with equality if and only if the complexified Lie algebra of G is equal to the Lie algebra of a separating G -invariant bilinear form on E . *This is the case if and only if the neutral component G_0 of G is a maximal compact subgroup of the group of automorphisms of such a bilinear form.*

We shall use in the proof the following lemmas.

Lemma 5. — Let π be a unitary representation of G in a Hilbert space E of finite dimension. Let $(\varrho_i)_{i \in I}$ be a family of nonzero unitary representations of G and $(n_i)_{i \in I}$ a family of integers ≥ 1 such that the representation $\overline{\pi} \otimes \pi$ of G (resp. the representation $\pi \otimes \pi$) is isomorphic to the direct sum $\bigoplus_{i \in I} \varrho_i^{n_i}$. Then we have

$$M_4(\pi) \geq \sum_{i \in I} n_i^2$$

with equality if and only if the representations ϱ_i are irreducible and pairwise non-isomorphic.

Denote by χ_i the character of ϱ_i for $i \in I$. We have

$$|\chi_\pi|^2 = \chi_{\bar{\pi} \otimes \pi} = \sum_{i \in I} n_i \chi_i, \quad (\text{resp. } \chi_\pi^2 = \chi_{\pi \otimes \pi} = \sum_{i \in I} n_i \chi_i).$$

By the definition and cor. 3 of V, p. 466, we have

$$M_4(\pi) = \int_G |\chi_\pi|^4 d\mu,$$

whence, in both cases, the formula

$$M_4(\pi) = \sum_{i,j} n_i n_j \int_G \chi_i \bar{\chi}_j d\mu = \sum_{i,j} n_i n_j \dim \text{Hom}_G(\varrho_i, \varrho_j)$$

(cor. 4, b) of V, p. 459). Let $i \in I$. Since the representation ϱ_i is not zero, the space $\text{Hom}_G(\varrho_i, \varrho_i)$ is not zero, and we deduce the lower bound

$$M_4(\pi) \geq \sum_{i \in I} n_i^2.$$

Moreover, since $n_i \geq 1$, there is equality if and only if we have $\dim \text{Hom}_G(\varrho_i, \varrho_j) = 0$ if $i \neq j$ and $\dim \text{Hom}_G(\varrho_i, \varrho_i) = 1$ for every i . The second condition holds if and only if the representations ϱ_i are irreducible (loc. cit.). The first is then satisfied if and only if the representations ϱ_i are pairwise non-isomorphic, by Schur's lemma (V, p. 387, cor. 2).

If E is a module over a commutative ring A , we will call the symmetric square (resp. exterior square) of E the A -module $S^2(E)$ (resp. $\wedge^2 E$).

Lemma 6. — Let q be a separating bilinear form on a finite-dimensional complex vector space E of dimension ≥ 3 .

- a) If q is symmetric, then the adjoint representation of the orthogonal Lie algebra $\mathfrak{so}(q)$ is isomorphic to the exterior square $\wedge^2 E$ of the natural representation $\mathfrak{so}(q) \rightarrow \mathfrak{gl}(E)$;
- b) If q is alternating, then the adjoint representation of the symplectic Lie algebra $\mathfrak{sp}(q)$ is isomorphic to the symmetric square $S^2(E)$ of the natural representation $\mathfrak{sp}(q) \rightarrow \mathfrak{gl}(E)$.

Endow C with the identity involutive automorphism. The bilinear form q is then ε

-Hermitian (A, IX, § 3, no. 1, def. 1) with $\varepsilon = 1$ in case a) and $\varepsilon = -1$ in case b). For every $u \in \text{End}(E)$, we denote by ${}^t u$ the adjoint of u with respect to q . We therefore have $q(x, u(y)) = q({}^t u(x), y)$ for all $(x, y) \in E \times E$. Let v denote the unique automorphism of $E \otimes E$ such that $v(x \otimes y) = y \otimes x$ for all $(x, y) \in E^2$.

There exists a unique linear map w from $E \otimes E$ into $\text{End}(E)$ such that $w(a \otimes b)$ is the linear map defined by $x \mapsto q(a, x)b$ for all $(a, b, x) \in E^3$. The map w is an isomorphism. For every $s \in E \otimes E$, we have

$${}^t(w(s)) = \varepsilon w(v(s)). \quad (3)$$

Indeed, for all (x, y) and (a, b) in $E \times E$, it follows that

$$q(x, w(a \otimes b)y) = q(a, y)q(x, b) = \varepsilon q(b, x)q(a, y) = q(\varepsilon w(b \otimes a)x, y).$$

Finally, denote by $x \mapsto x^*$ the isomorphism from E to E' deduced from q , that is, such that $\langle x^*, y \rangle = q(x, y)$ for all $(x, y) \in E^2$.

With these notations established, let H denote the orthogonal (resp. symplectic) group of q . It is a Lie subgroup of $\mathbf{GL}(E)$ whose Lie algebra $\mathfrak{h}$ is the subspace of $\text{End}(E)$ consisting of the $u \in \text{End}(E)$ such that ${}^t u = -u$ (LIE, III, p. 146, cor. 1). Since the form q is H -invariant, it follows that

$$\langle (ga)^*, b \rangle = q(ga, b) = \langle a, g^{-1}b \rangle$$

for all $g \in H$ and every $(a, b) \in E \times E$.

Equip $\text{End}(E)$ with the adjoint representation Ad of H . The linear map w is a morphism of representations of H : indeed, for every $g \in H$ and every $(a, b, x) \in E^3$, we have

$$\begin{aligned} w(g(a \otimes b))(x) &= \langle (ga)^*, x \rangle gb = \langle a, g^{-1}x \rangle gb \\ &= g(\langle a, g^{-1}x \rangle b) = \text{Ad}(g)(w(a \otimes b))x, \end{aligned}$$

whence the conclusion by linearity.

Since v is also a morphism of representations of H , the same holds for the linear map $\theta =$

$w - \varepsilon w \circ v$; this is therefore a morphism of $\mathfrak{h}$ -modules.

Denote by $F \subset E \otimes E$ the subspace of elements $s \in E \otimes E$ such that $v(s) = -\varepsilon s$. If $\varepsilon = -1$, the restriction to F of the canonical map from $E \otimes E$ into the symmetric square of E is an isomorphism of $\mathbb{C}$ -vector spaces (A, III, p. 72); if $\varepsilon = 1$, the restriction to F of the canonical map from $E \otimes E$ into the exterior square of E is an isomorphism of $\mathbb{C}$ -vector spaces (loc. cit., p. 82).

The image of F under θ is contained in $\mathfrak{h}$: indeed, for every $s \in F$ the formula (3) implies

$${}^t\theta(s) = {}^tw(s) - \varepsilon {}^tw(v(s)) = \varepsilon w(v(s)) - w(s) = -\theta(s).$$

By definition of F , the restriction of the map θ to F coincides with that of $2w$ and the linear map θ is therefore injective on F . Since $\dim(F) = \dim(\mathfrak{h})$. From (A, III, p. 75, th. 1 and p. 87, cor. 1 and LIE, VIII, p. 192 and p. 201), we conclude that θ induces, by passage to subspaces, an isomorphism of $\mathfrak{h}$ -modules from F onto $\mathfrak{h}$, whence the lemma.

Lemma 7. — Let H_2 be a real Lie group and let H_1 be a closed subgroup of H_2 . The complexified Lie algebra of H_1 is identified with a subrepresentation of the adjoint representation of H_1 on the complexified Lie algebra of H_2 .

Indeed, the restriction to H_1 of the adjoint representation of H_2 on its Lie algebra identifies with the adjoint representation of H_1 .

Lemma 8. — Let E be a complex Hilbert space of finite dimension n .

- a) The groups $\mathbf{SU}(E)$ and $\mathbf{U}(E)$ are connected;
- b) The derived group of $\mathbf{SU}(E)$ is equal to $\mathbf{SU}(E)$;
- c) The derived subgroup of $\mathbf{U}(E)$ is equal to $\mathbf{SU}(E)$.

We may assume that $n \geq 1$ and that $E = \mathbb{C}^n$.

Let A be the subgroup of $\mathbf{U}(E)$ formed by diagonal matrices; it is homeomorphic to U^n and therefore connected (TG, VI, p. 11, cor. 2 and I, p. 83, prop. 8). Its intersection with $\mathbf{SU}(E)$ is homeomorphic to U^{n-1} , hence it is also connected.

By th. 1 of IV, p. 149, the group $\mathbf{U}(E)$ (resp. $\mathbf{SU}(E)$) is the union of the connected subsets

gAg^{-1} for $g \in G$ (resp. of the connected subsets $g(A \cap \mathbf{SU}(E))g^{-1}$); since the identity element belongs to each of these sets, the space $\mathbf{U}(E)$ (resp. $\mathbf{SU}(E)$) is connected (TG, I, p. 81, prop. 2).

Let us prove *b*). The assertion is true when $n = 1$ since $\mathbf{SU}(E)$ is then reduced to the identity element. Suppose therefore that $n \geq 2$. The Lie algebra of $\mathbf{SU}(E)$ is then simple (LIE, IX, p. 20, § 3, n°4), so the derived group of $\mathbf{SU}(E)$ has finite index in $\mathbf{SU}(E)$. From this we deduce the result since $\mathbf{SU}(E)$ is connected by *a*).

The assertion *c* follows from *b* since the derived group of $\mathbf{U}(E)$ is contained in $\mathbf{SU}(E)$.

We now prove Theorem 3. Let n denote the dimension of the Hilbert space E . Since the representation π is faithful, one may assume that G is a compact subgroup of $\mathbf{U}(E)$. The group G is closed in the real Lie group $\mathbf{U}(E)$, hence is a compact real Lie group (LIE, III, §8, no. 2, th. 2). By hypothesis, G is infinite, hence of dimension ≥ 1 . Let $\mathfrak{g}$ be its Lie algebra; it is nonzero.

Let us prove *a*). Suppose that the derived group $D(G)$ is infinite. We have the G -invariant decomposition $\text{End}(E) = \mathbb{C}1_E \oplus \mathfrak{sl}(E)$. The representations of G on $\mathbb{C}1_E$ and on $\mathfrak{sl}(E)$ are not isomorphic, since $\dim \mathfrak{sl}(E) \geq 2$. By Lemma 5, we therefore have $M_4(\pi) \geq 2$ with equality if and only if the representation of G in $\mathfrak{sl}(E)$ is irreducible. Now, the derived group of G is contained in $\mathbf{SL}(E)$, and the complexification of the Lie algebra of $D(G)$ therefore identifies with a subspace of $\mathfrak{sl}(E)$, which is a subrepresentation of the representation of G in $\mathfrak{sl}(E)$ (Lemma 7). This subrepresentation is nonzero, since $D(G)$ is infinite. We therefore have $M_4(\pi) = 2$ if and only if $(\mathcal{D}\mathfrak{g})_{(\mathbb{C})} = \mathfrak{sl}(E)$. Since $D(G)$ is contained in $\mathbf{SU}(E)$, and since $\mathbf{SU}(E) = D(\mathbf{SU}(E))$ (Lemma 8), this condition is equivalent to $\mathbf{SU}(E) \subset G$.

For the proof of assertion *b*), we resume the notation of LIE, VIII, §13, nos. 2–4.

Suppose that π is of symplectic type and that $\dim(E) \geq 3$. Let q be a nonzero G -invariant alternating bilinear form on E ; it is separating (LIE, IX, p. 103, App. 2, prop. 3) and G is a compact subgroup of the symplectic group $\mathbf{Sp}(q)$. Denote by $q^* \in \wedge^2 E$ the element to which the alternating form q identifies. The complexified Lie algebra $\mathfrak{g}_{(\mathbb{C})}$ of G is contained in $\mathfrak{sp}(q)$. Now, since $\dim(E) \geq 3$, the representation of $\mathfrak{sp}(q)$ in $E \otimes E$ admits the direct sum decomposition

$$E \otimes E = S^2(E) \oplus \wedge^2 E = S^2(E) \oplus E_2 \oplus Cq^*,$$

where E_2 is the second fundamental representation of $\mathfrak{sp}(q)$ (LIE, VIII, p. 202, §13, no. 3, (IV)). The representations E_2 and Cq^* of $\mathfrak{sp}(q)$ are irreducible (loc. cit.) and the representation $S^2(E)$ is nonzero.

By Lemma 5, we therefore have $M_4(\pi) \geq 3$ with equality if and only if the representations of G in $S^2(E)$, E_2 and Cq^* are irreducible and pairwise non-isomorphic.

Suppose that $M_4(\pi) = 3$. Since the representation $S^2(E)$ identifies with the adjoint representation of $\mathfrak{sp}(q)$ (Lemma 6), it contains as a subrepresentation the complexification of the adjoint representation of G (Lemma 7). We therefore have $\mathfrak{g}_{(C)} = \mathfrak{sp}(q)$.

Conversely, suppose that $\mathfrak{g}_{(C)} = \mathfrak{sp}(q)$. The adjoint representation of $\mathfrak{sp}(q)$ and the representation E_2 of $\mathfrak{sp}(q)$ are irreducible, of dimensions $n(n+1)/2$ and $n(n-1)/2 - 1$ respectively (LIE, VIII, p. 202, §13, no. 3, (IV)), which are different and ≥ 2 since $n \geq 3$, whence $M_4(\pi) = 3$.

Finally, suppose that π is of real type and that $\dim(E) \geq 3$ and $\dim(E) \neq 4$. Let q be a nonzero symmetric bilinear form on E , invariant under G ; it is separating (LIE, IX, p. 103, App. 2, prop. 3) and G is a compact subgroup of the orthogonal group $O(q)$. Denote by $q^* \in S^2(E)$ the element to which q identifies.

The complexified Lie algebra $\mathfrak{g}_{(C)}$ of G is contained in $\mathfrak{so}(q)$. The representation of $\mathfrak{so}(q)$ in $E \otimes E$ admits the direct sum decomposition

$$E \otimes E = \wedge^2 E \oplus S^2(E) = \wedge^2 E \oplus S_0^2(E) \oplus Cq^*,$$

where $S_0^2(E)$ is the orthogonal complement of q^* in $S^2(E)$. These representations are of dimension at least 2 since $\dim(E) \geq 3$. By Lemma 5, we have $M_4(\pi) \geq 3$ with equality if and only if the representations of G in $\wedge^2 E$ and $S_0^2(E)$ are irreducible and non-isomorphic.

Suppose that $M_4(\pi) = 3$. Since the representation $\wedge^2 E$ identifies with the adjoint representation of $\mathfrak{so}(q)$ (Lemma 6), it contains as a subrepresentation the complexification of the adjoint representation of G (Lemma 7). The condition $M_4(\pi) = 3$ therefore implies that $\mathfrak{g}_{(C)} = \mathfrak{so}(q)$.

Conversely, suppose that $\mathfrak{g}_{(C)} = \mathfrak{so}(q)$. The adjoint representation of $\mathfrak{so}(q)$ is irreducible,

since $\dim(E) \neq 4$ (LIE, VIII, § 13, p. 193, (I) and p. 206, (I)), and of dimension $n(n-1)/2$. The representation $S_0^2(E)$ of $\mathfrak{so}(q)$ has as highest weight $2\varpi_1$. By comparing its dimension with that of the irreducible representation of $\mathfrak{so}(q)$ of highest weight $2\varpi_1$, computed by the Weyl formula (cf. LIE, VIII, §9, no. 2, th. 2 and LIE, VI, plates II and IV), one verifies that $S_0^2(E)$ is irreducible. We therefore conclude that $M_4(\pi) = 3$ if $\mathfrak{g}_{(\mathbb{C})} = \mathfrak{so}(q)$.

Remarks. — 1) The condition “G is infinite” is necessary in Theorem 3 (exercise 9 of V, p. 507).

2) Let $n \in \mathbb{N}$. For every unitary representation ϱ of a compact group in a Hilbert space of dimension n , we have $M_4(\varrho) \leq n^2$; equality is possible (exercise 15 of V, p. 508).

3) One can prove (cf. R. GURALNICK and P.H. TIEP, Decomposition of small tensor powers and Larsen’s conjecture, Representation Theory 9 (2005), 138–208) that the condition that G is infinite can be omitted if one assumes that E has dimension ≥ 7 and if one replaces the hypothesis $M_4(\pi) = 2$ (resp. $M_4(\pi) = 3$) by $M_8(\pi) = 24$ (resp. $M_8(\pi) = 105$).

Exercises

Exercises for Section 1

- 1) Give an example of a group G , a field K , and a faithful representation of G such that the associated $K[G]$ -module is not faithful.
- 2) Let π be a unitary representation of G on a Hilbert space E and $x \in E$ of norm 1 such that

$$\sup_{g \in G} \|\pi(g)x - x\| \leq 1.$$

There exists a nonzero G -invariant vector $y \in E$.

- 3) Let E be a Hilbert space. We denote by $\mathbf{U}(E)$ the group of unitary endomorphisms of E and $\mathbf{GL}(E)$ the group of invertible endomorphisms of E .
 - a) The group $\mathbf{U}(E)$ endowed with the topology induced by the topology of simple convergence on $\mathcal{L}(E)$ is a topological group, denoted $\mathbf{U}(E)_s$; if E is infinite-dimensional, the group $\mathbf{GL}(E)$ is not a topological group with the topology induced by the topology of simple convergence.
 - b) The topology of $\mathbf{U}(E)_s$ coincides with the topology induced by the topology of compact convergence.
 - c) If E is countably generated, the group $\mathbf{U}(E)_s$ is metrizable and complete.
 - d) Let L be the space of $u \in \mathcal{L}(E)$ such that $u^* = -u$; it is the Lie algebra of the real Lie group $G = \mathbf{U}(E)$ endowed with the topology induced by the Banach space topology of

$\mathcal{L}(E)$ (LIE, III, p. 146, cor. 2). The exponential map from L to $U(E)_s$ is continuous.

e) If E is of infinite dimension, then $U(E)_s$ is not a real Lie group of type $\{L\}$. (Consider the subgroup K of $U(E)_s$ formed by the unitary endomorphisms diagonal in a fixed orthonormal basis of E , and note that K is compact.)

f) The exponential map is not a local homeomorphism from L to $U(E)_s$. (Consider its restriction to the diagonal endomorphisms in a fixed orthonormal basis.)

4) Let G be a topological group. We say that G is amenable if there exists a linear form $\lambda: \mathcal{C}_b(G; \mathbf{R}) \rightarrow \mathbf{R}$ such that

(i) If $f \geq 0$, then $\lambda(f) \geq 0$.

(ii) One has $\lambda(1) = 1$.

(iii) For every $g \in G$ and every $f \in \mathcal{C}_b(G; \mathbf{R})$, we have $\lambda(\gamma(g)f) = \lambda(\delta(g)f)$, where

$$\gamma(g)f(x) = f(g^{-1}x), \quad \delta(g)f(x) = f(xg).$$

If G is a discrete group, this definition coincides with that of EVT, IV, p. 73, exerc. 4.

a) The topological group G is amenable if and only if for every integer $n \in \mathbf{N}$, every finite family $(f_i)_{1 \leq i \leq n}$ in $\mathcal{C}_b(G; \mathbf{R})$, all families $(g_i)_{1 \leq i \leq n}$ and $(h_i)_{1 \leq i \leq n}$ in G , every real number a , the condition

$$\sum_{i=1}^n (f_i(x) - f_i(g_i x h_i)) \geq a$$

for all $x \in G$ implies that $a \leq 0$. (To show that this condition is sufficient, consider the set of functions of the form

$$x \mapsto \sum_{i=1}^n (f_i(x) - f_i(g_i x h_i))$$

and apply the Hahn–Banach theorem.)

- b) Every compact topological group is amenable, and every abelian topological group is amenable.
- c) Let G be a non-commutative free group. The discrete group G is not amenable.
- d) For every Hilbert space E and every continuous bounded representation of G in E , there exists a Hilbert norm q on E such that the representation of G in the Hilbert space E endowed with q is unitary.
- 5) Let G be a topological group and H an open subgroup of G . We denote by p the canonical projection $G \rightarrow G/H$.
- a) Let $\sigma : G/H \rightarrow G$ be a map such that $p \circ \sigma = \text{Id}_{G/H}$. We denote by $\beta : G \times G/H \rightarrow H$ the map defined by $\beta(g, x) = \sigma(gx)^{-1}g$. The map β is continuous.
- b) Let π be a bounded representation of H in a Hilbert space E . We denote by F the Hilbert space $\ell^2(G/H; E)$. The formula

$$\varrho(g)f(x) = \pi(\beta(g^{-1}, x)^{-1})f(g^{-1}x)$$

defines a continuous bounded representation of G in F .

- c) Suppose there exists a Hilbert norm q on F such that the representation ϱ is a unitary representation of G in (F, q) . Then there exists a Hilbert norm $\tilde{q}$ on E such that π is a unitary representation of H in $(E, \tilde{q})$.
- 6) a) Let E be a complex Banach space and let $u \in \mathcal{L}(E)$. If there exists $n \in \mathbb{N}$ such that $n \geq 1$ and $u^n = 1_E$ then the spectrum of u is contained in the group of n -th roots of unity and consists of eigenvalues of u . b) Let G be a commutative topological group such that every element of G has finite order. Every continuous irreducible representation of G in a Banach space has dimension 1, and takes values in the group $U \subset C^*$ of complex numbers of modulus 1.
- c) Let G be a commutative topological group such that the set of elements of finite order of G is dense in G . Every continuous irreducible representation of G in a Banach space has

dimension 1, and takes values in the group $U \subset C^*$ of complex numbers of modulus 1.

7) For every integer $n \geq 1$, we denote by s_n the function $Z \rightarrow R$ defined by

$$s_n(k) = \begin{cases} (1-k)^{n(1-k)} & \text{if } k \leq -1 \\ 1 & \text{if } k = 0 \\ (k+1)^{-(k+1)/n} & \text{if } k \geq 1. \end{cases}$$

We denote by S the set of functions s_n .

a) For every integer $n \geq 1$ and every $(k_1, k_2) \in \mathbb{Z}^2$, we have $s_{4n}(k_1)s_{4n}(k_2) \geq s_n(k_1 + k_2)$.

b) For $f \in \mathbf{C}(T)$, we denote by $(a_k(f))_{k \in \mathbb{Z}}$ the coefficients of the Laurent expansion of f at

0. For every $s \in S$ and every $f \in \mathbf{C}(T)$, the series

$$\sum_{k \in \mathbb{Z}} |a_k(f)|s(k)$$

converges, and the sets

$$\mathcal{V}(s, \varepsilon) = \{f \in \mathbf{C}(T) \mid \sum_{k \in \mathbb{Z}} |a_k(f)|s(k) < \varepsilon\}$$

defined for $s \in S$ and $\varepsilon > 0$, form a fundamental system of open neighborhoods of 0 for a locally convex topology on $\mathbf{C}(T)$. The multiplication $\mathbf{C}(T) \times \mathbf{C}(T) \rightarrow \mathbf{C}(T)$ is continuous for this topology.

c) The topology defined in the preceding question is metrizable but not complete.

d) For $f \in \mathbf{C}(T)^*$, we denote by $\varrho(f)$ the map $g \mapsto fg$ from $\mathbf{C}(T)$ into $\mathbf{C}(T)$. The map ϱ is a continuous representation of $\mathbf{C}(T)^*$ in the locally convex space $\mathbf{C}(T)$.

e) The representation ϱ is irreducible.

Exercises for Section 2

In the exercises of this section, unless otherwise stated, G denotes a locally compact topological group, equipped with a left Haar measure which is denoted by μ .

- 1) Let π be a unitary representation of G on a complex Hilbert space E . The representation π is irreducible if and only if the image of $\mathcal{H}(G)$ in $\mathcal{L}(E)$ is dense for the topology of pointwise convergence. (Use Exercise 24 of IV, p. 326.)
- 2) Let H be a closed subgroup of G . Let ν be a quasi-invariant measure on G/H (INT, VII, p. 56, § 2, n° 5, th. 2), and let ϱ be the function on G with values in $\mathbf{R}_+^*$ associated with it. The maps $\varrho(g)$ defined by

$$\varrho(g)f(x) = \left(\frac{\varrho(g^{-1}x)}{\varrho(x)} \right)^{1/2} f(g^{-1}x)$$

define a continuous linear representation of G on $\mathcal{L}^2(G/H, \nu)$, and, by passing to the quotient, define a unitary representation of G on $L^2(G/H, \nu)$, called the quasi-regular representation of G/H .

- 3) Let $G = \mathbf{R}$ and let H be the group $\mathbf{R}$ equipped with the discrete topology and with the counting measure, which is a Haar measure.
 - a) For every $t \in G$, and every $f \in L^2(H)$, the map $\varrho(t)f : x \mapsto f(x+t)$ belongs to $L^2(H)$; the map $\varrho(t)$ is a unitary map on $L^2(H)$. b) For all $f, g \in L^2(H)$, the map from G to $\mathbf{C}$ defined by $t \mapsto \langle f | \varrho(t)g \rangle$ is measurable.
 - b) Let f and g be $L^2(H)$. If there exists $t \in \mathbf{R}$ such that $\langle f | \varrho(t)g \rangle$ is nonzero, then the map $t \mapsto \langle f | \varrho(t)g \rangle$ from $\mathbf{G}$ into $\mathbf{C}$ is not continuous. (In particular, the hypothesis that E is countably generated cannot be omitted from cor. 2 of INT, VIII, p. 171, §4, n° 7.)
 - d) There exists $f \in L^2(H)$ such that the subspace of $L^2(H)$ generated by the elements $\varrho(t)f$, where $t \in \mathbf{R}$, is dense in $L^2(H)$.
- 4) Let $p \geq 1$. The kernel of the biregular representation of G on $L^p(G, \mu)$ is the image of the center of G under the homomorphism $g \mapsto (g, g)$.

5) Let G be a locally compact unimodular group.

a) If there exists a finite-dimensional square-integrable representation of G , then G is compact.

b) If there exists a square-integrable representation of G , then the center of G is compact.

6) We assume that the group G is unimodular. We denote by $\widehat{G}_d \subset \widehat{G}$ the set of classes of irreducible unitary square-integrable representations of G . For $\pi \in \widehat{G}_d$, we denote by p_π the orthogonal projector on $L^2(G)$ whose image is the π -isotypic component of the right regular representation of G .

a) Let $f \in \mathcal{L}^2(G)$ and let $\pi \in \widehat{G}_d$. We denote by E the representation space of π . The map $g \mapsto f(g)\pi(g)$ belongs to $L^1(G; \mathcal{L}(E))$; the endomorphism

$$v = \int_G f(g)\pi(g)d\mu(g)$$

is a Hilbert–Schmidt endomorphism, denoted $v = \pi(f)$.

b) Let $f \in \mathcal{L}^2(G)$. We have $\|\pi(\bar{f})\|_2 = \|p_\pi(f)\|$.

c) Let $L_d^2(G)$ be the closed subspace of $L^2(G)$ generated by the π -isotypic components of $L^2(G)$ for $\pi \in \widehat{G}_d$. It is the Hilbert sum of the images of the orthogonal projectors p_π for $\pi \in \widehat{G}_d$. For every $f \in L_d^2(G)$, we have

$$\|f\|^2 = \sum_{\pi \in \widehat{G}_d} c(\pi) \|\pi(f)\|_2^2$$

where $c(\pi)$ is the formal degree of π relative to $\{e\}$.

7) Let $n \in \mathbb{N}$. Determine the unitary automorphic representations of R^n (that is, the representations Γ -automorphic for every discrete subgroup Γ of R^n).

¶ 8) We assume that G is a locally compact unimodular group. Let Γ be a discrete subgroup of G such that the quotient space $X = \Gamma \backslash G$ is compact. Let μ be a Haar measure on G and β the counting measure on Γ ; we denote by $\tilde{\mu} = \mu / \beta$.

Let ϱ be the unitary representation of G on $L^2(X, \tilde{\mu})$ defined in Example 3 of V, p. 404.

For $\gamma \in \Gamma$, we denote by Γ_γ (resp. $\mathbf{G}_\gamma$) the centralizer of γ in Γ (resp. $\mathbf{G}$). a) The group $\mathbf{G}_\gamma$ is unimodular.

b) Let $\varphi \in \mathcal{H}(G)$. For every Γ -automorphic unitary representation π , the endomorphism $\pi(\varphi)$ has finite trace.

c) One has

$$\mathrm{Tr}(\varrho(\varphi)) = \sum_{\pi \in \widehat{G}} m_\pi(\varrho) \mathrm{Tr}(\pi(\varphi))$$

where $m_\pi(\varrho)$ is the multiplicity of π in ϱ (“spectral side of the trace formula”).

d) Let $\Gamma^\#$ be the set of conjugacy classes of Γ . For $\gamma \in \Gamma^\#$, we denote by β_γ the counting measure on Γ_γ . We have

$$\mathrm{Tr}(\varrho(\varphi)) = \sum_{\gamma \in \Gamma^\#} \int_{\Gamma_\gamma \backslash G} \varphi(x^{-1}\gamma x) d(\mu/\beta_\gamma)(x).$$

e) We fix a Haar measure μ_γ on G_γ for all $\gamma \in \Gamma$. We have

$$\mathrm{Tr}(\varrho(\varphi)) = \sum_{\gamma \in \Gamma^\#} \int_{G_\gamma \backslash G} \int_{\Gamma_\gamma \backslash G_\gamma} \varphi(x^{-1}y^{-1}\gamma yx) d(\mu_\gamma/\beta_\gamma)(y) d(\mu/\mu_\gamma)(x).$$

f) For every $\gamma \in \Gamma$, the quotient measure μ_γ/β_γ of $\Gamma_\gamma \backslash G_\gamma$ is finite. g) One has

$$\mathrm{Tr}(\varrho(\varphi)) = \sum_{\gamma \in \Gamma^\#} (\mu_\gamma/\beta_\gamma)(\Gamma_\gamma \backslash G_\gamma) \int_{G_\gamma \backslash G} \varphi(x^{-1}\gamma x) d(\mu/\mu_\gamma)(x)$$

(“geometric side of the trace formula”; the formula expressing the equality of this expression with that obtained in question c) is called the “trace formula” for Γ -automorphic representations).

h) Set $G = \mathbf{R}$ and $\Gamma = \mathbf{Z}$. Verify that the trace formula is equivalent in this case to the Poisson formula (cf. II, p. 239).

9) We assume that G is a perfect real semisimple Lie group without compact factor. There is no nontrivial finite-dimensional unitary representation of G .

10) a) Every matrix coefficient of the left regular representation of G on $L^2(G)$ belongs to $\mathcal{C}_0(G)$.

b) If G is not compact, the left regular representation of G on $L^2(G)$ contains no nonzero finite-dimensional subrepresentation.

11) Let E be a Hilbert space.

a) Let ϱ be a unitary representation of the group $\mathbf{R}$ in E . Let D be a dense subspace of E stable under ϱ such that, for every $x \in D$, the map $\psi_x : t \mapsto \varrho(t)x$ is differentiable on $\mathbf{R}$. Let u be the partial operator with domain D such that $u(x) = i^{-1}\psi'_x(0)$ for all $x \in D$. Then u is essentially self-adjoint and $\bar{u}$ is the infinitesimal generator of ϱ .

b) Let u be a self-adjoint partial operator on E and let D be a dense subspace of $\text{dom}(u)$. Suppose that for every $t \in \mathbf{R}$, we have $\exp(itu)(D) \subset D$. Then the space D is dense in E_u .

c) Let u be a self-adjoint partial operator on E . Let F be a Hilbert space and v a self-adjoint partial operator on F . The partial operator $u \otimes 1_F + 1_E \otimes v$ on $E \otimes F$ defines an essentially self-adjoint partial operator on $\widehat{E \otimes_2 F}$.

12) Let G be a real Lie group and let ϱ be a unitary representation of G in a Hilbert space E . We say that an element $x \in E$ is a vector of class C^∞ if the map from G to E defined by $g \mapsto \varrho(g)x$ is of class C^∞ . a) Let π be a unitary representation of G in F and $u : E \rightarrow F$ a G -morphism. For every vector $x \in E$ of class C^∞ , the vector $u(x)$ is of class C^∞ in F .

We denote by $C^\infty(\varrho)$ the space of vectors of class C^∞ in E . By question a), every G -morphism from ϱ into π defines, by passing to subspaces, a linear map from $C^\infty(\varrho)$ into $C^\infty(\pi)$.

b) The space $C^\infty(\varrho)$ is dense in E .

13) Let π be an irreducible unitary representation of G in a complex Hilbert space E . Let ϱ be a unitary representation of G in a complex Hilbert space F . Suppose there exists a closed densely defined partial operator u from E into F whose domain is a G -stable subspace of

E and which satisfies $u \circ \pi(g) = \varrho(g) \circ u$ for all $g \in \mathbf{G}$.

The partial operator u belongs to $\mathcal{L}(E; F)$, and there exists $\lambda > 0$ such that λu is isometric.

- 14) Let Z be a subgroup of the center C of G such that C/Z is compact and let ν be a Haar measure on G/Z . Let π be an irreducible unitary representation of G in a complex Hilbert space E . We assume that π is square-integrable modulo the center. Let $u \in \mathcal{L}(E)$ be a nuclear endomorphism (def. 4 of IV, p. 169).

- a) For every x and y in E , the function

$$g \mapsto \langle x | \pi(g) u \pi(g^{-1}) y \rangle$$

belongs to $\mathcal{F}(G/Z)$.

- b) For every x and y in E , we have

$$\int_{G/Z} \langle x | \pi(g) u \pi(g^{-1}) y \rangle d\nu(g) = \frac{1}{c_Z(\pi)} \langle x | y \rangle \text{Tr}(u).$$

- 15) Let Z be a subgroup of the center C of G such that C/Z is compact and let ν be a Haar measure on G/Z . Let π be an irreducible unitary representation of G in a complex Hilbert space E . Denote by χ the restriction to Z of the central character of π .

We denote by $f_{x,y}$ the matrix coefficient $g \mapsto \langle x | \pi(g) y \rangle$ for $(x, y) \in E^2$. Let $p \geq 1$ be a real number and let q be the conjugate exponent of p .

- a) Let $y \in E$ such that the matrix coefficients $f_{x,y}$ belong to $\mathcal{L}_\chi^p(G)$. The map $x \mapsto f_{x,y}$ is a continuous G -morphism from π into $\delta_{G,\chi}^{(p)}$.
- b) Let ϱ be an irreducible unitary representation of G in a complex Hilbert space F whose central character restricts to χ on Z . One assumes that $p > 1$ and that the matrix coefficients of π belong to $\mathcal{L}_\chi^p(G)$ and those of ϱ belong to $\mathcal{L}_\chi^q(G)$. Let $x_0 \in E$ and $y_0 \in F$. For every $x \in E$ there exists a continuous linear form ℓ_x from F to $\mathbb{C}$ such that

$$\ell_x(y) = \int_{G/Z} \overline{\langle x_0 | \pi(g)x \rangle} \langle y_0 | \varrho(g)y \rangle d\nu(g)$$

for all $y \in \mathbf{F}$.

c) There exists a continuous linear map u from E into F such that $\ell_x(y) = \langle u(x) | y \rangle$ for all $(x, y) \in E \times F$.

d) One has

$$\int_{G/Z} \overline{\langle x | \pi(g)x' \rangle} \langle y | \varrho(g)y' \rangle d\nu(g) = 0$$

for $(x, x', y, y') \in E^2 \times F^2$

e) We assume that $p = 1$. The matrix coefficients of π are orthogonal to those of ϱ .

16) Let $\mathbf{G} = \mathbf{SL}(2, \mathbf{R})$ and let H be the open subset of $\mathbf{C}$ formed by the complex numbers z such that $\mathcal{I}(z) > 0$. Let μ be the Lebesgue measure on $\mathbf{C}$. For every element

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

of $\mathbf{G}$ and every $z \in \mathbf{H}$ we set $j(g, z) = cz + d$.

a) The group $\mathbf{G}$ acts continuously on the left on H by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

The measure $\nu = y^{-2} \cdot \mu$ is $\mathbf{G}$ -invariant.

b) Let n be an integer such that $n \geq 2$. We denote by E_n the vector space of holomorphic functions $f : H \rightarrow \mathbf{C}$ such that the function $z \mapsto |f(z)|^2 \mathcal{I}(z)^n$ belongs to $\mathcal{L}^2(\mathbf{H}, \nu)$. The space E_n endowed with the sesquilinear form

$$\langle f | g \rangle = \int_{\mathbf{H}} \overline{f(z)} g(z) \mathcal{T}(z)^n d\nu(z)$$

is a Hilbert space of countable type and of infinite dimension.

c) For every $g \in G$ and every $f \in E_n$ we set

$$\pi_n(g)f(z) = j(g^{-1}, z)^{-n} f(g^{-1} \cdot z).$$

The map π_n defines a unitary representation of G in E_n . d) The function f_n defined by $f_n(z) = (z+i)^{-n}$ belongs to E_n ; it satisfies $\pi_n(k)f_n = \chi_{-n}(k)f_n$ for all $k \in \mathbf{SO}(\mathbf{R}^2)$, where χ_n is the character such that

$$\chi_n \left(\begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \right) = e^{in\theta}$$

for all $\theta \in \mathbf{R}$.

e) Let F be a subrepresentation of E_n . The orthogonal projector of E_n whose image is the χ_{-n} -isotypic component of the restriction of π_n to $\mathbf{SO}(\mathbf{R}^2)$ is given by $f \mapsto -(2i)^n f(i)f_n$. (Use cor. 2 of V, p. 465 and the Cauchy integral formula, FVC, to appear.)

f) The representation π_n is irreducible.

g) The matrix coefficient $g \mapsto \langle f_n | \pi_n(g)f_n \rangle$ is nonzero and belongs to $\mathcal{L}^2(G)$.

h) The representation π_n is square-integrable and there exists a constant $c > 0$ such that the formal degree of π_n is $c(n-1)$.

17) Let G be a connected real Lie group. There exists a faithful finite-dimensional unitary representation of G if and only if the group G is locally isomorphic to a compact group.

18) Let X be a locally compact space equipped with a bounded positive measure μ . For every Banach space F , we denote by $S(X, \mu; F)$ the space of classes of functions μ -measurable from X into F modulo the relation of equality μ -almost everywhere, endowed with the topology of convergence in measure (INT, IV, p. 194, § 5, no. 11). We denote by A the

commutative subalgebra of $\mathcal{L}(L^2_{\mathbb{C}}(X, \mu))$ formed by the multiplication endomorphisms by functions $f \in \mathcal{L}^{\infty}_{\mathbb{C}}(X, \mu)$.

We denote by $e: \mathbf{R} \rightarrow \mathbf{C}$ the homomorphism $t \mapsto \exp(2i\pi t)$.

Let E be a topological vector space.

- a) Let $f \in S(X, \mu; \mathbf{R})$ such that $|f(x)| \geq 1$ for μ -almost all $x \in X$. There exists a real number $t \in]0, 1[$ such that

$$\mu(\{x \in X \mid tf(x) \in [1/4, 3/4] + \mathbf{Z}\}) \geq \frac{2}{5}\mu(X).$$

- b) Let u be a linear map from E into $S(X, \mu; \mathbf{R})$. For every $x \in E$, we denote by $\varrho_u(x)$ the element of A given by the endomorphism of multiplication by the function $e \circ u(x)$, which belongs to $\mathcal{L}^{\infty}_{\mathbb{C}}(X, \mu)$. The map ϱ_u is a continuous representation of G into $L^2_{\mathbb{C}}(X, \mu)$ if and only if the map u is continuous. (If u is not continuous, use the previous question to verify that ϱ is not continuous.)
- c) Let ϱ be a unitary representation of E in $L^2(X, \mu)$ whose image is contained in $\mathcal{A}$. There exists a unique continuous linear map $u: E \rightarrow S(X, \mu; \mathbf{R})$ such that $\varrho = \varrho_u$. (First consider the case where $E = \mathbf{R}$, and then apply the Stone theorem.)

- 19) Let G be a finite group and H a subgroup of G . For every unitary representation π of H in a finite-dimensional complex Hilbert space E , prove that the induced unitary representation $\text{Ind}_H^G(\pi)$ is isomorphic to the induced representation defined in A, VIII, p. 392, § 21, n° 3.
- 20) Let G be a locally compact group which is countable at infinity, and let H be a closed subgroup of G . Let π be a unitary representation of H in a Hilbert space E and let ϱ be the unitary representation of G induced by π . We denote by F the representation space of ϱ .

We fix Haar measures μ on G and β on H , and we denote by $\kappa: \mathbf{G} \rightarrow \mathbf{R}_+^*$ a continuous function such that $\kappa(xh) = \Delta_H(h)\Delta_G(h)^{-1}\kappa(x)$ for $(x, h) \in \mathbf{G} \times H$ and such that $\kappa(e) = 1$.

- a) If G/H admits a bounded G -invariant measure, and if there exists a nonzero vector $x \in E$ which is H -invariant, then there exists a nonzero G -invariant vector $y \in F$.
- b) Conversely, suppose that $y \in F$ is a nonzero and G -invariant vector of ϱ . Show that there exists a function $f \in L^2_{\overline{\pi}}$ nonzero such that

$$f(x) = \left(\frac{\kappa(g^{-1}x)}{\kappa(x)} \right)^{1/2} f(g^{-1}x)$$

for all $(g, x) \in G^2$. Deduce that $\kappa = 1$, and then the measure μ/β on G/H is defined and bounded, and finally that $f(e) \in E$ is a nonzero H -invariant vector.

Exercises for Section 3

- 1) Let G be a locally compact group. Let π be a unitary representation of G in a complex Hilbert space E and $x \in E$. We have $\pi(g)x = x$ for all $g \in G$ if and only if the positive-definite function $g \mapsto \langle x | \pi(g)x \rangle$ is constant.
- 2) Let G be a locally compact group. For every positive-definite function φ on G , we denote by $(\pi_{\varphi}, x_{\varphi})$ a cyclic Hilbert representation of φ .
 - a) Let φ_1, φ_2 and φ_3 be positive-definite functions on G . If $\varphi_3 = \varphi_1 + \varphi_2$, then there exists an injective G -morphism from π_{φ_1} into π_{φ_3} .
 - b) If φ is a constant nonzero positive-definite function, then π_{φ} is trivial of dimension 1.
 - c) For every unitary representation π of G in a Hilbert space E , and for every $x \in E$, there exists an injective G -morphism from π_{φ} into π , where $\varphi(g) = \langle x | \pi(g)x \rangle$.
 - d) Let H be a closed subgroup of G and let φ be a positive-definite function on G . Denote by ψ the restriction of φ to H ; this is a positive-definite function on H , and there exists an injective H -morphism from π_{ψ} into $\text{Res}^H_{\mathbb{H}}(\pi)$.
- 3) a) Give an example of an involutive Banach algebra admitting a positive linear form which is not continuous.

- b) Give an example of an involutive Banach algebra A , a continuous positive linear form λ on A , and Hilbert realizations (E_1, e_1, φ_1) and (E_2, e_2, φ_2) of λ such that (E_1, e_1, φ_1) is a cyclic Hilbert realization and such that the unique linear map u of prop. 6 of V, p. 440 does not satisfy $u(e_1) = e_2$.
- 4) Let A and B be star algebras. A linear map u from A to B is said to be positive if $u(A_+) \subset B_+$, and completely positive if, for every integer $k \in \mathbb{N}$, the map $u_k = 1_{C^k} \otimes u : C^k \otimes A \rightarrow C^k \otimes B$ is positive.
- a) Let A be a unital star algebra, E and F complex Hilbert spaces, v a linear map from E to F , and φ a unital morphism of involutive algebras from A to $\mathcal{L}(F)$ such that $\|\varphi\|^2 = \|\varphi(1)\|$. The map $u : A \rightarrow \mathcal{L}(E)$ defined by $u(a) = v^* \circ \varphi(a) \circ v$ is completely positive.

The goal of the exercise is to prove the converse of this assertion (“Stinespring theorem”). Let A be a unital star algebra, E a complex Hilbert space, and $u : A \rightarrow \mathcal{L}(E)$ a completely positive linear map.

- b) Let $F = A \otimes E$. There exists a positive sesquilinear form q on F such that

$$q(a \otimes x, b \otimes y) = \langle x | \varphi(a^* b) y \rangle$$

for all $(a, b) \in A^2$ and $(x, y) \in E^2$.

Let $\widehat{F}$ be the separated completion Hilbert space of F for the sesquilinear form q .

- c) Let $b \in A$. The endomorphism of F such that $a \otimes x \mapsto ba \otimes x$ for all $a \in A$ and $x \in E$ induces by passage to quotients an endomorphism $\varphi(b)$ of $\widehat{F}$ such that $\|\varphi(b)\| \leq \|b\|$; the map $\varphi : A \rightarrow \mathcal{L}(F)$ is a unital morphism of involutive algebras.
- d) Let v be the linear map from E to $\widehat{F}$ such that $v(x)$ is the class of $1 \otimes x$ for all $x \in E$. We have $\|v\|^2 = \|\varphi(1)\|$, and $u(a) = v^* \circ \varphi(a) \circ v$ for all $a \in A$.
- 5) Let X be a separated topological space. A continuous function $f : X \times X \rightarrow \mathbb{R}$ is said to be of conditionally negative type if it satisfies the conditions
- (i) For every $x \in X$, we have $f(x, x) = 0$;

- (ii) For all x and y in X , we have $f(x, y) = f(y, x)$;
- (iii) For every integer $n \in \mathbb{N}$, every family $(x_i)_{0 \leq i \leq n}$ of elements of X and every family $(t_i)_{0 \leq i \leq n}$ of real numbers such that

$$\sum_{i=0}^n t_i = 0,$$

we have

$$\sum_{i=0}^n \sum_{j=0}^n t_i t_j f(x_i, x_j) \leq 0.$$

We denote by $\text{Noy}_-(X)$ the set of functions of conditionally negative type on $X \times X$.

- a) Let E be a real Hilbert space and $\varphi: X \rightarrow E$ a continuous map. The function $f(x, y) = \|\varphi(x) - \varphi(y)\|^2$ is of conditionally negative type.
- b) Let $f \in \text{Noy}_-(X)$ and $x_0 \in X$. There exists a real Hilbert space E and a continuous map $\varphi: X \rightarrow E$ such that
- a) For every $(x, y) \in X \times X$, we have $f(x, y) = \|\varphi(x) - \varphi(y)\|^2$;
 - b) The family $(\varphi(x) - \varphi(x_0))_{x \in X}$ is total in E .

We say that (E, φ) is a cyclic Hilbert representation of f .

- c) Let $f \in \text{Noy}_-(X)$ and $x_0 \in X$. Let (E, φ) and (E', φ') be cyclic Hilbert representations of f . There exists a unique affine isometry $u: E \rightarrow E'$ such that $\varphi' = u \circ \varphi$.
- d) The set $\text{Noy}_-(X)$ is a convex cone in $\mathcal{C}(X \times X; \mathbb{R})$. For every universally positive kernel $g \in \text{Noy}_+(X)$ such that g is real-valued, the map $f(x, y) = g(x, x) - g(x, y)$ belongs to $\text{Noy}_-(X)$.
- e) Let $\mathfrak{F}$ be a filter on $\text{Noy}_-(X)$ which converges simply to $f: X \times X \rightarrow \mathbb{R}$. If f is continuous, then $f \in \text{Noy}_-(X)$.
- f) Let $f: X \times X \rightarrow \mathbb{R}$ be a continuous function satisfying conditions (i) and (ii) above. Let

$x_0 \in X$. Let $g : X \times X \rightarrow \mathbf{R}$ be such that $g(x, y) = f(x, x_0) + f(y, x_0) - f(x, y)$. Then $f \in \text{Noy}_-(X)$ if and only if $g \in \text{Noy}_+(X)$.

- g) Suppose X is nonempty. Let $f : X \times X \rightarrow \mathbf{R}$ be a continuous function satisfying conditions (i) and (ii) above. Then $f \in \text{Noy}_-(X)$ if and only if we have $\exp(-tf) \in \text{Noy}_+(X)$ for all $t \in \mathbf{R}_+$ ("Schoenberg theorem"). (To prove that $\exp(-f)$ is of positive type when $f \in \text{Noy}_-(X)$, fix $x_0 \in X$; show that

$$g(x, y) = \exp(f(x, x_0) + f(y, x_0) - f(x, y))$$

and

$$h(x, y) = \exp(-f(x, x_0)) \exp(-f(y, x_0))$$

belong to $\text{Noy}_+(X)$.)

If G is a topological group, we say that a function $\varphi \in \mathcal{C}(G)$ is of negative type if the function f on $G \times G$ defined by $f(g, h) = \varphi(h^{-1}g)$ is a conditionally negative-definite kernel.

- 6) Let G be a discrete topological group. Let A be the involutive subalgebra of $\mathcal{L}(\ell^2(G))$ generated by the image of the left regular representation λ . This is a unital star algebra.

For every function of positive type $\varphi \in \text{Pos}(G)$, there exists a unique completely positive linear map u_φ from A to A (cf. exercise 4) such that $u_\varphi(\lambda(g)) = \varphi(g)\lambda(g)$ for all $g \in G$ (Use a cyclic Hilbertian representation of φ .)

- 7) Let I be a finite set and $G = F(I)$ the free group constructed on I (A, I, p. 84). Let A be the involutive subalgebra of $\mathcal{L}(\ell^2(G))$ generated by the image of the left regular representation λ .

- a) For every $g \in G$, we denote by $|g|$ the length of the unique reduced decomposition of g (A, I, p. 146, exercise 19). The map $d(g, h) = |h^{-1}g|$ is a metric on G .
- b) Let $\lambda \in \mathbf{R}_+^*$. The function on G defined by $\psi_\lambda(g) = \exp(-\lambda|g|)$ is a positive-definite function. (Show that the function $g \mapsto |g|$ is a function of negative type, cf. exercise 5.)

- c) For every integer $k \in \mathbf{N}$, let G_k be the set of $g \in G$ such that $|g| = k$ and let φ_k be the characteristic function of G_k . Let f and g be functions on G with support in G_k and G_l respectively. For every $m \in \mathbf{N}$, we have $(f * g)\varphi_m = 0$ unless $k + l - m$ is even and $|k - l| \leq m \leq k + l$; in this case, we have

$$\|(f * g)\varphi_m\| \leq \|f\|\|g\|.$$

- d) Let f be a function on G with finite support. We have

$$\|\lambda(f)\| \leq \sum_{n \in \mathbf{N}} (n+1) \|f\varphi_n\|,$$

and

$$\|\lambda(f)\| \leq 2 \left(\sum_{g \in G} (1 + |g|)^4 |f(g)|^2 \right)^{1/2}.$$

- e) Let ψ be a function on G such that $M = \sup_{g \in G} |\psi(g)|(1 + |g|)^2$ is finite. There exists a unique linear map u_ψ from A to A such that $u_\psi(\lambda(g)) = \psi(g)\lambda(g)$ for all $g \in G$. We have $\|u_\psi\| \leq 2M$.
- f) Let X be the set of functions ψ on G with finite support such that $\|u_\psi\| \leq 1$. There exists a filter $\mathfrak{F}$ on X such that

$$\lim_{\psi, \mathfrak{F}} u_\psi = 1_A$$

in the space $\mathcal{L}(A)$ endowed with the topology of simple convergence. In particular, the Banach space underlying A has the approximation property. (Consider the functions $\psi_{\lambda, n}(g) = (\varphi_0 + \cdots + \varphi_n)\psi_\lambda$ for $\lambda \in \mathbf{R}_+^*$ and $n \in \mathbf{N}$, and use the preceding exercise.)

- 8) One identifies the group $\mathbf{R}$ and its dual group by the map $(x, y) \mapsto \exp(i\pi xy)$. Let $\alpha \in \mathbf{R}_+$.

- a) There exists a positive measure on $\mathbf{R}$ whose Fourier transform is the function $f_\alpha(x) = \exp(-|x|^\alpha)$ if and only if $\alpha \in]0, 2]$. When this is the case, one denotes by μ_α this measure,

which is unique; it is called the stable measure of parameter α .

b) The measure μ_α has total mass 1; the identity function f on $\mathbb{R}$ is integrable with respect to μ_α if and only if $\alpha = 2$.

9) The goal of this exercise is to present another proof of Theorem 5 of V, p. 455. Let G be a locally compact commutative topological group, equipped with a Haar measure μ . We denote by $\hat{f}$ the Fourier transform of $f \in L^1(G)$, and we denote by $\mathcal{F}(G) \subset \mathcal{C}_0(\widehat{G})$ the image of the Fourier transform on $L^1(G)$.

a) The space $\mathcal{F}(G)$ is a dense subspace of $\mathcal{C}_0(G)$.

b) For every $f \in L^1(G)$, the spectral radius of f in the Banach algebra $L^1(G)$ is equal to $\|\hat{f}\|_\infty$.

c) Let $\varphi \in \text{Pos}(G)$. It is a uniformly continuous and bounded function on G ; for every $f \in L^1(G)$, we have

$$\int_G \int_G f(x) \overline{f(y)} \varphi(x-y) d(\mu \otimes \mu)(x, y) \geq 0.$$

d) Let λ be the continuous linear form on $L^1(G)$ determined by φ and we set $b(f, g) = \lambda(f^* * g)$ for all $(f, g) \in L^1(G) \times L^1(G)$. The map b is a positive hermitian form on $L^1(G)$.

e) For every $f \in L^1(G)$, we have $|\lambda(f)|^2 \leq b(f, f)$.

f) Let $f \in L^1(G)$ and $g = f^* * f \in \mathcal{C}_b(G)$. One defines $g_1 = g$ and $g_{n+1} = g_n * g$ for every integer $n \geq 1$. For every integer $n \geq 1$, we have $|\lambda(f)| \leq \|g_n\|_1^{2^{-n}}$.

g) For every $f \in L^1(G)$, we have $|\lambda(f)| \leq \|\hat{f}\|_\infty$. There exists a continuous linear map $\hat{\lambda}$ on $\mathcal{F}(G)$ such that $\lambda(f) = \hat{\lambda}(\hat{f})$ for all $f \in L^1(G)$.

h) There exists a bounded measure ν on $\widehat{G}$ such that $\lambda(f) = \nu(\hat{f})$ for every $f \in L^1(G)$; the measure ν is positive and one has

$$\varphi(x) = \int_{\hat{G}} \langle \chi, x \rangle d\nu(\chi)$$

for all $x \in G$.

10) a) Let $G = \mathbf{R}$. The set $\text{Pos}_1(G)$ is not compact for the weak topology.

b) Let $G = \mathbf{R}/\mathbf{Z}$. The weak topology on $\text{Pos}_{\leq 1}(\mathbf{G})$ does not coincide with the topology of compact convergence.

11) Let G be a locally compact group.

Let ϱ and π be unitary representations of G . One says that ϱ is weakly contained in π , and one writes $\varrho \leq \pi$, if the set of diagonal matrix coefficients of ϱ is contained in the closure of the set of finite sums of diagonal matrix coefficients of π in the space $\mathcal{C}(G)$ endowed with the topology of compact convergence.

a) The relation « ϱ and π are unitary representations of G and $\varrho \leq \pi$ » is a preorder relation.

If ϱ' (resp. π') is isomorphic to ϱ (resp. π) then $\varrho \leq \pi$ if and only if $\varrho' \leq \pi'$.

b) If ϱ is isomorphic to a subrepresentation of π , then $\varrho \leq \pi$. If I and J are nonempty finite sets, and if $\varrho \leq \pi$, then $\bigoplus_{i \in I} \varrho \leq \bigoplus_{j \in J} \pi$.

c) Let X be a subset of the space E of ϱ which generates π . We have $\varrho \leq \pi$ if and only if every matrix coefficient $g \mapsto \langle x | \varrho(g)x \rangle$ with $x \in X$ belongs to the closure of the set of finite sums of diagonal matrix coefficients of π in the space $\mathcal{C}(G)$ endowed with the topology of compact convergence. (Show that the set Y of elements of E satisfying this property is a closed vector subspace of E .)

d) Let H be a closed subgroup of G . Let π and ϱ be unitary representations of G such that $\varrho \leq \pi$; we then have $\text{Res}_H^G(\varrho) \leq \text{Res}_H^G(\pi)$.

e) Let H be a closed subgroup of G and let ν be a Haar measure on H . Let π and ϱ be unitary representations of H such that $\varrho \leq \pi$; we then have $\text{Ind}_H^G(\varrho) \leq \text{Ind}_H^G(\pi)$. (Let $\varrho' = \text{Ind}_H^G(\varrho)$ and $\pi' = \text{Ind}_H^G(\pi)$; using part (c), reduce to approximating matrix coefficients $g \mapsto \langle f | \varrho'(g)f \rangle$ for f of the form

$$f(x) = \int_H f(xh) \varrho(h) y d\nu(y)$$

where $\varphi \in \mathcal{H}(G)$ and y belongs to the space of ϱ .)

- 12) Let $G = \mathbb{R}$ and let π be the regular representation of G in $L^2(\mathbb{R})$. Prove that $\chi \leq \pi$ for every character $\chi \in \widehat{G}$.
- 13) Let ϱ be an irreducible unitary representation of G in a Hilbert space E , and π a unitary representation of G in F . Let $x \in E$ be of norm 1. One assumes that $\varrho \leq \pi$ (cf. Exercise 11).

We denote by $A \subset \mathcal{C}(G)$ the set of normalized diagonal matrix coefficients of π , and $\tilde{A}$ the closed convex hull of A in $L^\infty(G)$ for the weak topology.

Let $x \in E$ be of norm 1 and let $f : g \mapsto \langle x \mid \varrho(g)x \rangle$ be the associated diagonal matrix coefficient.

- a) The set $\tilde{A}$ is a compact convex set in $L^\infty(G)$ and f is an extreme point of $\tilde{A}$.
- b) The function f belongs to the closure of A in $\tilde{A}$ for the weak topology.
- c) The function f belongs to the closure of A in $\mathcal{C}(G)$ endowed with the topology of compact convergence.
- d) The trivial representation of G in $\mathbb{C}$ is weakly contained in π if and only if, for every compact subset C of G and every real $\varepsilon > 0$, there exist $x \in E$ of norm 1 such that

$$\sup_{g \in C} \|\pi(g)x - x\| < \varepsilon.$$

- 14) Let G be a locally compact group. One says that G has the property]T[of Kazhdan ⁽¹⁾ if a unitary representation π of G has a nonzero trivial subrepresentation ⁽²⁾ if and only if the trivial representation $\mathbf{1}_G$ of G on $\mathbb{C}$ is weakly contained in π .
- a) Let G be a locally compact group having property]T[. Let H be the set of open subgroups of G that are generated by a compact subset of G . The group G is the union of the subgroups

$H \in H$.

b) Let π be the representation of G that is the Hilbert sum of the quasi-regular representations of G on G/H for $H \in H$ (cf. exercise 2 of V, p. 486). We have $1_G \leq \pi$.

c) The group G is generated by a compact subset of G . In particular, if G is discrete, then G is generated by a finite subset of G .

15) Let G be a locally compact group, C a subset of G , and $\varepsilon > 0$. One says that (C, ε) is a Kazhdan pair for G if, for every unitary representation π of G in a Hilbert space E , the existence of a vector $x \in E$

such that

$$\sup_{g \in C} \|\pi(g)x - x\| < \varepsilon \|x\|,$$

implies the existence of a nonzero trivial subrepresentation of π . We say that C is a Kazhdan set if there exists $\varepsilon > 0$ such that (C, ε) is a Kazhdan pair.

a) Let (C, ε) be a Kazhdan pair of G . Let π be a unitary representation of G in a Hilbert space E . Let $x \in E$ be nonzero and $\delta > 0$ such that

$$\sup_{g \in C} \|\pi(g)x - x\| < \delta \varepsilon \|x\|.$$

We then have $\|p(x) - x\| \leq \delta \|x\|$, where p is the orthogonal projector of E whose image is the space of G -invariant vectors.

b) A locally compact group G has property]T[if and only if there exists a Kazhdan pair (C, ε) .

(To prove that property]T[implies the assertion, proceed by contraposition by considering the unitary representation that is the Hilbert sum of representations $\pi_{C, \varepsilon}$ parametrized by a compact subset C of G and $\varepsilon > 0$ such that $\pi_{C, \varepsilon}$ contains no nontrivial subrepresentation but there exists $x_{C, \varepsilon}$ of norm 1 with $\sup_{g \in C} \|\pi(g)x_{C, \varepsilon} - x_{C, \varepsilon}\| < \varepsilon$.

c) Let G be a locally compact group having property]T[. Every compact subset C of G that generates G is a Kazhdan set.

d) Let G be a locally compact group having property]T[and let C be a compact subset of G with nonempty interior that is a Kazhdan set. Then C generates G . In particular, if G is discrete, then the Kazhdan sets are precisely the finite generating subsets of G . (The subgroup H of G generated by C is open; consider the quasi-regular representation π of G on G/H and show that the projection of the characteristic function φ of $H \in G/H$ onto the space of invariant vectors of π is equal to φ .)

16) Let G be a locally compact group that is countable at infinity and let H be a closed subgroup of G . If G has property]T[, and if G/H admits a bounded G -invariant measure, then H has property]T[. (Use Exercise 20 of V, p. 492.)

17) Let $n \geq 2$ be an integer. We consider the real Lie group $\mathbf{G} = \mathbf{R}^n \rtimes \mathbf{SL}(n, \mathbf{R})$, and we denote by H the subgroup $\mathbf{R}^n \times \{e\}$ of $\mathbf{G}$, identified with $\mathbf{R}^n$.

Let π be a unitary representation of G in a complex Hilbert space E . Suppose that for every compact subset C of G and every $\varepsilon > 0$, there exists $x \in E$ nonzero such that

$$\sup_{g \in C} \|\pi(g)x - x\| < \varepsilon \|x\|.$$

a) There exists a sequence (x_m) of vectors of norm 1 in E such that the functions $\varphi_m(g) = \langle x_m | \pi(g)x_m \rangle$ converge to the constant function 1 in $\mathcal{C}(G)$ endowed with the topology of compact convergence. (Consider a sequence $(C_m)_{m \in \mathbf{N}}$ of compact subsets whose union is equal to G , and take x_m satisfying the above condition for $(C_m, 1/(m+1))$.)

b) For every $m \in \mathbf{N}$, there exists a positive measure μ_m of total mass 1 on $\mathbf{R}^n$ whose Fourier transform is identified with the restriction of φ_m to H .

c) Suppose that $\mu_m(\{0\}) = 0$ for all $m \in \mathbf{N}$. We then identify μ_m with a positive measure of total mass 1 on $\mathbf{R}^n - \{0\}$, and we denote by ν_m the image measure of μ_m on projective space $\mathbf{P}_{n-1}(\mathbf{R})$. There exists a subsequence $(\nu_{n_k})_{k \in \mathbf{N}}$ converging weakly to a positive measure ν of total mass 1 on $\mathbf{P}_{n-1}(\mathbf{R})$; the measure ν is $\mathbf{SL}(n, \mathbf{R})$ -invariant. Deduce a contradiction.

d) Let $m \in \mathbf{N}$ be such that $\mu_m(\{0\}) > 0$. Let ψ be the restriction of φ_m to H . There exists c

> 0 such that $\psi = c + \tilde{\psi}$ where $\tilde{\psi}$ is a function of positive type on R^n .

- e) The representation π contains H-invariant vectors. (Use Exercise 2.) One says that the pair (G, H) has the relative]T[property.

18) Let $G = \mathbf{SL}(2, \mathbf{R})$. Let N and P denote the subgroups

$$N = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \mid t \in \mathbf{R} \right\}, \quad P = \left\{ \begin{pmatrix} a & t \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbf{R}_*, t \in \mathbf{R} \right\}$$

of G. Let π be a unitary representation of G in a complex Hilbert space E. Suppose that there exists $x \in E$ such that $\pi(n)x = x$ for all $n \in N$.

- a) The homogeneous space $\mathbf{SL}(2, \mathbf{R})/N$ is identified with $\mathbf{R}^2 - \{0\}$ by the mapping $g \mapsto ge_1$ where e_1 is the first vector of the canonical basis of $\mathbf{R}^2$, and the homogeneous space $\mathbf{SL}(2, \mathbf{R})/P$ is identified with the subspace $(\mathbf{R} - \{0\}) \times \{0\}$ of $\mathbf{SL}(2, \mathbf{R})/N$.
- b) Every continuous function f on G such that $f(ngn') = f(g)$ for all $(n, n') \in N^2$ satisfies $f(pgp') = f(g)$ for all $(p, p') \in P^2$. (Interpret f as a continuous function φ on $\mathbf{R}^2 - \{0\}$ invariant under the natural action of N and verify that it is constant on $\mathbf{SL}(2, \mathbf{R})/P$.)
- c) One has $\pi(g)x = x$ for all $g \in G$.

19) Let $n \geq 2$ be an integer. We consider the real Lie group $G = \mathbf{SL}(n, \mathbf{R})$. Let N and H denote the closed subgroups of G whose elements are of block form

$$\begin{pmatrix} \text{Id}_{n-1} & x \\ 0 & 1 \end{pmatrix}, \quad (x \in \mathbf{R}^{n-1}), \quad \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}, \quad (g \in \mathbf{SL}(n-1, \mathbf{R}))$$

respectively. The group N is identified with $\mathbf{R}^{n-1}$ and the group H is identified with $\mathbf{SL}(n-1, \mathbf{R})$.

- a) Let π be a unitary representation of G in a complex Hilbert space E. An element x of E is invariant under G if and only if it is invariant under N.
- b) If $n \geq 3$, then the group G has property]T[. (Combine the result of the previous question with the previous exercise.)

- c) If $n \geq 3$, then the group $\widetilde{G} = R^n \rtimes G$ has property]T[. (Let π be a unitary representation such that $1_{\widetilde{G}} \leq \pi$; there exists a vector $x \neq 0$ invariant under the subgroup G of $\widetilde{G}$ by the preceding question; show that the corresponding matrix coefficient is constant.)
- d) If $n \geq 3$, the group $\mathbf{SL}(n, \mathbf{Z})$ has property]T[. (Use Exercise 7 of INT, VII, p. 116.)
- e) Let $n \geq 3$. The group $\mathbf{SL}(n, \mathbf{Z})$ is finitely generated. Let H be the family of normal subgroups of finite index of $\mathbf{SL}(n, \mathbf{Z})$; it is infinite. For every finite generating subset S of $\mathbf{SL}(n, \mathbf{Z})$ the family of Cayley graphs $(\mathcal{G}(\mathbf{SL}(n, \mathbf{Z})/H, SH/H))_{H \in \mathcal{H}}$ is a family of expander graphs (cf. TA, 2, p. 221, exercise 7 and exercise 16 of IV, p. 323).

20) Let G be a locally compact group and μ a left Haar measure on G . We denote by Δ the modulus of G . For any function f on G , we will denote by $\tilde{f} = \Delta f^*$.

A measure ν on G is said to be of positive type if $\langle \nu, \tilde{f} * f \rangle \geq 0$ for every function $f \in \mathcal{K}(G)$. A locally integrable function φ on G is said to be of positive type if $\langle \varphi, f^* * f \rangle \geq 0$ for every function $f \in \mathcal{K}(G)$.

- a) Suppose that ν is bounded. The measure ν is of positive type if and only if $\gamma(\nu) \geq 0$.
- b) Let $\varphi \in \mathcal{K}(G)$. The function φ is of positive type as a locally integrable function if and only if it is so according to Definition 4 of V, p. 442.
- c) Let ν be a measure of positive type on G . The formula

$$(f | g)_{\nu} = \langle \nu, \tilde{f} * g \rangle$$

defines a positive Hermitian form on $\mathcal{K}(G)$. We denote by E_{ν} the separated completion Hilbert space of $\mathcal{K}(G)$ equipped with this Hermitian form.

- d) For $f \in \mathcal{K}(G)$, the function $\sigma(g)f: g \mapsto (\varepsilon_g * f)\Delta^{1/2}$ belongs to $\mathcal{K}(G)$ and $\|\sigma(g)f\|_{\nu} = \|f\|_{\nu}$. The map $g \mapsto \sigma(g)$ extends to a unitary representation of G in E_{ν} .
- e) Let U be an open neighborhood of e in G , and let ν be a positive-type measure on G such that there exists a real number $c \geq 0$ so that

$$\langle \nu, f * \tilde{f} \rangle \leq c \left(\int_G f d\mu \right)^2$$

if $f \in \mathcal{K}(G)$ is positive and vanishes outside U . There exists a continuous function of positive type φ on G such that $\nu = \varphi \Delta^{-1/2} \cdot \mu$. (Let j be the canonical morphism from $\mathcal{K}(G)$ into E_ν ; if $\mathfrak{B}$ is a basis for the filter of neighborhoods of e whose elements are contained in U and $f_V \in \mathcal{K}_+(G)$ is a function with support in $V \in \mathcal{B}$ of integral 1, prove that $j(f_V)$ converges weakly according to $\mathcal{B}$ to an element y_0 of E_ν such that

$$(j(f) | y_0) = \int_G f^* d\mu$$

for $f \in \mathcal{K}(G)$; setting $\varphi(g) = (y_0 | \sigma(g)y_0)$, verify that φ has the required properties.)

- f) Let φ_1 be a continuous positive-definite function on G and φ_2 a locally integrable function of positive type. If $\varphi_1 - \varphi_2$ is of positive type, then φ_2 is equal locally almost everywhere to a continuous function of positive type. (Verify that one can apply the previous question.)

21) We retain the notation of the preceding exercise.

- a) Let $f \in \mathcal{L}^2(G)$ that is of positive type as a locally integrable function. The partial operator on $L^2(G)$ with domain $\mathcal{K}(G)$ defined by $\varphi \mapsto \gamma(\varphi)f$ is positive. We denote by $\varrho(f)$ the Friedrichs extension of f (example 8 of IV, p. 295); it is a positive self-adjoint partial operator on $L^2(G)$.
- b) Let $f \in \mathcal{L}^2(G)$. One says that f is sober if there exists a real number $c \geq 0$ so that $\|\gamma(\varphi)f\| \leq c\|\varphi\|$ for all $\varphi \in \mathcal{K}(G) \subset \mathcal{L}^2(G)$. We then denote by $\varrho(f)$ the endomorphism of $L^2(G)$ which extends $\varphi \mapsto \gamma(\varphi)f$. This definition coincides with the previous one if f is of positive type, and this condition then amounts to saying that $\varrho(f)$ is positive.
- c) Let f_1 and f_2 be sober functions in $\mathcal{L}^2(G)$. If $\varrho(f_1) = \varrho(f_2)$, then f_1 and f_2 are equal in $L^2(G)$.
- d) Let $f \in \mathcal{L}^2(G)$ be of positive type. For all $g \in G$, we have $\varrho(f) \circ \gamma(g) = \gamma(g) \circ \varrho(f)$. Moreover, for every $h \in \text{dom}(\varrho(f))$, we have $\varrho(f)h = h * f$.

- e) Let f_1 and f_2 be sober elements of $\mathcal{L}^2(G)$ such that $\varrho(f_1)$ and $\varrho(f_2)$ commute. The function $f_1 * f_2$ is a continuous positive-definite function on G ; it is sober, and we have

$$\varrho(f_1 * f_2) = \varrho(f_1)\varrho(f_2), \quad \langle f_1 | f_2 \rangle \geq 0.$$

- f) With the notations and hypotheses of the previous question, if $f_2 - f_1$ is of positive type, then

$$\|f_2 - f_1\|^2 \leq \|f_2\|^2 - \|f_1\|^2.$$

- g) Let $(f_n)_{n \in \mathbf{N}}$ be a sequence of sober positive-type elements of $\mathcal{L}^2(G)$, pairwise commuting, and such that $f_{n+1} - f_n$ is of positive type for all $n \in \mathbf{N}$. If the sequence $(\|f_n\|)$ is bounded, then the sequence (f_n) converges in $\mathcal{L}^2(G)$.

22) We retain the notation of the preceding exercise.

Let f be a continuous function of positive type belonging to $\mathcal{L}^2(G)$.

- a) There exists an increasing sequence $(p_n)_{n \in \mathbf{N}}$ of polynomial functions on $[0,1]$ which converges uniformly on $[0,1]$ to the function $t \mapsto \sqrt{t}$.
- b) Suppose that f is sober and that $0 \leq f \leq 1$ in $\mathcal{L}(\mathcal{L}^2(G))$. There exists a sequence (h_n) in $\mathcal{L}^2(G)$ satisfying the following conditions:
- (i) Each h_n is sober and of positive type;
 - (ii) The function $h_{n+1} - h_n$ is of positive type for all $n \in \mathbf{N}$;
 - (iii) The endomorphisms $\varrho(h_n)$ commute pairwise;
 - (iv) One has $\varrho(h_n)^2 \leq \varrho(f)$ for all n .
- c) The sequence $(h_n)_{n \in \mathbf{N}}$ converges in $\mathcal{L}^2(G)$; if h is a limit of (h_n) , then h is of positive type and sober, and satisfies $\varrho(h)^2 = \varrho(f)$; we have $f = h * h$.
- d) We no longer assume that f is sober. For $t \in \mathbf{R}$, we denote by p_t the spectral projection of

the partial self-adjoint operator $\varrho(f)$ defined by $[-\infty, t]$. We have $p_t \circ \gamma(g) = \gamma(g) \circ p_t$ for all $t \in \mathbf{R}$ and $g \in G$.

- e) For $t \in \mathbf{R}$, set $f_t = p_t(f)$; this is the class in $L^2(G)$ of a continuous positive-definite sober function in $\mathcal{L}^2(G)$ (Check that $\gamma(\varphi)f_t = \varrho(f)p_t(g)$ for $g \in \mathcal{H}(G)$, and deduce that $\varrho(f_t) = \varrho(f) \circ p_t$, then apply the results of Exercise 20.)
- f) There exists $h \in \mathcal{L}^2(G)$ such that $f = h * h$ (Consider $f_n = p_n(f)$ for $n \in \mathbf{N}$ and apply part c).)

23) We denote $G = R_+^*$; it is a locally compact commutative group and the measure $\mu = dx/x$ is a Haar measure on G . The group G and the group $i\mathbf{R}$ are in duality via the mapping $(x, it) \mapsto x^{it}$; the dual measure of μ is Lebesgue measure on $i\mathbf{R}$ (cf. Cor. 4 of II, p. 236). We denote by F the Fourier transform of G , and $\overline{F}$ the Fourier cotransform of $\widehat{G}$.

We denote by X the identity function on G .

- a) Let ν be a positive measure on G . We denote by I_ν the set of $\sigma \in \mathbf{R}$ such that the function X^σ is ν -integrable, and B_ν the set of $s \in \mathbf{R}$ such that $\mathcal{R}(s) \in I_\nu$. The set I_ν is an interval of $\mathbf{R}$ and the function $s \mapsto \nu(X^s)$ is defined and holomorphic in the interior of B_ν .
- b) Let f be a locally integrable function on G . We denote by $I_f = I_{f \cdot \mu}$ and $B_f = B_{f \cdot \mu}$. For every $s \in B_f$ we set

$$\widehat{f}(s) = \int_G f(x) x^s d\mu(x).$$

If there exists $\delta > 0$ and $c \geq 0$ such that $|\widehat{f}(s)| \leq c(1 + |s|)^{-1-\delta}$ for $s \in B_f$, then for all σ in the interior of B_f , we have

$$f(x) = \frac{1}{2\pi} \int_{\mathbf{R}} \widehat{f}(\sigma + it) x^{-\sigma - it} dt.$$

- c) Let $f \in \mathcal{L}^2(G)$. We assume that 0 belongs to the interior of I_f . We then have $\overline{\mathcal{F}f}(it) = \widehat{f}(it)$ for almost every $t \in \mathbf{R}$.
- d) Let f and g be measurable functions on G . Let σ be a real number such that $X^{-2\sigma}f \in$

$L^2(G)$ and $X^{2\sigma}g \in L^2(G)$. We then have

$$\int_G f g d\mu = \frac{1}{2\pi} \int_{\mathbf{R}} \hat{f}(-\sigma - it) \hat{g}(\sigma + it) dt.$$

e) Let f and g be measurable functions on G . One assumes that

- (i) There exists $c \in \mathbf{R}$ such that $|f(x)| \leq cx/(1+x)$ for all $x \in G$;
- (ii) One has $]0,1[\subset I_g$ and for every σ such that $0 < \sigma < 1$, the function $X^\sigma g$ is bounded;
- (iii) There exists $c' \in \mathbf{R}$ such that $|\hat{g}(s)| \leq c'(1+|s|)^{-2}$ for all $s \in \mathbf{C}$ such that $\Re(s) \in]0,1[$.

Then we have

$$\int_G f g d\mu = \frac{1}{2\pi} \int_{\mathbf{R}} \hat{f}(-\sigma - it) \hat{g}(\sigma + it) dt$$

for all $\sigma \in \mathbf{R}$ such that $0 < \sigma < 1$.

- f) Let $\sigma \in \mathbf{R}$ and let $f \in \mathcal{C}(\sigma + i\mathbf{R})$. There exists a positive measure ν on G such that $\sigma \in I_\nu$ and $\hat{\nu}(\sigma + it) = f(\sigma + it)$ for all $t \in \mathbf{R}$ if and only if for every integer $n \geq 1$, every family $(t_i)_{0 \leq i \leq n}$ of complex numbers and any family $(z_i)_{0 \leq i \leq n}$ of complex numbers such that $\Re(z_i) = \sigma$ for all i , we have

$$\sum_{i=0}^n \sum_{j=0}^n \bar{t}_i t_j f\left(\frac{1}{2}(\bar{z}_i + z_j)\right) \geq 0.$$

Exercises for Section 4

- 1) Let G be a compact topological group and let B be the family of characters of irreducible unitary representations of G .
 - a) The pair $(R(G), \mathcal{B})$ is a natural unital basal algebra (cf. exercise 6 of III, p. 130).
 - b) Assume that G is finite. The algebra $R(G)$ is then of finite rank; compute $p_f(\pi)$ for every $\pi \in \widehat{G}$, and identify the representation of G corresponding to the element r_0 defined in question j) of loc. cit.

2) Let $n \geq 1$ and $p \in [1, +\infty]$. We equip $\mathbf{R}^n$ with the norm

$$\|x\| = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \quad \text{if } p \neq +\infty,$$

and $\|x\| = \sup_i |x_i|$ if $p = +\infty$. Let G be the group of isometries of the metric space $\mathbf{R}^n$ equipped with this norm.

- a) Let σ be a permutation of $\{1, \dots, n\}$ and $\varepsilon \in \{-1, 1\}^n$. Let (e_i) be the canonical basis of $\mathbf{R}^n$. The unique endomorphism of $\mathbf{R}^n$ such that $e_i \mapsto \varepsilon_i e_{\sigma(i)}$ belongs to G . The set of isometries of this form is a subgroup W of G ; it is an extension of $\mathfrak{S}_n$ by $(\mathbf{Z}/2\mathbf{Z})^n$.
 - b) The group G is compact.
 - c) The action of the group G on $\mathbf{R}^n$ is irreducible; the group G is contained in the orthogonal group of the quadratic form $\sum x_i^2$.
 - d) If $p \neq 2$, then $G = W$.
- 3) We recall that a group Γ is said to be residually finite if the intersection of the finite-index subgroups of Γ is reduced to $\{e\}$ (A, I, p. 129, exerc. 5, b)).

For a group Γ , we denote by $\widehat{\Gamma}$ the projective limit $\widehat{\Gamma} = \varprojlim \Gamma/H$, where H ranges over the set of finite-index normal subgroups of Γ ordered by inclusion. We say that $\widehat{\Gamma}$ is the profinite completion of Γ .

- a) A group Γ is residually finite if and only if, for every $g \in \Gamma$ different from the identity element, there exists a finite group F and a homomorphism $\varphi : \Gamma \rightarrow F$ such that $\varphi(g) \neq e$.
- b) Show that Γ is residually finite if and only if the canonical homomorphism from Γ into $\widehat{\Gamma}$ is injective.
- c) Let K be a field. Every finitely generated subgroup (cf. A, I, p. 145, exerc. 16) of $\mathbf{GL}(n, K)$ is residually finite ("Mal'cev's theorem"). (Show that there exists an integer $m \geq 0$, a ring A equal to $\mathbf{Z}$ or to a finite field and an injective homomorphism of such a subgroup

into $\mathbf{GL}(m, R)$, where R is of the form $A[X_1, \dots, X_r, 1/f]$.)

d) Let Γ be a finitely generated discrete group. Suppose that there exists an injective homomorphism from Γ into a compact topological group. For every $\gamma \in \Gamma$ different from the identity element, there exists a finite-dimensional unitary irreducible representation of Γ such that the image of γ by π is not the identity map.

e) A finitely generated group is residually finite if and only if it is isomorphic to a subgroup of a compact group.

¶ 4) Let G be a group and let A and B be subgroups of G . Let $\varphi: A \rightarrow B$ be an isomorphism.

Let r' be a family in the free group $F(G)$ such that G is isomorphic to the group $F(G, r')$ (A, I, p. 87, n°6). One calls the HNN extension of G according to φ the group $G^\varphi = F(X, r)$ where X is the sum of the sets G, A, B and an element t , and where r is the sum family of r' and of the relators $t^{-1}at = \varphi(a)$ for all $a \in A$.

We denote by $j: G \rightarrow G^\varphi$ the unique homomorphism such that $j(g)$ is the element represented by g in G^φ .

a) Let $k \geq 1$ an integer and let $(g_0, \varepsilon_1, g_1, \dots, \varepsilon_k, g_k)$ be a finite family such that $g_i \in G$ and $\varepsilon_i \in \{-1, 1\}$ for every i . Suppose that there does not exist an integer j such that $1 \leq j \leq k-1$ and

$$(\varepsilon_j, \varepsilon_{j+1}) = (-1, 1), \quad g_i \in A$$

or

$$(\varepsilon_j, \varepsilon_{j+1}) = (1, -1), \quad g_i \in B.$$

Then the element

$$j(g_0)t^{\varepsilon_1}j(g_1) \dots t^{\varepsilon_k}j(g_k)$$

of G^φ is not the identity element ("Britton's Lemma".)

- b) The homomorphism j is injective.
- c) Let a and b be distinct symbols. Let G be the group $F(\{a, b\}, a^{-1}b^2ab^{-3})$. We denote by $b_1 = a^{-1}ba$. Let H be a normal subgroup of finite index of G and $\pi : G \rightarrow G/H$ the canonical projection. One has $[b_1, b] \in \text{Ker}(\pi)$.
- d) The group G admits no faithful finite-dimensional representation. (Show that G is not residually finite.)
- 5) Let G be a compact group and ϱ a faithful finite-dimensional unitary representation of G . Let $\pi \in \widehat{G}$.
- a) There exist integers a and b such that π is isomorphic to a subrepresentation of $\varrho^{\otimes a} \otimes \overline{\varrho}^{\otimes b}$. (One may use, for example, the Weierstrass–Stone theorem.)

We denote by $w(\pi)$ the minimum of $a + b$ when (a, b) ranges over the set of integers satisfying this property.

- b) Let Λ be the weight lattice of G^0 and let q be a norm on $\Lambda \otimes \mathbf{R}$. There exist real numbers C and D such that for every representation $\pi \in \widehat{G}$ whose restriction to G^0 has highest weights $(\lambda_1, \dots, \lambda_k)$, we have

$$w(\pi) \leq C \sup(q(\lambda_1), \dots, q(\lambda_k)) + D.$$

(Consider first the case of a connected semisimple group and apply a) to a suitably chosen finite family of representations of G).

- 6) Let G be a compact group and let ϱ be an irreducible representation of G . There exists a central function $f \in \Theta(G^\#)$ such that

$$f = \sum_{\pi \in \widehat{G}} c(\pi) \chi_\pi$$

where the integers $c(\pi)$ satisfy

- (i) $c(\pi) \geq 0$ for all $\pi \in \widehat{G}$;

(ii) $c(1) \leq c(\varrho)$, and the inequality is strict except possibly if ϱ is of dimension 1 and satisfies

$$\varrho^2 = 1 ;$$

(iii) $\mathcal{R}(f) \geq 0$.

¶ 7) For every integer $n \geq 1$, we denote by μ_n the positive measure on $\mathbf{C}$ image of the normalized Haar measure of the group $\mathbf{U}(\mathbf{C}^n) \subset \mathbf{GL}(\mathbf{C}^n)$ under the trace map $g \mapsto \text{Tr}(g)$; it is a compactly supported measure of total mass 1.

a) Let a and b be natural integers such that $a + b \leq n$. We have

$$\int_{\mathbf{C}} z^a \bar{z}^b d\mu_n(z) = \begin{cases} 0 & \text{if } a \neq b \\ a! & \text{if } a = b. \end{cases}$$

b) The sequence of measures $(\mu_n)_{n \geq 1}$ converges narrowly to the Gaussian measure on the real vector space $\mathbf{C}$ whose covariance is $Q(z) = |z|^2$.

8) Let G be a compact group and let ϱ_1 and ϱ_2 be irreducible unitary representations of G . Let $n \geq 2$ be an integer. If $S^n \varrho_1$ is isomorphic to $S^n \varrho_2$, then there exists a subgroup H of finite index of G such that the restriction of ϱ_1 to H is isomorphic to the restriction of ϱ_2 to H .

9) Let k be a finite field with 5 elements and $G = \mathbf{SL}(2, k)$.

a) The conjugacy classes of the group G are represented by e , $-e$ and the seven elements

$$g_3 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad g_4 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad g_5 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \quad g_6 = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}$$

$$g_7 = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \quad g_8 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad g_9 = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

Determine the cardinality of each conjugacy class.

- b) Let $\alpha = \frac{1}{2}(1 + \sqrt{5})$. There exists a faithful irreducible unitary representation π of dimension 2 of G whose character χ satisfies $\chi(-e) = -2$ and

$$\begin{aligned}\chi(g_4) &= \alpha, & \chi(g_6) &= -\alpha, & \chi(g_3) &= -\alpha - 1, & \chi(g_5) &= \alpha + 1 \\ \chi(g_7) &= 0, & \chi(g_8) &= 1, & \chi(g_9) &= -1.\end{aligned}$$

(One may consider the restriction to G of the irreducible representations of dimension 4 of $\mathbf{GL}(2, k)$, cf. A, VIII, p. 422, exercise 27.)

- c) One has $M_4(\pi) = 2$.

- 10) a) There exists an irreducible representation π of orthogonal type of the alternating group $\mathfrak{A}_5$ such that $M_4(\pi) = 3$.

- b) Let R be a root system of type E_6 (resp. of type E_7 , of type E_8) and W the Weyl group of R . Let π be the geometric representation of W (LIE, V, § 4, no. 3). It is irreducible, of orthogonal type, and satisfies $M_4(\pi) = 3$.

(One may use software such as GAP or MAGMA to determine the list of conjugacy classes of W and their images under π .)

- 11) Let G be a compact group and π a finite-dimensional unitary representation of G in a Hilbert space E . If $M_4(\pi) \leq 5$ then the representation π is irreducible.

- 12) Let G be a connected compact semisimple Lie group whose complexified Lie algebra is isomorphic to the simple Lie algebra e_6 .

- a) There exists a faithful irreducible representation ϱ of G of dimension 27; such a representation is of complex type.

- b) The fourth moment of the representation ϱ is equal to 3.

- 13) Let G be a connected compact semisimple Lie group and let $\mathfrak{g}$ be its Lie algebra. Suppose there exists a faithful irreducible representation ϱ of dimension 27 of G . We assume that the fourth moment of ϱ is equal to 3 and that the representation ϱ is of complex type.

- a) The Lie algebra $\mathfrak{g}$ is simple.

b) The Lie algebra $\mathfrak{g}$ is isomorphic to $\mathfrak{e}_6$.

14) Let G be a connected compact semisimple Lie group. Suppose there exists a faithful irreducible representation ϱ of dimension 27 of G . The complexified Lie algebra of G is isomorphic to $\mathfrak{e}_6$ if and only if there exists an irreducible subrepresentation of dimension 78 in the representation $\text{End}(\varrho)$.

15) Let q be a power of an odd prime number and F a finite field with q elements. Consider the Heisenberg group $H_3(F)$ of matrices of size 3 with coefficients in F that are upper triangular and unipotent. For $(a, b, c) \in F^3$, we denote

$$\sigma(a, b, c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \in H_3(F).$$

The map σ is therefore a bijection.

a) The center C of $H_3(F)$ is identified with F by the group homomorphism sending $\sigma(a, b, c)$ to c .

b) There exist q^2 one-dimensional unitary characters of $H_3(F)$; these characters are trivial on C .

c) Let X be the subgroup of index q of F consisting of the elements $\sigma(0, b, c)$ for $(b, c) \in F^2$. The subgroup X is identified with F^2 . Let ψ_1 and ψ_2 be unitary characters of F , and set $\psi(\sigma(0, b, c)) = \psi_1(b)\psi_2(c)$. The unitary representation ϱ of $H_3(F)$ induced by the character ψ of X is irreducible if and only if ψ_2 is nontrivial; it depends only on ψ_2 up to isomorphism, and its central character is equal to ψ_2 . The support of the character of ϱ is equal to C .

d) The irreducible representations constructed in the two preceding questions are all the irreducible unitary representations of $H_3(F)$.

e) Let ϱ be one of the irreducible unitary representations of dimension q of $H_3(F)$. The representation ϱ is faithful and satisfies $M_4(\varrho) = q^2$.

16) Let $r \geq 1$ be an integer and let G be a connected compact subgroup of $U(C^r)$ acting irreducibly on C^r . We denote by D the subgroup of diagonal matrices in $U(C^r)$.

a) Let A be a subgroup of D contained in the normalizer of G in $U(C^r)$. Let S be the subset of $\widehat{A}$ formed by the diagonal coefficients $a = (a_{i,j}) \mapsto a_{i,i}$, where $1 \leq i \leq r$. We assume that S contains r elements and that the equation

$$\chi_1 \chi_2^{-1} = \chi_3 \chi_4^{-1}$$

with $\chi_i \in S$ admits only the solutions of the form $(\chi_1, \chi_1, \chi_3, \chi_3)$ and $(\chi_1, \chi_2, \chi_1, \chi_2)$ (one then says that S is a Sidon set).

If G is semisimple, prove that $G = SU(C^r)$ (Consider the representation of A on $\text{End}(C^r)$, and deduce from the hypothesis that the complexified Lie algebra of G is stable under conjugation by D .)

b) The group G contains $SU(C^r)$ if and only if there exists an element $g \in G$ such that the eigenvalues of g are distinct and form a Sidon set in C^* .

17) Let G be a compact group and H a closed subgroup of G . Let ϱ be a unitary representation of H in a finite-dimensional Hilbert space E .

a) Let $\pi \in \widehat{G}$. The space $\text{Hom}_G(\pi, \text{Ind}_H^G(\varrho))$ is of finite dimension.

b) For every $u \in \text{Hom}_G(\pi, \text{Ind}_H^G(\varrho))$, the image of u is contained in the image of $\mathcal{C}(G; E)$ in $L^2(G; E)$.

c) There exists a linear map Φ from the space $\text{Hom}_G(\pi, \text{Ind}_H^G(\varrho))$ into the space $\text{Hom}_H(\text{Res}_H^G(\pi), \varrho)$ such that, for every $u \in \text{Hom}_G(\pi, \text{Ind}_H^G(\varrho))$ and every $x \in E_\pi$, $\Phi(u)(x)$ is the image at e of the continuous function representing $u(x)$.

d) The linear map Φ is an isomorphism.

e) Suppose that ϱ is irreducible. The multiplicity of π in $\text{Ind}_H^G(\varrho)$ is equal to the multiplicity of ϱ in $\text{Res}_H^G(\pi)$ (Frobenius reciprocity formula, cf. A, VIII, p. 392).

18) Assume that G is compact. Let ϱ and ϱ' be unitary representations of G . The representation ϱ is weakly contained in ϱ' (cf. exercise 11 of V, p. 497) if and only if for every irreducible unitary representation π of G , the condition $m_\pi(\varrho) \geq 1$ implies that $m_\pi(\varrho') \geq 1$. (Reduce to the case where ϱ' is irreducible; for every normalized diagonal matrix coefficient f of ϱ , prove by applying Exercise 13 of V, p. 498 that there exists a diagonal matrix coefficient of ϱ' which is not orthogonal to f , and conclude using the orthogonality relations.)

19) Let G be a compact group equipped with its normalized Haar measure μ and let $f \in \mathcal{C}(G)$ a continuous function on G . There exists an irreducible representation $\pi \in \widehat{G}$ such that $f = \dim(\pi)^{-1} \chi_\pi$ if and only if

$$\int_G f(xgx^{-1}h) d\mu(x) = f(g)f(h)$$

for all $(g, h) \in G^2$

20) Let H be a locally compact topological group equipped with a left Haar measure ν . Let G be a compact subgroup of H . Let ϱ be an irreducible unitary representation of H in a Hilbert space E .

We assume that the multiplicity of π in the restriction of ϱ to G is finite for every irreducible representation $\pi \in \widehat{G}$. We identify a function on G with a compactly supported measure on H .

a) For every $\pi \in \widehat{G}$, the endomorphism $\varrho(f * \chi_\pi)$ of E is of finite rank.

b) For every $f \in L^1(H)$, the endomorphism $\varrho(f)$ is compact.

21) Let $n \geq 1$ be an integer.

a) For every integer $r > n^2$ and every family $(a_1, \dots, a_r)$ of elements of $\mathcal{L}(\mathbb{C}^n)$, we have

$$\sum_{\sigma \in \mathfrak{S}_r} \varepsilon(\sigma) a_{\sigma(1)} \dots a_{\sigma(r)} = 0. \quad (4)$$

b) We denote by $r(n)$ the smallest integer $r \geq 1$ such that formula (4) is valid for every family $(a_i)_{1 \leq i \leq r}$. We have $r(n) \geq r(n-1) + 2$ (Consider a family $(a_1, \dots, a_{r(n-1)-1})$ in $\mathcal{L}(\mathbb{C}^{n-1})$)

such that the expression above, denoted b , is nonzero; interpreting the elements of $\mathcal{L}(\mathbf{C}^n)$ as matrices, choose i and j such that $b_{i,j}$ is nonzero; view a_i and b as elements of $\mathcal{L}(\mathbf{C}^n)$ by adding a row and a column of zeros, then define matrices $a_{r(n-1)}$ and $a_{r(n-1)+1}$ so that expression (4) for $(a_1, \dots, a_{r(n-1)+1})$ is nonzero.)

22) Let $r \geq 1$ be an integer and let $H \subset \mathbf{GL}(r, \mathbf{C})$ be a closed subgroup.

- a) The set of functions on H of the form $g \mapsto \langle \varrho(g)x, \lambda \rangle$, where ϱ is a continuous linear representation of H in a finite-dimensional complex vector space E , $x \in E$ and $\lambda \in E'$, is a dense involutive subalgebra of $\mathcal{C}(H)$.
- b) For any function $f \in \mathcal{K}(H)$ nonzero, there exists a continuous linear representation ϱ of H in a finite-dimensional complex vector space such that $\varrho(f)$ is nonzero.
- c) Assume that H is a connected semisimple real Lie group. For every $f \in \mathcal{K}(H)$ nonzero, there exists an irreducible continuous linear representation ϱ of H in a finite-dimensional complex vector space such that $\varrho(f)$ is nonzero.

23) We retain the hypotheses and notation of the preceding exercise, and assume that H is a connected semisimple real Lie group.

Let G be a compact subgroup of H . Assume that there exists a connected solvable closed subgroup N of H such that $H = GN$. (For every semisimple real Lie group, this condition is satisfied when G is a maximal compact subgroup of H ; see for example INT, VII, p. 91, § 3, n° 3, prop. 7 when $G = \mathbf{SL}(n, \mathbf{R})$.)

Let $\pi \in \widehat{\mathbf{G}}$ and $e_\pi = \dim(\pi) \overline{\chi}_\pi$. Let ϱ be the irreducible representation of H given by the last question of the preceding exercise; we denote by E the space of ϱ .

- a) There exists a cyclic vector for the restriction of ϱ to G . (Use Lie's theorem, LIE, III, p. 241, cor., to find a nonzero $x \in E$ such that Cx is invariant under the action of N .) The multiplicity of π in the restriction of ϱ to G is at most equal to $\dim(\pi)$.
- b) Let r be the integer $r(\dim(\pi)^2)$ defined in Exercise 22; the identity

$$\sum_{\sigma \in \mathfrak{S}_r} \varepsilon(\sigma) a_{\sigma(1)} \dots a_{\sigma(r)} = 0$$

is valid for every family $a_i = e_\pi * g_i * e_\pi$ of elements of the algebra $\mathcal{M}_c(H)$ of measures with compact support on H , where $g_i \in \mathcal{K}(H)$ for $1 \leq i \leq r$.

- c) Let ϖ be an irreducible unitary representation of H in a Hilbert space F . Let F_π be the π -isotypic component of the restriction of ϖ to G and let p be the orthogonal projector of F with image F_π . For every family $(u_i)_{1 \leq i \leq r}$ in $\mathcal{L}(F_\pi)$, we have

$$\sum_{\sigma \in \mathfrak{S}_r} \varepsilon(\sigma) u_{\sigma(1)} \dots u_{\sigma(r)} = 0$$

(use Exercise 1 of V, p. 486).

- d) The multiplicity of π in the restriction of ϖ to G is at most equal to $\dim(\pi)$. (Use the result of Exercise 22 of V, p. 510.)
- 24) A topological group G is said to be minimally almost periodic if every continuous homomorphism from G into a compact group is trivial.
- a) A topological group G is minimally almost periodic if and only if G admits no non-trivial finite-dimensional unitary representation. If G is commutative, then G is minimally almost periodic if and only if every continuous unitary character of G is trivial.
- b) Let G be a topological group and $g \in G$ such that, for every integer $k \geq 1$, there exists an integer $m \geq 1$ such that g^{km} is conjugate to g . Then one has $\varrho(g) = 1_E$ for every unitary representation ϱ of finite dimension of G in a Hilbert space E .
- c) Let k be a subfield of $\mathbb{C}$. The group $\mathbf{SL}(2, k)$ endowed with the discrete topology is minimally almost periodic. (Verify that the preceding question applies to $\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}$ for all $z \in k$.)
- 25) Let G be a locally compact group endowed with a left Haar measure ν and $K \subset G$ a compact subgroup of G endowed with the normalized Haar measure μ .

We denote by $\mathcal{H}(G, K)$ the space of functions $\varphi \in \mathcal{H}(G)$ such that $\varphi(k_1 g k_2) = \varphi(g)$ for all $g \in G$ and every $(k_1, k_2) \in K \times K$.

The space $\mathcal{H}(G, K)$ is a unital subalgebra of the algebra $\mathcal{M}_c(G)$; one says that (G, K) is a Gelfand pair if the algebra $\mathcal{H}(G, K)$ is commutative. a) The map u^K which to f associates the function f^K defined by

$$f^K(g) = \int_{K \times K} f(k_1 g k_2) d(\mu \otimes \mu)(k_1, k_2)$$

is a projector in $\mathcal{H}(G)$ whose image is $\mathcal{H}(G, K)$.

b) Let $f \in \mathcal{H}(G)$. We have

$$\int_G f(g) d\nu(g) = \int_G f^K(g) d\nu(g)$$

and $(f\varphi)^K = f^K\varphi$ for every function $\varphi \in \mathcal{H}(G, K)$.

c) Let $C \subset G$ be a compact subset. There exists a function $\varphi \in \mathcal{H}(G, K)$ such that $\varphi(x) = 1$ for all $x \in C$.

d) Let (G, K) be a Gelfand pair. The group G is unimodular. (Note that the modulus of G belongs to $\mathcal{H}(G, K)$ and interpret the integral of $\varphi \in \mathcal{H}(G, K)$ as the value at e of a suitably chosen convolution.)

e) Suppose that there exists a continuous involution $\theta: G \rightarrow G$ such that $K\theta(x)K = Kx^{-1}K$ for all $x \in G$. Then (G, K) is a Gelfand pair. (Verify that the map $f \mapsto \nu(f \circ \theta)$ on $\mathcal{H}(G)$ is a left Haar measure on G , then that it is equal to $\underline{\nu}$; deduce that $(f * g) \circ \theta = (f \circ \theta) * (g \circ \theta)$ and compare with the formula $f * g = \check{g} * f$.)

f) If G is compact, then $(G \times G, \Delta)$ is a Gelfand pair, where Δ is the diagonal subgroup of $G \times G$ image of the homomorphism $x \mapsto (x, x)$ from G into $G \times G$.

26) We resume the notation of the preceding exercise. We denote by φ_K the characteristic function of K .

a) Let π be a unitary representation of G in a complex Hilbert space E . The representation

of $\mathcal{M}_c(G)$ in E defines, by passage to subspaces, a representation of the algebra $\mathcal{H}(G, K)$ in the space E^K of K -invariant vectors of E .

- b) If E admits a cyclic vector x such that $x \in E^K$, then π is irreducible.
- c) Suppose that π is irreducible. If the space E^K is not reduced to 0, the representation of $\mathcal{H}(G, K)$ in E^K is irreducible.
- d) Let φ be a positive-definite function on G and let (ϱ, x) be a cyclic Hilbertian realization of φ . We have $\varphi \in \mathcal{H}(G, K)$ if and only if $x \in E^K$.

27) We resume the notation of the preceding exercise. We assume that (G, K) is a Gelfand pair.

We say that $\varphi \in \mathcal{H}(G, K)$ is a K -spherical function if the map

$$f \mapsto \int_G f \check{\varphi} d\nu$$

is a non-zero character of the algebra $\mathcal{H}(G, K)$.

a) Let $\varphi \in \mathcal{H}(G, K)$. Prove that the following conditions are equivalent:

- (i) The function φ is a K -spherical function;
- (ii) For all x and y in G , we have

$$\int_K \varphi(xky) d\mu(k) = \varphi(x) \varphi(y);$$

(iii) One has $\varphi(e) = 1$ and for every function $\psi \in \mathcal{H}(G, K)$, there exists $c \in \mathbb{C}$ such that $\psi * \varphi = c\varphi$.

b) We denote by $\mathcal{H}^1(G, K)$ the space of $f \in L^1(G)$ such that $\varrho_G(k_1, k_2)f = f$ for all $(k_1, k_2) \in K \times K$. This is a Banach subalgebra of the involutive Banach algebra $L^1(G)$; it is commutative.

c) The algebra $\mathcal{H}(G, K)$ is identified with a dense subalgebra of $\mathcal{H}^1(G, K)$.

d) The map which to $\varphi \in \mathcal{H}(G, K)$ associates the linear form

$$f \mapsto \int_G f \check{\varphi} d\nu$$

on $\mathcal{H}^1(G, K)$ defines a bijection from the space of bounded K -spherical functions on G onto the space of non-zero characters of $\mathcal{H}^1(G, K)$. (Note that every character of $\mathcal{H}^1(G, K)$ is continuous and of norm ≤ 1 , cf. prop. 2 of I, p. 104.)

28) Let G be a locally compact group and K a compact subgroup of G .

- a) The pair (G, K) is a Gelfand pair if and only if, for every unitary representation π of G in a complex Hilbert space E , the dimension of E^K is at most 1. (To prove that the condition is necessary, prove that an irreducible representation of a commutative involutive Banach algebra in a Hilbert space is of dimension 1; to prove that it is sufficient, use the cor. of th. 4 of V, p. 454 to verify that the characters of the algebra $\mathcal{H}(G, K)$ separate its points.)
- b) We assume that (G, K) is a Gelfand pair. Let $\varphi \in \mathcal{H}(G, K)$. Assume that φ is a positive-definite function and that $\varphi(e) = 1$. Show that φ is a K -spherical function if and only if φ has a Hilbert representation (ϱ, x) such that ϱ is irreducible.
- c) Suppose that (G, K) is a Gelfand pair. The set of positive-definite K -spherical functions on G is in bijection with the set of irreducible representations $\pi \in \widehat{G}$ such that E_π^K is nonzero.

29) Let G be a compact group and K a closed subgroup of G . Suppose that (G, K) is a Gelfand pair. We denote by ν (resp. μ) the normalized Haar measure on G (resp. on K).

Let φ be a K -spherical function on G .

- a) We denote by E_φ the subspace of $\mathcal{C}(G)$ functions of the form $\alpha * \varphi$, where α is a measure on G with finite support. For every $\psi \in \mathcal{H}(G, K)$, the endomorphism $v_\psi : f \mapsto f * \psi$ of $\mathcal{C}(G)$ is compact and E_φ is contained in an eigenspace of v_ψ .
- b) The space E_φ is of finite dimension. It identifies with a subspace of G -finite vectors of the regular representation γ_G containing φ .

- c) The space of K-invariant vectors of E_φ is of dimension 1 and the representation of G in E_φ is irreducible.
- d) Let $(f_i)_{i \in I}$ be an orthonormal basis of E_φ and $k: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{C}$ the function defined by

$$k(x, y) = \sum_{i \in I} \overline{f_i(x)} f_i(y).$$

One has

$$f(x) = \int_{\mathbf{G}} k(x, y) f(y) d\nu(y)$$

for all $f \in E_\varphi$ and $x \in \mathbf{G}$.

- e) One has $k(x, y) = \dim(E_\varphi) \varphi(y^{-1}x)$ for all $(x, y) \in \mathbf{G} \times \mathbf{G}$.
- f) The function φ is of positive type.
- g) Let φ be a K-spherical function on G and let (π, x) be a Hilbert realization of φ . The representation π is irreducible; we have

$$\varphi(x) = \int_{\mathbf{K}} \chi_\pi(x^{-1}k) d\mu(k)$$

for all $x \in \mathbf{G}$, and $\int_{\mathbf{G}} |\varphi|^2 d\nu = \dim(\pi)^{-1}$.

- 30) Let $n \in \mathbf{N}$. Let us denote by G the subgroup of the affine group $A(\mathbf{R}^n)$ of the space $\mathbf{R}^n$ formed by the elements $g \in A(\mathbf{R}^n)$ such that the linear map g_0 associated with g belongs to $\mathbf{SO}(\mathbf{R}^n)$; it is a real Lie group (cf. A, II, p. 131 and LIE, III, p. 102, example). We denote by $g = (g_0, a)$ the elements of G, with $g_0 \in \mathbf{SO}(\mathbf{R}^n)$ and $a \in \mathbf{R}^n$.

- a) The pair $(\mathbf{G}, \mathbf{SO}(\mathbf{R}^n))$ is a Gelfand pair.
- b) Let $\varphi \in \mathcal{H}(\mathbf{G}, \mathbf{SO}(\mathbf{R}^n))$. Denote by $f: \mathbf{R}^n \rightarrow \mathbf{C}$ the function defined by $f(r) = \varphi(1_{\mathbf{R}^n}, g(e_1))$. The map $u: \varphi \mapsto f$ is an injective morphism of complex algebras from $\mathcal{H}(\mathbf{G}, \mathbf{K})$ into $\mathcal{H}(\mathbf{R}^n)$; its image is the subspace of $\mathcal{H}(\mathbf{R}^n)$ consisting of functions f such that $f(g_0(x)) = f(x)$ for all $g_0 \in \mathbf{SO}(\mathbf{R}^n)$ and every $x \in \mathbf{R}^n$.

c) We denote by Δ the Laplacian on $\mathbf{R}^n$. A function $\varphi \in \mathcal{K}(G)$ is a spherical function if and only if the function $f = u(\varphi)$ is of class C^∞ , satisfies $f(0) = 1$, and if there exists $\lambda \in \mathbf{C}$ such that $\Delta f = \lambda f$ (Use the fact that $\Delta(f \circ g_0) = (\Delta f) \circ g_0$ for all $g_0 \in \mathbf{SO}(\mathbf{R}^n)$.)

31) Let n be an integer ≥ 1 . We denote by N the Euclidean norm on $\mathbf{R}^n$, and we denote by λ_n the Lebesgue measure on $\mathbf{R}^n$. We denote by ω_n the unique positive measure of total mass 1 on the sphere $\mathbf{S}_n \subset \mathbf{R}^{n+1}$ invariant under the action of the orthogonal group $\mathbf{O}(n+1, \mathbf{R})$ (cf. INT, VIII, p. 116, exercise 8).

We identify R^n with the complement of a projective hyperplane H in the projective space $\mathbf{P}_n(\mathbf{R})$ (cf. TG, VI, p. 15, prop. 5), and we shall interpret the measures on R^n as measures on $\mathbf{P}_n(\mathbf{R})$ supported on the complement of H .

For every $g \in \mathbf{GL}(n+1, \mathbf{R})$, we denote by $\tilde{g}$ the homeomorphism of $\mathbf{P}_n(\mathbf{R})$ defined by g .

a) The positive measure c_n on $\mathbf{R}^n$ defined by

$$c_n = \pi^{-(n+1)/2} \Gamma\left(\frac{n+1}{2}\right) (1 + N^2)^{-(n+1)/2} \lambda_n$$

(“Cauchy measure”) is bounded and has total mass 1. If $n = 1$ this is the stable measure of parameter 1 (cf. exercise V, p. 496). For every projective hyperplane H' of $\mathbf{P}_n(\mathbf{R})$, we have $c_n(H') = 0$.

b) The canonical map from $\mathbf{R}^{n+1} - \{0\}$ into $\mathbf{P}_n(\mathbf{R})$ defines, by passing to subspaces, a continuous surjective map $\sigma : \mathbf{S}_n - (\mathbf{R}^n \times \{0\}) \rightarrow \mathbf{R}^n$. We have $\sigma(\omega_n) = c_n$.

c) Let $g \in \mathbf{GL}(n+1, \mathbf{R})$. There exists $a \in \mathbf{GL}(n, \mathbf{R})$ and $b \in \mathbf{R}^n$ such that the image measure $\tilde{g}(c_n)$ is equal to the image of c_n by the affine map $x \mapsto a(x) + b$.

In what follows, we propose to prove a converse of this assertion.

d) Let μ be a bounded positive measure on $\mathbf{P}_n(\mathbf{R})$ such that every projective hyperplane is μ -negligible. The set of $g \in \mathbf{GL}(n+1, \mathbf{R})$ such that $g(\mu) = \mu$ and $|\det(g)| = 1$ is a compact subgroup of $\mathbf{GL}(n+1, \mathbf{R})$.

e) Let μ be a bounded positive measure on $\mathbf{P}_n(\mathbf{R})$ of total mass 1 such that, for all $g \in$

$\mathbf{GL}(n+1, \mathbf{R})$ there exists an affine map f of $\mathbf{R}^n$ such that $g(\mu) = f(\mu)$. Every projective hyperplane is μ -negligible.

- f) Let K_μ be the subgroup of $g \in \mathbf{GL}(n+1, \mathbf{R})$ such that $g(\mu) = \mu$. There exists $h \in \mathbf{GL}(n+1, \mathbf{R})$ such that $K_\mu \subset h\mathbf{O}(n+1, \mathbf{R})h^{-1}$.
- g) Let $K = h^{-1}K_\mu h$. We have $\mathbf{O}(n+1, \mathbf{R}) = \mathbf{G} \cdot \mathbf{K}$, where $\mathbf{G}$ is the subgroup of $\mathbf{O}(n+1, \mathbf{R})$ consisting of the matrices

$$\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}$$

with $x \in \mathbf{O}(n, \mathbf{R})$

- h) The measure $h^{-1}(\mu)$ coincides with c_n . (Observe that the group $\mathbf{K}$ acts transitively on S_n and that the image measure $\sigma(h^{-1}(\mu))$ is a $\mathbf{K}$ -invariant measure on S_n .)
- 32) Let $\mathbf{G}$ be a locally compact group. Suppose there exists a subgroup $\mathbf{H}$ of $\mathbf{G}$ that is compact and open in $\mathbf{G}$. We denote by μ the unique left Haar measure on $\mathbf{G}$ such that $\mu(\mathbf{H}) = 1$ and φ the characteristic function of $\mathbf{H}$.
- a) Let π be a square-integrable representation of $\mathbf{G}$ in a complex Hilbert space $\mathbf{E}$. Show that the endomorphism $\pi(\varphi)$ is a Hilbert-Schmidt endomorphism.
- b) The space $\mathbf{E}^{\mathbf{H}}$ is finite-dimensional and satisfies

$$\dim(\mathbf{E}^{\mathbf{H}}) \leq \frac{1}{c(\pi)}$$

where $c(\pi)$ is the formal degree of π relative to $\{e\}$ (Use Exercise 6 of V, p. 487.)

- c) A discrete infinite group has no square-integrable representation; if $n \geq 2$ is an integer, the group $\mathbf{SL}(n, \mathbf{Z})$ has no square-integrable representation modulo the center.

HISTORICAL NOTE

(Chapters I to V)

The notions of eigenvalue and Fourier series were already well known in 1830, permeating many fields of analysis throughout the nineteenth century (cf. ÉHM, pp. 114, 260, 275). But spectral theory and harmonic analysis, in the sense in which we understand them in this book, did not begin to take their present form until about a century later — at the same time as their junction with the study of group representations around 1930, and then with that of normed algebras around 1940.

We confine ourselves here to sketching the main lines of their evolution in the period from 1906—when Hilbert introduced the notion of the “continuous spectrum” in his work on integral equations—to the years 1945–1950, when the theories presented here essentially acquired the form they have retained to the present day.

1. Discovery of the continuous spectrum

In the Note on topological vector spaces (cf. ÉHM, pp. 262–263), we described how I. Fredholm, around 1900 — continuing the work of C. Neumann, V. Volterra, and H. Poincaré — came to consider the equation

$$u(x) - \lambda \int_I K(x, y)u(y)dy = f(x) \quad (1)$$

with unknown function $u: I \rightarrow \mathbf{C}$, in the case where the interval $I \subset \mathbf{R}$ is compact and where the kernel K is continuous on $I \times I$. A skillful use of “infinite determinants,” based on ideas of Poincaré and H. von Koch, allowed him to obtain families of solutions whose dependence on

the complex variable λ is meromorphic, then to prove the famous “alternative” regarding the existence of solutions (cf. III, p. 81, th. 5). But Fredholm immediately specializes his results to the case $\lambda = -1$ without exploiting the fact that equation (1) is an eigenvalue problem ⁽¹⁾ .

When the kernel K is symmetric, we also saw how Hilbert, in his first memoir on integral equations [31], recognized the kinship between the Fredholm problem and the search for the “principal axes” of a quadratic form. Starting from this reduction for finite “sections” (discretizations) of the kernel K , he obtained by passage to the limit the formula

$$\int_I K(s, t)x(s)x(t)dt = \sum_{n=1}^{\infty} \frac{1}{\lambda_n} \int_I \varphi_n(s)x(s)ds, \quad (2)$$

where the λ_n are the eigenvalues of the kernel K , the φ_n form an orthonormal system of associated eigenvectors ⁽²⁾ and where the equality holds as soon as x is square-integrable on I (cf. ÉHM, p. 264).

The principal obstacles facing Hilbert concerned the passage from finite sums to the integrals of formula (2). He deduced from this formula the existence of the eigenvalues — however, he gave for the latter a variational characterization independent of the passage to the limit, inspired by the classical method for determining eigenvalues in finite dimension (cf. no. 4 of IV, p. 153). E. Schmidt would show in 1905, drawing on the work of H. Schwarz, how to obtain (2) without going through the “infinite determinants” on which the methods of Fredholm and Hilbert depended (cf. ÉHM, p. 265).

But already in 1906, Hilbert [32] turns to the more general problem of reducing a quadratic form

$$B(x, x) = \sum_{p, q=0}^{\infty} b_{pq}x_p x_q, \quad x = (x_p)_{p \in \mathbb{N}}$$

on $E = \ell^2_{\mathbb{C}}(\mathbb{N})$. He certainly has in mind that by passing to coefficients in an orthonormal basis of E , problem (1) reduces to the search for solutions of the “infinite linear system”

$$x_p + \sum_{q=0}^{\infty} b_{pq}x_q = f_p \quad (p \in \mathbb{N}).$$

He notes, however, that for the Fredholm problem one deals with very particular quadratic

forms, for which $B(x_n, x_n)$ tends to $B(x, x)$ as soon as x_n converges weakly to x ; he calls them “completely continuous” (vollstetig).

Hilbert insists on the essential difference that distinguishes them from quadratic forms bounded on E , and decides to undertake the more general reduction of the latter. Again, his method is to start from the reduction of the “finite sections.”

$$(x_0, \dots, x_n) \mapsto \sum_{p,q=0}^n b_{pq} x_p x_q$$

of the form B , and then to let n tend to infinity. For this passage to the limit, he takes advantage of the ideas on integration introduced by T. Stieltjes in his work on continued fractions [71]. He shows that every bounded quadratic form can, after an orthogonal transformation of the variables, be put in the form

$$\sum_{p \in \mathbf{N}} \frac{1}{\lambda_p} x_p^2 + \int_{s \in \mathbf{R}} \frac{1}{s} d\sigma(s, x). \quad (3)$$

In this formula, the λ_p are the eigenvalues of B and $x \mapsto \sigma(s, x)$ is (for a fixed $s \in \mathbf{R}$) a positive definite separating quadratic form on E , while $s \mapsto \sigma(s, x)$ is ⁽³⁾ (for a fixed $x \in E$) a continuous increasing function that tends to 0 at $-\infty$ and toward $\|x\|^2$ at $+\infty$ (cf. ÉHM, p. 284, for Stieltjes notation).

Let S denote the set of points of $\bar{R}$ that do not admit a neighborhood in which $s \mapsto \sigma(s, x)$ remains constant for all x ; then the integral in (3) can naturally be taken over S . Hilbert calls the set S the “continuous spectrum” (Streckenspektrum), the set of eigenvalues of B the “point spectrum” (Punktspektrum), and their union the “spectrum” (Spektrum). This term, coming from optics, had been used in 1897 by W. Wirtinger [91] in the study of differential equations with periodic coefficients.

Hilbert would quickly draw from his “spectral” methods a profound renewal of several problems of classical analysis, pertaining to the study of ordinary differential equations (in particular of Sturm–Liouville type), partial differential equations, or functions of a complex variable. We will not report here on the profusion of results that followed in the first quarter of the twentieth century and refer to the synthesis by E. Hellinger and O. Toeplitz [29]. We shall nevertheless

mention some of these works on integral equations, in which the germs of later developments of the abstract theory are to be found.

For Hilbert and his students, faithful to a German tradition in linear algebra, the model remains the principal axes theorem; thus it is always a matter of studying the spectra of bilinear or quadratic forms, in particular on the “Hilbert” space $E = \ell_{\mathbb{C}}^2(\mathbb{N})$. The fact that one can associate any continuous bilinear form B on E with the continuous endomorphism A of E characterized by $\langle Ax, y \rangle = B(x, y)$ was known to the Hilbert school, particularly after the works of E. Schmidt, M. Fréchet, and F. Riesz in 1907–1908. But it was F. Riesz who showed, in an admirable book [59] from 1913, the interest of placing endomorphisms at the forefront in this context.

Riesz gives the modern definition of the spectrum of an element A of $\mathcal{L}(E)$ [59, p. 139], notes that this is a closed, bounded set, and shows that the resolvent $\lambda \mapsto (\lambda 1_E - A)^{-1}$ is holomorphic on the complement of the spectrum of A . Above all, he assigns a central role to the algebra structure of $\mathcal{L}(E)$: Drawing on the work of Volterra, he developed a first version of the functional calculus for symmetric endomorphisms, which allowed him to recover simply the reduction results of Hilbert and Schmidt. If A is a symmetric element of $\mathcal{L}(E)$, Riesz shows how to define $f(A)$ for every function f semicontinuous (from below or from above) on $\mathbf{R}$ in $\mathbf{R}$, and notes that $f \mapsto f(A)$ is what we today call an algebra morphism [59, p. 129–130]. Applying this idea to the characteristic function of an interval $[\xi, +\infty[$, one obtains for every ξ a symmetric endomorphism A_{ξ} of E ; using the Stieltjes integral, Riesz then proves a version of Hilbert’s relation (3):

$$\langle Ax, y \rangle = \int_{\mathbf{R}} \xi d\langle A_{\xi} x, y \rangle$$

for $x, y \in E$, an equality that he even writes

$$A = \int_{\mathbf{R}} \xi dA_{\xi}.$$

The spectrum of A is then the set of values ξ_0 such that A_{ξ} does not remain constant in any neighborhood of ξ_0 and the point spectrum that of the points of discontinuity of $\xi \mapsto A_{\xi}$.

Shortly before concluding with the applications of the theory, Riesz notes [59, p. 146] that

its results take a particularly simple form when the endomorphism A arises from one of Hilbert's "completely continuous" bilinear forms: the spectrum reduces to the set of eigenvalues (to which 0 must possibly be added), the eigenvalues are arranged in a sequence tending to 0, and each nonzero eigenvalue has finite multiplicity (III, p. 90, prop. 5).

2. Compact operators

Riesz considerably amplifies these remarks less than three years later, in his memoir on functional equations [60] which develops, with a clarity undimmed after more than a century, the essentials of the spectral theory of compact operators on Banach spaces. The stated aim of this text is to reformulate Fredholm theory for equation (1) on a compact interval I , by considering the space $E = \mathcal{C}(I)$ equipped with the topology of uniform convergence. But as we have already indicated in an earlier note (ÉHM, p. 268), Riesz is clearly aware that his results apply to any Banach space — a notion that would only be introduced in the following decade.

Inspired by Fréchet's ideas on general topology, Riesz now considers the endomorphisms of E that map every bounded sequence to a relatively compact sequence. Like Hilbert, he still calls them "completely continuous" and notes that this new notion is equivalent, in the case of $E = \ell_{\mathbb{C}}^2(\mathbb{N})$, to the notion of complete continuity that he had used in his 1913 book ⁽⁴⁾.

Riesz studies endomorphisms of the form $B = 1_E - A$ where A is a completely continuous endomorphism of E , and derives all their spectral properties from the fact that a locally compact normed vector space must be finite-dimensional — a fact apparently discovered for the occasion. He proves easily that the kernel of B is finite-dimensional and that its image is closed and of finite codimension. As E. Weyr [87] had done it in finite dimension, he then considers the iterates B^k . He shows that the sequence of their kernels and that of their images are stationary, then introduces what we today call the nilspace and conilspace of B , efficiently showing that E is their topological direct sum. Applying this observation to endomorphisms $B_{\lambda} = 1_E - \lambda A$ for $\lambda \in \mathbb{C}$, he deduces without difficulty the spectral properties of completely continuous endomorphisms of a Banach space, and this in a more or less definitive form.

Among the main results on the spectrum of these operators, only those that would require a more mature view of the notion of function space, particularly of duality, seem to escape Riesz.

For example, in the late 1920s, T. Hildebrandt [33] and J. Schauder [63] will show (within the now well-established framework of Banach spaces) that the adjoint of a completely continuous endomorphism is completely continuous.

Moreover, the notion of Riesz complete continuity still depends on the use of sequences; Banach [4] will free himself from it in his 1932 book to consider mappings that transform every bounded set into a relatively compact set. The term “compact” linear mapping would only gradually establish itself in the second half of the 20th century, presumably following E. Hille’s book [34] on semigroups of operators.

Several questions raised in the 1930s would remain open for a long time. In this direction, we note the resolution [18] in 1973 of the “approximation problem” made famous by Banach and S. Mazur in 1932 and 1938 (see Remark 6 of III, p. 16 and Exercise 25 of III, p. 112).

At the same time, Lomonosov [40] proves the existence of nontrivial closed invariant subspaces for an endomorphism commuting with a nonzero compact endomorphism in a locally convex space (cor. 2 of III, p. 13). This result was one of the most significant advances on the problem of the existence of such a subspace for an arbitrary continuous endomorphism (the “invariant subspace problem”). This problem is still open at present in the case of continuous endomorphisms of separable Hilbert spaces; P. Enflo [19] constructed in 1987 the first example of an endomorphism of a Banach space admitting no nontrivial closed invariant subspace.

3. Fredholm index and perturbations

The notion of index appears in 1920 in a work by Fritz Noether on integral equations. In his lecture at the 1904 International Congress, Hilbert considered a problem posed by Riemann in potential theory: given a bounded open domain Ω of the plane whose boundary is a smooth simple closed curve Γ , as well as three functions a, b, f on Γ , the problem is to find a function $z : \overline{\Omega} \rightarrow \mathbb{C}$, holomorphic on Ω and continuous on $\overline{\Omega}$, satisfying

$$a\Re(z) + b\Im(z) = f \quad \text{on } \Gamma.$$

Parametrizing Γ by a real number $s \in [0, 2\pi]$, Hilbert showed that the problem reduces to

solving an integral equation “with singular kernel” for $\varphi = \mathcal{R}(z)$, namely

$$a(s)\varphi(s) + \int_0^{2\pi} K(s,t)\varphi(t)dt = f(s). \quad (4)$$

Here the kernel $K(s,t)$ takes the form $b(s) \cot\left(\frac{t-s}{2}\right) + A(s,t)$ for a continuous function A . It is therefore singular along the diagonal, and the above integral is to be taken in the sense of the Cauchy principal value.

Noether [53] observes that, unlike integral equations satisfying the Fredholm alternative, in the case of a kernel K as above and a nonzero right-hand side f , it is possible that (4) admits families of nontrivial solutions. He shows that the space of solutions is governed by the integer

$$n = \frac{1}{2\pi} \int_{\Gamma} d(\log(a - ib)). \quad (5)$$

If $n < 0$, there is no nontrivial solution, and if $n \geq 0$, equation (4) admits a family of solutions depending on $2n$ parameters. Noether uses the term “index” (Index) for the integer n and recognizes in it a winding number, a notion classical since the end of the nineteenth century in the study of functions of a complex variable.

Later works would go on to study examples of equations requiring an extension of Fredholm theory. But it would take more than twenty years before the notion of a Fredholm map was systematically extracted, at the same time as the study of perturbations by compact operators matured.

In 1909, H. Weyl [81] had shown that if two quadratic forms bounded on $\ell_C^2(\mathbf{N})$ have a completely continuous difference, then their essential spectra coincide (cf. III, p. 89, th. 2). Moreover, Riesz had noted in his 1916 memoir, naturally without the modern language, that the compact operators on a Banach space E form a two-sided ideal of $\mathcal{L}(E)$. It seems, however, that the path indicated by these two remarks was not explored until the 1940s. In 1941, J. Calkin, motivated by the work of J. von Neumann on operator algebras (which we shall have occasion to discuss shortly, see p. 537), pointed out the interest of studying congruences modulo this ideal in the framework of Hilbert spaces [10]. He shows how the structure of the algebra that now bears his name (cf. III, p. 75) makes it possible to recover Weyl’s result simply.

Meanwhile, around 1942, J. Dieudonné [15] became interested in the idea of studying perturbations of a continuous linear map u between normed spaces, in the case where the perturbation is “small” in the sense of the operator norm. He restricts himself to the case where u is a strict morphism whose kernel is finite-dimensional or of finite codimension; he proves that the same then holds for any “small perturbation” of u , and shows that in the case of Fredholm maps (cf. n°2 from III, p. 40), the index is locally constant (th. 1 of III, p. 58), as are several of the results presented in § 4 of III, p. 55.

These two ideas converge around 1950, when F. Atkinson [3], I. Gohberg [26], S. Mikhlin [46], and B. Yood [93] study perturbations by compact operators. They explicitly make the connection with the Noether index and bring out the notions of Fredholm map [3, p. 8] and of Riesz map [93, §5]. The latter are the subject of systematic research in the following decade, where the results presented in Chapter III take on more or less their current form.

The usual framework for these theories is that of Banach spaces; however, various applications motivate their generalization to broader classes of topological vector spaces. Thus, the case of Fréchet spaces appears naturally in 1954 in works of H. Cartan and J-P. Serre [13] which demonstrate the finiteness of the cohomology of a compact complex analytic manifold with coefficients in a coherent sheaf, the proof invoking deformation results proved by L. Schwartz [66] (cf. th. 2 of III, p. 73).

The invariance of the index under deformation is perfectly illuminated, in the case of formula (5), if one observes that the right-hand side (winding number) is invariant under homotopy. Such topological expressions of the index would soon become famous for Fredholm applications arising from differential geometry. If D is a differential operator on a compact C^∞ manifold X (cf. VAR, §14), and if D is “elliptic” (cf. [2, §1]), then an extension of D to suitable Sobolev spaces defines a Fredholm map. The search for a formula giving a topological expression for its index using the “characteristic classes” of X , suggested among others by the work of I. Gelfand around 1960 [23, vol. I, p. 65], will lead in 1963 to the “index theorem” of M. Atiyah and I. Singer [2]. This latter would give rise to extraordinary developments at the confluence of analysis, topology, and differential geometry. We cannot describe here the vast fields they opened, which, still very fertile today, have not ceased to draw nourishment from Fredholm theory.

4. Partial operators and the spectral theorem

General Hilbertian spectral theory was created in large part to address the problem of the mathematical foundations of quantum mechanics, notably under the impetus of von Neumann.

We refer to the work of M. Jammer [35] for a detailed description of the development of quantum theory up to the fundamental paper by W. Heisenberg that introduced quantum mechanics [28]. Less than five years elapsed between its publication in the summer of 1924 and that of the article [50] in 1929, where von Neumann presents in a perfectly lucid and rigorous manner all the fundamental results of the theory of partial (often called "unbounded") Hermitian operators on Hilbert spaces.

By giving the first axiomatic presentation of Hilbert spaces [49], von Neumann crowned an idea long in gestation (cf. ÉHM, p. 267), and illuminated various isomorphisms between spaces of sequences or functions well known to Schmidt, Fréchet, Fischer, and Riesz before 1910. This rise in abstraction allowed von Neumann to make considerable conceptual advances.

Such is the case with the idea of a partial operator, which seems to hover, still formless, over several of the works that, after Hilbert, took up Fredholm equations again (see above, no. 1). Hilbert had noticed that many differential equations of Sturm–Liouville type, in particular in the case of an unbounded interval, could be reduced to Fredholm equations; but to apply the integral methods, it was necessary to weaken the conditions imposed on the kernel and to leave the framework in which the results of Fredholm, Hilbert, and Schmidt could be applied. Over the years, more and more singular kernels were studied, until equations were considered for which the left-hand side makes sense only if the unknown belongs to a subspace of $L^2(I)$. In this direction, one must mention works of E. Hilb [30] in 1908, of H. Weyl on Sturm–Liouville theory [82] in 1910, and above all of T. Carleman [11] in 1923, in a very general framework.

By unifying these problems, von Neumann cast entirely new light on these classical works. He emphasized in particular the fundamental distinction between symmetric and self-adjoint operators, and interpreted the self-adjoint extensions of a symmetric operator in terms of abstract "boundary conditions," generalizing the well-known boundary conditions that appeared, for example, in Dirichlet and Neumann problems for the Laplace operator on a bounded open set of R^n .

At the same time, von Neumann [51] brought to the fore the notion of normal operator ⁽⁵⁾ as the natural framework for spectral theory, insisting on the role of the commutative algebra generated by the operator and its adjoint.

Relying on this precise mathematical formalism, which also revealed the equivalence between the viewpoints of Heisenberg and Schrödinger, and on the Copenhagen interpretation of experimental procedures and results, non-relativistic quantum mechanics from then on reached a state of refinement that has hardly evolved since; the validity of this physical theory, despite its surprising, even paradoxical-looking consequences, has since been constantly confirmed by experiment, and with remarkable precision.

From a formal point of view, the foundations of spectral theory were laid by von Neumann. In parallel, after the presentation of the latter's first ideas, M. Stone had pursued similar research between 1928 and 1930 and obtained closely related results. The clear and complete presentation of the known results that Stone published in 1932 [72] had an important influence on the diffusion of Hilbertian spectral theory.

Nevertheless, other improvements in the presentation have been important for the diffusion of these results. Indeed, the spectral theorem (cf. th. 1 of IV, p. 266), as presented by von Neumann, was long considered very difficult, largely because it was stated in the framework of vector-valued measures.

Later, P. Halmos [27], by emphasizing the interpretation of the spectral theorem by means of multiplication operators on L^2 spaces, revealed its essentially elementary aspect in the case of bounded operators. Quite recently, S. Woronowicz (in the somewhat different framework of regular operators on Hilbert modules over C^* -algebras [92]) brought a simplification by introducing

the "bornification" (cf. no. 2 of IV, p. 265), which allows one to treat normal partial operators as simply as self-adjoint operators.

Moreover, if D is a formally symmetric scalar differential operator on an open set U of $\mathbb{R}^n$, the study of the self-adjoint extensions of D to $L^2(U)$ is naturally linked to the study of distributions on U (in the case of the Laplacian, cf. no. 6 of IV, p. 242). We refer to the work of J. Dieudonné [16, ch. VII] for the history of the latter — a sinuous history that unravels shortly

after the period we have just evoked, with decisive contributions of S. Sobolev [69] in 1936, and of L. Schwartz [65] ten years later.

From a purely mathematical point of view, spectral theory interacts in a particularly fruitful way with Riemannian geometry. Thus, taking up a question of S. Bochner, M. Kac gave in 1966 a strikingly vivid presentation [36] of the problem of determining which geometric properties of a Riemannian manifold can be deduced from the spectral properties of the Laplace operator. This question had been anticipated by the physicist A. Schuster [64] in 1882: To find out the different tunes sent out by a vibrating system is a problem which may or may not be solvable in certain special cases, but it would baffle the most skillful mathematician to solve the inverse problem and to find out the shape of a bell by means of the sounds which it is capable of giving out. (“Determining the harmonics emitted by a vibrating system is a problem which may or may not be solvable in certain special cases, but the most skillful mathematician would be baffled if he had to solve the inverse problem and determine the shape of a bell from the sounds it can emit”).

5. Connection between harmonic analysis and group theory

We have described, in the historical notes of the Book on Integration (cf. ÉHM, p. 275), how questions related to the heat equation led Fourier to the revolutionary conception of representing an arbitrary function as a sum of trigonometric functions.

We cannot trace here the development of the theory of Fourier series and integrals throughout the nineteenth century, nor the profound influence that the questions raised by this theory had on the evolution of Analysis — to the point of overturning the conception of real numbers, contributing to the birth of set theory (cf. ÉHM, p. 42). But in order to understand the form of the ideas set forth in this Book, it is appropriate to describe the manner in which harmonic analysis was connected after 1925 to group theory and to Hilbertian spectral theory.

Recognition of the link between Fourier’s ideas and the group structures of $\mathbb{R}/\mathbb{Z}$ and $\mathbb{R}$, as well as the generalization of these ideas to other groups, came rather late.

One can naturally, in the early stages of the theory of finite groups and their characters, identify results that today appear to contain the rudiments of finite Fourier analysis. For example,

the dual of the group of invertible elements of $\mathbb{Z}/q\mathbb{Z}$, as well as the orthogonality relation expressing the characteristic function of an element as a linear combination of characters, appear implicitly in 1837 in Dirichlet's article [17] proving that there are infinitely many prime numbers $p \equiv a \pmod q$ if a is relatively prime to q . He relies on ideas of Gauss, who had introduced the term “character” in his study of binary quadratic forms with integer coefficients and fixed discriminant (cf. [22, § 230]).

But neither Gauss nor Dirichlet uses the language of groups. It was Dedekind, presenting these works in 1879, who recognized their latent role and defined the notion of character for commutative groups. Dedekind's remarks were considerably amplified by H. Weber, first in 1882 [77], then in 1886 [78, p. 112], where he introduced the dual group of a finite abelian group A and noted that the latter is isomorphic to A . His purpose was then to construct abelian extensions of the field $\mathbb{Q}$ with given Galois group, and he did not consider the problem of harmonic analysis on such a group.

In the case of noncommutative groups, the foundations of the theory of representations of finite groups were solidly established around the turn of the twentieth century, following the works of G. Frobenius, W. Burnside, and I. Schur (cf. ÉHM, p. 154). After 1905, it was clear that the orthogonality relations of characters played a crucial role in the organization of the theory. But their kinship with harmonic analysis remained hidden.

It is to Hermann Weyl that we owe the connection between these fields, when he conceived the idea of a general theory of representations for compact groups. In several major texts published between 1925 and 1927, he recognized the links that would unite such a theory with Fourier analysis and Hilbertian spectral theory, and he laid the foundations for many later discoveries.

It was a letter from Schur that set Weyl on this path, thanks to their common interest in invariant theory. We described, in the note on Haar measure (ÉHM, pp. 289–291), how A. Hurwitz had conceived as early as 1897 the idea of using an “invariant integration” to determine the polynomials on R^n invariants under the orthogonal group. It seems that Schur was not aware of Hurwitz's results before 1924, when he suddenly understood how invariant integration could be used to extend to the group $\mathbf{O}(n)$ character theory and the orthogonality relations he had

established for finite groups. Weyl immediately saw that Schur's methods, combined with the work of Élie Cartan, made it possible to construct the irreducible representations of compact semisimple Lie groups. The series of three memoirs in which Weyl blended the ideas of Cartan and Schur, published in 1925 [83], already contained the essential results of LIE, IX, §6–7; cf. ÉHM, p. 328–330.

But it was the following year, in an equally famous text written with his student F. Peter [54], that Weyl brought to light the connections between the representation theory of compact groups, Fourier analysis, and Hilbertian spectral theory. This text of fewer than twenty pages contains in embryo a great many future developments.

Peter and Weyl free themselves of any algebraic hypothesis on the structure of the group under study, assuming only that G is a compact topological group equipped with an invariant measure. The existence of such a measure was clear for connected Lie groups; soon A. Haar would establish it in general, partly motivated by the results presented here (cf. ÉHM, p. 291).

The starting point of their study is the orthogonality of matrix coefficients. Schur had observed that by choosing a representative $\pi \in \widehat{G}$ for each equivalence class of irreducible representations, a G -invariant inner product on the space E of π , and an orthonormal basis of E , one obtains a family of matrix coefficients that is orthonormal in $L^2(G)$. Peter and Weyl intend to show that such a family is complete.

In the case of finite groups, the analogous result had been proved by Frobenius by studying the group algebra $C[G]$. Peter and Weyl then consider the space $\mathcal{C}(G)$ of continuous functions on G and equip it with the convolution product. This appears to be the first use of the convolution product as an abstract operation in explicit connection with the group structure (cf. ÉHM, p. 295). Peter and Weyl moreover call *Gruppenzahl* ("group number") an element of the algebra $\mathcal{C}(G)$, and denote by xy the (convolution) product of two *Gruppenzahlen*. They also equip $\mathcal{C}(G)$ with the involution $f \mapsto \tilde{f}$, where $\tilde{f}(g) = \overline{f(g^{-1})}$ for $g \in G$.

Having thus introduced the involutive algebra $\mathcal{C}(G)$, the first observation made by Peter and Weyl is that every unitary representation π of G gives rise to a representation $f \mapsto \pi(f)$ of $\mathcal{C}(G)$ (cf. V, p. 400). The notion of Hermitian element of $\mathcal{C}(G)$ is explicitly defined, as is (implicitly) that of positive element, which opens the way to the application of Hilbert space

techniques.

Peter and Weyl no longer hesitate to call the operator $\pi(f)$ the "Fourier coefficient" of f , and continue the parallel with Fourier theory by noting that Schur's orthogonality relations imply Bessel's inequality

$$\sum_{\pi \in \widehat{G}} \dim(\pi) \operatorname{Tr}(\pi(f) \pi(f)^*) \leq (f * \tilde{f})(e) = \|f\|_2^2. \quad (6)$$

The Hilbert–Schmidt spectral theory then allows them to show that this is an equality ("Plancherel formula"). To every element φ of $\mathcal{C}(G)$, Peter and Weyl associate the continuous kernel $K_\varphi: G \times G \rightarrow \mathbb{C}$ defined by $K_\varphi(x, y) = \varphi(xy^{-1})$. They then observe that if φ is of the form $f * \tilde{f}$ with $f \in \mathcal{C}(G)$, ⁽⁶⁾ then K_φ is a positive Hermitian kernel in the sense of Hilbert–Schmidt theory. A variant of a Schmidt algorithm ⁽⁷⁾ then allows them to construct an orthonormal basis of eigenfunctions of the kernel K_φ , and then to reduce the proof of the equality in (6) to the fact that the trace of the operator defined by K_φ is equal to the sum of its eigenvalues.

Naturally, only the representations π satisfying $\pi(f) \neq 0$ can occur in (6); the spectral theory for K_φ shows the existence of such a representation if f is not identically zero. Peter and Weyl easily deduce from this that the irreducible representations separate the points of G — a special case of the Gelfand–Raikov theorem, which will be discussed later.

In the celebrated works of Frobenius, an equality analogous to (6) made it possible to show that the regular representation of G contains all irreducible representations. But this involved applying this equality by taking for f the identity element of $C[G]$; one then obviously had $\pi(f) \neq 0$ for all π . According to Peter and Weyl, the absence of an identity element for convolution explains many difficulties in their proof; they nevertheless observe, borrowing from the theory of Fourier series an idea destined for a great future, that one can deduce the analogue of Frobenius's result by applying (6) to a sequence of functions forming an approximate identity for convolution (cf. no. 10 of I, p. 120).

Likewise, starting from the equality arising from (6), the polarized version

$$\sum_{\pi \in \widehat{G}} \dim(\pi) \operatorname{Tr}(\pi(x)\pi(y)) = (x * y)(e) \quad (7)$$

makes it possible to obtain, by taking for y an approximate identity for convolution, a uniform approximation of every element of $\mathcal{C}(G)$ by linear combinations of matrix coefficients of irreducible representations. These methods had just enabled Weyl to give a new proof [84] of the fundamental result of the theory of "almost periodic functions" of H. Bohr, to which we shall return shortly. Peter and Weyl point out that one can read therein the first analytic application of the theory of representations of a noncompact group, here that of the translations of $\mathbf{R}$.

The young quantum theory very quickly provided Weyl with the opportunity to return to the representations of R . At the end of 1927, he noted the interest of groups (finite or not) and of representations for clarifying the foundations of this theory [85]. He would return to this the following year, in a resounding book [86].

In his 1927 article, Weyl observes that if u is a continuous self-adjoint operator on a Hilbert space, then $t \mapsto e^{itu}$ is a unitary representation of R . But there exist unitary representations of R that are not of this form; taking up von Neumann's idea on the role of symmetric partial operators in representing observable physical quantities, Weyl suggests that every unitary representation of R is of the form $t \mapsto e^{itu}$ where u is a partial self-adjoint operator on a Hilbert space. This point of view allows him to reduce the analysis of the "canonical relations" of Heisenberg and Schrödinger, already studied in detail by von Neumann, to that of the connections between two unitary representations of R on the same Hilbert space. Stone [73] shows in 1932 the result hoped for by Weyl (V, p. 428, th. 1), in the wake of his study of the spectral properties of self-adjoint operators.

6. Locally compact commutative groups

The interaction between harmonic analysis and group representations was thus firmly established by the early 1930s. It was greatly strengthened in the following decade, with each of the two subjects prompting rapid progress in the other.

A very fertile ground for this interaction was the study of almost periodic functions, intro-

duced by H. Bohr in 1924 in the case of functions of a real variable [9]. By one of the fundamental theorems of the latter, these are the bounded continuous functions that are uniform limits on $\mathbf{R}$ of linear combinations of functions $x \mapsto e^{i\lambda x}$, $\lambda \in \mathbf{R}$ (cf. II, p. 292, exerc. 54). Bochner [6] had proved in 1926 that a function $f \in \mathcal{C}_b(\mathbf{R})$ is almost periodic if and only if its translates $x \mapsto f(x - a)$, for $a \in \mathbf{R}$, form a relatively compact subset of $\mathcal{C}_b(\mathbf{R})$ for the topology of uniform convergence. Bochner and von Neumann observed that if G is a topological group, this characterization visibly provides a notion of almost periodic function (on the right or on the left) on G . In the early 1930s, the theory promised to develop rapidly, in particular under the impetus of von Neumann [52] and the methods of Peter and Weyl.

Another fertile area appears on the periphery of N. Wiener's work. He introduced, in a famous paper [88] from 1932, algebraic methods in the study of so-called Tauberian problems—that is, in the study of the asymptotic behavior of a function (or of a sequence) along a filter, given information concerning the behavior of certain weighted means. His approach was motivated by an idea of R. Schmidt.

One of the auxiliary results of Wiener's work ([88, Lemma IIe]) will particularly strike the minds and inspire numerous developments: if f is a periodic function whose Fourier series is absolutely convergent, and if f does not vanish, then the Fourier series of $1/f$ is absolutely convergent (cf. I, p. 38, example). As early as 1932, R. Paley and N. Wiener [89] sketched the link between this convergence result, the notion of dual group for a discrete commutative group, and the theory of almost periodic functions.

Against this analytic backdrop, it is algebraic topology that leads L. Pontryagin to give the decisive impetus for the study of characters of locally compact abelian groups. He announces as early as 1932 [55] the interest of the notion of the character group lies in placing in duality the homology groups of a compact subset of Euclidean space and those of its complement. In a paper [58] published in 1934, he shows that the character group of a discrete group is compact, and then establishes the duality theorems in this framework. Although Pontryagin uses the results of Peter and Weyl, as well as the Haar measure whose existence had just been established in general, he aims mainly at applications to topology [57] and is not explicitly concerned with harmonic analysis ⁽⁸⁾: the duality theorems are proved by a fine study of the structure of compact abelian

groups.

Pontryagin's work immediately aroused great interest. On a suggestion of von Neumann, E. van Kampen extended the duality the following year to all locally compact commutative groups [37]. Shortly thereafter, he gives a spectacular application of the new invariant analysis to the study of almost periodic functions [38], discovered independently by A. Weil during the same period [79]: If G is a locally compact abelian group, one obtains a compact group $\tilde{G}$ beginning by endowing the dual group $\widehat{G}$ with the discrete topology, and then defining $\tilde{G}$ as the compact dual group of the discrete group thus obtained. The group G is then canonically embedded in $\tilde{G}$ (cf. II, p. 292, exerc. 54), and a simple application of the Peter–Weyl theory for continuous functions on $\tilde{G}$ yields all the results known at the time about almost periodic functions on G .

At the same time, functions of positive type, well known in classical analysis, appear at this period in the study of unitary representations of R . The notion of a universally positive kernel had been studied by J. Mercer [45] shortly before 1910, in connection with the theory of integral equations, while its analogue for sequences gave rise to a considerable number of works during the same period, related in particular to the moment problem (cf. IV, p. 359, exerc. 21). Meanwhile, functions of positive type on R , already studied systematically by M. Mathias [43] in 1923, had become, in Bochner's hands, one of the fundamental tools of Fourier analysis and of probability theory [7]. In 1933, Bochner [8] and Riesz [61] independently observed that every diagonal matrix coefficient of a unitary representation of R is a function of positive type on R . This observation enabled them to give a simpler proof of the Stone theorem, and was to have a decisive influence on the development of the general theory (cf. n° 7).

The results of this period were organized by A. Weil in a book completed before 1937 and published in 1940 [80]. There he sets out in detail the results of Peter–Weyl, Pontryagin, and van Kampen. But in the case of commutative groups, whereas Pontryagin and van Kampen emphasized the structure of groups and duality theorems, Weil systematically develops harmonic analysis, introducing the Fourier transform and proving the Plancherel formula and Bochner's theorem (V, p. 455, th. 5). He insists in particular on the role of convolution (under the name "composition product") and on that of functions of positive type, now defined on an arbitrary group and related to the diagonal matrix coefficients of representations, at least in the case of

finite-dimensional representations.

The generality thus achieved in the commutative case would open, a little later, new horizons in number theory. As early as 1936, with a view to introducing algebraic methods into class field theory, C. Chevalley [14] introduced the group of idèles of a number field, which would later appear as the group of invertible elements of the ring of adèles (cf. AC, VII, p. 221–222, and ÉHM, p. 143). In 1950, J. Tate [75] would show that harmonic analysis for groups of adèles and idèles makes it possible to recover easily the functional equations of Hecke L-functions, foreshadowing extraordinary developments linking harmonic analysis on adelic groups to the study of automorphic forms.

But one can say that from the end of the 1930s, the principal results of invariant harmonic analysis had been established for compact groups and for locally compact commutative groups. In the commutative case, Weil's proofs still depend (like those of Pontryagin and van Kampen) on a fine knowledge of the structure of the group, that is, roughly speaking, on the classification results of § 2 of II, p. 244. The following decade would show how to dispense with this, thanks to the new methods of the Gelfand school (cf. H. Cartan and R. Godement [12]).

7. Operator algebras

We have seen the role that Riesz, in expounding the work of the Hilbert school in 1913, assigned to the algebra $\mathcal{L}(E)$ of endomorphisms of a Hilbert space, as well as to holomorphic functional calculus and the abstract notion of spectrum. In his 1916 memoir, he already seems to announce normed algebras as the natural framework for spectral theory; after the work of Banach and his school during the decade

of the 1920s, it is hardly surprising that this idea took a central place. The notion of Banach algebra was introduced by M. Nagumo [48] in 1936, while S. Mazur [44], in 1938, showed that the only normed algebra over $\mathbb{C}$ which is a field is $\mathbb{C}$ itself (I, p. 26, cor. 2).

Toward the middle of the 1930s, in connection with spectral theory, opportunities multiplied for considering algebras of operators on Hilbert spaces abstractly. F. Murray and J. von Neumann, in a series of profound and influential works [47], systematically studied the involutive subalgebras of $\mathcal{L}(E)$ equal to their bicommutant (“von Neumann algebras”). On the other hand,

in 1936, S. Steen proposed introducing an axiomatic notion of “algebra of operators” and studying the associated structures in depth [70]. Moreover, Stone [74], motivated by the study of spectral projectors, studied in 1937 the unital algebras all of whose elements are idempotent (“Boolean algebras”: cf. ÉHM, p. 146). He noted that if X is a locally compact topological space, the characteristic functions of open sets and those of complements of everywhere dense sets generate, in $\mathcal{C}(X)$, a Boolean algebra A . He then proved that the ideals of A correspond naturally to the closed subsets of the space X , thereby introducing an idea that would prove particularly fruitful.

Beginning in 1939, Gelfand’s viewpoint opened new perspectives on these relatively scattered results. His work seems to have been motivated in part by Wiener’s ideas, in particular by his theorem on absolutely convergent Fourier series, as well as by the methods of Peter and Weyl. Between 1939 and 1946, in collaboration with M. Naimark, D. Raikov, and G. Shilov, Gelfand established the foundations of the theory of commutative Banach algebras and of the theory of stellar algebras (cf. [23], vol. I, p. 169–400). In particular, he gave the classical proof of Wiener’s theorem — generalized by P. Lévy [39] — which is found in this book (I, p. 38, example).

In Gelfand’s approach, the essential object associated with a commutative Banach algebra A is the space X of maximal ideals of A . Indeed, via the Gelfand–Mazur theorem (I, p. 26, cor. 2), every element of A defines a function on X . It is well known how important this viewpoint would become in the subsequent evolution of algebraic geometry (cf. ÉHM, p. 146–148). This approach was itself motivated by Stone’s work on Boolean algebras mentioned above.

It seems that this is L. Loomis, in a course published in 1953 [41, p. 53], who first presented Gelfand theory with emphasis on the space of characters of A , defining the Gelfand transform in the sense in which we understand it in this book. The perhaps clearest advantage of this viewpoint is that the topological properties of the space of characters follow immediately from the compactness of the unit ball of the dual of a Banach space in the weak topology.

From his earliest works, Gelfand had constructed the holomorphic functional calculus in one variable in a Banach algebra by means of the Cauchy integral, and had deduced from it that an involutive Banach algebra admits a nontrivial decomposition as a product of rings when the

space of its maximal ideals is not connected. The case of an arbitrary Banach algebra was treated only in 1953, when Shilov developed a form of the functional calculus in several variables. [68].

As for star algebras, which Gelfand and Naimark had shown as early as 1942 to be realizable as algebras of operators on a Hilbert space (cf. V, p. 442, th. 2), they soon became the object of abundant and profound studies, spurred on by applications to group representations, by the work of Murray and von Neumann, and by the mathematical formalization of quantum mechanics. Let us mention in particular the contributions of I. Segal, who was one of the pioneers of the systematic study of representations of star algebras, in connection with the study of the unitary dual of locally compact groups, which we shall discuss in a few lines.

Although Gelfand, Naimark, Raikov, and Shilov had laid out the essential foundations of the theory of C^* -algebras before 1946, certain technical points were improved later. Thus, R. Arens [1] showed how to avoid invoking the existence of the “Shilov boundary” (cf. I, p. 171, exerc. 26) in order to prove that the Gelfand transform is an involutive morphism. Moreover, the Gelfand–Naimark theory for unital C^* -algebras initially added as an axiom the fact that $1 + x^*x$ is invertible for all x . They conjectured, however, that this axiom was unnecessary; the proof of this assertion, elementary but delicate, was only provided around 1952, following the work of M. Fukamiya [21] taken up by I. Kaplansky [62].

8. Representations of locally compact groups

Until 1939, the study of representations of topological groups was limited to the compact case and the commutative case. Physicists, for their part, had found in quantum mechanics many reasons to study irreducible unitary representations in order to seek Hilbert spaces capable of modeling the states of elementary particles. P. Dirac, building on work by E. Majorana, had pointed out around 1936 the interest, in this perspective, of the Lorentz group $SO(3,1)$ and the Poincaré group $SO(3,1) \times R^4$. Their methods remained far removed from those of the mathematicians of the time. But in 1939, E. Wigner [90] opened a breach when he classified the irreducible unitary representations of the Poincaré group, building on the work of Murray and von Neumann.

It is to Gelfand that we owe, it seems, having glimpsed a general path toward the study of

representations of locally compact groups. In a memoir written in 1942 (cf. [23], vol. II, p. 3–17), he notes with Raikov that such a theory is made possible by the link between unitary representations and functions of positive type. They prove the existence of a Hilbertian realization for every function of positive type (V, p. 432, th. 1) and deduce from it that irreducible unitary representations separate the points of G (V, p. 454, th. 4). The positive linear forms on the involutive algebra $L^1(G)$ play an essential role: Gelfand and Raikov show the link that connects them to functions of positive type (cf. V, p. 448, prop. 13). The enveloping C^* -algebra of an involutive algebra is presented (without a name) in [24, § 48], although it is not yet directly a question of associating with G the algebra $\text{Stell}(G)$ (cf. 9, def. of I, p. 125).

After 1945, the study of unitary representations developed by leaps and bounds. The interaction with C^* -algebras, as well as with the analysis of the foundations of quantum theory, quickly provided valuable general results: around 1947, Segal associated with a locally compact group various ⁽⁹⁾ stellar algebras [67] and wove the link between their representations and those of the group under consideration. Shortly thereafter, G. Mackey extracts the notion of induced representation [42] which will be omnipresent in the concrete results on unitary representations.

At the same time, the theory is considerably stimulated by the study of examples that reveal its variety and depth. Gelfand and Naimark, studying the example of $SL(2, \mathbb{C})$, isolated in 1947 a special case of the notion of induced representation, then showed that it is the key to the study of the representations of this group [23, vol. II, p. 41–124]. In the same year, V. Bargmann [5] studied the group $SL(2, \mathbb{R})$ and discovered the existence of a countable family of square-integrable representations, called the "discrete series," as well as the orthogonality relations between their matrix coefficients (cf. prop. 8 of V, p. 424). As Godement immediately realized [25], the Bargmann orthogonality relations are satisfied by all square-integrable representations of locally compact groups.

Soon the study of particular classes of groups would again, and for a long time, take precedence over abstract theory. Harish-Chandra, in particular, would undertake the general study of representations of reductive Lie groups — an epic task that we cannot describe here, any more than the prodigious consequences that Langlands would foresee for number theory.

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INDEX OF NOTATIONS

$\mathcal{L}^c(E; F)$ (compact linear maps) 2 $E_{(C)}$ (complexified vector space) 2 $u_{(C)}$ (complexified linear map) 3 $\mathcal{L}^f(E; F)$ (finite-rank linear maps) 4 $\mathcal{C}_0(X; K)$, $\mathcal{C}_0(X)$ (continuous functions tending to 0 at infinity) 20 $\mathcal{N}^{p,q}(X \times Y)$ (kernel space) 27 u_k (integral operator with kernel) 29 $k_\circ$ (map $y \mapsto k_y$ where $k_y(x) = k(x, y)$) 30 $\mathcal{L}^{p',q'}(X, Y, \mu, \nu)$ (kernel space) 31 $u \equiv v$ (congruence modulo $\mathcal{L}^f(E; F)$) 40 $\mathcal{F}(E; F)$ (Fredholm maps) 41
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 $((u))$ (conorm of u) 61 $\mathcal{M}(E; F)$ (strict injective morphisms) 67 $\mathcal{QM}(E; F)$ (strict morphisms with finite-dimensional kernel) 67 $\mathcal{E}(E; F)$ (continuous surjective morphisms) 70 $\mathcal{QE}(E; F)$ (morphisms whose image has finite codimension) 70 $\mathcal{Calk}(E)$ (Calkin algebra of a Banach space E) 75 N_λ (nilspace) 78 I_λ (conilspace) 78 $e_\lambda(u)$ (spectral projector) 82 $m_\lambda(u)$ (spectral multiplicity) 82 $\text{Sp}_s(u)$ (sensitive points of the spectrum) 83 $\text{Sp}_e(u)$ (essential spectrum) 83 $\text{Sp}_n(u)$ (elements of the spectrum such that $u - \lambda 1_E$ has index n) 86 $\text{Sp}_\omega(u)$ (complement of the union of the $\text{Sp}_n(u)$) 86 $\mathcal{D}_B(E)$ (endomorphisms diagonal in the basis B) 146 I_E (dimensions of finite-dimensional subspaces $F \neq E$) 151 $\lambda_n(u)$ (decreasing sequence of eigenvalues of u) 152
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 222 $W_0^{k,p}(U)$ (closure of $\mathcal{D}(U)$ in $W^{k,p}(U)$) 222 $H^k(U)$ (Sobolev space of index
 k and exponent 2) 222 $H_0^k(U)$ (closure of $\mathcal{D}(U)$ in $H^k(U)$) 222 $\mathcal{P}(E; F)$ (partial
 operators from E to F) 224 $\mathcal{P}(E)$ (partial operators on E) 224 $\text{dom}(u)$ (domain of u
) 224 1_D (identical partial operator with domain D) 225 $u \circ v$ (composition of partial
 operators u and v) 225 $u \subset v$ (v is an extension of u) 226 $(u_i)_{i \in I}$ (product partial
 operator) 226 $u + v$ (sums of partial operators u and v) 226 $\overline{u}$ (closure of u) 228
 E_u (normed space defined by the domain of u) 230 $\|x\|_u$ (norm on the space E_u) 230
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 242 D_- (differential operator with domain $\mathcal{D}(U)$) 242 D_+ (differential operator with
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SELF-ADJOINTNESS CRITERIA

We summarize here the criteria stated in the text that allow one to prove that certain partial operators are self-adjoint.

We fix a complex Hilbert space E and a partial operator u on E . This operator is self-adjoint in each of the following cases:

- (i) There exists $v \in \mathcal{L}(E)$ injective and Hermitian such that $u = v^{-1}$. (Proposition 9 of IV, p. 239);
- (ii) The operator u is symmetric and $\text{dom}(u) = E$ (Proposition 10 of IV, p. 240);
- (iii) The operator u is symmetric and induces a bijection of $\text{dom}(u)$ onto E (corollary of Proposition 10 of IV, p. 240);
- (iv) There exists $\lambda \in \mathbb{C}$ such that $u + \lambda 1_E$ and $u + \bar{\lambda} 1_E$ are surjective (Proposition 11 of IV, p. 240);
- (v) There exists a partial operator v , closed with dense domain, such that $u = v^* \circ v$ (Proposition 12 of IV, p. 241);
- (vi) The operator u is symmetric, closed, and its deficiency index is $(0, 0)$ (corollary of Proposition 26 of IV, p. 257), that is, $\text{Ker}(u^* - i) = \text{Ker}(u^* + i) = \{0\}$;
- (vii) The space E is of the form $L^2(X, \mu)$, where X is a locally compact topological space and μ is a measure on X , and u is the operator of multiplication by a function μ -measurable on X which is locally μ -almost everywhere real-valued (corollary of Proposition 23);
- (viii) There exists a normal partial operator v on E and a universally measurable function $f \in$

$\mathcal{L}_u(\text{Sp}(v))$ with real values such that $u = f(v)$ (Corollary 1 of IV, p. 273);

- (ix) There exists a partial Hermitian form q on E such that u is the partial operator representing q (Proposition 14 of IV, p. 293);
- (x) One has $u = v + w$ where v is a self-adjoint partial operator and w is a symmetric partial operator bounded relative to v satisfying $\|w\|_v < 1$ (Theorem 5 of IV, p. 306);
- (xi) The operator u is the infinitesimal generator of a unitary representation ϱ of $\mathbf{R}$ in E (Theorem 1 of V, p. 428).